OSTERWALDER–SCHRADER AXIOMS FOR THREE-DIMENSIONAL SCALAR QUANTUM ELECTRODYNAMICS AT FIXED PHOTON MASS

Abstract. King constructed the continuum, infinite-volume limit of three-dimensional scalar quantum electrodynamics, the $U(1)$ (abelian) Higgs model, along Neumann–Dirichlet cubes at block factor 5 [15, 16]. For his printed state at fixed photon mass we prove the Osterwalder–Schrader axioms: regularity E0″ with absolute constants, invariance under the full Euclidean group, reflection positivity about every plane, symmetry, and the measure with its ergodic decomposition, on whose components the cluster axiom holds in all its forms. The existence of King's cube limits, the existence clause of his Theorem 2.1, is proved from his renormalisation-group analysis in a ratio form in which the vacuum energies cancel; rotation invariance is proved by comparing King's cubes with periodic tori carrying a smooth mass wall on two lattices whose orientations differ by arctan(3/4). With real scalar sources the same holds jointly for the curvature and the renormalised Wick square on a tube of small Wick amplitudes, without a Euclidean clause. The curvature results hold on King's whole coupling range; the Wick results hold under a one-sided sign clause of the Wick leg, which we prove on explicit coupling regions. The limit of vanishing photon mass is left to later work. Complete proofs of all technical estimates are given in the supplemental material.

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1. Introduction and main results

1.1. **King's construction.** King [15, 16] constructed three-dimensional scalar quantum electrodynamics with a massive photon (the $U(1)$ Higgs model in King's terminology) as a continuum and infinite-volume limit of lattice models. The lattice theory lives on $\Lambda_k = \varepsilon_k \mathbb{Z}^3$ with spacings $\varepsilon_k = 5^{-k}$; its action has a photon A in Feynman form with mass $\mu_0 > 0$, an N-component

scalar ϕ with quartic coupling $\lambda > 0$ and bare mass $m > 0$, minimally coupled with charge $e > 0$, and one mass counterterm $\delta m^2_{\varepsilon_k}$. The counterterm is King's free-boundary one: it is computed with infinite-lattice propagators, so it is one number per spacing, the same at every site of every box and every torus. The ultraviolet problem is settled in [15] with Bałaban's renormalisation group [1, 2, 3, 4] at block factor 5: on a finite volume the effective actions converge as $k \to \infty$. The infinite-volume problem is treated in [16] with the correlation inequalities of Brydges, Fröhlich and Seiler [7]. King's finite-volume states are Neumann–Dirichlet states on the cubes $C_J = [-MJ, MJ)^3$, $M \geq 1$ his block integer: the photon gradient terms are kept only inside the cube (Neumann), and the scalar field vanishes on a boundary layer (Dirichlet). The limit is iterated, first $k \to \infty$ at fixed J, then $J \to \infty$; the second limit exists because the cube states are monotone in J on a cone of observables, by [7].

This paper proves the Osterwalder–Schrader axioms for King's printed state at fixed photon mass, with full Euclidean invariance, and the corresponding statements for the joint field of the curvature and the renormalised Wick square on a tube of small Wick amplitudes. It is self-contained in the following sense. The existence of King's cube limits, the existence clause of his Theorem 2.1 [16], is proved in the existence part of the paper (Section 5, and Section 6 for the version with scalar sources): King's route, the comparison of two spacings at a common block scale, is run in a ratio form in which the vacuum energies cancel, so that the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube, is not needed. The passage from the cube limits to the axioms in [16] is asserted at three places, and at each we proceed differently. First, King's Neumann–Dirichlet cube form is not reflection symmetric about any plane, and its natural symmetrisation is not reflection positive (an exact counterexample at $e = 0$, Remark 9.7), so his reflection positivity goes through a modified box state; that this box state is reflection positive is his Proposition 3.3, which he derives from [7, Cor. 5.4], whose hypotheses do not cover the finite-box Gaussian of the modified action. Proposition 3.3 is true, and it is proved here (Lemma 9.3) rather than derived from [7, Cor. 5.4]. Second, the inequality by which he passes from the box states back to the cube limit holds in the reverse direction for the class of observables his argument needs: weakly, at every coupling, for the box state with the site-independent counterterm, and strictly at $e = 0$ (Lemma 9.6); it is replaced by a two-sided sandwich between cube states with a margin of order μ_0^{-1} (Lemmas 9.9 and 9.10). Third, the continuum subsequences through which he reaches the limit theory are built here (Remark 14.15). With scalar sources, finally, his Theorem 2.1 needs a vacuum counterterm that his Wick ordering does not contain; we work with a scheme a and prove the existence of the cube limits with sources in that scheme.

The authors' earlier paper [27] constructs torus limits of the same model at block factor 2 and proves the Osterwalder–Schrader axioms with cubic symmetry for them; its abstract measure theory (the reconstruction of the measure, its ergodic decomposition and the weak clustering of the components) is used here as it stands, cited by label. Rotation invariance is proved in the rotation part of this paper, whose main theorem (Theorem 15.1) precedes the main theorem of the paper: King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared, and King's state is identified as the lower envelope of the walled tori through a heavy-shell decoupling estimate on King's block-spin flow with the shell mass inside the matter covariance.

1.2. **The two cube conventions.** King's cube state $\mathrm{ND}^K_{k,J}$ (Definition 2.2) has its photon variables on $X_J = C_J \cap \Lambda_k$ and its scalar alive on $\{x \in X_J : \ x_\nu < MJ - \varepsilon_k \text{ for all } \nu\}$, that is, killed on the three top face layers T_J of the cube. The comparison cube state $\mathrm{ND}^{(a)}_{k,J}$ (Definition 2.1) has the scalar alive on all of X_J. The Wick-tube lemmas and the lattice comparisons of the rotation part are written for the comparison convention, while King's printed state is on his own. The two are interlaced: for observables of the cone class, with supports at ℓ^∞-distance > 1 from the complement of the cube,

$$\langle\Theta\rangle^K_{k,J} \ \leq \ \langle\Theta\rangle^{(a)}_{k,J} \ \leq \ \langle\Theta\rangle^K_{k,J+1},$$

and reversely for the opposite class (Lemma 4.2). Hence both families have the same J-limit at fixed k; the comparison family's fixed-cube limits inferior and superior are squeezed to King's printed state Z^K_∞ as $J \to \infty$, and every continuum-subsequence limit of the comparison family equals Z^K_∞ (Corollary 4.4). What does not transfer is the existence of the fixed-J limit $k \to \infty$ itself: on one convention it does not imply it on the other. This is why the existence theorems are proved on King's convention, where the printed state lives and where King's block-spin flow runs; the Wick tube, proved on the comparison convention, is carried to King's convention by a transfer theorem (Theorem 21.3).

1.3. **The main theorem.** Sources are real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3;\Lambda^2)$ with coefficients z_j and real scalar test functions $h_j \in C^\infty_c(\mathbb{R}^3)$ with coefficients w_j; $F(f)$ is the curvature paired with f, $W^a(h)$ the renormalised Wick square in the scheme a, and $g(z) = \sum_j (\operatorname{Re} z_j) f_j$, $k(\zeta) = \sum_j (\operatorname{Re} w_j) h_j = k_+ - k_-$, with the norms $\|\cdot\|_2$, $|\cdot|_4$, $|\cdot|_w$ and the tube $\mathcal{D}_{\sigma_+,H}$ of Definitions 2.3 and 2.5. The Schwinger functions S_n, $S_{n,p}$ are the Taylor coefficients of the limit functionals at 0. A scheme a is admissible if $a = z$, the reference scheme, or $z_k - a_k$ converges (Definition 3.7). The following summary is restated with every clause in full and proved as Theorem 21.1.

Theorem 1.1 (Main Theorem, summary)**.** *Let $e > 0$, $\lambda, m, \mu_0 > 0$ and $M \geq 1$ be arbitrary.*

(A) The curvature field. *King's printed state $Z^K_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)} \rangle^K_{k,J}$ exists for every real 2-form f, extends to an entire function of the coefficients of every finite family, and satisfies:*

(i) (OS0) $|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$ *for all $z \in \mathbb{C}^N$, with no dependence on μ_0, e, λ, m or M, and $|Z^K_\infty(\sum_j z_j f_j, 0)| \leq \exp\big(\frac{1}{2}\mu_0^{-2}\|\sum_j (\operatorname{Re} z_j) d^* f_j\|_2^2\big)$; for real test functions $|S_n(f_1,\dots,f_n)| \leq (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \leq (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$; the S_n are tempered distributions and satisfy the regularity axiom* E0″ *with $s = 4$, the constant $\sigma_n = (e\pi)^n (n!)^{1/2}$ for complex $f_j \in \mathcal{S}$, and every $\beta > \frac{1}{2}$.*

(ii) (OS1) *Invariance under the full Euclidean group $\mathbb{R}^3 \rtimes O(3)$: under all translations of $\mathbb{R}^3$, under the cubic group O_h about every point, and under all rotations, F transforming as a 2-form; the S_n are Euclidean covariant.*

(iii) (OS2) *Reflection positivity about every plane $\{x \cdot n = c\}$, n a unit vector, $c \in \mathbb{R}$, in matrix form for exponentials with arbitrary complex curvature coefficients and in polynomial form.*

(iv) (OS3) *S_n is real and symmetric, and $S_n = 0$ for odd n.*

(v) (The measure) *For every even $r \geq 14$ there is exactly one Borel probability measure μ_F on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ with moments S_n and characteristic functional $Z^K_\infty(\mathrm{i}\,\cdot, 0)$ on $\mathcal{S}$; its image on H^{-r}_{loc} is determined by the characteristic functional and by the moments. The laws of the localised lattice fields χF_{Λ_k} under $\mathrm{ND}^K_{k,J}$ converge weakly on $H^{-r}_{\mathrm{per}}(Q')$, Q' a box containing $\operatorname{supp}\chi$, along the full sequence $k \to \infty$ at every fixed $J \geq J_0(\chi)$, and then as $J \to \infty$, to the law of χF under μ_F.*

(vi) (OS4 on the ergodic components) *μ_F is $\mathbb{R}^3 \rtimes O(3)$-invariant and $\mu_F = \int \mu_\xi \, d\nu(\xi)$ with translation-ergodic components; ν-almost every component is $\mathbb{R}^3 \rtimes O(3)$-invariant, reflection positive about every plane, clusters in the mean and under diagonal separation, has a unique joint vacuum in every direction, satisfies* OS4 *in all its forms for every choice of the time direction, and has real symmetric moments with a factorial growth bound with component constants; the decomposition is O_h-invariant, each component being a mixture of $k \in \{1,2,3,6\}$ ergodic measures. The vanishing of the odd moments is not claimed for a component.*

(B) The joint field on the Wick tube. *Let the couplings satisfy the sign clause* (T3) *of Hypothesis 3.12, $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$; in particular, let them lie in one of the explicit regions of Proposition 22.15. Let a be an admissible scheme. Then for every family of sources supported in $Q_R = [-R,R]^3$ the joint functional of (F, W^a) has a limit G_∞ on the tube $\mathcal{D}_{\sigma_+,H}$, with*

$$\sigma_+ = s_1, \qquad H = \tfrac{s_1}{16}, \qquad \kappa = \tfrac{C_U}{s_1},$$

and K_- determined by the cell constant B_{cell} (Equation (30)), where $s_1 \in (0, \sigma_1(M_1))$ is the amplitude of the extensive bound and C_U its constant (Equation (28) and Proposition 12.6); the constants depend on $(e, \lambda, m, \mu_0, M, a)$ and on the auxiliary choices M_1, χ_0, s_1, and on nothing else, and they are not uniform in μ_0. The joint Schwinger functions $S_{n,p}$ satisfy (i)–(vi) *in the following form:*

(i) $|G_\infty(\zeta)| \le \exp\big(\|g\|_2^2 + \kappa\|k_+\|_{L^1} + K_-|k_-|_w\big)$ *on $\mathcal{D}_{\sigma_+,H}$, and for real test functions $|S_{n,p}| \le e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w$ with $\sigma_0 = H/4$ and $K' = \kappa\pi^2 + K_-$ (a factor 2^{n+p} for complex test functions); $\mathrm{E0}''$ holds with $s = 4$ and every $\beta > 1$.*

(ii) *Translations and O_h about every point hold as identities on the tube, W^a transforming as a scalar; there is no Euclidean clause.*

(iii) *Reflection positivity about every coordinate plane for families whose Wick amplitudes satisfy $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$, in particular for all purely imaginary Wick coefficients, and the polynomial form.*

(iv) *$S_{n,p} = 0$ for odd n.*

(v), (vi) *As in* (A) *for the measure on $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2 \oplus \mathbb{R})$, $r \ge 14$ even, with the scheme-shifted lattice field in the convergence clause, and with the weak form of* OS4 *(ergodicity under $\mathbb{R}^3$ and clustering in coordinate directions) on ν-almost every component; there is no Euclidean clause.*

The same statements hold for the tube object G_∞^K of King's printed state on his own convention (Theorem 21.3 and Proposition 21.4).

The μ_0-free bound in (A)(i) is absolute: it depends on none of the parameters of the model. The μ_0^{-2} bound is the finite-volume Gaussian bound carried to the limit. Part (B) is the headline form of the Wick leg; the extension of (B) to arbitrary Wick amplitudes, under the hypothesis (W2) at every amplitude, with the only Euclidean clause of the Wick leg, is stated separately as Theorem 23.7 among the open problems.

1.4. **The massless limit.** The photon mass μ_0 is fixed throughout: King's construction, and every statement above, is at $\mu_0 > 0$, and the constants of the Wick leg are not uniform in μ_0. The limit of vanishing photon mass, along a family whose counterterm and scheme are frozen at a reference mass, is not treated in this paper; it is listed among the open problems as later work.

1.5. **The coupling range.** The curvature leg holds on all of King's range: $e > 0$ and $\lambda, m, \mu_0 > 0$ arbitrary. Every smallness condition in [15, 16, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale that is chosen after the point (e, λ, m, μ_0) and the datum, so the quantifier over the couplings is the one over the sources, and no sentence "e, λ sufficiently small" occurs (Lemma 22.1; Remark 22.16 of the supplemental material). No relation $\mu_0 \le m$ is used or needed. The only coupling condition in the paper is the one-sided sign clause (T3) of the Wick leg, $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$; it fails exactly when m minimises King's free-boundary bare coefficient α_∞ on $(0, m]$, and to one-loop order it holds if and only if

$$e^2 < e_c^2(\lambda, m, \mu_0) = \max\Big\{ \frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1 + \frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1 + \frac{\lambda(N+2)}{\pi m}\Big)\Big\}$$

(Equation (59)), so that a half-line of couplings fails it at one loop. Beyond one loop, (T3) is proved on explicit coupling regions, at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$ and at every $\mu_0 \in (0, m]$ on explicit boxes (Proposition 22.15); the Wick leg is stated on those regions, and more generally wherever (T3) holds. The mass clause in Bałaban's upper bound, a cap on the dimensionless scalar mass at the stopping scales where his Proposition 3.1 is used, is part of that proposition's statement and is one more one-sided cap on the stopping scale, not a condition on the couplings (Remark 22.2 of the supplemental material).

1.6. **What is proved where.** The inputs of the main theorem are collected in Section 3: the existence theorems for King's cube limits without and with scalar sources (Theorems 3.3 and 3.6), proved in the existence part; the rotation theorem (Theorem 3.14), proved in the

rotation part; Bałaban's torus stability bounds and the c-number lemma in the form the Wick leg uses (Proposition 3.9 and Lemma 3.10); and the coupling condition (T3). Everything else is proved in the sections between. Open are only the optional extensions: the Wick sector at arbitrary amplitudes ((W2), Form (A), Theorem 23.7), the relation of King's state to the torus states of [27] and its uniqueness and ergodicity, the limit of vanishing photon mass, and non-triviality; they are collected in the final section.

1.7. **Structure of the paper.** The model, King's cube, the comparison cube, the sources and the standing lemmas (the correlation inequalities of [7], with the two inequalities that are only sketched there proved in full, Gaussian domination, cutting and the elementary facts) come first, then the inputs with their provenance. The interlacing of the two cube conventions (Lemma 4.2 and Corollary 4.4) and its three geometry margins follow, and then the existence part: King's existence theorem on the cube (Section 5) and its version with scalar sources. The curvature sector is then built up in the order of the axioms: regularity with the absolute constants (Theorems 7.4 and 7.6), translations, the cubic group and the rotation clause (Theorems 8.5 and 8.7), reflection positivity through the modified box and its passage to the limit with OS3 and the polynomial form (Lemma 9.3, Theorem 10.6, and Propositions 10.9 and 10.10), and the measure with its ergodic decomposition (Theorem 11.1). The Wick tube is treated in two sections, the torus inputs and the extensive bound (Propositions 12.2 and 12.6) and the uniform tube bound with its limits (Theorem 13.10 and Lemma 13.11); continuum subsequences and the states they define, with no existence input, form a section of their own (Theorem 14.9). The rotation part follows: the geometry of the two lattices and the walled tori, the admissible schemes and the full object, the torus comparison and its reduction to one-sided bounds at one spacing, the closing estimate and the invariance theorems, and the heavy shell in two sections; the corrections to King's papers that it uses are collected in the supplemental material. The main theorem is assembled in Theorem 21.1, and the coupling range is established in Lemma 22.1 and Proposition 22.15. The paper ends with the open problems.

2. The model, King's cube and the standing lemmas

2.1. **Couplings, lattices and the action.** Throughout, $e > 0$ and $\lambda, m, \mu_0 > 0$ are the charge, the quartic coupling, the bare scalar mass and the photon mass; no further condition on the couplings is imposed, since every smallness condition of [15, 16, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale chosen after the point (e, λ, m, μ_0) and after the datum (the section on the coupling range). The block factor is 5: the spacings are $\varepsilon_k := 5^{-k}$, the lattices $\Lambda_k := \varepsilon_k \mathbb{Z}^3$ with the standard frame (e_α), and $M \in \mathbb{N}$ is King's large-block integer, fixed once and for all. King's cubes are $C_J := [-MJ, MJ)^3$, $J \in \mathbb{N}$, and $X_J := C_J \cap \Lambda_k$ (the spacing understood from the context). A *King torus* is $\mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in M\mathbb{N}$ even; Λ_k descends to it, with $\ell\, 5^k$ sites per side.

The scalar field is $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and the photon field a real $A_\alpha(x)$ for every site x and direction α; the bond $\langle x, x + \varepsilon e_\alpha\rangle$ carries $A_\alpha(x)$ and the reversed bond $-A_\alpha(x)$. King's action [16, (2.5)] at spacing ε is

$$S = \tfrac{1}{2}\sum_{\alpha,\beta}\sum_{x} \varepsilon \left(A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\right)^2 + \frac{\mu_0^2}{2}\sum_x \varepsilon^3 |A(x)|^2 + \tfrac{1}{2}\sum_{\beta}\sum_{x} \varepsilon \left|e^{\mathrm{i}e\varepsilon A_\beta(x)}\phi(x+\varepsilon e_\beta) - \phi(x)\right|^2 + \sum_x \varepsilon^3 \Big[\tfrac{1}{2}\big(m^2 + \delta m_\varepsilon^2\big) r_x^2 + \lambda r_x^4\Big],$$

the photon in Feynman form with mass μ_0, and the hopping term of a bond $\langle x, y\rangle$, $y = x + \varepsilon e_\beta$, equal to $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\vartheta_x - e\varepsilon A_\beta(x) - \vartheta_y)$: a *diagonal part* and a *cosine part*. The counterterm δm_ε^2 is King's free-boundary counterterm [16, p. 330], fixed by his two-point graphs of order at most four in (e, λ) at zero momentum with the propagators of the infinite lattice $\varepsilon\mathbb{Z}^3$: it is *one number per spacing*, the same at every site of every box and torus. The Wick constant is the free one, $\mathsf{W}_\varepsilon := (-\Delta_{\varepsilon\mathbb{Z}^3} + m^2)^{-1}(0,0)$. This *free convention* is used on cubes and tori

alike unless said otherwise; King's own torus convention appears only inside the torus inputs of the Wick leg.

2.2. **The states.**

Definition 2.1 (Neumann–Dirichlet boxes, killed sites, the comparison cube)**.** For a finite set $X \subset \Lambda_k$ the *Neumann–Dirichlet state* $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ is the normalised measure $Z^{-1}e^{-S}\prod_{x\in X} d^2\phi(x)\prod_{x\in X,\alpha} dA_\alpha(x)$ with $\phi = 0$ at every site outside X (Dirichlet condition: a bond leaving X contributes only its diagonal part $\frac{\varepsilon}{2}r_x^2$ at its end $x \in X$), photon gradient terms only for pairs $x, x+\varepsilon e_\beta$ both in X, and the photon mass term on X (Neumann condition). For $U \subset X$ the state *with the matter killed on* U replaces the single-site measure $d^2\phi(x)$ times its weight by the point mass at $\phi(x) = 0$ for $x \in U$; every hopping term with an endpoint in U then reduces to its diagonal part at the other endpoint, and the photon variables are unchanged. The *comparison cube* is $\mathrm{ND}^{(a)}_{k,J} := \langle\cdot\rangle^{\mathrm{ND}}_{k,X_J}$, written $\langle\cdot\rangle^{(a)}_{k,J}$: the scalar is alive on all of X_J, with Dirichlet planes $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ$.

Definition 2.2 (King's RG cube)**.** Let $\mathrm{alive}^K_J := \{x \in X_J : \ x_\mu < MJ - \varepsilon_k \ \forall\mu\}$ and $T_J := X_J \setminus \mathrm{alive}^K_J$, the three top face layers. *King's RG cube* $\mathrm{ND}^K_{k,J}$, written $\langle\cdot\rangle^K_{k,J}$, is the Neumann–Dirichlet state on X_J with the matter killed on T_J: the photon variables $A_\alpha(x)$, $x \in X_J$, their gradient terms for pairs in X_J and their mass term on X_J are those of the comparison cube; the scalar is alive on alive^K_J and $\phi = 0$ on every other site of Λ_k, in particular on T_J. Its Dirichlet planes are $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, at separation $2MJ$; the Neumann image planes of the photon, $x_\mu = -MJ - \varepsilon_k/2$ and $x_\mu = MJ - \varepsilon_k/2$, are the same in both conventions.

The two conventions differ exactly by the matter on T_J; the photon Gaussian is the same. King's RG cube is the box [16, (3.3)–(3.4)] with the Dirichlet planes of [16, (3.9)] and the photon reflection planes of [16, (3.7)], the geometry of his renormalisation group; his Theorem 3.2 and his modified box [16, (3.38)] are written in the comparison convention. The two families are interlaced in Section 4; the existence theorems of this paper are posed on ND^K, while the rotation part and the cell lemmas of the Wick leg are written for the comparison family. The *periodic state* $\langle\cdot\rangle_{k,T}$ on a King torus T has the action summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, in the free convention.

BFS measures. A *BFS measure* on a finite set V of sites is a probability measure

$$Z^{-1}\exp\Big(\sum_{\{x,y\}\in B} c_{xy}\, r_x r_y \cos\big(\vartheta_x - \vartheta_y - \ell_{xy}(A)\big)\Big)\prod_{x\in V} d\rho_x(r_x)\, d\vartheta_x\, d\gamma(A),$$

with B a set of pairs of sites, $c_{xy} \geq 0$, each ρ_x a finite measure on $[0,\infty)$ with $\int e^{\alpha r^2} d\rho_x < \infty$ for all $\alpha \geq 0$ (point masses at 0, that is killed sites, included), γ a non-degenerate centred Gaussian measure on $\mathbb{R}^m$ (the photon variables), and linear functionals ℓ_{xy}. Every state above is a BFS measure: $c_{xy} = \varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$ and the diagonal parts $\frac{\varepsilon}{2}r_x^2$ of all bonds at x (those leading out of the box or into a killed site included); the photon form is positive definite because $\mu_0 > 0$, and the exponential moments are finite because $\lambda > 0$. So is every measure obtained from such a state by multiplying some c_{xy} or some photon gradient terms by $t \in [0,1]$, adding $s|\ell(A)|^2$ ($s \geq 0$) to the action, multiplying single-site measures by $e^{\varepsilon^3 q(x) r^2}$ with real q, or killing sites. A BFS measure is *of King type* if $c_{xy} \in [0,\varepsilon]$ and each ρ_x is a point mass at 0 or has the density $r\, e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]-\varepsilon^3 w(x) r^2 - \frac{\varepsilon}{2}n_x r^2}$ with $w(x) \geq 0$, $n_x \in \{0,\dots,6\}$; every state above, and every measure obtained from one by the operations just listed with $q \leq 0$, is of King type. Every state above is invariant under the *charge conjugation* $(A,\vartheta) \mapsto (-A,-\vartheta)$, which changes the sign of every field strength and fixes r.

2.3. **Sources, schemes and the objects.**

Definition 2.3 (sources, schemes, classes, transport)**.** A *curvature source* is a real 2-form $f \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, a *scalar source* a real $h \in C_c^\infty(\mathbb{R}^3)$. For the plaquette $p = (x;\alpha<\beta)$ with centre

$c(p)$ let $F(p) := \varepsilon^{-1}(A_\alpha(x) + A_\beta(x+\varepsilon e_\alpha) - A_\alpha(x+\varepsilon e_\beta) - A_\beta(x))$ and

$$F_{\Lambda_k}(f) := \sum_p \varepsilon^3 f_{c(p)}(e_\alpha, e_\beta) F(p), \qquad \mathsf{S}_{\Lambda_k}(h) := \sum_x \varepsilon^3 h(x), \qquad :\phi^2:(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big),$$

the sums over the plaquettes (corners in the box), respectively the sites, of the region. A *scheme* is a real sequence $a = (a_k)$, a_k depending only on ε_k and the couplings; $W^a(h) := \sum_x \varepsilon^3 h(x)(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon - a_k)$. For a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with complex coefficients put $\Theta(\zeta) := \exp(\sum_j z_j F(f_j) + \sum_j w_j :\phi^2:(h_j))$ and, for a Neumann–Dirichlet state on a box X (killed sites allowed),

$$Z^a_{k,X}(\zeta) := \Big\langle \exp\Big(\sum_j z_j F(f_j) + \sum_j w_j W^a(h_j)\Big)\Big\rangle^{\mathrm{ND}}_{k,X} = e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)} \big\langle \Theta(\zeta)\big\rangle^{\mathrm{ND}}_{k,X};$$

$Z^a_{k,J} := Z^a_{k,X_J}$ for the comparison cube, $Z^{a,K}_{k,J}$ for King's RG cube, and $N^a_{k,X}(0,q) := e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)} \langle e^{\sum_x \varepsilon^3 q(x) r_x^2}\rangle^{\mathrm{N}}_{k,}$ for one real scalar source. At $h = 0$ the multiplier is 1 and a is dropped: $Z_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$, $Z^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ for $z \in \mathbb{C}^n$. The *cone* is $\mathcal{K} := \{(f,g) : f \text{ a real 2-form},\ g \geq 0\}$ with the observable $\Theta(\mathrm{i}f, g)$ (*class* $\mathcal{I}$); *class* $\mathcal{D}$ consists of the observables $\Theta(f,-g)$, $(f,g) \in \mathcal{K}$. Transport: $(O\cdot h)(x) := h(O^{-1}x)$, $(O\cdot f)_x(u,v) := f_{O^{-1}x}(O^{-1}u, O^{-1}v)$ for $O \in O(3)$; $\tau_b h := h(\cdot - b)$, $(\tau_b f)_x := f_{x-b}$. For $\nu \in \{1,2,3\}$, $c \in \mathbb{R}$ let $\rho = \rho_{\nu,c}$, $x \mapsto x - 2(x_\nu - c)e_\nu$; $\theta h := h\circ\rho$, $\theta f := \rho^* f$, $(\rho^* f)_x(u,v) := f_{\rho x}(\rho u, \rho v)$ (the components $f_{\mu\nu}$, $\mu \neq \nu$, change sign); $\theta\zeta^* := (\bar z_j, \rho^* f_j; \bar w_j, h_j \circ \rho)_j$, and $\zeta \sqcup \zeta'$ is the union of two families. The *amplitude* is $\mathsf{a}(\zeta) := \sum_j |\operatorname{Re} w_j|\, \|h_j\|_\infty$. The *reference scheme* is $z = a^\sharp$, $a^\sharp_k := e^2 X_{\varepsilon_k}$, X_ε the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ on $\varepsilon\mathbb{Z}^3$ at $\lambda = 0$ with the order-e^2 part of δm^2_ε inserted.

By charge conjugation $\langle\Theta(\mathrm{i}f,g)\rangle = \langle \cos F(f)\, e^{:\phi^2:(g)}\rangle$ is real for every state above, and $Z^a_{k,X}(-z,w) = Z^a_{k,X}(z,w)$.

Remark 2.4 (schemes). The Wick leg is stated for the reference scheme z, or for an admissible scheme a (Definition 3.7: $z_k - a_k$ converges, hence is bounded). If the fixed-cube limits exist in $(0,\infty)$ for both a and z at a real scalar source, the scheme is admissible: for $0 \leq h \in C^\infty_c$ with $\int h > 0$ supported well inside some C_J, $Z^z_{k,J}(0,h)/Z^a_{k,J}(0,h) = e^{-(z_k-a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$. What the Wick leg uses is only $\sup_k |z_k - a_k| < \infty$.

Definition 2.5 (norms and the tube)**.** For a family with sources $f_1,\dots,f_n$, $h_1,\dots,h_p$ supported in $Q_R := [-R,R]^3$ and $\zeta = (z,w) \in \mathbb{C}^{n+p}$ put $g(\zeta) := \sum_i (\operatorname{Re} z_i) f_i$ and $k(\zeta) := \sum_j (\operatorname{Re} w_j) h_j = k_+ - k_-$ (pointwise positive and negative parts). Lattice norms: $\|g\|^2_{2,k} := \sum_p \varepsilon_k^3 g_{c(p)}(e_\alpha, e_\beta)^2$ over the plaquettes of Λ_k, $\|q\|_{1,k} := \sum_{x\in\Lambda_k} \varepsilon_k^3 |q(x)|$. Continuum norms: $\|g\|^2_2 := \sum_{\alpha<\beta} \|g_{\alpha\beta}\|^2_{L^2}$; $|q|_w := \sup_x (1+|x|^2)^2 |q(x)|$; $|q|_4 := \sup_{x,|\gamma|\leq 4} (1+|x|^2)^2 |D^\gamma q(x)|$ and $|g|_4 := (\sum_{\alpha<\beta} |g_{\alpha\beta}|^2_4)^{1/2}$, as in [27]. For $\sigma, H > 0$ the *tube* is

$$\mathcal{D}_{\sigma,H} := \big\{\zeta \in \mathbb{C}^{n+p} :\ \|k_+(\zeta)\|_\infty < \sigma/2,\ \ \|k_-(\zeta)\|_\infty < H/2\big\}.$$

Since $s \mapsto \|(\sum_j s_j h_j)_\pm\|_\infty = \sup_x (\pm\sum_j s_j h_j(x))_+$ is a supremum of convex functions of $s = \operatorname{Re} w$, hence convex and continuous, $\mathcal{D}_{\sigma,H}$ is the preimage under $\zeta \mapsto \operatorname{Re} w$ of the intersection of two open convex sublevel sets: open, convex, hence connected; it contains 0 and every ζ with purely imaginary w; $\|k(\zeta)\|_\infty < \min(\sigma, H)/2$ implies $\zeta \in \mathcal{D}_{\sigma,H}$; its real points form an open subset of $\mathbb{R}^{n+p}$. The tube of the Wick leg is $\mathcal{D}_{\sigma_+,H}$, with σ_+ and H fixed there.

The objects. For real 2-forms $f_1,\dots,f_n$ let $J_0(f)$ be the least J such that every support lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$. King's *cube limits* and *printed state* are

$$Z^K_J(z) := \lim_{k\to\infty} Z^K_{k,J}(z) \quad (J \geq J_0(f)), \qquad Z^K_\infty\Big(\sum_j z_j f_j, 0\Big) := \lim_{J\to\infty} Z^K_J(z),$$

whenever the limits exist: their existence at real z is Theorem 3.3, and their existence at every $z \in \mathbb{C}^n$, entire in z, is derived from it in the curvature-sector section. The object is scheme-free.

For general sources in a scheme a the *tube object* is $G_J(\zeta) := \lim_k Z^a_{k,J}(\zeta)$, $G_\infty(\zeta) := \lim_J G_J(\zeta)$, $\zeta \in \mathcal{D}_{\sigma_+,H}$, whenever these limits exist (the Wick-leg sections); where the full object $Z^a_\infty := \lim_J \lim_k Z^a_{k,J}$ of the optional extension (Theorem 23.7) exists it agrees with G_∞ on the tube, being the same iterated limit.

2.4. **Elementary facts.**

Lemma 2.6 (elementary facts)**.** (E1) *For $\varepsilon \le 1$ and any translate Γ of $\varepsilon\mathbb{Z}^3$: $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-4} \le C_0 := (5/2)^4\pi^2/8 < 49$ and $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-2} \le C_0' := (5/2)^2\pi^2 < 62$.* (E2) *$\|q\|_{L^1} \le \pi^2|q|_w$, $\|q\|_{L^2} \le \frac{\pi}{2\sqrt{2}}|q|_4$, $\|q\|_\infty \le |q|_w \le |q|_4$, $\|g\|_2 \le \frac{\pi}{2\sqrt{2}}|g|_4$; $|\cdot|_w$ is a norm, so $|k(\zeta)|_w \le \sum_j |\operatorname{Re} w_j||h_j|_w$ and $|k_\pm|_w \le |k|_w$.* (E3) *Sampling is evaluation at points: it commutes with pointwise positive and negative parts, and a sampled sup is at most the continuum sup, so $\zeta \in \mathcal{D}_{\sigma,H}$ for the continuum sources implies the same bounds for the sampled ones. For continuous compactly supported g, q: $\|g\|_{2,k} \to \|g\|_2$, $\|q\|_{1,k} \to \|q\|_{L^1}$, and for every k: $\|g\|^2_{2,k} \le C_0|g|^2_4$, $\|q\|_{1,k} \le C_0'|q|_w$.* (E4) *$n^{n/2} \le e^{n/2}(n!)^{1/2}$.*

Proof. (E1) For y in the cube of side ε centred at $x \in \Gamma$, $|x-y|^2 \le \frac{3}{4}$ and $|y|^2 \le 2|x|^2+2|x-y|^2 \le 2|x|^2+\frac{3}{2}$, so $1+|y|^2 \le \frac{5}{2}(1+|x|^2)$ and $(1+|x|^2)^{-r} \le (5/2)^r(1+|y|^2)^{-r}$. Integrate over the cube and sum over the disjoint cubes: $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-r} \le (5/2)^r \int_{\mathbb{R}^3}(1+|y|^2)^{-r}dy$, with $\int(1+|y|^2)^{-4} = 4\pi\cdot\frac{\pi}{32} = \frac{\pi^2}{8}$ and $\int(1+|y|^2)^{-2} = 4\pi\cdot\frac{\pi}{4} = \pi^2$. (E2) $\|q\|_{L^1} \le |q|_w \int(1+|y|^2)^{-2}$, $\|q\|^2_{L^2} \le |q|^2_4 \int(1+|y|^2)^{-4}$; the rest is immediate. (E3) Riemann sums for the convergences; the plaquette centres of one orientation form a translate of $\varepsilon_k\mathbb{Z}^3$, so (E1) bounds $\sum_{p:(\alpha,\beta)} \varepsilon_k^3 g_{c(p)}(e_\alpha, e_\beta)^2$ by $C_0|g_{\alpha\beta}|^2_4$, and the three orientations sum to $C_0|g|^2_4$; the last bound is (E1) with $r = 2$. (E4) is $n^n \le e^n n!$. □

Lattice-to-continuum. On a box X let $\langle A, B\rangle_k := \sum_{x\in X}\sum_\alpha \varepsilon^3 A_\alpha(x)B_\alpha(x)$, and for a real 2-form g whose non-vanishing plaquettes have their corners in X define $d^*_k g$ by $F_{\Lambda_k}(g) = \langle d^*_k g, A\rangle_k$. Each component $(d^*_k g)_\gamma$ is a backward difference quotient of the sampled components $g_{\gamma\beta}$, $\beta \ne \gamma$; for smooth compactly supported g it converges uniformly to $(d^*g)_\gamma$, up to the orientation sign convention of the components, at the rate $O(\varepsilon_k)$ of Taylor's theorem. Hence

$$\|d^*_k g\|_{2,k} \to \|d^*g\|_{L^2}, \qquad \sup_k \|d^*_k g\|_{2,k} < \infty, \qquad \|g\|_{2,k} \to \|g\|_2 \qquad (k \to \infty). \tag{1}$$

2.5. **Correlation inequalities.** For a BFS measure, a real linear functional F of A and $g \ge 0$ on V put $\Theta_{F,g} := e^{\mathrm{i}F}e^{\sum_x g_x r_x^2}$, $\Theta^-_{F,g} := e^F e^{-\sum_x g_x r_x^2}$, and $\langle X; Y\rangle := \langle XY\rangle - \langle X\rangle\langle Y\rangle$. Since $F(f)$ is a real linear functional of A and $e^{:\phi^2:(g)}$ a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$, class $\mathcal{I}$ consists of multiples of observables $\Theta_{F,g}$ and class $\mathcal{D}$ of multiples of observables $\Theta^-_{F,g}$.

Lemma 2.7 (correlation inequalities)**.** *Let $\langle\cdot\rangle$ be a BFS measure, F a real linear functional of A, $g \ge 0$, $\Theta = \Theta_{F,g}$, $\Theta^- = \Theta^-_{F,g}$. Then $\langle\Theta\rangle$ is real, and*

- (a) *$\langle\Theta; |\ell(A)|^2\rangle \le 0$ for every complex linear functional ℓ of A; hence adding $s|\ell(A)|^2$ to the action, $s \ge 0$, does not decrease $\langle\Theta\rangle$, and deleting a photon gradient term does not increase it;*
- (b) *$\langle\Theta; r_x r_y \cos(\vartheta_x - \vartheta_y - \ell_{xy}(A))\rangle \ge 0$; hence decreasing c_{xy} does not increase $\langle\Theta\rangle$;*
- (c) *$\langle\Theta; r_x^2\rangle \ge 0$; hence multiplying ρ_x by e^{-sr^2}, $s \ge 0$, does not increase $\langle\Theta\rangle$, and killing a set U of sites does not increase it, provided $g = 0$ on U or the measure is of King type;*
- (d) *for Θ^- the three inequalities and the three monotonicity statements hold with the opposite signs;*
- (e) *if the pairs in B and the Gaussian γ do not couple the fields at the sites of $X \subset V$ (and the photon variables attached to them) to the others, the measure is a product, and for an observable depending only on the fields in X the expectation equals the expectation in the X-factor.*

Proof. (a)–(c) are [7, Thms. 6.1, 6.2(1),(2)], whose hypotheses are those of a BFS measure: an arbitrary finite lattice, $c_{xy} \ge 0$, finite single-site measures on $[0,\infty)$ with $\int e^{\alpha r^2}d\rho_x < \infty$ for all

$\alpha \geq 0$ (a point mass at 0 is one), an arbitrary centred Gaussian with covariance $C \geq 0$, and $g \geq 0$ (so $g = 0$ is included). Theorem 6.1 states $\langle P(r)\cos(n\cdot\vartheta + f\cdot A); Q(r)\cos(m\cdot\vartheta + h\cdot A)\rangle \geq 0$ for polynomials P, Q in the r_x with positive coefficients; (a), (b) follow as in the printed proof of 6.2(1),(2): $e^{\sum g_x r_x^2}$ is a limit of such polynomials, $e^{\mathrm{i}F}$ may be replaced by $\cos F$ because the expectation and both observables are invariant under $(A, \vartheta) \mapsto (-A, -\vartheta)$ (which also gives the realness), and $|\ell(A)|^2 = \ell_1(A)^2 + \ell_2(A)^2$ with $\ell_r(A)^2 = \lim_{t\downarrow 0} t^{-2}[1 - \cos(t\ell_r(A))]$; (c) is Theorem 6.1 with $Q = r_x^2$ and $m = 0$, $h = 0$. Each monotonicity statement follows by differentiating in the parameter, every interpolating measure being a BFS measure. Killing: if $g = 0$ on U, set $c_{xy} = 0$ on every bond at U (not increasing, by (b)); the measure is then a product (e) of the matter at U, which carries no observable and cancels, and of the state with the matter killed on U (the diagonal parts at the other ends of the bonds into U are already in their single-site measures). For King type in general, multiply ρ_x, $x \in U$, by $e^{-s\varepsilon^3 r^2}$ (not increasing) and let $s \to \infty$ by Lemma 2.9.

(d) is stated in [7, Thm. 6.2(3),(4)] for the photon and the hopping inequality with a sketched proof, and its r^2-inequality is [16, (3.28)], stated there without proof; we prove all three. Let μ be the unnormalised measure and $\omega' = (r', \vartheta', A')$ an independent copy of $\omega = (r, \vartheta, A)$; then $(\int d\mu)^2\langle \Theta^-; Y\rangle = \iint \Theta^-(\omega)(Y(\omega) - Y(\omega'))\, d\mu(\omega) d\mu(\omega')$. Write $r = u - v$, $r' = u + v$ (so $u_x \geq |v_x|$), $\vartheta = \vartheta_+ + \vartheta_-$, $\vartheta' = \vartheta_+ - \vartheta_-$, $A = a_+ + a_-$, $A' = a_+ - a_-$. Then $a_\pm = (A \pm A')/2$ are independent with the common centred Gaussian law of covariance $C/2$, and the radial law $\prod_x d\rho_x(u_x - v_x) d\rho_x(u_x + v_x)$ on $\{u \geq |v|\}$ is invariant under $v_x \mapsto -v_x$ at each site. With $\chi_{xy} := \vartheta_{+,x} - \vartheta_{+,y} - \ell_{xy}(a_+)$ and $\psi_{xy} := \vartheta_{-,x} - \vartheta_{-,y} - \ell_{xy}(a_-)$ the unprimed phase is $\chi + \psi$, the primed one $\chi - \psi$, and

$$r_x r_y \cos(\chi + \psi) + r'_x r'_y \cos(\chi - \psi) = 2(u_x u_y + v_x v_y)\cos\chi\cos\psi + 2(u_x v_y + v_x u_y)\sin\chi\sin\psi,$$
$$r_x r_y \cos(\chi + \psi) - r'_x r'_y \cos(\chi - \psi) = -2(u_x v_y + v_x u_y)\cos\chi\cos\psi - 2(u_x u_y + v_x v_y)\sin\chi\sin\psi,$$

$r_x^2 - r_x'^2 = -4u_x v_x$, $\ell(A)^2 - \ell(A')^2 = 4\ell(a_+)\ell(a_-)$ for real ℓ, and $\Theta^-(\omega) = e^{F(a_+)} e^{F(a_-)} e^{-\sum_x g_x(u_x^2 + v_x^2)} e^{2\sum_x g_x u_x v_x}$. Expand $\exp(\sum_b c_b[\text{first line}])$ and $e^{2\sum g_x u_x v_x}$ in power series (absolutely convergent at fixed spacing). Every term of the integrand is a sign σ times a monomial in (u, v) with non-negative coefficient, times the even radial factor $e^{-\sum g_x(u_x^2 + v_x^2)}$, times $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ with one real function V (a product of $e^{F(\cdot)}$, of factors $\cos\chi_b$ or $\sin\chi_b$, and of $\ell(\cdot)$ for the photon slot). The photon integral of $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ factorises because a_+, a_- are independent and identically distributed. The angular integral factorises by the following argument: per site $d\vartheta\, d\vartheta' = 2\, d\vartheta_+ d\vartheta_-$, and $[0, 2\pi)^2$ in the (ϑ, ϑ')-variables is a fundamental domain of the lattice L generated by (π, π) and $(\pi, -\pi)$ in the $(\vartheta_+, \vartheta_-)$-plane, which contains $(2\pi\mathbb{Z})^2$ with index 2; $V(\vartheta_+, \cdot)V(\vartheta_-, \cdot)$ is L-invariant, because the simultaneous shift of $\vartheta_{+,x}$ and $\vartheta_{-,x}$ by π at one site changes every χ_b and ψ_b with $b \ni x$ by π and leaves $\cos\chi_b\cos\psi_b$ and $\sin\chi_b\sin\psi_b$ unchanged; hence $\int_{[0,2\pi)^2} VV\, d\vartheta\, d\vartheta' = 2\int_{L\text{-domain}} VV\, d\vartheta_+ d\vartheta_- = \int_{[0,2\pi)^2} VV\, d\vartheta_+ d\vartheta_- = (\int_0^{2\pi} V)^2$. The radial integral of a monomial vanishes if its degree in some v_x is odd, by the $v \mapsto -v$ symmetry, and is non-negative otherwise, since $u_x \geq |v_x|$. So every term has the sign σ: $\sigma = -1$ for $Y = r_x^2$ (from $-4u_x v_x$) and for the hopping term (second line), $\sigma = +1$ for $Y = \ell(A)^2$ (for complex ℓ, real and imaginary parts). Hence $\langle \Theta^-; r_x^2\rangle \leq 0$, $\langle \Theta^-; r_x r_y \cos(\cdot)\rangle \leq 0$, $\langle \Theta^-; |\ell(A)|^2\rangle \geq 0$. Monotonicity and killing follow as in (a)–(c), with (d)'s hopping inequality in place of (b) and Lemma 2.9, which covers Θ^-, for King type. (e) is clear. □

Lemma 2.8 (current representation and Gaussian domination)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C. Integrating out the matter at fixed A, the matter integral is $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$, q running over the integer-valued divergence-free currents on B, $w(q) \geq 0$, $\sum_q w(q) < \infty$, $\langle\kappa(q), A\rangle := \sum_{xy} q_{xy}\ell_{xy}(A)$. Let $\mathbb{E}^\sharp$ be the probability measure on currents with weights proportional to $w(q)e^{-\langle\kappa(q), C\kappa(q)\rangle/2}$. Then for $v \in \mathbb{R}^m$, $s \in \mathbb{R}$,*

$$\langle e^{s\langle v, A\rangle}\rangle = e^{s^2\langle v, Cv\rangle/2}\, \mathbb{E}^\sharp \cos\big(s\langle Cv, \kappa\rangle\big) \leq e^{s^2\langle v, Cv\rangle/2}.$$

Proof. Each bond factor expands as $e^{c_{xy} r_x r_y \cos t} = \sum_{q\in\mathbb{Z}} I_q(c_{xy} r_x r_y) e^{\mathrm{i}qt}$ with $I_q \geq 0$, $I_{-q} = I_q$, $\sum_q I_q(c) = e^c$; after the radial integrals the multiple series converges absolutely, dominated

by $\prod_x e^{\frac{1}{2}\sum_y c_{xy} r_x^2}$, integrable against the single-site measures. Integrating the angles forces $\sum_y q_{xy} = 0$ at every site (at a killed site $r_x = 0$ and $I_q(0) = \delta_{q0}$, so the currents vanish on its bonds); pairing q with $-q$ gives Ψ with $w(q) \geq 0$. For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a, A\rangle} d\gamma = e^{\langle a, Ca\rangle/2}$; with $a = sv \pm \mathrm{i}\kappa$ and the average of the two signs, $\int e^{s\langle v, A\rangle} \cos\langle \kappa, A\rangle \, d\gamma = e^{s^2\langle v, Cv\rangle/2 - \langle \kappa, C\kappa\rangle/2} \cos(s\langle v, C\kappa\rangle)$. Sum over q with the weights $w(q)$, divide by the same sum at $s = 0$ (positive), and use $|\cos| \leq 1$. □

The lemma applies to every state of this paper at every k, J, $\mu_0 > 0$, with or without a real scalar source absorbed in the single-site measures (the photon covariance is unchanged).

Lemma 2.9 (killing). *Let $\langle\cdot\rangle$ be a BFS measure of King type, U a set of alive sites, $\chi : U \to (0, \infty)$, and $\langle\cdot\rangle_{U,s}$ the measure with ρ_x multiplied by $e^{-s\varepsilon^3\chi(x) r_x^2}$ for $x \in U$. Let Θ be continuous with $|\Theta(A, \phi)| \leq e^{\ell(A)} e^{C\sum_x r_x^2}$ for a real linear functional ℓ and $C \geq 0$. Then $\langle\Theta\rangle_{U,s} \to \langle\Theta\rangle_U$ as $s \to \infty$, $\langle\cdot\rangle_U$ the measure with ρ_x replaced by the point mass at 0 for $x \in U$ (the state with the matter killed on U).*

Proof. The unnormalised density is $e^{-Q(A)/2} dA \prod_{x \text{ alive}} w_x(r_x) r_x dr_x d\vartheta_x \prod_b e^{c_b r_x r_y \cos(\cdot)} \prod_{x \in U} e^{-s\varepsilon^3\chi(x) r_x^2}$, Q positive definite, killed sites carrying no variables. Substitute $r_x = (s\varepsilon^3\chi(x))^{-1/2} u_x$ for $x \in U$; the Jacobians cancel between numerator and denominator. *Domination for $s \geq s_0$:* at $x \in U$, $w_x(r_x) \leq e^{\varepsilon^3 |\delta m_\varepsilon^2| r_x^2/2}$ (the other exponents in w_x are ≤ 0), $|\Theta|$ contributes $e^{C r_x^2}$, and each of the at most six bond weights at x is at most $e^{\frac{\varepsilon}{2} r_x^2 + \frac{\varepsilon}{2} r_y^2}$ ($0 \leq c_b \leq \varepsilon$); in the variable u_x these factors together are $\leq e^{u_x^2/2}$ once $s \geq s_0(\varepsilon, m, \delta m_\varepsilon^2, C, \min\chi)$, so the U-factor is at most $e^{-u_x^2/2} u_x$, integrable on $[0, \infty) \times [0, 2\pi)$; at an alive $y \notin U$ the factor is at most $w_y(r_y) e^{(C+3\varepsilon) r_y^2} r_y$, integrable since $\lambda > 0$; the photon factor $e^{\ell(A)} e^{-Q(A)/2}$ is integrable since Q is positive definite. *Pointwise limit:* at fixed $(u, \vartheta, A, r_{V\setminus U})$, $r_x \to 0$ for $x \in U$, so $\Theta \to \Theta|_{\phi_U = 0}$ by continuity, every bond weight with an endpoint in U and every $w_x(r_x)$, $x \in U$, tends to 1, and the U-factor tends to $e^{-u_x^2} u_x \, du_x d\vartheta_x$, whose integral cancels between numerator and denominator. By dominated convergence the ratio tends to $\int \Theta|_{\phi_U = 0} d\mu_U / \int d\mu_U$, where μ_U has the cosines at U removed, the weights w_y (which already contain all diagonals at y) and the photon unchanged, and the point mass at U: that is $\langle\cdot\rangle_U$, with a positive denominator. □

Lemma 2.10 (cutting). *Let $\langle\cdot\rangle_Y$ be a Neumann–Dirichlet box state on Λ_k, possibly with killed sites, or a periodic state on a King torus, and let X be a set of sites of Y such that X and its nearest neighbours lie in one fundamental cell (a clause in torus language: for a box state it has no content beyond $X \subset Y$). Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (killed sites and the diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let $(f, g) \in \mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,*

$$\langle\Theta(\mathrm{i}f, g)\rangle_X^{\mathrm{ND}} \leq \langle\Theta(\mathrm{i}f, g)\rangle_Y, \qquad \langle\Theta(f, -g)\rangle_X^{\mathrm{ND}} \geq \langle\Theta(f, -g)\rangle_Y,$$

and the same for Z^a. The first inequality uses Lemma 2.7(a),(b),(e) only; the second uses (d),(e). If no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}}$ is the state of Definition 2.1.

Proof. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2 + r_y^2)$. By Lemma 2.7(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The result is a product (Lemma 2.7(e)) whose X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$: the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms the Neumann ones of X; the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier are the same on both sides. □

With $Y = X_{J+1}$, $X = X_J$ this is King's monotonicity in the volume [16, Thm. 3.2, (3.31)–(3.33)] between comparison cubes, in the form his proof proves: with one Wick constant for all J, the free subtraction King himself switches to at the start of that proof [16, p. 333]. With the Dirichlet Wick constant of [16, (2.13)], which increases with J, the c-number factor moves

against the monotone factor and the monotonicity does not follow; that form is not used. The box version with killed sites and the monotonicity of King's RG cubes are in Section 4.

2.6. **Algebra of the scheme, covariance, reflection, continuity.**

Lemma 2.11 (algebra of the scheme)**.** *Let a be a scheme and Z^a the functional of a state above on a box at spacing ε_k.*

(a) *For real coefficients w the multiplier $e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ is a positive constant depending only on k, w and $\mathsf{S}_{\Lambda_k}(h_j)$. Hence the inequalities of Lemmas 2.7 and 2.10 between two states on the same lattice with the same sources hold for Z^a in the same direction.*

(d) $|Z^a(\zeta)| \le Z^a(\operatorname{Re}\zeta)$, and $\ln Z^a$ is convex in the real coefficients.

(e) $Z^a(f,h)\,Z^a(-f,-h) \ge 1$ for real (f,h).

(f) For real f and real $k_1, k_2 \ge 0$: $Z^a(f, k_1-k_2) \le Z^a(2f,-2k_2)^{1/2}\, Z^a(0,2k_1)^{1/2}$.

Proof. (a) is immediate. (d) $|\Theta(\zeta)| = \Theta(\operatorname{Re}\zeta)$ and the multiplier has modulus $e^{-\sum_j (\operatorname{Re} w_j) a_k \mathsf{S}(h_j)}$; convexity is Hölder's inequality. (e) $\langle e^X\rangle\langle e^{-X}\rangle \ge 1$ by Cauchy–Schwarz, and the multipliers cancel. (f) Cauchy–Schwarz gives $\langle e^{F+:\phi^2:(k_1)-:\phi^2:(k_2)}\rangle \le \langle e^{2F-2:\phi^2:(k_2)}\rangle^{1/2}\langle e^{2:\phi^2:(k_1)}\rangle^{1/2}$, and $e^{-a_k\mathsf{S}(k_1-k_2)} = (e^{2a_k\mathsf{S}(k_2)})^{1/2}(e^{-2a_k\mathsf{S}(k_1)})^{1/2}$. □

Lemma 2.12 (passive covariance)**.** *Let $O \in O(3)$, Λ a lattice of spacing ε with frame (f_α), and X a box or a torus for Λ. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions. Then $\langle\Theta(z\,O\cdot f, w\,O\cdot h)\rangle_{O\Lambda,OX} = \langle\Theta(zf,wh)\rangle_{\Lambda,X}$ and $\mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h)$, likewise for families and for Z^a. On a torus the periodic model does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, π a permutation.*

Proof. The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving Lebesgue measure; the action is written in the frame, and δm_ε^2, W_ε, a_k depend on ε only; plaquettes, centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$. A permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables preserving the action, which changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ and preserves the set of plaquette centres. □

Lemma 2.13 (the reflection sandwich at fixed spacing, comparison convention)**.** *Let $\sigma_3 = \operatorname{diag}(1,1,-1)$ and let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J-1}$. Then for every $k \ge 0$*

$$Z^a_{k,J-1}(\mathrm{i}f, g) \le Z^a_{k,J}(\mathrm{i}\,\sigma_3\cdot f, \sigma_3\cdot g) \le Z^a_{k,J+1}(\mathrm{i}f, g),$$

and the reversed inequalities hold for $Z^a(f,-g)$.

Proof. σ_3 preserves Λ_k and S_{Λ_k} and maps X_J onto $X_+ := (C_J + \varepsilon_k e_3)\cap\Lambda_k$. By Lemma 2.12, $Z^a_{k,J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g)$ is the functional with sources (f,g) of the Neumann–Dirichlet state on X_+ in the frame $(e_1, e_2, -e_3)$. Written in the frame (e_α), with $A'_3(x) = -A_3(x-\varepsilon e_3)$, this is the BFS measure R in which ϕ, A_1, A_2 live on X_+ and A_3 on $X_+ - \varepsilon e_3 = X_J$, with Dirichlet conditions for ϕ on X_+, photon gradient and mass terms within these sets, and the hopping term in direction 3 the standard one of the bond $\langle y-\varepsilon e_3, y\rangle$; the plaquette terms $(x;\alpha,3)$ pick up two sign changes and keep their centres, so the observable is $\Theta(\mathrm{i}f,g)$ in the standard frame. R is obtained from the Neumann–Dirichlet state on X_{J+1} by deleting photon gradient terms and hopping cosines and factorising, since $X_+ \cup X_J \subset X_{J+1}$, and the state on X_{J-1} is obtained from R in the same way, since $X_{J-1} \subset X_J \cap X_+$; the diagonal parts of the cut bonds are the Dirichlet terms of the smaller state, and the sources stay inside X_{J-1}. Lemma 2.7(a),(b),(e), respectively (d),(e), give the directions, and Lemma 2.11(a) carries them to Z^a. □

Lemma 2.14 (continuity of convex functions)**.** *Let U be a non-empty open convex subset of a Fréchet space and $\Phi : U \to \mathbb{R}$ a convex Borel function. Then Φ is continuous on U.*

Proof. U is a Baire space and $U = \bigcup_{j\in\mathbb{N}}\{\Phi \le j\}$, so some $S := \{\Phi \le j\}$ is non-meagre in U; being Borel, S has the Baire property [13]: $O \setminus \mathcal{M} \subset S$ for a non-empty open $O \subset U$ and a meagre $\mathcal{M}$. Pick $x_0 \in O$ and a balanced open neighbourhood W of 0 with $x_0 + W + W \subset O$.

For $w \in W$ and $y := x_0 + w$, $y + W \subset O$, and the point reflection $\iota(x) := 2y - x$ maps $y + W$ onto itself and preserves meagre sets; so $S \cap (y + W)$ and $\iota(S \cap (y + W))$ are comeagre in $y + W$ and intersect: there is $u \in S$ with $2y - u \in S$, and $\Phi(y) \le \frac{1}{2}(\Phi(u) + \Phi(2y - u)) \le j$. Hence $\Phi \le j$ on $x_0 + W$. A convex function on an open convex set bounded above near one point is bounded above near every point $y \in U$ (choose $z \in U$ with y strictly between x_0 and z, and take the convex combinations of $x_0 + W$ with z), and a convex function locally bounded above is continuous [24]. □

3. The inputs

This section states, in the form in which the later sections consume them, the results on which the theorems of this paper rest, and says where each is proved. Three are theorems of this paper proved in later parts: the existence of King's cube limits, without and with scalar sources (Theorems 3.3 and 3.6; the existence part, Sections 5 and 6), and the rotation theorem (Theorem 3.14; the rotation part, Section 15, whose main theorem is Theorem 15.1). Two are the torus inputs of the Wick leg, Bałaban's stability bounds and the c-number lemma (Proposition 3.9 and Lemma 3.10), established in the supplemental material. One is a condition on the couplings, the sign clause (T3) (Hypothesis 3.12), proved on explicit coupling regions in Proposition 22.15; it is the only condition on the couplings anywhere in the paper. The optional extension of the Wick leg to arbitrary amplitudes, together with the hypotheses it needs ((W2), (R_W), (FA^K)), is stated in the final section (Section 23); no clause of the main theorem uses them. From the literature, the paper uses the correlation inequalities of [7], the chessboard estimate of [9], the reconstruction theorems of [19, 20], Bałaban's torus analysis [1, 2, 3, 4] and King's papers [15, 16], each at the place where it is applied; the general lemmas on the measure, its ergodic decomposition and weak clustering are taken from the authors' earlier paper on the torus limits [27] (Remark 3.17).

3.1. **Objects and conventions.** The two cube conventions of Definitions 2.2 and 2.1 are King's convention ND^K (the scalar field killed on the three top face layers T_J of the cube C_J, the convention of the state King prints) and the comparison convention $\mathrm{ND}^{(a)}$ (the scalar alive on all of X_J), in which the tube results (Theorem 13.10, Lemma 13.11) and the lattice comparisons of the rotation part are written. The two are interlaced by Lemma 4.2. Which convention a statement is posed on is part of it:

- The existence theorems Theorems 3.3 and 3.6 are posed on King's convention ND^K, on which King's renormalisation-group flow runs (Remark 5.3). By Corollary 4.4 the existence of the fixed-cube limits on one convention does not imply it on the other; the comparison family's fixed-cube limits are reached through that corollary, and the statements proved for the comparison family under the assumption that they exist (Definition 3.1; on the Wick tube Definition 3.2) serve as templates for King's convention.
- The rotation theorem Theorem 3.14 is proved in the rotation part for King's printed state Z_∞^K on King's convention (Theorem 15.1) and is carried to the limit states of the comparison family through Corollary 4.4; its Wick-sector form is posed on the comparison family (Theorems 18.9 and 18.11).
- Proposition 3.9 and Lemma 3.10 and (T3) are statements about King's tori (Definition 12.1) and about c-numbers.

Throughout, $Z^a_{k,J}(z, w)$ denotes the generating functional of Definition 2.3 with curvature coefficients z and Wick coefficients w in the scheme a, on the convention in question (written $Z^{a,K}_{k,J}$ when the convention ND^K is to be stressed); $z = a^\sharp$ is the reference scheme of Remark 2.4; $\langle\cdot\rangle^K_{k,J}$ is King's cube expectation at spacing ε_k.

Definition 3.1 (convergence of the comparison family at fixed cubes)**.** The comparison family is *convergent at fixed cubes* if for every finite family $f_1, \dots, f_n$ of real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$, every $J \ge J_0(f)$ and every $x \in \mathbb{R}^n$ the limit $\lim_k \langle e^{\sum_j x_j F(f_j)} \rangle^{(a)}_{k,J}$ exists and lies in $(0, \infty)$.

This is the clause of Theorem 3.3 read for the family $\mathrm{ND}^{(a)}$. It is not a result of this paper and is not needed for the main theorem: the statements below that assume it are proved for the comparison family as templates, and their forms for King's printed state are proved under Theorem 3.3; the continuum-subsequence section (Theorem 14.9) removes the dependence on the fixed-cube limits altogether.

Definition 3.2 (convergence of the comparison family at fixed cubes on the tube)**.** The comparison family is *convergent at fixed cubes on the tube* (in a scheme a) if the clause of Theorem 3.6 holds for it: for every finite family of real 2-forms f_j and real functions h_i in C_c^∞, every $J \geq J_0(f,h)$ and every real coefficient vector (z,w) whose positive scalar part $k_+ = (\sum_i w_i h_i)_+$ satisfies $2\|k_+\|_\infty < s_W$, the limit $\lim_k Z^a_{k,J}(z,w)$ on the convention $\mathrm{ND}^{(a)}$ exists and lies in $(0,\infty)$.

This is the Wick-sector counterpart of Definition 3.1, with the same status: it is not a result of this paper and is not needed for the main theorem, and the two conventions' clauses do not imply one another by the inputs of this paper (Corollary 4.4, Remark 5.10). The tube statements that assume it (Theorem 13.10(i), Lemma 13.11, Proposition 10.5, Theorem 10.7, Proposition 14.12(c)) are templates for the comparison family; their forms for King's convention, on which the limits exist by Theorem 3.6, are Theorem 21.3 and Proposition 21.4.

3.2. **Existence of King's cube limits.**

Theorem 3.3 (King's cube limits, curvature)**.** *For every finite family $f_1,\dots,f_n$ of real 2-forms in $C_c^\infty(\mathbb{R}^3;\Lambda^2)$, every $J \geq J_0(f)$, where $J_0(f)$ is the least J such that every* $\operatorname{supp} f_j$ *lies at ℓ^∞-distance greater than 1 from $\mathbb{R}^3 \setminus C_J$, and every $z \in \mathbb{R}^n$, the limit*

$$\lim_{k\to\infty} \langle e^{\sum_j z_j F(f_j)} \rangle^K_{k,J}$$

exists and lies in $(0,\infty)$.

Proved in Section 5 (Theorem 5.9), from King's renormalisation-group analysis of the cube.

Remark 3.4 (on Theorem 3.3). (i) *Existence clause only.* Theorem 3.3 is used only through the existence of the fixed-cube limit at real coefficients: no constants clause, no uniformity in J and no locally uniform convergence in z enter from it. Every bound used on the limit is proved here, with absolute constants, in Theorem 7.4; the monotonicity in the cube, the J-limit, the entire extension and the translations are proved in Lemma 4.2, Theorem 7.4, Lemma 7.7 and Theorem 8.7. (ii) *Origin.* This is the existence clause of King's Theorem 2.1 [16] at zero scalar source, for the state he prints. The proof in Section 5 follows King's route, the comparison of two spacings at a common block scale, in a ratio form in which the vacuum energies cancel; what it uses of Bałaban's block-factor-5 analysis [1, 2, 3, 4] is stated there. The mass clause of [2, Prop. 3.1], upstream of it, is part of that proposition's statement and is cited under it (Remark 22.2 of the supplemental material); it is not a condition of this paper. (iii) *Consumers.* The whole curvature sector (Theorems 7.4, 8.7, 10.6, 11.1), the identification of the LIM states with King's state (Corollary 14.13) and the main theorem (Theorem 21.1).

Remark 3.5 (the cube). The cube on which the existence theorem is proved is King's RG cube C_J^K of Definition 2.2, with King's killing of the scalar on the top layers T_J; it is the only cube on which King's block-spin flow exists (Remark 5.3). Every placement of a comparison-cube result on King's cube goes through Lemma 4.2 and its corollaries.

Theorem 3.6 (King's cube limits with scalar sources)**.** *Let $z = a^\sharp$ be the reference scheme and let $s_W := s_1$ be the amplitude of Proposition 12.6. For every finite family of real 2-forms $f_j \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and real functions $h_i \in C_c^\infty(\mathbb{R}^3)$, every $J \geq J_0(f,h)$ and every real coefficient vector (z,w) whose positive scalar part $k_+ := \big(\sum_i w_i h_i\big)_+$ satisfies $2\|k_+\|_\infty < s_W$, the condition of the tube $\mathcal{D}_{\sigma_+,H}$ with $\sigma_+ = s_W$ (the negative scalar part is unrestricted), the limit $\lim_{k\to\infty} Z^z_{k,J}(z,w)$ exists and lies in $(0,\infty)$.*

Proved in Section 6 (Theorem 6.17 and Corollary 6.18), by the route of Section 5 with the scheme c-number carried along.

Definition 3.7 (admissible schemes)**.** A scheme a (Remark 2.4) is *admissible* if $a = z$ or $\lim_k (z_k - a_k)$ exists. For an admissible scheme Theorem 3.6 holds for $Z^a_{k,J}$ in place of $Z^z_{k,J}$ (Lemma 2.11(a)), and $\sup_k |z_k - a_k| < \infty$.

Remark 3.8 (on Theorem 3.6). (i) At zero Wick coefficient Theorem 3.6 is Theorem 3.3. (ii) *Origin.* Theorem 3.6 is *not* King's Theorem 2.1 with scalar sources: as printed, in King's Wick ordering, that statement fails, and the state with scalar sources exists only after a vacuum counterterm, which is what the scheme supplies (Remark 2.4). The threshold on the positive scalar part is the one at which the positive-source bound of the Wick leg is available (Proposition 13.2); what happens above it is the subject of the optional extension (Hypothesis 23.1). (iii) One instance of Theorem 3.6 is named separately because the statements on continuum subsequences need it alone: ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4; one cube, one negative Wick source). (iv) *Consumers.* The Wick tube (Proposition 12.6, Theorem 13.10), its transfer to King's convention (Theorem 21.3), the Wick sector of Theorem 14.9, and Form (B) of the main theorem (Theorem 21.1, Proposition 21.4).

3.3. **The torus inputs of the Wick leg.**

Proposition 3.9 (torus stability at two scalar masses)**.** *For the scalar masses $M = m$ and $M = M_1$ (the auxiliary mass fixed in Proposition 12.6) there are constants $E_\pm(M) < \infty$, a tail K_1 and, for each $k \geq K_1$, a $J_*(k)$ such that the source-free partition functions of King's tori T (Definition 12.1) at spacing ε_k and at scalar mass M satisfy, in King's torus convention,*

$$e^{-E_-(M)|T|} \leq Z^{(M)}_{k,T} \leq e^{E_+(M)|T|} \qquad (k \geq K_1,\ J \geq J_*(k)).$$

These are Bałaban's lower and upper stability bounds, [1, (1.14)] and [2, (1.3)], at a fixed scalar mass, read at the two masses m and M_1 and in King's torus normalisation of the counterterm; the form used here is established in the supplemental material (Theorem F.1 there). Proposition 3.1 of [2], behind the upper bound, is stated under a cap on the dimensionless scalar mass at the scales at which it is applied; it is cited under that cap, one more one-sided cap on the stopping scale and no condition on the bare mass (Remark 22.2 of the supplemental material). Only a tail in k and, for each k, large J are used (Proposition 12.6, Theorem 13.10).

Lemma 3.10 (the c-number input)**.** *Let $\alpha^\infty_k(M) = M^2 + \delta m^2_k(M)$ be King's free-boundary bare coefficient at scalar mass M and spacing ε_k, in the reference scheme z. For the auxiliary mass M_1 of Proposition 12.6, the quantities*

$$\sigma_1(M_1) := \liminf_{k\to\infty} \tfrac{1}{2}\big(\alpha^\infty_k(m) - \alpha^\infty_k(M_1)\big) \qquad \text{and} \qquad \beta_1(M_1)$$

are finite, where $\beta_1(M_1)$ is the second constant entering Proposition 12.6.

The proof is given in the supplemental material (Theorem F.159 there).

Remark 3.11 (on Lemma 3.10). (i) This is a statement about lattice Feynman integrals: the difference of two mass counterterms at the same photon mass. The limit $\alpha_\infty(M) = \lim_k \alpha^\infty_k(M)$ and its description as a function of the couplings are discussed in Lemma 22.1. (ii) *Consumers.* Proposition 12.6 (the constants $\sigma_+ = s_1$, H, κ of the tube) and the coupling range, Lemma 22.1.

Hypothesis 3.12 ((T3): the coupling condition)**.** $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$. The couplings (e, λ, m, μ_0) are said to lie in the *coupling region* of this paper when (T3) holds. By Proposition 22.15 the coupling region contains the explicit regions listed there: at $\mu_0 = m$ a region containing $0 \leq \lambda, e^2 \leq 0.1m$, and at every $\mu_0 \in (0, m]$ explicit boxes.

This is the condition on the couplings under which the Wick leg of the main theorem is stated; the curvature leg carries no condition on the couplings.

Remark 3.13 (on (T3)). (i) (T3) is the only condition on the couplings anywhere in this paper, and it is one-sided: it fails exactly on the set $E_W^<$ of couplings (e,λ,m,μ_0) at which m minimises $M\mapsto\alpha_\infty(M)$ on $(0,m]$ (Lemma 22.1). At one loop it holds if and only if

$$e^2<e_c^2(\lambda,m,\mu_0)=\max\Big\{\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1+\frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1+\frac{\lambda(N+2)}{\pi m}\Big)\Big\},$$

N the number of real components of the scalar field (equation (59)); so at one loop a half-line of values of e^2 fails. Beyond one loop, (T3) is *proved* on explicit coupling regions (Proposition 22.15): at $\mu_0=m$ on a region containing $0\le\lambda,e^2\le 0.1m$, and at every $\mu_0\in(0,m]$ on explicit boxes; the main theorem is stated on the region where it holds. The two-sided variant and the set $E_W\subseteq E_W^<$ are described in Lemma 22.1. (ii) *Consumers.* Proposition 12.6 and Lemma 22.1.

3.4. **The rotation theorem.** For a family f of real 2-forms and $J\ge J_0(f)$ let L_J^f be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a_{k,J}^{(a)})_k$, $a_{k,J}^{(a)}(z)=\langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$ (the comparison family; by Lemma 7.3 and Montel's theorem the set is non-empty).

Theorem 3.14 (rotation invariance of the comparison family's limit states)**.** *For every finite family f of real 2-forms in $C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and every choice $\zeta_J\in L_J^f$, $J\ge J_0(f)$, the limit $Z_\infty^{\mathrm{LIM}}(\sum_j z_j f_j,0):=\lim_{J\to\infty}\zeta_J(z)$ exists locally uniformly on $\mathbb{C}^N$, is entire and does not depend on the choices. The functional $Z_\infty^{\mathrm{LIM}}(\cdot,0)$ is invariant under the Euclidean group $\mathbb{R}^3\rtimes O(3)$: $Z_\infty^{\mathrm{LIM}}(\gamma\cdot f,0)=Z_\infty^{\mathrm{LIM}}(f,0)$ for every Euclidean motion γ and every complex coefficient vector, F transforming as a 2-form.*

Proved in the rotation part: Theorem 15.1 gives the $\mathbb{R}^3\rtimes O(3)$-invariance of King's printed state $Z_\infty^K(\cdot,0)$, and by Corollary 4.4(iii), with the fixed-cube limits of Theorem 3.3, every J-limit of subsequential fixed-cube limits of the comparison family exists and equals $Z_\infty^K(\cdot,0)$, which is the statement as posed. King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared (Theorem 17.1), and King's state is identified as the lower envelope of the walled tori (Theorem 18.4) through a heavy-shell decoupling estimate on King's block-spin flow (Theorem 20.2).

Remark 3.15 (on Theorem 3.14). (i) The theorem is posed for the comparison family and for the J-limit of its subsequential fixed-cube limits; its proof goes through King's printed state, whose invariance Theorem 15.1 proves directly, and Corollary 4.4 identifies $Z_\infty^{\mathrm{LIM}}(\cdot,0)$ with King's printed state $Z_\infty^K(\cdot,0)$, which is thereby $\mathbb{R}^3\rtimes O(3)$-invariant (Theorem 8.7). (ii) *Consumers.* The $O(3)$ clause of the curvature sector (Theorem 8.7), the Euclidean clause of the measure's ergodic components (Theorem 11.1(c)), and the $O(3)$ form of the LIM theorem (Theorem 14.9).

Remark 3.16 (the Wick tube on King's convention). Theorem 13.10 is proved for the comparison convention $\mathrm{ND}^{(a)}$. Its transfer to King's convention — the (k,J)-uniform tube bound (31) for $Z_{k,J}^{a,K}$ and the fixed-cube limits on the tube — is Theorem 21.3, with the kill step of Remark 4.8, the cell statement of Lemma 4.7 and the direction check for the negative leg as its convention-sensitive steps; Form (B) of the main theorem then holds on King's convention (Proposition 21.4).

3.5. **The optional extension.** The extension of the Wick leg to arbitrary amplitudes (Form (A), Theorem 23.7) is stated in the final section, with the three statements it needs: (W2) (Hypothesis 23.1; the fixed-cube limits and the positive-source bound at every amplitude), (R_W) (Hypothesis 23.3; its Euclidean clause) and (FA^K) (Hypothesis 23.5; its King-convention form). No clause of Theorem 21.1 uses them. What this paper proves of (W2) is its cone instance below the threshold $\sigma_1(M_1)$ (Corollary 13.3), which is all that Form (B) needs.

3.6. **Auxiliary statements of the section on continuum subsequences.** The statements on continuum subsequences (Theorem 14.9) are formulated in terms of three named properties proved in that section: the cell bound ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4; the boundedness form of Theorem 3.6 at one cube and one negative Wick source), the scheme bound (SCH) (Lemma 14.5; $a=z$ or $\sup_k|z_k-a_k|<\infty$) and the independence of the subsequence (UNIQ) (Proposition 14.6; any

two continuum subsequences give the same LIM state). The first two hold for every admissible scheme under (T3), and the third follows from the rotation theorem; the three are collected in Proposition 14.12. The limit of vanishing photon mass is not treated in this paper (Section 23).

3.7. **Results of the earlier paper used as inputs.**

Remark 3.17 (results of the torus paper used as inputs). The following results of the authors' earlier paper on the torus limits [27] are used under the hypotheses they carry there, which are matched to King's state at the places where they are applied (Theorem 11.1, Proposition 22.15): its abstract chain from a limit functional with the properties of Theorem 7.4 and the reflection positivity of Theorem 10.6 to the measure, its ergodic decomposition and weak clustering — Theorems 10.10, 10.19, 10.20, Propositions 10.4, 10.16, 10.17(iii), Lemmas 10.3, 10.5, Corollaries 10.43(iv), 10.44, Theorem 10.12, Definition 2.12, Proposition 8.3, Proposition 7.9(a), Lemmas 7.3, 7.7 and the lemmas of its Appendix G (the two-loop majorants behind Proposition 22.15). Of [7], Theorems 6.1 and 6.2(1),(2) are cited as printed; items (3),(4) of its Theorem 6.2, a sketch in print, are a lemma of this paper (Lemma 2.7). The chessboard estimate is Theorem 4.1/4.2 of [9], applied after its hypotheses are checked (Proposition 12.2). Nothing of [15, 16] is used as a result: King's existence theorem is proved in Sections 5 and 6, and his reflection-positivity argument for the modified box and his passage to subsequences are replaced by proofs in this paper.

4. The two cube conventions: interlacing and margins

King's printed state lives on King's convention $\mathrm{ND}^{K}_{k,J}$ (Definition 2.2); the rotation part and the Wick-tube estimates of this paper use the comparison convention $\mathrm{ND}^{(a)}_{k,J}$ (Definition 2.1). We show that the two families are interlaced in J (Lemma 4.2), draw the consequences for the limits (Corollaries 4.3 and 4.4), and show that the comparisons of a cube with a reflected cube, a larger box and a tiling by cells carry over to King's convention with the printed margins (Lemmas 4.5 to 4.7).

4.1. **The two conventions and the support condition.**

Throughout, $\varepsilon = \varepsilon_k = 5^{-k}$, $X_J = C_J \cap \Lambda_k = \{-MJ, \dots, MJ-\varepsilon\}^3$, $T_J = \{x \in X_J : \ x_\mu = MJ - \varepsilon \text{ for some } \mu\}$ and $\mathrm{alive}^K_J = X_J \setminus T_J = \{-MJ, \dots, MJ-2\varepsilon\}^3$. Both states carry the same photon variables $A_\alpha(x)$, $x \in X_J$, with the Neumann form of X_J, so their photon Gaussians are identical and every plaquette observable is the same function of the shared variables. They differ exactly by the matter on T_J: alive in the comparison convention (scalar Dirichlet planes $x_\mu = -MJ - \varepsilon$, $x_\mu = MJ$), killed in King's (planes $x_\mu = -MJ - \varepsilon$, $x_\mu = MJ - \varepsilon$, $2MJ$ apart, a multiple of every block scale; cf. Remark 3.5). A bond with one end alive and the other killed or outside the site set contributes its diagonal part $\frac{\varepsilon}{2} r_x^2$ at the alive end, so an alive site carries six diagonals in either convention. In the proof of his Theorem 3.2 and in his box form (3.38), King [16] works in the comparison convention: every site of the region is alive there.

For (f, g) with f a real 2-form and $g \geq 0$ a scalar, both in C_c^∞, we write as in Definition 2.3 $\Theta(\mathrm{i}f, g) = e^{\mathrm{i}F(f) + W(g)}$ (class $\mathcal{I}$), $\Theta(f, -g) = e^{F(f) - W(g)}$ (class $\mathcal{D}$), $W(g) = \sum_x \varepsilon^3 g(x)(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon)$ (free Wick constant), and $Z^a_{k,J}(\mathrm{i}f, g) = e^{-a_k \mathsf{S}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$, $Z^a_{k,J}(f, -g) = e^{a_k \mathsf{S}(g)} \langle \Theta(f, -g) \rangle$, $\mathsf{S}(g) = \sum_x \varepsilon^3 g(x)$; a superscript K ($Z^{a,K}_{k,J}$) marks King's convention. All these expectations are real. The sources lie inside X_J if

(S) every plaquette $p = (x; \alpha, \beta)$ of Λ_k with $f_{c(p)}(e_\alpha, e_\beta) \neq 0$ has its four corners in X_J, and every site $x \in \Lambda_k$ with $g(x) \neq 0$ lies in $\mathrm{alive}^K_J = X_J \setminus T_J$.

If $\operatorname{supp} f \cup \operatorname{supp} g \subset \{y : \operatorname{dist}_\infty(y, \mathbb{R}^3 \setminus C_J) > 1\}$, as every hypothesis of this paper demands of its sources, then (S) holds: a site with $g(x) \neq 0$ has $-MJ + 1 < x_\mu < MJ - 1 \leq MJ - \varepsilon$, so $x \in \mathrm{alive}^K_J$; a plaquette with $f_{c(p)} \neq 0$ has its centre at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and its corners within $\varepsilon/2 \leq \frac{1}{2}$ of it, hence in X_J. Also (S) for X_{J_0} implies (S) for every $J \geq J_0$.

4.2. **Cut-outs and the interlacing lemma.**

Lemma 4.1 (cut-outs from a box with killed sites)**.** *Let $\langle\cdot\rangle_Y$ be the state of Definition 2.1 on a finite set $V \subset \Lambda_k$ of sites with the matter killed on $T \subset V$, and $X \subset V$ any subset. Let (f, g), $g \geq 0$, satisfy* (S_X)*: every plaquette with $f \neq 0$ has its four corners in X and every site with $g \neq 0$ lies in X (g on $X \cap T$ is immaterial). Then deleting from $\langle\cdot\rangle_Y$ the photon gradient terms coupling X to $V \setminus X$ and the cosine parts of the hopping terms between X and $V \setminus X$ yields a product whose X-factor is $\langle\cdot\rangle_X^{(a),X\cap T}$, the state of Definition 2.1 on X with the matter killed on $X \cap T$; consequently*

$$\langle\Theta(\mathrm{i}f, g)\rangle_X^{(a),X\cap T} \;\leq\; \langle\Theta(\mathrm{i}f, g)\rangle_Y, \qquad \langle\Theta(f, -g)\rangle_X^{(a),X\cap T} \;\geq\; \langle\Theta(f, -g)\rangle_Y,$$

and the same for Z^a. In particular, (a) *if $X\cap T = \emptyset$ the factor is $\langle\cdot\rangle_X^{(a)}$;* (b) *if $X = \prod_\mu\{a_\mu, \dots, b_\mu\}$ is a box and $X\cap T = \{x \in X :\ x_\mu = b_\mu \text{ for some } \mu\}$, the factor is the state of King's convention on X.*

Proof. Each deletion is in the cone direction: deleting a photon gradient term does not increase $\langle\Theta_{F,g}\rangle$ and does not decrease $\langle\Theta^-_{F,g}\rangle$ (Lemma 2.7(a),(d)), and setting a hopping coefficient c_{xy} to 0 likewise (Lemma 2.7(b),(d)); every intermediate measure is a BFS measure. After all deletions nothing couples the X-variables to the rest, so the measure is a product (Lemma 2.7(e)). In the X-factor the photon is the Neumann photon of X; a site of $X \cap T$ carries its point mass; an alive $x \in X \setminus T$ carries the single-site weight times $\frac{\varepsilon}{2}r_x^2$ for each of its six bonds, since in $\langle\cdot\rangle_Y$ every bond at an alive x contributes $\frac{\varepsilon}{2}r_x^2$ at x whether its other end is alive in V (the diagonal part of $\frac{\varepsilon}{2}|e^{\mathrm{i}e\varepsilon A}\phi_y - \phi_x|^2$), killed, or outside V ($\phi = 0$ there), and the cut keeps all diagonals; the cosines are those of the bonds within X between alive sites. That is the state of Definition 2.1 on X killed on $X \cap T$. By (S_X) the observable is a function of the X-factor's variables (at $x \in X\cap T$ the factor $e^{\varepsilon^3 g(x) r_x^2}$ is 1 in both states), and the c-numbers $e^{\mp 2\mathsf{W}_\varepsilon \mathsf{S}(g)}$, $e^{\mp a_k \mathsf{S}(g)}$ are the same positive constants on both sides (Lemma 2.11(a); the counterterm is site-independent). Hence $\langle\Theta\rangle_X^{(a),X\cap T}$ is the expectation of Θ in the fully deleted measure, which is $\leq \langle\Theta\rangle_Y$ on $\mathcal{I}$ and $\geq$ on $\mathcal{D}$. □

The clause of Lemma 2.10 that X and its neighbours lie in one fundamental cell is a torus clause; for a box it has no content, since a bond leaving V already carries its Dirichlet diagonal in $\langle\cdot\rangle_Y$. Every sandwich $C_{J-1} \subset \cdot \subset C_{J+1}$ of this paper, read in King's convention, is an instance of Lemma 4.1.

Lemma 4.2 (interlacing of the two cube families)**.** *Fix $k \geq 0$, $J \geq 1$ and (f, g), $g \geq 0$, satisfying* (S) *for X_J. Then*

$$\langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^K \;\leq\; \langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^{(a)} \;\leq\; \langle\Theta(\mathrm{i}f, g)\rangle_{k,J+1}^K, \qquad \langle\Theta(f, -g)\rangle_{k,J}^K \;\geq\; \langle\Theta(f, -g)\rangle_{k,J}^{(a)} \;\geq\; \langle\Theta(f, -g)\rangle_{k,J+1}^K,$$

and the same for the scheme functionals $Z^a_{k,J}(\mathrm{i}f, g)$, $Z^a_{k,J}(f, -g)$ of the three states.

Proof. Step 0 (one observable, one multiplier). By (S) the plaquettes with non-zero coefficient have all corners in X_J, whose photon variables are present in all three states, so $F(f)$ is one real linear functional F of the shared variables; the sites with $g \neq 0$ lie in alive_J^K, alive in all three states; W_ε, δm_ε^2, $\mathsf{S}(g)$ are the same for the three regions. Hence, with $g'_x := \varepsilon^3 g(x) \geq 0$, $\Theta(\mathrm{i}f, g) = e^{-2\mathsf{W}_\varepsilon \mathsf{S}(g)}\Theta_{F,g'}$ and $\Theta(f, -g) = e^{2\mathsf{W}_\varepsilon \mathsf{S}(g)}\Theta^-_{F,g'}$, where $\Theta_{F,g'} = e^{\mathrm{i}F} e^{\sum_x g'_x r_x^2}$, $\Theta^-_{F,g'} = e^F e^{-\sum_x g'_x r_x^2}$ are the observables of Lemma 2.7, and the scheme multiplier $e^{\mp a_k \mathsf{S}(g)}$ is one positive constant for all three states (Lemma 2.11(a)). It suffices to treat $\langle\Theta_{F,g'}\rangle$, $\langle\Theta^-_{F,g'}\rangle$.

Step 1 (kill step on $\mathcal{I}$). For $s \geq 0$ let $\langle\cdot\rangle_s$ be $\langle\cdot\rangle_{k,J}^{(a)}$ with the single-site measure at each $x \in T_J$ multiplied by $e^{-s\varepsilon^3 r_x^2}$, a BFS measure, and $\varphi(s) := \langle\Theta_{F,g'}\rangle_s = N(s)/D(s)$, $N(s) = \int \Theta\, e^{-s\varepsilon^3 \sum_{T_J} r_x^2} d\mu_0$, $D(s) = \int e^{-s\varepsilon^3 \sum_{T_J} r_x^2} d\mu_0$, μ_0 the unnormalised measure at $s = 0$. The s-derivative of either integrand is bounded in modulus by $\varepsilon^3 \sum_{T_J} r_x^2$ times the integrand at $s = 0$, integrable against the quartic weight $e^{-\lambda\varepsilon^3 r^4}$ and the Gaussian; so differentiation under

the integral sign is justified, $D(s) > 0$, and

$$\varphi'(s) = -\varepsilon^3 \sum_{x\in T_J} \big(\langle \Theta\, r_x^2\rangle_s - \langle\Theta\rangle_s \langle r_x^2\rangle_s\big) = -\varepsilon^3 \sum_{x\in T_J} \langle \Theta; r_x^2\rangle_s \;\le\; 0$$

by Lemma 2.7(c). So $\varphi(s) \le \varphi(0) = \langle\Theta\rangle_{k,J}^{(a)}$. Since $\Theta_{F,g'}$ is continuous with $|\Theta_{F,g'}| = e^{\sum_x g'_x r_x^2}$ and $\langle\cdot\rangle_{k,J}^{(a)}$ is of King type, Lemma 2.9 gives $\varphi(s) \to \langle\Theta\rangle_{T_J}$, the state with the matter killed on T_J, which is $\langle\cdot\rangle_{k,J}^K$. Hence $\langle\Theta\rangle_{k,J}^K \le \langle\Theta\rangle_{k,J}^{(a)}$.

Step 2 (cut step). Apply Lemma 4.1 with $V := X_{J+1}$, $T := T_{J+1}$ (so $\langle\cdot\rangle_Y = \langle\cdot\rangle_{k,J+1}^K$), $X := X_J$: $X_J \subset X_{J+1}$, and $X_J \cap T_{J+1} = \emptyset$ since every site of T_{J+1} has a coordinate $M(J+1)-\varepsilon \ge MJ > MJ - \varepsilon$. (At $k = 0$, $M = 1$ the outward neighbours of X_J lie on T_{J+1} and are killed in $\langle\cdot\rangle_Y$; harmless, since the bond from an alive $x \in X_J$ to such a neighbour already carries $\frac{\varepsilon}{2} r_x^2$ at x, the Dirichlet diagonal of the comparison cube.) By Lemma 4.1(a) the X_J-factor is $\langle\cdot\rangle_{k,J}^{(a)}$, and (S) is ($\mathrm{S}_{X_J}$): $\langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^{(a)} \le \langle\Theta(\mathrm{i}f,g)\rangle_{k,J+1}^K$, $\langle\Theta(f,-g)\rangle_{k,J}^{(a)} \ge \langle\Theta(f,-g)\rangle_{k,J+1}^K$.

Step 3 (kill step on $\mathcal{D}$). With $\langle\cdot\rangle_s$ as in Step 1 and $\varphi^-(s) := \langle\Theta^-_{F,g'}\rangle_s$ (differentiation justified as before, now with $|\Theta^-| \le e^{F(A)}$ integrable against the Gaussian), $(\varphi^-)'(s) = -\varepsilon^3 \sum_{x\in T_J} \langle\Theta^-; r_x^2\rangle_s \ge 0$ by Lemma 2.7(d), so $\varphi^-(s) \ge \varphi^-(0) = \langle\Theta^-\rangle_{k,J}^{(a)}$; as $|\Theta^-_{F,g'}| \le e^{F(A)}$ with F real linear, Lemma 2.9 (with $\ell = F$, $C = 0$) gives $\varphi^-(s) \to \langle\Theta^-\rangle_{k,J}^K$. Hence $\langle\Theta(f,-g)\rangle_{k,J}^K \ge \langle\Theta(f,-g)\rangle_{k,J}^{(a)}$.

Step 4 (Z^a). The multiplier of Step 0 is one positive constant. □

Nothing in the proof uses a reflection symmetry of either cube (neither has one), the counterterm beyond its site-independence, or the renormalisation group. Since the observable vanishes at the killed sites in every kill step of this section, each such step also follows with no limit: set $c_{xy} = 0$ on every bond at the set U to be killed, one bond at a time (Lemma 2.7(b),(d)); the measure is then a product (Lemma 2.7(e)) whose U-factor cancels, and the other factor is the state with U killed.

4.3. **Monotonicity of the K family and the limits.**

Corollary 4.3 (the K family is monotone in J)**.** *Under* (S) *for X_J: $\langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^K \le \langle\Theta(\mathrm{i}f,g)\rangle_{k,J+1}^K$ and $\langle\Theta(f,-g)\rangle_{k,J}^K \ge \langle\Theta(f,-g)\rangle_{k,J+1}^K$, and the same for $Z^{a,K}$; hence for all $J \ge J_0$ whenever the sources satisfy* (S) *for X_{J_0}.*

Proof. Chain the two inequalities of Lemma 4.2. (For the comparison family this is Lemma 4.1(a) with $V = X_{J+1}$, $X = X_J$, i.e. the first inequality of Lemma 2.10.) □

The proof of Theorem 3.2 of [16] first replaces the J-dependent Dirichlet Wick subtraction (2.13) by the free, J-independent one and then proves the monotonicity in J in the comparison convention (Lemma 4.1(a)); by Corollary 4.3 that statement holds between cubes of King's convention too, with the kill step (Step 1) not printed there. The form with the Dirichlet constant follows in neither convention and is not used here; at $g = 0$ no Wick constant enters.

Corollary 4.4 (the limits)**.** *Let $f = (f_1, \dots, f_N)$ be real 2-forms in C_c^∞ with $\operatorname{supp} f_j \subset \{\operatorname{dist}_\infty(\cdot, \mathbb{R}^3 \setminus C_{J_0}) > 1\}$; for $J \ge J_0$, $z \in \mathbb{C}^N$ put $a_{k,J}^K(z) := \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^K$, $a_{k,J}^{(a)}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$, $g(z) := \sum_j (\operatorname{Re} z_j) f_j$. At the cone points $z = \mathrm{i}t$, $t \in \mathbb{R}^N$, both are real with modulus ≤ 1, and* (S) *holds for X_J.*

(i) (One J-limit at fixed k; no hypothesis.) *For every k and t, $J \mapsto a_{k,J}^K(\mathrm{i}t)$ and $J \mapsto a_{k,J}^{(a)}(\mathrm{i}t)$ are nondecreasing on $J \ge J_0$, bounded by 1 and interlaced; hence $\lim_J a_{k,J}^K(\mathrm{i}t) = \lim_J a_{k,J}^{(a)}(\mathrm{i}t) \in [-1,1]$. The same holds for $\langle\Theta(\mathrm{i}f,g)\rangle$, $g \ge 0$, with a common limit in $(-\infty,\infty]$.*

(ii) (King's cube limits.) *By Theorem 3.3, posed on King's convention, for every such family, every $J \ge J_0$ and every $t \in \mathbb{R}^N$, $\lim_k a_{k,J}^K(t)$ exists in $(0,\infty)$. Then: (α) $a_{k,J}^K \to Z_J^K$ locally*

uniformly on $\mathbb{C}^N$, Z_J^K *is entire and* $|Z_J^K(z)| \le e^{\|g(z)\|_2^2}$; *(β)* $Z_J^K \to Z_\infty^K$ *locally uniformly on* $\mathbb{C}^N$, Z_∞^K *is entire and* $|Z_\infty^K(z)| \le e^{\|g(z)\|_2^2}$; *(γ) for all* $J \ge J_0$ *and* $t \in \mathbb{R}^N$,

$$Z_J^K(\mathrm{i}t) \;\le\; \liminf_k a_{k,J}^{(a)}(\mathrm{i}t) \;\le\; \limsup_k a_{k,J}^{(a)}(\mathrm{i}t) \;\le\; Z_{J+1}^K(\mathrm{i}t),$$

hence $\lim_J \liminf_k a_{k,J}^{(a)}(\mathrm{i}t) = \lim_J \limsup_k a_{k,J}^{(a)}(\mathrm{i}t) = Z_\infty^K(\mathrm{i}t)$.

(iii) (The subsequential limits of the comparison family.) *Let* $\mathcal{S} = (k_m)$ *be an increasing sequence of spacings along which* $a_{k_m,J}^{(a)} \to Z_J^{\mathcal{S}}$ *locally uniformly on* $\mathbb{C}^N$ *for every* $J \ge J_0$ *and every such family (a continuum subsequence of the comparison family, Definition 14.1; they exist with no hypothesis, Lemma 14.7). Then* $Z_J^{\mathcal{S}} \to Z_\infty^K$ *locally uniformly on* $\mathbb{C}^N$ *as* $J \to \infty$*: every subsequential fixed-cube limit of the comparison family, followed by* $J \to \infty$*, is King's printed state. In particular, if the fixed-cube limits of the comparison family exist along the full sequence,* $\lim_J \lim_k a_{k,J}^{(a)} = Z_\infty^K$.

Proof. (i) Monotonicity is Corollary 4.3 for the K family and Lemma 4.1(a) for the comparison family; Lemma 4.2 interlaces them; two nondecreasing sequences with $a_J \le b_J \le a_{J+1}$ have the same limit, finite or $+\infty$.

(ii)(α) This is the first step of the proof of Theorem 7.4(b) run on the K family. (1) $a_{k,J}^K$ is entire (Gaussian decay in A, $e^{-\lambda\varepsilon^3 r^4}$ decay in r at the alive sites). (2) The local bound of Lemma 7.3 holds for the K family with the same constant: $\mathrm{Re}\sum_j z_j F(f_j) = F(g(z))$, so $|a_{k,J}^K(z)| \le \langle e^{F(g(z))}\rangle_{k,J}^K$; Lemma 2.8(a) with $s = 1$ and $v := d_k^* g(z)$ (defined by $F_{\Lambda_k}(g(z)) = \langle v, A\rangle_k$) applies to any BFS measure whose Gaussian has covariance C and gives $\langle e^{\langle v,A\rangle}\rangle \le e^{\langle v,Cv\rangle/2}$; the K cube is a BFS measure whose Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so its covariance is the C of Lemma 7.1, a statement about this C and the plaquettes with corners in X_J: $\langle v, Cv\rangle_k \le 2\|g(z)\|_{2,k}^2$. Hence $|a_{k,J}^K(z)| \le e^{\|g(z)\|_{2,k}^2} \le e^{3C_0|g(z)|_4^2}$ uniformly in $k \ge 0$, $J \ge J_0$, by (E3) of Lemma 2.6: the plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, each contributing at most $C_0|g|_4^2$ to $\|g\|_{2,k}^2$ by (E1), and the three orientations give $3C_0$. Nothing in this chain sees the matter beyond the state being a BFS measure. (3) Theorem 3.3 gives convergence at every real point. (4) $\mathbb{R}^N$ is a set of uniqueness for entire functions on $\mathbb{C}^N$; by Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence, all limits are entire and agree on $\mathbb{R}^N$, hence everywhere [11]. So $a_{k,J}^K \to Z_J^K$ locally uniformly, Z_J^K is entire, and the bound passes to the limit, $|Z_J^K(z)| \le \lim_k e^{\|g(z)\|_{2,k}^2} = e^{\|g(z)\|_2^2}$ (Lemma 2.6), a J-free bound.

(β) By Corollary 4.3 at every k and (α), $J \mapsto Z_J^K(\mathrm{i}t)$ is nondecreasing and ≤ 1, so $Z_\infty^K(\mathrm{i}t) := \lim_J Z_J^K(\mathrm{i}t)$ exists; $\{Z_J^K\}_{J\ge J_0}$ is locally uniformly bounded by the J-free bound of (α); $(\mathrm{i}\mathbb{R})^N$ is a maximal totally real subspace, hence a set of uniqueness for entire functions; Montel's theorem and the identity theorem give $Z_J^K \to Z_\infty^K$ locally uniformly, Z_∞^K entire, $|Z_\infty^K(z)| \le e^{\|g(z)\|_2^2}$. (This is the proof of Theorem 7.4(b) with Lemma 2.10 replaced by Corollary 4.3.)

(γ) Lemma 4.2 at every k, with $k \to \infty$ at the two ends (limits by (α)) and lim inf, lim sup in the middle, gives the sandwich; (β) gives the limits.

(iii) Lemma 4.2 along $\mathcal{S}$, with the full-sequence limits of (α) at the ends, gives $Z_J^K(\mathrm{i}t) \le Z_J^{\mathcal{S}}(\mathrm{i}t) \le Z_{J+1}^K(\mathrm{i}t)$, so $\lim_J Z_J^{\mathcal{S}}(\mathrm{i}t) = Z_\infty^K(\mathrm{i}t)$ by (β). $\{Z_J^{\mathcal{S}}\}_J$ is locally bounded by the J-free bound $e^{\|g(z)\|_2^2}$ (Lemma 7.3 passed to the limit), so by Montel's theorem and the uniqueness set $(\mathrm{i}\mathbb{R})^N$ it converges locally uniformly to an entire function (Lemma 14.8), which agrees with Z_∞^K on $(\mathrm{i}\mathbb{R})^N$, hence on $\mathbb{C}^N$. If the fixed-cube limits exist along the full sequence, that sequence is a continuum subsequence. □

Statements (ii)–(iii) also hold with the two conventions exchanged, for $J \ge J_0 + 1$, when the comparison family is convergent at fixed cubes (Definition 3.1; read Lemma 4.2 as $a_{k,J-1}^{(a)} \le a_{k,J}^K \le a_{k,J}^{(a)}$); the exchanged (iii) needs Lemmas 14.7 and 14.8 for the K family, whose proofs use

only the local bound and the photon-only bound $\langle F(u-u')^2\rangle \le 2\|u-u'\|^2_{2,k}$, both transferred by the covariance identity in (α).

What does not follow. At fixed J, Lemma 4.2 yields exactly (γ) and nothing on whether $\lim_k a^{(a)}_{k,J}(it)$ exists: the bounds $Z^K_J(it)$, $Z^K_{J+1}(it)$ differ in general (the cubes differ by M, so the gap does not shrink with k), and nothing constrains the oscillation of $a^{(a)}_{k,J}(it)$ in k between them. So the existence of the fixed-cube limits on one convention does not imply it on the other by these inputs, and likewise with scalar sources: the existence theorem must be proved on the family on which the renormalisation-group construction is carried out; the other is a comparison family with the same infinite-volume state by (ii)–(iii). No counterexample is claimed.

4.4. **Three margins in King's convention.** Let $\sigma_3 = \mathrm{diag}(1,1,-1)$, $X := X_J$, $X_+ := \sigma_3 X = (C_J + \varepsilon e_3)\cap\Lambda_k$ (third coordinates $\{-MJ+\varepsilon,\dots,MJ\}$), $\sigma_3 T_J = \{x \in X_+ :\ x_1 = MJ-\varepsilon$ or $x_2 = MJ-\varepsilon$ or $x_3 = -MJ+\varepsilon\}$ and $A_\sigma := \sigma_3(\mathrm{alive}^K_J) = \{x :\ x_1, x_2 \in \{-MJ,\dots,MJ-2\varepsilon\},\ x_3 \in \{-MJ+2\varepsilon,\dots,MJ\}\}$.

Lemma 4.5 (the reflection sandwich)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.* (a) *If $M \ge 2\varepsilon_k$ (i.e. $k \ge 1$, or $M \ge 2$), then for every $J \ge J_0 + 1$*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f, g) \ \le\ Z^{a,K}_{k,J}(\mathrm{i}\,\sigma_3\cdot f, \sigma_3\cdot g) \ \le\ Z^{a,K}_{k,J+1}(\mathrm{i}f, g).$$

(b) *For every $k \ge 0$, $M \ge 1$ and $J \ge J_0 + 2$ the same holds with $J \mp 2$ in place of $J \mp 1$.* (c) *For $k = 0$, $M = 1$ the one-step sandwich* (a) *is not obtainable by deletions and kills: A_σ contains alive sites ($x_3 = MJ$) killed in King's cube of index $J+1$, and* alive^K_{J-1} *contains sites ($x_3 = -MJ+\varepsilon$) killed in the reflected state. Whether the inequality itself holds there is left open.*

Proof. Step 1 (transport). By Lemma 8.4, $Z^{a,K}_{k,J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g) = e^{-a_k \mathsf{S}(g)}\langle\Theta(\mathrm{i}f,g)\rangle_R$, with R the state on X_+ for the frame $(e_1, e_2, -e_3)$ killed on $\sigma_3 T_J$ ($\mathsf{S}(\sigma_3\cdot g) = \mathsf{S}(g)$). In the frame (e_α), as in Lemma 2.13, R is the BFS measure with A_1, A_2 on X_+ and A_3 on $X_+ - \varepsilon e_3 = X$ (gradient pairs within these sets, masses on them), matter alive exactly on A_σ with point masses on $\sigma_3 T_J$ and $\phi = 0$ outside X_+, all six diagonals at each alive site, and the standard cosine on every bond with both ends in A_σ (the flipped bond with $A'_3(x) = -A_3(x - \varepsilon e_3)$ gives the standard cosine of $\langle x - \varepsilon e_3, x\rangle$); $F(f)$, $W(g)$ are the standard ones (the plaquettes $(x;\alpha,3)$ pick up two sign changes and keep their centres). R is of King type. The sources satisfy (S) for X_{J-1} (resp. X_{J-2}), so by the inclusions below Step 0 of Lemma 4.2 applies to all states compared.

Step 2 (upper end). If $A_\sigma \subset \mathrm{alive}^K_{J'}$ and $X_+ \cup X \subset X_{J'}$: from $\langle\cdot\rangle^K_{k,J'}$ kill every site of $\mathrm{alive}^K_{J'} \setminus A_\sigma$ (Lemma 2.7(c), Lemma 2.9, which allows the already killed $T_{J'}$), leaving exactly the cosines of bonds within A_σ, i.e. R's; then delete every photon gradient term except the pairs within X_+ for A_1, A_2 and within X for A_3 (Lemma 2.7(a); these are gradient pairs of $X_{J'}$). The result is a product (Lemma 2.7(e)) whose factor on R's variables is R (photon, alive set with all six diagonals by the bookkeeping of Lemma 4.1, cosines); the other factor cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle_R \le \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J'}$. The inclusions hold iff $MJ \le MJ' - 2\varepsilon$ (direction 3; the rest is trivial): for $J' = J+1$ iff $M \ge 2\varepsilon$, for $J' = J+2$ always.

Step 3 (lower end). If $\mathrm{alive}^K_{J''} \subset A_\sigma$ and $X_{J''} \subset X \cap X_+$: from R kill every site of $A_\sigma \setminus \mathrm{alive}^K_{J''}$ (Lemma 2.7(c), Lemma 2.9) and delete every photon gradient term coupling an $X_{J''}$-variable to another variable of R (Lemma 2.7(a)). The $X_{J''}$-factor has the Neumann photon of $X_{J''}$ (gradient pairs within $X_+ \cap X_{J''} = X \cap X_{J''} = X_{J''}$) and the matter alive on $\mathrm{alive}^K_{J''}$ with all diagonals, killed on $T_{J''}$: it is $\langle\cdot\rangle^K_{k,J''}$; the rest cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J''} \le \langle\Theta(\mathrm{i}f,g)\rangle_R$. The inclusions hold iff $-MJ'' \ge -MJ + 2\varepsilon$: for $J'' = J-1$ iff $M \ge 2\varepsilon$, for $J'' = J-2$ always.

Step 4. Steps 2–3 with $J\pm1$ give (a), with $J\pm2$ give (b); the multipliers agree (Lemma 2.11(a)). For (c): at $k = 0$, $M = 1$ the top of A_σ in direction 3 is J while that of alive^K_{J+1} is $J-1$, and the bottom of alive^K_{J-1} is $-J+1 = -MJ+\varepsilon$, the killed layer of R; deletions and kills never revive a killed site. □

In the curvature sector ($g = 0$) the limit sentences of Theorem 8.5 run on (a) for $k \geq 1$ or on (b) for all k, with $k \to \infty$ by Corollary 4.4(ii) and $J \to \infty$ by Corollary 4.3; with scalar sources they need the K-family tube limits, supplied by Theorem 21.3 (Remark 4.9).

Lemma 4.6 (the upper box needs no change)**.** *Let $\Omega \subset B$ be lattice boxes at spacing ε such that B contains every site within ℓ^∞-distance N lattice steps of Ω, $N \geq 1$ (the hypothesis of Lemma 9.10, whose mass condition forces $N \geq 1$), and let $\langle\cdot\rangle_B^K$ be the state of King's convention on B (killed on its top face layers T_B). Then* (a) *$\Omega \cap T_B = \emptyset$;* (b) *the second part of Lemma 9.10 holds with $\langle\cdot\rangle_B^K$ in place of the comparison state on B, with the same proof, and its first part is a photon statement, unchanged;* (c) *consequently the sandwich in the proof of Theorem 10.6 holds for King's cubes with the printed cubes and margins (20): for every k and every ζ admitted there, $Z_{k,J}^{a,K}(\zeta) \leq \Phi_k(\zeta) \leq Z_{k,J+2n}^{a,K}(\zeta)$, Φ_k the functional of the box state of Definition 9.2 on $\Omega_{J,k}$; the lower end is Lemma 9.9 with Step 1 of Lemma 4.2 prepended. (By Corollary 4.3 the cube of index $J + 2n + 1$ would also do; it is not needed.)*

Proof. (a) For $x \in \Omega$ and each μ, $x + N\varepsilon e_\mu \in B$, so $x_\mu < \max_B x_\mu$ and $x \notin T_B$. (b) By (a) every site of Ω is alive, so once the hopping cosines touching $B \setminus \Omega$ are removed the matter on Ω is the Dirichlet matter of Ω, and the photon law of B is the same in both conventions; the complete proof is given in Lemma J.3 of the supplemental material. (c) Lower end: Lemma 9.9 gives $\langle Y\rangle_{C_J}^{(a)} \leq \langle Y\rangle_{\Omega_{J,k}}$ on $\mathcal{I}$ (its proof lands on the Dirichlet matter of C_J, the comparison convention), and Step 1 of Lemma 4.2 gives $\langle Y\rangle_{C_J}^{K} \leq \langle Y\rangle_{C_J}^{(a)}$, (S) holding since the sources of the enlarged family and their reflections are supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $C_{J_0} \subset C_J$. Upper end: (b) with $B = X_{J+2n}$, whose hypothesis is the second margin of (20), a statement about photon sites. The multipliers agree (Lemma 2.11(a)). □

For the cell bound we use the cells of Definition 13.4: $\ell_c = 2ML_c \geq 10$; the cells $c = \prod_\mu [w_\mu, w_\mu + \ell_c)$ of the tiling $\mathcal{T}_s$, $s \in \{0,1\}^3$, $w_\mu \in W_\mu(s_\mu)$, $W_\mu(0) = \ell_c\mathbb{Z}$, $W_\mu(1) = ML_c + \ell_c\mathbb{Z}$ (so $w_\mu, w_\mu + \ell_c, MJ \in \mathbb{Z}$); the good sets $G_s = \{x : \operatorname{dist}(x_\mu, W_\mu(s_\mu)) > 2\ \forall\mu\}$ and cutoffs χ_s with $\operatorname{supp}\chi_s \subset G_s$; $X_c := c \cap X_J$; $J_1(R) = \min\{J : MJ \geq R + \ell_c\}$ for sources supported in $Q_R = [-R, R]^3$; B_{cell}, defined on the comparison cell C_{L_c}. N_{k,C_J}^K is the normalised scalar-source functional of $\langle\cdot\rangle_{k,J}^K$ and $N_{k,X}^{(a),T}$ that of the comparison state on X killed on $T \subset X$.

Lemma 4.7 (the cell lemmas hold as printed)**.** (a) *For a tiling $\mathcal{T}_s$ and any lattice function $q \geq 0$ on X_J: $N_{k,C_J}^K(0,-q) \leq \prod_i N_{k,X_i}^{(a),X_i\cap T_J}(0,-q|_{X_i})$, over the site sets $X_i = c \cap X_J$ of the cells meeting C_J.* (b) *For a full cell $c \subset C_J$: $X_c \cap T_J \neq \emptyset$ iff $w_\mu + \ell_c = MJ$ for some μ.* (c) *Let $q \geq 0$ be any lattice function supported in Q_R, $J \geq J_1(R)$, $s \in \{0,1\}^3$ and $c \in \mathcal{T}_s$ with $(\chi_s q)|_{X_c} \not\equiv 0$. Then $-MJ < w_\mu$ and $w_\mu + \ell_c < MJ - 2$ for every μ: c is a full cell inside C_J with $X_c \cap T_J = \emptyset$, and $(\chi_s q)|_{X_c}$ is supported at ℓ^∞-distance > 2 from $\mathbb{R}^3 \setminus c$.* (d) *Hence Lemmas 13.6 to 13.8 are unchanged and Proposition 13.9 holds for $\langle\cdot\rangle_{k,J}^K$ as printed, under its hypotheses and with its constant K_-: $\ln N_{k,C_J}^K(0,-q) \leq K_-|q|_w$ for every k, every $J \geq J_1(R)$ and every lattice $0 \leq q \leq H$ supported in Q_R.* (e) *(Direction.) For a cell c and $\emptyset \neq T \subset X_c$: $N_{k,c}^{(a),T}(0,-\tilde q) \geq N_{k,c}^{(a)}(0,-\tilde q)$ for $\tilde q \geq 0$ vanishing on T. So B_{cell}, a bound on a comparison cell, does not bound a source-carrying cell of King's type, and* (c) *is load-bearing.*

Proof. (a) The proof of Lemma 13.5 deletes the cross-cell photon gradient terms and hopping cosines, each in the $\mathcal{D}$ direction (Lemma 2.7(d)); by Lemma 2.7(e) and the factor identification of Lemma 4.1 with $V = X_J$, $T = T_J$, $X = X_i$, the product has the factors $N_{k,X_i}^{(a),X_i\cap T_J}$; the observable $e^{-\sum \varepsilon^3 q r^2}$ and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise, and at the sites of T_J the observable factor is 1 in every state. (b) The sites of a full cell have $x_\mu \in \{w_\mu, \dots, w_\mu + \ell_c - \varepsilon\}$, and $x \in T_J$ iff $w_\mu + \ell_c - \varepsilon = MJ - \varepsilon$ for some μ. (c) Let $x \in X_c$ with $(\chi_s q)(x) \neq 0$. Then $x \in G_s$; since $w_\mu, w_\mu + \ell_c \in W_\mu(s_\mu)$ and $w_\mu \leq x_\mu < w_\mu + \ell_c$, this gives $w_\mu + 2 < x_\mu < w_\mu + \ell_c - 2$; and $q(x) \neq 0$ gives $|x_\mu| \leq R$. Hence $w_\mu + \ell_c < R + \ell_c - 2 \leq MJ - 2$ (as $MJ \geq R + \ell_c$) and $w_\mu > -R - \ell_c + 2 \geq -MJ + 2$. So $c \subset (-MJ, MJ)^3$, $w_\mu + \ell_c \neq MJ$, hence $X_c \cap T_J = \emptyset$ by (b), and $\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus c) = \min_\mu \min(x_\mu - w_\mu, w_\mu + \ell_c - x_\mu) > 2$. (d) In the chain of Proposition 13.9

the Hölder step of Lemma 13.8 is an inequality within one state, convention-free; then (a) with the tiling $\mathcal{T}_s$ and $q_s := 8\chi_s q$. A factor with $q_s|_{X_i} \equiv 0$ equals 1 whatever is killed on X_i (this covers the clipped cells at the faces of C_J and any cell flush with them); a factor with $q_s|_{X_i} \not\equiv 0$ is, by (c), the comparison-cell functional $N^{(a)}_{k,c}(0, -q_s|_c)$ of a full cell $c \subset C_J$ with $q_s|_c$ supported at distance > 2 from $\mathbb{R}^3 \setminus c$ and $0 \le q_s \le 8H = H'$: exactly the hypotheses of Lemma 13.7 (hence of Lemma 13.6: translation to C_{L_c}, B_{cell}, Proposition 13.2 on the comparison box), as printed; Lemma 13.8 sums the cells. (A cell flush with a face, $w_\mu + \ell_c = MJ$, can meet $\operatorname{supp} q$ but never $\operatorname{supp}(\chi_s q)$; the chain feeds the cutoff sources, not q, to (a).) (e) Step 3 of Lemma 4.2 applied to the comparison cell with $U = T$: $s \mapsto \langle \Theta^- \rangle_s$ is nondecreasing (Lemma 2.7(d)) and tends to the killed state (Lemma 2.9). □

Remark 4.8 (one-line kill steps). Proposition 13.1 (cube $\le$ torus on $\mathcal{I}$), Proposition 13.2 (box $\le$ cube $\le$ torus), the lower end of Lemma 9.9, and the comparison of the cube with the box state of Definition 9.2 on the cone (Remark 9.8) are each proved for the comparison cube; for King's cube prepend Step 1 of Lemma 4.2 ($\langle\cdot\rangle^K_{k,J} \le \langle\cdot\rangle^{(a)}_{k,J}$ on $\mathcal{I}$), and for the $\mathcal{D}$ versions (cube $\ge$ torus for negative sources) Step 3. The same applies to a box of King's type (a killed face layer) in Proposition 13.2. Nothing beyond Lemma 4.2 is needed.

Remark 4.9 (the Wick sector). With scalar sources $g \ge 0$, Corollary 4.4(i) holds as stated. The analogues of Corollary 4.4(ii)–(iii) on the Wick tube need the fixed-cube limits and the (k, J)-uniform bound (31) for both families. For King's family the bound is the chain of Theorem 13.10(i) run on ND^K, whose convention-sensitive steps are the cube-versus-torus comparison (Corollary 13.3, via Remark 4.8) and the cell bound (Lemma 4.7); the assembly of that chain and the existence of the K-family tube limits are carried out in Theorem 21.3, after which the tube statements hold on King's convention (Remark 13.12, Proposition 21.4).

The existence part

5. King's existence theorem on the cube

The existence theorem for King's cube limits (Theorem 3.3), and its form with scalar sources that the Wick leg consumes (Theorem 3.6), are proved in this section and the next from estimates on King's renormalisation-group integrals on the cube C^K_J. The argument is the one King gives for his Theorem 2.1 [16, Sect. 3.1] and [15, Sect. 3.2]: the generating functional is approximated, up to a small-field remainder, by an integral of the effective action after j renormalisation steps, and two spacings whose effective actions live on the same unit lattice are compared. We write it as a Cauchy property of the *normalised* functional, in a ratio form in which every field-independent constant of the effective action cancels; this removes from King's route the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube.

The inputs are of two kinds. Estimates taken from Bałaban's block-factor-5 analysis [1, 2, 3, 4] on tori, and their transfer to King's cube, are stated as propositions, each followed by a description of its content; their proofs are given in the supplemental material. Three of these statements are used at several places and are therefore stated once, with a precise formulation, in Section 5.2 ((BAL_5)[int], $(\mathrm{REN}_{\mathrm{orb}})$) and Section 6.1 ($(\mathrm{K}_h)$). Everything else in the two sections is proved in place.

5.1. **Renormalisation-group integrals on King's cube.** Fix the couplings, a cube C_J and King's RG cube C^K_J of Definition 2.2. In this section and the next the index k is the spacing index ($\varepsilon_k = 5^{-k}$, as throughout), and j counts renormalisation steps, $0 \le j \le k - J_*$ for a fixed J_* (King's "J" of [15, Thm. 3.1]); the block scale after j steps is $\eta := 5^j \varepsilon_k = 5^{j-k} \le 1$, and the unit lattice after j steps is $\eta^{-1}(C_J \cap \eta\mathbb{Z}^3)$. Since the side $2MJ$ of C_J is an integer, the faces of C_J lie on block planes at every scale.

Sources. A real source is $s = (u, h)$ with $u = u_x := \sum_i x_i f_i$ a real combination of 2-forms $f_i \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$ sampled at the plaquette centres in the lattice's own frame, and $h \in C^\infty_c(\mathbb{R}^3)$ real; all supports lie at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ (so $J \ge J_0$). Put $X(s) := F(u) + {:}\phi^2{:}(h)$,

with the free Wick constant of Definition 2.3; X is real and linear in s. The unnormalised functional is

$$Z_k(s) \;:=\; \int e^{-S}\, e^{X(s)} \prod dA \prod d^2\phi$$

over the variables of C_J^K at spacing ε_k (any fixed, s-independent normalisation of the measure; it cancels everywhere below), and $N_k(s) := Z_k(s)/Z_k(0) = \langle e^{X(s)}\rangle_{k,J}^K$. At $h = 0$ we write $a_k(x) := N_k(u_x, 0)$, which is the $a_{k,J}^K(x)$ of Corollary 4.4.

King's flow on the cube. The block-spin transformations of [15, Sect. 2] are run on C_J^K with block factor 5: the photon field is averaged over the $5^j\varepsilon_k$-blocks that tile X_J, with the Neumann condition of Definition 2.2; the scalar field is averaged with King's Dirichlet weight [16, (3.8)–(3.9)] (in the corrected form in which the averaging blocks have their top layers on the Dirichlet planes) at the finest step and with the uniform weight above it. The Dirichlet planes of C_J^K are $2MJ$ apart, a multiple of every block side, which is what makes this flow possible on C_J^K and not on the comparison cube (Remark 5.3). Let $S_k^{(j)}(A_j, \phi_j; s)$ be King's effective action after j steps [15, (3.14)], [16, (3.12)], with the source vertices [15, (3.20)–(3.21)]; let χ_j be King's small-field indicator, [15, (3.2)] times the face factor [16, (5.41)], a function of (A_j, ϕ_j) alone; and

$$I_k^{(j)}(s) \;:=\; \int dA_j\, d\phi_j\; \chi_j\; e^{-S_k^{(j)}(A_j,\phi_j;s)}.$$

The *vacuum constant* $v_{k,j} := S_k^{(j)}(0, 0; 0)$ is finite; it contains King's E_0, E_1, the normalisation constants $\ln Z_j$, $\ln Z_j(0)$ of the block-spin kernels and every field- and source-free graph. The *normalised action* is $S_{N,k}^{(j)}(\cdot; s) := S_k^{(j)}(\cdot; s) - v_{k,j}$ for every s (the field-free, source-dependent part is kept), and $\hat I_k^{(j)}(s) := \int \chi_j e^{-S_{N,k}^{(j)}(\cdot;s)} = e^{v_{k,j}} I_k^{(j)}(s)$. The flow is a device of the proof: the state $\langle\cdot\rangle_{k,J}^K$ does not depend on it.

Diagonal pairs. Fix $\alpha \in (0, \frac12)$ and put $j(k) := \lfloor (1-\alpha)k \rfloor$, $\eta_k := 5^{j(k)-k}$; then $\eta_k \le 5^{-\alpha k}$ and $k - j(k) \le \alpha k + 1$. The *diagonal pair* at k consists of the k-model after $j(k)$ steps and the $(k+1)$-model after $j(k)+1$ steps. Both have the block scale η_k ($5^{j+1}\varepsilon_{k+1} = 5^j\varepsilon_k$), hence the same unit lattice and the same indicator $\chi_{j(k)}$; this is the situation of [15, p. 656] ("the fields A_k, ϕ_k have the same bounds in each integral"). No relation between $j(k+1)$ and $j(k)+1$ is used anywhere.

Bałaban's decomposition. In Bałaban's block-spin representation [1, 2], $Z_k(s) = \int d\mu_{k,s}$ with $d\mu_{k,s} = e^{X(s)} d\mu_{k,0}$ a positive measure on the fields and on the block variables of the scales $\le j$, and $P = P_j \in [0, 1]$ is the product of all small-field factors of those scales, a function of the block and fluctuation variables alone. The identity $Z_k(s) = \int P\, d\mu_{k,s} + \int (1-P)\, d\mu_{k,s}$ is exact; the first term is the *small-field part*, the second the *large-field remainder*.

5.2. **Two statements from the renormalisation-group analysis.**

Lemma 5.1 ((BAL$_5$)[int]: Bałaban's interaction-term bound at a nonzero background)**.** *In the setting of* [2, Sect. 2] *at block factor $L = 5$, in unit-lattice form at the step k, $t = 5^k\varepsilon$: let $\Omega_A = B^k(\Lambda_2^{(k-1)\prime}) \supset \Omega_C = B^k(\Lambda_2^{(k)})$ be the two propagator domains of* [2, Prop. 2.6]*, let $\tilde B$ be the background field of* [2, p. 579]*, regular on Ω_A, let $G_k(\Omega, \tilde B)$ be the scalar propagator of the step on Ω in the background $\tilde B$, and write the interaction terms of the stage of* [2, p. 579]*, localised in $B^k(\Lambda_7^{(k)})$, as a sum of per-label full renormalised classes with scalar lines $G_k(\Omega_A, \tilde B)$, external scalar fields $E_A = a_k G_k(\Omega_A, \tilde B) f$ with $\|f\|_\infty \le Cp\lambda(t)^{-1/4}$, and other external fields of norms $\le Cp^a\lambda(t)^{-b}$ (see the supplemental material). Replace $G_k(\Omega_A, \tilde B)$ by $G_k(\Omega_C, B^{(k+1),\eta})$ in every scalar line and in every E_A. For every $\kappa > 0$ there are $t_0(\kappa) > 0$ and $C_\kappa < \infty$, depending on κ, on Bałaban's parameters $(R, \varrho, b_0, \tilde p, \bar n, \alpha_0)$ and on the couplings only, such that for $t \le t_0(\kappa)$ the sum over all localisations of the absolute values of all terms containing at least one difference of propagators or of external fields is at most $C_\kappa\, t^\kappa\, |\Lambda_7^{(k)}|$. The constants do not depend on k, on the domains or on $\tilde B$, and $\tilde B = 0$ is included.*

The proof is given in the supplemental material (Theorem G.15 there). This is the interaction-term half of Bałaban's domain-change estimate [2, Prop. 2.6], at every background. The terms with exactly one difference line, grouped line by line, carry their counterterm partners but not the complete renormalised classes of the cut graph, and a lone divergent two-photon-leg subgraph is not small per graph; the proof expands every line and every external field in the difference and regroups the terms exactly into contractions of *full* classes of the cut components, to which the renormalisation bound [3, Prop. 1, (1.33)] applies with constants free of the domains and of the background. Every use of the upper stability bound [2, (1.3)] and of [15, Thm. 3.1, (3.4)] at block factor 5 passes through this statement.

Lemma 5.2 ((REN$_{\mathrm{orb}}$): orbit regrouping of cut renormalised families)**.** *Let* $\mathcal{R} = \sum_{\Gamma\in\mathcal{G}} c_\Gamma E(\Gamma)$ *be a renormalised family of* [3, Sects. 1–3] *with* $\mathcal{G}$ *a finite union of renormalised classes (Definition G.1 of the supplemental material), in which the objects (lines or external legs) of certain kinds are decomposed as* $o = o' - \Delta o$ *with symmetric kernels* Δo*, and expand multilinearly,* $\mathcal{R} = \sum_\tau 2^{-|S_L|} c_\Gamma \prod_{o\in S}(-\Delta o)\prod_{o\notin S} o'$ *over the oriented cut terms* $\tau = (\Gamma, S, \theta)$ *of Definition G.2 of the supplemental material. Call two cut terms equivalent when one arises from the other by finitely many of Bałaban's three completions of* [3, pp. 420, 429–432] *(the counterterm partner; the two-vector-leg class; the permutation of the external scalar legs of a divergent* $\phi\phi A$ *subgraph) applied at divergent subgraphs containing no line of* S*, the objects of* S *carried along to the image slots. Then:* (i) *the equivalence classes (orbits) partition the cut terms, and every move keeps the label multiset and the number and kinds of the objects of* S*;* (ii) *a connected line set of* Γ *avoiding* S *is a subgraph of the cut graph* $\Gamma\setminus S$ *with the same vertices, slots, slot types, order and degree* [3, (2.1)–(2.3)]*, and cutting commutes with the moves, so that the cut graphs of one orbit form exactly one renormalised class of the cut graph, the product of the classes of its connected components;* (iii) *at fixed labels, after the sum over the separable indices of the cut kernels, the sum of the cut terms of an orbit* O *is* $(-1)^{|S|}2^{-|S_L|}r_O^{-1}c_*\prod_K E(\mathrm{Cl}(K);$ *its fields), the product over the components* K *of the class sums with the cut kernels as external fields,* c_* *a constant and* $r_O \geq 1$ *the number of relabellings of identical* S*-objects preserving the class;* (iv) *the sum over all orbits is the full expansion;* (v) *if* $\mathcal{G}$ *is a per-label full class, the cut terms with a given cut datum (components with labels and orders, stubs and fields at each component, incidence of the* S*-objects) correspond* s_δ*-to-one to the tuples of members of the per-label full classes of the components, members without internal line and of order* 0 *included; each such group is a union of orbits and is closed under the exchange of the scalar legs of every two-scalar-leg or* $\phi\phi A$ *subgraph.*

The proof is given in the supplemental material (Lemmas G.3 and G.5 there). This is Bałaban's regrouping step in the proof of [3, Prop. 1, p. 432] ("if we remove a line in any graph of the class, then for the obtained graph we can find in this class all graphs necessary to form a new renormalized class"), made explicit for decomposed objects, with the sharper conclusion that one orbit gives exactly one class of the cut graph. It is needed wherever the renormalisation bound is applied to a family some of whose lines have been cut, in particular in the comparison of two propagator domains and in the telescoping of the cube's large-field remainders; at a nonzero background, where the bound of [3, Prop. 1] is available for per-label full classes, it is used in the form (v).

Remark 5.3 (the cube). The propositions below are stated for King's RG cube C_J^K, on which King's Dirichlet weight is compatible with the image group of the two Dirichlet planes (they are $2MJ$ apart). On the comparison cube ND$^{(a)}$ the planes are $2MJ + \varepsilon_k$ apart, no partition into consecutive blocks is invariant under the image group, and the flow does not exist; this is why Theorem 3.3 is posed on King's convention, and why the fixed-cube limits of the comparison family are reached only through Corollary 4.4. The cube of the existence theorem is C_J^K itself (Remark 3.5).

5.3. **The inputs at zero scalar source.**

Proposition 5.4 ((H0rel)$_C$: the small-field representation on King's cube, zero scalar source)**.** *Fix* $J \geq J_0$*, a family* $f_1, \dots, f_N$ *and* $x \in \mathbb{R}^N$*, and let* $s \in \{u_x, 0\}$*. There are* $\eta_1 > 0$*,* k_0 *and*

$C, c, \sigma > 0$, depending on the couplings, on J, on f and x, and on nothing else (not on k, j), such that for all $k \geq k_0$ and all j with $J_* \leq j \leq k - J_*$ and $\eta = 5^{j-k} \leq \eta_1$:

$$e^{-C\eta^\sigma} I_k^{(j)}(s) \;\leq\; Z_k(s) \;\leq\; e^{C\eta^\sigma} I_k^{(j)}(s) + \rho(\eta)\, Z_k(0), \qquad \rho(\eta) := e^{-cp(\eta)^2 + C}, \tag{2}$$

where $p(\eta) = b_0(1 + \ln \eta^{-1})^p$, with $p = 6$, compatible with the conditions [1, (3.54)]*, is Bałaban's small-field threshold. Moreover, at $s = 0$ the two terms of the upper line are the two parts of Bałaban's decomposition: $\int P\, d\mu_{k,0} \leq e^{C\eta^\sigma} I_k^{(j)}(0)$ and $\int (1 - P)\, d\mu_{k,0} \leq \rho(\eta) Z_k(0)$ ((R_0)).*

The proof is given in the supplemental material. This is [15, Thm. 3.1] on the cube, which [16, Sect. 5.2] asserts ("the proof follows by modifying suitably the material in [Ba 1] and [Ba 2]"), in the form in which the remainder is measured against $Z_k(0)$ rather than against an absolute constant, and with constants uniform in the step j (the form the diagonal pairs need, since $\eta_k \to 0$). The lower line is the weaker of the two readings of the sign in [15, (3.3)]. No lower bound on $Z_k(0)$ appears. Of the statements of Section 5.2, (BAL_5)[int] (through Bałaban's torus bounds) and $(\mathrm{REN}_{\mathrm{orb}})$ (through the comparison of propagator domains and the telescoping of the large-field remainders) enter.

Proposition 5.5 ($(\text{K-pos})_N$: one-step bound on the normalised action, zero scalar source)**.** *With J, f, x and $s \in \{u_x, 0\}$ as above there are $\gamma \in (0,1)$, $\beta, \sigma > 0$ and $C_3 < \infty$, depending on the couplings, J, f, x only, such that for all $k \geq k_0$, all $J_* \leq j \leq k - J_*$ with $\eta = 5^{j-k} \leq \eta_1$, and all (A_j, ϕ_j) in the support of χ_j,*

$$\big|S_{N,k}^{(j)}(A_j, \phi_j; s) - S_{N,k+1}^{(j+1)}(A_j, \phi_j; s)\big| \;\leq\; C_3\big(5^{-\gamma j}\eta^{-\beta} + \eta^\sigma\big). \tag{3}$$

The proof is given in the supplemental material. This is [15, Thm. 3.4] at $n = 1$ on the cube, for the field- and source-dependent part of the effective action: the two members of a diagonal pair live on one unit lattice with one indicator χ_j, and the statement compares them there. It is assembled from six pieces: the photon quadratic form (an even-restriction identity reduces the cube to the doubled torus, where [15, Prop. 3.10] applies); the scalar quadratic form at zero background (the Dirichlet symbol) and with background (a two-leg tree graph); the background determinant (one-loop graphs with a remainder, [15, (3.94)–(3.98)], of which the middle line of (3.98) is false as printed and is replaced); the graphs with curvature vertices; the source-free field-dependent graphs away from the faces and near them (the photon face terms and the scalar boundary remainders of [16, pp. 331–332]); and the block-vertex classes. For the curvature-vertex and the source-free field-dependent classes, the localisations whose tree length R exceeds the small-field threshold p are bounded by cutting the localised graph into clusters of mutual distance $> 3\sqrt{d}$ and reducing each cluster in its own box of side $O(p)$, with the cut lines as external lines of both clusters (Corollary B.37(b) and Lemma B.43; the zone-B tails by Theorem B.52(a) and Theorem B.59(d), all in the supplemental material). The vacuum constant $v_{k,j}$ is not part of the statement.

Lemma 5.6 (bounds for free)**.** *Put $\beta_f := C_0 \sum_i |f_i|_4^2$ with C_0 as in Lemma 2.6. For all $k \geq 0$, $J \geq J_0$ and $x \in \mathbb{R}^N$,*

$$e^{-\beta_f |x|^2} \;\leq\; a_k(x) \;\leq\; e^{\beta_f |x|^2}.$$

Proof. By Theorem 7.6(a), $a_k(x) \leq \exp(\|u_x\|_{2,k}^2)$, and $\|u_x\|_{2,k} \leq \sum_i |x_i| \|f_i\|_{2,k} \leq |x| (\sum_i \|f_i\|_{2,k}^2)^{1/2} \leq |x|\, \beta_f^{1/2}$ by Lemma 2.6(E3). The lower bound is $a_k(x) a_k(-x) \geq 1$, Lemma 2.11(e). □

5.4. **The Cauchy property in the spacing.**

Theorem 5.7 (the Cauchy property, ratio form)**.** *Fix $J \geq J_0$, f, $x \in \mathbb{R}^N$, and assume (2) and (3) for $s \in \{u_x, 0\}$ with the constants of Propositions 5.4 and 5.5. Choose $\alpha \in (0, \min\{\frac{1}{2}, \gamma/(\gamma + \beta)\})$ and the diagonal pairs $j(k) = \lfloor (1 - \alpha)k \rfloor$, $\eta_k = 5^{j(k)-k}$. Put*

$$b_k := C\eta_k^\sigma, \qquad \theta_k := C_3\big(5^{-\gamma j(k)}\eta_k^{-\beta} + \eta_k^\sigma\big), \qquad \rho_k := \rho(\eta_k), \qquad r_k(0) := 2\rho_k, \qquad r_k(u_x) := 2\rho_k e^{\beta_f |x|^2}.$$

Then there is k_1 such that for all $k \geq k_1$

$$\big|\ln a_{k+1}(x) - \ln a_k(x)\big| \;\leq\; 4b_k + 2\theta_k + r_k(u_x) + r_k(0). \tag{4}$$

The right side is summable in k, and its tail from k on is at most $C'5^{-\kappa k}$ for some $\kappa > 0$. Hence $a_k(x)$ converges, and its limit lies in $[e^{-\beta_f|x|^2}, e^{\beta_f|x|^2}]$.

Proof. Fix $k \geq k_0$ so large that $j(k) \geq J_*$, $\eta_k \leq \eta_1$ and $r_k(u_x) \leq 1$ (possible by Step 6), and put $j := j(k)$, $b := b_k$, $\theta := \theta_k$. For $s \in \{u_x, 0\}$ write $Z := Z_k(s)$, $I := I_k^{(j)}(s)$, $\hat{I} := \hat{I}_k^{(j)}(s)$ for the k-model after j steps, and $Z' := Z_{k+1}(s)$, $I' := I_{k+1}^{(j+1)}(s)$, $\hat{I}' := \hat{I}_{k+1}^{(j+1)}(s)$ for the $(k+1)$-model after $j+1$ steps. Both members have the block scale η_k and the indicator χ_j, and (2) applies to both, since $j+1 \leq (k+1) - J_*$; their remainders are $R := \rho_k Z_k(0)$ and $R' := \rho_k Z_{k+1}(0)$.

Step 1 (the remainders relative to Z). For $s = 0$, $R/Z = \rho_k = r_k(0)/2$ and $R'/Z' = \rho_k$. For $s = u_x$, $Z = a_k(x)Z_k(0) \geq e^{-\beta_f|x|^2} Z_k(0)$ by Lemma 5.6, so $R/Z \leq \rho_k e^{\beta_f|x|^2} = r_k(u_x)/2$, and likewise $R'/Z' \leq r_k(u_x)/2$. In both cases $R/Z,\ R'/Z' \leq r_k(s)/2 \leq \frac{1}{2}$. No lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond Lemma 5.6 is used.

Step 2 (one model). From $e^{-b}I \leq Z \leq e^b I + R$: $I \leq e^b Z$ and $I \geq e^{-b}(Z - R) = e^{-b}Z(1 - R/Z)$. So $I \in [e^{-b}Z(1 - R/Z),\ e^b Z]$, and in the same way $I' \in [e^{-b}Z'(1 - R'/Z'),\ e^b Z']$. In particular $I, I' > 0$.

Step 3 (across the pair). On the support of χ_j, (3) gives $|S_{N,k}^{(j)}(\cdot; s) - S_{N,k+1}^{(j+1)}(\cdot; s)| \leq \theta$. The two integrals $\hat{I}$, $\hat{I}'$ run over the same variables with the same χ_j and have non-negative integrands, so $\hat{I}'/\hat{I} \in [e^{-\theta}, e^{\theta}]$. Since $I = e^{-v_{k,j}}\hat{I}$ and $I' = e^{-v_{k+1,j+1}}\hat{I}'$,

$$I'/I \ \in \ e^{\Delta}\,[e^{-\theta}, e^{\theta}], \qquad \Delta := v_{k,j} - v_{k+1,j+1},$$

and Δ does not depend on s: the vacuum constant is the same number in the numerator and in the denominator of a_k.

Step 4 (ratio of partition functions). By Steps 2 and 3,

$$\frac{Z'}{Z} = \frac{Z'}{I'} \cdot \frac{I'}{I} \cdot \frac{I}{Z} \ \in \ e^{\Delta}\Big[e^{-2b-\theta}\big(1 - \tfrac{R}{Z}\big),\ e^{2b+\theta}\big(1 - \tfrac{R'}{Z'}\big)^{-1}\Big].$$

With $|\ln(1-t)| \leq 2t$ for $0 \leq t \leq \frac{1}{2}$ and Step 1, $|\ln(1 - R/Z)|$, $|\ln(1 - R'/Z')| \leq r_k(s)$, hence

$$\big|\ln(Z'/Z) - \Delta\big| \ \leq \ 2b + \theta + r_k(s).$$

Step 5 (ratio of ratios). $\ln \dfrac{a_{k+1}(x)}{a_k(x)} = \Big[\ln \dfrac{Z_{k+1}(u_x)}{Z_k(u_x)} - \Delta\Big] - \Big[\ln \dfrac{Z_{k+1}(0)}{Z_k(0)} - \Delta\Big]$, and Step 4 at $s = u_x$ and at $s = 0$ gives (4). (If the constants of (2), (3) differ between the two sources, take the larger ones.)

Step 6 (summability). Since $k - j(k) \geq \alpha k$, $\eta_k \leq 5^{-\alpha k}$, so b_k and $C_3\eta_k^{\sigma}$ decay geometrically. Since $j(k) \geq (1-\alpha)k - 1$ and $k - j(k) \leq \alpha k + 1$,

$$5^{-\gamma j(k)}\eta_k^{-\beta} \ \leq \ 5^{\gamma+\beta}\,5^{-(\gamma(1-\alpha)-\beta\alpha)k},$$

which decays geometrically because $\alpha < \gamma/(\gamma+\beta)$. Finally $p(\eta_k)^2 \geq b_0^2(1 + \alpha k \ln 5)^2 \geq c_1 k^2$ for $p \geq 1$, so $\rho_k \leq e^{-cc_1k^2 + C}$; thus $r_k(u_x) \leq 1$ for large k, $\sum_k r_k(s) < \infty$, and the tail of $\sum_k \rho_k$ is eventually below any geometric tail. The series in (4) converges with a geometric tail, so $(\ln a_k(x))_k$ is Cauchy, $a_k(x)$ converges, and the limit lies in the closed interval of Lemma 5.6. □

Remark 5.8 (what the argument does not use). (i) No convergence, and no bound, of the vacuum constants $v_{k,j}$ (King's E_0, E_1, the kernel normalisations, the face vacuum energies) enters: they cancel in Step 3. King's printed route [16, p. 329] goes through the convergence of Z_k itself, which requires exactly those constants to converge. (ii) No upper stability bound, no lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond Lemma 5.6 is used; the remainder of (2) is compared with $Z_k(0)$ only. (iii) No reflection positivity, no Ward identity, and no subsequence. (iv) The all-n form of [15, (3.13)] (comparing the k-model with the $(k+n)$-model at a fixed stopping scale η for every n) also gives the Cauchy property, provided the constants of (2) and (3) are taken at fixed η with no growth in k: then $|\ln a_{k+n} - \ln a_k| \leq 4b(\eta) + 2\theta_k(\eta) + 4\rho(\eta)(e^{\beta_f|x|^2} + 1)$ for every n, and one lets $k \to \infty$ and then $\eta \to 0$. The diagonal form above uses $n = 1$ only, at the price of constants uniform in j; the two forms need different readings of the inputs, and neither implies the other.

Theorem 5.9 (King's existence theorem on the cube: the curvature field)**.** *For every finite family $f_1, \dots, f_N$ of real 2-forms in $C_c^\infty(\mathbb{R}^3; \Lambda^2)$, every $J \geq J_0(f)$ and every $x \in \mathbb{R}^N$ the limit*

$$Z_J^K(x) = \lim_{k \to \infty} \langle e^{\sum_i x_i F(f_i)} \rangle_{k,J}^K$$

exists and lies in $[e^{-\beta_f |x|^2}, e^{\beta_f |x|^2}] \subset (0, \infty)$, with β_f as in Lemma 5.6; the convergence is geometric. That is, Theorem 3.3 holds. Consequently Z_J^K and Z_∞^K exist locally uniformly on $\mathbb{C}^N$ and are entire, with the bounds and the monotonicity of Theorem 7.6(b).

Proof. Propositions 5.4 and 5.5 supply the hypotheses of Theorem 5.7 at every x; the theorem gives the limit and its bounds; Theorem 7.6(b) (that is, Corollary 4.4(ii)) gives the rest. The constants k_0, η_1 and the rates may depend on x and on f; existence needs no uniformity in x. □

Remark 5.10 (the comparison family). Theorem 5.9 says nothing about the fixed-cube limits of the comparison family $\mathrm{ND}^{(a)}$ (Remark 5.3); for that family Corollary 4.4(ii)(γ) squeezes the limits inferior and superior at fixed J between Z_J^K and Z_{J+1}^K, and Corollary 4.4(iii) identifies every subsequential limit, followed by $J \to \infty$, with Z_∞^K. At fixed J every inequality available for the comparison family bounds its oscillation in k only by the difference of two distinct limits, which tends to zero only as $J \to \infty$; a Cauchy statement at fixed J for that family would need the small-field analysis at its faces, where no flow of the kind used above exists.

6. King's existence theorem on the cube with scalar sources

This section proves Theorem 3.6, the existence theorem with scalar sources in the form the Wick leg consumes: in the reference scheme z, the fixed-cube limits of the generating functional with curvature and scalar sources exist, locally uniformly and holomorphically, on the domain of coefficient vectors whose positive scalar part is below a threshold, and they are positive at real points; the negative scalar part is unrestricted. The route is the Cauchy argument of Section 5 with the scheme c-number carried along (Theorem 6.14), applied at real points with a small non-negative scalar source, followed by a normal-family argument on the domain. The domain is bounded by Hölder's inequality from three one-sided estimates: the curvature bound of Lemma 5.6, a positive-source bound (Proposition 6.10), and a negative-source bound (Theorem 6.16), the last obtained from positive-source statements at a heavier scalar mass by an exact mass shift and a correlation inequality. The notation is that of Section 5.1.

6.1. **Setting; the source-dependent graphs.** Throughout, the scheme is the reference scheme $z = a^\sharp$ of Definition 2.3; write z_k for its k-th term. Put $\mathsf{S}_k(h) := \sum_{x \in \Lambda_k \cap C_J} \varepsilon_k^3 h(x)$ and $c_k(h) := z_k \mathsf{S}_k(h)$, which is linear in h. For a real source $s = (u, h)$ the normalised functional in the scheme is

$$n_k(s) \;:=\; e^{-c_k(h)}\, N_k(s) \;=\; e^{-z_k \mathsf{S}_k(h)}\, \langle e^{F(u) + :\phi^2:(h)} \rangle_{k,J}^K \;=\; Z_{k,J}^{z,K}(s),$$

the object of Theorem 3.6. For a finite family $f_1, \dots, f_n$ of real 2-forms and $h_1, \dots, h_p$ of real functions, all supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and $\zeta = (z, w) \in \mathbb{C}^{n+p}$, $n_k(\zeta) := e^{-\sum_j w_j c_k(h_j)} \langle e^{\sum_i z_i F(f_i) + \sum_j w_j :\phi^2:(h_j)} \rangle_{k,J}^K$ (here z_i are coefficients and z_k the scheme term; no confusion will arise). Every n_k is entire in ζ, $|n_k(\zeta)| \leq n_k(\mathrm{Re}\, \zeta)$, and $\ln n_k$ is convex on real points (Lemma 2.11(d)): a log-Laplace transform minus a linear function.

A second scalar mass $M > 0$ will be used with the *same* e, λ, μ_0, lattice, cube and boundary conditions; $\langle \cdot \rangle^{(M)}$ denotes King's cube state at mass M, $\alpha_k(M) := M^2 + \delta m_{\varepsilon_k}^2(M)$ King's free-boundary bare coefficient at mass M (as in Lemma 3.10), $u_k(M) := \frac{1}{2}(\alpha_k(M) - \alpha_k(m))$, and $s_-(M) := \liminf_k u_k(M)$. Every statement of Section 5 and of this section holds at every point of King's range with point-dependent constants, in particular at the point (e, λ, M, μ_0); we use this at $M = m$ and at one heavier M_2.

Lemma 6.1 ((K_h): one-step convergence of the source-dependent graphs on King's cube)**.** *Let $P_k^{(j),1}$ be the interaction part of King's effective action after j steps on C_J^K [15, (3.17)], written as a sum over connected renormalised graphs H, and let $\mathcal{N}_h$ be the family of its members that*

depend on the scalar source h: the field-dependent part of every graph with an h-vertex, and every field-independent h-dependent graph except the h-linear ones of the three lowest orders $(\alpha,\beta)\in\{(0,0),(0,1),(2,0)\}$ in (e^2,λ). Then for every fixed real $h\in C_c^\infty$ supported at distance >1 from $\mathbb{R}^3\setminus C_J$ and every member of $\mathcal{N}_h$, for $J_*\le j\le k-J_*$: the localisation bound [15, (3.38)]*, with the factor $(1+k-j)^{p_0}$, holds for its localisations at distance from the faces larger than the small-field scale; the bound* [15, (3.38)] *with the restricted degree holds with the factor $(1+k-j)^{p_0}$, summed over the localisations sharing the cube of an h-vertex, and for each single localisation at distance >1 from the faces; and the sum over positions of the difference between the member at (k,j) and the member at $(k+1,j+1)$, on the common unit lattice, is bounded by $C\big[5^{-\gamma' j}\eta^{-\beta'}+\eta^{\sigma'}\big]\,|C_J|$ with C independent of j and k.*

This is the cube form, on C_J^K with King's Dirichlet weight, of the corresponding statement on King's tori; the h-free graphs, those with a curvature vertex and no h-vertex and the source-free ones, are covered on the cube by Proposition 5.5. The localisations of a member of $\mathcal{N}_h$ with tree length $R\le p$ lie at distance >1 from the faces, where the bounds reduce to those of the free flow on the torus (Proposition 6.4); a localisation with $R>p$ carries the factor $e^{-\delta R}$ once the localised graph is cut into clusters of mutual distance $>3\sqrt{d}$, each reduced in its own box of side $O(p)$, the clusters containing the source vertex lying away from the faces and the clusters meeting the Dirichlet faces being source-free (Proposition 6.5). The proof is given in Section 6.2.

6.2. **Proof of Lemma 6.1.** We work in the units of depth j of the proof of Proposition 5.5 in the supplemental material: the live sites are T_η, F denotes the scalar Dirichlet planes, $t(x):=\operatorname{dist}(x,F)$, and a localised graph H carries King's boxes $\square\subset\square''\subset\square'$ and $\Omega=\square'\cap T_\eta$, in zone I if $\square'\cap F=\emptyset$ and in zone B otherwise. The localisation is by centres, as in the proof of Proposition 5.5: a tuple τ has the point set $P(\tau)$, $N:=|P(\tau)|\le N_*$ with N_* depending on the order $\bar n$ of the expansion only, and the tree length $R=R(\tau):=\operatorname{dist}(P(\tau))$. The cut by clusters is that of Lemma B.33 of the supplemental material: $c_0:=\frac{13}{3}\sqrt{3}$, $R_0:=3\sqrt{3}$, the clusters K are the components of the c_0-proximity graph on $P(\tau)$, every line whose ends lie in different clusters is cut and kept whole as an external line of both cluster graphs H_K, and H_K is reduced with its own boxes $\square_K\subset\square''_K\subset\square'_K$ and gauge $B_K=A_j(x_0^K)$; put $\Omega_K:=\square'_K\cap T_\eta$. A cluster is in zone I if $\Omega_K\cap F=\emptyset$ (then $\square'_K$ is not truncated and $\Omega_K=\square'_K$) and in zone B otherwise. The two members of the diagonal pair, the k-model at depth j and the $(k+1)$-model at depth $j+1$, live on the same unit lattice; every bound below holds for each of them separately with the same constants, and the two are compared only in zone I.

The h-vertex is King's vertex [15, (3.21)]: the smooth function h sampled at the points of the lattice's own frame, the coupling $\theta\eta^2$ in the units of depth j, and at most two fields $\phi^{(j)}$. In King's count of the coupling power $\rho(H)=\sum_v\rho_v$ of a graph [15, p. 665], every vertex has $\rho_v-s_v/4\ge 0$, s_v its number of external scalar fields, and the h-vertex has $\rho_h-s_h/4\ge 2-\frac12=\frac32$; hence

$$\rho(H)-\tfrac14 s(H)\ \ge\ \tfrac32\qquad\text{for every member of }\mathcal{N}_h\text{ with external fields,}\tag{5}$$

$s(H)$ its number of external scalar legs, and $\rho(H)\ge 0$ for the field-independent members. The Wick constant of [15, (3.21)] is field-free and h-linear of the lowest order, so it enters only the excluded family $\mathcal{T}_h$ of Definition D.1 of the supplemental material and no member of $\mathcal{N}_h$ depends on it.

The range. Let $\eta_2\le\eta_1$ be so small that for $\eta\le\eta_2$, with $p=p(\eta)$ and M the block integer of C_J,

$$\eta^{-1}\ \ge\ 4p+8M+2c_0N_*^2+8,\qquad p\ \ge\ c_0N_*^2,\qquad C_\flat\,e\,\eta^{1/2}\,p^2\ \le\ \tfrac12,\tag{6}$$

where $C_\flat$ is the constant C_{box} of Lemma 6.2(e) below times $\sqrt{3}$ and the constants of the difference bound [15, (3.45)] and of the operator bound in Lemma J.27(b) of the supplemental material. All three hold for η small because $p(\eta)$ grows slower than every power of η^{-1} and faster than every logarithm. The first condition contains the corresponding condition of that proof. Since $C_{\mathrm{box}}p$ bounds the side of every cluster box $\square'_K$ and also the side $\le 5p+4M+2$ of the one box $\square'$ of a localisation with $R\le p$ (Lemma 6.2(e)), the third condition contains the convergence

condition $Ce\eta^{1/2}p\sqrt{3}\,(5p+4M+2) \le \frac{1}{2}$ of the expansion in $\tilde{A}$ on that one box (Lemma J.27 of the supplemental material with $\Omega_K \to \Omega$), as well as the one on every cluster box. J_* is taken so large that $\eta = 5^{-(k-j)} \le \eta_2$ and $C_{\mathrm{PW}}e\eta^{1/2}p(\eta) \le \theta_{\mathrm{PW}}$ (Lemma J.25 of the supplemental material) whenever $k-j \ge J_*$; the least such integer is the largest of the thresholds of the two geometric conditions of (6), of the convergence conditions of the expansions in the cluster fields and on the one box, and of the smallness of the bounds at the background.

Lemma 6.2 (the source vertices and the clusters)**.** *Let $\eta \le \eta_2$, let H be a member of $\mathcal{N}_h$, s an h-vertex of H with centre z_s, and τ a tuple of H. Then:* (a) $t(z_s) \ge \eta^{-1} - 3$. (b) *If τ is in zone B, then $R(\tau) \ge \eta^{-1}/2$.* (c) *If τ is in zone B, then τ has at least two clusters.* (d) *Every cluster K containing an h-vertex has $\operatorname{dist}(\square_K, F) \ge 2p+M+1$, $\Omega_K = \square'_K$ and $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$: it is a zone-I cluster. Hence every zone-B cluster is source-free.* (e) *Every $\square'_K$ has side $\le c_0N_* + 4p + 6M$, and the one box $\square'$ of a localisation with $R \le p$ has side $\le R+4p+4M+2 \le 5p+4M+2$; both are $\le C_{\mathrm{box}}\,p$, where $C_{\mathrm{box}} := \max(c_0N_*+6M+4,\, 4M+7)$. The same holds with a curvature vertex in place of the h-vertex.*

Proof. (a) $h(x) \ne 0$ forces ℓ^∞-$\operatorname{dist}(x, \mathbb{R}^3 \setminus C_J) > 1$ in physical units, that is $> \eta^{-1}$ unit lengths; the planes F lie within one fine spacing $5^{-j} \le 1$ of ∂C_J, and the vertex lies in $\operatorname{supp}\chi_{z_s}$, within $\frac{2}{3}\sqrt{3} < 2$ of its centre. (b) A tree of total length R joins the points of $P(\tau)$, so all of them lie within R of z_s; the vertices lie within $\frac{2}{3}$ of these points in each coordinate, so $\square$ lies in the sup-norm cube of half-side $R+M+1$ about z_s, and $\square'$, which adds at most $2p+M$ on each face (as in (e)), in that of half-side $R+2p+2M+1 \le R+2p+3M$. If $\square'$ meets F, then $t(z_s) \le R+2p+3M$, and with (a) and (6) $R \ge \eta^{-1} - 2p - 3M - 3 \ge \eta^{-1}/2$. (c) With one cluster $\operatorname{diam} P(\tau) \le c_0(N-1)$ (Lemma B.33(c) of the supplemental material), so $R(\tau) \le (N-1)\operatorname{diam} P(\tau) \le c_0(N_*-1)^2 < \eta^{-1}/2$ by (6), against (b). (d) If $s \in K$, every point of P_K lies within $c_0(N-1)$ of z_s, so $\square_K$ lies in the cube of half-side c_0N+M about z_s and $\square'_K$ in that of half-side $c_0N+2p+3M$. By (a) and (6), $t(z_s) - (c_0N+2p+3M) \ge \eta^{-1} - 3 - c_0N - 2p - 3M > 2p+M+1$. So $\square'_K$ misses F, it is not truncated, it contains every large block within $2p$ of $\square_K$, and $\operatorname{dist}(\square_K, F) \ge t(z_s) - c_0N - M \ge 2p+M+1$. (e) $\square_K$ has side $\le \operatorname{diam} P_K + \frac{4}{3} + 2M \le c_0N_* + 2M$ (the vertices lie within $\frac{2}{3}$ of the points in each coordinate, $\operatorname{diam} P_K \le c_0(N_*-1)$ by Lemma B.33(c) of the supplemental material, and $c_0 > \frac{4}{3}$) and $\square'_K$ adds at most $2p+M \le 2p+2M$ per side, as for the one box below; truncation at ∂T_η only shrinks it. For the one box: the points of $P(\tau)$ lie within R of one another, and every vertex lies in the support of its localisation weight, within $\frac{2}{3}$ of its centre in each coordinate; so $\square$ has side $\le R + \frac{4}{3} + 2M \le R+2M+2$, and $\square'$ adds on each face at most the large blocks within $2p$ of $\square$, that is at most $2p+M$; hence its side is $\le R+4p+4M+2$. Since $p \ge 1$, $c_0N_*+4p+6M \le (c_0N_*+6M+4)p$ and $5p+4M+2 \le (4M+7)p$. The curvature source f is supported at the same distance from $\mathbb{R}^3 \setminus C_J$, so the same arithmetic applies to its vertex. $\square$

Lemma 6.3 (cut lines vanish on the Dirichlet planes)**.** *Let ℓ be a scalar, dressed or external cut line of Lemma B.33 of the supplemental material with its end u in a cluster K, in either model, at the background $A^{(j)}$ or at zero background. Put $m(u') := \min(1, t(u'))$ for a scalar or an external line and $m(u') := m_\alpha(u';1) = \min(1,t(u'))^\alpha$ for a dressed line (Lemma B.50 of the supplemental material); in every case $m(u') \le m_\alpha(u';1)$.* (a) *For every evaluation point u' of the u-end with $t(u') \le \frac{1}{4}$ and every operation O_v at the far end (a difference or Hölder quotient of order $\le 1+\alpha$, or the identity) whose evaluation set E_v lies at distance $\ge \frac{1}{2}$ from u',*

$$|O_v\ell(u',\cdot)| \;\le\; C\,m(u')\,e^{-\delta\operatorname{dist}(u',E_v)}.$$

(b) *For a scalar or dressed cut line and evaluation points u', v' of its two ends, both undifferentiated, with $|u'-v'| \ge 1$ and $t(u'), t(v') \le \frac{1}{4}$,*

$$|\ell(u',v')| \;\le\; C\,m(u')\min(1,t(v'))\,e^{-\delta|u'-v'|} \;\le\; C\,m_\alpha(u';1)\,m_\alpha(v';1)\,e^{-\delta|u'-v'|}.$$

The constants depend on those of Lemma B.34(b) and, for dressed lines, also on those of Lemma B.50(i), Lemma J.25 and (114) *of the supplemental material.*

Proof. Each of the three kinds of line vanishes when its u-end lies on F: a scalar line by the Dirichlet condition of the cube propagator, a dressed line and an external line through their first factor, which is such a propagator.

(a) for scalar and external lines. Let $u'_F \in F$ be the foot of u' and telescope along the $t(u')/5^{-j}$ bonds of the normal segment from u'_F to u'; the parallel transporters along a bond are unimodular, so $|O_v\ell(u',\cdot)| \le t(u')\max_b |O_vD_b\ell(\cdot,\cdot)|$, the maximum over the bond differences D_b on that segment. Every such bond lies within $\frac14$ of u', hence at distance $\ge \frac14$ from E_v, and the mixed bound of Lemma B.34(b) of the supplemental material ($C^{1,\alpha}\otimes C^{1,\alpha}$ at unit scale in both arguments, decay $e^{-\delta_1|u-v|}$, no distance condition) bounds each by $Ce^{-\delta\,\mathrm{dist}(u',E_v)}$.

(a) for dressed lines. Write the line as $\sum_x \vartheta^3 G_A(u',x)\,c_{\ge j}\,G_A(x,\cdot)$, with ϑ the fine spacing (chains iterated likewise). Its first factor is not cut off near u', so the face factor is taken from the slices of the first factor at all scales. Each slice $G_{(i)}$ obeys the face clause of Lemma B.50(i) of the supplemental material with $\mathsf{q}=0$, $|G_{(i)}(u',x)| \le C\rho_i^{-1}m_\alpha(u';\rho_i)e^{-\delta|u'-x|/\rho_i}$: its proof uses only the factorisation $G_{(i)} = a_i^2\rho_i^{-4}(G_iQ_i^*)C^{(i)}(Q_iG_i)$ [15, (2.17)], the Hölder clause of the outer factor $G_iQ_i^*$ up to the faces and its vanishing on F, and the exponential decay of the coarse factor $C^{(i)}$. At the background $A^{(j)}$ the outer-factor clauses hold at every level i by [16, (5.2)–(5.3)], the background being regular at level i. The coarse factor $C^{(i)}(A) = K_i(A)^{-1}$ (defined in the supplemental material) is the lower right block of the inverse of the block operator with diagonal entries $M(A)$ and $D_i := a_i + a5^{-2}Q^*Q$ and off-diagonal entries $-a_iQ_i^*$ and $-a_iQ_i$, which is positive, its quadratic form being $\langle g_1, M_0(A)g_1\rangle + a_i\|g_2 - Q_ig_1\|^2 + a5^{-2}\|Qg_2\|^2$. The Schur complement of D_i, with $QQ^*=I$, $QQ_i = Q_{i+1}$ and $a_{i+1}^{-1} = a^{-1} + a_i^{-1}5^{-2}$, gives $C^{(i)} = D_i^{-1} + a_i^2D_i^{-1}Q_iG'_{i+1}Q_i^*D_i^{-1}$ with $G'_{i+1} := (M_0(A) + a_{i+1}5^{-2}Q_{i+1}^*Q_{i+1})^{-1}$, the propagator of level $i+1$ in the units of level i. Since $D_i^{-1} = a_i^{-1}(I-Q^*Q) + (a_i + a5^{-2})^{-1}Q^*Q$ is block-local and bounded, the decay of $C^{(i)}$ is the outer-factor bound at level $i+1$, applied to the block-local bounded sources $Q_i^*D_i^{-1}\delta_y$. Since $m_\alpha(u';\rho_i) \le \rho_i^{-\alpha}m_\alpha(u';1)$ for $\rho_i \le 1$, the sum over the slices gives

$$|G_A(u',x)| \;\le\; C\,m_\alpha(u';1)\,(|u'-x|+\vartheta)^{2-d-\alpha}\,e^{-\delta|u'-x|/2},$$

a kernel that is locally summable in $d=3$. The sum over x against the constant $c_{\ge j}$ and the last factor $O_vG_A(x,\cdot)$, which (PW1) of Lemma J.25 of the supplemental material bounds, is the convolution of two kernels of the form of (PW1), and (114) of the supplemental material bounds it by $C\,m_\alpha(u';1)\,e^{-\delta\,\mathrm{dist}(u',E_v)}$, since $\mathrm{dist}(u',E_v) \ge \frac12$.

(b). Put $f(w) := \ell^\flat(u',w) = \bar\psi(u'-w)\ell(u',w)$; then $f(v') = \ell(u',v')$, since $|u'-v'| \ge 1$. Let $v'_F \in F$ be the foot of v'; $f(v'_F) = 0$ by the Dirichlet condition in the second variable (for a dressed line through its last factor). Since $|u'-v'| \ge 1$ (on the supports even $|u'-v'| \ge R_0 - 4\cdot5^{-j} > 1$, Lemma B.34(a) of the supplemental material) and $t(v') \le \frac14$, every point of the segment from v'_F to v' lies at distance $\ge \frac34$ from u'. Telescope along its $t(v')/5^{-j}$ bonds, with unimodular transporters: $|f(v')| \le t(v')\max_b|D_bf|$. On each bond, D_bf is $\bar\psi(u'-\cdot)D_b\ell(u',\cdot)$ plus the difference of the fixed smooth cutoff $\bar\psi$, of size $\le \|\nabla\bar\psi\|_\infty$, times $\ell(u',\cdot)$ at an end of the bond; $\bar\psi$ is differentiated once only. Every point involved lies at distance $\ge \frac34 - 5^{-j} \ge \frac12$ from u' and within $\frac14 + 5^{-j}$ of v', so (a), with $O_v := D_b$ and with O_v the identity, bounds both terms by $C\,m(u')e^{-\delta|u'-v'|}$. Hence $|\ell(u',v')| \le C\,m(u')\min(1,t(v'))e^{-\delta|u'-v'|}$, and the second inequality of (b) is $m(u') \le m_\alpha(u';1)$ and $\min(1,t) \le \min(1,t)^\alpha$. Where $t > \frac14$ at an end, the factor of that end is $\ge \frac14$: if $t(v') > \frac14$, (a) with O_v the identity gives the bound; if $t(u') > \frac14$, the telescoping at v' above, with $D_bf = D_b\ell^\flat(u',\cdot)$ bounded by Lemma B.34(b) of the supplemental material, gives it; if both, Lemma B.34(b) alone. □

Proposition 6.4 (the branch $R \le p$: zone I)**.** *Let H be a member of $\mathcal{N}_h$ and τ a localisation of H in zone I; by Lemma B.20(i) of the supplemental material this covers every localisation with $R \le p$. Then, for each model, on* $\operatorname{supp}\chi_j$*,*

$$|E^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta R}, \tag{7}$$

which is [15, (3.38)] *with the factor* $(1+k-j)^{p_0}$*, and across the diagonal pair, on the common unit lattice,*

$$|E'^{(j+1)}(H;\tau) - E^{(j)}(H;\tau)| \;\le\; C\,\eta^{\rho(H)-1/2}\,p^{n_1}\big[5^{-\gamma j} + \eta^{3+\sigma+1/2}\big]\,e^{-\delta R}. \tag{8}$$

The constants depend on the point of King's range, J*,* $\bar n$*,* $\|h\|_{C^1}$ *and, for members with curvature vertices,* $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*; not on* j*,* k*,* τ*.*

Proof. The complete proof is given in the supplemental material, where the statement is restated as Proposition J.36. □

Proposition 6.5 (the branch $R > p$: the cut by clusters)**.** *Let* $\eta \le \eta_2$*, let* H *be a member of* $\mathcal{N}_h$ *and* τ *a tuple of* H *with at least two clusters. By Lemma 6.2(c) this covers every zone-B tuple, and by Lemma B.33(c) of the supplemental material and* (6) *every tuple with* $R > p$*. Then, for each model separately, on* $\operatorname{supp}\chi_j$*,*

$$|E_N^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta\,\operatorname{dist}(P(\tau))}, \tag{9}$$

E_N *the field-dependent part (the whole value for a field-independent member); this is* [15, (3.38)] *at the tuple. Moreover, for every unit cube* y_* *and* $0 < \delta' \le \delta/2$*,*

$$\sum_{\substack{\tau \text{ in zone } B\\ \text{with an } h\text{-vertex in } y_*}} |E_N^{(j)}(H;\tau)|\,e^{\delta' R(\tau)} \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta/(8\eta)}. \tag{10}$$

The constants depend on the point of King's range, J*,* $\bar n$*,* $\|h\|_\infty$ *and, for members with curvature vertices,* $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*.*

Proof. The complete proof is given in the supplemental material; in outline, at a tuple with at least two clusters every connected component of every cluster graph carries a cut-line leg (Corollary B.37(c) of the supplemental material), so the vacuum members and the component-wise cancellation of [15, p. 667] do not arise, and $E(H;A^{(j)};\tau)$ and $E(H;0;\tau)$ are bounded separately; at zero background the cut lines are the cube propagators at $A = 0$ and $B_K = 0$. Fix one of the two terms.

Step 0: regularised cut lines. Replace every cut line ℓ by $\ell^\flat$; by Lemma B.34(a) of the supplemental material this changes nothing.

Step 1: reduction per cluster. Each H_K is reduced by [15, (3.43)–(3.55)] with its own box Ω_K, centre x_0^K, gauge B_K and field $\tilde A_K = A^{(j)} - B_K$ on Ω_K (Lemma B.33 of the supplemental material), its cut lines keeping the background and receiving the unitary factors [15, (3.46)] at their ends. The expansion of the reduced propagator in $\tilde A_K$ converges on Ω_K: $e_j \sup_{\Omega_K}|\tilde A_K| \le Ce_j p\,\operatorname{diam}\Omega_K \le C_\flat e\eta^{1/2}p^2 \le \frac12$ by the difference bound [15, (3.45)], Lemma 6.2(e) and (6): this is the reduction of the branch $R \le p$ on a box of side $O(p)$, run inside each cluster. By Lemma B.33(a) of the supplemental material the partition, the boxes, the gauges and the cut status of every line are the same for every member of every family met in the nesting and for every position, so renormalisation families are formed inside each H_K. A zone-I cluster, in particular every cluster containing a source vertex (Lemma 6.2(d)), satisfies $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$ and $\Omega_K \cap F = \emptyset$, so Lemma B.21 of the supplemental material applies to H_K: it is its free f_D-flow version plus terms carrying an image line of any prescribed degree, with the cut lines as external legs; the vertices of the expansion in $\tilde A_K$ that lie in $\square'_K \setminus \square_K$ are joined to a localised vertex of K by a path of lines one of which is long, and that line supplies a credit of any order, as in step (2) of the proof of Theorem B.30 run inside H_K (Theorem B.36 of the supplemental material). A zone-B cluster is source-free and is rewritten by Definition B.45 of the supplemental material within Ω_K.

Step 2: the cut-line factors. Every evaluation point of a cut-line end in an output term is joined to a localised anchor of its own cluster by at most $C(\bar n)$ lines of that cluster's rewritten units (Theorem B.36 of the supplemental material), and by Lemma B.34(b) of the supplemental material the phase-dressed cut line satisfies $|O_u O_v \ell^{\flat,\mathrm{ph}}| \le Ce^{-\delta_2 d(K,K')}\times$(transfer factors on the lines of those paths), for all operations O_u, O_v of order $\le 1+\alpha$ at its ends, and $\le C$ crudely. The quotient graph (clusters as nodes, cut lines as edges) is connected because H is (Lemma B.33(d)

of the supplemental material); bound the cut lines of one spanning tree T of it by the first bound and the others by C.

Step 3: factorisation. The output term is bounded by $\prod_K \mathfrak{E}_K \times \prod_{\ell\in T} Ce^{-\delta_2 d(K,K')} \times C^{\#\text{cut lines}}$, where $\mathfrak{E}_K$ depends on the positions of K only; the transfer factors cost one extra decay draw per line of a cluster, within the budget of Theorem B.36 of the supplemental material.

Step 4: the cluster factors. Zone-I clusters. The cluster graph is a free f_D-flow graph with its cut lines as undifferentiated external legs of unit scale (Lemma B.34(b) of the supplemental material). For the cluster containing the h-vertex, the census is that of Proposition 6.4; since the cluster carries a cut leg, every component has an external leg and the excepted field-independent family does not occur. For the source-free zone-I clusters, the degree count holds at every tuple by Theorems B.24 and B.36, and block points by Lemma B.42 of the supplemental material. The multiscale bound Theorem B.41(a) of the supplemental material with the root held, summed over the root in Ω_K (volume $\le Cp^3$ by Lemma B.33(c) of the supplemental material), gives $\sum_{\text{positions of } K} \mathfrak{E}_K \le C\eta^{\rho_K} p^{n_K} (1+k-j)^{p_{0,K}}$, with ρ_K the coupling power of the vertices of K. *Zone-B clusters.* The cluster is source-free (Lemma 6.2(d)); apply Theorem B.52(a) and Theorem B.59 of the supplemental material with no vertex held to the renormalised family of H_K, localised at P_K with the box Ω_K of side $O(p)$; the hypothesis "at least one external field" is the cut leg. The proof of Theorem B.52 of the supplemental material uses an external scalar leg only in the following ways, each of which a cut line satisfies: through its sup and $(1,\alpha)$-norms on Ω_K up to the faces, in the Taylor and transport steps, the Hölder splits and the ordered induction (Lemma B.34(b) of the supplemental material); and, when the charging rule of Lemma B.51 of the supplemental material charges the leg at a $D'=-1$ remainder vertex v, through the vanishing factor $m_\alpha(v;1)$ of Lemma B.50(ii) of the supplemental material, which a cut line supplies by Lemma 6.3(a); only scalar, dressed and external cut lines are ever charged (Lemma J.46 of the supplemental material, (T2)). A cut line joining two zone-B clusters may be charged at both ends, by the charging rules of the two clusters; then Lemma 6.3(b) supplies both factors. After the borrowing of Lemma B.51 of the supplemental material the cut-line factor left in the output term is $\ell/[m_\beta(u';1)m_\beta(v';1)]$ (or with one factor), an undifferentiated evaluation bounded by $C\, m_{\alpha-\beta}(u';1)m_{\alpha-\beta}(v';1)e^{-\delta\,\mathrm{dist}} \le Ce^{-\delta\,\mathrm{dist}}$ by Lemma 6.3, since its face factors are $\le m_\alpha$ and $m_\alpha/m_\beta = m_{\alpha-\beta} \le 1$, which Steps 2–3 treat like every other cut-line factor. The resulting bound is again $\le C\eta^{\rho_K} p^{n_K}(1+k-j)^{p_{0,K}}$, the logarithm coming from the infrared sums inside Ω_K.

Step 5: the tree. By Lemma B.33(d) of the supplemental material, $\prod_{\ell\in T} e^{-\delta_2 d(K,K')} \le e^{\delta_2 c_0 N} e^{-\delta_2\,\mathrm{dist}(P(\tau))}$.

Step 6: exponents. $\rho(H) = \sum_K \rho_K$, the count being vertex-local (an unlocalised mass-counterterm vertex on a cut dressed line keeps its own coupling power); $n_1 := \sum_K n_K$ and $p_0 := \sum_K p_{0,K}$ plus the dressed-line factor of Lemma B.34(b) of the supplemental material; the number of clusters is $\le N_*$ and $\delta := \delta_2$. This is (9).

The root sum. On the tuples of (10), $R(\tau) \ge \eta^{-1}/2$ by Lemma 6.2(b). The number of tuples of H with a point in y_* and $R \in [n, n+1)$ is $\le C(n+1)^{3N_*}$, and $e^{\delta' R} e^{-\delta R} \le e^{-\delta R/2}$. So the sum is $\le C(1+k-j)^{p_0}\eta^{\rho(H)} p^{n_1} \sum_{n\ge \eta^{-1}/2-1} (n+1)^{3N_*} e^{-\delta n/2} \le C'(1+k-j)^{p_0}\eta^{\rho(H)} p^{n_1} e^{-\delta/(8\eta)}$, since $(n+1)^{3N_*} e^{-\delta n/4} \le C$ and $\eta^{-1} \ge 8$. Members with curvature vertices: the curvature vertex lies in a zone-I cluster (Lemma 6.2(d)), is invariant under the reduction (Lemma B.22(b)) and is bounded through Lemma B.22(c) of the supplemental material. □

Proof of Lemma 6.1. The complete proof, with the two propositions above proved in full, is given in the supplemental material; in outline, for a member H of $\mathcal{N}_h$ and $J_* \le j \le k - J_*$ (so that $\eta \le \eta_2$), it runs as follows. *First clause.* (7) is [15, (3.38)] with the factor $(1+k-j)^{p_0}$ for the localisations in zone I, which are those at distance from the faces larger than the small-field scale. *Second clause.* A tuple with one cluster has $R \le c_0(N_*-1)^2 \le p$, lies in zone I by Lemma B.20(i) of the supplemental material, and obeys (7); a tuple with at least two clusters obeys (9). Summed over the localisations sharing the cube y_* of an h-vertex: the zone-I tuples by $\sum_n (n+1)^{3N_*} e^{-(\delta-\delta')n} \le C$, the zone-B tuples by (10). This is [15, (3.38)] with the factor

$(1+k-j)^{p_0}$. *Third clause.* Split the localisations at $R=p$. Those with $R\le p$ are in zone I and are reduced on one box of side $\le 5p+4M+2$; there (8) is the comparison of [15, pp. 663–665] made element by element, in which the only element depending on h is the coefficient of the h-vertex, replaced by sampling at the cost $C\varepsilon_k\|\nabla h\|_\infty$, the points of the block products being never source vertices. Sum (8) by Lemma B.43 of the supplemental material (branch $R\le p$): the result is $\le C\eta^{\rho(H)-1/2}p^{n_1}(1+k)^{p_0}\eta^{-3}|C_J|[5^{-\gamma j}+\eta^{3+\sigma+1/2}]$, whose first term is $\le C5^{-\gamma' j}\eta^{-\beta'}|C_J|$ for every $\gamma'<\gamma$ and $\beta'>\frac{7}{2}-\rho(H)$, the powers $p^{n_1}(1+k)^{p_0}$ being absorbed on the diagonal sequence as in Lemma B.43 of the supplemental material, and whose second term is $\le C\eta^{\rho(H)+\sigma}p^{n_1}(1+k)^{p_0}|C_J|\le C\eta^{\sigma'}|C_J|$ for every $\sigma'<\sigma$. Those with $R>p$ have at least two clusters; bound the two members of the pair separately by (9), with the roots at the h-vertex, $\le C\eta^{-3}|C_J|$ of them. For the zone-I tuples among them the factor $e^{-(\delta-\delta')p/2}$ of Lemma B.43 of the supplemental material (branch $R>p$) is $\le C_N\eta^N$ for every N, in particular $\le C\eta^{3+\sigma}$, since the exponent of the logarithm in $p(\eta)$ is $6\ge 2$, and the sum is $\le C\eta^{-3}|C_J|\eta^{\rho(H)}p^{n_1}(1+k-j)^{p_0}\eta^{3+\sigma}\le C\eta^{\sigma'}|C_J|$; for the zone-B tuples (10) gives $\le C\eta^{-3}|C_J|\eta^{\rho(H)}p^{n_1}(1+k-j)^{p_0}e^{-\delta/(8\eta)}\le C_N\eta^N|C_J|$ for every N. In both cases $\rho(H)\ge 0$, and the external fields on $\operatorname{supp}\chi_j$ are accounted for in $\rho(H)$ by (5). Adding the three parts gives the third clause with C independent of j and k. The constants depend on the point of King's range, so the lemma holds at every point of the range, in particular at the second mass M_2 of Section 6.1. □

6.3. **The inputs with scalar sources.** For a source $s=(u,h)$ put $\mathsf{S}(s):=\sum_i|x_i|\max_{\mu\nu}\|f_{i,\mu\nu}\|_{C^{2,\alpha}}+\|h\|_\infty$, a size; a compact family of sources is contained in $\{\mathsf{S}(s)\le S_0\}$.

Proposition 6.6 ((U^+): the small-field lines at a non-negative scalar source)**.** *Fix $J\ge J_0$, $\alpha\in(0,\frac12)$ and $S_0>0$. There are k_0 and $C,\sigma>0$, depending on the couplings, J, α, S_0 and on nothing else, such that for every real source $s=(u,h)$ with $h\ge 0$ and $\mathsf{S}(s)\le S_0$, every $k\ge k_0$ and both members $(k',j')\in\{(k,j(k)),(k+1,j(k)+1)\}$ of the diagonal pair,*

$$Z_{k'}(s)\ \ge\ e^{-C\eta_k^\sigma}\,I_{k'}^{(j')}(s)\qquad\text{and}\qquad\int P\,d\mu_{k',s}\ \le\ e^{C\eta_k^\sigma}\,I_{k'}^{(j')}(s).\tag{11}$$

The proof is given in the supplemental material. These are the two small-field lines of (2) with the scalar source inserted into the small-field analysis through the vertex [15, (3.21)]; the large-field remainder at $s\neq 0$ is not part of the statement, and is supplied instead by Lemma 6.12. The source enters the small-field steps only through its vertex, and the sign $h\ge 0$ is used only through the sign of the Wick-ordered source term in the large-field region, which is not examined here.

Proposition 6.7 ((B^h): one-step bound with the scheme c-number)**.** *Fix $J\ge J_0$ and a compact family Σ of real sources. For $\sigma\in\Sigma\cup\{0\}$ and h_σ its scalar part, there is a sequence $\theta_k(\sigma)\le\theta_k$, with θ_k independent of $\sigma\in\Sigma$ and decaying geometrically, such that for $k\ge k_0$, on the support of $\chi_{j(k)}$,*

$$\big|S_{N,k}^{(j(k))}(\cdot;\sigma)+c_k(h_\sigma)-S_{N,k+1}^{(j(k)+1)}(\cdot;\sigma)-c_{k+1}(h_\sigma)\big|\ \le\ \theta_k(\sigma).\tag{12}$$

The proof is given in the supplemental material. The field-dependent and the h-nonlinear parts are Proposition 5.5, the graphs with curvature vertices at $s=(u,h)$, and (K_h). The field-independent h-linear part of the three lowest orders is a c-number $\bar\beta_k^{(j)}\int h$ up to a localisation error, and $c_k(h)$ is exactly the c-number that makes $\bar\beta_k^{(j)}+z_k$ converge along the diagonal pairs at the rate $5^{-\gamma' j}\eta^{-\beta'}+\eta^{\sigma'}$; this is the sense in which z is the scheme adapted to King's flow. At $\sigma=0$, (12) is (3) with $j=j(k)$.

Lemma 6.8 (c-numbers)**.** (a) $|z_k|\le C(1+k)$. (b) *For every $M>0$, $\Delta_k(M):=2(\mathsf{W}_{\varepsilon_k}(m)-\mathsf{W}_{\varepsilon_k}(M))+z_k(m)-z_k(M)$ is bounded in k, where $\mathsf{W}_\varepsilon(M)=(-\Delta_{\varepsilon\mathbb{Z}^3}+M^2)^{-1}(0,0)$ and $z_k(M)$ is the reference scheme at mass M.*

The proof is given in the supplemental material. Both are statements about lattice Feynman integrals: (a) is the logarithmic growth of the order-e^2 coefficient of $\langle|\phi|^2\rangle$ with the counterterm

inserted; in (b) the Wick difference is a convergent lattice integral in three dimensions, and the scheme difference converges.

Proposition 6.9 ((NEG$_\infty$): the bare coefficient grows with the mass)**.** *For every point of King's range ($e, \lambda \geq 0$, $m, \mu_0 > 0$) and every $M > m$, $u_\infty(M) := \lim_k u_k(M)$ exists and*

$$2u_\infty(M) \;\geq\; (M-m)\Big(M+m+\frac{4\lambda}{\pi}-\frac{e^2}{4\pi}\Big)-\Big(\frac{e^4}{4\pi^2}+\frac{8\lambda^2}{\pi^2}\Big)\ln\frac{M}{m};$$

in particular $s_-(M) = u_\infty(M) \to \infty$ as $M \to \infty$, and $s_-(M_2) \geq \frac{1}{2}$ for $M_2 := \max\{m,\ \frac{e^2}{4\pi} + (\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2})/m\} + 1$.

The proof is given in the supplemental material. The content is the exact continuum limit of King's free-boundary counterterm at all five of its orders and the k-uniform bounds on its derivatives in the mass; only $\sup_M s_-(M) = \infty$ is used below.

Proposition 6.10 ((W$^\sharp$)[s_W]: the positive-source bound)**.** *Assume* (I2) *and* (I3) *(Propositions 12.2 and 12.6;* (I2) *follows from* (T)*). Then with $s_W := s_1$ (the amplitude of Proposition 13.2) and $\kappa = C_U/s_1$: for every k, J and every lattice function $0 \leq q \leq s_W$ on X_J,*

$$\ln n_k(0,q) \;\leq\; \kappa\,\|q\|_{1,k}.$$

In particular $\sup_k n_k(0,q) < \infty$ for every $0 \leq q \in C_c^\infty$ with $\sup q \leq s_W$ and $\operatorname{supp} q$ at distance > 1 from $\mathbb{R}^3 \setminus C_J$.

Proof. $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is a class-$\mathcal{I}$ observable, so by Step 1 of Lemma 4.2 (killing the matter on T_J does not increase it; Remark 4.8) $n_k(0,q) \leq N^z_{k,C_J}(0,q)$, the comparison-cube functional, which Proposition 13.2 at $X = X_J$ bounds by $e^{\kappa\|q\|_{1,k}}$; the scheme multiplier is common (Lemma 2.11(a)). For $q \in C_c^\infty$, $\|q\|_{1,k} \leq C_0'|q|_w$ by Lemma 2.6(E3). □

Any other threshold with the same property serves below in place of s_1; the supplemental material contains a second instance, obtained from Bałaban's torus bounds through the comparison of the cube with King's tori.

6.4. **Three lemmas on positive and negative scalar sources.** For a mass M let $Y_k^{(M)}(u,h) := e^{-c_k(h)}\langle e^{F(u)+:\phi^2:(h)}\rangle^{(M)}$ denote the functional of the mass-M state *in the conventions of the mass m* (free Wick constant $\mathsf{W}_{\varepsilon_k}(m)$ and scheme $z_k = z_k(m)$), so that $Y_k^{(m)} = n_k$ and $Y_k^{(M)}(u,h) = e^{-\Delta_k(M)\mathsf{S}_k(h)} n_k^{(M)}(u,h)$, with $n_k^{(M)}$ the functional in the mass-M conventions and $\Delta_k(M)$ as in Lemma 6.8(b). Write $r^2(q) := \sum_x \varepsilon_k^3 q(x) r_x^2$ over the live sites.

Lemma 6.11 (mass shift and the correlation inequality)**.** *Let $M > 0$, $k \geq 0$, and let $0 \leq q$ be a lattice function on the live sites with $q(x) \leq u_k(M)$ for all x. Then*

$$n_k(0,-q) \;\leq\; \frac{1}{Y_k^{(M)}(0,q)}, \qquad \text{and} \qquad \frac{1}{n_k(0,q)} \;\leq\; n_k(0,-q) \;\leq\; \frac{1}{Y_k^{(M)}(0,q)}, \quad \text{hence} \quad Y_k^{(M)}(0,q) \leq n_k(0,q).$$

Proof. Mass shift. In King's action the scalar mass enters only through $\sum_x \varepsilon^3 \frac{1}{2}(M^2 + \delta m_\varepsilon^2(M)) r_x^2$: δm_ε^2 is one number per spacing and mass (Definition 2.3), the Dirichlet terms $\frac{\varepsilon}{2} r_x^2$ do not depend on M, and the photon does not see M. Hence, with $u := u_k(M)$ and $r^2(1) := \sum_x \varepsilon_k^3 r_x^2$, $e^{-S^{(m)}} = e^{-S^{(M)}} e^{u\, r^2(1)}$, and for every Φ integrable against both measures $\langle\Phi\rangle^{(m)} = \langle \Phi e^{u r^2(1)}\rangle^{(M)} / \langle e^{u r^2(1)}\rangle^{(M)}$; the quartic term makes $e^{\pm r^2(\cdot)}$ integrable. *Correlation inequality.* Put $g_1 := u\mathbf{1} - q \geq 0$ and $g_2 := q \geq 0$. The mass-M state is a BFS measure, and so is the state with its single-site measures multiplied by $e^{t\, r^2(g_1)}$, $t \in [0,1]$ (Definition 2.1 and the paragraph on BFS measures following it: the exponential moments stay finite because $\lambda > 0$). By Lemma 2.7(c), multiplying the single-site measures by $e^{-t r^2(g_1)}$ does not increase $\langle e^{r^2(g_2)}\rangle$; read from $t = 1$ down to $t = 0$, $\langle e^{r^2(g_2)}\rangle_{t=1}^{(M)} \geq \langle e^{r^2(g_2)}\rangle_{t=0}^{(M)}$, that is

$$\langle e^{r^2(g_1)} e^{r^2(g_2)}\rangle^{(M)} \;\geq\; \langle e^{r^2(g_1)}\rangle^{(M)} \langle e^{r^2(g_2)}\rangle^{(M)}.$$

Conclusion. With $\Phi = e^{-r^2(q)}$: $\langle e^{-r^2(q)}\rangle^{(m)} = \langle e^{r^2(g_1)}\rangle^{(M)}/\langle e^{r^2(g_1+g_2)}\rangle^{(M)} \le 1/\langle e^{r^2(q)}\rangle^{(M)}$. Multiply both sides by $e^{(2\mathsf{W}_{\varepsilon_k}(m)+z_k)\mathsf{S}_k(q)}$: the left side becomes $n_k(0,-q)$ and the right side $1/Y_k^{(M)}(0,q)$, both in the conventions of the mass m. The left inequality of the second display is Lemma 2.11(e), and the last inequality follows. □

Lemma 6.12 (the large-field remainder at a source, from the one at zero source)**.** *Let σ be a real source and $p>1$, and assume (11) at σ and (R_0) of Proposition 5.4 for the member (k,j), with $a := C\eta^\sigma$ and $\rho := \rho(\eta)$. Then*

$$e^{-a}\, I_k^{(j)}(\sigma) \;\le\; Z_k(\sigma) \;\le\; e^{a}\, I_k^{(j)}(\sigma) + \rho^{1-1/p}\,\langle e^{X(p\sigma)}\rangle_k^{1/p}\, Z_k(0).$$

Proof. The lower line is the first inequality of (11). For the upper line, $Z_k(\sigma) = \int P\, d\mu_{k,\sigma} + \int (1-P) e^{X(\sigma)} d\mu_{k,0}$. The first term is at most $e^a I_k^{(j)}(\sigma)$ by (11). For the second, Hölder's inequality for the positive measure $(1-P)d\mu_{k,0}$ with the functions 1 and $e^{X(\sigma)}$, then $1-P\le 1$ in the second factor and $pX(\sigma) = X(p\sigma)$ (linearity), give

$$\int (1-P)e^{X(\sigma)}d\mu_{k,0} \;\le\; \Big[\int (1-P)d\mu_{k,0}\Big]^{1-1/p}\Big[\int e^{X(p\sigma)}d\mu_{k,0}\Big]^{1/p} \;\le\; \big[\rho Z_k(0)\big]^{1-1/p}\big[Z_k(0)\langle e^{X(p\sigma)}\rangle_k\big]^{1/p},$$

using (R_0) for the first bracket. □

At a negative scalar source the same step would need a bound on $\langle e^{X(p\sigma)}\rangle$ at a negative source of larger amplitude, which is what is to be proved; Theorem 6.16 avoids this through the mass shift.

Lemma 6.13 (positivity of the limit at a positive source)**.** *Fix a mass M, a function $0 \le q \in C_c^\infty$ as in Proposition 6.10, and $\bar w \in (0,1]$. Assume that at mass M, in its own conventions, the hypotheses of Theorem 6.14 below hold at $\sigma = (0, wq)$ for every $w \in [0,\bar w]$ and at $\sigma = 0$, with sequences b_k, $\theta_k(wq)$, $r_k(0)$ and $e^{-c_k(wq)}r_k(wq)$ dominated by w-independent summable sequences, for all $k \ge k_0$ with k_0 independent of w. Then $\inf_{k\ge k_0} n_k^{(M)}(0,q) > 0$.*

Proof. Steps 1–4 of the proof of Theorem 6.14 use the hypotheses at $\{wq, 0\}$ only and give the lower half of (15) with w-independent summable $\bar\delta_k$, $\bar\epsilon_k$: $n_{k+1}(wq) \ge (1-\bar\delta_k)n_k(wq) - \bar\epsilon_k$. By induction, $n_k(wq) \ge \prod_{i=k_1}^{k-1}(1-\bar\delta_i)\, n_{k_1}(wq) - \sum_{i\ge k_1}\bar\epsilon_i$ for $k \ge k_1$. Choose $k_1 \ge k_0$ with $\prod_{i\ge k_1}(1-\bar\delta_i) \ge \frac12$ and $\sum_{i\ge k_1}\bar\epsilon_i \le \frac18$. At this fixed k_1, $w \mapsto n_{k_1}(wq)$ is continuous (a finite-dimensional integral) and equals 1 at $w=0$; choose $w_0 \in (0,\bar w]$ with $n_{k_1}(w_0 q) \ge \frac12$. Then $n_k(w_0 q) \ge \frac18$ for all $k \ge k_1$. Since $\ln n_k(wq)$ is convex in w and vanishes at $w=0$, $n_k(w_0 q) \le n_k(q)^{w_0}$, so $n_k(q) \ge 8^{-1/w_0}$ for $k \ge k_1$; the finitely many $k_0 \le k < k_1$ have $n_k(q) > 0$. □

6.5. **The Cauchy argument with a *c*-number.**

Theorem 6.14 (the Cauchy property with the scheme *c*-number)**.** *Fix a real source $s = (u,h)$ and let σ range over $\{s, 0\}$. Suppose there are k_0 and non-negative sequences b_k, $r_k(\sigma)$, $\theta_k(\sigma)$ such that for all $k \ge k_0$:*

(U) *for every σ and both members (k',j') of the diagonal pair at k, $e^{-b_k} I_{k'}^{(j')}(\sigma) \le Z_{k'}(\sigma) \le e^{b_k} I_{k'}^{(j')}(\sigma) + r_k(\sigma) Z_{k'}(0)$, with $\sum_k b_k < \infty$, $r_k(0) \le \frac12$, $\sum_k r_k(0) < \infty$ and $\sum_k e^{C_c(h)(k+2)} r_k(s) < \infty$, where $|c_k(h)| \le C_c(h)(1+k)$ (Lemma 6.8(a));*

(B) *on the support of $\chi_{j(k)}$, (12) holds at σ with $\sum_k \theta_k(\sigma) < \infty$.*

Put $\Theta_k := \theta_k(s) + \theta_k(0)$, $\delta_k := (4b_k + \Theta_k + 2r_k(0))\, e^{4b_k+\Theta_k+2r_k(0)}$ and $\epsilon_k := (e^{-c_k(h)} + e^{-c_{k+1}(h)})\, r_k(s)$. Then for all $k \ge k_0$:

(i) $n_k(s) \le \mathsf{M}(s) := \exp\big(\sum_{i\ge k_0}\delta_i\big)\big(n_{k_0}(s) + \sum_{i\ge k_0}\epsilon_i\big) < \infty$;

(ii) $|n_{k+1}(s) - n_k(s)| \le \delta_k \mathsf{M}(s) + \epsilon_k$, *so $n_k(s)$ converges to a limit $n(s) \in [0, \mathsf{M}(s)]$;*

(iii) *if* (U) *and* (B) *hold at $-s$ as well, $n_k(s) \ge 1/\mathsf{M}(-s)$ and $n(s) \in [1/\mathsf{M}(-s), \mathsf{M}(s)] \subset (0,\infty)$.*

If b_k, Θ_k decay geometrically and $r_k(\sigma) \le e^{-ck^2+Ck}$, then $|n_k(s) - n(s)| \le C' 5^{-\kappa k}$ for some $\kappa > 0$.

Proof. Fix $k \geq k_0$, $j := j(k)$, and let primes denote the $(k+1)$-model after $j+1$ steps.

Step 1 (one model, (U) *only).* For one member write $Z(\sigma)$, $I(\sigma)$, $b = b_k$, $r(\sigma) = r_k(\sigma)$. From (U) at $\sigma = 0$, $(1 - r(0))Z(0) \leq e^b I(0)$ and $Z(0) \geq e^{-b} I(0)$, so $I(0) > 0$. With $\varrho(\sigma) := I(\sigma)/I(0)$ and $N(\sigma) = Z(\sigma)/Z(0)$,

$$e^{-2b}(1 - r(0))\,\varrho(\sigma) \;\leq\; N(\sigma) \;\leq\; e^{2b}\varrho(\sigma) + r(\sigma)\,: \tag{13}$$

the upper bound from $Z(\sigma) \leq e^b I(\sigma) + r(\sigma)Z(0)$ and $1/Z(0) \leq e^b/I(0)$, the lower one from $Z(\sigma) \geq e^{-b} I(\sigma)$ and $1/Z(0) \geq e^{-b}(1 - r(0))/I(0)$. No lower bound on $Z(\sigma)$ for $\sigma \neq 0$ is used: the remainder enters relative to $Z(0)$ only.

Step 2 (the vacuum constant). $\varrho(\sigma) = \hat{I}(\sigma)/\hat{I}(0)$, since $I = e^{-v}\hat{I}$ with one v for all σ; likewise for the primed member.

Step 3 (across the pair, (B) *only).* On the support of χ_j, (B) at $\sigma = s$ reads $S'_N(\cdot\,; s) = S_N(\cdot\,; s) + c_k(h) - c_{k+1}(h) + E$ with $|E| \leq \theta_k(s)$; the integrals $\hat{I}(s)$, $\hat{I}'(s)$ run over the same variables with the same χ_j, so $\hat{I}'(s) \in e^{c_{k+1}(h) - c_k(h)}[e^{-\theta_k(s)}, e^{\theta_k(s)}]\,\hat{I}(s)$. At $\sigma = 0$ ($c_k(0) = 0$), $\hat{I}'(0) \in [e^{-\theta_k(0)}, e^{\theta_k(0)}]\,\hat{I}(0)$. With $\varrho := \varrho(s)$ for the k-model and ϱ' for the $(k+1)$-model,

$$e^{-c_{k+1}(h)}\varrho' \;\in\; [e^{-\Theta_k}, e^{\Theta_k}]\; e^{-c_k(h)}\varrho. \tag{14}$$

Step 4 (two-sided recursion). Write $n := n_k(s)$, $n' := n_{k+1}(s)$, $r_0 := r_k(0)$, $r_s := r_k(s)$, $\Theta := \Theta_k$. Apply (13) to the primed member, multiply by $e^{-c_{k+1}(h)}$, use (14), and then (13) for the k-model in the forms $\varrho \leq e^{2b}(1 - r_0)^{-1} N(s)$ and $\varrho \geq e^{-2b}(N(s) - r_s)$:

$$\begin{aligned} n' &\leq e^{2b}\, e^{-c_{k+1}}\varrho' + e^{-c_{k+1}} r_s \;\leq\; e^{2b+\Theta} e^{-c_k}\varrho + e^{-c_{k+1}} r_s \;\leq\; e^{4b+\Theta}(1 - r_0)^{-1}\, n + e^{-c_{k+1}} r_s, \\ n' &\geq e^{-2b}(1 - r_0)\, e^{-c_{k+1}}\varrho' \;\geq\; e^{-2b-\Theta}(1 - r_0)\, e^{-c_k}\varrho \;\geq\; e^{-4b-\Theta}(1 - r_0)\big(n - e^{-c_k} r_s\big). \end{aligned}$$

For $t \leq \frac{1}{2}$, $(1 - t)^{-1} \leq e^{2t}$ and $1 - t \geq e^{-2t}$; with $e^y - 1 \leq y e^y$ and $1 - e^{-y} \leq y$ this gives $e^{4b+\Theta}(1 - r_0)^{-1} - 1 \leq \delta_k$ and $1 - e^{-4b-\Theta}(1 - r_0) \leq \delta_k$, hence

$$(1 - \delta_k)\, n_k(s) - \epsilon_k \;\leq\; n_{k+1}(s) \;\leq\; (1 + \delta_k)\, n_k(s) + \epsilon_k. \tag{15}$$

Step 5 (boundedness and convergence). By induction from the upper half of (15), $n_k \leq \prod_{i=k_0}^{k-1}(1 + \delta_i)\big(n_{k_0} + \sum_{i<k} \epsilon_i\big) \leq \mathsf{M}(s)$. Here $\sum_i \delta_i < \infty$ because b_i, Θ_i, $r_i(0)$ are summable (so bounded, and $\delta_i \leq C(4b_i + \Theta_i + 2r_i(0))$), and $\sum_i \epsilon_i \leq 2\sum_i e^{C_c(h)(i+2)} r_i(s) < \infty$ by (U) and $e^{-c_i(h)} \leq e^{C_c(h)(1+i)}$. Then (15) gives $|n_{k+1} - n_k| \leq \delta_k n_k + \epsilon_k \leq \delta_k \mathsf{M}(s) + \epsilon_k$, summable; so $(n_k(s))$ is Cauchy. This is (i), (ii).

Step 6 (positivity). $X(-s) = -X(s)$ and $c_k(-h) = -c_k(h)$, so $n_k(s) n_k(-s) = \langle e^X \rangle \langle e^{-X} \rangle \geq 1$ (Lemma 2.11(e)). Steps 1–5 at $-s$ give $n_k(-s) \leq \mathsf{M}(-s)$, hence $n_k(s) \geq 1/\mathsf{M}(-s)$, and the limit inherits both bounds. The rate follows by summing the bound of (ii) from k on. □

Remark 6.15 (why the ratio is not taken). At $h = 0$ and $c_k \equiv 0$ this is Theorem 5.7 with Lemma 5.6 replaced by Step 6. The logarithmic form of Theorem 5.7 needs $r_k(\sigma)Z(0)/Z(\sigma)$ small, that is a lower bound on $Z_k(s)$ uniform enough in k; at $h \neq 0$ the only candidates are Lemma 2.11(e), which needs the upper bound at $-s$ the theorem is to produce, and Jensen's inequality, which needs a renormalised one-point bound that is not available. The additive comparison (13) avoids both: the remainder is compared with $Z(0)$, and the lower bound on $n_k(s)$ is produced at the end. Nothing in the proof uses the cube or any relation between $j(k+1)$ and $j(k) + 1$.

6.6. **The negative-source bound.**

Theorem 6.16 ((UB$^-$): uniform bound at a negative scalar source)**.** *Assume* (I2) *and* (I3). *Let* $0 \leq q \in C_c^\infty(\mathbb{R}^3)$ *be supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \backslash C_J$. *Then* $\sup_k n_k(0, -q) < \infty$. *Moreover, for every real* 2*-form* u *with the same support condition,* $n_k(u, -q) \leq n_k(2u, 0)^{1/2} n_k(0, -2q)^{1/2}$, *so* $\sup_k n_k(u, -q) < \infty$.

Proof. By Proposition 6.9 choose $M_2 > m$ with $s_-(M_2) > \sup q$. Propositions 5.4, 6.6 and 6.7 and Lemma 6.8 hold at the point (e, λ, M_2, μ_0) as well; $\Delta := \sup_k |\Delta_k(M_2)| < \infty$.

Step 1. Choose $\bar w \in (0, \frac12]$ with $2\bar w \sup q < s_W$, and $k_1 \ge k_0$ with $u_k(M_2) \ge \sup q$ for $k \ge k_1$ (possible since $\sup q < \liminf_k u_k(M_2)$). Then for $k \ge k_1$ and $w \in [0, \bar w]$, both $q \le u_k(M_2)$ and $2wq \le u_k(M_2)$, so Lemma 6.11 applies at q and at $2wq$.

Step 2 (an a priori bound at M_2). For $k \ge k_1$, $w \in [0, \bar w]$: by Lemma 6.11, $Y_k^{(M_2)}(0, 2wq) \le n_k(0, 2wq)$; by convexity of $t \mapsto \ln n_k(0, tq)$ with value 0 at $t = 0$, $n_k(0, 2wq) \le \max\{1, n_k(0, 2\bar w q)\}$; and by Proposition 6.10, $n_k(0, 2\bar w q) \le e^{\kappa \|2\bar w q\|_{1,k}} \le e^{\kappa C_0' |2\bar w q|_w}$. Hence, in the conventions of the mass M_2,

$$n_k^{(M_2)}(0, 2wq) = e^{\Delta_k(M_2)\mathsf{S}_k(2wq)}\, Y_k^{(M_2)}(0, 2wq) \;\le\; e^{\Delta\, \mathsf{S}_k(2q)} \max\{1, e^{\kappa C_0' |2\bar w q|_w}\} \;=:\; \bar{\mathsf{M}},$$

uniformly in w and k ($\mathsf{S}_k(2q) \le C_0' |2q|_w$).

Step 3 (the hypotheses of Lemma 6.13 at M_2). The family $\{wq : 0 \le w \le \bar w\}$ is compact. At $\sigma = (0, wq)$: (U) holds by Lemma 6.12 at mass M_2 with $p = 2$, from (11) (Proposition 6.6 at M_2; $wq \ge 0$) and (R_0) (Proposition 5.4 at M_2), with $b_k = C\eta_k^\sigma$ and, in the conventions of M_2, $e^{-c_k(wq)} r_k(wq) = \rho_k^{1/2}\, n_k^{(M_2)}(0, 2wq)^{1/2} \le \rho_k^{1/2} \bar{\mathsf{M}}^{1/2}$ by Step 2, a w-independent sequence; since $\rho_k \le e^{-ck^2 + C}$ and $e^{c_k(wq)} \le e^{C_c(1+k)}$ with C_c uniform on the family, $\sum_k e^{C_c(k+2)} r_k(wq) < \infty$ uniformly in w. At $\sigma = 0$, (U) is (2) with $r_k(0) = \rho_k \le \frac12$ for $k \ge k_0$. (B) at $\sigma \in \{wq, 0\}$ is Proposition 6.7 at M_2 on the compact family, with w-independent θ_k.

Step 4 (positivity at M_2). Lemma 6.13 gives $c := \inf_{k \ge k_1} n_k^{(M_2)}(0, q) > 0$, hence $Y_k^{(M_2)}(0, q) \ge e^{-\Delta \mathsf{S}_k(q)} c$ for $k \ge k_1$.

Step 5 (the mass shift). By Lemma 6.11 at q, $n_k(0, -q) \le 1/Y_k^{(M_2)}(0, q) \le e^{\Delta \mathsf{S}_k(q)}/c \le e^{\Delta C_0' |q|_w}/c$ for $k \ge k_1$; the finitely many $k < k_1$ are finite numbers. The mixed bound is Cauchy–Schwarz, $\langle e^{F(u) - :\phi^2:(q)} \rangle \le \langle e^{2F(u)} \rangle^{1/2} \langle e^{-2:\phi^2:(q)} \rangle^{1/2}$, with the c-numbers distributed ($e^{c_k(q)} = (e^{c_k(2q)})^{1/2}$); the first factor is bounded by Lemma 5.6. □

6.7. **Existence on the cube with scalar sources.**

Theorem 6.17 (King's existence theorem on the cube with scalar sources, reference scheme)**.** *Assume* (I2) *and* (I3) *(so that Proposition 6.10 holds with $s_W = s_1$). Let $f_1, \dots, f_n$ be real 2-forms and $h_1, \dots, h_p \ge 0$ non-negative real functions, all in C_c^∞ and supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $J \ge J_0$. For $\zeta = (z, w) \in \mathbb{C}^{n+p}$ put $k_1(\zeta) := \sum_j (\operatorname{Re} w_j)_+ h_j$, $k_2(\zeta) := \sum_j (\operatorname{Re} w_j)_- h_j$ (both ≥ 0, smooth, with $\sum_j (\operatorname{Re} w_j) h_j = k_1 - k_2$), and*

$$D_J(s_W) \;:=\; \big\{\zeta \in \mathbb{C}^{n+p} : \sup_x 2k_1(\zeta)(x) < s_W\big\}.$$

Then $D_J(s_W)$ is an open convex tube, containing every ζ with $\operatorname{Re} w_j \le 0$ for all j and every ζ with w purely imaginary, and

(i) *$n_k \to F_J$ locally uniformly on $D_J(s_W)$, and F_J is holomorphic there;*
(ii) *$F_J(\zeta) \in (0, \infty)$ for every real $\zeta \in D_J(s_W)$.*

Proof. The domain. For each x, $\operatorname{Re} w \mapsto 2k_1(\zeta)(x) = \sum_j 2(\operatorname{Re} w_j)_+ h_j(x)$ is convex, since $h_j(x) \ge 0$ and $t \mapsto t_+$ is convex; so $\operatorname{Re} w \mapsto \sup_x 2k_1(\zeta)(x)$ is a finite convex function on $\mathbb{R}^p$, hence continuous, and $D_J(s_W)$ is the preimage under $\zeta \mapsto \operatorname{Re} w$ of an open convex set: open, convex, connected. Write $D := D_J(s_W)$ and $D_{\mathbb{R}} := D \cap \mathbb{R}^{n+p}$, an open convex subset of $\mathbb{R}^{n+p}$.

Step 1 (local boundedness). Let $Q \subset D$ be compact. $\operatorname{Re} Q$ is a compact subset of the open set $D_{\mathbb{R}}$, so it is covered by finitely many closed coordinate boxes $B \subset D_{\mathbb{R}}$. For $\zeta \in Q$ with $\operatorname{Re}\zeta \in B$, $|n_k(\zeta)| \le n_k(\operatorname{Re}\zeta) \le \max_{v \in \mathrm{vert}(B)} n_k(v)$, by Lemma 2.11(d) and the convexity of $\ln n_k$ on B. At a vertex $v = (x, y) \in D_{\mathbb{R}}$ put $k_1 := k_1(v)$, $k_2 := k_2(v)$; then $X(v) = F(u_x) + :\phi^2:(k_1) - :\phi^2:(k_2)$ and the c-number is $e^{-c_k(k_1) + c_k(k_2)}$. Hölder's inequality with exponents $(4, 2, 4)$, with the c-number distributed as $e^{-c_k(k_1)} = (e^{-c_k(2k_1)})^{1/2}$, $e^{c_k(k_2)} = (e^{-c_k(-4k_2)})^{1/4}$, gives

$$n_k(v) \;\le\; a_k(4x)^{1/4}\, n_k(0, 2k_1)^{1/2}\, n_k(0, -4k_2)^{1/4}.$$

The three factors are bounded uniformly in k: the first by Lemma 5.6; the second by Proposition 6.10, since $2k_1$ is a non-negative smooth function with $\sup 2k_1 < s_W$ ($v \in D_{\mathbb{R}}$) and the required support; the third by Theorem 6.16 at $q = 4k_2$. Hence $\sup_k \sup_Q |n_k| < \infty$.

Step 2 (convergence on a real open set). Choose $\delta > 0$ with $4\delta \sup_x \sum_j h_j(x) < s_W$ and put $U := \{\zeta = (x,y) \in \mathbb{R}^{n+p} : \ 0 < y_j < \delta \ \forall j\}$, a non-empty open subset of $\mathbb{R}^{n+p}$ contained in $D_{\mathbb{R}}$. At $\zeta \in U$, $k_2(\zeta) = 0$ and $s := (u_x, k_1(\zeta))$ has a non-negative scalar part; the sources $\{k_1(\zeta) : \zeta \in \bar{U}\}$ form a compact family. Apply Theorem 6.14(i),(ii) at s. Hypothesis (U) at $\sigma = 0$ is (2) on both members, with $r_k(0) = \rho_k$. Hypothesis (U) at $\sigma = s$ is Lemma 6.12 with $p = 2$, from (11) (Proposition 6.6) and (R_0), with $r_k(s) = \rho_k^{1/2} \langle e^{X(2s)} \rangle_k^{1/2}$; here $\langle e^{X(2s)} \rangle_k = e^{c_k(2k_1)} n_k(2u_x, 2k_1) \le e^{c_k(2k_1)} a_k(4x)^{1/2} n_k(0, 4k_1)^{1/2}$ by Cauchy–Schwarz, bounded by $e^{C_c(1+k)}$ times a constant (Lemma 5.6; Proposition 6.10, as $\sup 4k_1 < s_W$), so $\sum_k e^{C_c(k+2)} r_k(s) \le C \sum_k e^{2C_c(k+2)} \rho_k^{1/2} < \infty$ since $\rho_k \le e^{-ck^2+C}$. Hypothesis (B) at $\sigma \in \{s, 0\}$ is Proposition 6.7. Hence $n_k(\zeta)$ converges for every $\zeta \in U$.

Step 3 (Vitali). By Step 1 the family $\{n_k\}$ is locally bounded on the connected domain D; by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a holomorphic function; any two such limits agree on U (Step 2), hence on D, since a holomorphic function vanishing on a non-empty open subset of $\mathbb{R}^{n+p}$ has all its derivatives equal to zero at a point of it and vanishes on the connected D. So the whole sequence converges locally uniformly on D to a holomorphic F_J: (i).

Step 4 (positivity at real points). On $D_{\mathbb{R}}$ every n_k is positive and $\ln n_k$ is convex, so $F_J \ge 0$ there and $g := \ln F_J : D_{\mathbb{R}} \to [-\infty, \infty)$ satisfies $g(\lambda a + (1-\lambda)b) \le \lambda g(a) + (1-\lambda) g(b)$ (the limit of the inequalities for $\ln n_k$; all values are $< \infty$). For $\zeta \in U$ the point $-\zeta$ lies in $D_{\mathbb{R}}$ ($k_1(-\zeta) = 0$), both $n_k(\zeta)$ and $n_k(-\zeta)$ converge by (i), and $n_k(\zeta) n_k(-\zeta) \ge 1$ (Lemma 2.11(e)), so $F_J(\zeta) F_J(-\zeta) \ge 1$ and $g > -\infty$ on U. If $g(t) = -\infty$ for some $t \in D_{\mathbb{R}}$, pick $t' \in U$ and, $D_{\mathbb{R}}$ being open, a point $t'' \in D_{\mathbb{R}}$ beyond t' on the ray from t through t', so that $t' = \lambda t + (1-\lambda) t''$ with $\lambda \in (0,1)$; then $g(t') \le \lambda g(t) + (1-\lambda) g(t'') = -\infty$, a contradiction. Hence $F_J > 0$ on $D_{\mathbb{R}}$: (ii). □

Corollary 6.18 (King's cube limits with scalar sources in the reference scheme, where they are consumed)**.** *Under the hypotheses of Theorem 6.17:*

(a) *For every finite family of real 2-forms f_i and real functions h_j (of either sign) in C_c^∞ with the support condition, and every real coefficient vector (x,y) with $2\|(\sum_j y_j h_j)_+\|_\infty < s_W$, the limit $\lim_k Z_{k,J}^{z,K}(x,y)$ exists and lies in $(0,\infty)$. In particular Theorem 3.6 for the reference scheme holds at every real point of the tube $\mathcal{D}_{\sigma_+,H}$ with $\sigma_+ = s_1$ (there $2\|k_+\|_\infty < \sigma_+ = s_W$), and at every real coefficient vector whose scalar source $\sum_j y_j h_j$ is non-positive, with no restriction on its amplitude.*

(b) *For $0 \le q$ as in Theorem 6.16, the comparison-convention functional satisfies $\sup_k N_{k,C_J}^z(0,-q) \le \sup_k n_k(0,-q) < \infty$.*

Proof. (a) Put $\mathsf{k} := \sum_j y_j h_j$ and let k_+ be its pointwise positive part. Choose $\chi \in C_c^\infty$ with $0 \le \chi \le 1$, $\chi = 1$ on a neighbourhood of $\operatorname{supp} \mathsf{k}$ and $\operatorname{supp} \chi$ at distance > 1 from $\mathbb{R}^3 \setminus C_J$, and $\delta > 0$ with $2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; put $h^+ := (\|\mathsf{k}_+\|_\infty + \delta)\chi$ and $h^- := h^+ - \mathsf{k}$. Both are smooth, non-negative ($h^+ \ge \mathsf{k}_+ \ge \mathsf{k}$ on $\operatorname{supp} \mathsf{k}$, and $h^+ \ge 0 = \mathsf{k}$ off it), compactly supported with the support condition, and $\mathsf{k} = h^+ - h^-$. Then $Z_{k,J}^{z,K}(x,y)$ for the given family equals $n_k(\zeta')$ for the family $(f_1, \dots, f_n; h^+, h^-)$ at $\zeta' = (x; 1, -1)$, for which $k_1(\zeta') = h^+$ and $\sup 2k_1(\zeta') = 2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; so $\zeta' \in D_J(s_W)$ and Theorem 6.17 applies. (b) $e^{-:\phi^2:(q)}$ is a class-$\mathcal{D}$ observable; killing the matter on T_J does not decrease its expectation (Step 3 of Lemma 4.2, the $\mathcal{D}$ direction), and the scheme multiplier is common. □

Remark 6.19 (other schemes). For a scheme a, $Z_{k,J}^{a,K}(x,y) = e^{-(a_k - z_k)\sum_j y_j \mathsf{S}_k(h_j)} Z_{k,J}^{z,K}(x,y)$. At a coefficient vector with $\sum_j y_j \int h_j \ne 0$ the limit for a therefore exists if and only if $a_k - z_k$ converges: on the domain of Corollary 6.18(a), Theorem 3.6 for a is the convergence of the scheme difference, in agreement with Remark 2.4, and Corollary 6.18(b) holds for a whenever $\sup_k |a_k - z_k| < \infty$.

Remark 6.20 (the cell bound). ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4) is the boundedness of $N_{k,C_{L_c}}^a(0, -2H'\chi_0)$ on one comparison-convention cell. When the cell is a King cube, that is $\ell_c = 2MJ'$ for some

J' up to a translation by a lattice vector (Lemma 2.12), and χ_0 is supported at ℓ^∞-distance > 1 from the cell's complement, Corollary 6.18(b) and Remark 6.19 give ($\mathrm{B_{cell}}$) for every scheme with $\sup_k |a_k - z_k| < \infty$, and Corollary 6.18(a) gives the convergence of the King-convention cell functional $Z^{z,K}_{k,C_{L_c}}(0, -2H'\chi_0)$ as well; for the comparison-convention cell functional only the boundedness is asserted (Remark 5.10).

Remark 6.21 (what is and is not claimed). Beyond the statements of Section 5.2 and Lemma 6.1, the only input of this section is the threshold s_W of Proposition 6.10, supplied by (I2) and (I3). Nothing is claimed at a non-negative scalar source of amplitude $\geq s_W/2$: there the positive-source bound is the content of (W2) above the threshold (Remark 23.2). The negative scalar sources are unrestricted, because the mass shift of Lemma 6.11 turns a negative source of the mass-m model into a positive source of a heavier model, where the small-field analysis applies with no threshold on the amplitude, and Proposition 6.9 supplies a heavy enough mass for every amplitude.

The axioms

7. Regularity in the curvature sector

This section proves the regularity axiom E0″ for the curvature field, for the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) and for King's convention ND^K (Definition 2.2), on which King's printed state Z^K_∞ lives. Gaussian domination (Lemma 2.8) and a bound on the Neumann photon form of the cube (Lemma 7.1) give a bound on every lattice generating functional, uniform in the spacing, the cube and the scalar sector, with absolute constants (Remark 7.5); under the existence hypothesis it passes to the limits by Montel's theorem, and Cauchy's estimates turn it into E0″ with $s = 4$. The section closes with the curvature-sector properties of the fixed-cube limits of the comparison family (locally uniform convergence, monotonicity in the cube, positivity, continuity and translations), proved with cone-direction correlation inequalities only (Lemma 7.7).

Throughout, $f_1, \dots, f_n$ are real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$ with supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \geq J_0$, $z \in \mathbb{C}^n$, and $g(z) := \sum_j (\mathrm{Re}\, z_j) f_j$, so that $\mathrm{Re} \sum_j z_j F(f_j) = F(g(z))$ (Lemma 2.11(d)). For a box X, $\langle A, B\rangle_k$ and $d^*_k g$ are as in (1): $d^*_k g$ is the lattice 1-form with $F_{\Lambda_k}(g) = \langle d^*_k g, A\rangle_k$, defined whenever every plaquette at which $g \neq 0$ has its four corners in X. The support condition implies this for $X = X_J$: the plaquette centre is at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and the corners are within $\varepsilon/2 \leq \frac{1}{2}$ of it.

7.1. The Neumann cube photon form.

Lemma 7.1 (the Neumann cube photon form). *Let $X = C_J \cap \Lambda_k$, let C be the photon covariance of the cube states on X (the same for $\mathrm{ND}^{(a)}_{k,J}$ and $\mathrm{ND}^K_{k,J}$) with respect to $\langle\cdot,\cdot\rangle_k$, let g be a real 2-form on the plaquettes $p = (x; \alpha < \beta)$ all four of whose corners lie in X, and $v := d^*_k g$. Then*

$$\langle v, Cv\rangle_k \;\leq\; \min\big\{\, 2\|g\|^2_{2,k},\ \mu_0^{-2}\|v\|^2_{2,k} \,\big\}.$$

Proof. The Gaussian density of the photon field is $e^{-Q(A)/2}$ with

$$Q(A) = \langle A, C^{-1}A\rangle_k = \sum_{\alpha,\beta} \sum_{x,\, x+\varepsilon e_\beta \in X} \varepsilon\, (\partial_\beta A_\alpha(x))^2 + \mu_0^2 \|A\|^2_{2,k}, \qquad \partial_\beta A_\alpha(x) := A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\,:$$

the Neumann rule keeps exactly the gradient terms of pairs inside X. By Legendre duality, $\langle v, Cv\rangle = \sup_A \big[2\langle v, A\rangle - Q(A)\big]$, attained at $A = Cv$. *Second bound.* $Q(A) \geq \mu_0^2 \|A\|^2$, so $C \leq \mu_0^{-2}$ as forms. *First bound.* Let P_g be the set of plaquettes on which $g \neq 0$. For $p = (x; \alpha < \beta) \in P_g$ both gradient terms $(\partial_\alpha A_\beta(x))^2$ (pair $(x, x + \varepsilon e_\alpha)$, component β) and $(\partial_\beta A_\alpha(x))^2$ (pair $(x, x + \varepsilon e_\beta)$, component α) are present in Q, because the pairs lie in X. With $F(p) = \varepsilon^{-1}(\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))$,

$$\|dA\|^2_{P_g} := \sum_{p \in P_g} \varepsilon^3 F(p)^2 = \sum_{p \in P_g} \varepsilon\, (\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))^2 \leq 2 \sum_{p \in P_g} \varepsilon \big[(\partial_\alpha A_\beta(x))^2 + (\partial_\beta A_\alpha(x))^2\big] \leq 2Q(A),$$

where the last step is a counting statement: the ordered triple (site x, component β, direction $\alpha \neq \beta$) determines the plaquette $(x;\{\alpha,\beta\})$, so distinct plaquettes use distinct gradient terms of Q and no term is counted twice. Also $\langle v, A\rangle_k = F_{\Lambda_k}(g) = \sum_{p\in P_g} \varepsilon^3 g_p F(p) =: \langle g, dA\rangle_{P_g}$. Hence

$$\langle v, Cv\rangle \le \sup_A \big[2\langle g, dA\rangle_{P_g} - \tfrac12 \|dA\|^2_{P_g}\big] \le \sup_{B\in\mathbb{R}^{P_g}} \big[2\langle g, B\rangle_{P_g} - \tfrac12 \|B\|^2_{P_g}\big] = 2\|g\|^2_{2,k},$$

the last supremum being attained at $B = 2g$. The inequality $Q(A) - \frac12\|dA\|^2_{P_g} \ge 0$ used in the middle is in fact the identity

$$Q(A) - \tfrac12\|dA\|^2_{P_g} = \tfrac12 \sum_{p\in P_g} \varepsilon\,(\partial_\alpha A_\beta(x) + \partial_\beta A_\alpha(x))^2 + \big(\text{gradient terms of } Q \text{ used by no } p\in P_g\big) + \mu_0^2\|A\|^2_{2,k}.$$

□

Remark 7.2 (on the constant 2). On a torus the Hodge decomposition gives the constant 1 in place of 2 ([27, Lemma 7.3]); on the Neumann cube no Hodge argument is available, and the constant 2 appears to be nearly sharp: the exact quantity $\rho := \sup_g \operatorname{Var} F(g)/\|g\|^2_{2,k}$, which depends on ε and μ_0 only through $\mu_0\varepsilon$, exceeds 1.94 on a cube of 28^3 sites at $\mu_0\varepsilon = 10^{-4}$ and increases with the cube at small $\mu_0\varepsilon$. Whether it tends to 2 is not proved; nothing below depends on it. The second bound is the cube form of the two-point bound $0 \le S_2(f) \le \mu_0^{-2}\|d^*f\|^2_{L^2}$; it is the worse of the two at small μ_0.

7.2. **The comparison convention.**

Lemma 7.3 (standalone local bound)**.** *For either convention and every compact $K \subset \mathbb{C}^n$,*

$$\sup_{k\ge0}\ \sup_{J\ge J_0}\ \sup_{z\in K} \big|\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle_{k,J}\big| \ \le\ \exp\Big(3C_0 \sup_{z\in K}|g(z)|_4^2\Big) < \infty,$$

with C_0 the constant of Lemma 2.6(E1)*. No hypothesis and no correlation inequality is used.*

Proof. $|\langle e^{\sum z_j F(f_j)}\rangle| \le \langle e^{F(g(z))}\rangle$. Both cube states are BFS measures whose Gaussian is the Neumann photon Gaussian of X_J with covariance C (the matter on T_J does not touch the photon variables), so Lemma 2.8 with $s = 1$ and $v = d_k^* g(z)$ gives $\langle e^{F(g(z))}\rangle \le e^{\langle v, Cv\rangle_k/2} \le e^{\|g(z)\|^2_{2,k}}$ by Lemma 7.1. The plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, and $|g_{c(p)}(e_\alpha, e_\beta)| \le (1+|c(p)|^2)^{-2}|g|_4$, so by (E1) each orientation contributes at most $C_0|g|_4^2$ to $\|g\|^2_{2,k}$ and $\|g(z)\|^2_{2,k} \le 3C_0|g(z)|_4^2$ for every k (this is the count behind (E3) of Lemma 2.6). Finally $|g(z)|_4 \le \sum_j |\operatorname{Re} z_j||f_j|_4$ is bounded on K. □

The existence theorem Theorem 3.3 is posed on King's convention. For the comparison family the analogous clause is its convergence at fixed cubes (Definition 3.1): for every such family, every $J \ge J_0$ and every real z, $\lim_k \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ exists in $(0,\infty)$. It is not needed for the main theorem (the two clauses do not imply one another by the inputs of this paper, Corollary 4.4); Theorem 7.4 is stated under it because the rotation part and the tube estimates are written for the comparison family, and Theorem 7.6 is its form for King's printed state, where the limits exist by Theorem 3.3.

Theorem 7.4 (regularity, comparison convention)**.** *Let $f_1,\dots,f_n$, J_0, $g(z)$ be as above.*

(a) (Finite volume; no hypothesis.) *For all $k \ge 0$, $J \ge J_0$, $\mu_0 > 0$, $z \in \mathbb{C}^n$, and in the presence of any real scalar source $h \in C_c^\infty$,*

$$\left|\frac{\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}{\langle e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}\right| \ \le\ \frac{\langle e^{F_{\Lambda_k}(g(z))} e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}{\langle e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}} \ \le\ \exp\Big(\min\big\{\|g(z)\|^2_{2,k},\ \tfrac12\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}\big\}\Big).$$

(b) (Limits; if the comparison family is convergent at fixed cubes.) *$Z_J(\sum_j z_j f_j, 0) := \lim_k \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $Z_\infty(\sum_j z_j f_j, 0) := \lim_J Z_J(\sum_j z_j f_j, 0)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; and for $J \ge J_0$*

$$|Z_J(\textstyle\sum_j z_j f_j, 0)|,\ |Z_\infty(\sum_j z_j f_j, 0)| \ \le\ \exp\Big(\min\big\{\|g(z)\|^2_{2},\ \tfrac12\mu_0^{-2}\|d^* g(z)\|^2_{L^2}\big\}\Big).$$

(c) (Schwinger functions; if the comparison family is convergent at fixed cubes.) $S_n(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty(\sum_j z_j f_j, 0)\big|_{z=0}$ *satisfies, for real* f_i,

$$|S_n(f_1,\dots,f_n)| \;\le\; (\sqrt{2}\,e)^n (n!)^{1/2} \prod_{i=1}^n \|f_i\|_2 \;\le\; (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n |f_i|_4.$$

S_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$ and defines a tempered distribution; for complex test functions the bound carries the factor 2^n. This is the axiom E0″ *of* [20] *with $s=4$ and $\sigma_n := (e\pi)^n(n!)^{1/2} \le \alpha_\beta (n!)^\beta$ for every $\beta > \frac12$ (and for no α at $\beta = \frac12$).*

Proof. (a) The first inequality is $\operatorname{Re}\sum_j z_j F(f_j) = F(g(z))$. The measure $e^{:\phi^2:(h)} d\langle\cdot\rangle^{(a)}_{k,J}$, normalised, is a BFS measure whose single-site weights carry the extra factor $e^{\varepsilon^3 h(x) r_x^2}$ and whose photon covariance is the same C; Lemma 2.8 with $s=1$ and $v = d_k^* g(z)$ bounds the middle expression by $e^{\langle v, Cv\rangle_k/2}$, and Lemma 7.1 gives both terms of the minimum.

(b) *Step 1: $k\to\infty$ at fixed $J \ge J_0$.* The functions $\Phi_k(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ are entire on $\mathbb{C}^n$: finite-dimensional integrals with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda\varepsilon^3 r^4}$ decay in r, converging absolutely and locally uniformly in z. By Lemma 7.3 they are locally uniformly bounded; by the convergence at fixed cubes they converge at every real point. $\mathbb{R}^n$ is a set of uniqueness for entire functions on $\mathbb{C}^n$ (a holomorphic function vanishing on $\mathbb{R}^n$ has all its Taylor coefficients at 0 equal to 0). By Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence; all subsequential limits are entire and agree on $\mathbb{R}^n$, hence everywhere [11]; so $\Phi_k \to Z_J(\cdot,0)$ locally uniformly on $\mathbb{C}^n$ and $Z_J(\cdot,0)$ is entire. The bound of (a) at $h=0$ passes to the limit: $|Z_J(\sum z_j f_j,0)| \le \lim_k e^{\|g(z)\|^2_{2,k}} = e^{\|g(z)\|^2_2}$ by (E3) of Lemma 2.6, and likewise $|Z_J(\sum z_j f_j,0)| \le \lim_k \exp(\frac12\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}) = \exp(\frac12\mu_0^{-2}\|d^* g(z)\|^2_{L^2})$ by (1). Both bounds are free of J.

Step 2: monotonicity in J on the imaginary axes. For $t\in\mathbb{R}^n$ and $f := \sum_j t_j f_j$, $\langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ is real (the state is invariant under $(A,\vartheta)\mapsto(-A,-\vartheta)$, which maps $F(f)$ to $-F(f)$) and of modulus ≤ 1, and $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ is nondecreasing on $J \ge J_0$ at fixed k: this is the first inequality of Lemma 2.10 with $Y = C_{J+1}\cap\Lambda_k$, $X = C_J\cap\Lambda_k$ and the class-$\mathcal{I}$ observable $\Theta(\mathrm{i}f,0)$, whose plaquettes lie in X by the support condition (its proof uses Lemma 2.7(a),(b) only, and the site-independence of the counterterm). Letting $k\to\infty$ (Step 1), $J\mapsto Z_J(\mathrm{i}f,0)$ is nondecreasing and bounded by 1, hence convergent, for every $t\in\mathbb{R}^n$.

Step 3: $J\to\infty$. $\{Z_J(\cdot,0)\}_{J\ge J_0}$ is locally uniformly bounded on $\mathbb{C}^n$ by the J-free bound of Step 1, and converges on $(\mathrm{i}\mathbb{R})^n$ (Step 2), a totally real subspace of maximal dimension, hence a set of uniqueness for entire functions (at a point $\mathrm{i}t_0$ the restriction to $\mathrm{i}t_0 + \mathrm{i}\mathbb{R}^n$ determines the Taylor series). Montel's theorem and the identity theorem as in Step 1 give $Z_J(\cdot,0)\to Z_\infty(\cdot,0)$ locally uniformly on $\mathbb{C}^n$, $Z_\infty(\cdot,0)$ entire, and both bounds of Step 1 pass to the limit.

(c) By multilinearity normalise $\|f_i\|_2 = 1$. On the polydisc $|z_i|\le r$, $\|g(z)\|_2 \le \sum_i |\operatorname{Re} z_i| \le nr$, so $|Z_\infty(\sum z_j f_j,0)| \le e^{n^2 r^2}$ by (b). Cauchy's estimate for the mixed first derivative in the n distinct variables gives $|S_n| \le r^{-n}\sup_{|z_i|=r}|Z_\infty| \le r^{-n} e^{n^2 r^2}$; at $r^2 = 1/(2n)$ this is $(2en)^{n/2} = (2e)^{n/2} n^{n/2} \le (\sqrt{2}\,e)^n (n!)^{1/2}$ by (E4) of Lemma 2.6. Then (E2), $\|f_i\|_2 \le \frac{\pi}{2\sqrt{2}}|f_i|_4$, gives the second bound. S_n is symmetric and multilinear (derivatives of an entire function of a linear combination) and continuous for the L^2 norms, so it extends by density from C_c^∞ to $\mathcal{S}$; for complex $f_i = f_i' + \mathrm{i}f_i''$ with real parts, multilinearity writes $S_n(f_1,\dots,f_n)$ as a sum of 2^n terms $\mathrm{i}^m S_n(f_1^\sharp,\dots,f_n^\sharp)$, each bounded by $(e\pi/2)^n(n!)^{1/2}\prod|f_i^\sharp|_4 \le (e\pi/2)^n(n!)^{1/2}\prod|f_i|_4$, since $|\operatorname{Re} f|_4, |\operatorname{Im} f|_4 \le |f|_4$; the kernel theorem gives the tempered distribution. The axiom E0″ of [20, p. 287] asks for $|S_n(f_1,\dots,f_n)| \le \sigma_n \prod_i |f_i|_s$ on $\mathcal{S}$ with $\sigma_n \le \alpha(n!)^\beta$; here $s=4$ ($|\cdot|_4$ is the norm that [27] identifies with that of [20] at $s=4$), and $\sigma_n = (e\pi)^n(n!)^{1/2} \le \alpha_\beta (n!)^\beta$ exactly when $(e\pi)^n/(n!)^{\beta-1/2}$ is bounded, that is for every $\beta > \frac12$. □

Remark 7.5 (the constants are absolute). The bound $e^{\|g\|_2^2}$ of Theorem 7.4 and the constants $(\sqrt{2}\,e)^n(n!)^{1/2}$, $(e\pi/2)^n(n!)^{1/2}$, σ_n of E0″ depend on none of e, λ, m, μ_0 and M: the only inputs are Gaussian domination, the Neumann form's constant 2 and the elementary facts (E1)–(E4). The second bound of (a) and (b) is the one that carries μ_0^{-2}. The proof of (b) uses, besides the existence clause, only [7, Thm. 6.2(1),(2)] through Lemma 2.10, Gaussian domination and Lemma 7.1; it uses neither [7, Thm. 6.2(3),(4)] (Lemma 2.7(d)) nor any local bound resting on that reversed inequality.

7.3. **King's convention.** For the K family write $a^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ as in Corollary 4.4, and Z^K_J, Z^K_∞ for its limits.

Theorem 7.6 (regularity of King's printed state)**.** *Let $f = (f_1, \dots, f_n)$, J_0, $g(z)$ be as above.*

(a) (No hypothesis.) *For all $k \ge 0$, $J \ge J_0$, $z \in \mathbb{C}^n$,*

$$|a^K_{k,J}(z)| \;\le\; \langle e^{F(g(z))}\rangle^K_{k,J} \;\le\; \exp\Big(\min\big\{\|g(z)\|^2_{2,k},\ \tfrac12\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}\big\}\Big),$$

and for every real 2-form u whose plaquettes have their corners in X_J: $\langle F(u)\rangle^K_{k,J} = 0$ and $\langle F(u)^2\rangle^K_{k,J} \le \langle v, Cv\rangle_k \le 2\|u\|^2_{2,k}$, $v = d_k^* u$.

(b) (Limits.) *$Z^K_J(z) := \lim_k a^K_{k,J}(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $Z^K_\infty(z) := \lim_J Z^K_J(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $|Z^K_J(z)|, |Z^K_\infty(z)| \le e^{\|g(z)\|_2^2}$ and also $\le \exp(\frac12\mu_0^{-2}\|d^*g(z)\|^2_{L^2})$; $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing on $J \ge J_0$ with values in $[0,1]$ for $t \in \mathbb{R}^n$; and $Z^K_\infty(x) \ge 1$ at every real $x \in \mathbb{R}^n$.*

(c) (Schwinger functions.) *The Schwinger functions $S^K_n(f_1, \dots, f_n) := \partial_{z_1}\cdots\partial_{z_n} Z^K_\infty(0)$ satisfy $|S^K_n| \le (\sqrt{2}\,e)^n(n!)^{1/2}\prod_i\|f_i\|_2 \le (e\pi/2)^n(n!)^{1/2}\prod_i|f_i|_4$ for real f_i; the lattice Schwinger functions $S^{K,(k,J)}_n := \partial_{z_1}\cdots\partial_{z_n} a^K_{k,J}(0)$ satisfy the same bound at every (k,J) with $\|f_i\|_{2,k}$ in place of $\|f_i\|_2$, and $S^{K,(k,J)}_n \to S^K_n$ as $k \to \infty$ and then $J \to \infty$. Each S^K_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$ with the same bound, carrying the factor 2^n for complex test functions: E0″ holds with $s = 4$ and $\sigma_n = (e\pi)^n(n!)^{1/2} \le \alpha_\beta(n!)^\beta$ for every $\beta > \frac12$.*

Proof. (a) The two inequalities are the bound of Lemma 7.3 for the K family, with Lemma 7.1 kept in both forms: $\mathrm{Re}\sum z_j F(f_j) = F(g(z))$; $\langle\cdot\rangle^K_{k,J}$ is a BFS measure (the killed sites carry point masses, which Lemma 2.8 allows: at a killed site $I_q(0) = \delta_{q0}$ and the currents vanish on its bonds), and its Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so Lemma 2.8 with $s = 1$, $v = d_k^* g(z)$ and Lemma 7.1 give $\langle e^{F(g(z))}\rangle^K_{k,J} \le e^{\langle v,Cv\rangle_k/2} \le \exp(\frac12\min\{2\|g\|^2_{2,k}, \mu_0^{-2}\|v\|^2_{2,k}\})$. *Second-moment clause.* $\langle F(u)\rangle^K_{k,J} = 0$ because the state is invariant under $(A,\vartheta) \mapsto (-A,-\vartheta)$ and $F(u)$ is odd. For real s, $e^x + e^{-x} \ge 2 + x^2$ pointwise and $e^{\pm sF(u)}$ are integrable, so $2 + s^2\langle F(u)^2\rangle^K_{k,J} \le \langle e^{sF(u)}\rangle^K_{k,J} + \langle e^{-sF(u)}\rangle^K_{k,J} \le 2e^{s^2\langle v,Cv\rangle_k/2}$ (Lemma 2.8 at $\pm s$; $\pm v$ give the same quadratic form). Hence $\langle F(u)^2\rangle^K_{k,J} \le 2(e^{s^2\langle v,Cv\rangle_k/2} - 1)/s^2 \to \langle v, Cv\rangle_k$ as $s \downarrow 0$, and Lemma 7.1 applies since the corners lie in X_J.

(b) The two limits, their entireness and the J-free bound $e^{\|g(z)\|_2^2}$ are Corollary 4.4(ii)(α),(β): the proof of Theorem 7.4(b) run on the K family, with the local bound of (a) (uniform in k, J by $\|g\|^2_{2,k} \le 3C_0|g|^2_4$), Theorem 3.3 for the convergence at real points, Montel's theorem with the uniqueness set $\mathbb{R}^n$, and then, for $J \to \infty$, the monotonicity of Corollary 4.3 on the cone in place of Lemma 2.10 and the uniqueness set $(\mathrm{i}\mathbb{R})^n$. Three additions. (1) *Values in $[0,1]$.* $a^K_{k,J}(\mathrm{i}t)$ is real by the symmetry used in (a) and of modulus ≤ 1; it is ≥ 0 by the current representation of Lemma 2.8: with $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$, $w(q) \ge 0$, and the Gaussian identity at $a = \mathrm{i}v \pm \mathrm{i}\kappa$,

$$\langle e^{\mathrm{i}\langle v,A\rangle}\rangle^K_{k,J} = e^{-\langle v,Cv\rangle_k/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle \;\ge\; 0, \qquad v = d_k^*\textstyle\sum_j t_j f_j,$$

the killed sites being covered as in (a). Monotonicity in J is Corollary 4.3 at every k (condition (S) holds for X_{J_0} by the support condition), passed to the limit $k \to \infty$. (2) *The μ_0^{-2} bound.* $\|d_k^* g\|_{2,k} \to \|d^*g\|_{L^2}$ by (1), so the second bound of (a) passes to Z^K_J and then to Z^K_∞ exactly

as the first. (3) $Z^K_\infty(x) \geq 1$. For real x, $a^K_{k,J}(x) = \langle e^{F(g(x))}\rangle^K_{k,J} \geq e^{\langle F(g(x))\rangle^K_{k,J}} = 1$ by Jensen's inequality and (a); the inequality is closed under both limits.

(c) Let $\|f_j\|_{2,k} > 0$ for all j (otherwise $F_{\Lambda_k}(f_j) = 0$ and $S^{K,(k,J)}_n = 0$). Cauchy's estimates on the polydisc $|z_j| \leq \rho_j := \rho/(n\|f_j\|_{2,k})$ with the bound (a): there $\|g(z)\|_{2,k} \leq \sum_j |z_j|\|f_j\|_{2,k} \leq \rho$, so $|a^K_{k,J}| \leq e^{\rho^2}$ and $|S^{K,(k,J)}_n| \leq e^{\rho^2}\prod_j \rho_j^{-1} = e^{\rho^2} n^n \rho^{-n} \prod_j \|f_j\|_{2,k}$; at $\rho^2 = n/2$ the prefactor is $e^{n/2}2^{n/2}n^{n/2} \leq e^{n/2}2^{n/2}e^{n/2}(n!)^{1/2} = (\sqrt{2}\,e)^n (n!)^{1/2}$ by (E4). The Taylor coefficients at 0 converge under the locally uniform convergences of (b) (Weierstrass), and $\|f_j\|_{2,k} \to \|f_j\|_2$ by (E3), which gives the bound for S^K_n; (E2) gives the $|\cdot|_4$ form. The 2^n factor for complex test functions, the extension to $\mathcal{S}$ and the growth class σ_n are as in Theorem 7.4(c). □

7.4. **The curvature sector of the comparison family's limits.** For the comparison family and an admissible scheme a, the general-source functionals $Z^a_J(\zeta) = \lim_k Z^a_{k,J}(\zeta)$, $\zeta \in \mathbb{C}^{2N}$, have, under (W2) (the optional extension), the following properties: (i) locally uniform convergence to an entire Z^a_J; (ii) on the cone, $J \mapsto Z^a_J(\mathrm{i}f, g)$ nondecreasing and $J \mapsto Z^a_J(f, -g)$ nonincreasing; (iii) the limit $Z^a_\infty = \lim_J Z^a_J$, locally uniform and entire, positive at real sources, with $Z^a_J \leq Z^a_\infty$ on the cone; (iv) $\Phi^a = \ln Z^a_\infty$ finite, convex and continuous on the Fréchet space $E_\rho = C^\infty_c(\bar{B}_\rho; \Lambda^2) \times C^\infty_c(\bar{B}_\rho)$; (v) translation invariance; (vi) the change of scheme. At $h = 0$ these conclusions hold under the existence clause alone; the natural local bound, $|Z_J(\zeta)| \leq Z_J(\mathrm{Re}\,\zeta) \leq Z_{J_0}(\mathrm{Re}\,\zeta)$, would use the nonincreasing clause of (ii), that is Lemma 2.7(d). The following lemma states the curvature sector with that use removed; the Wick sector on the tube is Lemma 13.11.

Lemma 7.7 (the comparison family's limits, curvature sector, with cone-direction inequalities only)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). For every finite family $f_1, \dots, f_N$ of real 2-forms in C^∞_c supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, with $Z_{k,J}(\sum_j z_j f_j, 0) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ and $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$:*

(i) *$Z_{k,J}(\cdot, 0) \to Z_J(\cdot, 0)$ locally uniformly on $\mathbb{C}^N$ as $k \to \infty$, and $Z_J(\cdot, 0)$ is entire, for every $J \geq J_0$;*

(ii) *$J \mapsto Z_J(\mathrm{i}f, 0)$ is nondecreasing on $J \geq J_0$ for every real $f = \sum_j t_j f_j$;*

(iii) *for all $k \geq 0$, $J \geq J_0$ and $z \in \mathbb{C}^N$,*

$$|Z_{k,J}(\textstyle\sum_j z_j f_j, 0)| \leq \langle e^{F(g(z))}\rangle^{(a)}_{k,J} \leq e^{\frac{1}{2}\langle v, Cv\rangle_k} \leq \exp\left(\tfrac{1}{2}\mu_0^{-2}\|d^*_k g(z)\|^2_{2,k}\right) \leq \exp\left(\tfrac{1}{2}\mu_0^{-2}\sup_{k'}\|d^*_{k'} g(z)\|^2_{2,k'}\right), \quad v = c$$

a bound uniform in k and J and locally uniform in z; consequently $Z_\infty(\cdot, 0) := \lim_J Z_J(\cdot, 0)$ exists locally uniformly on $\mathbb{C}^N$, is entire, satisfies $Z_\infty(f, 0) > 0$ for real f, and $Z_J(\mathrm{i}f, 0) \leq Z_\infty(\mathrm{i}f, 0)$ for real f;

(iv) *for every $\rho > 0$, $\Phi(f) := \ln Z_\infty(f, 0)$ is finite, convex and continuous on the Fréchet space $C^\infty_c(\bar{B}_\rho; \Lambda^2)$;*

(v) *$Z_\infty(\tau_b f, 0) = Z_\infty(f, 0)$ for all $b \in \mathbb{R}^3$ and all complex coefficient vectors, τ_b acting on every f_j.*

Clauses (i), (iii)–(v) *use, besides the convergence at fixed cubes, only* [7, Thm. 6.2(1),(2)] *through Lemma 2.10, Lemma 2.8, Lemma 2.11*(d), (1), *Lemma 2.14, Montel's theorem and the identity theorem. The nonincreasing clause* (ii) *at $h = 0$, $J \mapsto Z_J(f, 0)$ nonincreasing for real f, holds as well and uses Lemma 2.7*(d)*; it is not consumed by the curvature sector of this paper. Clause* (vi) *is void at $h = 0$, no scheme entering.*

Proof. (i) is Step 1 of the proof of Theorem 7.4(b), which uses Lemma 7.3, the convergence at fixed cubes and Montel's theorem. (ii) is its Step 2. (iii) The first inequality is Lemma 2.11(d), the second is Lemma 2.8 with $s = 1$ (the cube state is a BFS measure), the third is $C \leq \mu_0^{-2}$ as forms (the photon form is bounded below by $\mu_0^2\|A\|^2_{2,k}$, the second bound of Lemma 7.1), and the supremum over k' is finite by (1). The bound passes to Z_J by (i) and makes $\{Z_J(\cdot, 0)\}_{J \geq J_0}$ locally bounded on $\mathbb{C}^N$; with the convergence on $(\mathrm{i}\mathbb{R})^N$ from (ii) and the bound $|Z_J(\mathrm{i}f, 0)| \leq 1$, Montel's theorem and the identity theorem give the locally uniform convergence to an entire $Z_\infty(\cdot, 0)$ as in Step 3 of Theorem 7.4(b). Positivity: $Z_{k,J}(f, 0) \geq e^{\langle F(f)\rangle^{(a)}_{k,J}} = 1$ for real f by

Jensen's inequality and $\langle F(f)\rangle^{(a)}_{k,J} = 0$ (the symmetry $(A,\vartheta)\mapsto(-A,-\vartheta)$), closed under both limits. The last clause is (ii) in the limit.

(iv) Φ is finite by (iii). Convexity: for $f, f' \in C^\infty_c(\bar B_\rho;\Lambda^2)$ and $t\in[0,1]$ apply (i) and (iii) to the family (f,f') at the real coefficients $(t,1-t)$; at fixed k,J, $\ln Z_{k,J}(\cdot,0)$ is convex in the coefficients by Hölder's inequality, and convexity survives the two pointwise limits. Borel measurability: fix J with $MJ>\rho+1$, so that every $f\in C^\infty_c(\bar B_\rho;\Lambda^2)$ is supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_J$; at each k, $\ln Z_{k,J}(f,0)$ depends on f only through the finitely many sampled values $f_{c(p)}(e_\alpha,e_\beta)$, continuous functionals of f in the C^∞ topology, and the finite-dimensional integral is continuous in them (dominated convergence); so $\Phi=\lim_J\lim_k$ of continuous functions is Borel (of Baire class ≤ 2). $C^\infty_c(\bar B_\rho;\Lambda^2)$ is a closed subspace of $C^\infty(\bar B_{\rho+1};\Lambda^2)$, hence Fréchet. Lemma 2.14 gives the continuity.

(v) is Proposition 8.1, whose proof uses the lattice sandwich between C_{J-1} and C_{J+1} (first inequality of Lemma 2.10), $k\to\infty$ by (i) (the convergence at fixed cubes covers the translated family), $J\to\infty$ by (iii), the identity theorem with the uniqueness set $(\mathrm{i}\mathbb{R})^N$, the continuity of (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$. □

For King's convention the same clauses hold by Theorem 3.3: (i) and (iii) are Theorem 7.6(b), with the local bound of Theorem 7.6(a); (ii) is Corollary 4.3; (iv) follows from Theorem 7.6(b) by the argument above, word for word; (v) is Corollary 8.3. In every clause the inequalities used are the cone directions of Lemma 2.7(a),(b) and of Lemma 4.2; the reversed inequalities of Lemma 2.7(d) enter the curvature sector of this paper nowhere.

8. Translations, the cubic group, and rotations

This section proves the clause OS1 of King's printed state as far as the lattice allows: invariance under all translations and under the cubic group O_h about every point; full $O(3)$ invariance comes from the rotation theorem (Theorem 3.14, proved in the rotation part, Theorem 15.1) and is stated at the end. No cube is invariant under a non-trivial translation or a reflection; what replaces exactness is a sandwich between a smaller and a larger cube of the same family, in the direction of the cone inequalities of Lemmas 2.10 and 4.1, followed by the limits $k\to\infty$, $J\to\infty$. Coordinate permutations alone are exact (Lemma 8.4).

Conventions. We use the notation of Definition 2.3 and Corollary 4.4: for a finite family $f=(f_1,\dots,f_N)$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$ and $z\in\mathbb{C}^N$, $a^K_{k,J}(z)=\langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ and $a^{(a)}_{k,J}(z)=\langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ are the lattice functionals, Z^K_J, Z^K_∞ and Z_J, Z_∞ their limits where these exist, and $g(z)=\sum_j(\mathrm{Re}\, z_j)f_j$. Since F_{Λ_k} is linear in the source (extended $\mathbb{C}$-linearly), a lattice functional depends on a family only through its total source, and we use the same symbol for a coefficient vector and for its total source: $a^K_{k,J}(z)=Z^K_{k,J}(\sum_j z_j f_j)$ with $Z^K_{k,J}(u):=\langle e^{F(u)}\rangle^K_{k,J}$ for a complex-valued 2-form u; in particular, for a real 2-form u, $Z^K_{k,J}(\mathrm{i}u)$ is the one-member functional at the coefficient i, and the same convention is carried to Z^K_J, Z^K_∞ (so $Z^K_\infty(\mathrm{i}u_t)=Z^{K,f}_\infty(\mathrm{i}t)$ for $u_t=\sum_j t_j f_j$). When several families are in play we write $a^{K,f}_{k,J}$, $Z^{K,f}_J$, $Z^{K,f}_\infty$. $J_0(f)$ is the least J such that every $\operatorname{supp} f_j$ lies at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_J$; for $J\ge J_0(f)$ the support condition (S) holds for X_J. The transport $(O\cdot f)_x(u,v)=f_{O^{-1}x}(O^{-1}u,O^{-1}v)$ and the translation $(\tau_b f)_x=f_{x-b}$ are those of Definition 2.3; $O\cdot(O'\cdot f)=(OO')\cdot f$, and both are linear in f. Translations and the cubic group act on the Schwinger functions by this signed push-forward of 2-forms, as in the torus paper [27]: for an improper O no extra sign appears.

8.1. Translations on the comparison convention.

Proposition 8.1 (translations, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1), so that by Lemma 7.7(i),(iii) the limits $Z_J(z)=\lim_k a^{(a)}_{k,J}(z)$ and $Z_\infty(z)=\lim_J Z_J(z)$ exist locally uniformly on $\mathbb{C}^N$ for every family and are entire. Then for every finite family f of real 2-forms, every $b\in\mathbb{R}^3$ and every $z\in\mathbb{C}^N$,*

$$Z_\infty\Big(\sum_j z_j\tau_b f_j\Big)=Z_\infty\Big(\sum_j z_j f_j\Big).$$

Proof. Step 1 (lattice covariance). Let $k_0 \geq 0$, $b \in \Lambda_{k_0}$ and $k \geq k_0$. Then τ_b maps Λ_k onto itself, and the substitution $A'_\alpha(x) := A_\alpha(x+b)$, $\phi'(x) := \phi(x+b)$ is a bijection of the configurations of the comparison state on X_J onto those of the comparison state on $X_J - b$ (Definition 2.1 on the translated box) which carries the Lebesgue measure and the action to themselves: the action is a sum of local terms with site-independent coefficients (δm_ε^2, W_ε depend on ε only), and the boundary conditions are transported. The plaquette p corresponds to $p-b$ with the same orientation and $F(p)(A) = F(p-b)(A')$, so $F_{\Lambda_k}(\tau_b f)(A) = F_{\Lambda_k}(f)(A')$. Hence, for all coefficients,

$$Z_{k,X_J}\Big(\sum_j z_j \tau_b f_j\Big) = Z_{k,X_J-b}\Big(\sum_j z_j f_j\Big) \qquad (b \in \Lambda_{k_0},\ k \geq k_0), \tag{16}$$

the right side being the functional of the comparison state on $X_J - b$.

Step 2 (interlacing of the translated box). For $|b|_\infty \leq M$: $X_{J-1} \subset X_J - b \subset X_{J+1}$, since $-MJ - b_\mu \leq -M(J-1)$, $MJ - b_\mu \geq M(J-1)$, $-MJ - b_\mu \geq -M(J+1)$ and $MJ - b_\mu \leq M(J+1)$ for every μ.

Step 3 (the sandwich on the imaginary axes). Let $J \geq J_0(f)+1$, $t \in \mathbb{R}^N$ and $u_t = \sum_j t_j f_j$, so that $(u_t, 0) \in \mathcal{K}$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J-1}$. Apply the first inequality of Lemma 2.10 twice: with $Y = X_J - b$ and $X = X_{J-1}$, and with $Y = X_{J+1}$ and $X = X_J - b$. The support condition of the lemma holds because the plaquettes carrying u_t have their corners in X_{J-1}; no site is killed, so each cut-out factor is the comparison state on the smaller box (the Dirichlet terms of the smaller state are the diagonal parts of the cut bonds, and where the boundaries of two of the boxes touch they coincide). With (16),

$$a^{(a)}_{k,J-1}(\mathrm{i}t) \ \leq\ a^{(a),\tau_b f}_{k,J}(\mathrm{i}t) \ \leq\ a^{(a)}_{k,J+1}(\mathrm{i}t) \qquad (k \geq k_0,\ J \geq J_0(f)+1).$$

Let $k \to \infty$: the two ends converge by Lemma 7.7(i) for the family f at $J \mp 1 \geq J_0(f)$, and the middle term by Lemma 7.7(i) for the family $\tau_b f$, which is admissible at J because $\operatorname{supp} f_j + b$ lies at distance > 1 from $\mathbb{R}^3 \setminus C_J$ (as $C_{J-1} + b \subset C_J$). Then let $J \to \infty$ by Lemma 7.7(iii): $Z^{\tau_b f}_\infty(\mathrm{i}t) = Z^f_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$.

Step 4 (all coefficients). $D(z) := Z^{\tau_b f}_\infty(z) - Z^f_\infty(z)$ is entire on $\mathbb{C}^N$ and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension N; every partial derivative of D at 0 vanishes ($\partial_{y_j} D(\mathrm{i}y) = \mathrm{i}\,\partial_{z_j} D$), so $D \equiv 0$. Thus the identity holds for $b \in \Lambda_{k_0}$, $|b|_\infty \leq M$, and all $z \in \mathbb{C}^N$.

Step 5 (all lattice translations). A vector $b \in \bigcup_k \Lambda_k$ lies in some Λ_{k_0} and is a finite sum $b_1 + \cdots + b_n$ of vectors $b_i \in \Lambda_{k_0}$ with $|b_i|_\infty \leq M$ (each coordinate splits into steps $\pm M$ and a remainder in $\varepsilon_{k_0}\mathbb{Z}$ of modulus $< M$). Steps 3–4 hold for every family, in particular for the translated families $\tau_{b_1+\cdots+b_{i-1}} f$; composing gives the identity for b.

Step 6 (all $b \in \mathbb{R}^3$ at real coefficients). Fix $\rho > 0$ and let $E_\rho := C^\infty_c(\bar{B}_\rho; \Lambda^2)$ with its C^∞ topology, a Fréchet space. On E_ρ the function $\Phi(u) := \ln Z_\infty(u)$ at real u is finite ($Z_\infty(u) \geq 1$ by Jensen's inequality, as $\langle F(u)\rangle = 0$ by charge conjugation), convex (a pointwise limit of the convex functions $u \mapsto \ln\langle e^{F(u)}\rangle^{(a)}_{k,J}$) and Borel (each lattice functional depends continuously on finitely many values of u), hence continuous by Lemma 2.14; this is Lemma 7.7(iv). For $|b| \leq \rho' - \rho$ the map $b \mapsto \tau_b u \in E_{\rho'}$ is continuous, $\bigcup_k \Lambda_k$ is dense, and Φ on $E_{\rho'}$ restricted to E_ρ is the same pointwise limit; so Step 5 gives $Z_\infty(\tau_b u) = Z_\infty(u)$ for all $b \in \mathbb{R}^3$ and real $u \in E_\rho$, and exhausting over ρ for all real $u \in C^\infty_c$.

Step 7 (complex coefficients). For a family f and $b \in \mathbb{R}^3$, both sides of the asserted identity are entire in z and agree on $\mathbb{R}^N$ by Step 6 (applied to $u = \sum_j x_j f_j$), a set of uniqueness; hence they agree on $\mathbb{C}^N$. □

Only the first inequality of Lemma 2.10 enters: the sandwich is taken on the cone, and the passage off it is the identity theorem. The translation clause for the joint field on the tube, by the same sandwich at the cone points of the tube, is Lemma 13.11(v).

8.2. **The translation sandwich on King's convention.** On King's convention the translated cube has its killed sites on $T_J - b$, which neither contains nor is contained in $T_{J\pm1}$, so Lemma 2.10

does not compare it with $C^K_{J\pm1}$ directly; one kill step (Lemma 2.7(c) through Lemma 2.9) and one cut-out with killed sites (Lemma 4.1) at each end do.

Lattice covariance on King's convention. For $b \in \Lambda_{k_0}$ and $k \geq k_0$ let $\langle\cdot\rangle^{K,-b}_{k,J}$ denote the comparison state on $X_J - b$ with the matter killed on $T_J - b$ (King's cube translated by $-b$). The substitution of Step 1 above is a bijection of the configurations on X_J killed on T_J onto those on $X_J - b$ killed on $T_J - b$, carrying the point masses to the point masses and the action to the action (the kill set translates with the box). Hence, for every finite family and all coefficients,

$$Z^{a,K}_{k,J}\Big(\sum_j z_j\tau_b f_j, \sum_j w_j\tau_b h_j\Big) = Z^{a,K,-b}_{k,J}\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) \qquad (b \in \Lambda_{k_0},\ k \geq k_0), \tag{17}$$

the right side being the scheme functional of $\langle\cdot\rangle^{K,-b}_{k,J}$ with the same multiplier ($\mathsf{S}_{\Lambda_k}(\tau_b h) = \mathsf{S}_{\Lambda_k}(h)$).

Lemma 8.2 (translation sandwich, King's convention)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and let $b \in \Lambda_{k_0}$ with $|b|_\infty \leq M$. Then for every $k \geq k_0$, every $M \geq 1$ and every $J \geq J_0 + 1$,*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f, g) \ \leq\ Z^{a,K}_{k,J}(\mathrm{i}\,\tau_b f, \tau_b g) \ \leq\ Z^{a,K}_{k,J+1}(\mathrm{i}f, g),$$

and the reversed inequalities hold for $(f, -g)$.

Proof. By (17) the middle term is $Z^{a,K,-b}_{k,J}(\mathrm{i}f, g)$; the multipliers of the three terms are the same positive constant (Lemma 2.11(a); S is translation invariant), so it suffices to prove the inequalities for $\langle\Theta(\mathrm{i}f, g)\rangle$, and the reversed ones for $\Theta(f, -g)$; all these expectations are real.

Geometry. Here $b_\mu \in \varepsilon_{k_0}\mathbb{Z}$, $|b_\mu| \leq M$, and all sets are subsets of Λ_k. (G1) $\mathrm{alive}^K_J - b = \prod_\mu\{-MJ - b_\mu, \ldots, MJ - 2\varepsilon - b_\mu\} \subset \mathrm{alive}^K_{J+1} = \{-MJ - M, \ldots, MJ + M - 2\varepsilon\}^3$, since $-MJ - b_\mu \geq -MJ - M$ and $MJ - 2\varepsilon - b_\mu \leq MJ + M - 2\varepsilon$. (G2) $X_J - b \subset X_{J+1}$: the same arithmetic with ε in place of 2ε. (G3) $\mathrm{alive}^K_{J-1} = \{-MJ + M, \ldots, MJ - M - 2\varepsilon\}^3 \subset \mathrm{alive}^K_J - b$, since $-MJ + M \geq -MJ - b_\mu$ and $MJ - M - 2\varepsilon \leq MJ - 2\varepsilon - b_\mu$. (G4) $X_{J-1} \subset X_J - b$, likewise. All four inclusions hold with margin $M - |b_\mu| \geq 0$ on every face, at every spacing, including $k = 0$, $M = 1$, $|b_\mu| = M$; non-strict inclusions are all that Lemma 4.1 needs.

Sources. By (S) for X_{J-1}, every plaquette carrying f has its corners in X_{J-1} and every site carrying g lies in alive^K_{J-1}; by (G1)–(G4) these lie in the site sets and alive sets of all three states. So the observable is one function of variables present and alive in all three states, and its scalar factor equals 1 at every site killed below.

Upper end. Start from $\langle\cdot\rangle^K_{k,J+1}$, the comparison state on X_{J+1} killed on T_{J+1}, a BFS measure of King type (Definition 2.2). (i) Kill $U_+ := \mathrm{alive}^K_{J+1} \setminus (\mathrm{alive}^K_J - b)$. These are alive sites of the state, $g = 0$ on U_+ (by (G3), $U_+ \cap \mathrm{alive}^K_{J-1} = \emptyset$), and $\Theta(\mathrm{i}f, g)$ is continuous with $|\Theta| = e^{\sum_x g_x r_x^2}$; so by Lemma 2.7(c), through Lemma 2.9 (which allows the already killed sites T_{J+1}), killing U_+ does not increase $\langle\Theta(\mathrm{i}f, g)\rangle$, and by Lemma 2.7(d) with Lemma 2.9 it does not decrease $\langle\Theta(f, -g)\rangle$. The result is the comparison state on X_{J+1} killed on $T' := T_{J+1} \cup U_+ = X_{J+1} \setminus (\mathrm{alive}^K_J - b)$ (by (G1); nothing is revived). (ii) Apply Lemma 4.1 with $V = X_{J+1}$, $T = T'$ and $X = X_J - b \subset V$ (by (G2)); the lemma allows any $T \subset V$ and any $X \subset V$. The cut-out factor is the comparison state on $X_J - b$ killed on $(X_J - b) \cap T' = (X_J - b) \setminus (\mathrm{alive}^K_J - b) = T_J - b$, that is $\langle\cdot\rangle^{K,-b}_{k,J}$; the support condition (S_X) holds for $X = X_J - b$ by *Sources*. Hence $\langle\Theta(\mathrm{i}f, g)\rangle^{K,-b}_{k,J} \leq \langle\Theta(\mathrm{i}f, g)\rangle^{(a),T'}_{X_{J+1}} \leq \langle\Theta(\mathrm{i}f, g)\rangle^K_{k,J+1}$, and $\geq$ for class $\mathcal{D}$.

Lower end. Start from $\langle\cdot\rangle^{K,-b}_{k,J}$, the comparison state on $X_J - b$ killed on $T_J - b$ (of King type, a translate of a state of Definition 2.2). (i) Kill $U_- := (\mathrm{alive}^K_J - b) \setminus \mathrm{alive}^K_{J-1}$, alive sites at which $g = 0$ by *Sources*; as above, $\langle\Theta(\mathrm{i}f, g)\rangle$ does not increase and $\langle\Theta(f, -g)\rangle$ does not decrease, and the result is the comparison state on $X_J - b$ killed on $T'' := (X_J - b) \setminus \mathrm{alive}^K_{J-1}$ (by (G3)). (ii) Lemma 4.1 with $V = X_J - b$, $T = T''$ and $X = X_{J-1} \subset V$ (by (G4)): the factor is the comparison state on X_{J-1} killed on $X_{J-1} \cap T'' = X_{J-1} \setminus \mathrm{alive}^K_{J-1} = T_{J-1}$, that is $\langle\cdot\rangle^K_{k,J-1}$; ($\mathrm{S}_X$) for X_{J-1} is the hypothesis. Hence $\langle\Theta(\mathrm{i}f, g)\rangle^K_{k,J-1} \leq \langle\Theta(\mathrm{i}f, g)\rangle^{K,-b}_{k,J}$, and $\geq$ for class $\mathcal{D}$.

There is no edge case because a translation by $b \in \Lambda_{k_0}$, $|b|_\infty \le M$, moves King's cube rigidly, with the alive sets nested at margins $M - |b_\mu| \ge 0$ at every spacing; the reflection of Lemma 4.5 shifts the cube by one lattice layer instead, which is why its one-step form needs $M \ge 2\varepsilon_k$. □

The class-$\mathcal{I}$ clause uses Lemma 2.7(a),(b),(c),(e) and Lemma 2.9 only; the class-$\mathcal{D}$ clause uses Lemma 2.7(d) in addition. Only the class-$\mathcal{I}$ clause is used in this section.

Corollary 8.3 (translations of King's printed state)**.** *For every finite family f of real 2-forms, every $b \in \mathbb{R}^3$ and every $z \in \mathbb{C}^N$, $Z^K_\infty(\sum_j z_j \tau_b f_j) = Z^K_\infty(\sum_j z_j f_j)$. Moreover, for all real 2-forms $u, u' \in C^\infty_c$,*

$$|Z^K_\infty(\mathrm{i}u) - Z^K_\infty(\mathrm{i}u')| \le \sqrt{2}\,\|u - u'\|_2,$$

the L^2 norm of a 2-form being computed in a basis (e_c) of Λ^2 orthonormal for the norm formula.

Proof. Write $u_t = \sum_j t_j f_j$ for $t \in \mathbb{R}^N$. (1) *$b \in \Lambda_{k_0}$, $|b|_\infty \le M$, on the imaginary axes.* For $J \ge J_0(f) + 1$ and $k \ge k_0$, Lemma 8.2 at $(u_t, 0) \in \mathcal{K}$ gives $a^K_{k,J-1}(\mathrm{i}t) \le a^{K,\tau_b f}_{k,J}(\mathrm{i}t) \le a^K_{k,J+1}(\mathrm{i}t)$. Let $k \to \infty$: the ends converge by Theorem 7.6(b) for f at $J \mp 1 \ge J_0(f)$; the middle term by Theorem 7.6(b) for the family $\tau_b f$, which is admissible at J since $\operatorname{supp} f_j + b$ lies at distance > 1 from $\mathbb{R}^3 \setminus C_J$ ($C_{J-1} + b \subset C_J$ for $|b|_\infty \le M$), so that Theorem 3.3, posed for every finite family, applies to it. Hence $Z^K_{J-1}(\mathrm{i}t) \le Z^{K,\tau_b f}_J(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t)$, and $J \to \infty$ (Corollary 4.4(ii)(β) at both ends) gives $Z^{K,\tau_b f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$. (2) *All coefficients, same b.* $D(z) := Z^{K,\tau_b f}_\infty(z) - Z^{K,f}_\infty(z)$ is entire on $\mathbb{C}^N$ (Theorem 7.6(b) for both families) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension, so every partial derivative of D at 0 vanishes and $D \equiv 0$. (3) *All $b \in \bigcup_k \Lambda_k$*: as in Step 5 of Proposition 8.1, by composition over the translated families. (4) *The Lipschitz bound.* For real 2-forms u, u' whose plaquettes have corners in X_J, at every (k, J), $|\langle e^{\mathrm{i}F(u)}\rangle^K_{k,J} - \langle e^{\mathrm{i}F(u')}\rangle^K_{k,J}| \le \langle |F(u-u')|\rangle^K_{k,J} \le \langle F(u-u')^2\rangle^{K\,1/2}_{k,J} \le \sqrt{2}\|u - u'\|_{2,k}$, by $|e^{\mathrm{i}x} - e^{\mathrm{i}y}| \le |x - y|$, the linearity of F_{Λ_k} and the second-moment clause of Theorem 7.6(a). Both sides have limits as $k \to \infty$ and $J \to \infty$ (Theorem 7.6(b) for the one-member families at the coefficient i; $\|u - u'\|_{2,k} \to \|u - u'\|_2$ by (1)). (5) *All $b_0 \in \mathbb{R}^3$.* Pick $b_n \to b_0$ with $b_n \in \bigcup_k \Lambda_k$. For real $u \in C^\infty_c$, $\tau_{b_n} u \to \tau_{b_0} u$ in L^2 (uniform continuity and compact support), so by (4) and (3), $Z^K_\infty(\mathrm{i}\tau_{b_0} u) = \lim_n Z^K_\infty(\mathrm{i}\tau_{b_n} u) = Z^K_\infty(\mathrm{i}u)$. With $u = u_t$ this is $Z^{K,\tau_{b_0} f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all t, and (2) extends it to all $z \in \mathbb{C}^N$. □

Nothing here uses a reflection or any property of the limit beyond Theorem 7.6; King's derivation of translation invariance from the uniqueness of his limit [16, Thm. 2.3(i)] is not used.

8.3. **The cubic group.**

Lemma 8.4 (passive covariance with killed sites)**.** *Let O, Λ, (f_α) and X be as in Lemma 2.12, and let $U \subset X$ be a set of killed sites. Then the identities of Lemma 2.12 hold between the state on X with the matter killed on U and the state on OX (frame (Of_α), transported boundary conditions and wall) with the matter killed on OU:*

$$\big\langle \Theta(z\,O\cdot f, w\,O\cdot h)\big\rangle_{O\Lambda, OX; OU} = \big\langle \Theta(zf, wh)\big\rangle_{\Lambda, X; U}, \qquad \mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h).$$

Proof. The configuration-space bijection $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$ of Lemma 2.12 carries $\prod_{x \in U} \delta_0(d\phi(x))\,d\nu$ to $\prod_{x \in OU} \delta_0(d\phi'(Ox))\,d\nu$ (the point mass at $\phi(x) = 0$ goes to the point mass at $\phi'(Ox) = 0$), and the diagonal parts $\frac{\varepsilon}{2} r^2$ at the alive ends of the bonds into U are frame-free; the rest of that proof is unchanged. □

Theorem 8.5 (cubic symmetry of King's printed state)**.** *For all real 2-forms $f_1, \dots, f_N \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, all $z \in \mathbb{C}^N$, all $O \in O_h$ (the signed permutation matrices) and all $b \in \mathbb{R}^3$,*

$$Z^K_\infty\Big(\sum_j z_j\, \tau_b O\cdot f_j\Big) = Z^K_\infty\Big(\sum_j z_j f_j\Big).$$

Hence the Schwinger functions S^K_n of Theorem 7.6(c) are $\mathbb{R}^3 \rtimes O_h$-covariant, F transforming as a 2-form.

Proof. "(k, J) exact" means an identity of the lattice functionals for all complex coefficients and all k, J for which both families have their plaquettes in X_J.

Step 1 (permutations are exact). Let P be a permutation matrix, $Pe_\alpha = e_{\pi(\alpha)}$. Then $P\Lambda_k = \Lambda_k$, $PX_J = X_J$ and $PT_J = T_J$ (the condition "$x_\mu = MJ - \varepsilon$ for some μ" is permutation invariant), hence P alive^K_J = alive^K_J. By Lemma 8.4 with $O = P$, $X = X_J$, $U = T_J$, $\langle\Theta(zP \cdot f)\rangle_{P\Lambda_k, PX_J; PT_J}$ in the frame $(e_{\pi(\alpha)})$ equals $\langle\Theta(zf)\rangle^K_{k,J}$, and the state on the left is $\langle\cdot\rangle^K_{k,J}$ itself: the Dirichlet rule is a condition on the site set X_J, the kill set is T_J, the Neumann rule keeps the same set of (bond, direction) pairs relabelled, the mass term runs over the same bonds, and the hopping and single-site terms are frame-free. The plaquette set and the centres are the same; for a transposed pair $(x; \beta, \alpha)$ both $F(p)$ and $f_{c(p)}(e_\beta, e_\alpha)$ change sign by antisymmetry, so their product is unchanged. Thus $a^{K, P\cdot f}_{k,J}(z) = a^{K,f}_{k,J}(z)$ is (k, J) exact. The distance-> 1 interior $(-MJ + 1, MJ - 1)^3$ of C_J is P-invariant, so $J_0(P \cdot f) = J_0(f)$, and Theorem 7.6(b) for both families gives $Z^{K,P\cdot f}_\infty = Z^{K,f}_\infty$ on $\mathbb{C}^N$.

Step 2 (the reflection σ_3 on the imaginary axes). Let $\sigma_3 = \mathrm{diag}(1, 1, -1)$, $t \in \mathbb{R}^N$, $u_t = \sum_j t_j f_j$, $J_0 = J_0(f)$. Lemma 4.5(b), for every $k \geq 0$, every $M \geq 1$ and every $J \geq J_0 + 2$, gives

$$a^{K,f}_{k,J-2}(\mathrm{i}t) \;\leq\; a^{K,\sigma_3\cdot f}_{k,J}(\mathrm{i}t) \;\leq\; a^{K,f}_{k,J+2}(\mathrm{i}t),$$

since $\sigma_3 \cdot u_t = \sum_j t_j \sigma_3 \cdot f_j$. The family $\sigma_3 \cdot f$ is admissible at J, the interior being σ_3-symmetric, so $J_0(\sigma_3 \cdot f) = J_0(f)$. Let $k \to \infty$ (Theorem 7.6(b) for f at $J \mp 2$ and for $\sigma_3 \cdot f$ at J), then $J \to \infty$ (Corollary 4.4(ii)(β) at both ends): $Z^{K,\sigma_3\cdot f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$. The one-step sandwich of Lemma 4.5(a) would serve whenever $M \geq 2\varepsilon_k$, that is for all but finitely many k, which the k-limit ignores; the two-step form is used because it needs no case distinction, and the J-limit absorbs the extra step. This is the use King makes of the correlation inequalities in the proof of his Theorem 3.2 [16], with the kill steps added; no reversed inequality enters.

Step 3 (σ_1, σ_2). $\sigma_\nu = P\sigma_3 P^{-1}$ with P the transposition of the axes ν and 3. Since the transport is a left action, for every real 2-form u,

$$Z^K_\infty(\mathrm{i}\sigma_\nu \cdot u) = Z^K_\infty(\mathrm{i}P \cdot (\sigma_3 P^{-1} \cdot u)) = Z^K_\infty(\mathrm{i}\sigma_3 \cdot (P^{-1} \cdot u)) = Z^K_\infty(\mathrm{i}P^{-1} \cdot u) = Z^K_\infty(\mathrm{i}u),$$

by Step 1 for the family $\sigma_3 P^{-1} \cdot u$, Step 2 for the family $P^{-1} \cdot u$, and Step 1 again; each step is a statement about every family.

Step 4 (the group, on the imaginary axes). Every $O \in O_h$ is a product of a permutation matrix and some of $\sigma_1, \sigma_2, \sigma_3$, and if $Z^K_\infty(\mathrm{i}O \cdot u) = Z^K_\infty(\mathrm{i}u)$ for all real u holds for O_1 and O_2, it holds for $O_1 O_2$. By Steps 1–3, $Z^{K,O\cdot f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for every $O \in O_h$, every family f and every $t \in \mathbb{R}^N$.

Step 5 (all coefficients). $D(z) := Z^{K,O\cdot f}_\infty(z) - Z^{K,f}_\infty(z)$ is entire on $\mathbb{C}^N$ (Theorem 7.6(b); $J_0(O \cdot f) = J_0(f)$) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a set of uniqueness; so $D \equiv 0$. No splitting of sources is needed, there being no scalar source.

Step 6 (about every point). O_h about b is $\tau_b O \tau_{-b}$, and translations are Corollary 8.3 (every family, every $b \in \mathbb{R}^3$, all coefficients): $Z^K_\infty(\sum_j z_j \tau_b O \cdot f_j) = Z^K_\infty(\sum_j z_j O \cdot f_j) = Z^K_\infty(\sum_j z_j f_j)$.

Step 7 (Schwinger functions). Differentiating the identity of entire functions at $z = 0$ gives $S^K_n(\tau_b O \cdot f_1, \ldots, \tau_b O \cdot f_n) = S^K_n(f_1, \ldots, f_n)$ for all n and $(b, O) \in \mathbb{R}^3 \rtimes O_h$. □

Remark 8.6 (what cubic symmetry needs). King's cube is exactly symmetric under the coordinate permutations (Step 1) but not under the reflections; there the sandwich of Lemma 4.5 replaces exactness, as that of Lemma 2.13 does on the comparison convention. So O_h-invariance of King's printed state follows from the uniqueness of the monotone cube limit (Corollary 4.3 with Theorem 7.6(b)) together with the cut sandwich, and does not need the uniqueness of the infinite-volume state. The remark of the torus paper [27] that O_h-invariance "would follow from the uniqueness of the infinite-volume limit, which is not known" is superseded by Theorem 8.5. Only class-$\mathcal{I}$ clauses enter, and no scheme or Wick constant.

8.4. **Rotations.**

Theorem 8.7 (rotations)**.** *Z^K_∞ is $\mathbb{R}^3 \rtimes O(3)$-invariant: for every finite family f, every $z \in \mathbb{C}^N$ and every $(b,R) \in \mathbb{R}^3 \rtimes O(3)$, $Z^K_\infty(\sum_j z_j \tau_b R \cdot f_j) = Z^K_\infty(\sum_j z_j f_j)$, and the Schwinger functions S^K_n are Euclidean covariant, F transforming as a 2-form.*

Proof. By Theorem 18.12 (the rotation theorem Theorem 3.14 on King's convention: the torus comparison Theorem 17.1 with the closing estimate Theorem 18.4), $Z^K_\infty(\cdot,0)$ is $O(3)$-invariant for all complex coefficients; translations are Corollary 8.3; the Schwinger functions follow by differentiation at $z=0$. By Corollary 4.4(iii), with the fixed-cube limits of Theorem 3.3, the limit state $Z^{\mathrm{LIM}}_\infty(\cdot,0)$ of the comparison family equals $Z^K_\infty(\cdot,0)$ and is therefore $\mathbb{R}^3 \rtimes O(3)$-invariant as well. □

The rotation theorem is proved in the rotation part (Section 15, Theorem 15.1). Nothing in Corollary 8.3 and Theorem 8.5 depends on it. Its consequences for reflection positivity about every plane and for the components of the measure are drawn in Theorem 11.1(c).

9. Reflection positivity I: the modified box

King's Neumann–Dirichlet cube is not reflection symmetric about any plane, and the natural symmetrisation of its photon form is not reflection positive on gauge-invariant functionals (Remark 9.7). Reflection positivity is therefore proved first for the modified box state of [16, (3.38)] and then carried to the cube states by a two-sided comparison on the cone: the cube C_J sits inside the box (Lemma 9.9), and the box sits inside a larger cube once a margin proportional to μ_0^{-1} is allowed (Lemma 9.10). The passage to the limits is in Theorem 10.6. Throughout, $\varepsilon = \varepsilon_k$, the lattice is $\Lambda_k \subset \mathbb{R}^d$ with $d=3$ (the certificates of Lemma 9.6 are also given for $d=2$), a *lattice box* is a product $\Omega = \prod_\mu \{a_\mu, a_\mu+\varepsilon, \dots, b_\mu\} \subset \Lambda_k$, Ω^* is the set of its internal bonds (both ends in Ω) and Ω^{**} the set of its plaquettes (all four corners in Ω).

9.1. The lattice reflection and the modified box.

Definition 9.1 (mid-planes and the lattice reflection)**.** The *mid-planes* are the planes $\{x_\nu = c\}$ with $c \in D := \bigcup_{k\ge 0} \varepsilon_k(\mathbb{Z}+\frac12)$. If $c \in \varepsilon_{k_0}(\mathbb{Z}+\frac12)$ then $c \in \varepsilon_k(\mathbb{Z}+\frac12)$ for every $k \ge k_0$, because $5^j(m+\frac12) = 5^j m + \frac{5^j-1}{2} + \frac12$ and 5^j is odd; the same set D is obtained from any tail $k \ge k_1$ of the spacings. D is dense in $\mathbb{R}$ but not all of $\mathbb{R}$.

Fix ν, a mid-plane $c \in \varepsilon_k(\mathbb{Z}+\frac12)$ and a lattice box $\Omega \subset \Lambda_k$ with $\rho\Omega = \Omega$, $\rho = \rho_{\nu,c}$ the reflection of Definition 2.3. On the fields put $(\hat\rho\phi)(x) := \phi(\rho x)$ and $(\hat\rho A)(b) := A(\rho b)$, the image bond taken with its orientation: $(\hat\rho A)_\alpha(x) = A_\alpha(\rho x)$ for $\alpha \ne \nu$ and $(\hat\rho A)_\nu(x) = -A_\nu(\rho x - \varepsilon e_\nu)$. Then $\hat\rho$ is a signed permutation of the variables attached to Ω and Ω^* (it preserves Lebesgue measure and maps Ω^*, Ω^{**} to themselves); the *crossing bonds* $B_0 := \{\langle x^-, x^- + \varepsilon e_\nu\rangle : \ x^-_\nu = c - \varepsilon/2\}$ are mapped to their reversals, $(\hat\rho A)(b) = -A(b)$ for $b \in B_0$. Let $\Omega_\pm := \{x \in \Omega : \ \pm(x_\nu - c) > 0\}$ (no site lies on the plane), B_+ the bonds with both ends in Ω_+, $\mathfrak{B}_+$ the Borel functions of $(\phi|_{\Omega_+}, A|_{B_+})$, and

$$\theta G := \overline{G \circ \hat\rho} \qquad (G \in \mathfrak{B}_+).$$

On the smeared fields, $F_{\Lambda_k}(f)\circ\hat\rho = F_{\Lambda_k}(\rho^* f)$ and $W^a_{\Lambda_k}(h)\circ\hat\rho = W^a_{\Lambda_k}(h\circ\rho)$. Indeed, $F(p)\circ\hat\rho = \sigma_p F(\rho p)$ with $\sigma_p = -1$ exactly when $\nu \in \{\alpha,\beta\}$ (the orientation of the image plaquette, whose base point is shifted back one layer), the centres satisfy $c(\rho p) = \rho c(p)$, and $(\rho^* f)_{c(\rho p)}(e_\alpha, e_\beta) = f_{c(p)}(\rho e_\alpha, \rho e_\beta) = \sigma_p f_{c(p)}(e_\alpha, e_\beta)$; reindexing $p \mapsto \rho p$ gives the first identity. The scalar case is a relabelling of sites, and the c-numbers $2\mathsf{W}_\varepsilon\mathsf{S}(h)$ and $a_k\mathsf{S}(h)$ are invariant because ρ permutes Λ_k ($2(x_\nu - c) \in \varepsilon\mathbb{Z}$). Consequently, for families ζ_i whose lattice observables $\Theta(\zeta_i)$ lie in $\mathfrak{B}_+$,

$$\theta(\Theta(\zeta_i))\,\Theta(\zeta_j) = \Theta(\theta\zeta_i^* \sqcup \zeta_j), \qquad m(\theta\zeta_i^* \sqcup \zeta_j) = \overline{m(\zeta_i)}\, m(\zeta_j) \tag{18}$$

for the scheme multiplier $m(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ (a_k real), so that for any state $\langle\cdot\rangle$ on the box

$$\sum_{i,j} \bar\gamma_i\gamma_j\, m(\theta\zeta_i^* \sqcup \zeta_j)\langle\Theta(\theta\zeta_i^* \sqcup \zeta_j)\rangle = \sum_{i,j} \overline{(\gamma_i m(\zeta_i))}\,(\gamma_j m(\zeta_j))\,\big\langle\theta(\Theta(\zeta_i))\,\Theta(\zeta_j)\big\rangle. \tag{19}$$

Support. If the sources of ζ_i are supported in $\{x_\nu \ge c+\delta\}$ with $\delta > 0$, then for $\varepsilon_k < 2\delta$ every site carrying a sampled h and every corner of every plaquette carrying a sampled f (centre in the

support, corners at $x_\nu \geq c+\delta-\varepsilon_k/2 > c$) lies in Ω_+, so $\Theta(\zeta_i) \in \mathfrak{B}_+$. The splitting $h = h^+ - h^-$ of Definition 10.1 is taken with cutoffs supported in $\{x_\nu \geq c+\delta/2\}$, so that $h^\pm$ are supported in $\{x_\nu > c\}$ as well.

Definition 9.2 (the modified box state)**.** Let $\Omega \subset \Lambda_k$ be a lattice box, $F(p) = \varepsilon^{-1}\sum_{b\in\partial p} A(b)$, $\Omega(y)$ the indicator of Ω, and x^* the nearest neighbours of x in Λ_k. King's modified action [16, (3.38)] on Ω is

$$\begin{aligned} S^\varepsilon_\Omega(A,\phi) = {}& \tfrac12 \sum_{p\in\Omega^{**}} \varepsilon^d |F(p)|^2 + \tfrac12\mu_0^2 \sum_{b\in\Omega^*} \varepsilon^d |A(b)|^2 + \tfrac12 \sum_{x\in\Omega} \varepsilon^d \Big| \sum_{y\in x^*} \varepsilon^{-1}\Omega(y) A(\langle x,y\rangle)\Big|^2 \\ & + \tfrac12 \sum_{\langle x,y\rangle} \varepsilon^{d-2}\big[(\Omega r)(x)^2 + (\Omega r)(y)^2 - 2(\Omega r)(x)(\Omega r)(y)\cos(\vartheta_x - e\varepsilon A(\langle x,y\rangle) - \vartheta_y)\big] \\ & + \sum_{x\in\Omega} \varepsilon^d\big[\tfrac12 m^2 r_x^2 + \lambda r_x^4 + \tfrac12 \delta m_\varepsilon^2\, r_x^2\big], \end{aligned}$$

with the *constant* free-boundary counterterm δm_ε^2 of Definition 2.2 at every site, and with the free Wick constant W_ε in the Wick observables. (King prints a site-dependent counterterm in the last term; it is not used here, and the statements below are made for the constant one.) The photon variables are the internal bonds Ω^* only. The matter part is exactly the Dirichlet matter of the Neumann–Dirichlet state on $X = \Omega$: cosines on the internal bonds, and the diagonal $\frac12\varepsilon^{d-2} r_x^2$ for every bond leaving Ω. The *modified box state* $\langle\cdot\rangle^{(3.38)}_{k,\Omega}$ is the normalised measure $e^{-S^\varepsilon_\Omega}\prod_{x\in\Omega} d^2\phi_x \prod_{b\in\Omega^*} dA_b$; it is a BFS measure (the Gaussian is non-degenerate since $\mu_0 > 0$, the single-site measures have all exponential moments since $\lambda > 0$). We write $Z^{a,(3.38)}_{k,\Omega}(\zeta) := m(\zeta)\langle\Theta(\zeta)\rangle^{(3.38)}_{k,\Omega}$ with the same observables and multiplier as in Definition 2.3.

The third term is the graph divergence of (Ω,Ω^*): with $d_0 : \mathbb{R}^\Omega \to \mathbb{R}^{\Omega^*}$, $(d_0\chi)(\langle x, x+\varepsilon e_\alpha\rangle) = \varepsilon^{-1}(\chi(x+\varepsilon e_\alpha) - \chi(x))$, one has $\sum_{y\in x^*}\Omega(y)A(\langle x,y\rangle) = -\varepsilon(d_0^T A)(x)$ (only internal bonds contribute). With $d_1 A := (\varepsilon F(p))_{p\in\Omega^{**}}$, the *unitary form* on the box and the modified form are

$$Q^u_\Omega(A) := \tfrac12 \sum_{p\in\Omega^{**}} \varepsilon^d|F(p)|^2 + \tfrac12\mu_0^2\sum_{b\in\Omega^*}\varepsilon^d|A(b)|^2, \qquad Q^{(3.38)}_\Omega = Q^u_\Omega + \tfrac12\varepsilon^d\|d_0^T A\|^2,$$

i.e. the modified form is the unitary form plus a gauge-fixing term depending on A only through $d_0^T A$. The modified action is invariant under $\hat\rho$ for every mid-plane c with $\rho\Omega = \Omega$: every term is written through $A(\langle x,y\rangle)$ and nearest-neighbour pairs, which ρ permutes, $(d_0^T\hat\rho A)(x) = (d_0^T A)(\rho x)$, and δm_ε^2 is site independent.

The boxes. Fix $c \in D$, $k \geq k_0(c)$, $J \geq J_0$ such that every source of the enlarged family of Definition 10.1 and its reflection is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3\setminus C_{J_0}$, an integer $n \geq 1$, and put $a_k := c + \varepsilon_k/2 \in \varepsilon_k\mathbb{Z}$ and

$$\Omega_{J,k} := (C_{J+n} + a_k e_\nu)\cap\Lambda_k, \qquad Mn \;\geq\; |c| + 2 + R(\mu_0), \qquad R(\mu_0) := \frac{\pi\sqrt{3}}{2\mu_0}. \tag{20}$$

$C_{J+n}\cap\Lambda_k$ is symmetric in direction ν about $-\varepsilon_k/2$ (sites $x_\nu \in \{-M(J+n),\dots,M(J+n)-\varepsilon_k\}$, and $M(J+n)\in\varepsilon_k\mathbb{Z}$), so $\Omega_{J,k}$ is a lattice box symmetric about $\{x_\nu = c\}$ with an even number $2M(J+n)5^k$ of layers. Under the margin condition of (20), for every k: (i) C_J plus one layer is contained in $\Omega_{J,k}$ (this needs $Mn \geq |c| + \frac32\varepsilon_k$); (ii) C_{J+2n} contains every site at ℓ^∞-distance $\leq R(\mu_0) + \varepsilon_k$ from $\Omega_{J,k}$ (this needs $Mn \geq |c| + R(\mu_0) + \frac32\varepsilon_k$). We fix such an $n = n(\mu_0, c)$, independent of k and J. The two margins are exactly the hypotheses of Lemmas 9.9 and 9.10 below; by Lemma 4.6 they serve King's convention as they stand.

9.2. **Reflection positivity of the modified box.**

Lemma 9.3 (lattice reflection positivity of the modified box)**.** *Let $\mu_0 > 0$, c a mid-plane and Ω a lattice box with $\rho\Omega = \Omega$, $\rho = \rho_{\nu,c}$.*

(a) *For every gauge-invariant $G \in \mathfrak{B}_+$ that is bounded, or of the form $\Theta(\zeta)$ with sources supported in Ω_+, or more generally with $\langle |G|^2\rangle^{(3.38)}_{k,\Omega} < \infty$: $\langle \theta(G)G\rangle^{(3.38)}_{k,\Omega} \geq 0$.*

(b) *$(G, G') \mapsto \langle \theta(G')G\rangle^{(3.38)}_{k,\Omega}$ is a Hermitian positive semidefinite form on such G; hence $\sum_{i,j} \bar\gamma_i\gamma_j \langle \theta(G_i)G_j\rangle^{(3.38)}_{k,\Omega} \geq 0$, and by (19) and the support clause of Definition 9.1, $\sum_{i,j} \bar\gamma_i\gamma_j Z^{a,(3.38)}_{k,\Omega}(\theta\zeta_i^* \sqcup \zeta_j) \geq 0$ for families ζ_i with sources supported in $\{x_\nu \geq c + \delta\}$, $\varepsilon_k < 2\delta$.*

Proof. Step 1: gauge-invariant expectations of the modified form equal those of the unitary form. The graph (Ω, Ω^*) is connected, $\ker d_0$ consists of the constants, $\mathbb{R}^{\Omega^*} = \operatorname{ran} d_0 \oplus \ker d_0^T$ orthogonally, and $d_1 d_0 = 0$. Write $A = A^T + d_0\chi$ with $d_0^T A^T = 0$ and $\chi \perp 1$; $(A^T, \chi) \mapsto A$ is a linear bijection with constant Jacobian. Then $Q^u_\Omega(A) = Q^u_\Omega(A^T) + \frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2$ (orthogonality and $d_1 d_0 = 0$) and $Q^{(3.38)}_\Omega(A) = Q^u_\Omega(A^T) + q(\chi)$ with $q(\chi) = \frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2 + \frac{\varepsilon^d}{2}\|d_0^T d_0\chi\|^2$; both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. A gauge transformation is $A \mapsto A + d_0\chi$ together with $\vartheta \mapsto \vartheta - e\chi$: the hopping phase $\vartheta_x - e\varepsilon A_b - e(\chi(y) - \chi(x)) - \vartheta_y$ equals $(\vartheta_x + e\chi_x) - e\varepsilon A_b - (\vartheta_y + e\chi_y)$ and the Dirichlet diagonals carry no A, so for gauge-invariant G the matter integral $\Psi(A) := \int \prod_x d^2\phi_x\, (Ge^{-S_m})(A, \phi)$ satisfies $\Psi(A^T + d_0\chi) = \Psi(A^T)$ (rotate ϕ site by site; $\prod d^2\phi$ is invariant). Hence $\int dA\, e^{-Q^{(3.38)}_\Omega}\Psi = \int dA^T e^{-Q^u_\Omega(A^T)}\Psi(A^T) \int d\chi\, e^{-q(\chi)}$, and the same with Q^u_Ω in place of $Q^{(3.38)}_\Omega$ and a different finite χ-factor; the factors cancel in the normalised expectation. Fubini applies, since $|G| \leq Ce^{a\sum_b |A(b)|}e^{C\sum_x r_x^2}$ for $G = \Theta(\zeta)$ (trivially for bounded G) is dominated by the Gaussian and by $e^{-\lambda\varepsilon^d r^4}$. This is the mechanism of [7, Thm. 5.1] and of [27, Lemma 7.7], carried to the box graph.

Step 2: the unitary box measure is reflection positive on all of $\mathfrak{B}_+$. Let $S^u_\Omega := Q^u_\Omega + S_m$, write $\omega = (\omega_+, \omega_-, a)$ with $a = (a_b)_{b\in B_0}$ the crossing-bond variables, and put $\mathsf{M} := \mu_0^2\varepsilon^d$. The terms of S^u_Ω are of four kinds. Single-site terms (local matter and Dirichlet diagonals), hopping cosines on bonds of $B_\pm$, masses of bonds of $B_\pm$, and plaquettes with all corners in Ω_+ or all in Ω_- involve one side only; by $\hat\rho$-invariance the minus-side terms are the $\hat\rho$-transport of the plus-side terms. The crossing masses are $\frac{\mathsf{M}}{2}a_b^2$, $b \in B_0$. A crossing plaquette $p = (x^-; \alpha, \nu)$, $\alpha \neq \nu$, is present iff $x^-, x^- + \varepsilon e_\alpha \in \Omega$ (then all four corners are in Ω by symmetry); with $x^+ := x^- + \varepsilon e_\nu$, $b = \langle x^-, x^+\rangle$ and $b' = \langle x^- + \varepsilon e_\alpha, x^+ + \varepsilon e_\alpha\rangle$ it reads $\varepsilon F(p) = \pm[A_\alpha(x^-) + a_{b'} - A_\alpha(x^+) - a_b]$, the overall sign immaterial after squaring. A crossing hopping term is $-\varepsilon^{d-2} r_{x^-} r_{x^+} \cos(\vartheta_{x^-} - e\varepsilon a_b - \vartheta_{x^+})$, $b \in B_0$.

Split each crossing variable: $\int_{\mathbb{R}} g(a_b) e^{-\frac{\mathsf{M}}{2}a_b^2} da_b = \sqrt{2\pi/\mathsf{M}}\, \mathbb{E}\, g(\alpha_b - \alpha'_b)$ with α_b, α'_b independent $N(0, \frac{1}{2\mathsf{M}})$, whose difference is $N(0, \frac{1}{\mathsf{M}})$. Put $\xi := (\omega_+, \alpha)$ and $\xi' := (\omega_-^\rho, \alpha')$, where ω_-^ρ is the $\hat\rho$-image of the minus data transported to the plus side ($(\omega_-^\rho)_\alpha(x^+) = A_\alpha(x^-)$ for the tangential bonds, $\phi^\rho(x^+) = \phi(x^-)$, the ν-bonds of B_- carried with the sign of Definition 9.1); by $\hat\rho$-invariance the minus-side one-sided terms are the plus-side terms evaluated on ω_-^ρ, and the a priori measures of ξ and ξ' coincide. For a crossing plaquette put $X_p(\xi) := A_\alpha(x^+) - \alpha_{b'} + \alpha_b$, a function of plus variables and α (note $A_\alpha(x^+) \in B_+$ since $x^+, x^+ + \varepsilon e_\alpha \in \Omega_+$), and the same function on ξ', $X_p(\xi') = A_\alpha(x^-) - \alpha'_{b'} + \alpha'_b$; then $\varepsilon F(p) = \pm[X_p(\xi') - X_p(\xi)]$ and

$$\tfrac{\varepsilon^d}{2}F(p)^2 = \tfrac{\varepsilon^{d-2}}{2}X_p(\xi)^2 + \tfrac{\varepsilon^{d-2}}{2}X_p(\xi')^2 - \varepsilon^{d-2}X_p(\xi)X_p(\xi').$$

For a crossing hopping term put $\psi_b(\xi) := \phi(x^+)e^{\mathrm{i}e\varepsilon\alpha_b}$, $\psi_b(\xi') = \phi(x^-)e^{\mathrm{i}e\varepsilon\alpha'_b}$; then $r_{x^-}r_{x^+}\cos(\vartheta_{x^-} - e\varepsilon(\alpha_b - \alpha'_b) - \vartheta_{x^+}) = \operatorname{Re}[\psi_b(\xi')\overline{\psi_b(\xi)}] = \sum_{i=1,2}\psi_b^i(\xi)\psi_b^i(\xi')$. Hence, with S' the action minus the crossing masses,

$$e^{-S'} = \mathcal{G}(\xi)\,\mathcal{G}(\xi')\,\exp\Big(\sum_\iota \gamma_\iota Y_\iota(\xi)Y_\iota(\xi')\Big), \qquad \mathcal{G}(\xi) := e^{-S_+(\xi) - \frac{\varepsilon^{d-2}}{2}\sum_p X_p(\xi)^2},$$

where S_+ collects the one-sided plus terms and $\{(\gamma_\iota, Y_\iota)\} = \{(\varepsilon^{d-2}, X_p)\}_p \cup \{(\varepsilon^{d-2}, \psi_b^i)\}_{b,i}$: all $\gamma_\iota > 0$, all Y_ι real. For $G \in \mathfrak{B}_+$ we have $G(\omega) = G(\xi)$ and $\theta G(\omega) = \overline{G(\xi')}$, the same function on

the transported data (G depends neither on a nor on α, α'). Expanding the exponential,

$$Z\,\langle\theta(G)G\rangle^{u}_{k,\Omega} = K\sum_{n\in\mathbb{N}^I}\prod_{\iota}\frac{\gamma_\iota^{n_\iota}}{n_\iota!}\Big|\int d\xi\, G(\xi)\mathcal{G}(\xi)\prod_{\iota}Y_\iota(\xi)^{n_\iota}\Big|^2 \;\geq 0, \qquad K>0,$$

the two integrals of each term being complex conjugates because measures and functions coincide on the two copies. The interchange of sum and integral is justified by $\sum_n\prod_\iota\gamma_\iota^{n_\iota}|Y_\iota Y_\iota'|^{n_\iota}/n_\iota! \leq \exp(\frac{1}{2}\sum_\iota\gamma_\iota(Y_\iota^2+Y_\iota'^2))$: the plaquette parts cancel against $\mathcal{G}$ exactly, $|\psi_b| = r_{x^+}$, and $\int|G|e^{-S_+(\xi)+\frac{\varepsilon^{d-2}}{2}\sum_b r_{x^+}^2}d\xi <$ ∞ by $\lambda>0$ (a quartic term at every plus site), $\mu_0>0$ (a mass on every B_+ bond and the α-Gaussians) and the growth bound on G. This is the mechanism of [27, Lemma 7.8(ii)] with one plane in place of a pair and the Dirichlet diagonals as single-site terms; the classical form is the positive cross-term criterion of [10, 9]. The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle(G'\circ\hat\rho)\bar G\rangle = \langle G'\,\overline{G\circ\hat\rho}\rangle = \langle\theta(G)G'\rangle$ follows from the $\hat\rho$-invariance of the measure.

Step 3: combination and L^2 extension. For gauge-invariant $G\in\mathfrak{B}_+$, bounded or equal to $\Theta(\zeta)$, the product $\theta(G)G$ is gauge invariant ($\hat\rho$ maps gauge transformations to gauge transformations: $\hat\rho\, d_0\chi = d_0(\chi\circ\rho)$), so $\langle\theta(G)G\rangle^{(3.38)}_{k,\Omega} = \langle\theta(G)G\rangle^{u}_{k,\Omega}\geq 0$ by Steps 1–2. For gauge-invariant $G\in\mathfrak{B}_+$ with $\langle|G|^2\rangle<\infty$ (for instance polynomials in the plus-side smeared fields, Proposition 10.10), $G_n := G\,\mathbf{1}\{|G|\leq n\}$ is bounded, gauge invariant and in $\mathfrak{B}_+$, so $\langle\theta(G_n)G_n\rangle\geq 0$; moreover $|\theta(G_n)G_n|\leq|\theta G||G|$ and $\langle|\theta G||G|\rangle\leq\langle|G|^2\rangle^{1/2}\langle|\theta G|^2\rangle^{1/2} = \langle|G|^2\rangle$ by Cauchy–Schwarz and $\hat\rho$-invariance, so dominated convergence gives (a). The modified measure is itself $\hat\rho$-invariant (Definition 9.2), so the computation of the Hermitian property in Step 2 holds verbatim for it on L^2; the form is sesquilinear, Hermitian and nonnegative on the diagonal, hence positive semidefinite (apply (a) to $G=\sum_i\gamma_i G_i$). With (19) and the support clause this is (b). □

Remark 9.4 (on King's Proposition 3.3). Lemma 9.3 is the statement of [16, Prop. 3.3], which King derives from [7, Cor. 5.4]. That corollary (with [7, Lemma 5.2, Thm. 5.3]) concerns the lattice actions of [7, §5], whose photon Gaussian is translation invariant with a reflection-positive boundary term; King's finite-box Gaussian with its masked divergence is not one of them, the obstruction being the box geometry of the gauge-fixing term. So his Proposition 3.3 is true, and it is proved here rather than derived from [7, Cor. 5.4]. The gauge-fixing term $\|d_0^T A\|^2$ is not itself compatible with the mechanism of Step 2 (at a site next to the plane it produces cross terms $Y(\xi)Z(\xi')+Z(\xi)Y(\xi')$ of indefinite sign); on the torus it drops out for gauge-invariant observables, on King's cube it does not reduce to a divergence at the boundary (Lemma 9.5), and this is why the photon form has to be changed before reflecting (Remark 9.7).

9.3. **The same box: the boundary identity and the reversed inequality.** In this subsection we use lattice units, $\varepsilon=1$ (every term of the three photon forms carries the same weight $\frac{1}{2}\varepsilon^{d-2}$), and write $\mu := \mu_0\varepsilon$. For a lattice box Ω and $A\in\mathbb{R}^{\Omega^*}$ let $K^{\mathrm{int}}_\Omega(A) := \sum_\mu\sum_\nu\sum|A_\mu(x+e_\nu)-A_\mu(x)|^2$, the sum over pairs with both bonds $(x,\mu),(x+e_\nu,\mu)$ in Ω^* (the componentwise Neumann gradient form on the internal bonds), and $N_\partial(A) := \sum|A_b|^2$, the sum over the normal crossing pairs: for $b=(x,\mu)\in\Omega^*$ one term for each of $(x\pm e_\mu,\mu)$ that is not in Ω^*. Split $N_\partial = N_\partial^{\mathrm{up}}+N_\partial^{\mathrm{low}}$ according to whether the missing partner is $(x+e_\mu,\mu)$ (i.e. $x+2e_\mu\notin\Omega$) or $(x-e_\mu,\mu)$ (i.e. $x-e_\mu\notin\Omega$).

Lemma 9.5 (the boundary identity)**.** *For every lattice box Ω and every $A\in\mathbb{R}^{\Omega^*}$,*

$$\|d_1A\|^2_{\Omega^{**}} + \|d_0^TA\|^2 = K^{\mathrm{int}}_\Omega(A) + N_\partial(A).$$

Consequently $Q^{(3.38)}_\Omega = \frac{1}{2}\big[K^{\mathrm{int}}_\Omega + N_\partial + \mu^2\|\cdot\|^2_{\Omega^*}\big]$: the modified photon form is the internal Neumann–Feynman form plus the positive boundary term N_∂. (By the proof of Lemma 9.6(a), the marginal of the Neumann–Dirichlet Gaussian of Ω on the internal bonds has the precision $M^{(3.38)} - N_\partial^{\mathrm{low}} - P_{12}P_{22}^{-1}P_{21}$: the modified form minus the squares of the near-normal bonds and minus the Schur term of the far-face variables.)

Proof. Extend A by 0 to a 1-form a on $\mathbb{Z}^d$. For finitely supported 1-forms, $\sum_p|da(p)|^2 + \sum_x|d^*a(x)|^2 = \sum_\mu\sum_\nu\sum_x|a_\mu(x+e_\nu)-a_\mu(x)|^2$ (the Hodge Laplacian on $\mathbb{Z}^d$ is the componentwise Laplacian; summation by parts). On the left, $d^*a(x)$ is the masked divergence for $x\in\Omega$ and

0 outside, since no internal bond has an end outside Ω; a plaquette $p \notin \Omega^{**}$ meeting Ω^* has exactly one internal bond b (two parallel internal edges put all four corners in Ω; two adjacent ones put three corners in Ω, and a product of intervals then contains the fourth), so $|da(p)|^2 = |A_b|^2$. On the right, pairs with both bonds internal give K_Ω^{int}; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond contributes $|A_b|^2$; for $\nu \neq \mu$ the partner is non-internal iff $x+e_\nu \notin \Omega$ (as $x, x+e_\mu \in \Omega$, and in a box $x+e_\nu \in \Omega \iff x+e_\mu+e_\nu \in \Omega$), i.e. iff the plaquette spanned by b and e_ν lies outside Ω^{**}; these pairs are in bijection with the plaquettes counted on the left and cancel; for $\nu = \mu$ they are the terms of N_∂. □

Lemma 9.6 (the same box: the second inequality of King's (3.42) is reversed)**.** *Let Ω be a lattice box, $\langle\cdot\rangle_\Omega^{\mathrm{ND}}$ the Neumann–Dirichlet state on $X = \Omega$ (variables $A_\mu(x)$, $x \in \Omega$, all outgoing bonds; photon form $Q_\Omega^{\mathrm{ND}} = \frac{1}{2}[K_\Omega + \mu^2\|\cdot\|^2]$ of Definition 2.1) and $\langle\cdot\rangle_\Omega^{(3.38)}$ the modified box state of Definition 9.2 (variables Ω^*); both carry the same Dirichlet matter on Ω (constant δm_ε^2, free Wick constant), coupled to A through the internal bonds only. Let $\Sigma_\Omega^{\mathrm{ND}}$ and Σ_Ω^u be the covariances of $F = (d_1 A(p))_{p\in\Omega^{**}}$ under the two photon Gaussians. Then:*

(a) $\Sigma_\Omega^u = \Sigma_\Omega^{(3.38)} \leq \Sigma_\Omega^{\mathrm{ND}}$ *on* $\mathbb{R}^{\Omega^{**}}$*;*

(b) *for every gauge-invariant $Y \in \mathcal{I}$ with sources in Ω and every $e \geq 0$: $\langle Y\rangle_\Omega^{\mathrm{ND}} \leq \langle Y\rangle_\Omega^{(3.38)}$, and $\langle X\rangle_\Omega^{\mathrm{ND}} \geq \langle X\rangle_\Omega^{(3.38)}$ for $X \in \mathcal{D}$. This is the reverse of the second inequality of* [16, (3.42)]*, whose middle term is the modified box state on C_{J+1}; it holds weakly, for the modified state with the constant δm_ε^2 (nothing is asserted for King's printed site-dependent counterterm);*

(c) *King's inequality fails strictly in the Gaussian sector: at $e = 0$ and $Y = e^{\mathrm{i}t\langle v,F\rangle}$ it reads $v^T\Sigma_\Omega^{\mathrm{ND}}v \leq v^T\Sigma_\Omega^u v$, while $\Delta_\Omega := \Sigma_\Omega^{\mathrm{ND}} - \Sigma_\Omega^u \geq 0$ is nonzero. In exact rational arithmetic the inertia (negative, zero, positive) of Δ_Ω is $(0,1,8)$ on the 4×4 box ($d=2$) for $\mu^2 \in \{\frac{1}{20},\frac{1}{4},1\}$ and $(0,3,13)$ on the $4\times2\times2$ box ($d=3$) for $\mu^2 \in \{\frac{1}{20},\frac{1}{4}\}$; single-plaquette certificates are $\Delta_{pp} = \mathrm{Var}^{\mathrm{ND}} F(p) - \mathrm{Var}^{(3.38)} F(p) = \frac{14285248}{44314767}$ at $p = ((0,0);0,1)$ on 4×4, $\mu^2 = \frac{1}{4}$, and $\frac{121897601600}{395580222961}$ at $p = ((0,0,0);1,2)$ on $4\times2\times2$, $\mu^2 = \frac{1}{20}$.*

Proof. (a) Write the Neumann–Dirichlet precision $P := K_\Omega + \mu^2$ in blocks, the internal variables Ω^* first and the outgoing boundary variables second. Its internal block is $P_{11} = K_\Omega^{\mathrm{int}} + N_\partial^{\mathrm{up}} + \mu^2$: pairs of K_Ω with both bonds internal give K_Ω^{int}; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond is impossible for $\nu \neq \mu$ (box convexity, as in Lemma 9.5) and for $\nu = \mu$ is (x,μ) internal with $(x+e_\mu,\mu)$ outgoing, i.e. $x+2e_\mu \notin \Omega$ (the other assignment would need $x+e_\mu \notin \Omega$ and $x+e_\mu \in \Omega$ at once); its diagonal contribution to the internal block is $|A_\mu(x)|^2$, which is N_∂^{up}. Hence, by Lemma 9.5, $M^{(3.38)} := K_\Omega^{\mathrm{int}} + N_\partial + \mu^2 = P_{11} + N_\partial^{\mathrm{low}} \geq P_{11}$. The marginal covariance of the internal variables under the Neumann–Dirichlet Gaussian is the inverse of the Schur complement, $(P^{-1})_{11} = (P_{11} - P_{12}P_{22}^{-1}P_{21})^{-1} \geq P_{11}^{-1} \geq (M^{(3.38)})^{-1}$, by operator monotonicity of the inverse on positive definite matrices and $P_{12}P_{22}^{-1}P_{21} \geq 0$. Since $F = d_1 A$ on Ω^{**} uses internal bonds only, $\Sigma_\Omega^{\mathrm{ND}} = D_1(P^{-1})_{11}D_1^T \geq D_1(M^{(3.38)})^{-1}D_1^T = \Sigma_\Omega^{(3.38)}$, and $\Sigma_\Omega^{(3.38)} = \Sigma_\Omega^u$ by Step 0 in the proof of Lemma 9.10 (the field strength depends only on the $\ker d_0^T$-component, whose law is the same under both Gaussians, Step 1 of Lemma 9.3). (b) Both states carry the Dirichlet matter of Ω coupled to Ω^*, and their field-strength laws on Ω^{**} are ordered by (a); the complete proof is given in Lemma J.2 of the supplemental material. (c) At $e = 0$ the matter decouples from A and is identical in the two states, so for $Y = e^{\mathrm{i}t\langle v,F\rangle}$ both sides equal $\exp(-\frac{t^2}{2}v^T\Sigma v)$ with the respective Σ; King's $\leq$ becomes $v^T\Sigma_\Omega^{\mathrm{ND}}v \leq v^T\Sigma_\Omega^u v$, which is false for every v with $v^T\Delta_\Omega v > 0$, for instance $v = e_p$ at the plaquettes listed. The inertias and the certificates are obtained from the two covariance matrices in exact rational arithmetic (symmetric elimination with diagonal pivoting over $\mathbb{Q}$). □

Scope. (i) King's passage from the box states to the cube limit on [16, p. 336] needs the sandwich for the box functionals with purely imaginary curvature coefficients, i.e. for the whole class $\mathcal{I}$; (c) refutes the second inequality on that class exactly. (ii) For the scalar observable $\Theta(0,2h_+)$ of the printed display, (b) shows that the inequality can hold only with equality;

at $e = 0$ it does (identical decoupled matter), and at $e > 0$ a strict failure is to be expected (the Wick square couples to F at order e^2) but is not certified here. Both clauses are for the modified state with the constant δm_ε^2: if King's site-dependent counterterm differs from the free-boundary number, the two matter measures differ, the mechanism of (b) does not apply, and neither clause is proved for his printed objects; (c) alone is independent of the counterterm, the matter decoupling from A at $e = 0$.

Remark 9.7 (what is false in the printed route). (i) King's Neumann–Dirichlet cube state is not reflection symmetric about any plane: the site set $C_J \cap \Lambda_k$ is symmetric in direction ν about $x_\nu = -\varepsilon_k/2$, the set of ν-bonds about $x_\nu = 0$, and no reflection maps the variable set to itself (Lemma 2.13 accordingly obtains the reflection symmetry of the limit from a cut sandwich, not from a lattice symmetry). (ii) The natural symmetrisation, the Feynman gradient form on the internal bonds $K_\Omega^{\mathrm{int}} + \mu^2$ (the modified form without N_∂; reflection symmetric), is *not* reflection positive on gauge-invariant functionals. This is exact at $e = 0$: on the 4×4 box at $\mu^2 = \frac{1}{20}$ the reflection kernel $R_{pq} = \sigma_p \Sigma[\rho p, q]$ of the plus-side plaquettes is negative definite (inertia $(3, 0, 0)$), and on $4 \times 2 \times 2$ at $\mu^2 = \frac{1}{20}$ it has inertia $(2, 2, 2)$ with $R_{pp} = \mathrm{Cov}(\theta F(p), F(p)) < 0$ for $p = ((2, 0, 0); 0, 1)$. Indeed, at $e = 0$ the matter decouples, and the gauge-invariant plus-side functionals $G_1 = e^{\mathrm{i}t\langle v, F_+\rangle}$, $G_2 = 1$ give the matrix $(\langle \theta(G_i) G_j\rangle)$ with determinant $e^{-t^2 v^T \Sigma_{++} v}(e^{t^2 v^T R v} - 1) < 0$ for every $t > 0$ when $v^T R v < 0$, since $\langle \theta(G_1) G_1\rangle = \exp(-t^2 v^T \Sigma_{++} v + t^2 v^T R v)$ and $\langle \theta(G_1)\rangle = \langle G_1\rangle = \exp(-\frac{1}{2} t^2 v^T \Sigma_{++} v)$ by the reflection symmetry $\Sigma_{--} = \Sigma_{++}$. By continuity at fixed spacing the symmetrised cube state is not reflection positive for small $e > 0$ either. A proof must work at every coupling, so reflection positivity of King's own cube form, or of this symmetrisation, at fixed spacing is not available as a route: any route through cube states at fixed spacing has to change the photon form, as the modified box does (a reflection-symmetric, reflection-positive modification whose values converge to the same limit, Theorem 10.6). Nothing is claimed about the symmetrised form at a given larger e or about other proofs; the route taken here needs the comparison of Lemmas 9.9 and 9.10, and its reflecting plane must be a lattice mid-plane for the half-bond mechanism of Lemma 9.3 (King's cubes have an even number of layers). (iii) Accordingly, the second inequality of [16, (3.42)] is false as printed in the following sense: false for the class of observables that the argument on [16, p. 336] needs (exactly, Lemma 9.6(c)), and never strictly true for the modified state with the constant counterterm (Lemma 9.6(b)). Its valid first half is Lemma 9.9; the invalid half is replaced by Lemma 9.10 with the margin $R(\mu_0)$ of (20), $n(\mu_0, c)$ cube steps instead of one.

Remark 9.8 (the box form is the larger one). King describes the modified box action as obtained from the cube action by omitting some terms. By Lemma 9.5 the relation is the reverse: on one box the modified photon form is the internal Neumann–Dirichlet form P_{11} plus the positive term $N_\partial^{\mathrm{low}}$ and without the outgoing boundary variables, so the modified state is the one with the *larger* form, the smaller field-strength covariance, and the larger $\mathcal{I}$-expectations: $\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_C^{(3.38)}$ on the cone for every box C (Lemma 9.6(b)), the direction that Lemma 2.10 requires for consistency with Lemmas 9.9 and 9.10. For King's convention the same comparison holds with Step 1 of Lemma 4.2 prepended (Remark 4.8): $\langle Y\rangle_{k,J}^{K} \le \langle Y\rangle_{k,J}^{(a)} \le \langle Y\rangle_{k,X_J}^{(3.38)}$ on $\mathcal{I}$.

9.4. **The sandwich.**

Lemma 9.9 (the lower half)**.** *Let $C \subset \Omega$ be lattice boxes with C plus one layer contained in Ω, so that the variables $A_\mu(x)$, $x \in C$, of the Neumann–Dirichlet state on C (the outgoing bonds) are internal bonds of Ω. Then*

(a) $Q_\Omega^{(3.38)} \ge Q_C^{\mathrm{ND}} \oplus \frac{\mu^2}{2} I_{\mathrm{rest}}$ *as quadratic forms on* $\mathbb{R}^{\Omega^*} = \mathbb{R}^{C\text{-variables}} \oplus \mathbb{R}^{\mathrm{rest}}$;

(b) *for sources such that every plaquette and every site at which they do not vanish lies in C,*

$$\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_\Omega^{(3.38)} \quad (Y \in \mathcal{I}), \qquad \langle X\rangle_C^{\mathrm{ND}} \ge \langle X\rangle_\Omega^{(3.38)} \quad (X \in \mathcal{D}),$$

and the same for Z^a (equal positive multipliers, Lemma 2.11(a)). The $\mathcal{D}$ half uses Lemma 2.7(d); the $\mathcal{I}$ half only (a), (b), (e).

Proof. (a) $K_\Omega^{\rm int}$ contains every gradient pair of K_C (both base points in C, both bonds internal to Ω); the remaining pairs and N_∂ are nonnegative; the mass term splits. (b) Start from $\langle\cdot\rangle_\Omega^{(3.38)}$ and remove the hopping cosines across ∂C and all cosines not internal to C; by Lemma 2.7(b),(d) $\langle Y\rangle$ does not increase and $\langle X\rangle$ does not decrease, every intermediate measure being a BFS measure. The matter is now the Dirichlet matter of C: the diagonal parts $\frac12 r^2$ of the cut bonds stay and are exactly the Dirichlet terms, as in the proof of Lemma 2.10; it is coupled to A only through bonds internal to C, which are C-variables, and the matter of $\Omega\setminus C$ is decoupled and cancels (Lemma 2.7(e)). Then pass from the form $Q_\Omega^{(3.38)}$ to the smaller form $Q_C^{\rm ND}\oplus\frac{\mu^2}{2}I_{\rm rest}$: by (a) the difference is positive semidefinite, a finite sum $\sum_i s_i|\ell_i(A)|^2$ with $s_i\ge0$, and adding each term does not decrease $\langle Y\rangle$ and does not increase $\langle X\rangle$ (Lemma 2.7(a),(d)). So the state with the larger form, the modified one, has the larger $\langle Y\rangle$ and the smaller $\langle X\rangle$. The state with the smaller form factorises as $\langle\cdot\rangle_C^{\rm ND}$ times an independent Gaussian on the rest, which cancels. □

For King's convention the lower end is used in the form $\langle Y\rangle_{k,J}^K\le\langle Y\rangle_{k,J}^{(a)}\le\langle Y\rangle_\Omega^{(3.38)}$ on $\mathcal I$ (Step 1 of Lemma 4.2, then Lemma 9.9; Remark 4.8), and the upper end needs no change (Lemma 4.6).

Lemma 9.10 (the upper half, with a margin)**.** *Let $\Omega\subset B$ be lattice boxes at spacing ε such that B contains every site within ℓ^∞-distance N lattice steps of Ω, and let $\mu=\mu_0\varepsilon$ satisfy $d(\pi/2N)^2\le\mu^2$, i.e.*

$$N\varepsilon\ \ge\ \frac{\pi\sqrt d}{2\mu_0}\qquad\big(=R(\mu_0)=\tfrac{\pi\sqrt3}{2\mu_0}\ \text{in } d=3\big).$$

Let Σ_X denote the covariance of $(\varepsilon F(p))_{p\in\Omega^{**}}$ under the photon Gaussian of the state on X. Then

(a) $\Sigma_B^{\rm ND}\le\Sigma_\Omega^{(3.38)}=\Sigma_\Omega^u$ *on* $\mathbb R^{\Omega^{**}}$;

(b) *for sources supported in Ω as in Lemma 9.9,*

$$\langle Y\rangle_\Omega^{(3.38)}\le\langle Y\rangle_B^{\rm ND}\quad(Y\in\mathcal I),\qquad\langle X\rangle_\Omega^{(3.38)}\ge\langle X\rangle_B^{\rm ND}\quad(X\in\mathcal D),$$

and the same for Z^a. Only the $\mathcal I$ half is used in the sequel.

Proof. (a), in lattice units. Step 0: $\Sigma_\Omega^{(3.38)}=\Sigma_\Omega^u$, because $d_1A=d_1A^T$ depends only on the $\ker d_0^T$-component, whose law $\propto e^{-Q_\Omega^u(A^T)}$ is the same under both Gaussians (Step 1 of Lemma 9.3). Step 1 (variational identity): for a Gaussian with density $\propto e^{-\frac12A^TM_XA}$, $\langle g,\Sigma_Xg\rangle=\mathrm{Var}\langle g,d_1A\rangle=\sup_A[2\langle g,d_1A\rangle-A^TM_XA]$, the supremum attained at $A=M_X^{-1}d_1^Tg$. So it suffices to show that for every A on the B-variables there is $A'\in\mathbb R^{\Omega^*}$ with $d_1A'=d_1A$ on Ω^{**} and $\|d_1A'\|_{\Omega^{**}}^2+\mu^2\|A'\|_{\Omega^*}^2\le K_B(A)+\mu^2\|A\|_B^2$; take $A':=A|_{\Omega^*}$ (the plaquettes of Ω^{**} use only Ω^*-bonds). Step 2 (cutoff): let $\psi(x)=\prod_\nu\phi_\nu(x_\nu)$ with $\phi_\nu=1$ on the ν-range of Ω and on the first layer outside, $\phi_\nu(t)=\cos(\pi(t-1)/2N)$ at the t-th layer outside for $1\le t\le N+1$ (so $\phi_\nu(N+1)=0$), and 0 beyond; then $0\le\psi\le1$, $\psi=1$ on Ω and its neighbours, $\mathrm{supp}\,\psi\subset B$. Put $a:=\psi A$ on $\mathbb Z^d$ (the bond $\langle x,x+e_\mu\rangle$ weighted by $\psi(x)$; $A:=0$ off the B-variables). Step 3 (Hodge on $\mathbb Z^d$): $\|d_1A\|_{\Omega^{**}}^2=\|d_1A'\|_{\Omega^{**}}^2\le\sum_p|da(p)|^2\le\sum_p|da|^2+\sum_x|d^*a|^2=\sum_\mu\sum_\nu\sum_x|a_\mu(x+e_\nu)-a_\mu(x)|^2$, since the bonds of plaquettes in Ω^{**} have base points in Ω, where $\psi=1$. Step 4 (discrete IMS identity): for finitely supported ψf and a direction e,

$$\sum_x|\psi(x+e)f(x+e)-\psi(x)f(x)|^2=\sum_x\psi(x)\psi(x+e)|f(x+e)-f(x)|^2+\sum_x\psi(x)\big(2\psi(x)-\psi(x+e)-\psi(x-e)\big)f(x)^2,$$

exactly (per bond the difference of the two sides is $\psi'(\psi'-\psi)f'^2+\psi(\psi-\psi')f^2$; collect by site). With $f=A_\mu$ and $e=e_\nu$: $\psi\psi'\le1$, and $\psi\psi'>0$ only for pairs with both base points in B, which are gradient pairs of K_B; so the first sums, over μ,ν, total at most $K_B(A)$. The potential is $V(x)=\sum_\nu\big(\prod_{\kappa\ne\nu}\phi_\kappa^2\big)\phi_\nu(-\Delta\phi_\nu)(x_\nu)$. Step 5 (the profile): with $\vartheta:=\pi/2N$, $\phi(-\Delta\phi)=0$ on the range of Ω (all neighbours equal 1), $=1-\cos\vartheta\le\vartheta^2/2$ at $t=1$, $=2\cos^2(\vartheta(t-1))(1-\cos\vartheta)\le\vartheta^2$ for $2\le t\le N$ (using $\phi(N+1)=\cos(\vartheta N)=0$ at $t=N$), and $=0$ for $t\ge N+1$; hence $V=0$ on Ω and $V\le d\vartheta^2$ everywhere. Step 6: $\|d_1A'\|_{\Omega^{**}}^2\le K_B(A)+d\vartheta^2\sum_{x\in B\setminus\Omega}|A(x)|^2$; the variables

with base point in $B\setminus\Omega$ and those of Ω^* (base point in Ω) are disjoint subsets of the B-variables, so $\|d_1A'\|^2_{\Omega^{**}}+\mu^2\|A'\|^2_{\Omega^*}\le K_B(A)+\mu^2\|A\|^2_B$ as soon as $d\vartheta^2\le\mu^2$.

(b). Start from $\langle\cdot\rangle_B^{\mathrm{ND}}$ and remove all hopping cosines touching $B\setminus\Omega$ (Lemma 2.7(b): $\mathcal{I}$ does not increase; (d): $\mathcal{D}$ does not decrease). The matter is now the Dirichlet matter of Ω (the diagonals of the bonds leaving Ω kept; every such bond ends in B since $N\ge1$), coupled to A on Ω^*, and the sites of $B\setminus\Omega$ decouple. This matter is identical to that of the modified box state: the same cosines on Ω^*, the same diagonals $\frac12 r_x^2$ for every bond leaving Ω, the same single-site weights with the constant δm_ε^2 and the same Wick constant. For a gauge-invariant observable with sources in Ω the matter integral at fixed A, $\Psi_Y(A)=\int\prod_{x\in\Omega}d^2\phi_x\,Y\,e^{-S_m}$, is invariant under $A\mapsto A+d_0\chi$, $\chi\in\mathbb{R}^\Omega$ (as in Step 1 of Lemma 9.3), hence a function of the class of $A|_{\Omega^*}$ modulo $\operatorname{ran}d_0=\ker d_1$ (a box has $H^1=0$: the cycle space of a grid graph is generated by its 4-cycles), hence a function $\tilde\Psi_Y(d_1A|_{\Omega^{**}})$ of the field strengths on Ω^{**}. So $\langle Y\rangle=\mathbb{E}[\tilde\Psi_Y(F)]/\mathbb{E}[\tilde\Psi_1(F)]$ depends on the photon law only through the centred Gaussian law of $F=(d_1A(p))_{p\in\Omega^{**}}$, with covariance Σ_B^{ND} on the Neumann–Dirichlet side after the removal and Σ^u_Ω on the modified side (Step 0). Both laws live on $\operatorname{ran}d_1$. Let $T:\mathbb{R}^{\Omega^{**}}\to\mathbb{R}^{\Omega^*}$ be a linear map with $d_1T=\mathrm{id}$ on $\operatorname{ran}d_1$, extended linearly to all of $\mathbb{R}^{\Omega^{**}}$ (any extension serves, both laws being supported on $\operatorname{ran}d_1$), and realise each law as the photon Gaussian of $A''\coloneqq TF\in\mathbb{R}^{\Omega^*}$, covariance $T\Sigma T^T$: then $d_1A''=F$ almost surely, the gauge-invariant expectations are reproduced, and $T\Sigma_B^{\mathrm{ND}}T^T\le T\Sigma^u_\Omega T^T$ by (a). These covariances are degenerate; regularise to $T\Sigma T^T+\delta I$, $\delta>0$, which are non-degenerate, so the regularised states are BFS measures, the ordering of the covariances is preserved, and the law of $F''=d_1A''$ is $N(0,\Sigma+\delta d_1d_1^T)$, still ordered. Lemma 2.7(a) by interpolation (the difference of the precision matrices is positive semidefinite by operator monotonicity of the inverse, hence a finite sum of terms $s|\ell(A)|^2$, $s\ge0$) gives $\langle Y\rangle_{\Sigma^u+\delta}\le\langle Y\rangle_{\Sigma^{\mathrm{ND}}+\delta}$ for $Y\in\mathcal{I}$, and the reverse for $\mathcal{D}$ by Lemma 2.7(d). *The limit* $\delta\downarrow0$. For the $\mathcal{I}$ half, $\tilde\Psi_Y$ is bounded and continuous: $|\Psi_Y(A)|\le\int e^{r^2(g)}e^{\sum\varepsilon^{d-2}r_xr_y}\prod_x d\rho_x<\infty$ independently of A ($|e^{\mathrm{i}F}|=1$; the hopping bound $e^{\varepsilon^{d-2}r_xr_y}\le e^{\frac12\varepsilon^{d-2}(r_x^2+r_y^2)}$ cancels the diagonals and $\lambda>0$ does the rest), and so is $\tilde\Psi_1$, with $\tilde\Psi_1>0$; the Gaussians $N(0,\Sigma+\delta d_1d_1^T)$ converge weakly to $N(0,\Sigma)$, so numerator and denominator converge and the inequality passes to the limit. For the $\mathcal{D}$ half (not used below), $X=\Theta(f,-g)$ gives $\tilde\Psi_X(F)=e^{\langle v,F\rangle}\tilde\Psi^0_X(F)$ with v the plaquette vector of f and $0<\tilde\Psi^0_X\le e^{2\mathsf{W}_\varepsilon\mathsf{S}(g)}\tilde\Psi_1\le C$ bounded and continuous, so $\tilde\Psi_X$ is continuous but unbounded; under $N(0,\Sigma+\delta d_1d_1^T)$ one has $\mathbb{E}[e^{2\langle v,F''\rangle}]=e^{2v^T(\Sigma+\delta d_1d_1^T)v}\le e^{2v^T\Sigma v+2\|d_1^Tv\|^2}$ for $\delta\le1$, so $\{\tilde\Psi_X(F'')\}_{0<\delta\le1}$ is bounded in L^2, hence uniformly integrable, and weak convergence with uniform integrability and continuity gives $\mathbb{E}[\tilde\Psi_X(F'')]\to\mathbb{E}[\tilde\Psi_X(F)]$. Altogether $\langle Y\rangle_\Omega^{(3.38)}\le\langle Y\rangle^{\text{after removal}}\le\langle Y\rangle_B^{\mathrm{ND}}$, and the reverse for $\mathcal{D}$. □

Remark 9.11 (why a margin, and why a fixed photon mass). Part (a) of Lemma 9.10 fails without a margin: for $B=\Omega$ the comparison is reversed (Lemma 9.6), and with a single layer ($N=1$) the difference $\Sigma^u_\Omega-\Sigma_B^{\mathrm{ND}}$ has a negative eigenvalue already at moderate μ. No fixed margin works at $\mu_0=0$: there Σ^u_Ω is the projector onto exact 2-forms, while $\mathrm{Var}_B\langle g,dA\rangle>\|g\|^2$ for a Coulomb potential whose gradient leaks out of B. So Lemma 9.10 is a statement at fixed $\mu_0>0$ with $R(\mu_0)\to\infty$ as $\mu_0\to0$; a limit of vanishing photon mass inherits reflection positivity, if it exists, as a pointwise limit of positive semidefinite matrices. On one box the three states are ordered on $\mathcal{I}$ as $\langle Y\rangle_\Omega^{\mathrm{ND}}\le\langle Y\rangle_\Omega^{(3.38)}\le\langle Y\rangle_B^{\mathrm{ND}}$ (Lemmas 9.6 and 9.10), as the monotonicity of Lemma 2.10 requires. The sandwich with the boxes (20) is assembled in the proof of Theorem 10.6; for King's convention its lower end carries the kill step of Remark 4.8 and its upper end is unchanged (Lemma 4.6).

10. Reflection positivity II: the limit, OS3 and the polynomial form

The lattice statements of the previous section are now carried to the limits $k\to\infty$ and $J\to\infty$. The scheme of proof is the same for the curvature sector and for the Wick tube: uniform local bounds for the modified-box functionals at complex coefficients (Lemma 10.3), Montel's theorem in k at fixed J, the two-sided sandwich of Lemmas 9.9 and 9.10 on the cone as

$J \to \infty$ with the identity theorem off it, and a translation for the planes that are not mid-planes. The sector-specific input is the convergence of the cube functionals themselves: the convergence at fixed cubes (Definition 3.1; on King's convention Theorem 3.3) in the curvature sector, and on the Wick tube the positive-leg bound of Corollary 13.3 together with the convergence at fixed cubes on the tube (Definition 3.2; on King's convention Theorem 3.6) (Proposition 10.5). The arbitrary-amplitude form of the Wick sector, which needs (W2), is the optional extension of Theorem 23.8; the one lemma written for it here, Lemma 10.2, is marked as such and is not used by the main theorem.

10.1. **The enlarged family.**

Definition 10.1 (the enlarged family, its tube and its bounded domain)**.** Fix $\nu \in \{1,2,3\}$, a mid-plane $c \in D$ (Definition 9.1), and finite families $\zeta_1, \dots, \zeta_N$ of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu \ge c + \delta\}$ for some $\delta > 0$, with complex coefficients z_{il}, w_{il}. Split each scalar source as $h_{il} = h_{il}^+ - h_{il}^-$ with $h_{il}^- := \|h_{il}\|_\infty \chi_{il}$, where $\chi_{il} \in C_c^\infty$, $0 \le \chi_{il} \le 1$, $\chi_{il} = 1$ on $\operatorname{supp} h_{il}$ and $\operatorname{supp} \chi_{il} \subset \{x_\nu \ge c + \delta/2\}$, and $h_{il}^+ := h_{il} + h_{il}^- \ge 0$. The *enlarged family* is the family of real sources

$$\mathcal{F} := \{\rho^* f_{il}\} \cup \{f_{il}\} \cup \{h_{il}^\pm \circ \rho\} \cup \{h_{il}^\pm\}, \qquad \text{coefficient space } \mathbb{C}^m,\ m = m_F + m_h,$$

whose scalar members $\tilde{h}_j \ge 0$ are all nonnegative. For a real coefficient vector $\xi \in \mathbb{R}^m$ the total source is $(f_\xi, k_1(\xi) - k_2(\xi))$ with $k_1(\xi) := \sum_j (\xi_j)_+ \tilde{h}_j$, $k_2(\xi) := \sum_j (\xi_j)_- \tilde{h}_j$, both smooth and nonnegative ($x_\pm := \max(\pm x, 0)$). Let J_0 be the least J such that every member of $\mathcal{F}$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; for $J \ge J_0$, $n = n(\mu_0, c)$ and the boxes $\Omega_{J,k}$ of (20) put

$$\Phi_k(\zeta) := Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta), \qquad \zeta \in \mathbb{C}^m,$$

an entire function (a finite-dimensional integral with Gaussian decay in A and $e^{-\lambda \varepsilon^3 r^4}$ decay in r). The *matrix points* $\zeta^*_{ij} \in \mathbb{C}^m$ carry $\bar{z}_{il}$ on $\rho^* f_{il}$, z_{jl} on f_{jl}, $\pm\bar{w}_{il}$ on $h_{il}^\pm \circ \rho$, $\pm w_{jl}$ on $h_{jl}^\pm$ and 0 elsewhere; their total source is that of $\theta\zeta^*_i \sqcup \zeta_j$, so $\Phi_k(\zeta^*_{ij})$ is the (i,j) entry of Lemma 9.3(b). Since reflected and unreflected members have disjoint supports, $\|h^+\|_\infty \le 2\|h\|_\infty$ and $\|h^-\|_\infty = \|h\|_\infty$,

$$\|k_1(\operatorname{Re}\zeta^*_{ij})\|_\infty \ \le\ 2\max\big(\mathsf{a}(\zeta_i), \mathsf{a}(\zeta_j)\big). \tag{21}$$

(i) The *Wick tube* of $\mathcal{F}$ is $T_\mathcal{F} := \{\zeta \in \mathbb{C}^m :\ 2\|k_1(\operatorname{Re}\zeta)\|_\infty < \sigma_+/2\}$, with $\sigma_+ = s_1$ the amplitude of (I2) (Definition 12.5, (30)). It is open, convex ($\xi \mapsto \|k_1(\xi)\|_\infty$ is a supremum of convex functions), connected, invariant under $z \mapsto -z$ on the curvature coordinates, contains 0, and by (21) contains every ζ^*_{ij} as soon as $\mathsf{a}(\zeta_i) < \sigma_+/8 =: \sigma_W$ for every i.

(ii) The *bounded domain*: let $B_1 \subset \mathbb{R}^m$ be the open convex hull of the unit ℓ^∞-balls about 0 and about the real parts $\operatorname{Re}\zeta^*_{ij}$; let $\beta_j := 2\sup_{\xi \in B_1} |\xi_j|$ for the scalar coordinates (so $\beta_j = 2(|\operatorname{Re} w_{il}| + 1)$ for the coordinate of $h_{il}^\pm$ or of $h_{il}^\pm \circ \rho$), $B'' := \prod_j [0, \beta_j]$ over the scalar coordinates, and $q_v := \sum_j v_j \tilde{h}_j$ for $v \in \operatorname{vert}(B'')$, the *vertex sources*. All q_v are pointwise $\le q_\zeta := \sum_{i,l} 2(|\operatorname{Re} w_{il}| + 1)(h_{il}^+ + h_{il}^- + h_{il}^+ \circ \rho + h_{il}^- \circ \rho)$. Put $\mathcal{T}_1 := \{\zeta \in \mathbb{C}^m : \operatorname{Re}\zeta \in B_1\}$: open, convex, connected, containing every ζ^*_{ij} and the unit polydisc about 0, so that $\mathcal{T}_1 \cap T_\mathcal{F}$ is a non-empty open set. For $\xi \in B_1$ both $2\xi_+$ and ξ_+ lie in B''.

(iii) The *uniqueness set* $U \subset \mathbb{C}^m$ is the set of coefficient vectors whose curvature coefficients are purely imaginary and whose scalar coefficients are positive reals. It is a non-empty, relatively open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^{m_F} \times \mathbb{R}^{m_h}$ of maximal real dimension m (it is not open in $\mathbb{C}^m$), hence a set of uniqueness for holomorphic functions on any connected open set meeting it: a holomorphic function vanishing on a relatively open piece of a maximal totally real subspace has all derivatives zero there [11]. $U \cap \mathcal{T}_1 \ne \emptyset$ and $U \cap T_\mathcal{F} \ne \emptyset$ (small positive scalar coefficients). On U the total source is $(\mathrm{i}f_\zeta, g_\zeta) \in \mathcal{K}$, so $\Theta(\zeta) \in \mathcal{I}$; the scheme multipliers are positive reals, equal for the Neumann–Dirichlet and the modified states; and every lattice value is real (charge conjugation, Definition 2.3: $\overline{Z^a(\mathrm{i}f, g)} = Z^a(-\mathrm{i}f, g) = Z^a(\mathrm{i}f, g)$).

Every lattice functional, and hence every limit below, depends on a family only through its total sources; the splitting is therefore immaterial at the matrix points and at the Taylor coefficients at 0.

10.2. **Uniform local bounds.** The following local bound is used only in the optional extension (Theorem 23.8); it is stated here because its domain $\mathcal{T}_1$ is what makes "finitely many instances of (W2)" literally true there. The fixed-cube limits $Z^a_J := \lim_k Z^a_{k,J}$ exist for all complex coefficients and are entire under (W2)(a) alone: it gives convergence at the real points, the vertex argument of Lemma 10.3 with Lemma 2.11(d) gives local boundedness in k, and Montel's theorem with the uniqueness set $\mathbb{R}^m$ does the rest.

Lemma 10.2 (uniform local bound; optional extension)**.** *Assume* (W2)*(a) and* (W2)*(b) at the finitely many sources q_v, $v \in \mathrm{vert}(B'')$. Then $\{Z^a_J\}_{J\ge J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z^a_J \to Z^a_\infty$ locally uniformly on $\mathcal{T}_1$, Z^a_∞ is holomorphic on $\mathcal{T}_1$, and $Z^a_J \le Z^a_\infty$ on $U \cap \mathcal{T}_1$.*

Proof. For real $\xi \in B_1$, Lemma 2.11(f) gives $Z^a_J(\xi) \le Z^a_J(2f_\xi, -2k_2(\xi))^{1/2} Z^a_J(0, 2k_1(\xi))^{1/2}$. The first factor is nonincreasing in J: at fixed k this is the second inequality of Lemma 2.10 with Lemma 2.11(a), and it passes to $k \to \infty$ by (W2)(a); (W2)(b) is not used there. Hence it is $\le Z^a_{J_0}(2f_\xi, -2k_2(\xi))$, and $\xi \mapsto \ln Z^a_{J_0}(2f_\xi, -2k_2(\xi))$ is bounded on the compact $\bar B_1$: the coefficient map is piecewise linear, so the image of $\bar B_1$ lies in a real box B', on which $\ln Z^a_{J_0}$ is convex (Lemma 2.11(d)) and so bounded by its maximum over the vertices of B', each finite by (W2)(a). For the second factor, $2\xi_+ \in B''$ and $\ln Z^a_J(0,\cdot)$ is convex in the coefficients, so $Z^a_J(0, 2k_1(\xi)) \le \max_{v\in\mathrm{vert}(B'')} Z^a_J(0, q_v) \le \max_v \sup_{J'\ge J_{q_v}} Z^a_{J'}(0, q_v) =: C_{B''} < \infty$ by (W2) at the q_v (note $J \ge J_0 \ge J_{q_v}$). By Lemma 2.11(d), $|Z^a_J(\zeta)| \le Z^a_J(\mathrm{Re}\,\zeta)$, so $\sup_{J\ge J_0} \sup_{\mathcal{T}_1} |Z^a_J| < \infty$. On $U \cap \mathcal{T}_1$ the sources lie in $\mathcal{K}$, so $J \mapsto Z^a_J(\zeta)$ is nondecreasing (first inequality of Lemma 2.10 at fixed k, then $k \to \infty$) and bounded, hence convergent. $U \cap \mathcal{T}_1$ is a set of uniqueness for holomorphic functions on the connected $\mathcal{T}_1$ (Definition 10.1(iii)); Montel's theorem gives locally uniform convergence on $\mathcal{T}_1$ to a holomorphic limit, which is Z^a_∞ wherever the latter is defined (the same J-limit), and the monotone limit on $U \cap \mathcal{T}_1$ gives $Z^a_J \le Z^a_\infty$ there. Under (W2)(b) for every $0 \le q \in C^\infty_c$ the same holds on all of $\mathbb{C}^m$. □

Lemma 10.3 (uniform local bounds for the modified-box functionals)**.** *Let $\mathcal{F}$, Φ_k, $\mathcal{T}_1$, $T_{\mathcal{F}}$ be as in Definition 10.1.*

(A) (Optional extension.) *Under* (W2)*(a) and* (W2)*(b) at the q_v, $v \in \mathrm{vert}(B'')$: there is $C < \infty$, independent of k and of $J \ge J_0$, with $\limsup_k \sup_{\mathcal{T}_1} |\Phi_k| \le C$; and at fixed J, $\sup_k \sup_Q |\Phi_k| < \infty$ for every compact $Q \subset \mathbb{C}^m$ (this part uses* (W2)*(a) only).*

(A_∞) (Optional extension.) *Under* (W2) *for every $0 \le q \in C^\infty_c$: the same with $\mathcal{T}_1$ replaced by any compact $Q \subset \mathbb{C}^m$, with $C = C_Q$.*

(T) (The tube.) *Under Definition 3.2 (on King's convention Theorem 3.6),* (I2), (I3) *and the correlation inequalities of Lemma 2.7(a)–(e) only (no* (W2)*): the same for every compact $Q \subset T_{\mathcal{F}}$ and all $J \ge J_0$, with no further threshold on J.*

(C) (Curvature sector, $h \equiv 0$; no hypothesis.) $|\Phi_k(z)| \le \langle e^{F(f_{\mathrm{Re}\,z})}\rangle^{(3.38)}_{k,\Omega_{J,k}} \le \exp(\tfrac12 \mathrm{Var}^{(3.38)} F(f_{\mathrm{Re}\,z}))$ *and* $\mathrm{Var}^{(3.38)} F(f) \le \sum_p \varepsilon^3 f^2_{c(p)} \to \|f\|^2_{L^2}$, *uniformly in k and J.*

Proof. $|\Phi_k(\zeta)| \le \Phi_k(\mathrm{Re}\,\zeta)$ by the proof of Lemma 2.11(d) ($|\Theta(\zeta)| = \Theta(\mathrm{Re}\,\zeta)$ and the modulus of the multiplier), which is valid for any probability measure with the same real multiplier, in particular for the modified box state, which is not one of the states of Definitions 2.1 and 2.2. At a real ξ the proof of Lemma 2.11(f) (Cauchy–Schwarz with the multiplier identity, again valid for any such measure) gives

$$\Phi_k(\xi) \;\le\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(2f_\xi, -2k_2(\xi))^{1/2}\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(0, 2k_1(\xi))^{1/2}. \tag{22}$$

First factor (class $\mathcal{D}$). By the $\mathcal{D}$ half of Lemma 9.9 with $C = C_J \cap \Lambda_k$ (its hypothesis is (20)(i)) it is $\le Z^a_{k,J}(2f_\xi, -2k_2) \le Z^a_{k,J_0}(2f_\xi, -2k_2)$, the second step being the second inequality of Lemma 2.10 at fixed k. The map $\xi \mapsto \eta(\xi) :=$ (coefficient vector of $(2f_\xi, -2k_2(\xi))$) is continuous, so $\eta(\mathrm{Re}\,Q)$, resp. $\eta(\bar B_1)$, lies in a real box B'; $\eta \mapsto \ln Z^a_{k,J_0}(\eta)$ is convex (Lemma 2.11(d)),

so $\sup_\xi Z^a_{k,J_0}(\eta(\xi)) \le \max_{v\in\mathrm{vert}(B')} Z^a_{k,J_0}(v)$, finitely many values, each convergent in k by Definition 3.2 (the sources here have negative Wick coefficients, on which the clause has no amplitude restriction; in versions (A), (A_∞) by (W2)(a)), and convergence in k implies boundedness in k. So the first factor is bounded uniformly in k, ξ and $J \ge J_0$. *Second factor* (class $\mathcal{I}$). By the $\mathcal{I}$ half of Lemma 9.10 with $B = C_{J+2n} \cap \Lambda_k$ and $N := \lceil R(\mu_0)/\varepsilon_k \rceil$ (hypothesis (20)(ii)) it is $\le Z^a_{k,J+2n}(0, 2k_1(\xi))$. (A): for $\xi \in B_1$, $2\xi_+ \in B''$, so by convexity it is $\le \max_{v\in\mathrm{vert}(B'')} Z^a_{k,J+2n}(0, q_v)$; each of these finitely many values converges in k by (W2)(a) (hence is bounded in k, which is the fixed-J bound Montel's theorem needs, and the $\limsup_k$ bound passes to any subsequential limit) with limit $Z^a_{J+2n}(0, q_v) \le \sup_{J'\ge J_{q_v}} Z^a_{J'}(0, q_v) < \infty$: this is (W2) at q_v, used directly ($J+2n \ge J_0 \ge J_{q_v}$), uniformly in J; the finitely many instances q_v are all of (W2) that version (A) uses. (A_∞): the same with a box $B'' \supset 2(\mathrm{Re}\, Q)_+$ depending on Q. (T): for $\xi \in \mathrm{Re}\, Q$, $Q \subset T_{\mathcal{F}}$, one has $0 \le 2k_1(\xi) \le \sigma_+/2 \le s_1$ on the lattice (the sampled supremum is at most the continuum one), so Corollary 13.3, $\ln N^a_{k,C_J}(0, q) \le \kappa\|q\|_{1,k}$ for $0 \le q \le s_1$ on $C_J \cap \Lambda_k$, uniformly in k and J, applied with $N^a_{k,X}(0, q) = Z^a_{k,X}(0, q)$ (the same functional with the same scheme multiplier, so no J-dependent factor intervenes), gives $Z^a_{k,J+2n}(0, 2k_1(\xi)) \le \exp(\kappa\|2k_1(\xi)\|_{1,k}) \le \exp\big(2\kappa \sum_j |\xi_j| \sup_{k'} \|\tilde h_j\|_{1,k'}\big)$, bounded uniformly in k, J and $\xi \in \mathrm{Re}\, Q$ because $\|\tilde h_j\|_{1,k'} \to \|\tilde h_j\|_{L^1}$ (Riemann sums, (1)). No lower bound on J beyond J_0 enters: Corollary 13.3 holds for every k and J. *(C).* At $h \equiv 0$ the first inequality is Lemma 2.11(d) as above and the second is Lemma 2.8 with $s = 1$, the modified box state being a BFS measure (Definition 9.2). Its photon Gaussian has the unitary form for gauge-invariant observables (Step 1 of Lemma 9.3), and $\mathrm{Var}^u\langle v, d_1 A\rangle = \varepsilon^{-1}\langle d_1^T v, (d_1^T d_1 + \mu_0^2\varepsilon^2)^{-1} d_1^T v\rangle \le \varepsilon^{-1}\|v\|^2$; with $v_p = \varepsilon^2 f_{c(p)}$ this is $\mathrm{Var}\, F(f) \le \sum_p \varepsilon^3 f^2_{c(p)}$. □

Remark 10.4 (one instance of (W2) in place of the vertex set). For $v \in \mathrm{vert}(B'')$ one has $0 \le q_v \le q_\zeta$ pointwise, and for $p = 1+\eta > 1$, $p' = p/(p-1)$, $q_v = \frac{1}{p}(pq_\zeta) - \frac{1}{p'}\big(p'(q_\zeta - q_v)\big)$. Hölder's inequality with the multiplier bookkeeping $e^{-a_k \mathsf{S}(q_v)} = (e^{-a_k \mathsf{S}(pq_\zeta)})^{1/p} (e^{a_k \mathsf{S}(p'(q_\zeta - q_v))})^{1/p'}$ gives, for every state, $Z^a(0, q_v) \le Z^a(0, pq_\zeta)^{1/p} Z^a(0, -p'(q_\zeta - q_v))^{1/p'}$. The second factor is of class $\mathcal{D}$, nonincreasing in J (Lemma 2.10, then $k \to \infty$ by (W2)(a)), hence $\le Z^a_{J_0}(0, -p'(q_\zeta - q_v)) < \infty$, one real point of Theorem 3.6; the first is bounded uniformly in J by (W2)(b) at the single source $(1+\eta)q_\zeta$. So (W2) at $(1+\eta)q_\zeta$, for any one $\eta > 0$, implies the finitely many instances used in Lemmas 10.2 and 10.3(A). Only this direction is proved; no converse is claimed. Pointwise domination $q_v \le q_\zeta$ alone gives nothing, since $Z^a(0, \cdot)$ is not monotone in q: the Wick square has no sign, and the scheme multiplier $e^{-a_k \mathsf{S}(q)}$ has no sign control.

10.3. **King's state on the Wick tube.**

Proposition 10.5 (the Wick-tube object: existence and translations)**.** *Let a be an admissible scheme (Definition 3.7), assume* (I2) *and* (I3)*, and assume that the comparison family is convergent at fixed cubes on the tube (Definition 3.2); the form on King's convention, where Theorem 3.6 supplies the limits, is Proposition 21.4(c). For a finite family $\mathcal{F}'$ of real sources with nonnegative scalar members $\tilde h_j \ge 0$ (any family is brought to this form by the splitting of Definition 10.1) let $T_{\mathcal{F}'}$ be its Wick tube, as in Definition 10.1(i).*

(a) *$Z^a_J \to Z^a_{\infty,T}$ locally uniformly on $T_{\mathcal{F}'}$ as $J \to \infty$; $Z^a_{\infty,T}$ is holomorphic on $T_{\mathcal{F}'}$; $Z^a_{\infty,T} = G_\infty$ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ and $Z^a_{\infty,T} = Z^a_\infty$ wherever the latter exists (the same J-limit of the same Z^a_J); $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\mathrm{Re}\,\zeta)$; $Z^a_{\infty,T}(\xi) > 0$ for real $\xi \in T_{\mathcal{F}'}$ with $-\xi \in T_{\mathcal{F}'}$; and $Z^a_{\infty,T}(-z, w) = Z^a_{\infty,T}(z, w)$.*

(b) (Translations.) *$Z^a_{\infty,T}(\tau_b \cdot \zeta) = Z^a_{\infty,T}(\zeta)$ for all $b \in \mathbb{R}^3$ and $\zeta \in T_{\mathcal{F}'}$, where τ_b acts on all sources; the translated family has the same tube.*

The proof uses, besides Definition 3.2, only Corollary 13.3 (hence (I2)*,* (I3) *and* [7, Thm. 6.2(1),(2)]*), the second inequality of Lemma 2.10 (hence Lemma 2.7(d)), Lemmas 2.11 and 2.14, Montel's theorem and the identity theorem;* (W2) *is not used.*

Proof. (a) *Local boundedness* of $\{Z_J^a\}_{J\ge J_0}$ on $T_{\mathcal{F}'}$: $|Z_J^a(\zeta)| \le Z_J^a(\operatorname{Re}\zeta) \le Z_J^a(2f,-2k_2)^{1/2} Z_J^a(0,2k_1)^{1/2} \le Z_{J_0}^a(2f,-2k_2)^{1/2}\exp(\kappa\|k_1\|_{L^1})$, by Lemma 2.11(d),(f), the nonincreasing $\mathcal{D}$-factor (second inequality of Lemma 2.10 at fixed k, then $k\to\infty$), and Corollary 13.3 in the limit $k\to\infty$ for the $\mathcal{I}$-factor, valid since the source $2k_1$ has $2\sup(2k_1) = 4\|k_1\|_\infty < \sigma_+ = s_1$; the first factor is bounded on compacts by the vertex argument of Lemma 10.3 with Definition 3.2 at C_{J_0}. *Convergence on $U\cap T_{\mathcal{F}'}$*: there the values are real, nondecreasing in J (first inequality of Lemma 2.10, then $k\to\infty$) and bounded, since $Z_J^a(\mathrm{i}f_\zeta, g_\zeta) \le Z_J^a(0,g_\zeta)$ by Lemma 2.11(d) and $Z_J^a(0,g_\zeta) \le \exp(\kappa\|g_\zeta\|_{L^1})$ by Corollary 13.3 in the limit $k\to\infty$ ($0\le g_\zeta \le \sigma_+/4$ there, so $2\sup g_\zeta < s_1$). Montel's theorem on the connected open $T_{\mathcal{F}'}$ with the uniqueness set $U\cap T_{\mathcal{F}'}$ gives locally uniform convergence to a holomorphic $Z^a_{\infty,T}$; on $T_{\mathcal{F}'}\cap\mathcal{D}_{\sigma_+,H}$ this is the limit G_∞ of Theorem 13.10(i), the same argument on the other tube. The bound $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\operatorname{Re}\zeta)$ and the evenness in z pass from the lattice (Lemma 2.11(d); the invariance $(A,\vartheta)\mapsto(-A,-\vartheta)$ of every state (charge conjugation, Definition 2.3), which maps $F(f)\mapsto -F(f)$ and fixes r; $T_{\mathcal{F}'}$ is invariant under $z\mapsto -z$). Positivity at symmetric real points is Lemma 2.11(e), $Z^a(\xi)Z^a(-\xi)\ge 1$, in the limit.

(b) *Lattice translations.* Let $b\in 5^{-k_1}\mathbb{Z}^3$, $|b|_\infty \le M$, $k\ge k_1$. τ_b preserves Λ_k and S_{Λ_k}, so $Z^a_{k,J}(\tau_b\cdot\zeta) = Z^a_{k,(C_J-b)\cap\Lambda_k}(\zeta)$ (the action is translation covariant on Λ_k). As sets of sites, $C_{J-1}\cap\Lambda_k \subset (C_J-b)\cap\Lambda_k \subset C_{J+1}\cap\Lambda_k$, so the first inequality of Lemma 2.10 (stated for any finite site set) gives, for $\zeta\in U$ and $J\ge J_0+1$ (so that every plaquette and site at which the sources do not vanish lies in C_{J-1}), $Z^a_{k,J-1}(\zeta) \le Z^a_{k,J}(\tau_b\cdot\zeta) \le Z^a_{k,J+1}(\zeta)$. Let $k\to\infty$ and then $J\to\infty$ on $U\cap T_{\mathcal{F}'}$, where the outer limits converge by (a): $Z^a_{\infty,T}(\tau_b\cdot\zeta) = Z^a_{\infty,T}(\zeta)$ there, hence on all of $T_{\mathcal{F}'}$ by the identity theorem (both sides are holomorphic on the connected $T_{\mathcal{F}'}$, the translated family having the same supremum norms). Iterating, the identity holds for all $b\in\bigcup_k 5^{-k}\mathbb{Z}^3$. *All $b\in\mathbb{R}^3$.* For $\rho>0$ let $E_\rho := C_c^\infty(\bar B_\rho;\Lambda^2)\times C_c^\infty(\bar B_\rho)$ (a Fréchet space) and $O_\rho := \{(f,h)\in E_\rho : 4\|h\|_\infty < \sigma_+/2\}$, open, convex and symmetric under $s\mapsto -s$. For $s=(f,h)\in O_\rho$ the one-member family with its splitting has the coefficient vector $(1,1,-1)$ in its tube ($2\|k_1\|_\infty = 2\|h^+\|_\infty \le 4\|h\|_\infty < \sigma_+/2$), so $\Phi_T(s) := \ln Z^a_{\infty,T}(f,h)$ is defined and finite on O_ρ by (a). Φ_T is convex (a pointwise limit of the functions $\ln Z^a_{k,J}$, convex by Hölder's inequality) and Borel (a pointwise limit of continuous functions of finitely many sampled values). By Lemma 2.14, Φ_T is continuous on O_ρ. For $s\in O_\rho$ and $b\in\mathbb{R}^3$ pick $b_n\to b$ in $\bigcup_k 5^{-k}\mathbb{Z}^3$; then $\tau_{b_n}s, \tau_b s\in O_{\rho+|b|+1}$, $\tau_{b_n}s\to\tau_b s$ in $E_{\rho+|b|+1}$ (translations act continuously on C_c^∞), and $\Phi_T(\tau_{b_n}s) = \Phi_T(s)$; so $\Phi_T(\tau_b s) = \Phi_T(s)$. *Complex coefficients.* For a family $\mathcal{F}'$ the two holomorphic functions $\zeta\mapsto Z^a_{\infty,T}(\tau_b\cdot\zeta)$ and $\zeta\mapsto Z^a_{\infty,T}(\zeta)$ on $T_{\mathcal{F}'}$ agree at the real points of $\{\xi\in\mathbb{R}^m : 4\|h_\xi\|_\infty < \sigma_+/2\}\cap T_{\mathcal{F}'}$, where both limits exist and both equal the one-member value $\exp\Phi_T(f_\xi,h_\xi)$ (the same J-limit of the same Z_J^a). This intersection is open in $\mathbb{R}^m$ and contains 0 (the first set alone need not lie in the tube: with $\tilde h_1=\tilde h_2$ and $\xi=(1,-1)$, $h_\xi=0$ while $k_1=\tilde h_1$), hence is a set of uniqueness in the connected $T_{\mathcal{F}'}$; so the two functions agree on $T_{\mathcal{F}'}$. □

10.4. **The two limits and the passage to every plane.** Throughout this subsection $\mathcal{F}$, Φ_k, ζ^*_{ij}, $\mathcal{T}_1$, $T_{\mathcal{F}}$, U are those of Definition 10.1 for a mid-plane $c\in D$, and "version" refers to Lemma 10.3: (C) is the curvature sector when the comparison family is convergent at fixed cubes (the uniform local bound being Theorem 7.4(b)), (T) the tube under Theorem 3.6, (I2), (I3), and (A), (A_∞) the optional extension under (W2).

Step 5 ($k\to\infty$ at fixed J). By the fixed-J bound of Lemma 10.3 and Montel's theorem there is a subsequence $k_m\to\infty$ along which $\Phi_{k_m}\to\Phi^J$ locally uniformly on $\mathbb{C}^m$, with Φ^J entire. No convergence of the full sequence is claimed or needed. By Lemma 9.3(b), $\sum_{ij}\bar\gamma_i\gamma_j\Phi_k(\zeta^*_{ij})\ge 0$ for every $k\ge k_0(c,\delta)$ and $\gamma\in\mathbb{C}^N$; positive semidefiniteness of a Hermitian matrix is closed under entrywise limits, so $\sum_{ij}\bar\gamma_i\gamma_j\Phi^J(\zeta^*_{ij})\ge 0$. The uniform bounds of Lemma 10.3 pass to Φ^J: $\sup_{\mathcal{T}_1}|\Phi^J|\le C$ in (A), $\sup_Q|\Phi^J|\le C_Q$ in (A_∞), (T) and (C), uniformly in $J\ge J_0$, whatever subsequence was taken. (The box $\Omega_{J,k}$ depends on J, so Φ^J does; the subsequence may depend on J.)

Step 6 ($J \to \infty$: sandwich on the cone, identity theorem off it). By the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10, whose hypotheses are (20)(i),(ii), for every k and every $\zeta \in U$,

$$Z^a_{k,J}(\zeta) \;\le\; \Phi_k(\zeta) \;\le\; Z^a_{k,J+2n}(\zeta). \tag{23}$$

Let $k \to \infty$ along the subsequence of Step 5. The outer terms converge along the full sequence (version (C): Theorem 7.4(b); (T): Definition 3.2; (A), (A_∞): (W2)(a); each with Montel's theorem), so $Z^a_J \le \Phi^J \le Z^a_{J+2n}$ on U for every subsequential Φ^J. As $J \to \infty$ the outer terms converge on the respective uniqueness set: in (C) on $U = (\mathrm{i}\mathbb{R})^{m_F}$ to $Z_\infty(\cdot, 0)$ (Theorem 7.4(b)); in (T) on $U \cap T_{\mathcal{F}}$ to $Z^a_{\infty,T}$ (Proposition 10.5(a); there $J \mapsto Z^a_J(\zeta)$ is real, nondecreasing and bounded by $Z^a_J(0, g_\zeta) \le \exp(\kappa \|g_\zeta\|_{L^1})$, since $0 \le g_\zeta \le \sigma_+/4$, so $2 \sup g_\zeta < s_1$); in (A) on $U \cap \mathcal{T}_1$ to Z^a_∞ (Lemma 10.2); in (A_∞) on U to Z^a_∞ (Lemma 10.2 with (W2)(b) for every q). Hence $\Phi^J \to Z^a_{\infty(,T)}$ pointwise on that set. By Step 5 the family $\{\Phi^J\}_J$ is locally bounded on the corresponding connected domain ($\mathbb{C}^m$, $T_{\mathcal{F}}$, $\mathcal{T}_1$, $\mathbb{C}^m$), on which the limit functional is holomorphic (Theorem 7.4(b), Proposition 10.5(a), Lemma 10.2, Lemma 10.2 on $\mathbb{C}^m$ respectively); by Montel's theorem every subsequence of (Φ^J) has a further subsequence converging locally uniformly to a holomorphic function, which agrees with the limit functional on the uniqueness set and hence on the whole domain [11]. So $\Phi^J \to Z^a_{\infty(,T)}$ locally uniformly on the domain, in particular at the finitely many matrix points ζ^*_{ij} (which lie in $\mathcal{T}_1$ by construction and in $T_{\mathcal{F}}$ by (21) when every $\mathsf{a}(\zeta_i) < \sigma_W$), where $\Phi^J(\zeta^*_{ij}) \to Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j)$ because the functionals depend on the family only through the total sources. Positivity passes to the limit:

$$\sum_{i,j} \bar\gamma_i \gamma_j \, Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j) \;\ge\; 0 \qquad \text{for the plane } \{x_\nu = c\},\ c \in D. \tag{24}$$

A side result: the modified-box values converge to the cube state, $\lim_J \lim_{k_m} Z^{a,(3.38)}_{k_m,\Omega_{J,k_m}}(\zeta) = Z^a_{\infty(,T)}(\zeta)$, on the respective domain and along any k-subsequence on which the fixed-J limits exist; this is the corrected form of King's passage from the modified boxes to the cube limit [16, (3.42)], with the margin $2n$ in place of one cube step (Lemma 9.6 and Remark 9.8). The reversal of King's second inequality is proved, weakly at every coupling and strictly at $e = 0$ (Lemma 9.6); no step of the passage to the limit uses that inequality in King's direction.

Step 7 (every plane). Let $c \in \mathbb{R}$ be arbitrary and the sources supported in $\{x_\nu > c\}$. If $c \in D$, (24) is the result. Otherwise pick $c' \in D$ and $b := (c' - c)e_\nu$. The translated sources $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$, and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, because $\rho_{c'}(x + b) = \rho_c(x) + b$ for this b and the linear parts of $\rho_c, \rho_{c'}$ agree. By (24) for the plane c' and the translation invariance of the limit functional,

$$\sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}(\theta_c\zeta_i^* \sqcup \zeta_j) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\tau_b(\theta_c\zeta_i^* \sqcup \zeta_j)\big) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\theta_{c'}(\tau_b\zeta_i)^* \sqcup \tau_b\zeta_j\big) \;\ge\; 0.$$

The translation invariance is supplied in (C) by Proposition 8.1, and in (T) by Proposition 10.5(b) on $T_{\mathcal{F}}$ (the amplitude a is translation invariant, so the matrix points stay in the tube). In the optional extension a plane $c \notin D$ needs a non-lattice translation, which finitely many instances of (W2)(b) do not supply; there the step is carried by Proposition 10.5(b) together with Lemma 10.2 for $\mathcal{F}$ and for $\tau_b\mathcal{F}$ and the identity theorem on $\mathcal{T}_1$ (Theorem 23.8).

10.5. **The theorems.**

Theorem 10.6 (reflection positivity, curvature sector, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). Let $\mu_0 > 0$, $\nu \in \{1, 2, 3\}$, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real 2-form sources $f_{il} \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$ supported in the open half-space $\{x_\nu > c\}$, with arbitrary complex coefficients z_{il} and no scalar sources. Then for all $\gamma \in \mathbb{C}^N$*

$$\sum_{i,j=1}^{N} \bar\gamma_i\gamma_j \, Z_\infty\big(\theta\zeta_i^* \sqcup \zeta_j,\, 0\big) \;\ge\; 0.$$

Besides the convergence at fixed cubes the proof uses only [7, Thm. 6.2(1),(2)] *(through the first inequality of Lemma 2.10 and the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10), Theorem 7.4(b) with*

Lemma 7.3, and Lemma 2.8, Lemma 2.11(d),(e), Lemma 2.7(e), Lemma 2.14; no inequality of class $\mathcal{D}$ enters.

Proof. At $h \equiv 0$ there is no Wick constant and no scheme, and $\mathcal{F}$ consists of the f_{il} and $\rho^* f_{il}$ with $U = (\mathrm{i}\mathbb{R})^m$. What the argument of Steps 5–7 needs of the cube functionals is: the existence of $Z_J(\cdot,0) = \lim_k Z_{k,J}(\cdot,0)$ locally uniformly on $\mathbb{C}^m$, entire; the monotonicity of $J \mapsto Z_J(\mathrm{i}f,0)$ on the cone (first inequality of Lemma 2.10); the existence of $Z_\infty(\cdot,0) = \lim_J Z_J(\cdot,0)$, entire, locally uniformly on $\mathbb{C}^m$; and translation invariance. The first three are Theorem 7.4(b), whose local bound is the μ_0-free bound of Theorem 7.4(a) and Lemma 7.3 rather than a local bound resting on the nonincreasing clause of Lemma 7.7 and so on [7, Thm. 6.2(3),(4)]; the fourth is Proposition 8.1, which uses Lemma 2.14 with $\ln Z_\infty$ finite on all of E_ρ (upper bound from Theorem 7.4(b), lower bound Lemma 2.11(e)) and the identity theorem. The lattice steps Lemmas 9.3, 9.9 and 9.10 include $g = 0$, and only their $\mathcal{I}$ halves are used; Lemma 10.3(C) is the fixed-J and J-uniform bound; Steps 5–7 in version (C) give the claim. □

Theorem 10.7 (reflection positivity on the Wick tube, comparison convention)**.** *Let a be an admissible scheme, and assume* (I2) *and* (I3) *(the hypotheses of Corollary 13.3) and, as in Proposition 10.5, that the comparison family is convergent at fixed cubes on the tube (Definition 3.2); the $\mathcal{D}$-class steps use in addition Lemma 2.7(d). Let $\sigma_+ = s_1$ be the amplitude of* (I2) *and $\sigma_W := \sigma_+/8$. Let $\mu_0 > 0$, ν, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu > c\}$, with complex coefficients z_{il}, w_{il} such that $\mathsf{a}(\zeta_i) < \sigma_W$ for every i; in particular, all purely imaginary w_{il} with arbitrary z_{il} are admitted. Then the values $Z^a_{\infty,T}(\theta\zeta_i^* \sqcup \zeta_j) = \lim_J Z^a_J(\theta\zeta_i^* \sqcup \zeta_j)$ exist (Proposition 10.5) and*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \, Z^a_{\infty,T}\big(\theta\zeta_i^* \sqcup \zeta_j\big) \;\geq\; 0 \qquad \text{for all } \gamma \in \mathbb{C}^N.$$

Where Z^a_∞ is defined, $Z^a_{\infty,T} = Z^a_\infty$ there. No lower bound on J beyond J_0 enters, and (W2) *is not used.*

Proof. With the splitting of Definition 10.1 the matrix points lie in $T_{\mathcal{F}}$ by (21). Lemma 10.3(T) gives the bounds; Steps 5 and 6 in version (T), with Proposition 10.5(a) for the limit on $U \cap T_{\mathcal{F}}$ and its holomorphy on $T_{\mathcal{F}}$, give (24) for $c \in D$; Step 7 with Proposition 10.5(b) gives every plane. The $\mathcal{D}$-class steps are the first factor of (22) (the $\mathcal{D}$ half of Lemma 9.9 and the second inequality of Lemma 2.10) and the nonincreasing factor in Proposition 10.5(a). □

The arbitrary-amplitude form, reflection positivity of Z^a_∞ at all complex coefficients under (W2), is Theorem 23.8 in the optional extension; its proof is Steps 5–7 in version (A) for $c \in D$, where (W2) enters only at the finitely many q_v, and in version (A) or (A′) for $c \notin D$.

Lemma 10.8 (reflection positivity of King's printed state)**.** *Theorem 10.6 holds for Z^K_∞: for ν, c, $\zeta_1, \dots, \zeta_N$ as there, $\sum_{i,j} \bar\gamma_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \geq 0$ for all $\gamma \in \mathbb{C}^N$; and Propositions 10.9 and 10.10 below hold for $Z^K_\infty(\cdot,0)$. The inputs are Theorem 3.3,* [7, Thm. 6.2(1),(2)]*, Theorem 7.6(a),(b) in place of Theorem 7.4, Lemma 4.6(c) and Remark 4.8 for the two places where the cube convention is seen, and Corollary 8.3 for the translations; no inequality of class $\mathcal{D}$ enters.*

Proof. Lemmas 9.3 and 9.5 concern the modified box state alone and the lattice reflection, and Lemma 10.3(C) bounds Φ_k on the modified photon form: these are convention-free, as is Step 5 (Montel in k; positivity passes). The two places where the Neumann–Dirichlet cube states enter are the ends of the sandwich (23), i.e. the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10; on King's convention they read $Z^K_{k,J}(\zeta) \leq \Phi_k(\zeta) \leq Z^K_{k,J+2n}(\zeta)$ for every k and $\zeta \in U = (\mathrm{i}\mathbb{R})^m$: the lower end is Lemma 9.9 with Step 1 of Lemma 4.2 prepended (Remark 4.8: $\langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{(a)}_{k,J} \leq \langle\cdot\rangle^{(3.38)}_{k,\Omega_{J,k}}$ on $\mathcal{I}$), the upper end is Lemma 9.10 with $\langle\cdot\rangle^K_{k,J}$ on X_{J+2n} in place of the comparison state (Lemma 4.6(b),(c): the box $\Omega_{J,k}$ does not meet the killed layers T_{J+2n}, and part (a) of Lemma 9.10 is photon-only), under the hypotheses of (20). Here J_0 is the J_0 of the enlarged family (the sources and their reflections), which exceeds that of the sources alone when $c \neq 0$;

Theorem 7.6(b) is applied to the enlarged family at J and $J+2n$, $J \geq J_0$. Along Step 5's subsequence the ends converge along the full sequence by Theorem 7.6(b), so $Z_J^K \leq \Phi^J \leq Z_{J+2n}^K$ on U for every subsequential Φ^J; as $J \to \infty$ both ends tend to Z_∞^K on U, $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ by Lemma 10.3(C), and U is a set of uniqueness, so $\Phi^J \to Z_\infty^K$ locally uniformly on $\mathbb{C}^m$ as in Step 6, in particular at the matrix points; positivity passes to the limit. Step 7 uses Corollary 8.3 in place of Proposition 8.1. Propositions 10.9 and 10.10 use only the existence, holomorphy near 0 and lattice symmetries of the limit, and the locally uniform convergences just established; their proofs below apply verbatim with Z_∞^K, the charge-conjugation invariance being exact for $\langle\cdot\rangle_{k,J}^K$ as well. □

10.6. OS3 **and the polynomial form.**

Proposition 10.9 (symmetry, realness and charge conjugation)**.** *Let Z be one of: $Z_\infty(\cdot,0)$ if the comparison family is convergent at fixed cubes (resp. $Z_\infty^K(\cdot,0)$); $Z_{\infty,T}^a$ under the hypotheses of Proposition 10.5; Z_∞^a under* (W2) *(optional extension). The Schwinger functions*

$$S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p) := \partial_{z_1}\cdots\partial_{z_n}\partial_{w_1}\cdots\partial_{w_p} Z\big(\textstyle\sum_i z_i f_i, \sum_l w_l h_l\big)\big|_0$$

(with $p=0$ and S_n in the curvature sector) are (a) symmetric multilinear in the f's and in the h's; (b) real for real test functions; (c) zero for odd n. The proof uses only that Z exists and is holomorphic near 0, with the stated symmetries of the lattice functionals; the existence is the input of the respective sector.

Proof. (a) Z depends on a family only through its total sources (so does every lattice functional, hence both limits), so its Taylor coefficients at 0 are symmetric multilinear forms in the f's and in the h's; $Z_{\infty,T}^a$ is holomorphic on a tube containing 0. The definition of $S_{n,p}$ does not depend on the family used to compute it (restrict a larger family to the given sources). (b) For real coefficients every lattice functional is real (real integrand and multiplier, Lemma 2.11(a)), so Z is real at the real points near 0 (for $Z_{\infty,T}^a$ on $\mathbb{R}^m \cap T$, an open piece of $\mathbb{R}^m$ containing 0); a holomorphic function real on the real points near 0 has real Taylor coefficients at 0. (c) Every cube state and every modified box state is invariant under $(A,\vartheta) \mapsto (-A,-\vartheta)$ (the photon terms are even in A, the masked divergence included; $\cos(\vartheta_x - e\varepsilon A - \vartheta_y)$ is invariant; W^a depends on r only; Lebesgue measure is preserved), which maps $F_{\Lambda_k}(f) \mapsto -F_{\Lambda_k}(f)$ and fixes $W_{\Lambda_k}^a(h)$. Hence $Z_{k,X}(-z,w) = Z_{k,X}(z,w)$ for all k, X, and in both limits $Z(-z,w) = Z(z,w)$ (on the $z \mapsto -z$ invariant tube for $Z_{\infty,T}^a$). So $\partial_z^{(n)}\partial_w^{(p)} Z(0) = 0$ for odd n. This is the argument of [27, Lemma 9.7, Thm. 9.8] for the cube states. □

Proposition 10.10 (the polynomial form of reflection positivity)**.** *Let $P = \sum_\alpha c_\alpha \prod_i F(f_{\alpha i}) \prod_l W^a(h_{\alpha l})$ be a polynomial in the smeared fields with real sources supported in $\{x_\nu > c\}$, and $\theta P := \sum_\alpha \bar{c}_\alpha \prod_i F(\rho^* f_{\alpha i}) \prod_l W^a(h_{\alpha l}\circ\rho)$. Then $S(\theta(P)P) \geq 0$, where S assigns to a monomial the corresponding Schwinger function of Proposition 10.9: the Taylor coefficient at 0 of $Z_{\infty,T}^a$ under the hypotheses of Theorem 10.7, resp. of $Z_\infty(\cdot,0)$ (of $Z_\infty^K(\cdot,0)$), where at $h \equiv 0$ the W^a factors are absent.*

Proof. Let all sources of P be supported in $\{x_\nu \geq c+\delta\}$ and let $\mathcal{F}'$ be the family of all sources of P and their reflections. Its scalar members have no sign, so Steps 4–6 are applied to the split family $\mathcal{F}'_{\text{split}}$ (members $h_{\alpha l}^{\pm}$, $h_{\alpha l}^{\pm}\circ\rho$, all ≥ 0); its tube contains 0 ($k_1(0)=0$), and the unsplit functional is the composition of the split one with the linear map $w_{\alpha l} \mapsto (w_{\alpha l}^+, w_{\alpha l}^-) = (w_{\alpha l}, -w_{\alpha l})$, so by the chain rule the derivatives at 0 of the unsplit Φ_k, Φ^J, $Z_{\infty(,T)}^a$ are finite linear combinations of those of the split ones, and locally uniform convergence near 0 passes through the linear map. (i) *Lattice.* For $k \geq k_0$ the sampled P is a gauge-invariant polynomial in the plus-side smeared fields, lies in L^2 of the modified box state (Gaussian moments; $\lambda > 0$) and in $\mathfrak{B}_+$ (Definition 9.1), and $\theta(P) = \overline{P\circ\hat{\rho}}$ by (18). By Lemma 9.3(a) (its L^2 extension), $\langle\theta(P)P\rangle_{k,\Omega_{J,k}}^{(3.38)} \geq 0$. (ii) *Moments as derivatives.* $\langle\theta(P)P\rangle^{(3.38)} = \sum_{\alpha,\beta}\bar{c}_\alpha c_\beta \langle\prod_i F(\rho^* f_{\alpha i})\prod_l W^a(h_{\alpha l}\circ\rho)\prod_i F(f_{\beta i})\prod_l W^a(h_{\beta l})\rangle^{(3.38)}$, and each mixed moment is a mixed partial derivative at 0 of Φ_k for the family $\mathcal{F}'$ (differentiate under the integral; the scheme multiplier $e^{-wa_k \mathsf{S}(h)}$ is the $-a_k$ inside W^a). (iii) *Limits.* By Steps

5–6 for $\mathcal{F}'_{\mathrm{split}}$, $\Phi_{k_m} \to \Phi^J$ and $\Phi^J \to Z^a_{\infty(,T)}$ locally uniformly on a neighbourhood of 0, so by Weierstrass' theorem all derivatives at 0 converge, and $\sum \bar{c}_\alpha c_\beta\, \partial^{(\alpha,\beta)} Z^a_{\infty(,T)}(0) \geq 0$; the left side is $S(\theta(P)P)$ by the family-independence of the Schwinger functions (Proposition 10.9(a)). Every plane: Step 7 (translation of the sources; S is translation invariant by Proposition 10.5(b), resp. Proposition 8.1 or Corollary 8.3). The curvature sector is the case $h \equiv 0$ with no splitting. This is the form OS2(b) of [27, Def. 2.10, Thm. 9.4] for the cube states. □

Remark 10.11 (which form is used, and the amplitude constants). The main theorem (Theorem 21.1) states reflection positivity of the Wick sector in the tube form of Theorem 10.7, with Propositions 10.9 and 10.10 for the Schwinger functions; its inputs for this clause are Theorem 3.6, (I2), (I3) and [7, Thm. 6.2(1)–(4)], without (W2). This suffices for the consumer: the weak form of OS4 and the reconstruction of Theorem 11.1 use reflection positivity for exponentials $e^{\mathrm{i}\Phi(f)}$ of the smeared fields, whose Wick coefficients are purely imaginary, so amplitude 0 only. Three positive, ε-free amplitude constants occur in the Wick sector: $\sigma_W = \sigma_+/8$ here; $H/2 = \sigma_+/32$, the negative-part radius of the tube $\mathcal{D}_{\sigma_+,H}$ of Theorem 13.10 ($H = s_1/16$); and $\sigma_0 = H/4$ in the bound E0″ of Theorem 13.10(ii). On $T_{\mathcal{F}} \cap \mathcal{D}_{\sigma_+,H}$ the objects $Z^a_{\infty,T}$ and G_∞ coincide (Proposition 10.5(a)). On King's convention the tube statements of this section are read under (TW^K) (Remark 13.12, Proposition 21.4); the curvature-sector statement needs no such hypothesis (Lemma 10.8). The margin $R(\mu_0)$ of Lemma 9.10 enters the proof, not the conclusion: reflection positivity is proved at every fixed $\mu_0 > 0$, and in a limit of vanishing photon mass it passes to the limit functionals as a pointwise limit of positive semidefinite matrices, the polynomial form by Weierstrass' theorem.

11. The measure and weak clustering

This section builds the measure of King's printed state in the curvature sector and draws from the authors' earlier paper on the torus limits [27] the consequences of reflection positivity: the ergodic decomposition, clustering, the unique joint vacuum on a component, the growth of the components' moments, and weak OS4 on almost every component. The joint field with the Wick square is the subject of the tube sections. The measure is built from the characteristic functional, not from the lattice laws, which converge only locally (Remark 11.2).

Conventions. $\mathcal{V} := \Lambda^2\mathbb{R}^3$ with its Euclidean fibre norm. For a real $u \in C^\infty_c(\mathbb{R}^3;\Lambda^2)$ put $\mathcal{Z}^K(u) := Z^K_\infty(\mathrm{i}u)$, the one-member functional at the coefficient i (Theorem 7.6(b), with the total-source convention of Theorem 8.5). Fix an even $r \geq 14$. $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ is the Hermite–Sobolev space, the dual of the completion of $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ under $\|u\|_r^2 = \sum_{\alpha,c} \lambda_\alpha^r \langle h_{\alpha,c}, u\rangle^2$, where $(h_{\alpha,c})_{\alpha\in\mathbb{N}^3,\, c\leq 3}$ is the Hermite basis of $L^2(\mathbb{R}^3;\Lambda^2)$ and $\lambda_\alpha = 2|\alpha| + 4$ are the eigenvalues of $-\Delta + |x|^2 + 1$; $|\cdot|_4$ is the weighted C^4 norm of Lemma 2.6, and $|u|_4 \leq C\|u\|_{10}$ on $\mathcal{S}$ [27, Lemma 10.5(a)]. $X := H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ carries the actions $(\tau_a\Phi)(u) = \Phi(\tau_{-a}u)$, $(g\Phi)(u) = \Phi(g^{-1}\cdot u)$ for $g \in O_h$ (or $O(3)$), $(\theta_j\Phi)(u) = \Phi(\sigma_j \cdot u)$ for the coordinate reflections σ_j, with the signed transport of 2-forms of Definition 2.3, and the local σ-algebras $\Sigma(U)$ generated by $\Phi \mapsto \Phi(u)$, $u \in C^\infty_c(U;\Lambda^2)$ real; $\theta_{j,c} := \tau_{ce_j}\theta_j\tau_{-ce_j}$ is the reflection in the plane $x_j = c$ and $\rho_{j,c*}u$ the reflected source (Definition 9.1). These are the space, actions and σ-algebras of the abstract part of [27], restricted from $\Lambda^2 \oplus \mathbb{R}$ to the curvature summand. The *lattice curvature field* of King's cube is the random distribution $\Phi^K_{k,J} : u \mapsto F_{\Lambda_k}(u)$ under $\langle\cdot\rangle^K_{k,J}$. $J_0(u)$ is the least J with $\operatorname{supp} u$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

Theorem 11.1 (the measure, the local lattice laws, weak clustering)**.** *The limits below exist by Theorem 3.3.*

(a) (The measure; uses Theorem 7.6 and Corollary 8.3 only.) *For every even $r \geq 14$ there is exactly one Borel probability measure $\mu^{K,H}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose mixed moments on C^∞_c are the Schwinger functions S^K_n of King's printed state. Its characteristic functional on C^∞_c, indeed on $\mathcal{S}$, is $\mathcal{Z}^K$ (extended by continuity); its moments on $\mathcal{S}$ are the continuous extensions of the S^K_n and satisfy the bounds of Theorem 7.6(c). Its image μ^K_S under the restriction map $\iota : \mathcal{H}^{-r} \to X$ is the unique Borel probability measure on X with characteristic functional*

$\mathcal{Z}^K$ on C_c^∞, and also the unique one whose mixed moments on C_c^∞ are the S_n^K. Local convergence of the lattice laws: *for every real $\chi \in C_c^\infty(\mathbb{R}^3)$ and every box Q containing* $\operatorname{supp}\chi$ *in its interior, the laws $\nu^K_{k,J,\chi}$ of the localised lattice fields $\chi\Phi^K_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$ converge weakly, along the full sequence $k\to\infty$ at every fixed $J \geq J_0(\chi)$, to a law $\nu^K_{J,\chi}$, and $\nu^K_{J,\chi} \Rightarrow \nu^K_\chi$ as $J\to\infty$, where ν^K_χ is the law of $\chi\Phi$ under μ^K_S. Nothing is asserted about the laws of the unlocalised lattice fields at fixed J (Remark 11.2).*

(b) (Invariance and weak clustering; uses also Theorem 8.5 and Lemma 10.8.) *μ^K_S satisfies the hypotheses (H1)–(H3) of the abstract part of* [27]*, is invariant under $\mathbb{R}^3 \rtimes O_h$ about every point, and the abstract theorems of* [27] *apply to it with their proofs: $\mu^K_S = \int \mu_\xi \, d\nu(\xi)$ with components that are translation-invariant and ergodic, θ_j-invariant, reflection positive about every coordinate plane and satisfy (H2)* [27, Thm. 10.10]*; clustering in the mean* [27, Thm. 10.19] *and under diagonal separation* [27, Thm. 10.20]*; the Osterwalder–Schrader reconstruction in each coordinate direction with a unique joint vacuum on every component* [27, Prop. 10.16, Prop. 10.17(iii)]*; the growth bound $|S^\xi_N(u_1,\dots,u_N)| \leq \sigma^\xi_N \prod_i \|u_i\|_{r+2}$, $\sigma^\xi_N \leq \alpha_\beta(\xi)(N!)^\beta$ for every $\beta > 1$, for the components' Schwartz moments* [27, Cor. 10.43(iv)]*; and the cubic-invariant decomposition into mixtures of $k(\zeta) \in \{1,2,3,6\}$ ergodic measures* [27, Thm. 10.12]*. The same holds on $\mathcal{H}^{-r}$ for $\mu^{K,H}$* [27, Lemma 10.5(d)]*. ν-almost every component satisfies* OS4 *in the weak form, ergodicity under the translation group $\mathbb{R}^3$* [27, Def. 2.12(iii)].

(c) (The rotation corollary.) *$\mathcal{Z}^K$ is $O(3)$-invariant, $\mathcal{Z}^K(R\cdot u) = \mathcal{Z}^K(u)$ for all $R \in O(3)$ and real u, by Theorem 8.7 (the rotation theorem Theorem 3.14); the clauses below use only this invariance. Then for ν-a.e. ξ: μ_ξ is invariant under $\mathbb{R}^3 \rtimes O(3)$; $\mathbb{E}_{\mu_\xi}[\overline{F\circ\theta_{n,c}}\,F] \geq 0$ for every unit vector n, every $c\in\mathbb{R}$ and every $F \in L^2(\Sigma(H^+_{n,c}),\mu_\xi)$, $H^+_{n,c} = \{x\cdot n > c\}$; and $\mathbb{E}_{\mu_\xi}[G'\,U(a)G] \to \mathbb{E}_{\mu_\xi}[G']\,\mathbb{E}_{\mu_\xi}[G]$ as $|a|\to\infty$ in any manner, for all $G, G' \in L^2(\mu_\xi)$. Consequently μ_ξ satisfies* OS4 *in each of the forms of* [27, Def. 2.11, Def. 2.12] *(the cluster property on the positive-time algebra, ergodicity of the time translations, ergodicity under $\mathbb{R}^3$) for every choice of the time direction, coordinate or not, and in the reconstruction from μ_ξ in any direction n the joint fixed space is* $\ker H_n = \mathbb{C}\Omega_n$ [27, Cor. 10.44]*. The moments of μ_ξ are real and symmetric; the vanishing of its odd curvature moments is* not *claimed for a component.*

11.1. **Proof of (a).** *Step 1 (moment bounds, and where they hold).* For real 2-forms $f_1,\dots,f_n$ at distance >1 from $\mathbb{R}^3\setminus C_{J_0}$, Theorem 7.6(c) gives, at every k and $J \geq J_0$, $|S_n^{K,(k,J)}| \leq (\sqrt{2}\,e)^n (n!)^{1/2} \prod_j \|f_j\|_{2,k}$ with $\|f\|^2_{2,k} \leq 3C_0|f|^2_4$ uniformly in k (Lemma 2.6), and in the limit the bound of Theorem 7.6(c); by Theorem 7.6(a), $\langle F(u)^2\rangle^K_{k,J} \leq 2\|u\|^2_{2,k}$ and hence $S_2^K(u,u) \leq 2\|u\|^2_2$. The lattice Schwinger functions are the Taylor coefficients at 0 of $a^K_{k,J}$, that is the moments of $\Phi^K_{k,J}$, and converge to the S_n^K. These bounds are uniform in k and $J\geq J_0$ for sources at distance >1 inside C_{J_0}; at a fixed J they are not available for sources reaching ∂C_J, and neither is the convergence in k: Theorem 3.3 and Theorem 7.6(a) are stated for sources whose plaquettes have their corners in X_J.

Step 2 (the measure). (i) *The functional on $\mathcal{S}$.* For real $u_1,\dots,u_N \in C_c^\infty$ and $c\in\mathbb{C}^N$, at every (k,J) with $J\geq J_0(u)$, $\sum_{l,m}\bar c_l c_m\, Z^K_{k,J}(\mathrm{i}(u_m-u_l)) = \langle|\sum_m c_m e^{\mathrm{i}F(u_m)}|^2\rangle^K_{k,J} \geq 0$ by the linearity of F_{Λ_k} in the source. The limits $k\to\infty$, then $J\to\infty$, exist at these points by Theorem 7.6(b) for the family u, so $\mathcal{Z}^K$ is positive definite on C_c^∞ with $\mathcal{Z}^K(0)=1$. By Corollary 8.3's Lipschitz bound and (E2), $|\mathcal{Z}^K(u)-\mathcal{Z}^K(u')| \leq \sqrt{2}\|u-u'\|_2 \leq \frac{\pi}{2}|u-u'|_4$. Since $|\cdot|_4$ is a continuous seminorm on $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ and C_c^∞ is dense in $\mathcal{S}$, $\mathcal{Z}^K$ extends uniquely to a continuous function on $\mathcal{S}$, still positive definite with $\mathcal{Z}^K(0)=1$. By the Bochner–Minlos theorem [10, 12] ($\mathcal{S}$ is nuclear) there is exactly one Borel probability measure μ' on $\mathcal{S}'(\mathbb{R}^3;\Lambda^2)$, with its cylinder σ-algebra, such that $\int e^{\mathrm{i}\Phi(u)}\,d\mu' = \mathcal{Z}^K(u)$ for all real $u\in\mathcal{S}$.

(ii) *Moments on C_c^∞, and on $\mathcal{S}$ by uniform integrability.* For real $u_1,\dots,u_N \in C_c^\infty$ the characteristic function of $(\Phi(u_l))_l$ under μ' is $\varphi(t) = \mathcal{Z}^K(\sum_l t_l u_l) = Z^{K,u}_\infty(\mathrm{i}t)$, the restriction to $\mathrm{i}\mathbb{R}^N$ of the entire function $Z^{K,u}_\infty$ (Theorem 7.6(b)). A characteristic function real-analytic near 0

has moments of all orders, with $\int \prod_l \Phi(u_l)\,d\mu' = (-\mathrm{i})^N \partial_{t_1}\cdots\partial_{t_N}\varphi(0)$ [18]; the chain rule gives $\partial_{t_1}\cdots\partial_{t_N}\varphi(0) = \mathrm{i}^N \partial_{z_1}\cdots\partial_{z_N} Z_\infty^{K,u}(0)$, so the factors cancel and

(25)
$$\int \prod_{l=1}^{N} \Phi(u_l)\,d\mu' = S_N^K(u_1,\dots,u_N); \qquad \int \Phi(u)^2\,d\mu' = S_2^K(u,u) \le 2\|u\|_2^2 \le \tfrac{\pi^2}{4}|u|_4^2 \le C_2''\|u\|_{10}^2$$

by Step 1, (E2) and $|u|_4 \le C\|u\|_{10}$. *Extension to* $\mathcal{S}$. By Theorem 7.6(c) each S_N^K extends to a continuous symmetric multilinear form on $\mathcal{S}^N$ with $|S_N^K| \le \sigma_N \prod_l |u_l|_4$. For $u_l \in \mathcal{S}$ take cut-offs $\chi_R = \chi(\cdot/R)$, $\chi \in C_c^\infty$, $\chi = 1$ on the unit ball, $0 \le \chi \le 1$, $R \ge 1$: $\chi_R u_l \to u_l$ in $\mathcal{S}$, so $\Phi(\chi_R u_l) \to \Phi(u_l)$ for every $\Phi \in \mathcal{S}'$; and $|\chi_R u|_4 \le C_\chi |u|_4$ uniformly in $R \ge 1$ with $C_\chi = 2^4\|\chi\|_{C^4}$, by Leibniz with $|R^{-|\beta|}(D^\beta\chi)(\cdot/R)| \le \|\chi\|_{C^4}$ (the norm $|\cdot|_4$ carries derivatives up to order 4, so the cut-off does not pass through it with constant 1). Hence $\int \prod_l \Phi(\chi_R u_l)^2\,d\mu' = S_{2N}^K(\chi_R u_1, \chi_R u_1, \dots) \le \sigma_{2N} C_\chi^{2N} \prod_l |u_l|_4^2$ uniformly in R: the products $\prod_l \Phi(\chi_R u_l)$ are bounded in $L^2(\mu')$, hence uniformly integrable, and converge pointwise, so $\int \prod_l \Phi(u_l)\,d\mu' = \lim_R S_N^K(\chi_R u_1,\dots) = S_N^K(u_1,\dots,u_N)$ by the continuity of S_N^K on $\mathcal{S}^N$.

(iii) *The Hermite carrier.* Put $M(\Phi) := \sum_{\alpha,c} \lambda_\alpha^{-r} \Phi(h_{\alpha,c})^2$. By (25) with $\|h_{\alpha,c}\|_{10}^2 = \lambda_\alpha^{10}$, $\int M\,d\mu' \le 3C_2'' \sum_\alpha \lambda_\alpha^{10-r} < \infty$ as soon as $r-10 > 3$ (there are $(n+1)(n+2)/2$ multi-indices with $|\alpha| = n$), that is for every even $r \ge 14$; so $\mu'(M < \infty) = 1$. For $\Phi \in \mathcal{S}'$ and $u \in \mathcal{S}$ the Hermite expansion $u = \sum_{\alpha,c} \langle h_{\alpha,c}, u\rangle\, h_{\alpha,c}$ converges in $\mathcal{S}$ [24, Thm. V.13], so $\Phi(u) = \sum_{\alpha,c} \langle h_{\alpha,c}, u\rangle\, \Phi(h_{\alpha,c})$ and, on $\{M < \infty\}$, $|\Phi(u)| \le M(\Phi)^{1/2}\|u\|_r$ by Cauchy–Schwarz: Φ extends to $\Psi(\Phi) \in \mathcal{H}^{-r}$ with $\|\Psi(\Phi)\|_{-r}^2 = M(\Phi)$ and $\Psi(\Phi) = \Phi$ on $\mathcal{S}$. The map $\Phi \mapsto \Psi(\Phi)$ (:= 0 off $\{M < \infty\}$) is measurable from the cylinder σ-algebra to the Borel σ-algebra of the separable Hilbert space $\mathcal{H}^{-r}$, its coordinates being evaluations. Put $\mu^{K,H} := \Psi_* \mu'$. Then $\int e^{\mathrm{i}\Psi(u)}\,d\mu^{K,H} = \mathcal{Z}^K(u)$ and $\int \prod_l \Psi(u_l)\,d\mu^{K,H} = S_N^K(u_1,\dots,u_N)$ for all $u, u_l \in \mathcal{S}$, by (ii). This is the carrier construction of [27, Lemma 10.5(b)] run on $\mathcal{S}'$.

(iv) *Uniqueness on* $\mathcal{H}^{-r}$. By [27, Lemma 10.5(c)], $\mu^{K,H}$ is the only Borel probability measure on $\mathcal{H}^{-r}$ whose mixed moments on C_c^∞ exist and equal the S_n^K. The proof there uses only that the moments m_n of $Z = \sum_l t_l \Psi(u_l)$ satisfy $|m_n| \le c_S(n+1)K^n n!$ (then, by Cauchy–Schwarz for the odd absolute moments and the Taylor remainder bound $|y|^n/n!$, the characteristic function of Z is the sum of its Taylor series on every interval of length $< 2/K$, hence determined by the moments on all of $\mathbb{R}$) and the Lusin–Souslin theorem on $\mathcal{H}^{-r}$ [13]. Our moments give the hypothesis with $c_S = 1$: $m_n = S_n^K(g,\dots,g)$ for $g = \sum_l t_l u_l$, and $|m_n| \le (\sqrt{2}\,e)^n (n!)^{1/2}\|g\|_2^n \le (n+1)K^n n!$ with $K = \sqrt{2}\,e\|g\|_2$.

(v) *The law on* X *and its two uniqueness statements.* The map $\iota : \mathcal{H}^{-r} \to X$, $\iota\Psi := \Psi|_{C_c^\infty}$, is continuous, linear and injective [27, Lemma 10.5(b)]: $|\Psi(\chi f)| \le \|\Psi\|_{-r}\|\chi f\|_r \le C_\chi \|\Psi\|_{-r}\|f\|_{H^r}$ for a cut-off χ, and C_c^∞ is dense in $\mathcal{S}$, which is dense in $\mathcal{H}^r$. Put $\mu_S^K := \iota_* \mu^{K,H}$; its characteristic functional on real C_c^∞ is $\mathcal{Z}^K$. By the uniqueness part of [27, Lemma 10.3], which applies to any Borel probability measure on X (for a countable dense $(f_n) \subset C_c^\infty$ the sets $\{\Phi : (\Phi(f_1),\dots,\Phi(f_N)) \in B\}$ form a π-system generating the Borel σ-algebra, by Lusin–Souslin, and the law of each such vector is determined by $t \mapsto \mathcal{Z}^K(\sum_k t_k f_k)$), μ_S^K is the only Borel probability measure on X with this characteristic functional. *Moment uniqueness on* X*:* if μ'' is any Borel probability measure on X whose mixed moments on C_c^∞ are the S_n^K, the argument of (iv) applies verbatim on X with this π-system [27, Prop. 10.4], so $\mu'' = \mu_S^K$. And $\mu^{K,H}$ is the only measure on $\mathcal{H}^{-r}$ with $\iota_*\mu^{K,H} = \mu_S^K$, since ι is a Borel isomorphism onto the Borel set $\iota\mathcal{H}^{-r}$. Nothing in Step 2 sees the lattice: the measure clauses of (a) depend on King's printed state only through $\mathcal{Z}^K$ and the S_n^K.

Step 2′ (local convergence of the lattice laws). Fix a real $\chi \in C_c^\infty(\mathbb{R}^3)$ and a box Q with $\operatorname{supp}\chi$ compact in the interior of Q (as in [27, Prop. 8.3]), and let $(e_{\alpha,c})_{\alpha\in\mathbb{Z}^3,\,c\le 3}$ be the Fourier basis of $L^2(Q;\Lambda^2)$, $|e_{\alpha,c}(x)| = |Q|^{-1/2}$. For a real Q-periodic smooth 2-form ψ put $\langle \chi\Phi_{k,J}^K, \psi\rangle := \Phi_{k,J}^K(\chi\psi) = F_{\Lambda_k}(\chi\psi)$, where $\chi\psi \in C_c^\infty$ is supported in $\operatorname{supp}\chi$ (for the complex basis vectors,

used only in (1), take real and imaginary parts); this identifies $\chi\Phi^K_{k,J}$ with an element of the real Hilbert space $H_\chi := H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$, and $\nu^K_{k,J,\chi}$ denotes its law.

(1) *Second moments, uniformly.* For $J \ge J_0(\chi)$ and every k, every plaquette carrying $\chi e_{\alpha,c}$ has its centre in $\operatorname{supp}\chi$, hence its corners in X_J, so the second-moment clause of Theorem 7.6(a) applies to the real and imaginary parts: $\mathbb{E}^K_{k,J}|F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \le 2\|\chi \operatorname{Re} e_{\alpha,c}\|^2_{2,k} + 2\|\chi \operatorname{Im} e_{\alpha,c}\|^2_{2,k} \le 4\|\chi e_{\alpha,c}\|^2_{2,k}$. *Riemann-sum bound:* the plaquettes of one orientation have their centres on a translate of $\varepsilon_k\mathbb{Z}^3$; those carrying $\chi e_{\alpha,c}$ have centres in $\operatorname{supp}\chi$; the cubes of side $\varepsilon_k \le 1$ about these centres are disjoint and lie in the $\frac12$-neighbourhood $N_{1/2}(\operatorname{supp}\chi)$; so $\|\chi e_{\alpha,c}\|^2_{2,k} \le 3\operatorname{vol}(N_{1/2}(\operatorname{supp}\chi))\,\|\chi\|^2_\infty/|Q| =: K_\chi$, uniformly in α, c, k and $J \ge J_0(\chi)$ (no derivative of χ enters). Hence, for $s_0 > \frac32$,

(26)
$$\mathbb{E}^K_{k,J}\|\chi\Phi^K_{k,J}\|^2_{H^{-s_0}_{\mathrm{per}}(Q)} = \sum_{\alpha,c}(1+|\alpha|^2)^{-s_0}\,\mathbb{E}^K_{k,J}|F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \;\le\; 12K_\chi \sum_{\alpha\in\mathbb{Z}^3}(1+|\alpha|^2)^{-s_0} =: C'_\chi < \infty,$$

uniformly in k and $J \ge J_0(\chi)$ (the constants 4 and $12K_\chi$ are valid but not sharp).

(2) *Tightness.* The embedding $H^{-s_0}_{\mathrm{per}}(Q) \hookrightarrow H^{-r}_{\mathrm{per}}(Q)$ is compact for $r > s_0$ (Rellich on the torus); by Chebyshev, $\nu^K_{k,J,\chi}(\|\cdot\|_{H^{-s_0}} > \rho) \le C'_\chi/\rho^2$, and the H^{-s_0}-ball of radius ρ is compact in H_χ; so $(\nu^K_{k,J,\chi})_{k\ge0,\,J\ge J_0(\chi)}$ is tight on the separable Hilbert space H_χ, and Prokhorov's theorem [8] gives weak limit points along k at fixed J.

(3) *Identification at fixed J: the full sequence, and where Theorem 3.3 enters.* For a real trigonometric polynomial ψ, $\mathbb{E}^K_{k,J}e^{\mathrm{i}\langle\chi\Phi^K_{k,J},\psi\rangle} = Z^K_{k,J}(\mathrm{i}\chi\psi) \to Z^K_J(\mathrm{i}\chi\psi)$ as $k\to\infty$, along the full sequence, by Theorem 7.6(b) (that is Corollary 4.4(ii)) for the one-member family $(\chi\psi)$ at $J \ge J_0(\chi) \ge J_0(\chi\psi)$: this is Theorem 3.3 for that family on King's cube, together with the local bound of Theorem 7.6(a) and Montel's theorem. A Borel probability measure on H_χ is determined by its characteristic functional on the dual $H^r_{\mathrm{per}}(Q;\Lambda^2)$, which is continuous there (dominated convergence), hence by its values on the dense set of real trigonometric polynomials. So every weak limit point of $(\nu^K_{k,J,\chi})_k$ is the one probability measure $\nu^K_{J,\chi}$ on H_χ with characteristic functional $\psi \mapsto Z^K_J(\mathrm{i}\chi\psi)$, and $\nu^K_{k,J,\chi} \Rightarrow \nu^K_{J,\chi}$ along the whole sequence $k\to\infty$ by tightness and the uniqueness of the limit point. Without Theorem 3.3 the same argument gives the convergence along every continuum subsequence (Theorem 14.9 and Corollary 14.14), not along the full sequence; the full-sequence clause is what Theorem 3.3 adds for King's printed state.

(4) $J\to\infty$. $\|\cdot\|^2_{H^{-s_0}_{\mathrm{per}}} = \sum_{\alpha,c}(1+|\alpha|^2)^{-s_0}|\langle\cdot,e_{\alpha,c}\rangle|^2$ is lower semicontinuous on H_χ (a supremum of finite sums of continuous functions), so by the portmanteau theorem [8] the bound (26) passes to $\nu^K_{J,\chi}$: $\int\|\cdot\|^2_{H^{-s_0}}\,d\nu^K_{J,\chi} \le C'_\chi$ for every $J \ge J_0(\chi)$; hence $(\nu^K_{J,\chi})_J$ is tight on H_χ by the argument of (2), and $Z^K_J(\mathrm{i}\chi\psi) \to Z^K_\infty(\mathrm{i}\chi\psi) = \mathcal{Z}^K(\chi\psi)$ (Theorem 7.6(b)) identifies every limit point: $\nu^K_{J,\chi} \Rightarrow \nu^K_\chi$, the measure with characteristic functional $\psi \mapsto \mathcal{Z}^K(\chi\psi)$.

(5) *ν^K_χ is the law of $\chi\Phi$ under μ^K_S.* For $\Phi \in X$, $\chi\Phi \in H^{-r}(\mathbb{R}^3;\Lambda^2)$ is supported in $\operatorname{supp}\chi$, compact in the interior of Q, so its periodisation is an element of H_χ with $\langle\chi\Phi,\psi\rangle = \Phi(\chi\psi)$ for Q-periodic smooth ψ; the map $\Phi \mapsto \chi\Phi$ is continuous from X to H_χ, since $|\Phi(\chi\psi)| \le \|\tilde\chi\Phi\|_{H^{-r}}\|\chi\psi\|_{H^r} \le c_\chi\|\tilde\chi\Phi\|_{H^{-r}}\|\psi\|_{H^r_{\mathrm{per}}(Q)}$ for a cut-off $\tilde\chi = 1$ on Q. And $\int e^{\mathrm{i}\langle\chi\Phi,\psi\rangle}\,d\mu^K_S = \int e^{\mathrm{i}\Phi(\chi\psi)}\,d\mu^K_S = \mathcal{Z}^K(\chi\psi)$ by Step 2(v).

(6) *Moments.* For every trigonometric polynomial ψ (indeed every $\psi \in H^r_{\mathrm{per}}(Q;\Lambda^2)$, by the uniform-integrability argument of Step 2(ii) with $\chi\psi_n \to \chi\psi$), all moments of $\langle\chi\Phi^K_{k,J},\psi\rangle = F_{\Lambda_k}(\chi\psi)$ are bounded uniformly in k and $J \ge J_0(\chi)$ (Step 1; $|\chi\psi|_4 \le d_\chi\|\psi\|_{C^4}$ by Leibniz), so by uniform integrability the moments of $\nu^K_{J,\chi}$ are the Taylor coefficients of Z^K_J at $\chi\psi$, and those of ν^K_χ are the $S^K_n(\chi\psi,\dots,\chi\psi)$, consistently with (25). This proves (a). □

Remark 11.2 (why the convergence clause is local). The convergence $Z^K_{k,J}(\mathrm{i}u) \to Z^K_J(\mathrm{i}u)$ is available only for sources u whose plaquettes have their corners in X_J, that is for u at distance > 1 inside C_J, where Theorem 3.3 speaks. For a u reaching ∂C_J the lattice functional is defined only after a convention for the plaquettes with a corner outside X_J (for instance, drop them),

and for it no bound uniform in k and no convergence is asserted: the second-moment bound of Theorem 7.6(a) carries the corner hypothesis, and the crude bound $\langle e^{F(\tilde u)}\rangle^K_{k,J} \le e^{\frac12\mu_0^{-2}\|d_k^*\tilde u\|^2_{2,k}}$ (Lemma 2.8 with $C \le \mu_0^{-2}$ as forms) is not uniform in k for a source $\tilde u$ truncated at ∂C_J: $d_k^*\tilde u$ is of order $\|u\|_\infty/\varepsilon_k$ on the $O(\varepsilon_k^{-2}|\partial C_J|)$ bonds of the truncation layer, so $\|d_k^*\tilde u\|^2_{2,k} = O(\varepsilon_k^{-1})$. In particular the fixed-J laws of the unlocalised lattice field on X are not asserted to be tight; nothing below depends on them.

11.2. **Proof of (b) and (c).** *Step 3 ((H1)–(H3)).* The hypotheses of the abstract part of [27] are: (H1) a Polish space with a jointly continuous $\mathbb{R}^3$-action and three involutive homeomorphisms θ_j with $\theta_j\tau_a = \tau_{\rho_j a}\theta_j$, and μ invariant under all of them; (H2) local σ-algebras $\Sigma(U)$ with monotonicity, $\Sigma(U) = \bigvee_{K\Subset U}\Sigma(K)$, $\bigvee_K \Sigma(K) = \mathcal{B}$, countable generation by bounded continuous functions, and covariance under $U(a)$ and Θ_j; (H3) $\mathbb{E}_\mu[\overline{F\circ\theta_{j,c}}\,F] \ge 0$ for $F \in L^2(\Sigma(H^+_{j,c}),\mu)$. On X with the actions of the Conventions, the structural parts of (H1) and all of (H2) are proved in [27, Prop. 10.4] by arguments that use only the space and the push-forward actions (joint continuity of the translations on H^{-s}_{loc}; the θ_j are linear homeomorphisms; Lusin–Souslin for $\bigvee_K\Sigma(K) = \mathcal{B}$; countable dense families $\arctan\Phi(f_n)$; $\operatorname{supp}\tau_a f = \operatorname{supp} f + a$, $\operatorname{supp}\sigma_j\cdot f = \sigma_j \operatorname{supp} f$); they hold verbatim for our X, that paper's space restricted to the curvature summand. *Invariance of μ_S^K:* by the uniqueness in Step 2(v) it suffices that $\mathcal{Z}^K(\tau_a u) = \mathcal{Z}^K(u)$ and $\mathcal{Z}^K(\sigma_j\cdot u) = \mathcal{Z}^K(u)$ for real u; the first is Corollary 8.3 at the coefficient i, the second Theorem 8.5 at $z = \mathrm{i}$ with $O = \sigma_j$, and the same theorem with $O = \tau_b g\tau_{-b}$ gives the O_h-invariance about every point. *(H3):* for $E_a = e^{\mathrm{i}\Phi(u_a)}$ with u_a real in $C_c^\infty(\{x_j > c\};\Lambda^2)$, $\overline{E_a\circ\theta_{j,c}}\,E_b = e^{\mathrm{i}\Phi(u_b - \rho_{j,c*}u_a)}$ and

$$\mathbb{E}_{\mu_S^K}\big[\overline{E_a\circ\theta_{j,c}}\,E_b\big] = \mathcal{Z}^K(u_b - \rho_{j,c*}u_a) = Z^K_\infty(\theta\zeta_a^*\sqcup\zeta_b), \qquad \zeta_a = (\mathrm{i}, u_a),$$

the conjugated coefficient $-\mathrm{i}$ sitting on the reflected source as in Definition 9.1. By Lemma 10.8, for the plane $x_j = c$ and every $c \in \mathbb{R}$, this Gram matrix is positive semidefinite. The exponential algebra is dense in $L^2(\Sigma(H^+_{j,c}),\mu_S^K)$ [27, Lemma J.2] (an algebra of bounded complex functions, closed under conjugation and containing the constants, is dense in L^2 of the σ-algebra it generates), and $\theta_{j,c}$ preserves μ_S^K, so (H3) follows as in [27, Prop. 10.4]. Finally [27, Lemma 10.5(d)] carries (H1)–(H3) and the O_h-invariance about every point from (X,μ_S^K) to $(\mathcal{H}^{-r},\mu^{K,H})$.

Step 4 (weak clustering). The standing hypothesis of the abstract part of [27] is that μ is any measure satisfying (H1)–(H3); so [27, Thm. 10.10] gives the ergodic decomposition $\mu_S^K = \int \mu_\xi\, d\nu$ with components translation-invariant and ergodic, θ_j-invariant, satisfying (H3) and (H2); [27, Thm. 10.19, Thm. 10.20] give clustering in the mean and under diagonal separation; [27, Prop. 10.16] is the reconstruction in direction j, and on an ergodic component the joint fixed space is $\mathbb{C}\Omega_j$ [27, Prop. 10.17(iii)]. The proof of [27, Cor. 10.43(iv)] uses only the E0″ bound in the form $\mathbb{E}_{\mu_S} Z^{2n}_{\alpha,c} \le c_S C_0^{2n}(2n)!\,|h_{\alpha,c}|_4^{2n}$ on $Z_{\alpha,c} = \Phi(h_{\alpha,c})$, together with [27, Lemma 10.5(a)], Jensen, Markov and Borel–Cantelli; Theorem 7.6(c) supplies that bound on $\mathcal{S}$ through (25), with $(e\pi/2)^{2n}((2n)!)^{1/2} \le (e\pi/2)^{2n}(2n)!$, so $c_S = 1$, $C_0 = e\pi/2$, and the growth bound of (b) holds for our components with the same proof. [27, Thm. 10.12] assumes in addition that O_h acts on X by homeomorphisms with $\rho_j = \theta_j$, $g\tau_a g^{-1} = \tau_{ga}$, μ invariant under every g, and $G \mapsto G\circ g^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(gU)$-measurable ones: structural for the push-forward action ($(g\Phi)(u) = \Phi(g^{-1}\cdot u)$ is a linear homeomorphism of X, the relations are identities on test functions, $\operatorname{supp} g\cdot u = g\operatorname{supp} u$) together with the invariance of Step 3; it gives the cubic-invariant decomposition with $k(\zeta) \in \{1,2,3,6\}$. Weak OS4 is [27, Def. 2.12(iii)], ergodicity for the action of $\mathbb{R}^3$, the ergodicity clause of the decomposition for ν-a.e. component.

Step 5 (the rotation corollary). [27, Cor. 10.44] assumes (H1)–(H3) and that $O(3)$ acts on X by homeomorphisms $x \mapsto Rx$, continuously in R for each x, with $\rho_j x = \theta_j x$ for the coordinate reflections, $R\tau_a R^{-1} = \tau_{Ra}$, $G \mapsto G\circ R^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(RU)$-measurable ones, and μ invariant under every $R \in O(3)$. The structural parts hold for $(R\Phi)(u) = \Phi(R^{-1}\cdot u)$ on X: the action is by linear homeomorphisms, continuous in R, and covariant for the local σ-algebras, as noted in [27, Rem. 10.45(1)] for $H^{-s}_{\mathrm{loc}}(\mathbb{R}^3;\mathcal{V})$, and the push-forward respects the restriction to the Λ^2 summand. The invariance of μ_S^K is, by Step 2(v), $\mathcal{Z}^K(R\cdot u) = \mathcal{Z}^K(u)$ for real u, the hypothesis of (c), that is the $O(3)$-invariance of Z^K_∞ at imaginary coefficients,

supplied by Theorem 8.7; this section does not derive it. The conclusions are those quoted in (c), proved there from the R-invariance and (H3) (reflection positivity about every plane), the argument of [27, Thm. 10.10] with a countable dense subgroup (the Euclidean invariance and reflection positivity of a.e. component), [27, Thm. 10.20] along $(1,1,1)/\sqrt{3}$ after a rotation (clustering in every manner), and [27, Prop. 10.41(a)] in a frame containing n ($\ker H_n = \mathbb{C}\Omega_n$). They contain nothing on OS3: the vanishing of the odd curvature moments of a component is not among them, and is not known for a single component.

Step 6 (curvature alone). Everything above runs on $X = H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ for the curvature field alone: the regularity input is Theorem 7.6, the reflection positivity input Lemma 10.8, the invariance input Theorem 8.5, the $O(3)$ input that of (c). The result is an ergodic decomposition of the law of the curvature field; nothing is claimed about the components of a joint law with the Wick square, and the single-axis statements of [27] that rest on a torus-specific domination are not claimed for King's state. □

Remark 11.3 (the hypotheses of the cited results, matched). For each result of [27] used above, the step that supplies its printed hypotheses. (H1)–(H3): Step 3 (structure from prop:discharge; invariance from Corollary 8.3 and Theorem 8.5; positivity from Lemma 10.8 and lem:A:mc). lem:D:realize, uniqueness part (a measure on X with a given characteristic functional; Lusin–Souslin, lem:A:LS): Step 2(v). prop:discharge (the space and the push-forward actions; "Determination by S"): Steps 3 and 2(v). lem:D:hermite: (a) no hypothesis; (b) the two-point bound $\mathbb{E}\Phi(f)^2 \le C\|f\|_{10}^2$, Step 2(ii); (c) $|m_n| \le c_S(n+1)K^n n!$, $r \ge 14$ even, Step 2(iv) with $K = \sqrt{2}\,e\|g\|_2$, $c_S = 1$; (d) (H1)–(H3) on X, Step 3. The standing hypothesis "μ is any measure satisfying (H1)–(H3)" of thm:D:decomp, thm:D:mean, thm:D:diag, prop:D:recon, prop:D:fixed: Step 3; the ergodicity needed in prop:D:fixed(iii) holds on a component by thm:D:decomp. add:Ohdecomp (the O_h action and invariance): Step 4 and Theorem 8.5. cor:D:schwinger(iv) (the E0″ bound on Hermite functions; lem:D:hermite(a)): Theorem 7.6(c) through Step 2(ii), $c_S = 1$, $C_0 = e\pi/2$. p:cor:rotcor (the $O(3)$ action and invariance): Step 5, the structure from add:rotcond-rem(1), the invariance from the hypothesis of (c). def:OS4, add:OS4(i)–(iii): definitions, Step 5; add:OS4(iii) is the ergodicity clause of thm:D:decomp. prop:C-tight, mechanism only (uniformly bounded second moments of $\Phi(\chi e_\alpha)$): Step 2′(1). lem:A:LS, lem:A:mc: Steps 2(iv),(v), 3. The results of [27] stated for its own limit states (its clustering theorem, its Schwinger-function corollary as stated, its remark on components) are not cited; their abstract content is in the items above, and the odd-moment clause is not transferred to the components (clause (c)).

Remark 11.4 (the comparison family). The same conclusions hold for the limit states of the comparison family $\mathrm{ND}^{(a)}$: for every continuum subsequence $\mathcal{K}$ (Definition 14.2) the limit state $Z^{\mathcal{K}}_\infty$ carries a measure with the properties of Theorem 11.1(a) and (b), the local lattice laws converging along $\mathcal{K}$ instead of the full sequence, and (c) (Theorem 14.9, clauses (v)–(vii)). By Theorem 3.3 every such limit state equals Z^K_∞ (Corollary 4.4(iii)), so for King's printed state the two routes to the measure agree. For the iterated limit of the comparison family itself, when it is convergent at fixed cubes (Definition 3.1), see Proposition 14.12(b); nothing beyond Corollary 4.4(iii) is asserted about it here, the two existence clauses not being interchangeable by the inputs of this paper (Corollary 4.4(iv)).

12. The Wick tube I: torus inputs and the extensive bound

The Wick leg controls the joint functional of the curvature and the Wick square on the tube $\mathcal{D}_{\sigma_+,H}$ of Definition 2.5. Its positive leg (sources $q \ge 0$) and negative leg (sources $-q$) are proved in the next section from three inputs on *King's tori*: lattice reflection positivity (Proposition 12.2), the chessboard estimate it yields (Lemma 12.4), and an *extensive bound* (I2) for constant positive scalar sources in the fixed-spacing volume limit (Definition 12.5). This section proves the first two and derives the third from the torus inputs Proposition 3.9 and Lemma 3.10 and the sign clause (T3) (Proposition 12.6); a variant with the auxiliary mass on either side of m (Proposition 12.11) serves the discussion of the coupling range; no continuum limit is taken here.

12.1. **King's tori, the two conventions and mid-plane pairs.**

Definition 12.1 (King's tori)**.** A *King torus* is $T = \mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in M\mathbb{N}$ even; $|T| := \ell^3$ is its volume, and $T_J := \mathbb{R}^3/(2MJ\mathbb{Z})^3$, $J \in \mathbb{N}$, is the torus with fundamental cell C_J. Since $\ell\mathbb{Z}^3 \subset \mathbb{Z}^3 \subset \Lambda_k$, the lattice Λ_k descends to T, with $\ell\, 5^k$ sites per side, an even number, and $|T|/\varepsilon_k^3$ sites in all. The *periodic state* $\langle\cdot\rangle_{k,T}$ is King's action of Section 2 summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, with periodic conditions for both fields [16, (4.1)], [15, (2.21)]: Feynman-form gradient terms for every pair of parallel neighbouring bonds, the photon mass μ_0^2, hopping terms on every bond, the single-site weights $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m^2)r^2+\lambda r^4]}$. Two normalisations of the counterterm are used:

(i) the *free convention* of Section 2 ($\delta m^2 = \delta m_\varepsilon^2$, Wick constant W_ε; the convention of every cube state of this paper), with $N_{k,T}^a(0,q) := e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)}\langle e^{\sum_x \varepsilon^3 q(x) r_x^2}\rangle_{k,T}$ for a real function q on the sites of T, exactly as on boxes (Definition 2.3), and

$$\alpha_{k,T}^a(u) := |T|^{-1} \ln N_{k,T}^a(0, u\mathbf{1}_T) \qquad (u \in \mathbb{R}),$$

the *free energy density* of a constant source;

(ii) *King's torus convention* [15, (2.4), (3.33), (3.35)]: the counterterm $\delta m_{k,T}^2(M)$ and the Wick constant $c_{k,T}(M)$ computed with the periodic propagators of T at scalar mass M, together with the field-independent constants $E(M) = E_0(M) + E_1(M)$ of King's torus action. $Z_{k,T}^{(M)} := \int e^{-S}$ is the source-free partition function of this model at scalar mass M (bare coefficient $M^2 + \delta m_{k,T}^2(M)$; Proposition 3.9 is a statement about these numbers), and $N_{k,T}^{z,\mathrm{tor}}(0,q)$ the normalised source functional of the torus model at mass m in the reference scheme z, with $c_{k,T} = c_{k,T}(m)$ in place of W_ε.

In either convention the coefficients are the same at every site of T. A *mid-plane pair* is $\pi = \{x_\nu = c\} \cup \{x_\nu = c + \ell/2\}$ with $\nu \in \{1,2,3\}$ and $c \in \varepsilon_k(\mathbb{Z} + \frac{1}{2})$ (both planes lie between lattice planes, since $\ell 5^k$ is even); θ denotes the reflection of T in π, which exchanges the two slabs of $T \setminus \pi$ ("the two sides of π"), acting on the fields as in Definition 9.1.

12.2. **Reflection positivity of King's tori.**

Proposition 12.2 (lattice reflection positivity of King's tori)**.** *Let T be a King torus, $k \geq 0$, π a mid-plane pair and θ the reflection in π. For every finite family $G_1, \dots, G_N$ of gauge-invariant Borel functions of the fields on one side of π with $\langle |G_i|^2\rangle_{k,T} < \infty$ and every $\gamma \in \mathbb{C}^N$,*

$$\sum_{i,l=1}^{N} \bar\gamma_i \gamma_l \, \langle \theta(G_i)\, G_l\rangle_{k,T} \;\geq\; 0.$$

The periodic state is invariant under θ and under all lattice translations. This holds in both conventions of Definition 12.1: reflection positivity does not see the value of the (site-independent) counterterm.

Proof. The proof is the mechanism of Lemma 9.3 with a pair of planes in place of one and without Dirichlet diagonals, in three steps.

Step 1: from the Feynman form to the Proca form. On the torus write $A = Z + d_0\chi$ with $d_0^T Z = 0$, $\chi \perp 1$ (Hodge decomposition on the finite periodic graph). The Feynman form equals $\frac{1}{2}\langle Z, (d_1^T d_1 + \mu_0^2)Z\rangle$ plus a quadratic form in χ, and the Proca form $\frac{1}{2}\langle A, (d_1^T d_1 + \mu_0^2)A\rangle$ equals the same Z-form plus another χ-form; both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. For gauge-invariant G the matter integral at fixed A is invariant under $A \mapsto A + d_0\chi$ (rotate ϕ site by site), so the χ-integrals factor and cancel in the normalised expectation: gauge-invariant expectations in the Feynman form equal those in the Proca form. Fubini's theorem applies by $\mu_0 > 0$, $\lambda > 0$ and a growth bound $|G| \leq Ce^{a\sum_b |A(b)|} e^{C\sum_x r_x^2}$, satisfied by bounded G and by the observables $\Theta(\zeta)$; so this step uses $\mu_0 > 0$, gauge invariance, $\lambda > 0$ and the growth bound, and nothing about the value of the counterterm.

Step 2: the Proca torus measure is reflection positive about π. In the Proca form the action couples the two sides of π only through the plaquettes and hopping terms crossing one of the two

planes of π (no gradient terms between parallel bonds across a plane; the mass term is local). Write each crossing bond variable as $a_b = \alpha_b - \alpha'_b$ with α_b, α'_b independent centred Gaussians of variance $(2\mu_0^2\varepsilon^3)^{-1}$, attached to the two sides. Then θ becomes the swap of the two data sets $\xi = (\omega_+, \alpha)$, $\xi' = (\omega_-^\theta, \alpha')$, whose a priori measures coincide by the θ-invariance of the one-sided terms, and exactly as in Step 2 of Lemma 9.3 every crossing plaquette term is $\frac{\varepsilon}{2}X_p(\xi)^2 + \frac{\varepsilon}{2}X_p(\xi')^2 - \varepsilon X_p(\xi)X_p(\xi')$ and every crossing hopping term is $-\varepsilon\sum_{i=1,2}\psi_b^i(\xi)\psi_b^i(\xi')$, with the same real functions on both copies. So the Boltzmann factor is $\mathcal{G}(\xi)\mathcal{G}(\xi')\exp(\sum_\iota \gamma_\iota Y_\iota(\xi)Y_\iota(\xi'))$ with all $\gamma_\iota = \varepsilon > 0$ and all Y_ι real, and for G a function of one side $\theta(G)G = \overline{G(\xi')}G(\xi)$. Expanding the exponential, $Z\langle\theta(G)G\rangle$ is a sum with nonnegative coefficients of squared moduli $|\int d\xi\, G(\xi)\mathcal{G}(\xi)\prod_\iota Y_\iota(\xi)^{n_\iota}|^2$, the interchange of sum and integral being justified as in Lemma 9.3 ($\lambda > 0$, $\mu_0 > 0$ and the growth bound). The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle\theta(G)G'\rangle$ follows from the θ-invariance of the measure, so $(G, G') \mapsto \langle\theta(G')G\rangle$ is a positive semidefinite Hermitian form on bounded gauge-invariant one-sided G.

Step 3: L^2 observables. For gauge-invariant one-sided G with $\langle|G|^2\rangle < \infty$ apply Step 2 to $G_n := G\mathbf{1}\{|G| \le n\}$; since $|\theta(G_n)G_n| \le |\theta G||G|$ and $\langle|\theta G||G|\rangle \le \langle|G|^2\rangle$ by Cauchy–Schwarz and θ-invariance, dominated convergence gives the inequality for $G = \sum_i \gamma_i G_i$, hence the form statement.

Invariance. Under θ and under every lattice translation each term of the periodic action, in either convention, goes to a term of the same kind with the same coefficient, and the a priori measure is invariant; the value of the site-independent bare coefficient enters nowhere above. □

Remark 12.3 (the torus paper and the literature). (i) The torus paper [27] proves Proposition 12.2 for its lattice model, King's action on the torus with a constant bare coefficient b_T, for every even side, every direction, both kinds of plane pair and every b_T [27, Prop. 7.9(a)]; its printed scope is $\mu_0 \in (0, \mu_*]$, where $\mu_* = m$ is the photon mass at which that paper freezes its counterterm. The restriction plays no role in the proof, which uses no bound on μ_0: the two lemmas behind it there, the passage to the Proca form [27, Lemma 7.7] and reflection positivity of the Proca torus measure in a pair of planes [27, Lemma 7.8(ii)], are stated for every photon mass $\mu > 0$, with the inputs site-independent coefficients, $\lambda > 0$ and even sides. So the statement lifts to every $\mu_0 > 0$, as the proof above also shows directly. (ii) [7, Cor. 5.4] (reflection positivity "in every gauge", for covariances $C_F = C_u + dFd^* \ge 0$ with F of either sign) is the same mechanism: on the torus the Feynman covariance has this form with $F = -\mu_0^{-2}(\Delta_0 + \mu_0^2)^{-1} \le 0$ on functions orthogonal to the constants. It is not a literal instance (the region of [7, §5] is a bounded subset of the lattice with a boundary condition, and reflection positivity of periodic conditions is asserted there in a remark), and Proposition 12.2 does not rest on it. (iii) The chessboard estimate uses Proposition 12.2 for functions of the r_x only; nothing in this section uses reflection positivity of a cube or of a limit state.

12.3. **The chessboard estimate on King's tori.**

Lemma 12.4 (chessboard estimate)**.** *Let T be a King torus at spacing $\varepsilon = \varepsilon_k$, a a scheme, and q a real function on the sites of T. In the free convention,*

$$\ln N^a_{k,T}(0, q) \;\le\; \sum_x \varepsilon^3\, \alpha^a_{k,T}(q(x)), \tag{27}$$

and $u \mapsto \alpha^a_{k,T}(u)$ is convex on $\mathbb{R}$ with $\alpha^a_{k,T}(0) = 0$. In King's torus convention $u \mapsto |T|^{-1}\ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}_T)$ is likewise convex with value 0 at $u = 0$.

Proof. We apply the chessboard estimate [9, Thm. 4.2] (lattice-gauge form: [7, Thm. 2.10]), whose hypotheses we check: the site set of T is a rectangular lattice box with sides of even length $\ell 5^k$; the periodic state is invariant under translations modulo the sides and reflection positive with respect to the reflections in all planes perpendicular to the coordinate axes running midway between neighbouring sites (Proposition 12.2; on the torus such a reflection is the reflection in a mid-plane pair, the antipodal plane $c + \ell/2$ being again a mid-plane). Take as the "spin at x"

the gauge-invariant variable r_x and

$$G_x := \exp\big(\varepsilon^3 q(x)(r_x^2 - 2\mathsf{W}_\varepsilon - a_k)\big), \qquad \text{so that} \qquad \Big\langle \prod_x G_x \Big\rangle_{k,T} = N^a_{k,T}(0,q).$$

The G_x are positive with all moments ($\lambda > 0$), and the Schwarz inequality for such unbounded gauge-invariant functions, all that [9, Thm. 4.1] uses beyond cyclic invariance, follows from the bounded case by truncation and monotone convergence. The conclusion is $|\langle\prod_x G_x\rangle| \le \prod_x \langle\prod_y G_x^{(y)}\rangle^{1/|\Lambda|}$, $G_x^{(y)}$ the copy of G_x carried to y by reflections, $|\Lambda| = |T|/\varepsilon^3$ the number of sites. Because W_ε, a_k and δm_ε^2 are the same at every site, $\prod_y G_x^{(y)} = \exp(q(x)\sum_y \varepsilon^3(r_y^2 - 2\mathsf{W}_\varepsilon - a_k))$ has expectation $N^a_{k,T}(0, q(x)\mathbf{1}_T) = e^{|T|\alpha^a_{k,T}(q(x))}$ in the same state; taking logarithms gives (27). Convexity: $|T|\alpha^a_{k,T}(u)$ is the logarithm of a Laplace transform in the linear parameter u plus a linear term, and vanishes at $u = 0$; the same computation gives the convexity statement in the torus convention. □

12.4. **Constant sources on large tori: the extensive bound.**

Definition 12.5 (the extensive bound (I2))**.** We say that (I2) (one-sided, in the volume limit at fixed spacing; also written (UVS$^+_\infty$)) holds for the scheme a in the free convention if there are $s_1 > 0$ and $C_U \in [0,\infty)$ such that

$$\sup_{k\ge 0} \limsup_{J\to\infty} \alpha^a_{k,T_J}(s_1) \;\le\; C_U. \tag{28}$$

The positive leg of the next section consumes exactly this, with $\kappa := C_U/s_1$ (by convexity and $\alpha(0) = 0$, $\limsup_J \alpha^a_{k,T_J}(u) \le \kappa u$ for $0 \le u \le s_1$). Throughout, $M_1 \in (0, m)$ is an auxiliary scalar mass, $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$ is King's free-boundary bare coefficient on Λ_k at scalar mass M (Lemma 3.10; it is the fixed-k volume limit of the torus coefficient $M^2 + \delta m^2_{k,T_J}(M)$, Lemma 12.9), and, in the reference scheme z,

$$s_k^\infty(M_1) := \tfrac12\big(\alpha_k^\infty(m) - \alpha_k^\infty(M_1)\big), \qquad \sigma_1 = \sigma_1(M_1) := \liminf_{k\to\infty} s_k^\infty(M_1),$$

$$B^z_{k,T}(M) := E(M) - E(m) - \tfrac12\big(2c_{k,T} + z_k\big)\big(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M)\big)|T|,$$

$$b_k^\infty(M_1) := \lim_{J\to\infty} |T_J|^{-1} B^z_{k,T_J}(M_1), \qquad \beta_1 = \beta_1(M_1) := \limsup_{k\to\infty} b_k^\infty(M_1),$$

where $\alpha^{\mathrm{tor}}_{k,T}(M) := M^2 + \delta m^2_{k,T}(M)$ is the bare coefficient of King's torus convention and the limit defining b_k^∞ exists by Lemma 12.9. The constant $\beta_1(M_1)$ is the second constant of Lemma 3.10, which says that $\sigma_1(M_1)$ and $\beta_1(M_1)$ are finite; (T3) says that $\sigma_1(M_1) > 0$.

Proposition 12.6 ((I2) from the torus inputs)**.** *Assume* (T3) *(Hypothesis 3.12); the torus inputs are Proposition 3.9 at the two masses m and M_1 and Lemma 3.10. Then* (I2) *holds in the reference scheme z for every $s_1 \in (0, \sigma_1)$, with a constant $C_U = C_U(s_1) < \infty$ depending on $(e, \lambda, m, \mu_0, M)$, on M_1 and on s_1. For a scheme $a \ne z$ with $\sup_k |z_k - a_k| < \infty$ (an admissible scheme, Definition 3.7) it holds with the constant $C_U + s_1 \sup_k |z_k - a_k|$.*

The proof assembles four lemmas on King's torus model at fixed spacing, stated in the form used and with their short proofs where elementary. Notation at fixed k (spacing ε) on $T = T_J$: $f^{\mathrm{tor}}(u) := \ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1})$ (King's torus model at mass m, Wick constant $c_{k,T}$, scheme z) and $f^F(u) := \ln N^z_{k,T}(0, u\mathbf{1})$ (free convention); both are convex with $f(0) = 0$ (Lemma 12.4). Write $\alpha^{\mathrm{tor}} := m^2 + \delta m^2_{k,T}(m)$, $\alpha^F := m^2 + \delta m^2_\varepsilon$, $\sigma_{k,T} := \frac12(\alpha^{\mathrm{tor}} - \alpha^F)$ and $\Delta c_T := \mathsf{W}_\varepsilon - c_{k,T}$ (the two *convention offsets*), and $s_{k,T}(M_1) := \frac12(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M_1))$.

Lemma 12.7 (mass shift)**.** *If $M'' > 0$ satisfies $\alpha^{\mathrm{tor}}_{k,T}(M'') = \alpha^{\mathrm{tor}}_{k,T}(m) - 2u$, then*

$$N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}) = e^B\, Z^{(M'')}_{k,T} / Z^{(m)}_{k,T}, \qquad B = E(M'') - E(m) - (2c_{k,T} + z_k)\, u\, |T|.$$

Proof. The source $u\sum_x \varepsilon^3 r_x^2$ shifts the bare coefficient of the torus model at mass m by $-2u$, that is to the bare coefficient of the model at mass M''; beyond the fixed terms, King's torus action [15, (2.4)] contains only the counterterm and the field-independent constants E_0, E_1, so the two Boltzmann factors differ by the constant $e^{E(M'')-E(m)}$, and the Wick and scheme multipliers $e^{-(2c_{k,T}+z_k)u|T|}$ of $N^{z,\mathrm{tor}}$ are bookkeeping. □

Lemma 12.8 (chord)**.** *If f is convex on $\mathbb{R}$ with $f(0)=0$, then $f(u)\le (u/s)f(s)$ for $0\le u\le s$, $s>0$.*

Proof. $u=(1-u/s)\cdot 0+(u/s)\cdot s$ and $f(0)=0$. □

Lemma 12.9 (the volume limit at fixed spacing)**.** *Fix k. As $J\to\infty$: $\alpha^{\mathrm{tor}}_{k,T_J}(M)\to\alpha^{\infty}_k(M)$ uniformly for $M\in[M_1,m]$; $s_{k,T_J}(M_1)\to s^{\infty}_k(M_1)$; $|T_J|^{-1}B^z_{k,T_J}(M_1)\to b^{\infty}_k(M_1)$, a finite limit; and the offsets vanish, $\sigma_{k,T_J}\to 0$ and $\Delta c_{T_J}\to 0$. Moreover (the* boundedness clause*) $\sup_{J\ge J_*(k)}\sup_{M\in[M_1,m]}\big(|\delta m^2_{k,T_J}(M)|+|c_{k,T_J}(M)|\big)<\infty$; in particular $\delta m^2_{k,T_J}(m)$ and c_{k,T_J} are bounded in $J\ge J_*(k)$.*

The complete proof, including the boundedness clause, is given in the supplemental material. No rate and no uniformity in the source are claimed or used. Of the clauses listed, Step 2 of the proof of Proposition 12.6 uses the convergence of $s_{k,T_J}(M_1)$ and the $\limsup_J$ of $|T_J|^{-1}B^z_{k,T_J}(M_1)$, and Step 3 the two offset clauses.

Lemma 12.10 (the fixed-spacing extensive bound)**.** *Fix k and $v>0$. There is $C_k(v)<\infty$ such that for every King torus $T=T_J$ with $J\ge J_*(k)$, in either convention, $f(\pm v)\le C_k(v)|T|$; hence, by convexity, $|f(w)|\le C_k(v)\,|w|\,|T|/v$ for $|w|\le v$.*

Proof. Let $I(b):=\int_{\mathbb{C}} e^{-\varepsilon^3[\frac12 b|\phi|^2+\lambda|\phi|^4]}d^2\phi<\infty$ for real b ($\lambda>0$), α the bare coefficient and c the Wick constant of the convention in use, and Z_γ the photon Gaussian integral, finite because $\mu_0>0$ (at $\mu_0=0$ the constant mode is in the kernel of the Feynman form). The hopping term is ≥ 0, so dropping it bounds the numerator: $\int e^{-S+w\sum\varepsilon^3 r^2}\le Z_\gamma I(\alpha-2w)^{|T|/\varepsilon^3}$. The hopping term of a bond is $\le\varepsilon(r_x^2+r_y^2)$, and each site lies on six bonds, so the hopping part of the action is $\le 6\varepsilon^{-2}\sum_x\varepsilon^3 r_x^2=\sum_x\varepsilon^3\cdot\frac12(12\varepsilon^{-2})r_x^2$ and $\int e^{-S}\ge Z_\gamma I(\alpha+12\varepsilon^{-2})^{|T|/\varepsilon^3}$. Hence $\ln\langle e^{w\sum\varepsilon^3 r^2}\rangle\le\varepsilon^{-3}|T|[\ln I(\alpha-2w)-\ln I(\alpha+12\varepsilon^{-2})]$ and

$$f(w)\le\big\{\varepsilon^{-3}[\ln I(\alpha-2w)-\ln I(\alpha+12\varepsilon^{-2})]+(2|c|+|z_k|)|w|\big\}|T|.$$

In the free convention α and c do not depend on T; in the torus convention they converge as $J\to\infty$ and are bounded for $J\ge J_*(k)$ (the boundedness clause of Lemma 12.9). So $C_k(v)$, the supremum over the sign and over $J\ge J_*(k)$ of the brace at $w=\pm v$, is finite. The last claim is convexity with $f(0)=0$: $f(w)\le(|w|/v)f(\pm v)$ and $f(w)\ge -f(-w)\ge -(|w|/v)f(\mp v)$. □

Proof of Proposition 12.6. Step 1: the exact convention bridge. On the same torus the free-convention action differs from the torus-convention action by $\frac12(\alpha^F-\alpha^{\mathrm{tor}})\sum_x\varepsilon^3 r_x^2=-\sigma_{k,T}\sum_x\varepsilon^3 r_x^2$ plus field-independent constants. Hence $\langle e^{u\sum\varepsilon^3 r^2}\rangle^F=\langle e^{(u+\sigma)\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}}/\langle e^{\sigma\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}}$ with $\sigma=\sigma_{k,T}$, and inserting the Wick and scheme multipliers,

$$f^F(u)=f^{\mathrm{tor}}(u+\sigma)-f^{\mathrm{tor}}(\sigma)-2\,\Delta c_T\,u\,|T|. \tag{29}$$

Step 2: the bound in the torus convention. Let $0\le u<\sigma_1$ and $\eta>0$. By the definitions of σ_1 and β_1 there is $K_1=K_1(u,\eta)$, which we take at least as large as the tail of Proposition 3.9, such that $s^{\infty}_k(M_1)>u$ and $b^{\infty}_k(M_1)<\beta_1+\eta$ for all $k\ge K_1$. Fix such a k. By Lemma 12.9 there is $J_*\ge J_*(k)$ with $s:=s_{k,T_J}(M_1)>u$ for $J\ge J_*$. The chord (Lemma 12.8) at this s gives $f^{\mathrm{tor}}(u)\le(u/s)f^{\mathrm{tor}}(s)\le[f^{\mathrm{tor}}(s)]_+$. Lemma 12.7 with $M''=M_1$ applies exactly, since $\alpha^{\mathrm{tor}}_{k,T_J}(M_1)=\alpha^{\mathrm{tor}}_{k,T_J}(m)-2s$ by the definition of s:

$$f^{\mathrm{tor}}(s)=B^z_{k,T_J}(M_1)+\ln Z^{(M_1)}_{k,T_J}-\ln Z^{(m)}_{k,T_J}\le B^z_{k,T_J}(M_1)+\big(E_+(M_1)+E_-(m)\big)|T_J|$$

by Proposition 3.9, used as an upper bound at M_1 and as a lower bound at m. Divide by $|T_J|$ and let $J \to \infty$ (Lemma 12.9 for the first term):

$$\limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \le \big[\beta_1 + \eta + E_+(M_1) + E_-(m)\big]_+ \qquad (k \ge K_1).$$

For each of the finitely many $k < K_1$, Lemma 12.10 with any $v \ge u$ gives $\limsup_J |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \le C_k(v)u/v < \infty$. Fix $\eta := 1$ (any $\eta > 0$ serves); then

$$C^{\mathrm{tor}}(u) := \sup_{k\ge 0} \limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) < \infty \qquad \text{for every } u \in [0,\sigma_1).$$

Step 3: transfer to the free convention. Fix $u \in [0,\sigma_1)$ and $p > 1$ with $pu < \sigma_1$, and let $p' = p/(p-1)$. Since $u+\sigma = \frac{1}{p}(pu) + \frac{1}{p'}(p'\sigma)$, convexity gives $f^{\mathrm{tor}}(u+\sigma) \le \frac{1}{p} f^{\mathrm{tor}}(pu) + \frac{1}{p'} f^{\mathrm{tor}}(p'\sigma)$, whatever the sign of $\sigma = \sigma_{k,T_J}$. Once $p'|\sigma_{k,T_J}| \le v$, which holds for all large J by Lemma 12.9, Lemma 12.10 gives $|f^{\mathrm{tor}}(p'\sigma)|, |f^{\mathrm{tor}}(\sigma)| \le C_k(v) p' |\sigma_{k,T_J}| |T_J|/v$. Inserting this into (29),

$$|T_J|^{-1} f^F_{k,T_J}(u) \;\le\; \tfrac{1}{p}|T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(pu) + \tfrac{2C_k(v)p'}{v}|\sigma_{k,T_J}| + 2|\Delta c_{T_J}|\, u \;\xrightarrow[J\to\infty]{\limsup}\; \tfrac{1}{p} C^{\mathrm{tor}}(pu),$$

because $\sigma_{k,T_J} \to 0$ and $\Delta c_{T_J} \to 0$ at fixed k (Lemma 12.9; the factor $\frac{1}{p'} \le 1$ in the middle term has been dropped). Hence $\sup_k \limsup_J \alpha^z_{k,T_J}(u) \le \frac{1}{p} C^{\mathrm{tor}}(pu) < \infty$ for every $u \in [0,\sigma_1)$. Take $u = s_1 \in (0,\sigma_1)$ and $C_U := [\inf_{p>1:\, ps_1<\sigma_1} \frac{1}{p} C^{\mathrm{tor}}(ps_1)]_+$; this is (28) in the scheme z. For a scheme $a \ne z$, $N^a_{k,T}(0,q) = e^{-(a_k - z_k)\mathsf{S}(q)} N^z_{k,T}(0,q)$ gives $\alpha^a_{k,T}(u) = \alpha^z_{k,T}(u) + (z_k - a_k)u \le \alpha^z_{k,T}(u) + |z_k - a_k|\, u$, so (28) holds with $C_U + s_1 \sup_k |z_k - a_k|$. □

Proposition 12.11 (the auxiliary mass on either side of m)**.** *Let $M_1 > 0$, $M_1 \ne m$. Assume Proposition 3.9 at m and at M_1, and Lemma 3.10 and* (T3) *read at this M_1: $s^\infty_k(M_1) = \frac{1}{2}(\alpha^\infty_k(m) - \alpha^\infty_k(M_1))$, of either sign, has $\sigma_1(M_1) = \liminf_k s^\infty_k(M_1) > 0$, and $\beta_1(M_1) < \infty$. Then* (I2) *holds in the scheme z for every $s_1 \in (0, \sigma_1(M_1))$, and for $a \ne z$ with the shifted constant, exactly as in Proposition 12.6.*

Proof. The proof of Proposition 12.6 is repeated with Lemma 12.9 stated on the interval $I := [\min(M_1, m), \max(M_1, m)]$, and we check that no step uses $M_1 < m$. Lemma 12.7 is an identity for any M'' with $\alpha^{\mathrm{tor}}(M'') = \alpha^{\mathrm{tor}}(m) - 2u$, whatever the order of M'' and m. Lemma 12.8 needs $0 \le u \le s$ only. Lemma 12.9: its proof uses only that the propagators are translation-invariant massive lattice covariances at fixed ε with masses in $I \cup \{\mu_0\}$, bounded below by $\min(M_1, m, \mu_0) > 0$, so it holds on I word for word; Step 2 uses it at the two masses m and M_1 only. Lemma 12.10 carries no mass order, and (29) is an identity. In Step 2 the sign clause is $s^\infty_k(M_1) > u \ge 0$, the chord is applied at $s = s_{k,T_J}(M_1) > u \ge 0$, Lemma 12.7 at $M'' = M_1$ is exact by the definition of s, and Proposition 3.9 is used as an upper bound at M_1 and a lower bound at m; none of this uses $M_1 < m$. Step 3 carries no mass. □

The two-sided variant matters exactly on the set $E^{\le}_W \setminus E_W$ of couplings at which m minimises α_∞ on $(0, m]$ but not on $(0, \infty)$ (Lemma 22.1; for instance when $\alpha'_\infty(m) < 0$ and m minimises α_∞ on $(0, m]$): there (T3) fails for every $M_1 < m$ while some $M_1 > m$ has $\sigma_1(M_1) > 0$. The main results use Proposition 12.6 with $M_1 \in (0, m)$, as (T3) is posed.

Remark 12.12 (the convention of Proposition 3.9 and how much of it is used). (i) Proposition 3.9 is posed in King's torus convention (counterterm and Wick constant computed with the periodic propagators of the torus itself), not in the normalisation of Bałaban's own papers. This is what the proof needs: the offsets σ_{k,T_J}, Δc_{T_J} between King's torus convention and the free convention vanish in the volume limit at fixed spacing (Lemma 12.9), and Step 3 of the proof of Proposition 12.6 transfers the bound through them; between other normalisations they need not vanish. (ii) Only a tail of Proposition 3.9 is used: Step 2 of that proof invokes it for $k \ge K_1$ and $J \ge J_*(k)$, the finitely many $k < K_1$ being handled by Lemma 12.10 at fixed spacing, and the constants $E_\pm(M)$ are independent of k and of the torus. (iii) In the scheme z, C_U depends on the couplings, on M and on the auxiliary choices M_1, s_1, and on nothing else; neither it nor any other bound of this section is uniform in μ_0.

13. The Wick tube II: the uniform bound and its limits

This section proves the (k, J)-uniform bound (31) on the general-source functionals of the comparison cubes on the tube $\mathcal{D}_{\sigma_+,H}$ from the extensive bound (I2) (Definition 12.5): the positive part of the Wick source is controlled through King's tori, the negative part through a tiling into cells of fixed size, each controlled by one instance of Theorem 3.6 and the positive bound. Then come the limits on the tube, E0″, and the continuity and translation invariance of the tube object. States are those of the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) unless King's convention is named; $N^a_{k,X}(0,q)$ is the functional of Definition 2.3 on a finite $X \subset \Lambda_k$ or a King torus (Definition 12.1), the torus with fundamental cell C_J being written $\mathbb{T}_J$ here (T_J is the top-layer set of King's RG cube), and $\alpha^a_{k,\mathbb{T}}(u) = |\mathbb{T}|^{-1} \ln N^a_{k,\mathbb{T}}(0, u\mathbf{1})$. $s_1 \in (0, \sigma_1)$ and C_U are the amplitude and constant of (I2), $\kappa := C_U/s_1$; the sources $f_1, \dots, f_n$, $h_1, \dots, h_p$ are in C^∞_c with supports in $Q_R = [-R, R]^3$; $g(\zeta)$, $k(\zeta) = k_+ - k_-$ and the norms are as in Definition 2.5. The scheme is the reference scheme z, or an admissible scheme a (Definition 3.7, so that $\sup_k |z_k - a_k| < \infty$); then C_U stands for $C_U + s_1 \sup_k |z_k - a_k|$ throughout (Proposition 12.6).

13.1. The positive leg on cubes.

Proposition 13.1 (cube $\le$ torus at one bare mass)**.** *For every k, J and every lattice function $q \ge 0$ on X_J: $N^a_{k,C_J}(0,q) \le N^a_{k,\mathbb{T}_J}(0,q)$, both in the free convention.*

Proof. The periodic state on $\mathbb{T}_J$ has the sites of X_J with opposite faces identified; its action is the Neumann–Dirichlet action on C_J plus, for every wrap-around pair $\{x, y\}$, the photon gradient terms $\varepsilon(A_\alpha(y) - A_\alpha(x))^2$ and the hopping term $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\cdot)$, whose diagonal part is exactly the pair of Dirichlet terms the cube action carries at x, y for their outward bonds; so the cube state is the torus state with the wrap-around gradient terms and cosine parts deleted. For the class-$\mathcal{I}$ observable $e^{\sum \varepsilon^3 q r^2}$ ($\Theta_{F,g}$, $F = 0$, $g = \varepsilon^3 q \ge 0$) deleting a gradient term, a quadratic $|\ell(A)|^2$, does not increase the expectation (Lemma 2.7(a)), nor does deleting a cosine part (Lemma 2.7(b)); each intermediate measure is a BFS measure, and δm^2_ε, W_ε, a_k, $\mathsf{S}(q)$ agree on both sides. □

Proposition 13.2 (the positive leg in L^1 form, on every Neumann–Dirichlet box)**.** *Assume* (I2) *and* (I3) *(Proposition 12.2, entering through the chessboard estimate Lemma 12.4). Let $X \subset \Lambda_k$ be finite, $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ the Neumann–Dirichlet state of Definition 2.1 on X, and $0 \le q \le s_1$ a lattice function on X. Then*

$$\ln N^a_{k,X}(0,q) \;\le\; \kappa\, \|q\|_{1,k}, \qquad \kappa = C_U/s_1.$$

Proof. Choose J' with X and all its nearest neighbours in $C_{J'}$. (1) *Box $\le$ cube:* the first inequality of Lemma 2.10 with $Y = X_{J'}$ and $(f, g) = (0, q)$ (Lemma 2.7(a),(b)); its X-state is the state of Definition 2.1 on X, since all six bonds at each site of X are present in Y and no site is killed; the scheme multiplier is common (Lemma 2.11(a)). (2) *Cube $\le$ torus:* Proposition 13.1 on $C_{J'}$, q extended by 0. (3) *Chessboard:* Lemma 12.4 on $\mathbb{T}_{J'}$, $\ln N^a_{k,\mathbb{T}_{J'}}(0,q) \le \sum_x \varepsilon^3 \alpha^a_{k,\mathbb{T}_{J'}}(q(x))$. (4) *Chord:* $\alpha = \alpha^a_{k,\mathbb{T}_{J'}}$ is convex with $\alpha(0) = 0$, so $\alpha(u) \le (u/s_1)[\alpha(s_1)]_+$ for $0 \le u \le s_1$, and (3) is at most $s_1^{-1}[\alpha(s_1)]_+ \|q\|_{1,k}$ (the ℓ^1 norm of q is the same on X, $C_{J'}$, $\mathbb{T}_{J'}$). (5) The left side does not depend on J'; let $J' \to \infty$ at fixed k and use (I2): $\limsup_{J'} \alpha^a_{k,\mathbb{T}_{J'}}(s_1) \le C_U$ for every k. □

Corollary 13.3 (the positive leg on the cubes)**.** *Under the hypotheses of Proposition 13.2, for $0 \le q \le s_1$ on X_J: $\ln N^a_{k,C_J}(0,q) \le \kappa \|q\|_{1,k}$, uniformly in k and J. Consequently, for every $0 \le q \in C^\infty_c$ with $\sup q \le s_1$ and every J with $\operatorname{supp} q$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $\limsup_k Z^a_{k,J}(0,q) \le e^{\kappa \|q\|_{L^1}}$, uniformly in J. If moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2), then* (W2)*(b) holds for $2 \sup q < s_1$: for every $0 \le q \in C^\infty_c$ with $2 \sup q < s_1$ and every J as above, $\lim_k Z^a_{k,J}(0,q)$ exists and $\sup_J \lim_k Z^a_{k,J}(0,q) \le e^{\kappa \|q\|_{L^1}}$.*

Proof. Proposition 13.2 at $X = X_J$. The sampled source satisfies $0 \le q \le s_1$ (Lemma 2.6(E3)) and $\|q\|_{1,k} \to \|q\|_{L^1}$, which gives the lim sup bound. The source q is the family $h_1 = q$ at the

coefficient $w = 1$, whose positive scalar part is q; Definition 3.2 gives the limit when $2\|q\|_\infty < s_W = s_1$, and there the limit equals the lim sup. □

13.2. **Cells and the constants of the negative leg.**

Definition 13.4 (cells, tilings, cutoffs)**.** Let $L_c := \min\{L \in \mathbb{N} : ML \geq 5\}$, $\ell_c := 2ML_c \geq 10$; the *cell* $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube. For $s \in \{0,1\}^3$ let $W_\mu(0) := \ell_c\mathbb{Z}$, $W_\mu(1) := ML_c + \ell_c\mathbb{Z}$, and let the tiling $\mathcal{T}_s$ consist of the cells $\prod_\mu [w_\mu, w_\mu + \ell_c)$, $w_\mu \in W_\mu(s_\mu)$; every cell is $C_{L_c} + b$ with $b \in ML_c\mathbb{Z}^3 \subset \Lambda_k$. The good sets $G_s := \{x : \operatorname{dist}(x, W_\mu(s_\mu)) > 2\ \forall \mu\}$ are open and cover $\mathbb{R}^3$ (the 2-neighbourhoods of $W_\mu(0)$, $W_\mu(1)$ are disjoint as $ML_c \geq 5 > 4$). Fix a smooth partition of unity $0 \leq \chi_s \leq 1$, $\sum_s \chi_s = 1$, $\operatorname{supp}\chi_s \subset G_s$; so χ_s vanishes within distance 2 of every wall of $\mathcal{T}_s$. Fix $\chi_0 \in C_c^\infty$, $0 \leq \chi_0 \leq 1$, $\chi_0 = 1$ on $\{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) \geq 2\}$, $\operatorname{supp}\chi_0 \subset \{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) > 1\}$, and put $\chi_c := \tau_b\chi_0$ for the cell $c = C_{L_c} + b$. For sources supported in Q_R let $J_1(R) := \min\{J \in \mathbb{N} : MJ \geq R + \ell_c\}$; a cell meeting Q_R has $w_\mu \in [-R - \ell_c, R]$, hence lies in $[-R - \ell_c, R + \ell_c)^3 \subset C_J$ for $J \geq J_1(R)$, so every cell meeting Q_R is a full cell inside C_J and the cells clipped by the faces of C_J carry no source.

The constants of the Wick leg are

(30)

$$\sigma_+ := s_1, \qquad H' := \frac{s_1}{2}, \qquad H := \frac{H'}{8} = \frac{s_1}{16}, \qquad \kappa := \frac{C_U}{s_1}, \qquad B_{\text{cell}} := \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0),$$

$$C_{\text{cell}} := \tfrac{1}{2}B_{\text{cell}} + \kappa H'\ell_c^3, \qquad C_\Sigma := \frac{(2 + 6\ell_c^2)^2\pi^2}{\ell_c^3}, \qquad K_- := \frac{8\,[C_{\text{cell}}]_+\, C_\Sigma}{H'}.$$

B_{cell} is finite by Lemma 14.4, that is by Corollary 6.18(b): at the single cube C_{L_c} and the single real source $-2H'\chi_0$ the comparison-convention functional is bounded by the King-convention one (the $\mathcal{D}$ direction of Lemma 4.2), which converges by Theorem 3.6 ($\operatorname{supp}\chi_0$ is at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$ as Theorem 3.6 requires, and the theorem puts no amplitude restriction on a negative Wick source, a position-dependent increase of the scalar mass squared). This boundedness statement is (B_{cell}) (Lemma 14.4) and the only place where the existence part enters the bound (31); no fixed-cube limit of the comparison family is used.

13.3. **The negative leg on cubes.**

Lemma 13.5 (wall insertion)**.** *Let $X_1, \dots, X_N$ be the site sets $c \cap X_J$ of the cells c of one tiling $\mathcal{T}_s$ that meet C_J, and $q \geq 0$ a lattice function on X_J. Then $N^a_{k,C_J}(0, -q) \leq \prod_{i=1}^N N^a_{k,X_i}(0, -q|_{X_i})$, with N^a_{k,X_i} the functional of the Neumann–Dirichlet state of Definition 2.1 on X_i.*

Proof. $e^{-\sum \varepsilon^3 q r^2}$ is the class-$\mathcal{D}$ observable $\Theta^-_{F,g}$, $F = 0$, $g = \varepsilon^3 q \geq 0$. Delete, for every nearest-neighbour pair with ends in different cells, the three photon gradient terms and the cosine part of the hopping term, one at a time: by Lemma 2.7(d) (the inequalities [7, Thm. 6.2(3),(4)], proved in Section 2) neither deletion decreases $\langle\Theta^-\rangle$, and each intermediate measure is a BFS measure. The result is a product over the cells (Lemma 2.7(e)) whose factors are the Neumann–Dirichlet states on the X_i: the kept diagonal parts of the deleted bonds are their Dirichlet terms, the remaining gradient terms their Neumann terms, and δm_ε^2, W_ε, a_k are site-independent. The observable and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise. □

Lemma 13.6 (one cell)**.** *Assume* (I2) *and* (I3). *Let $c = C_{L_c} + b$ be a cell ($b \in ML_c\mathbb{Z}^3$) and $0 \leq \tilde{q} \leq H'$ a lattice function supported in $\{x : \operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus c) \geq 2\}$. Then $\ln N^a_{k,c}(0, -\tilde{q}) \leq C_{\text{cell}}$ for every k.*

Proof. $\chi_c = 1$ on $\operatorname{supp}\tilde{q}$, so $-\tilde{q} = k_1 - k_2$ with $k_1 := H'\chi_c - \tilde{q} \geq 0$, $k_2 := H'\chi_c$, and Lemma 2.11(f) at $f = 0$ gives $N^a_{k,c}(0, -\tilde{q}) \leq N^a_{k,c}(0, -2H'\chi_c)^{1/2} N^a_{k,c}(0, 2k_1)^{1/2}$. *First factor:* exact lattice translation covariance ($b \in \Lambda_k$; the change of variables $x \mapsto x - b$ transports the boundary conditions, and the action, δm_ε^2, W_ε, a_k are site-independent) gives $N^a_{k,c}(0, -2H'\chi_c) = N^a_{k,C_{L_c}}(0, -2H'\chi_0) \leq e^{B_{\text{cell}}}$. *Second factor:* Proposition 13.2 on the box $c \cap \Lambda_k$ with $q' := 2k_1$,

$0 \le q' \le 2H' = s_1$, gives at most $\exp(\kappa\|q'\|_{1,k}) \le \exp(2\kappa H'\ell_c^3)$, the cell having volume ℓ_c^3. Hence $\ln N^a_{k,c}(0,-\tilde q) \le \frac12 B_{\mathrm{cell}} + \kappa H'\ell_c^3$. □

Lemma 13.7 (chord on the cell)**.** *Under the hypotheses of Lemma 13.6:* $\ln N^a_{k,c}(0,-\tilde q) \le ([C_{\mathrm{cell}}]_+/H')\sup\tilde q$.

Proof. Let $t := \sup\tilde q/H' \in (0,1]$, $\hat q := \tilde q/t$ ($0 \le \hat q \le H'$, same support). $\varphi(u) := \ln N^a_{k,c}(0,-u\hat q)$ is convex (Lemma 2.11(d)), $\varphi(0)=0$, $\varphi(1) \le C_{\mathrm{cell}}$ (Lemma 13.6); so $\varphi(t) \le t\varphi(1) \le t[C_{\mathrm{cell}}]_+$. □

Lemma 13.8 (eight families and the sum over cells)**.** (a) *For a lattice function $q \ge 0$ on X_J:* $N^a_{k,C_J}(0,-q) \le \prod_{s\in\{0,1\}^3} N^a_{k,C_J}(0,-8\chi_s q)^{1/8}$. *If q is supported in Q_R and $J \ge J_1(R)$, then for each s every cell of $\mathcal{T}_s$ meeting* $\mathrm{supp}(\chi_s q)$ *is a full cell inside C_J, on which $\chi_s q$ is supported at ℓ^∞-distance > 2 from the complement of the cell.* (b) *For any tiling of $\mathbb{R}^3$ by cubes of side ℓ_c:* $\sum_c \sup_{x\in c}(1+|x|^2)^{-2} \le C_\Sigma$.

Proof. (a) Hölder's inequality with eight exponents 8 in the one state $\langle\cdot\rangle^{\mathrm{ND}}_{k,C_J}$ for $e^{-\sum \varepsilon^3 qr^2} = \prod_s e^{-\sum \varepsilon^3 \chi_s qr^2}$; the multipliers match since $\mathsf{S}(q) = \sum_s \mathsf{S}(\chi_s q)$ at lattice points. Support: every cell meeting Q_R is a full cell inside C_J for $J \ge J_1(R)$ (Definition 13.4), and a point of G_s in the cell $\prod_\mu[w_\mu, w_\mu+\ell_c)$ has $\mathrm{dist}(x_\mu, \{w_\mu, w_\mu+\ell_c\}) > 2$ for every μ. (b) For x, y in one cell $|y| \le |x| + \sqrt{3}\ell_c$, so $1+|y|^2 \le (2+6\ell_c^2)(1+|x|^2)$ and $\sup_{x\in c}(1+|x|^2)^{-2} \le (2+6\ell_c^2)^2\ell_c^{-3}\int_c (1+|y|^2)^{-2}dy$; sum over the disjoint cells, with $\int_{\mathbb{R}^3}(1+|y|^2)^{-2}dy = \pi^2$. □

Proposition 13.9 (the negative leg)**.** *Assume* (I2) *and* (I3)*. For every k, every $J \ge J_1(R)$ and every lattice function $q \ge 0$ supported in Q_R with* $\sup q \le H$*:*

$$\ln N^a_{k,C_J}(0,-q) \;\le\; K_-\,|q|_w.$$

Proof. By Lemma 13.8(a), then Lemma 13.5 for each tiling $\mathcal{T}_s$ (the clipped cells carry no source and contribute $\ln N^a_{k,X_i}(0,0) = 0$), then Lemma 13.7, applicable since $8\chi_s q \le 8\sup q \le H'$ and by the support clause of Lemma 13.8(a):

$$\ln N^a_{k,C_J}(0,-q) \le \frac18 \sum_s \sum_{c\in\mathcal{T}_s} \ln N^a_{k,c}(0,-8\chi_s q|_c) \le \frac18 \sum_s \sum_c \frac{[C_{\mathrm{cell}}]_+}{H'}\, 8 \sup_c q \le \frac{[C_{\mathrm{cell}}]_+}{H'}\,|q|_w \sum_s \sum_c \sup_{x\in c}(1+|x|^2)^{-2}$$

using $\sup_c q \le |q|_w \sup_{x\in c}(1+|x|^2)^{-2}$ (a sampled supremum is at most the continuum one, Lemma 2.6(E3)) and Lemma 13.8(b) for each of the eight tilings. □

The output norm is $|q|_w$: the chord bound on a cell sees only $\sup_c q$, so no L^1 information inside a cell survives, and the tube bound carries $K_-|k_-|_w$ where the torus paper [27] has an L^1 norm.

13.4. **The uniform bound on the tube and its limits.** For a family with supports in Q_R and $\zeta = (z,w) \in \mathbb{C}^{n+p}$ put $G^{(k,C_J)}(\zeta) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j w_j h_j)$, and let $\mathcal{D}_{\sigma_+,H}$ be the tube of Definition 2.5 with $\sigma_+ = s_1$, $H = s_1/16$. Two facts from several complex variables [11] are used: (T2) a holomorphic function on a connected open $D \subset \mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D \cap \Sigma$, Σ an affine totally real subspace of real dimension m (such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1}\times\mathbb{R}^{m_2}$), vanishes on D (all its derivatives vanish at such a point); (T3) a locally bounded family of holomorphic functions on D is normal, and if it converges on a set as in (T2) it converges locally uniformly on D to a holomorphic limit (Montel, Vitali).

Theorem 13.10 (the tube bound)**.** *Assume* (I2) *(which Proposition 12.6 derives from* (T3) *and the torus inputs Proposition 3.9 and Lemma 3.10), for the reference scheme z or for an admissible scheme a (Definition 3.7), with* (I3) = *Proposition 12.2 and* (I4) = *Lemma 2.7. The constants σ_+, H, κ, K_- of* (30) *depend on (e,λ,m,μ_0,M,a) and on the fixed auxiliary choices M_1, χ_0, $s_1 \in (0,\sigma_1)$, and on nothing else; no uniformity in μ_0 is claimed. For every family with supports in Q_R, every $k \ge 0$, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$:*

$$|G^{(k,C_J)}(\zeta)| \;\le\; G^{(k,C_J)}(\mathrm{Re}\,\zeta) \;\le\; \exp\Big(\|g(\zeta)\|^2_{2,k} + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\Big). \tag{31}$$

Consequently, if moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2; its form on King's convention, where Theorem 3.6 supplies the limits, is Theorem 21.3):

(i) (Limits on the whole tube.) $G_J := \lim_k G^{(k,C_J)}$ *exists locally uniformly on* $\mathcal{D}_{\sigma_+,H}$ *for every* $J \geq J_1(R)$, $G_\infty := \lim_J G_J$ *exists locally uniformly on* $\mathcal{D}_{\sigma_+,H}$ *(the limit along* $J \geq J_1(R+\frac{1}{2})$*, hence along any tail), both are holomorphic there, and*

$$|G_\infty(\zeta)| \leq \exp\left(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\right) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}).$$

(W2) *is not used; where the full object* Z^a_∞ *of the optional extension exists,* $G_\infty = Z^a_\infty$ *on* $\mathcal{D}_{\sigma_+,H}$.

(ii) (E0″.) *With* $\sigma_0 := H/4$, $K' := \kappa\pi^2 + K_-$, *the mixed Schwinger functions* $S_{n,p}(f;h) := \partial_z^{(n)}\partial_w^{(p)}G_\infty(0)$ *satisfy, for real test functions,*

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \leq e^{K'\sigma_0}(\sqrt{2}e)^n(n!)^{1/2}(e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w \leq \sigma_{n+p} \prod_i |f_i|_4 \prod_j |h_j|_4,$$

$\sigma_N := e^{K'\sigma_0}C_1^N N!$, $C_1 := 2\max(e\pi/2, e/\sigma_0)$*; for complex test functions the first bound carries the factor* $2^{n+p} \leq 2^N$, $N = n+p$, *absorbed in* C_1. *This is* E0″ *with* $s = 4$ *and every* $\beta > 1$*; the* $S_{n,p}$ *are tempered distributions.*

σ_1, C_U depend on $(e, \lambda, m, \mu_0, M_1)$ through the torus stability constants of Proposition 3.9 at m, M_1, the c-number β_1 of Lemma 3.10 and finitely many fixed-spacing constants; B_{cell} depends on everything, including μ_0, M, a. The curvature term of (31) rests on the constant 2 of Lemma 7.1, numerically nearly sharp but not proved to be (Remark 7.2).

Proof. The bound (31). The first inequality is Lemma 2.11(d). Write $g = g(\zeta)$, $k = k(\zeta)$ (real). *Step 1* (curvature off): Theorem 7.4(a) in the presence of the real scalar source k gives $N^a(g,k) \leq e^{\|g\|^2_{2,k}}N^a(0,k)$; the supports of the f_i are at distance > 1 from $\mathbb{R}^3 \setminus C_J$ since $MJ \geq R + \ell_c > R+1$. *Step 2*: $N^a(0, k_+ - k_-) \leq N^a(0,2k_+)^{1/2}N^a(0,-2k_-)^{1/2}$ (Lemma 2.11(f) at $f = 0$). *Step 3* (positive leg): $\zeta \in \mathcal{D}_{\sigma_+,H}$ and (E3) give $0 \leq 2k_+ \leq s_1$ on the lattice, so $\frac{1}{2}\ln N^a(0,2k_+) \leq \kappa\|k_+\|_{1,k}$ by Corollary 13.3. *Step 4* (negative leg): $0 \leq 2k_- \leq H$, supported in Q_R, $J \geq J_1(R)$, so $\frac{1}{2}\ln N^a(0,-2k_-) \leq \frac{1}{2}K_-|2k_-|_w = K_-|k_-|_w$ by Proposition 13.9.

(i), $k \to \infty$ *at fixed* $J \geq J_1(R)$. Each $G^{(k,C_J)}$ is entire on $\mathbb{C}^{n+p}$. On a compact $K \subset \mathcal{D}_{\sigma_+,H}$ the family $\{G^{(k,C_J)}\}_k$ is uniformly bounded by (31) and Lemma 2.6(E1)–(E3) ($\|g\|^2_{2,k} \leq C_0|g|_4^2$, $\|k_+\|_{1,k} \leq C_0'|k_+|_w$, $|k_\pm|_w \leq \sum_j |\operatorname{Re} w_j||h_j|_w$). At the real points of $\mathcal{D}_{\sigma_+,H}$, a non-empty open subset of $\mathbb{R}^{n+p}$, it converges by Definition 3.2 (the positive scalar part is below $\sigma_+/2 < s_W$). By (T3) with $\Sigma = \mathbb{R}^{n+p}$, $G^{(k,C_J)} \to G_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G_J is holomorphic, and (31) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one: $|G_J|$ is bounded on $\mathcal{D}_{\sigma_+,H}$ by the right side of the display in (i), uniformly in $J \geq J_1(R)$.

(i), $J \to \infty$ *on a sub-tube.* Split $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty \chi_j$, $h_j^+ := h_j + h_j^- \geq 0$, with $\chi_j \in C_c^\infty$, $0 \leq \chi_j \leq 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$. Put $F_{k,J}(z,w^+,w^-) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{(z,w^+,w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the split family's scalar source has $\|k_\pm^{\text{sp}}\|_\infty \leq \|k^{\text{sp}}\|_\infty < H/2 \leq \sigma_+/2$, so (31) holds for it uniformly in k and $J \geq J_1(R+\frac{1}{2})$, and the previous paragraph gives $F_{k,J} \to F_J$ locally uniformly on $\mathcal{D}''$ with the limit bound. On $\Omega''_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the sources lie in the cone $\mathcal{K}$ (imaginary curvature coefficients, nonnegative total scalar source), so $J \mapsto F_J$ is nondecreasing there (first inequality of Lemma 2.10 with Lemma 2.11(a) at fixed k, then $k \to \infty$) and bounded, hence convergent. Ω''_+ is a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is convex, so (T3) gives $F_J \to F_\infty$ locally uniformly on $\mathcal{D}''$. Restricting to $w_j^+ = w_j$, $w_j^- = -w_j$ gives $G_J \to G_\infty$ locally uniformly on $\mathcal{D}' := \{(z,w) : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$ (there $\|k_\pm\|_\infty \leq \|k\|_\infty < \sum_j \delta_j\|h_j\|_\infty < H/2$).

(i), the whole tube. $\{G_J\}_{J \geq J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free limit bound; every Montel subsequential limit is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and agrees with G_∞

on the open set $\mathcal{D}'$; $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G_J \to G_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, and the bound passes to G_∞. Where Z^a_∞ exists it is the same iterated limit of the same Z^a_J.

(ii). Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $|w_j| \le \rho_j$, $z \in \mathbb{C}^n$: $\|k\|_\infty \le \sum_j |\operatorname{Re} w_j| \|h_j\|_\infty \le \sum_j \rho_j |h_j|_w = \sigma_0 < H/2 \le \sigma_+/2$, so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G_∞ exists with its bound by (i); the exponent there is $\le \kappa\pi^2|k_+|_w + K_-|k_-|_w \le K'|k|_w \le K' \sum_j |\operatorname{Re} w_j||h_j|_w \le K'\sigma_0$ by (E2). Cauchy's estimate for the mixed first w-derivatives at $w = 0$, z fixed: $|\partial_w^{(p)} G_\infty(z,0)| \le \prod_j \rho_j^{-1} \sup|G_\infty| \le (p/\sigma_0)^p \prod_j |h_j|_w \, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \le e^p p!$. The entire function $z \mapsto \partial_w^{(p)} G_\infty(z,0)$ is thus bounded by $Ae^{\|g(z)\|_2^2}$, and Cauchy's estimate in the z's with radius $(2n)^{-1/2}$, as in the proof of Theorem 7.4(c), gives $(\sqrt{2}e)^n (n!)^{1/2}$. Then (E2) and $(n!)^{1/2} p! \le (n+p)!$ give the second bound; the extension to $\mathcal{S}$, to complex test functions (the factor $2^{n+p} \le 2^N$) and to tempered distributions is as in Theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$. □

Lemma 13.11 (the tube object: limit, continuity, translations)**.** *Assume the bound* (31)*, which Theorem 13.10 derives from* (T3) *and the torus inputs (through* (I2)*), and, as in Theorem 13.10(i), that the comparison family is convergent at fixed cubes on the tube (Definition 3.2). Write $\mathcal{D} = \mathcal{D}_{\sigma_+,H}$ for the tube of a family with supports in Q_R.*

(iii) (The J-limit on the tube.) *For $J \ge J_1(R+\frac{1}{2})$, which exceeds the least J_0 with the supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ (as $MJ_1(R') - R' \ge \ell_c \ge 10 > 1$), $G_\infty = \lim_J G_J$ exists locally uniformly on $\mathcal{D}$, is holomorphic on $\mathcal{D}$, satisfies the bound of Theorem 13.10*(i)*, is > 0 at every real point of $\mathcal{D}$, and $G_J \le G_\infty$ at the cone points of $\mathcal{D}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \ge 0$: the real scalar source $k(\zeta) \ge 0$ sits in the slot of the cone).*

(iv) (Continuity on the real tube.) *Let $E_\rho := C_c^\infty(\bar{B}_\rho; \Lambda^2) \times C_c^\infty(\bar{B}_\rho)$ with its C^∞ topology, a Fréchet space, and $U_\rho := \{(f,h) \in E_\rho : \|h_+\|_\infty < \sigma_+/2,\ \|h_-\|_\infty < H/2\}$, open and convex ($h \mapsto \|h_\pm\|_\infty$ is continuous and convex). Then $\Phi^a(f,h) := \ln G_\infty(f,h)$ is finite, convex and continuous on U_ρ.*

(v) (Translations on the tube.) *For every family and every $b \in \mathbb{R}^3$: $G_\infty(\tau_b \cdot \zeta) = G_\infty(\zeta)$ for all $\zeta \in \mathcal{D}$; the tube of the translated family is that of the family, since $k(\tau_b \cdot \zeta) = \tau_b k(\zeta)$ has the same supremum norms.*

Besides Definition 3.2 and (31) *the proofs use only the first inequality of Lemma 2.10, Lemmas 2.11 and 2.14, lattice covariance and (T2), (T3): not* (W2)*, and Lemma 2.7(d) only inside* (31)*.*

Proof. (iii) Existence, holomorphy and the bound are the last three paragraphs of the proof of Theorem 13.10(i), which use (31), Definition 3.2 for the k-limits and the first inequality of Lemma 2.10. *Positivity at real points.* For real $\zeta \in \mathcal{D}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}$ ($\mathcal{D}$ is an open neighbourhood of 0). $\varphi_J(t) := \ln G_J(t\zeta)$ is convex on $[-\epsilon, 1]$ (Lemma 2.11(d) in the limit) with $\varphi_J(0) = 0$, so $0 \le \frac{\epsilon}{1+\epsilon}\varphi_J(1) + \frac{1}{1+\epsilon}\varphi_J(-\epsilon)$, i.e. $\varphi_J(1) \ge -\varphi_J(-\epsilon)/\epsilon \ge -C_\epsilon/\epsilon$, C_ϵ the limit bound of (31) at $-\epsilon\zeta$; hence $G_J(\zeta) \ge e^{-C_\epsilon/\epsilon}$ uniformly in J and $G_\infty(\zeta) > 0$. *Cone points.* There $G_J \le G_{J+1}$ (first inequality of Lemma 2.10 with Lemma 2.11(a) at fixed k, then $k \to \infty$; it needs $J \ge J_0$, which the range of (iii) ensures), so $G_J \le G_\infty$.

(iv) *Finite:* (iii) for the two-source family $(f; h)$ at $\zeta = (1,1)$, in $\mathcal{D}$ by the definition of U_ρ; the value is the pointwise iterated limit and does not depend on the family used. *Convex:* for $s, s' \in U_\rho$, $t \in [0,1]$, use the family $(f, f'; h, h')$ at the real point $(t, 1-t, t, 1-t)$, in $\mathcal{D}$ because $(th + (1-t)h')_\pm \le th_\pm + (1-t)h'_\pm$ pointwise; the convexity of $\ln Z^a_{k,J}$ in the real coefficients (Lemma 2.11(d)) passes to the limits. *Borel:* a pointwise limit of functions each depending continuously on finitely many values of the sources. *Continuous:* Lemma 2.14 on the open convex U_ρ.

(v) *Lattice translations at cone points.* Let $b \in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty \le M$, and ζ a cone point of $\mathcal{D}$ ($k_- = 0$, $q := k(\zeta) \ge 0$, $\|q\|_\infty < \sigma_+/2$). Steps 1–3 of the proof of Proposition 8.1 run unchanged with the real scalar source q in the slot of the cone: lattice covariance (τ_b preserves Λ_k and S_{Λ_k} for $k \ge k_0$) gives $Z^a_{k,C_J}(\tau_b \cdot \zeta) = Z^a_{k,C_J - b}(\zeta)$, the interlacing $X_{J-1} \subset (C_J - b) \cap \Lambda_k \subset X_{J+1}$ and the

first inequality of Lemma 2.10 twice (no killed site) give $Z^a_{k,C_{J-1}}(\zeta) \le Z^a_{k,C_J}(\tau_b\cdot\zeta) \le Z^a_{k,C_{J+1}}(\zeta)$ for $J \ge J_0+1$. Let $k\to\infty$ by Theorem 13.10(i) for the three families (the translated family is supported at distance >1 from $\mathbb{R}^3\setminus C_{J_0+1}$), then $J\to\infty$ by (iii), the three terms being cone points of their tubes: $G_\infty(\tau_b\cdot\zeta) = G_\infty(\zeta)$ at the cone points of $\mathcal{D}$. *All* $\zeta\in\mathcal{D}$. For the split family of the proof of Theorem 13.10(i) and its translate, $F^{\tau_b}_\infty - F_\infty$ is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω''_+, whose points are cone points of the split family's tube and which is an open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n\times\mathbb{R}^{2p}$; by (T2) it vanishes on $\mathcal{D}''$, and the diagonal gives $G^{\tau_b}_\infty = G_\infty$ on $\mathcal{D}'$. Then $G^{\tau_b}_\infty - G_\infty$ is holomorphic on the connected $\mathcal{D}$ ((iii) for both families) and vanishes on the open set $\mathcal{D}'$, hence on $\mathcal{D}$. *All* $b\in\bigcup_k 5^{-k}\mathbb{Z}^3$*:* iterate; the tube is the same at every step. *All* $b\in\mathbb{R}^3$ *at real points.* For real $\zeta\in\mathcal{D}$ the source pair $s := (\sum_i z_i f_i, \sum_j w_j h_j)$ lies in some U_ρ, $\tau_b s\in U_{\rho'}$ for $|b|\le\rho'-\rho$, and $b\mapsto\tau_b s$ is continuous into $E_{\rho'}$; by (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$, $\Phi^a(\tau_b s) = \Phi^a(s)$. *Complex* $\zeta\in\mathcal{D}$. $G^{\tau_b}_\infty - G_\infty$ is holomorphic on $\mathcal{D}$ and vanishes on the real points of $\mathcal{D}$, a non-empty open subset of $\mathbb{R}^{n+p}$; (T2) gives the claim. □

Remark 13.12 (King's convention). On King's RG cube $\mathrm{ND}^K_{k,J}$ the negative leg holds as printed: by Lemma 4.7, the wall insertion of Lemma 13.5 runs on $\mathrm{ND}^K_{k,J}$ with factors that are comparison box states killed on $X_i\cap T_J$, and every cell on which a cutoff source $8\chi_s q$ does not vanish is a full cell inside C_J meeting no killed site, because χ_s vanishes within distance 2 of every cell wall; so Lemmas 13.6 to 13.8 and Proposition 13.9 apply with the printed K_-. The direction matters: killing sites of a cell increases a class-$\mathcal{D}$ functional (Lemma 4.7(e)), so B_{cell} does not control a source-carrying cell with killed sites. For the positive leg, Remark 4.8 prepends the interlacing inequality $\langle\cdot\rangle^K_{k,J} \le \langle\cdot\rangle^{(a)}_{k,J}$ on class $\mathcal{I}$ (Step 1 of Lemma 4.2) to Propositions 13.1 and 13.2, so Corollary 13.3 holds for $N^K_{k,C_J}(0,q)$ at every (k,J) and, for the fixed-cube limits, in the form $\limsup_k Z^{a,K}_{k,J}(0,q) \le e^{\kappa\|q\|_{L^1}}$. The assembly of (31) on ND^K from these pieces, with the check that every step of the proof of Theorem 13.10 runs in the right direction on King's convention, and the existence of the fixed-cube limits of $Z^{a,K}_{k,J}$ on the tube (Theorem 13.10(i) on ND^K) are carried out in Theorem 21.3; the limit statements of this section and of the Wick sector of Sections 8, 9 and 12 then hold on King's convention and the Wick sector of the main theorem is stated for King's printed state (Proposition 21.4).

14. Continuum subsequences and LIM states

The axioms of the preceding sections rest on the existence of the fixed-cube limits $\lim_k$: on Theorem 3.3 in the curvature sector, on Theorem 3.6 on the Wick tube, both proved in the existence part on King's convention (for the comparison family, on which the tube statements are written, the corresponding clauses are the template hypotheses Definitions 3.1 and 3.2). This section shows that the axioms hold without that input, for the limit state along any continuum subsequence of spacings, and identifies every such state with the iterated limit. The k-uniform bounds of Lemma 7.3 and (31) make the fixed-cube functionals normal families, so one subsequence of spacings serves every cube and every family at once; along it the cube limit $J\to\infty$ exists because the cone functionals are monotone in the cube at fixed k (Lemma 2.10). The limit state of such a subsequence (a *LIM state*) satisfies every axiom proved above, with no hypothesis in the curvature sector and, on the Wick tube, under (T3) for every admissible scheme (through two properties of the scheme, the cell bound and the scheme bound, proved below); Theorem 3.3 (resp. Theorem 3.6) adds exactly the identification with the iterated limit and the independence of the subsequence. The section is written on the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1), in which the tube lemmas and the rotation part are written; King's convention is Corollary 14.14. Notation is that of Definitions 2.3, 2.5 and 13.4 and (30): $a^{(a)}_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ for a family $f=(f_1,\dots,f_N)$ of real 2-forms; $Z^a_{k,J}(\zeta)$ for $\mathcal{F}=(f_1,\dots,f_n;h_1,\dots,h_p)$ and $\zeta=(z,w)\in\mathbb{C}^{n+p}$ in the scheme a; $g(\zeta)$, $k(\zeta)=k_+-k_-$, $\mathsf{a}(\zeta)$; $U_J := \{x : \mathrm{dist}_\infty(x,\mathbb{R}^3\setminus C_J)>1\}$, $J_0(f)$ the least J with all supports in U_J, and $R(\mathcal{F})$ the least R with all supports in $Q_R=[-R,R]^3$. Two standard facts [11] are used by name: the *identity theorem* (a holomorphic function on a connected

open $D \subset \mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D \cap \Sigma$, Σ a maximal totally real affine subspace such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1} \times \mathbb{R}^{m_2}$, vanishes on D) and *Montel–Vitali* (a locally bounded family of holomorphic functions on D is normal, and converges locally uniformly on D if it converges on such a set).

14.1. **Definitions, the cell bound and the scheme bound.**

Definition 14.1 (continuum subsequences)**.** An increasing sequence $\mathcal{K} = (k_m)_{m\geq 1} \subset \mathbb{N}$ of spacings is a *continuum subsequence* (of the comparison family) if for every finite family f of real 2-forms and every $J \geq J_0(f)$ the sequence $(a^{(a)}_{k_m,J})_m$ converges locally uniformly on $\mathbb{C}^N$; the limit is written $Z^{\mathcal{K}}_J(\sum_j z_j f_j, 0)$ and depends on the family only through the total complex source, as each $a^{(a)}_{k,J}$ does. Fix a scheme a. $\mathcal{K}$ is a *continuum subsequence for the Wick sector* (Form (B)) if it is a continuum subsequence and, for every finite family $\mathcal{F}$ of real 2-forms and scalars and every J with $MJ > R(\mathcal{F}) + \ell_c$, the sequence $(Z^a_{k_m,J})_m$ converges locally uniformly on the tube $\mathcal{D}_{\sigma_+,H}$ of $\mathcal{F}$; the limit is $Z^{a,\mathcal{K}}_J$. (The strict inequality replaces the threshold $J_1(R)$ of Theorem 13.10 by at most one cube step; only tails in J are used.) Form (A), convergence on all of $\mathbb{C}^{n+p}$ for the same J, is the optional extension of Theorem 23.7 and is not used here.

Definition 14.2 (LIM states)**.** The *LIM state* of a continuum subsequence $\mathcal{K}$ is $Z^{\mathcal{K}}_\infty := \lim_{J\to\infty} Z^{\mathcal{K}}_J$ on $\mathbb{C}^N$ (curvature sector), resp. $Z^{a,\mathcal{K}}_\infty := \lim_J Z^{a,\mathcal{K}}_J$ on the tube (Wick sector); Lemma 14.8 gives existence. A subsequence of a continuum subsequence is a continuum subsequence with the same fixed-cube limits, hence the same LIM state.

Definition 14.3 (the limit state of the rotation theorem)**.** For a family f and $J \geq J_0(f)$ let L^f_J be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a^{(a)}_{k,J})_k$. The rotation theorem (Theorem 3.14) gives an entire $Z^{\mathrm{LIM}}_\infty(\cdot, 0)$ to which *every* choice $\zeta_J \in L^f_J$ converges locally uniformly as $J \to \infty$, the subsequence being allowed to depend on J. With J-dependent subsequences one only has $\liminf_k a^{(a)}_{k,J}(\mathrm{i}u) \leq \zeta_J(\mathrm{i}u) \leq \limsup_k a^{(a)}_{k,J}(\mathrm{i}u)$, and the J-limit is pinned only if the J-limits of the two bounds agree, which is the content of that theorem. A LIM state in the sense of Definition 14.2 uses one $\mathcal{K}$ common to all J and all families, along which the fixed-cube limits are monotone in J (Lemma 14.8), and needs no hypothesis; Proposition 14.12(a) relates the two.

The curvature sector needs no hypothesis. On the Wick tube the bound (31) of Theorem 13.10 is used; it rests on (T3) and the torus inputs (Proposition 3.9, Lemma 3.10, and Hypothesis 3.12, which give (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U through Proposition 12.6; (I3) is Proposition 12.2 and (I4) is Lemma 2.7) and on one instance of Theorem 3.6, isolated as follows.

We say that the scheme a has the *cell bound* ($\mathrm{B}_{\mathrm{cell}}$) if $B_{\mathrm{cell}} := \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$, for the one cell C_{L_c} (side $\ell_c \geq 10$) and the one negative Wick source $-2H'\chi_0$ of Definition 13.4, and that it has the *scheme bound* (SCH) if either $a = z$, the reference scheme in which Lemma 3.10 is computed, or $\sup_k |z_k - a_k| < \infty$. The cell bound is the boundedness form of Theorem 3.6 at one cube and one source, and the only place where that theorem enters the proof of (31) (through Lemmas 13.6 and 13.7 and Proposition 13.9; K_- is built from B_{cell}). Both properties hold for the schemes of the main theorem:

Lemma 14.4 (($\mathrm{B}_{\mathrm{cell}}$), the cell bound)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme (Definition 3.7). Then a has the cell bound: $B_{\mathrm{cell}} = \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$.*

Proof. By Definition 13.4 the cell $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube and $\operatorname{supp}\chi_0$ lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$; the source $q := 2H'\chi_0$ is a non-negative function of $C^\infty_c(\mathbb{R}^3)$ with this support condition. Under (T3) the inputs (I2) and (I3) of the existence theorem with scalar sources hold (Proposition 12.6, Proposition 12.2), so Corollary 6.18(b) applies and gives $\sup_k N^z_{k,C_{L_c}}(0, -q) < \infty$ for the comparison-convention functional in the reference scheme z. For a scheme a with $\sup_k |a_k - z_k| < \infty$ the two functionals differ by the factor $e^{-(a_k - z_k)\mathsf{S}_k(-q)}$ (Remark 2.4), bounded uniformly in k since $\mathsf{S}_k(q) \to \int q$; this is Remark 6.19, and an admissible

scheme has $\sup_k |z_k - a_k| < \infty$ by Definition 3.7. Hence $\sup_k N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$, and $B_{\mathrm{cell}} < \infty$ because each term is positive (Remark 6.20). □

Lemma 14.5 ((SCH), the scheme bound)**.** *An admissible scheme has the scheme bound. If $a \neq z$ has the scheme bound, the constants of (31) hold with C_U replaced by $C_U + s_1 \sup_k |z_k - a_k|$.*

Proof. For an admissible scheme either $a = z$ or $\lim_k(z_k - a_k)$ exists, so $\sup_k |z_k - a_k| < \infty$ (Definition 3.7). The replacement of the constant is the last clause of Proposition 12.6, carried through Theorem 13.10. □

Proposition 14.6 ((UNIQ), independence of the subsequence)**.** *Any two continuum subsequences $\mathcal{K}, \mathcal{K}'$ have the same LIM state: $Z^{\mathcal{K}}_\infty(\cdot, 0) = Z^{\mathcal{K}'}_\infty(\cdot, 0)$ for every family.*

Proof. This is Proposition 14.12(a) below, from the rotation theorem Theorem 3.14: every continuum subsequence's LIM state is the limit state $Z^{\mathrm{LIM}}_\infty(\cdot, 0)$ of that theorem. It also follows from Theorem 3.3 through Corollary 4.4(iii) (Corollary 14.14(b)): every LIM state of the comparison family is King's printed state Z^K_∞. □

The proofs of the axioms below hold for each LIM state separately and do not use (UNIQ); it enters only in the Euclidean clause (vii) of Theorem 14.9, through Proposition 14.12(a). Without the rotation theorem and the existence theorem one would only know that every LIM state lies on $(\mathrm{i}\mathbb{R})^N$ between $\lim_J \liminf_k a^{(a)}_{k,J}$ and $\lim_J \limsup_k a^{(a)}_{k,J}$ (both exist, by monotonicity in J), with equality of the LIM states if these coincide at every real argument, an entire function being determined on $(\mathrm{i}\mathbb{R})^N$.

14.2. **Existence of continuum subsequences and the cube limit.**

Lemma 14.7 (continuum subsequences exist)**.** (a) (Curvature sector; no hypothesis.) *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence $\mathcal{K}$, family f and $J \geq J_0(f)$: $Z^{\mathcal{K}}_J(\cdot, 0)$ is entire, $|Z^{\mathcal{K}}_J(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$, $Z^{\mathcal{K}}_J(x, 0)$ is real and ≥ 1 at real x, and $0 \leq Z^{\mathcal{K}}_J(\mathrm{i}t, 0) \leq 1$ at imaginary $\mathrm{i}t$.* Several models at once: *for every countable set $\mathcal{M}$ of lattice models of King's type (Neumann–Dirichlet cube states with a site-independent real bare scalar coefficient, photon mass $\mu > 0$, $\lambda > 0$; for instance the models at the photon masses of a countable set, with the counterterm fixed) one $\mathcal{K}$ is a continuum subsequence for every member of $\mathcal{M}$.*

(b) (Wick sector, Form (B).) *Let the couplings satisfy* (T3) *and let the scheme a have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(both hold for every admissible scheme, Lemmas 14.4 and 14.5). Then every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$, family $\mathcal{F}$ and J with $MJ > R(\mathcal{F}) + \ell_c$, $Z^{a,\mathcal{K}}_J$ is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and*

$$|Z^{a,\mathcal{K}}_J(\zeta)| \leq \exp\left(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\right) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}), \tag{32}$$

with the constants of (30). Several models at once: *for a countable set of models all satisfying (31) with one set of constants, one $\mathcal{K}$ serves all.*

Form (A), which needs in addition the existence of the limits at every real coefficient vector ((W2)(a)), is Theorem 23.7. The subsequence is chosen in Step 2 below and nowhere else (Remark 14.15).

Proof. (a). Inputs: Lemma 7.3 ($\sup_{k\geq 0,\, J\geq J_0(f)} \sup_K |a^{(a)}_{k,J}| < \infty$ for every family and compact $K \subset \mathbb{C}^N$); Lemma 7.1 at second order ($\langle F(u)^2\rangle^{(a)}_{k,J} \leq 2\|u\|^2_{2,k}$ for real u with plaquettes in X_J); charge conjugation $(A, \vartheta) \mapsto (-A, -\vartheta)$, a symmetry of every Neumann–Dirichlet state, so $\langle F(u)\rangle^{(a)}_{k,J} = 0$ and $\langle e^{\mathrm{i}F(u)}\rangle^{(a)}_{k,J}$ is real. *Step 1 (a countable core).* Let $\mathcal{B}$ be the countable set of finite unions of closed boxes with rational corners, for $B \in \mathcal{B}$ let $S_0(B)$ be a countable $\|\cdot\|_\infty$-dense subset of $C^\infty_c(\operatorname{int} B; \Lambda^2)$, and let $\mathcal{P}$ be the countable set of pairs (J, u'), u' a finite family from $\bigcup_B S_0(B)$, $J \geq J_0(u')$. *Step 2 (Montel and the diagonal).* For $(J, u') \in \mathcal{P}$, $(a^{u'}_{k,J})_k$ is locally bounded on $\mathbb{C}^N$ uniformly in k (Lemma 7.3), hence normal; enumerate $\mathcal{P}$, extract

successively, and let $\mathcal{K} = (k_m)$ be the diagonal sequence: $(a^{u'}_{k_m,J})_m$ converges locally uniformly for every $(J,u') \in \mathcal{P}$. *Step 3 (every family, imaginary coefficients).* Let f be a family, $J \ge J_0(f)$, and $B \in \mathcal{B}$ with $\bigcup_j \operatorname{supp} f_j \subset \operatorname{int} B \subset B \subset U_{J_0(f)}$ (so every $u' \in S_0(B)$ is admissible at every $J \ge J_0(f)$). For real 2-forms u, u', g supported in B and every k with $\varepsilon_k \le 1$,

$$|\langle e^{\mathrm{i}F(u)}\rangle^{(a)}_{k,J} - \langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J}| \le \langle F(u-u')^2\rangle^{(a)}_{k,J}{}^{1/2} \le \sqrt{2}\|u-u'\|_{2,k}, \qquad \|g\|^2_{2,k} \le V_B \|g\|^2_\infty, \quad V_B := 3\,\mathrm{vol}(N_{1/2}(B)),$$

the first by $|e^{\mathrm{i}x}-1| \le |x|$ and the second-order bound, the second because the cubes of side ε_k about the centres of the plaquettes of one orientation on which $g \ne 0$ are disjoint and lie in the closed ℓ^∞-neighbourhood $N_{1/2}(B)$. Given $t \in \mathbb{R}^N$ and $\eta > 0$ pick $u' \in S_0(B)$ with $\|\sum_j t_j f_j - u'\|_\infty \le \eta$; then $|a^f_{k_m,J}(\mathrm{i}t) - a^f_{k_{m'},J}(\mathrm{i}t)| \le 2\sqrt{2V_B}\,\eta + |\langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J k_m} - \langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J k_{m'}}|$, and the last term tends to 0 by Step 2. So $(a^f_{k_m,J}(\mathrm{i}t))_m$ converges for every t. *Step 4 (all complex coefficients).* $(a^f_{k_m,J})_m$ is locally bounded on $\mathbb{C}^N$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives locally uniform convergence to an entire $Z^{\mathcal{K}}_J(\cdot,0)$. The bound $|a^{(a)}_{k,J}(z)| \le e^{\|g(z)\|^2_{2,k}}$ (Theorem 7.4(a)) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$ (1); realness at real points passes; $a^{(a)}_{k,J}(x) = \langle e^{F(u_x)}\rangle^{(a)}_{k,J} \ge e^{\langle F(u_x)\rangle^{(a)}_{k,J}} = 1$ by Jensen, and $[1,\infty)$ is closed; $0 \le a^{(a)}_{k,J}(\mathrm{i}t) \le 1$ by the current representation of Lemma 2.8(a), $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle^{(a)}_{k,J} = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp \cosh\langle Cv,\kappa\rangle \ge 0$, and $|e^{\mathrm{i}F}| = 1$. Dependence only on the total source holds at every (k,J), hence in the limit. *Step 5 (several models).* The constant of Lemma 7.3, the bound $e^{\|g\|^2_{2,k}}$ and the Lipschitz constant $\sqrt{2}$ are model-free: Theorem 7.4(a) is Lemma 2.8(a) for a measure of the class of Lemma 2.7 plus Lemma 7.1 for the Neumann–Dirichlet photon covariance, valid for every real site-independent bare coefficient (the matter weight does not enter) and every $\mu > 0$. For countable $\mathcal{M}$ run Step 2 over $\mathcal{M} \times \mathcal{P}$ and Steps 3–4 member by member.

(b). Inputs: under (I2)–(I4) and $B_{\mathrm{cell}} < \infty$, for every family with supports in Q_R, every k, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$, (31) gives $|Z^a_{k,J}(\zeta)| \le Z^a_{k,J}(\operatorname{Re}\zeta) \le \exp(\|g\|^2_{2,k} + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)$, with (I2) from (T3) by Proposition 12.6 and (SCH) for the scheme; on compacts of the tube the right side is bounded uniformly in k (Lemma 2.6). *Steps 1–2.* As in (a), with $S_0^{\mathsf{W}}(B)$ a countable $\|\cdot\|_\infty$-dense subset of $C^\infty_c(\operatorname{int} B;\Lambda^2) \times C^\infty_c(\operatorname{int} B)$ and $\mathcal{P}^{\mathsf{W}}$ the countable set of pairs $(J,\mathcal{F}')$, $\mathcal{F}'$ a finite family from $\bigcup_B S_0^{\mathsf{W}}(B)$, $MJ > R(\mathcal{F}') + \ell_c$. For such a pair $J \ge J_1(R)$ for every $R \in (R(\mathcal{F}'), MJ - \ell_c]$, so (31) holds for $\mathcal{F}'$ at every k and $(Z^a_{k,J})_k$ is normal on the tube of $\mathcal{F}'$; the diagonal over $\mathcal{P}^{\mathsf{W}}$, together with that of (a), gives $\mathcal{K}$. *Step 3 (every family, imaginary coefficients).* Let $\mathcal{F} = (f_i; h_j)$, $MJ > R(\mathcal{F}) + \ell_c$, $R_J := MJ - \ell_c > R(\mathcal{F})$, so that all supports lie in the open cube $Q^\circ_{R_J}$ and $J \ge J_1(R_J)$; choose $B \in \mathcal{B}$ with the supports in $\operatorname{int} B \subset B \subset Q^\circ_{R_J}$, so that $(J,\mathcal{F}') \in \mathcal{P}^{\mathsf{W}}$ for every $\mathcal{F}'$ from $S_0^{\mathsf{W}}(B)$. For $(s,s') \in \mathbb{R}^{n+p}$ put $\sigma := (\sum_i s_i f_i, \sum_j s'_j h_j)$, so that $Z^a_{k,J}(\mathrm{i}s, \mathrm{i}s') = \langle e^{\mathrm{i}\Phi^a_k(\sigma)}\rangle^{(a)}_{k,J}$ with $\Phi^a_k(g,h) := F_{\Lambda_k}(g) + W^a_{\Lambda_k}(h)$. *Lipschitz bound.* For a real pair $q = (g_q, h_q)$ supported in Q_{R_J}, $(t_1,t_2) \mapsto Z^a_{k,J}(t_1 g_q, t_2 h_q)$ is holomorphic, and on the polydisc $|t_1| \le \|g_q\|^{-1}_{2,k}$, $|t_2| \le \sigma_0/|h_q|_w$ one has $\|k_\pm\|_\infty \le |\operatorname{Re} t_2|\,\|h_q\|_\infty \le \sigma_0 < H/2 < \sigma_+/2$, so the polydisc lies in the tube and (31) gives $|Z^a_{k,J}| \le e^{1+K'_{\mathrm{lat}}\sigma_0}$ there, $K'_{\mathrm{lat}} := \kappa C'_0 + K_-$ ($\|k_+\|_{1,k} \le C'_0|k_+|_w$ by Lemma 2.6). The Taylor coefficients at 0 are the second moments of $F(g_q)$ and $W^a(h_q)$, and Cauchy's estimates give

$$\langle \Phi^a_k(q)^2\rangle^{(a)}_{k,J}{}^{1/2} \le C\,(\|g_q\|_{2,k} + |h_q|_w), \qquad C := \sqrt{2}\,e^{(1+K'_{\mathrm{lat}}\sigma_0)/2}\max(1,\sigma_0^{-1}),$$

uniformly in k, J and R_J; hence $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \le C(\|g_{\sigma-\sigma'}\|_{2,k} + |h_{\sigma-\sigma'}|_w)$ for real pairs supported in Q_{R_J}. *Density.* Given $\eta > 0$ pick $\sigma' \in S_0^{\mathsf{W}}(B)$ with $\|g_\sigma - g_{\sigma'}\|_\infty, \|h_\sigma - h_{\sigma'}\|_\infty \le \eta$; with V_B and $|h|_w \le (1+3R_J^2)^2\|h\|_\infty$ for h supported in Q_{R_J}, $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \le C(\sqrt{V_B} + (1+3R_J^2)^2)\eta$ for every k, while $Z^a_{k_m,J}(\mathrm{i}\sigma')$, the core functional at the point (i,i) of its tube, converges by Step 2. So $(Z^a_{k_m,J})_m$ converges on $(\mathrm{i}\mathbb{R})^{n+p}$. *Step 4 (the tube).* $(Z^a_{k_m,J})_m$ is locally bounded on the connected open $\mathcal{D}_{\sigma_+,H}$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^{n+p} \subset \mathcal{D}_{\sigma_+,H}$; Montel–Vitali gives locally uniform convergence to a holomorphic $Z^{a,\mathcal{K}}_J$, and (31) passes to the

limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (Riemann sums; k_+ is continuous with compact support) and sampled $|k_-|_w \le$ continuum $|k_-|_w$: this is (32). *Step 5 (several models).* With one set of constants in (31) for all members, C is model-free and Step 2 runs over the product of the model set with $\mathcal{P}^{\mathsf{W}}$. □

Lemma 14.8 (the cube limit along $\mathcal{K}$ is a full limit)**.** *Let $\mathcal{K}$ be a continuum subsequence.* (Curvature sector.) *For every family, $Z_J^{\mathcal{K}}(\cdot,0) \to Z_\infty^{\mathcal{K}}(\cdot,0)$ locally uniformly on $\mathbb{C}^N$ as $J \to \infty$; the limit is entire, $|Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|_2^2}$, real and ≥ 1 at real points, in $[0,1]$ at imaginary points; $J \mapsto Z_J^{\mathcal{K}}(\mathrm{i}t,0)$ is nondecreasing, so $Z_J^{\mathcal{K}} \le Z_\infty^{\mathcal{K}}$ on $(\mathrm{i}\mathbb{R})^N$.* (Wick sector, Form (B).) *Under the hypotheses of Lemma 14.7(b), $Z_J^{a,\mathcal{K}} \to Z_\infty^{a,\mathcal{K}}$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$; the limit is holomorphic, obeys (32), is >0 at the real points of the tube, and $Z_J^{a,\mathcal{K}} \le Z_\infty^{a,\mathcal{K}}$ at the cone points ($z \in (\mathrm{i}\mathbb{R})^n$, w real, $k(\zeta) \ge 0$).*

Proof. Curvature sector. For $t \in \mathbb{R}^N$, $J \ge J_0(f)$ and every k: $a_{k,J}^{(a)}(\mathrm{i}t) \le a_{k,J+1}^{(a)}(\mathrm{i}t) \le 1$ (first inequality of Lemma 2.10 with $Y = C_{J+1} \cap \Lambda_k$, $X = X_J$; the cut-out factor is the Neumann–Dirichlet state on C_J because δm_ε^2 is site-independent; realness by charge conjugation). Both sides converge along $\mathcal{K}$, so $J \mapsto Z_J^{\mathcal{K}}(\mathrm{i}t,0)$ is nondecreasing, bounded by 1, convergent. $\{Z_J^{\mathcal{K}}\}_J$ is locally bounded on $\mathbb{C}^N$ by the J-free bound $e^{\|g(z)\|_2^2}$ and converges on $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives the entire limit with the same bound; realness, ≥ 1 at real points and $[0,1]$ at imaginary points pass to the limit.

Wick sector. (1) *Split family.* Write $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty \chi_j$, $h_j^+ := h_j + h_j^- \ge 0$, with $\chi_j \in C_c^\infty$, $0 \le \chi_j \le 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset N_{1/2}(\operatorname{supp} h_j)$; put $F_{k,J}(z, w^+, w^-) := Z_{k,J}^a(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{|\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$, a convex subset of the split family's tube. By Lemma 14.7(b) for the split family, $F_{k_m,J} \to F_J^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$ for $MJ > R(\mathcal{F}) + \frac{1}{2} + \ell_c$, with (32). (2) On $\Omega_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the observable has imaginary curvature coefficients and nonnegative total scalar source, so $J \mapsto F_{k,J}$ is nondecreasing at every k (first inequality of Lemma 2.10 with Lemma 2.11(a), the scheme multiplier being common); along $\mathcal{K}$, $J \mapsto F_J^{\mathcal{K}}$ is nondecreasing and bounded on Ω_+, hence convergent. (3) Ω_+ is a non-empty open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is connected: $F_J^{\mathcal{K}} \to F_\infty^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$. (4) At $w^+ = w$, $w^- = -w$ the lattice functionals equal $Z_{k,J}^a(\zeta)$ exactly (equal total sources), so $Z_J^{a,\mathcal{K}}$ converges locally uniformly on $\mathcal{D}' := \{|\operatorname{Re} w_j| < \delta_j\} \times \mathbb{C}^n \subset \mathcal{D}_{\sigma_+,H}$; $\{Z_J^{a,\mathcal{K}}\}_J$ is locally bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free (32) and converges on the open $\mathcal{D}'$; Montel–Vitali on the connected tube gives $Z_J^{a,\mathcal{K}} \to Z_\infty^{a,\mathcal{K}}$, holomorphic, with (32). (5) *Positivity at real tube points.* For real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$ and let $C_\epsilon := \sup_k(\|g\|_{2,k}^2 + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)|_{-\epsilon\zeta} < \infty$, the exponent of (31) at $-\epsilon\zeta$ (bounded in k by Lemma 2.6, free of J). At every k, $t \mapsto \ln Z_{k,J}^a(t\zeta)$ is convex on $\mathbb{R}$ (Hölder, Lemma 2.11(d)) and vanishes at $t = 0$, so $1 \le Z_{k,J}^a(\zeta)^{\epsilon/(1+\epsilon)} Z_{k,J}^a(-\epsilon\zeta)^{1/(1+\epsilon)} \le Z_{k,J}^a(\zeta)^{\epsilon/(1+\epsilon)} e^{C_\epsilon/(1+\epsilon)}$, i.e. $Z_{k,J}^a(\zeta) \ge e^{-C_\epsilon/\epsilon}$ for every k and J; the bound passes along $\mathcal{K}$ and then in J: $Z_\infty^{a,\mathcal{K}}(\zeta) \ge e^{-C_\epsilon/\epsilon} > 0$. (The product inequality of Lemma 2.11(e) is not used: the tube is asymmetric, $H = \sigma_+/16$, and for real $\xi \in \mathcal{D}_{\sigma_+,H}$ with $\|k_+(\xi)\|_\infty \ge H/2$ the point $-\xi$ lies outside it, where Definition 14.1 gives no limit.) (6) At a cone point of the original family $J \mapsto Z_{k,J}^a(\zeta)$ is nondecreasing at every k (first inequality of Lemma 2.10 with Lemma 2.11(a)); pass along $\mathcal{K}$. □

14.3. **The axioms for LIM states.**

Theorem 14.9 (LIM states of the comparison family at fixed photon mass)**.** *Let $e > 0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$, and the states $\mathrm{ND}_{k,J}^{(a)}$ of Definition 2.1.*

Curvature sector (no hypothesis). *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence $\mathcal{K}$ and every finite family of real 2-forms,*

$$Z_\infty^{\mathcal{K}}\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty}\lim_{m\to\infty} \Big\langle e^{\sum_j z_j F_{\Lambda_{k_m}}(f_j)} \Big\rangle_{k_m,J}^{(a)}$$

exists locally uniformly in $z \in \mathbb{C}^N$, is entire, and depends on the family only through the total source. Its Schwinger functions $S_n^{\mathcal{K}}(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)|_{z=0}$ satisfy:

(i) (OS0.) $|Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|_2^2}$ *(no dependence on μ_0, e, λ, m) and* $\le \exp(\frac{1}{2}\mu_0^{-2}\|\sum_j (\operatorname{Re} z_j) d^* f_j\|_2^2)$; $|S_n^{\mathcal{K}}| \le (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \le (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$; *the $S_n^{\mathcal{K}}$ are tempered distributions, and* E0″ *holds with $s = 4$ and every $\beta > \frac{1}{2}$.*

(ii) (OS1.) *$S_n^{\mathcal{K}}$ is invariant under all translations and under O_h about every point, F transforming as a 2-form.*

(iii) (OS2.) *Reflection positivity about every coordinate plane $\{x_\nu = c\}$: for exponentials $E_a = e^{\sum_l z_{al} F(f_{al})}$ with arbitrary complex z_{al} and f_{al} supported in $\{x_\nu > c\}$, $(S(\overline{\theta_c E_a} E_b))_{ab}$ is positive semidefinite, and $S(\theta_c(P)P) \ge 0$ for polynomials P in such fields.*

(iv) (OS3.) *$S_n^{\mathcal{K}}$ is real, symmetric, and zero for odd n.*

(v) (The measure.) *For every even $r \ge 14$ there is exactly one Borel probability measure $\mu_F^{\mathcal{K}}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose moments on C_c^∞ are the $S_n^{\mathcal{K}}$; its characteristic functional is $Z_\infty^{\mathcal{K}}(\mathrm{i}\,\cdot, 0)$ extended by continuity to $\mathcal{S}$, and its image on H^{-r}_{loc} is the unique measure there with this characteristic functional and the unique one with these moments. For every $\chi \in C_c^\infty$ and box Q containing* $\operatorname{supp}\chi$ *in its interior, the laws of $\chi F_{\Lambda_{k_m}}$ under $\mathrm{ND}^{(a)}_{k_m,J}$ converge weakly on $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$,* first *as $m \to \infty$* along *$\mathcal{K}$ at every fixed $J \ge J_0(\chi)$ and then as $J \to \infty$, to the law of $\chi\Phi$ under $\mu_F^{\mathcal{K}}$.*

(vi) (Weak OS4.) *$\mu_F^{\mathcal{K}}$ is $\mathbb{R}^3 \rtimes O_h$-invariant and satisfies the clauses of Theorem 11.1(b) with $\mu_F^{\mathcal{K}}$ in place of μ_S^K: it is an integral of translation-ergodic, reflection-positive, θ_j-invariant components clustering in the mean and under diagonal separation, with a unique joint vacuum per coordinate direction and the component growth bound; almost every component satisfies* OS4 *in the weak form; the O_h-invariant decomposition into mixtures of $k \in \{1,2,3,6\}$ ergodic measures exists.*

(vii) (Euclidean clause.) *$Z_\infty^{\mathcal{K}}$ does not depend on $\mathcal{K}$, equals the limit state Z_∞^{LIM} of Definition 14.3, is $\mathbb{R}^3 \rtimes O(3)$-invariant for all complex coefficients (Theorem 3.14), and the Euclidean forms of (ii) (full $\mathbb{R}^3 \rtimes O(3)$), (iii) (reflection positivity about every plane) and (vi) (almost every component $\mathbb{R}^3 \rtimes O(3)$-invariant, reflection positive about every plane, clustering in any manner, with a unique vacuum in every direction; the vanishing of a component's odd moments is not claimed) hold.*

Wick sector, Form (B). *Let the couplings satisfy* (T3) *and let the scheme a have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(Lemmas 14.4 and 14.5; every admissible scheme does). Every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$ and family the LIM state $Z_\infty^{a,\mathcal{K}}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, is holomorphic there, obeys (32), and (i)–(vi) hold for the joint Schwinger functions $S_{n,p}^{\mathcal{K}}(f;h) = \partial_z^{(n)}\partial_w^{(p)} Z_\infty^{a,\mathcal{K}}(0)$ with: in (i) the mixed bounds $|S_{n,p}^{\mathcal{K}}| \le e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$, and* E0″ *with $s = 4$ and every $\beta > 1$; in (ii) W^a a scalar, the identity holding on the tube; in (iii) exponentials with arbitrary complex curvature coefficients and Wick coefficient vectors $\zeta_1,\dots,\zeta_N$ of* amplitude $\mathsf{a}(\zeta_i) < H/2$ *(in particular all purely imaginary Wick coefficients), and the polynomial form; in (iv) $S_{n,p}^{\mathcal{K}} = 0$ for odd n; in (v) the measures on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2 \oplus \mathbb{R})$ and H^{-r}_{loc}, the localised lattice fields $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ converging along $\mathcal{K}$ and then in J; (vi) unchanged; and* no Euclidean clause. Schemes: *if a, a' both satisfy the hypotheses, every continuum subsequence has a further subsequence $\mathcal{K}'$ along which $a'_k - a_k \to \kappa^{\mathcal{K}'}$, and $Z_\infty^{a',\mathcal{K}'}(\zeta) = e^{-\kappa^{\mathcal{K}'}\sum_j w_j \int h_j} Z_\infty^{a,\mathcal{K}'}(\zeta)$: the two LIM states differ by the shift $W \mapsto W - \kappa^{\mathcal{K}'}\int(\cdot)$ of the Wick component, and joint cumulants of order ≥ 2 are scheme-free.*

Proof. The complete proof is given in the supplemental material; in summary, each clause is the corresponding proof for the iterated limit (Theorems 7.4, 8.5, 10.6, 10.7 and 11.1 and Propositions 8.1 and 10.9) with exactly one of three replacements: (α) a fixed-(k, J) identity or inequality between Neumann–Dirichlet functionals of explicit finite families is passed along $\mathcal{K}$ (both sides converge by Lemma 14.7, which covers *every* family); (β) Montel on a k-uniform bound, the

identification of the limit by convergence at real points being replaced by the choice of $\mathcal{K}$; (γ) a step about the limit object only, unchanged. The only uniqueness used is that of the monotone cube limit along $\mathcal{K}$ (Lemma 14.8).

(i). Lemmas 14.7 and 14.8 give existence, entireness and the first bound; the second, $\exp(\frac{1}{2}\mu_0^{-2}\|d_k^*g\|_{2,k}^2)$ at every (k,J) (Theorem 7.4(a)), passes along $\mathcal{K}$ and J with $\|d_k^*g\|_{2,k}\to\|d^*g\|_2$. The Schwinger bounds, the extension to $\mathcal{S}$, the kernel theorem and E0″ are the arguments of Theorem 7.4 (Wick sector: Theorem 13.10(ii), on the polydisc inside the tube) run on the limit function with its bound (γ).

(ii), translations. For $b\in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty\le M$, $k\ge k_0$: $Z^a_{k,C_J}(\tau_b\cdot\zeta)=Z^a_{k,C_J-b}(\zeta)$ (exact lattice covariance) and $C_{J-1}\cap\Lambda_k\subset(C_J-b)\cap\Lambda_k\subset C_{J+1}\cap\Lambda_k$, so for a cone point s supported in U_{J_0} and $J\ge J_0+1$, $Z^a_{k,C_{J-1}}(s)\le Z^a_{k,C_J}(\tau_b\cdot s)\le Z^a_{k,C_{J+1}}(s)$ (first inequality of Lemma 2.10, twice). The three terms are functionals of the families s and $\tau_b\cdot s$ (supports in Q_{R+M}, the same tube), so they converge along $\mathcal{K}$ (α), and Lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty(s)$: $Z^{\mathcal{K}}_\infty(\tau_b\cdot s)=Z^{\mathcal{K}}_\infty(s)$ at cone points, with no uniqueness of the k-limit. All coefficients (γ): the difference of the two limit functions is entire and vanishes on $(\mathrm{i}\mathbb{R})^N$ (curvature), resp. for the split family is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω_+, so the identity theorem, the diagonal (4) of Lemma 14.8 and the identity theorem on the connected tube give it on $\mathcal{D}_{\sigma_+,H}$. All $b\in\bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration; all $b\in\mathbb{R}^3$ at imaginary coefficients because the Lipschitz bounds of Lemma 14.7, Step 3, pass to the limits ($|Z^{\mathcal{K}}_\infty(\mathrm{i}u,0)-Z^{\mathcal{K}}_\infty(\mathrm{i}u',0)|\le\sqrt{2}\|u-u'\|_2$; $|Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma)-Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma')|\le C(\|g_{\sigma-\sigma'}\|_2+|h_{\sigma-\sigma'}|_w)$) and $\tau_{b_n}s\to\tau_{b_0}s$ in L^2 and in $|\cdot|_w$ for $s\in C_c^\infty$; complex coefficients by the identity theorem.

(ii), the cubic group. Coordinate permutations are exact at every (k,J) (Lemma 2.12); pass along $\mathcal{K}$ and J (α). For the reflection σ_3 and a cone point (f,g) supported in U_{J-1}, at every k, $Z^a_{k,C_{J-1}}(\mathrm{i}f,g)\le Z^a_{k,C_J}(\mathrm{i}\sigma_3\cdot f,\sigma_3\cdot g)\le Z^a_{k,C_{J+1}}(\mathrm{i}f,g)$ (the frame flip of Lemma 2.12 with the one-site shift $X_+=(C_J+\varepsilon_k e_3)\cap\Lambda_k$ and two deletions in the cone direction, Lemma 2.7(a),(b)); the middle term is the functional of the reflected family (same tube), so (α) and Lemma 14.8 give $Z^{\mathcal{K}}_\infty(\mathrm{i}\sigma_3\cdot f,\sigma_3\cdot g)=Z^{\mathcal{K}}_\infty(\mathrm{i}f,g)$ on the cone (all of it at $g=0$; in Form (B) at cone points with $\|g\|_\infty<\sigma_+/2$): the reflections are not exact lattice symmetries; the sandwich replaces exactness and survives along $\mathcal{K}$. Conjugation $\sigma_\nu=P\sigma_3P^{-1}$, group closure on the cone, the identity theorem off the cone (on $\mathbb{C}^N$ with $(\mathrm{i}\mathbb{R})^N$; on $\mathcal{D}''$ with Ω_+), invariance about every point by the translations, and differentiation are (γ), as in Theorem 8.5.

(iii), reflection positivity. Fix ν, a mid-plane $c\in D$ (Definition 9.1) and the enlarged family $\mathcal{F}$ of Definition 10.1 with the modified-box functionals $\Phi_k(\zeta)=Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta)$, $\zeta\in\mathbb{C}^m$, and support radius $R_{\mathcal{F}}$. The fixed-k steps (reflection positivity of the modified box, Lemma 9.3; the sandwich lemmas Lemmas 9.9 and 9.10) concern the photon form and the inequalities of Lemma 2.7 only and are unchanged. *Curvature sector.* At $h=0$, $|\Phi_k(z)|\le e^{\|f_{\mathrm{Re}\,z}\|^2_{2,k}}$ by Lemma 2.8(a) and the modified photon form, uniformly in k and J. (β) Extract from $\mathcal{K}$ a further $\mathcal{K}'$ along which $\Phi_k\to\Phi^J$ locally uniformly on $\mathbb{C}^m$; positive semidefiniteness of the Gram matrices at every $k\ge k_0(c,\delta)$ passes to Φ^J. (α) On U (imaginary curvature coefficients) $a^{(a)}_{k,C_J}(\zeta)\le\Phi_k(\zeta)\le a^{(a)}_{k,C_{J+2n}}(\zeta)$ at every k (Lemmas 9.9 and 9.10, cone halves); along $\mathcal{K}'$ the ends converge to $Z^{\mathcal{K}}_J, Z^{\mathcal{K}}_{J+2n}$, so $Z^{\mathcal{K}}_J\le\Phi^J\le Z^{\mathcal{K}}_{J+2n}$ on U, and Lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty$. $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ (J-free bound) and U is a set of uniqueness, so $\Phi^J\to Z^{\mathcal{K}}_\infty$ locally uniformly, in particular at the matrix points, where $\Phi^J(\zeta^*_{ij})\to Z^{\mathcal{K}}_\infty(\theta\zeta^*_i\sqcup\zeta_j)$ (dependence through the total source only); positivity passes. $\mathcal{K}'$ may depend on J; this is harmless, since the sandwich ends converge along all of $\mathcal{K}$ and the bound on Φ^J is J-free, so neither $\lim_J\Phi^J$ nor the positivity depends on the choice. Every plane: translate by $(c'-c)e_\nu$ and use (ii). The polynomial form: Weierstrass along $\mathcal{K}'$ and J (Proposition 10.10). *Wick sector, amplitude $<H/2$, without Theorem 3.6.* Let $\zeta_1,\dots,\zeta_N$ be families of real sources supported in $\{x_\nu>c\}$ with $\mathsf{a}(\zeta_i)<H/2$; the (i,j) Gram entry is $\Phi_k(\zeta^*_{ij})$, with total scalar source $\sum_l\bar w_{il}\,h_{il}\circ\rho+\sum_l w_{jl}h_{jl}$. The bound of Lemma 10.3, which depends on Theorem 3.6, is replaced by Lemma 14.10 below, after which the argument is completed.

(iv). Symmetry: Taylor coefficients of a function of the total source. Realness: real at real points at every (k, J), closed under limits (Form (B): at the real points of the tube). Charge conjugation gives $Z^a_{k,J}(-z, w) = Z^a_{k,J}(z, w)$ at every (k, J), hence for $Z^{\mathcal{K}}_J$ and $Z^{\mathcal{K}}_\infty$ (the tube is invariant under $z \mapsto -z$); odd z-derivatives at 0 vanish.

(v), (vi). The moment bounds at (k, J) (Theorem 7.4(a), resp. (31), with Cauchy's estimates) are uniform in k and J above the thresholds, and the lattice Schwinger functions converge along $\mathcal{K}$ then J (Weierstrass). The measure is built from the limit functional (γ): $\mathcal{Z}^{\mathcal{K}}(f) := Z^{\mathcal{K}}_\infty(\mathrm{i}f)$ on real $f \in C^\infty_c$ is normalised, positive definite (at every (k, J) the matrix $(Z^a_{k,J}(\mathrm{i}(f_m - f_l)))_{lm}$ is a Gram matrix; the entries converge along $\mathcal{K}$ then J) and Lipschitz for $\sqrt{2}\|\cdot\|_2$, resp. $C(\|\cdot\|_2 + |\cdot|_w)$, hence continuous on $\mathcal{S}$; Bochner–Minlos, moments by Lukacs [18] and uniform integrability, the Hermite carrier on $\mathcal{H}^{-r}$, $r \geq 14$ even, and the two uniqueness statements are the steps of Theorem 11.1(a) with [27, Lemma 10.5(c), Lemma 10.3, Prop. 10.4]. *Local laws along* $\mathcal{K}$ ((β); the one weakened clause): tightness of the laws of $\chi\Phi^a_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q; \mathcal{V})$ is uniform in k and $J \geq J(\chi)$ from the second-moment bounds (Prokhorov [8]); the characteristic functionals $Z^a_{k,J}(\mathrm{i}\chi\psi)$ converge *along* $\mathcal{K}$ to $Z^{a,\mathcal{K}}_J(\mathrm{i}\chi\psi)$; a probability measure on H^{-r}_{per} is determined by its characteristic functional on trigonometric polynomials; so the laws converge along $\mathcal{K}$, then in J (Lemma 14.8), to the law of $\chi\Phi$ under the measure just built (the periodisation identity of Theorem 11.1). (vi): $\mathbb{R}^3 \rtimes O_h$-invariance of the measure is (ii) at imaginary coefficients plus uniqueness of the characteristic functional; (H3) of [27] is (iii) at purely imaginary Wick coefficients ($\mathsf{a} = 0 < H/2$) with the density of the exponential algebra [27, Lemma J.2]; then [27, Thm. 10.10, Thm. 10.19, Thm. 10.20, Prop. 10.16, Cor. 10.43(iv), Thm. 10.12, Def. 2.12(iii)], stated for a measure satisfying (H1)–(H3) and matched as in Remark 11.3.

Schemes. $\sup_k |a'_k - a_k| \leq \sup_k |a'_k - z_k| + \sup_k |z_k - a_k| < \infty$ by (SCH) for both. For a continuum subsequence $\mathcal{K}$ for a, Bolzano–Weierstrass gives $\mathcal{K}' \subset \mathcal{K}$ with $a'_k - a_k \to \kappa^{\mathcal{K}'} \in \mathbb{R}$. At every (k, J), $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a_{k,J}(\zeta)$ (Remark 2.4) with $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$; so along $\mathcal{K}'$ the a'-functionals converge wherever the a-functionals do, $\mathcal{K}'$ is a continuum subsequence for a' with $Z^{a',\mathcal{K}'}_J = e^{-\kappa^{\mathcal{K}'} \sum_j w_j \int h_j} Z^{a,\mathcal{K}'}_J$, and then $J \to \infty$. (For admissible a, a', $a'_k - a_k$ converges along the full sequence and $\kappa^{\mathcal{K}'} = \kappa$.) The cumulant statement is that of Theorem 21.1. ($\mathrm{B}_{\mathrm{cell}}$) transfers between schemes with bounded difference: $B^{a'}_{\mathrm{cell}} \leq B^a_{\mathrm{cell}} + \sup_k |a'_k - a_k| \, \|2H'\chi_0\|_{1,k}$.

(vii). By Proposition 14.12(a), $Z^{\mathcal{K}}_\infty = Z^{\mathrm{LIM}}_\infty$, which is $\mathbb{R}^3 \rtimes O(3)$-invariant for all $z \in \mathbb{C}^N$ by Theorem 3.14. (ii) in Euclidean form follows by differentiation; for (iii), a plane P is $E\{x_\nu = c\}$ for a Euclidean motion E, $\theta_P = E\theta_{\nu,c}E^{-1}$, and $Z^{\mathcal{K}}_\infty(\theta_P \zeta_i^* \sqcup \zeta_j) = Z^{\mathcal{K}}_\infty(\theta_{\nu,c}(E^{-1}\zeta_i)^* \sqcup E^{-1}\zeta_j)$ by invariance at complex coefficients, the $E^{-1}\zeta_i$ being supported in $\{x_\nu > c\}$; (vi) is [27, Cor. 10.44] with $\mathcal{Z}^{\mathcal{K}}$ $O(3)$-invariant, as in Theorem 11.1(c), with its proviso on the odd moments of a component. □

Lemma 14.10 (a k- and J-uniform bound for the modified-box functionals on the tube of the enlarged family)**.** *Under* (T3), ($\mathrm{B}_{\mathrm{cell}}$), (SCH)*, for every* k*, every* $J \geq \max(J_0, J_1(R_{\mathcal{F}}))$ *and every* $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$ *(the tube of* $\mathcal{F}$ *in the sense of Definition 2.5),*

$$|\Phi_k(\zeta)| \leq \exp\big(2\|g(\zeta)\|^2_{2,k} + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\big).$$

The curvature constant is 2, not the 1 of (31), because the Cauchy–Schwarz split below doubles the curvature source and Theorem 7.4(a) is quadratic in it; nothing depends on its value.

Proof. $|\Phi_k(\zeta)| \leq \Phi_k(\xi)$, $\xi := \mathrm{Re}\,\zeta$ (Lemma 2.11(d)). Let $k(\xi)$ be the total real scalar source at ξ and $k_1 := k(\xi)_+$, $k_2 := k(\xi)_-$ its *pointwise* parts, sampled on the lattice. Cauchy–Schwarz at the lattice level (Lemma 2.11(f); the multiplier $e^{-a_k\mathsf{S}(k_1 - k_2)}$ splits by linearity of S): $\Phi_k(\xi) \leq Z^{a,(3.38)}(2f_\xi, -2k_2)^{1/2} Z^{a,(3.38)}(0, 2k_1)^{1/2}$. *First factor* (class $\mathcal{D}$): the $\mathcal{D}$-half of Lemma 9.9, which needs only that the sources vanish off X_J ($J \geq J_0$), gives $\leq Z^a_{k,C_J}(2f_\xi, -2k_2)$; Theorem 7.4(a) with the real scalar source $-2k_2$ gives $\leq e^{\|2f_\xi\|^2_{2,k}} N^a_{k,C_J}(0, -2k_2)$ (the same scheme multiplier on both sides); Proposition 13.9 with $q := 2k_2 \geq 0$, supported in $Q_{R_{\mathcal{F}}}$, $J \geq J_1(R_{\mathcal{F}})$ and $\sup q = 2\|k(\xi)_-\|_\infty < H$ (this is $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; the proposition allows $\sup q \leq H$) gives

$\ln N^a_{k,C_J}(0,-2k_2) \le K_-|2k_2|_w$. Hence $\frac12\ln$(first factor) $\le 2\|g(\zeta)\|^2_{2,k} + K_-|k_-(\zeta)|_w$, as $f_\xi = g(\zeta)$ and $k_2 = k_-(\zeta)$. *Second factor* (class $\mathcal{I}$): the $\mathcal{I}$-half of Lemma 9.10 with $B = C_{J+2n}\cap\Lambda_k$ gives $\le Z^a_{k,C_{J+2n}}(0,2k_1)$, and Corollary 13.3, uniform in k and in the cube, with $0 \le 2k_1 \le s_1$ on the lattice ($\|k_+(\zeta)\|_\infty < \sigma_+/2$, sampled $\le$ continuum) gives $\ln Z^a_{k,C_{J+2n}}(0,2k_1) \le \kappa\|2k_1\|_{1,k}$, so $\frac12\ln$(second factor) $\le \kappa\|k_+(\zeta)\|_{1,k}$. Add. On compacts of the tube the right side is bounded uniformly in k (Lemma 2.6). □

The lemma is (31) (with curvature constant 2) for the modified-box functional, transported through Lemmas 9.9 and 9.10; the two factors live on C_J and C_{J+2n}, which is harmless as Proposition 13.9 and Corollary 13.3 are uniform in the cube, and the amplitude $H/2$ is exactly the tube condition $\|k(\xi)_-\|_\infty \le H/2$ on the total source. *The matrix points lie in the tube:* at $\xi^*_{ij} = \mathrm{Re}\,\zeta^*_{ij}$ the two sums of the total scalar source have disjoint supports, so $\|k(\xi^*)_\pm\|_\infty \le \max(\mathsf{a}(\zeta_i),\mathsf{a}(\zeta_j)) < H/2 \le \sigma_+/2$, and by continuity a neighbourhood of the finitely many matrix points lies in $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$. *End of the proof of (iii), Wick sector.* (β) By Lemma 14.10, $(\Phi_k)_k$ is locally bounded on the open connected $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; extract $\mathcal{K}'\subset\mathcal{K}$ with $\Phi_k\to\Phi^J$ locally uniformly there; Gram positivity passes ($\mathcal{K}'$ may depend on J; harmless as in the curvature sector). (α) Let U be the set of coefficient vectors with purely imaginary curvature coefficients and positive real scalar coefficients on the nonnegative $\tilde h_j$; $U\cap\mathcal{D}^{\mathcal{F}}\neq\emptyset$ ($k_-=0$, k_+ small), and it is a relatively open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^{m_F}\times\mathbb{R}^{m_h}$, a set of uniqueness in $\mathcal{D}^{\mathcal{F}}$. On U, $Z^a_{k,C_J}(\zeta)\le\Phi_k(\zeta)\le Z^a_{k,C_{J+2n}}(\zeta)$ at every k (Lemmas 9.9 and 9.10, $\mathcal{I}$-halves); the ends are functionals of $\mathcal{F}$ at points of its tube, converging along $\mathcal{K}'$ (Lemma 14.7(b), J above the threshold of $\mathcal{F}$) to $Z^{a,\mathcal{K}}_J, Z^{a,\mathcal{K}}_{J+2n}$, which tend to $Z^{a,\mathcal{K}}_\infty$ (Lemma 14.8, locally uniformly on the tube of $\mathcal{F}$, in particular on $U\cap\mathcal{D}^{\mathcal{F}}$). $\{\Phi^J\}_J$ is locally bounded on $\mathcal{D}^{\mathcal{F}}$ (J-free), so $\Phi^J\to Z^{a,\mathcal{K}}_\infty$ locally uniformly there, in particular at the matrix points, where the limit is $Z^{a,\mathcal{K}}_\infty(\theta\zeta^*_i\sqcup\zeta_j)$; positivity passes. Every plane by (ii) (the amplitude is translation invariant); the polynomial form by Weierstrass.

Remark 14.11 (amplitudes in $[H/2,\sigma_+/8)$: the route through the fixed-cube limits, for a further subsequence). The proof of Lemma 10.3 bounds the first factor on the Wick tube $T_{\mathcal{F}'}$ of Definition 10.1 by the vertex argument at the cube C_{J_0}, which needs the fixed-cube limits in their boundedness form only. For amplitudes in $[H/2,\sigma_+/8)$ the matrix points lie in $T_{\mathcal{F}'}$ but in general outside $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$, the only domain on which Definition 14.1 gives convergence along $\mathcal{K}$; what follows is therefore stated here only for a further subsequence. Assume the boundedness in k of the functionals of the class-$\mathcal{D}$ families $(2f_\xi,-2k_2(\xi))$ at the one cube C_{J_0} (with the coefficient-sign split of Definition 10.1). The vertex argument then gives local boundedness of $(Z^a_{k,J})_k$ and $(\Phi_k)_k$ on $T_{\mathcal{F}'}$ (first factors at C_{J_0} by that boundedness, second factors by Corollary 13.3); Montel, with a diagonal over the countably many J, yields a further subsequence $\mathcal{K}'\subset\mathcal{K}$, depending on the family, along which $Z^a_{k,J}\to Z^{a,\mathcal{K}'}_J$ and $\Phi_k\to\Phi^J$ locally uniformly on $T_{\mathcal{F}'}$ (on $T_{\mathcal{F}'}\cap\mathcal{D}^{\mathcal{F}}$ the first limit is $Z^{a,\mathcal{K}}_J$). The sandwich step runs on $T_{\mathcal{F}'}$ with the uniqueness set $U\cap T_{\mathcal{F}'}$, the J-free local bound of Proposition 10.5(a) on $T_{\mathcal{F}'}$ serving the Vitali step in J, and Proposition 10.5(b) along $\mathcal{K}'$ for the remaining planes; so OS2 holds up to $\mathsf{a}(\zeta_i)<\sigma_W=\sigma_+/8$ for the LIM state of such a $\mathcal{K}'$ on $T_{\mathcal{F}'}$. Whether the convergence holds along $\mathcal{K}$ itself on $T_{\mathcal{F}'}$ is not claimed here; under the hypotheses of Form (A), Theorem 23.7 gives OS2 along $\mathcal{K}$ with no amplitude restriction. The remark is used nowhere else.

14.4. **Identification, and King's printed state.**

Proposition 14.12 (identification)**.** (a) *For every continuum subsequence $\mathcal{K}$ and every family, $Z^{\mathcal{K}}_\infty(\cdot,0) = Z^{\mathrm{LIM}}_\infty(\cdot,0)$, the limit state of the rotation theorem (Theorem 3.14); in particular all LIM states coincide,* (UNIQ) *holds, and they are $\mathbb{R}^3\rtimes O(3)$-invariant. Theorem 3.14 does* not *give Theorem 3.3.*

(b) *The following are equivalent: (i) the comparison family is convergent at fixed cubes (Definition 3.1), that is the existence in $(0,\infty)$ of the fixed-cube limits of the comparison family at real coefficients; (ii) for every J and family, $a^{(a)}_{k,J}\to Z_J$ locally uniformly on $\mathbb{C}^N$ along the*

full sequence; (iii) every increasing sequence of spacings is a continuum subsequence. Under them $Z_J^{\mathcal{K}} = Z_J$ and $Z_\infty^{\mathcal{K}} = Z_\infty := \lim_J Z_J$ for every $\mathcal{K}$: every LIM state is the iterated limit of the comparison family.

(c) *Assume* (T3) *and let a be admissible. Then* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *hold (Lemmas 14.4 and 14.5). If moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2), every increasing sequence of spacings is a continuum subsequence for the Wick sector, and for every $\mathcal{K}$, $Z_\infty^{a,\mathcal{K}} = G_\infty$ of Theorem 13.10 on $\mathcal{D}_{\sigma_+,H}$, which equals $Z^a_{\infty,T}$ of Proposition 10.5 on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ (Proposition 10.5 lives on the Wick tube $T_{\mathcal{F}'}$, a different domain). Once* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *are given, the argument uses Definition 3.2 for a alone; on King's convention, where Theorem 3.6 supplies the limits, the same argument is Step 7 of Theorem 21.3.*

Proof. (a) Let $\mathcal{K}$ be a continuum subsequence, f a family, $J \geq J_0(f)$. By Definition 14.1, $Z_J^{\mathcal{K}}(\cdot,0)$ is the locally uniform limit of $(a^{(a)}_{k_m,J})_m$, so $Z_J^{\mathcal{K}}(\cdot,0) \in L^f_J$. By Theorem 3.14 every choice $\zeta_J \in L^f_J$, $J \geq J_0(f)$, converges locally uniformly to $Z_\infty^{\mathrm{LIM}}(\cdot,0)$; take $\zeta_J := Z_J^{\mathcal{K}}(\cdot,0)$, which by Lemma 14.8 converges to $Z_\infty^{\mathcal{K}}(\cdot,0)$, and limits are unique. Hence any two continuum subsequences have the same LIM state, and invariance is the second clause of Theorem 3.14. That theorem pins the J-limit of the subsequential limits, not the fixed-cube limits themselves, so it does not give Theorem 3.3.

(b) (i)⇒(ii): $(a^{(a)}_{k,J})_k$ is locally bounded on $\mathbb{C}^N$ (Lemma 7.3) and converges at every real point; $\mathbb{R}^N$ is a set of uniqueness for entire functions, and Montel–Vitali gives locally uniform convergence of the whole sequence to an entire Z_J. (ii)⇒(iii): every subsequence converges to the same limit. (iii)⇒(i): the full sequence is itself an increasing sequence, hence a continuum subsequence, so $a^{(a)}_{k,J}$ converges locally uniformly, and at real x the limit lies in $[1,\infty) \subset (0,\infty)$ (Lemma 14.7, Step 4). The last clause is Lemma 14.8.

(c) (SCH) and ($\mathrm{B}_{\mathrm{cell}}$) are Lemmas 14.4 and 14.5; so (31) holds. $(Z^a_{k,J})_k$ is locally bounded on the tube by (31) and converges at the real points of the tube by Definition 3.2, a non-empty open subset of $\mathbb{R}^{n+p}$ and a set of uniqueness in the connected tube; Montel–Vitali: the whole sequence converges locally uniformly on the tube (Theorem 13.10(i)), so every increasing sequence is a continuum subsequence for the Wick sector with $Z_J^{a,\mathcal{K}} = G_J$, and $Z_\infty^{a,\mathcal{K}} = G_\infty$ by Lemma 14.8; on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ this is the same J-limit of the same G_J as $Z^a_{\infty,T}$ (Proposition 10.5(a)). □

Corollary 14.13 (the iterated limits of the comparison family)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). Then every increasing sequence of spacings is a continuum subsequence, all LIM states coincide with the iterated limit $Z_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^{(a)}_{k,J}$ of the comparison family, and the curvature sector (i)–(vi) of Theorem 14.9 holds for it with the lattice laws of (v) converging along the* full *sequence of spacings; (vii) holds as well. For general sources assume* (T3) *and let a be admissible; then every increasing sequence is a continuum subsequence for the Wick sector, every LIM state is G_∞ ($= Z^a_{\infty,T}$ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$), $\kappa^{\mathcal{K}} = \kappa$ is independent of $\mathcal{K}$, and Form (B) of Theorem 14.9 holds for G_∞ with the full-sequence lattice clause.*

Proof. Curvature sector: Proposition 14.12(b) makes every increasing sequence a continuum subsequence with $Z_\infty^{\mathcal{K}} = Z_\infty$; (i)–(vi) are those of Theorem 14.9, and in (v) the local laws converge along the full sequence because the characteristic functionals now do (step (β) with the full sequence); (vii) is Proposition 14.12(a). Wick sector: Proposition 14.12(c); the clauses are Form (B) of Theorem 14.9 for G_∞, with OS2 at amplitude $< H/2$ by the route of Lemma 14.10, which does not use Theorem 3.6, and at amplitude $< \sigma_W = \sigma_+/8$ by the route of Theorem 10.7; both hold ($H/2 = \sigma_+/32 < \sigma_+/8$: the route without the fixed-cube limits gives the smaller amplitude, and what it improves is the list of inputs); $\kappa^{\mathcal{K}} = \kappa$ by the scheme clause. □

Theorem 3.6 is the scheme-a analogue, with scalar sources, of the existence clause of King's theorem; it is not that theorem, which is false as printed with scalar sources in King's Wick ordering. With (W2) in addition one obtains Form (A) for the full object Z^a_∞ and, under (R_W), its Euclidean clause; this is the optional extension of Theorem 23.7.

Corollary 14.14 (King's convention)**.** *Define continuum subsequences and LIM states of the K-family* $a^K_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ *(Definition 2.2) by Definitions 14.1 and 14.2 with* $\langle\cdot\rangle^K_{k,J}$ *in place of* $\langle\cdot\rangle^{(a)}_{k,J}$*.*

(a) (No hypothesis.) *Every increasing sequence of spacings has a continuum subsequence of the K-family; for every such* $\mathcal{K}$*, family and* $J \geq J_0(f)$ *the fixed-cube limits* $Z^{K,\mathcal{K}}_J$ *and the LIM state* $Z^{K,\mathcal{K}}_\infty = \lim_J Z^{K,\mathcal{K}}_J$ *exist, are entire and obey* $|Z^{K,\mathcal{K}}_J|, |Z^{K,\mathcal{K}}_\infty| \leq e^{\|g(z)\|^2_2}$*, with the other clauses of Lemma 14.7(a) and Lemma 14.8. Nothing beyond this is asserted for a K-family LIM state from the bounds alone.*

(b) (With Theorem 3.3.) *Every increasing sequence of spacings is a continuum subsequence of the K-family, every K-family LIM state is King's printed state* Z^K_∞*, every LIM state of the comparison family is* Z^K_∞ *as well (Corollary 4.4(iii)), and the curvature-sector axioms of Theorem 21.1 hold for it, with the lattice laws of Theorem 11.1(a) converging along the full sequence.*

With scalar sources, the corresponding statements on King's convention are supplied by Theorem 21.3 and Proposition 21.4.

Proof. (a) The proof of Lemma 14.7(a) uses only the k- and J-uniform local bound of Lemma 7.3 (Step 2), the second-order bound $\langle F(u)^2\rangle \leq 2\|u\|^2_{2,k}$ of Lemma 7.1 (Step 3), and the current representation and Jensen's inequality (Step 4); that of Lemma 14.8 uses in addition the first inequality of Lemma 2.10. All of these hold on King's convention: the bounds by Theorem 7.6 (the second-order bound is photon-only and transfers by the covariance identity of Lemma 4.2), the cut inequality by Corollary 4.3. So the two lemmas hold for the K-family by the same proofs; the argument makes no use of Theorem 3.3. (b) By Theorem 3.3, Corollary 4.4(ii)(α) gives, for every $J \geq J_0$ and family, locally uniform convergence of $a^K_{k,J}$ along the full sequence, so every increasing sequence is a continuum subsequence of the K-family, and by (β) every K-family LIM state is Z^K_∞; Corollary 4.4(iii) is the statement for the comparison family; the axioms are the curvature sector of Theorem 21.1, and the full-sequence lattice laws are Theorem 11.1(a). □

Remark 14.15 (on the construction of the subsequences). Every subsequence of spacings used in this paper is constructed in Step 2 of the proof of Lemma 14.7 from the k-uniform bounds of Lemma 7.3 and (31) and Montel's theorem, or is a further extraction from one so constructed, and the identification of its limits uses only the monotone cube limit of Lemma 14.8; no subsequence statement is cited from [16].

Rotation invariance

15. The rotation part: lattices, walled tori and the route

This and the following six sections form the rotation part of the paper. Its object is King's printed state on curvature sources, $z \mapsto Z^K_\infty(\sum_j z_j f_j, 0)$, the entire function of the coefficients of real 2-forms $f_j \in C^\infty_c$ whose derivatives at $z = 0$ are the curvature Schwinger functions (Theorem 3.3 and the curvature-sector section); its result, Theorem 15.1, is that this function is invariant under every Euclidean motion.

Rotation invariance is not automatic, since all lattices in King's construction have the same orientation. Theorem 2.4 of [16] states that on the tori D_J the lattices $\varepsilon_k\mathbb{Z}^3$ and $R(\varepsilon_k\mathbb{Z}^3)$, with R the rotation by $\theta_0 = \arctan(3/4)$ in a coordinate plane, have the same continuum limit; since θ_0/π is irrational these rotations generate a dense subgroup of $SO(3)$. As printed the statement is false in two ways (Proposition H.2 of the supplemental material, Remark 16.1): the source it pairs with the plaquettes is one scalar function on both lattices, which couples the field to two different 2-forms, and the Wick constant it uses misses an order-e^2 logarithm. We therefore call the printed statement *King's torus conjecture*, and we use instead the corrected statement, for 2-form sources sampled in each lattice's own frame and for the two lattices at the same spacing; it is the torus comparison (K_0) of Theorem 17.1, proved in Section 17.3. Moreover, even granting the torus comparison, two steps remain between it and rotation invariance of King's state: the infinite-volume limit along tori, where correlation inequalities give no monotonicity, and its identification

with the limit along cubes, which King writes can be expected from a cluster expansion [16, (2.24)]. The ultraviolet analysis of [15, 16] needs boundaries that are lattice hyperplanes [16, p. 326], and no bounded region has that property for both lattices (Proposition 20.20). Where to place the two-lattice comparison is therefore the whole question. The rotation part places it on periodic tori carrying a smooth mass wall, and proves $O(3)$ invariance of King's printed state in the curvature sector from three inputs, each stated in one line and proved where it is stated: the existence theorem (E_C) of Theorem 3.3, the torus comparison (K_0) (Theorem 17.1) and the heavy-shell estimates (HS), (DS) (Theorem 20.2 and Proposition 20.1).

15.1. **The torus route.** On each lattice separately King's cubes are compared with tori carrying a *smooth mass wall*. Fix R in the generating set $\mathcal{R}_0 = \{R_{\mu\nu}(\pm\theta_0)\}$ of Definition 15.2, a pair $(f,g) \in \mathcal{K}$ (a real 2-form f and a function $g \geq 0$, both smooth with compact support) and a scheme a; at $g = 0$ nothing below depends on the scheme. For a walled torus $\mathfrak{c} = (J', w)$, the periodic model on $D_{J'}$ in which the scalar mass squared is raised by $2w$, with $w \geq 0$ smooth and vanishing near the supports (Definition 15.6, Definition 18.1), the chain is

(33)
$$Z^a_{k,L}(\mathrm{i}\,R\cdot f, R\cdot g) \;=\; Z^a_{R^{-1}\Lambda_k,\,R^{-1}X_L}(\mathrm{i}f,g) \;\leq\; T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \;\xrightarrow[k\to\infty]{}\; \mathcal{W}^a_{\mathfrak{c}}(f,g) \;\xleftarrow[k\to\infty]{}\; T^a_{\mathfrak{c},\Lambda_k}(f,g).$$

Here $Z^a_{k,L}$ is the functional of the comparison cube $X_L = C_L \cap \Lambda_k$ on the axis lattice (Definition 2.3); the first equality is a passive rotation (Lemma 15.9(b)); the inequality is a correlation inequality on the rotated lattice $R^{-1}\Lambda_k$ (Lemma 15.8), for which $R^{-1}C_L$ is a cube with faces on lattice hyperplanes; $T^a_{\mathfrak{c},\Lambda}$ is the walled ratio of Definition 15.6; and the two continuum limits exist and agree by the torus comparison (K_0) (Theorem 17.1; at $g \neq 0$ its scheme form $(\mathrm{K})_a$, Definition 17.2). The comparison cubes on the left are squeezed, as $k \to \infty$ and then $L \to \infty$, to King's printed state by the interlacing of the two cube families (the section on the interlacing). What is missing is an upper bound for walled tori on the axis lattice by King's printed state:

(34) for every $\eta > 0$ there is an admissible $\mathfrak{c}$ with $\mathcal{W}_{\mathfrak{c}}(f,0) \leq Z^K_\infty(\mathrm{i}f,0) + \eta$.

This is the closing estimate (TIGHT_0) of Theorem 18.4. It involves one lattice and one orientation, and the reverse inequality holds in the limit (Lemma 18.2), so (TIGHT_0) says that King's printed state is the lower envelope of the walled tori. With it, letting $L \to \infty$ in (33) gives $Z^K_\infty(\mathrm{i}\,R\cdot f,0) \leq Z^K_\infty(\mathrm{i}f,0)$, and applying this to $R^{-1} \in \mathcal{R}_0$ gives equality. The passage from the dense subgroup to $O(3)$ uses the Lipschitz bound of Corollary 8.3 at imaginary coefficients and the cubic symmetry of King's printed state (Theorem 18.12), so no regularity of the Schwinger functions beyond that corollary is needed.

The closing estimate (TIGHT_0) does not follow from the correlation inequalities: Remark 18.8 gives a family of Gaussian functionals with all the other properties used in the proof whose analogue of King's state is not rotation invariant. We therefore isolate the local estimate that implies (TIGHT_0). In the walled torus let the sources sit in a box H and let U be a thick shell around H inside the heavy plateau, where the wall is at least 1. Raise the scalar mass squared on the shell further by $2K$, through a second smooth profile, and let $\Psi_k(n,K)$ be the resulting expectation at lattice spacing ε_k, with $\Psi_k(n,\infty)$ the expectation with the matter removed on the shell (Definition 19.3). The decoupling property (RES_0) of Definition 19.8 says that for one such configuration and one finite K

$$\liminf_{k\to\infty} \big[\Psi_k(n,K) - \Psi_k(n,\infty)\big] \leq \eta,$$

with the width of the shell chosen so that a photon boundary term is below η as well. At every fixed spacing the bracket tends to 0 as $K \to \infty$ (Proposition 20.19); the content of (RES_0) is that the heavy-shell limit and the continuum limit can be interchanged, at one geometry and with no rate. A two-line sandwich (Lemma 19.7) shows that the heavy shell is bounded below by a higher wall, and once the matter on the shell is dead the photon on the far side can be pinned and then replaced by King's Neumann photon on a larger cube at a cost exponentially small in the width of the shell, uniformly in the spacing (Lemma 19.1); so (RES_0) implies (TIGHT_0) (Theorem 19.10); in the exterior-shell geometry the same chain is (DS) (Proposition 20.1). Under (K_0) the limit $\lim_k \Psi_k(n,K)$ exists for every finite K, and (RES_0) is equivalent to the statement that the

monotone limit $\lim_K \lim_k \Psi_k(n,K)$ lies within η of $\limsup_k \Psi_k(n,\infty)$ (Theorem 20.17); when the dead-shell limit $\lim_k \Psi_k(n,\infty)$ exists, which is a further existence statement not needed for (RES_0), this is the continuum form (RES^0), in which no lattice spacing appears. In the curvature sector (RES_0) is proved at one configuration (Theorem 20.14): the shell mass $\varkappa$ is put inside the matter covariance of King's block-spin flow, and two estimates on that flow, a two-sided bound on the stopping-scale integral uniformly in $\varkappa$ and a $\varkappa$-Cauchy property of the reduced small-field action on one test set, and a disc bound for the heavy shell, which is established there, give the tail bound at one configuration. The two estimates form (HS) (Theorem 20.2); they come from a renormalisation-group analysis of King's walled torus model with the shell mass inside the matter covariance (Propositions 20.3 and 20.13).

The torus comparison is reduced to one-sided bounds at a single lattice spacing (Theorem 17.6). The passive map $S = R^{-1}\sigma_\nu$ is an involution that exchanges the two lattices and maps the torus D_J to the torus $R^{-1}D_J$. A forward bound of King's type on each of the two tori therefore gives both directions of the comparison on D_J. Theorem 17.6 reduces (K_0) in the curvature sector to the one-sided bounds (F_0) and (V) of Definition 17.5 (bounds at one spacing, with a condition on their constants) together with the existence of the axis-lattice limits (Remark 17.9); these are what the proof of (K_0) along King's lines produces, with the large-field clause of King's upper bound in place of King's torus bound [16, (4.2)], which is vacuous at a non-zero wall (Remark 17.9, Remark 17.10 of the supplemental material). An alternative placement of the comparison, on a Neumann–Dirichlet box aligned with one lattice and a staircase for the other, is possible in principle and is not used here.

15.2. **The main theorem of the rotation part.** For $R \in O(3)$ and $b \in \mathbb{R}^3$, sources are transported as 2-forms, $R \cdot f$ and $\tau_b f$ (Definition 2.3).

Theorem 15.1 ($O(3)$ invariance of the curvature sector)**.** *King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty^K\Big(\sum_j z_j\, R\cdot f_j,\, 0\Big) = Z_\infty^K\Big(\sum_j z_j f_j,\, 0\Big),$$

so all curvature Schwinger functions are $O(3)$-covariant. With the translations of Corollary 8.3 the same holds for every Euclidean motion $x \mapsto Rx + b$, with $\tau_b R \cdot f_j$ in place of $R \cdot f_j$: the curvature sector is invariant under $\mathbb{R}^3 \rtimes O(3)$, and the curvature Schwinger functions are Euclidean covariant.

The proof is assembled at the end of the heavy-shell sections (after Theorem 20.14): the heavy-shell estimates (HS) (Theorem 20.2) give (RES_0) at one configuration (Theorem 20.14), (RES_0) gives the closing estimate (TIGHT_0) (Theorem 19.10, with the torus comparison (K_0) of Theorem 17.1 and the dead-shell chain (DS) of Proposition 20.1), and Theorem 18.12 gives the invariance from (E_C) (Theorem 3.3), (K_0) and (TIGHT_0). For the curvature two-point function the weaker form (TIGHT_2) of the closing estimate suffices (Corollary 18.13). The rotation part is organised as follows: this section sets up the lattices with a frame, the walled tori and the three lemmas (cutting, passive covariance, second moment) that every later argument uses; the next section treats Wick schemes and the full object on the comparison convention, which the scalar-source forms of the theorems need; then come the torus comparison, the closing hypothesis with the rotation theorems, the two heavy-shell sections, and the corrections to [15, 16] that the rotation part relies on.

15.3. **Geometry.**

Definition 15.2. Let $\theta_0 \in (0, \pi/2)$ with $\cos\theta_0 = \frac{4}{5}$, $\sin\theta_0 = \frac{3}{5}$. For $1 \le \mu < \nu \le 3$ let $R_{\mu\nu}(\theta)$ be the rotation with $Re_\mu = \cos\theta\, e_\mu + \sin\theta\, e_\nu$, $Re_\nu = -\sin\theta\, e_\mu + \cos\theta\, e_\nu$, $Re_\rho = e_\rho$ for $\rho \neq \mu, \nu$. Put $\mathcal{R}_0 := \{R_{\mu\nu}(\pm\theta_0) : 1 \le \mu < \nu \le 3\}$ and let G_0 be the group generated by $\mathcal{R}_0$. The spacings $\varepsilon_k = 5^{-k}$, the axis lattices $\Lambda_k = \varepsilon_k\mathbb{Z}^3$, King's large-block integer $M \ge 1$ [15, §2.3] and King's cubes $C_J = [-MJ, MJ)^3$, $J \in \mathbb{N}$ [16, (2.7)], are those of the first sections. The *King tori* of the rotation part are $D_J := \mathbb{R}^3/(50MJ\,\mathbb{Z})^3$ with fundamental cell $\Gamma_J := [-25MJ, 25MJ)^3$

[16, (4.1)]; D_J is a King torus in the sense of the section on the model ($\ell = 50MJ \in M\mathbb{N}$ is even), so Λ_k descends to it. We write $Q_b := [-b,b]^3$ (written Q_R in the first sections), $N_r(X) := \{y : \mathrm{dist}_\infty(y,X) \le r\}$, and $\bar{B}_\rho$ for the closed Euclidean ball of radius ρ about 0.

Lemma 15.3. (a) *For $R \in \mathcal{R}_0$ the matrix $5R$ has integer entries, $R^{-1} \in \mathcal{R}_0$, and $\sigma_\nu R_{\mu\nu}(\theta)\sigma_\nu = R_{\mu\nu}(-\theta)$, where σ_ν is the reflection $x_\nu \mapsto -x_\nu$.*
(b) *For $R \in \mathcal{R}_0$, $k \ge 0$ and $J \ge 1$ the lattices $R\Lambda_k$ and $R^{-1}\Lambda_k$ are invariant under translations by $50MJ\mathbb{Z}^3$, so they descend to D_J.*
(c) *For $R \in \mathcal{R}_0$, $|Rx|_\infty \le \frac{7}{5}|x|_\infty$, and $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ for $b > 0$.*
(d) *θ_0/π is irrational, and G_0 is dense in $SO(3)$.*

Proof. (a) The entries of R are $0, 1, \pm\frac{3}{5}, \pm\frac{4}{5}$, and $R_{\mu\nu}(\theta)^{-1} = R_{\mu\nu}(-\theta)$; conjugation by σ_ν reverses the orientation of the (x_μ, x_ν)-plane and fixes the third axis. (b) For $p \in 50MJ\mathbb{Z}^3$ we have $R^{\pm1}p = \frac{1}{5}(5R^{\pm1})p \in 10MJ\mathbb{Z}^3 \subset \Lambda_k$. Hence $R^{-1}\Lambda_k + p = R^{-1}(\Lambda_k + Rp) = R^{-1}\Lambda_k$, and likewise for $R\Lambda_k$. (c) The sums of the absolute values of the entries in each row of R are at most $\frac{7}{5}$. If $Rx \in Q_b$ then $|x|_\infty = |R^{-1}(Rx)|_\infty \le \frac{7}{5}b$, since $R^{-1} \in \mathcal{R}_0$; if $|x|_\infty \le \frac{5}{7}b$ then $|Rx|_\infty \le b$. (d) If $\theta_0 \in \pi\mathbb{Q}$, then $e^{\mathrm{i}\theta_0} = (4+3\mathrm{i})/5$ is a root of unity, hence an algebraic integer. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - \frac{8}{5}x + 1$, which does not have integer coefficients; this is a contradiction (compare [16, pp. 337–338]). Hence $\{n\theta_0\}_{n\in\mathbb{Z}}$ is dense modulo 2π, and the closure of $\{R_{\mu\nu}(\theta_0)^n\}_{n\in\mathbb{Z}}$ is the group of all rotations in the (x_μ, x_ν)-plane. Every element of $SO(3)$ is a product $R_{12}(\alpha)R_{23}(\beta)R_{12}(\gamma)$ (Euler angles), so $\overline{G_0} = SO(3)$. □

15.4. **Lattices with a frame.**

Definition 15.4 (lattices with a frame, plaquettes, sources)**.** A *lattice of spacing* ε_k is $\Lambda = O\Lambda_k$ with $O \in O(3)$, equipped with the frame $f_\alpha := Oe_\alpha$, $\alpha = 1,2,3$, and with the nearest-neighbour bonds $\langle x, x + \varepsilon_k f_\alpha\rangle$. (The frame vectors f_α and the 2-form sources f share a letter; the index tells them apart.) A field configuration on a set X of sites consists of $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and $A_\alpha(x) \in \mathbb{R}$, $x \in X$; the bond $\langle x, x+\varepsilon f_\alpha\rangle$ carries $A_{\langle x,x+\varepsilon f_\alpha\rangle} := A_\alpha(x)$ and the reversed bond carries $-A_\alpha(x)$ (Feynman form of the vector field, [16, (2.5)]). The action on Λ is King's action of the section on the model written in the frame, e_α replaced by f_α throughout, with the same free-boundary counterterm δm_ε^2 and the same free Wick constant W_ε, which depend on ε only. A plaquette is $p = (x;\alpha,\beta)$ with $\alpha < \beta$; its centre is $c(p) := x + \frac{\varepsilon}{2}(f_\alpha + f_\beta)$ and its field strength is

$$F(p) := \varepsilon^{-1}\big(A_\alpha(x) + A_\beta(x+\varepsilon f_\alpha) - A_\alpha(x+\varepsilon f_\beta) - A_\beta(x)\big) \qquad [16, (2.11)].$$

For a real 2-form f and a real function h, smooth and compactly supported (on a torus: smooth and periodic; a compactly supported source whose support lies in the interior of a fundamental cell is periodised from it), put

$$F_\Lambda(f) := \sum_p \varepsilon^3 f_{c(p)}(f_\alpha, f_\beta)\, F(p), \qquad \mathsf{S}_\Lambda(h) := \sum_x \varepsilon^3 h(x), \qquad W_\Lambda(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big),$$

the sums running over the plaquettes $p = (x;\alpha,\beta)$, respectively the sites, of the region (of one fundamental cell on a torus): the 2-form is *sampled in the lattice's own frame*. On Λ_k with the standard frame these are $F_{\Lambda_k}(f)$, $\mathsf{S}_{\Lambda_k}(h)$ and ${:}\phi^2{:}(h)$ of Definition 2.3. For a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with complex coefficients, $\Theta(\zeta) := \exp(\sum_j z_j F_\Lambda(f_j) + \sum_j w_j W_\Lambda(h_j))$ as there, and $O\cdot\zeta := (z_j, O\cdot f_j; w_j, O\cdot h_j)_j$ with the transport of Definition 2.3. The cone $\mathcal{K}$ of Definition 2.3 is invariant under $O(3)$, under translations, under $(f,g) \mapsto (tf, tg)$, $t \ge 0$, and under sums.

Remark 15.5 (the frame convention). In [15, 16] the curvature source is written $F(g) = \sum_p \varepsilon^3 g(p)F(p)$ with one function g on all plaquettes [16, (2.12)]. On one lattice a 2-form sampled as above is a function on plaquettes, and King's proofs treat the source terms componentwise ("the sources $g_{\mu\nu}(x'), h(x') \ldots$ can be replaced by $g_{\mu\nu}(x), h(x)$", [15, p. 665]; the sentence is stated there without an argument). King's single-lattice statements at $h = 0$ are used in this sense throughout: the existence theorem (E_C) is stated, in Theorem 3.3, for real smooth 2-form sources sampled at plaquette centres in the lattice's own frame, and this existence clause of King's Theorem 2.1

of [16] is proved in Section 5; the statements of [16, Thms. 2.2, 2.3, (3.40)–(3.44)] that the rotation part uses, all at $h = 0$, are established in the first sections from (E_C) and the correlation inequalities, [16, (3.42)–(3.43)] are not used, and the torus statement [15, Thm. 2.1] at a wall is part of the torus comparison (K_0) (Theorem 17.1). The two-lattice statement is different: Proposition H.2 of the supplemental material shows that it fails for a single scalar function on all plaquettes, and (K_0) is formulated for 2-forms sampled in each lattice's own frame.

15.5. **Tori and walled tori.**

Definition 15.6 (periodic and walled torus states)**.** Let Λ be a lattice of spacing $\varepsilon = \varepsilon_k$ with a frame, and $d\nu := \prod_x \prod_\alpha dA_\alpha(x)\, d^2\phi(x)$ over the sites in question.

(a) *Boxes.* For a finite set X of sites of Λ, $\langle\cdot\rangle^{\mathrm{ND}}_{\Lambda,X}$ is the Neumann–Dirichlet state of Definition 2.1 written in the frame of Λ (Dirichlet condition for the scalar field outside X, Neumann condition for the vector field), possibly with the matter killed on a set $U \subset X$ in the sense of that definition. On Λ_k these are the states $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ of Definition 2.1, among them the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ and King's RG cube $\langle\cdot\rangle^{K}_{k,J}$ (Definition 2.2).

(b) *Tori.* If Λ descends to D_J, $\langle\cdot\rangle_{\Lambda,D_J}$ is the normalised measure $Z^{-1}e^{-S}d\nu$ with the action S summed over the sites and bonds of the finite graph $\Lambda/(50MJ\,\mathbb{Z})^3$ (periodic conditions for both fields). On Λ_k this is the periodic state $\langle\cdot\rangle_{k,T}$ of the section on the model for the King torus $T = D_J$.

(c) *Walled tori.* A *walled configuration* is $\mathfrak{c} = (J', w)$ with $J' \in \mathbb{N}$ and $w \in C^\infty(D_{J'}; [0,\infty))$, the *wall*. If Λ descends to $D_{J'}$, $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ is the torus state of (b) with the density multiplied by $\prod_x \exp(-\varepsilon^3 w(x)|\phi(x)|^2)$. The wall raises the scalar mass squared by $2w$ and acts in the single-site measures. We write $(J', \psi, n) := (J', n\psi)$ for $n \geq 0$ and a fixed profile ψ.

(d) *Killed sites.* For any of the states (b), (c) and a set U of sites, the state *with the matter killed on* U is defined as in Definition 2.1: the single-site measure at $x \in U$ is replaced by the point mass at $\phi(x) = 0$, every hopping term with an endpoint in U reduces to its diagonal part at the other endpoint, and the photon variables are unchanged.

For every state $\langle\cdot\rangle_{\Lambda,X}$ above and a scheme a (Definition 2.3, Remark 2.4) the scheme functional is defined by the formula of Definition 2.3 with F_Λ, W_Λ, S_Λ in place of their axis-lattice versions,

$$Z^a_{\Lambda,X}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)} \big\langle \Theta(\zeta) \big\rangle_{\Lambda,X};$$

on Λ_k we keep the notation $Z^a_{k,X}$ of Definition 2.3, and on the rotated lattice we write $Z^a_{R^{-1}\Lambda_k,X}$. All these states are invariant under the charge conjugation $(A,\vartheta) \mapsto (-A,-\vartheta)$, which maps $F_\Lambda(f)$ to $-F_\Lambda(f)$ and fixes r; hence $\langle\Theta(\mathrm{i}f,g)\rangle = \langle\cos F_\Lambda(f)\, e^{W_\Lambda(g)}\rangle$ is real for $(f,g) \in \mathcal{K}$.

Definition 15.7 (the unnormalised torus functional and the walled ratio)**.** For a lattice Λ of spacing ε_k descending to a torus T, and real smooth periodic sources G (a 2-form) and h (a function), the *unnormalised* generating functional is

$$\mathsf{Z}_{\Lambda,T}(G,h) := \int e^{-S(A,\phi)+F_\Lambda(G)+W_\Lambda(h)}\, d\nu,$$

with no field-independent constant in the action; for a pair $s = (G,h)$ we also write $\mathsf{Z}_{\Lambda,T}(s)$. Two lattices of the same spacing descending to T have the same number of sites in a fundamental cell, so any field-independent constant per site multiplies both by the same factor. For a walled configuration $\mathfrak{c} = (J', w)$, a lattice Λ descending to $D_{J'}$, a scheme a and $(f,g) \in \mathcal{K}$ with $w = 0$ on the support of g, the *walled ratio* is

$$T^a_{\mathfrak{c},\Lambda}(f,g) := e^{-a_k \mathsf{S}_\Lambda(g)} \big\langle \Theta(\mathrm{i}f,g) \big\rangle_{\mathfrak{c},\Lambda} = \frac{Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g-w)}{Z^a_{\Lambda,D_{J'}}(0,-w)}.$$

The second expression agrees with the first because

$$e^{-W_\Lambda(w)} = e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)} \prod_x e^{-\varepsilon^3 w(x)|\phi(x)|^2},$$

so that the wall is the scalar source $-w$ up to the constant $e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)}$, which cancels between numerator and denominator, as do the multipliers $e^{a_k \mathsf{S}_\Lambda(w)}$ of both. The same formulas define the walled ratio for the periodic model of Λ on any torus $T = \mathbb{R}^3/P$ to which Λ descends (defined as in Definition 15.6(b), with the finite graph Λ/P) and any smooth wall $w \geq 0$ on T vanishing on the support of g; this is used only for the transported tori of Lemma 15.9(b).

Every state of Definition 15.6, with or without killed sites, is a BFS measure of King type in the sense of the section on the model: $c_{xy} = \varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$, the diagonal parts $\frac{\varepsilon}{2}r_x^2$ of all bonds at x (including, for boxes, those leading out of X, and those leading into a killed site) and the wall factor $e^{-\varepsilon^3 w(x) r_x^2}$ with $w \geq 0$; the photon form is positive definite because $\mu_0 > 0$, and the exponential moments are finite because $\lambda > 0$. The same holds for every measure obtained from such a state by multiplying some c_{xy} by $t \in [0,1]$, multiplying photon gradient terms by $t \in [0,1]$, adding $s|\ell(A)|^2$ ($s \geq 0$) to the action, multiplying single-site measures by e^{-sr^2} ($s \geq 0$), or killing sites. Curvature sources $F_\Lambda(f)$ are real linear functionals of A, and $e^{W_\Lambda(g)}$ is a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$; so Lemma 2.7, Lemma 2.8 and Lemma 2.9 apply to all of them, on every lattice with a frame.

15.6. **Cutting, passive covariance and the second moment.**

Lemma 15.8 (cutting, walled tori)**.** *Let $\langle\cdot\rangle_Y$ be a box, torus or walled-torus state of Definition 15.6 on a lattice Λ with a frame, possibly with killed sites, and let X be a set of sites of Λ such that X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (wall, killed sites and diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let $(f,g) \in \mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,*

$$\langle \Theta(\mathrm{i}f, g)\rangle_X^{\mathrm{ND}} \leq \langle \Theta(\mathrm{i}f, g)\rangle_Y, \qquad \langle \Theta(f, -g)\rangle_X^{\mathrm{ND}} \geq \langle \Theta(f, -g)\rangle_Y,$$

and the same for Z^a. If the wall of Y vanishes on X and no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}} = \langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the box state of Definition 15.6(a).

Proof. The wall and the killed sites are part of the single-site measures, and the frame plays no role, so the proof of Lemma 2.10 applies word for word. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2+r_y^2)$; since X and its neighbours lie in one fundamental cell, these are bonds between distinct sites of the finite graph. By Lemma 2.7(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The resulting measure is a product (Lemma 2.7(e)); its X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$, since the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms are the Neumann ones, and the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier $e^{\mp a_k \mathsf{S}_\Lambda(g)}$ are the same in both states. □

With $Y = X_{J+1}$ and $X = X_J$ on Λ_k this is King's monotonicity in the volume [16, Thm. 3.2, (3.31)–(3.33)] in the form his proof proves, with one Wick constant for all J, as discussed after Lemma 2.10; with Y a walled torus and X a cube of Λ inside the cell, it is the inequality of (33).

Lemma 15.9 (passive covariance and the algebra of the walled ratio)**.** *Let $\mathfrak{a}$ be a scheme.*

(a) *For real coefficients the multiplier $e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}$ is a positive constant depending only on k, the coefficients and the numbers $\mathsf{S}_\Lambda(h_j)$. Hence the inequalities of Lemma 2.7 and Lemma 15.8 between two states on the same lattice with the same sources hold for Z^a, and for $T^a_{\mathfrak{c},\Lambda}$, in the same direction.*

(b) *Let $O \in O(3)$, let Λ be a lattice of spacing ε with frame (f_α), and let X be a box, a torus or a walled torus for Λ, possibly with the matter killed on U. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions, the wall $O\cdot w$ and the killed set OU; for*

a torus $X = \mathbb{R}^3/P$ this means the torus $OX = \mathbb{R}^3/OP$, to which $O\Lambda$ descends. Then for every finite family ζ

$$\big\langle\Theta(O\cdot\zeta)\big\rangle_{O\Lambda,OX} = \big\langle\Theta(\zeta)\big\rangle_{\Lambda,X}, \qquad \mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h), \qquad Z^a_{O\Lambda,OX}(O\cdot\zeta) = Z^a_{\Lambda,X}(\zeta).$$

In particular, for a walled configuration $\mathfrak{c} = (J', w)$ and (f, g) as in Definition 15.7, $T^a_{O\cdot\mathfrak{c},O\Lambda}(O\cdot f, O\cdot g) = T^a_{\mathfrak{c},\Lambda}(f,g)$, where $O\cdot\mathfrak{c}$ denotes the transported walled torus, the torus $OD_{J'}$ with the wall $O\cdot w$, and the left side is the walled ratio on that torus in the sense of Definition 15.7; $O\cdot\mathfrak{c}$ is a walled configuration $(J', O\cdot w)$ of Definition 15.6(c) when $OD_{J'} = D_{J'}$, as for a signed permutation matrix O. Moreover, on every torus the periodic model on Λ does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, for a permutation π and any choice of signs.

(c) $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k\mathsf{S}_\Lambda(g)}\langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda}$ *is the normalised expectation of the walled torus state up to the multiplier of* (a)*, and at $g = 0$ the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,0) = \langle e^{\mathrm{i}F_\Lambda(f)}\rangle_{\mathfrak{c},\Lambda}$ does not depend on the scheme.*

Proof. (a) The first statement is immediate (compare the remark in [7, p. 281] that the inequalities are not affected by Wick ordering); for $T^a_{\mathfrak{c},\Lambda}$ the multiplier is $e^{-a_k\mathsf{S}_\Lambda(g)}$. (b) The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, with components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving $d\nu$; this is the proof of Lemma 2.12, which covers boxes and tori, and for a torus $X = \mathbb{R}^3/P$ it carries the finite graph Λ/P onto $O\Lambda/OP$. The action is written in the frame, and δm^2_ε, W_ε and a_k depend on ε only; the wall term $\sum_x \varepsilon^3 w(x)|\phi(x)|^2$ becomes $\sum_x \varepsilon^3 (O\cdot w)(Ox)|\phi'(Ox)|^2$ since $(O\cdot w)(Ox) = w(x)$, and a point mass at $\phi(x) = 0$ becomes a point mass at $\phi'(Ox) = 0$. Plaquettes, their centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$, $\sum_{Ox}\varepsilon^3(O\cdot h)(Ox) = \sum_x \varepsilon^3 h(x)$. The clause on the walled ratio is the first identity for $\zeta = (\mathrm{i}, f; 1, g)$ and the walled torus $X = D_{J'}$, together with $\mathsf{S}_{O\Lambda}(O\cdot g) = \mathsf{S}_\Lambda(g)$ and the first expression of Definition 15.7 read on $OD_{J'}$. For the last statement, a permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables that preserves the action, and changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ while preserving the set of plaquette centres. (c) The expression for $T^a_{\mathfrak{c},\Lambda}$ is its definition, and at $g = 0$ the multiplier is 1 and $W_\Lambda(0) = 0$. □

Lemma 15.10 (the characteristic function and the second moment of a photon functional)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C, and let $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$ and $\mathbb{E}^\sharp$ be the current representation and the current measure of Lemma 2.8. Then for $v \in \mathbb{R}^m$:*

(a) $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle$*; in particular $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle \in [0,1]$ is real;*

(b) $\langle\langle v,A\rangle\rangle = 0$ *and* $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 \in [0, \langle v,Cv\rangle]$.

Proof. (a) For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a,A\rangle}d\gamma = e^{\langle a,Ca\rangle/2}$. With $a = \mathrm{i}(v \pm \kappa)$ and the average of the two signs,

$$\int e^{\mathrm{i}\langle v,A\rangle}\cos\langle\kappa,A\rangle\,d\gamma = e^{-\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cosh\langle v,C\kappa\rangle,$$

since C is symmetric. Multiply by $w(q) \ge 0$, sum over the currents q (the series converges absolutely: $\cosh\langle v,C\kappa\rangle \le e^{|\langle v,C\kappa\rangle|} \le e^{\langle v,Cv\rangle/2+\langle\kappa,C\kappa\rangle/2}$ by the Cauchy–Schwarz inequality for the form C, so the terms are dominated by $w(q)$ and $\sum_q w(q) < \infty$), and divide by the same sum at $v = 0$, which is the positive normalisation $\sum_q w(q)e^{-\langle\kappa,C\kappa\rangle/2}$ of $\mathbb{E}^\sharp$; this is the formula. Its right side lies in $[0,\infty)$, and $|\langle e^{\mathrm{i}\langle v,A\rangle}\rangle| \le 1$. (b) Replace v by tv, $t \in \mathbb{R}$, in (a). The left side is the characteristic function of the real random variable $\langle v,A\rangle$, which has all moments because $\langle e^{s\langle v,A\rangle}\rangle \le e^{s^2\langle v,Cv\rangle/2}$ for every real s (Lemma 2.8); so it equals $1+\mathrm{i}t\langle\langle v,A\rangle\rangle - \frac{t^2}{2}\langle\langle v,A\rangle^2\rangle + O(t^3)$. The right side is $e^{-t^2\langle v,Cv\rangle/2}\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle)$, and $\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle) = 1 + \frac{t^2}{2}\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 + O(t^4)$, the expansion under $\mathbb{E}^\sharp$ being justified by the domination $\cosh(t\langle Cv,\kappa\rangle) \le e^{t^2\langle v,Cv\rangle/2+\langle\kappa,C\kappa\rangle/2}$ of (a), which shows that $\langle Cv,\kappa\rangle$ has all moments under $\mathbb{E}^\sharp$. The right side is real and even in t, so $\langle\langle v,A\rangle\rangle = 0$; comparing the terms of order t^2 gives $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$. The bounds follow since both $\langle\langle v,A\rangle^2\rangle$ and $\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$ are non-negative. □

The lemma applies to every state of Definition 15.6 and to every state of the first sections, with v the vector $d^*_\Lambda f$ defined by $F_\Lambda(f) = \langle d^*_\Lambda f, A\rangle$ for a real 2-form f whose plaquettes carry variables of the state; the photon Gaussian of a torus or walled torus is the periodic Feynman form with mass μ_0, and killed sites are covered as in Lemma 2.8.

16. Schemes, admissibility and the full object

This section fixes the scheme in which generating functionals with scalar sources are taken, defines W-admissibility of a scheme, and proves the full-object theorem (Theorem 16.6): the iterated limit of the comparison cubes in a W-admissible scheme exists for all curvature and scalar sources, is entire and translation invariant, and depends on the scheme only through one explicit factor. The curvature sector $h = 0$ of all this is scheme-free and is already in place (Theorem 7.6, Lemma 7.7); the rotation statements of the Wick sector (Theorem 18.9, Theorem 18.11) are stated for the object defined here.

16.1. **The Wick square.**

Remark 16.1 (a logarithm in the Wick square). With the free Wick constant W_ε alone, the expectation of the Wick square $:\phi^2:(h)$ of Definition 2.3 diverges at order e^2 in perturbation theory: for the model on $\varepsilon\mathbb{Z}^3$, at order $e^2\lambda^0$,

$$\big\langle :\phi^2:(h)\big\rangle = \frac{e^2}{4\pi^2}\,\ln(1/\varepsilon)\int h + O(1),$$

with a coefficient that depends neither on m nor on μ_0. We indicate the computation, in formal perturbation theory. In the normalisation of the action each real component of ϕ has propagator $(p^2+m^2)^{-1}$ at $e=0$. Write the one-loop correction to the inverse propagator as $\Sigma(p)$; the mass counterterm removes $\Sigma(0)$. The photon-exchange part is $-e^2\int \frac{d^3k}{(2\pi)^3}\frac{(2p-k)^2}{((p-k)^2+m^2)(k^2+\mu_0^2)}$; writing $(2p-k)^2 = 2p^2 + 2(p-k)^2 - k^2$ and using $\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(p-k)^2} = \frac{1}{8|p|}$, one finds $\Sigma(p)-\Sigma(0) = -\frac{e^2}{4}|p| + O(m+\mu_0)$ for $1 \ll |p| \ll \varepsilon^{-1}$, the lattice regularisation changing only momentum-independent terms and terms of order εp^2. Inserting,

$$\langle|\phi(0)|^2\rangle - 2\mathsf{W}_\varepsilon = -2\int \frac{d^3p}{(2\pi)^3}\frac{\Sigma(p)-\Sigma(0)}{(p^2+m^2)^2} + O(e^4) = \frac{e^2}{2}\cdot\frac{1}{2\pi^2}\ln(1/\varepsilon) + O(1)$$

at order e^2. Two consequences.

(a) For h with $\int h \neq 0$ the generating functionals with scalar sources in King's printed Wick ordering have no finite continuum limit beyond the curvature sector, at least in perturbation theory: for $h \geq 0$, $h \not\equiv 0$, Jensen's inequality gives $\langle e^{:\phi^2:(h)}\rangle \geq e^{\langle:\phi^2:(h)\rangle} \to \infty$. This affects [15, Thm. 2.1] and [16, Thms. 2.1–2.3, (3.43), (4.2)] for scalar sources with $\int h \neq 0$. In the cube King subtracts the Dirichlet constant [16, (2.13)] and, at distance > 1 from the boundary, the free one, which differ by a convergent amount only. The statement of [15, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$, $\alpha > 0$, in the small-field region does not hold for the marginal graph that produces the logarithm. We therefore allow an additional vacuum counterterm, linear in h and depending only on the lattice spacing: the schemes of Definition 16.2. Walled ratios (Definition 15.6) and the curvature sector $h = 0$ are unaffected, since the h-linear c-numbers cancel in them.

(b) King's proof of [16, Thm. 2.4] compares the effective action generated from spacing ε by a first block-spin step with rotated blocks with the effective action generated from the rotated lattice of spacing 5ε [16, (4.3)–(4.9)]. In a scheme a the two generating functionals carry the multipliers $e^{-wa_\varepsilon \mathsf{S}(h)}$ and $e^{-wa_{5\varepsilon}\mathsf{S}(h)}$. If a removes the logarithm, then $a_\varepsilon - a_{5\varepsilon} \to \frac{e^2}{4\pi^2}\ln 5$, so the comparison [16, (4.6)] must carry the c-number $\frac{e^2}{4\pi^2}\ln 5\int h$, and as printed it cannot hold for $\int h \neq 0$. Since a walled ratio contains the wall source $-w$ with $\int w > 0$, this enters the proof of walled comparisons even though their statement is scheme-free. In the equal-spacing formulation of the torus comparison (Theorem 17.1) the two lattices carry the same a_k, and the issue does not arise.

16.2. **Schemes.**

Definition 16.2 (schemes)**.** A *scheme* is a real sequence $a = (a_k)_{k\ge 0}$ such that a_k depends only on ε_k and on (e, λ, m, μ_0), and $Z^a_{k,X}(\zeta) = e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)} \langle \Theta(\zeta)\rangle^{\mathrm{ND}}_{k,X}$ is the functional of Definition 2.3, the expectation of $\exp(\sum_j z_j F(f_j) + \sum_j w_j \sum_x \varepsilon_k^3 h_j(x)[|\phi(x)|^2 - 2\mathsf{W}_{\varepsilon_k} - a_k])$; the same multiplier defines $Z^a_{\Lambda,T}(\zeta)$ for the periodic and walled states of Definition 15.6 on a lattice Λ of spacing ε_k. Three schemes are used.

(i) The *printed scheme* $a \equiv 0$.

(ii) The *reference scheme* $z = a^\sharp$ of Definition 2.3: $a^\sharp_k = e^2 X_{\varepsilon_k}$, where X_ε is the coefficient of e^2 in the perturbative expansion of $\langle |\phi(0)|^2\rangle$ for the model on $\varepsilon\mathbb{Z}^3$ at $\lambda = 0$, with the order-e^2 part of δm^2_ε inserted.

(iii) The *self-normalised scheme*: fix J_0 and $0 \le h_0 \in C^\infty_c(\mathbb{R}^3)$, $h_0 \not\equiv 0$, supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and put $a^{sn}_k := \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^0_{k,J_0}(0, h_0)$ when $\mathsf{S}_{\Lambda_k}(h_0) > 0$, and $a^{sn}_k := 0$ otherwise; here $Z^0_{k,J_0}(0,h_0) = \langle e^{:\phi^2:(h_0)}\rangle^{(a)}_{k,J}$ at $J = J_0$ is the functional of the comparison cube in the printed scheme.

At order e^2 the reference scheme removes the logarithm of Remark 16.1 identically on $\varepsilon\mathbb{Z}^3$: by translation invariance the order-$e^2\lambda^0$ part of $\langle :\phi^2:(h)\rangle$ there equals $e^2 X_\varepsilon \mathsf{S}(h)$.

16.3. **W-admissible schemes.**

Definition 16.3 ($(\mathrm{W})_a$: W-admissibility of the scheme a)**.** Let a be a scheme. $(\mathrm{W})_a$ is the statement (W2) of Hypothesis 23.1 for the scheme a, posed on the comparison convention $\mathrm{ND}^{(a)}$, in the following two clauses:

$(\mathrm{W1})^{(a)}$ For every King cube C_J, every finite family of curvature sources f_j and scalar sources h_j supported at ℓ^∞-distance > 1 from $\mathbb{R}^3\setminus C_J$, and every $(z, w) \in \mathbb{R}^{2N}$, the limit $\lim_{k\to\infty} Z^a_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0, \infty)$.

$(\mathrm{W2})^{(a)}$ For every $0 \le q \in C^\infty_c(\mathbb{R}^3)$, $\sup_{J\ge J_q} \lim_k Z^a_{k,J}(0, q) < \infty$, where J_q is the least J such that $\operatorname{supp} q$ has ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

A scheme with $(\mathrm{W})_a$ is called W-*admissible*. Below the tube threshold, $(\mathrm{W1})^{(a)}$ is the existence theorem with scalar sources, Theorem 3.6; at arbitrary amplitudes (W2) is a hypothesis of the optional extension of the Wick leg (Theorem 23.7), and so are the Wick-sector statements of this part that assume $(\mathrm{W})_a$. Two W-admissible schemes are compared in Lemma 16.4; by its part (b) they differ by a convergent sequence, so a W-admissible scheme is admissible in the sense of Definition 3.7 as soon as the reference scheme is.

$(\mathrm{W1})^{(a)}$ is the analogue in the scheme a of [16, Thm. 2.1], and $(\mathrm{W2})^{(a)}$ of the bound [16, (3.43)], which King asserts for the Neumann–Dirichlet cubes from (3.42), reflection positivity and ultra-violet stability by "a standard argument (see [GJ1])" (not printed) and which, in King's Wick ordering, fails for $\int h_+ > 0$ at order e^2 in perturbation theory (Remark 16.1). At $h = 0$ no multiplier enters: $(\mathrm{W1})^{(a)}$ at zero Wick coefficient is $(\mathrm{E}_C)^{(a)}$, the existence theorem (E_C) of Theorem 3.3 read for the comparison family, with 2-form sources read as in Remark 15.5, and $(\mathrm{W2})^{(a)}$ is not needed, the curvature sector being Lemma 7.7 on the comparison convention and Theorem 7.6 on King's.

Lemma 16.4 (comparison of schemes)**.** (a) *If a is* W*-admissible and $a'_k - a_k \to \kappa_{a'a} \in \mathbb{R}$, then a' is* W*-admissible, and for every family as in* $(\mathrm{W1})^{(a)}$, $\lim_k Z^{a'}_{k,J}(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} \lim_k Z^a_{k,J}(\zeta)$.

(b) *If a and a' are* W*-admissible, then $a'_k - a_k$ converges.*

(c) *If some scheme is* W*-admissible, then a^{sn} is* W*-admissible.*

Proof. (a) $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a_{k,J}(\zeta)$, and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$ (a Riemann sum of a smooth compactly supported function). (b) For $0 \le h \in C^\infty_c$ with $\int h > 0$, supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, the ratio $Z^{a'}_{k,J}(0,h)/Z^a_{k,J}(0,h) = e^{-(a'_k - a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$ by $(\mathrm{W1})^{(a)}$ and $(\mathrm{W1})^{(a')}$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$; this is the argument of

Remark 2.4. (c) If a is W-admissible, $a_k^{sn} - a_k = \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^a_{k,J_0}(0,h_0)$ for all large k, which converges by $(\mathrm{W1})^{(a)}$ and $\mathsf{S}_{\Lambda_k}(h_0) \to \int h_0 > 0$; apply (a). □

Remark 16.5 (the class of W-admissible schemes). By Lemma 16.4, $(\mathrm{W})_a$ holds for some scheme if and only if it holds for a^{sn}, and W-admissible schemes form a single class modulo convergent sequences; the reference scheme $a^\sharp$ is W-admissible if and only if some scheme is and $a^\sharp - a^{sn}$ converges. If the logarithm of Remark 16.1 is present beyond perturbation theory, the printed scheme $a \equiv 0$ is not W-admissible. A proof of $(\mathrm{W})_a$ would carry the counterterm $a_k\mathsf{S}(h)$ as an additional h-linear vacuum vertex through King's renormalisation group. $(\mathrm{W})_a$ enters none of the curvature-sector results of this part: at $g = 0$ every walled quantity is scheme-free (Lemma 15.9), and King's printed state on curvature sources is Z^K_∞ of Theorem 7.6, under (E_C) alone.

16.4. **The full object in a W-admissible scheme.** For $\rho > 0$ let $\mathcal{E}_\rho := C^\infty_c(\bar B_\rho;\Lambda^2) \times C^\infty_c(\bar B_\rho)$ be the space of pairs $s = (f,h)$ of smooth real curvature and scalar sources supported in the closed ball $\bar B_\rho$, with the C^∞ topology. It is a closed subspace of $C^\infty(\bar B_{\rho+1};\Lambda^2) \times C^\infty(\bar B_{\rho+1})$, hence a Fréchet space, and it is invariant under $O(3)$ and under the translations τ_b in the sense that τ_b maps $\mathcal{E}_\rho$ into $\mathcal{E}_{\rho+|b|}$. (The 2-form space $E_\rho = C^\infty_c(\bar B_\rho;\Lambda^2)$ of the translation section is the first factor.) For a W-admissible scheme a and a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with all supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ write

$$Z^a_J(\zeta) := \lim_{k\to\infty} Z^a_{k,J}(\zeta),$$

the limit of the comparison cubes at fixed J, which exists at real coefficients by $(\mathrm{W1})^{(a)}$. We use the total-source convention of the translation section: a functional depends on a family only through its total sources $(\sum_j z_j f_j, \sum_j w_j h_j)$, and we write $Z^a_J(s)$, $Z^a_\infty(s)$ for $s \in \mathcal{E}_\rho$.

Theorem 16.6 (the full-object theorem)**.** *Let a be a* W*-admissible scheme,* $(\mathrm{W})_a$.

(i) *For C_J and families as in* $(\mathrm{W1})^{(a)}$, *$Z^a_{k,J} \to Z^a_J$ locally uniformly on $\mathbb{C}^{2N}$ as $k \to \infty$, and Z^a_J is entire.*

(ii) *For $(f,g) \in \mathcal{K}$ supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \mapsto Z^a_J(\mathrm{i}f, g)$ is nondecreasing and $J \mapsto Z^a_J(f,-g)$ is nonincreasing on $J \ge J_0$.*

(iii) *For every finite family of curvature and scalar sources the limit $Z^a_\infty(\zeta) := \lim_{J\to\infty} Z^a_J(\zeta)$ exists locally uniformly on $\mathbb{C}^{2N}$ and is entire; $Z^a_\infty(f,h) > 0$ for real sources; and $Z^a_J \le Z^a_\infty$ on $\mathcal{K}$.*

(iv) *For every $\rho > 0$, $\Phi^a(s) := \ln Z^a_\infty(s)$ is finite, convex and continuous on $\mathcal{E}_\rho$.*

(v) *$Z^a_\infty(\tau_b \cdot \zeta) = Z^a_\infty(\zeta)$ for all $b \in \mathbb{R}^3$, where τ_b acts on all sources of the family.*

(vi) *If a' is another* W*-admissible scheme, then $a'_k - a_k \to \kappa_{a'a} \in \mathbb{R}$ and $Z^{a'}_\infty(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} Z^a_\infty(\zeta)$.*

Proof. (i) At fixed k the functionals are entire in the coefficients: a finite-dimensional integral with Gaussian decay in A and decay $e^{-\lambda\varepsilon^3 r^4}$ in r. Fix a compact $K \subset \mathbb{C}^{2N}$ and a real box $Q \supset \operatorname{Re} K$. By Lemma 2.11(d), $\sup_K |Z^a_{k,J}| \le \sup_Q Z^a_{k,J}$, and since $\ln Z^a_{k,J}$ is convex in the real coefficients the latter is the maximum over the vertices of Q, at which the sequences converge by $(\mathrm{W1})^{(a)}$, hence are bounded. So the family $\{Z^a_{k,J}\}_k$ is locally bounded; by Montel's theorem and since $\mathbb{R}^{2N}$ is a set of uniqueness for entire functions on $\mathbb{C}^{2N}$, it converges locally uniformly, and the limit is entire [11]. (Compare [16, (3.40)–(3.44)]: Lemma 2.11(d) plays the role of (3.40); King's (3.42)–(3.43) are not used, $(\mathrm{W2})^{(a)}$ replacing (3.43).)

(ii) The sources of (f,g) lie inside X_J for $J \ge J_0$: the corners of a plaquette lie within ℓ^∞-distance $\varepsilon/2 \le \frac12$ of its centre, so every plaquette and site at which (f,g) does not vanish lies in C_{J_0}. Lemma 2.10 with $Y = X_{J+1}$, $X = X_J$ (no killed sites, so the cut-out factor is the comparison cube) and Lemma 2.11(a) give both inequalities at fixed k for $Z^a_{k,J}$; let $k \to \infty$ by (i).

(iii) Split each $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$ and $h_j^+ := h_j + h_j^-$, where $\chi_j \in C^\infty_c$ takes values in $[0,1]$, equals 1 on $\operatorname{supp} h_j$ and is supported in $N_{1/2}(\operatorname{supp} h_j)$; both $h_j^\pm$ are smooth and

non-negative. Let J_0 be such that all supports (those of the f_j, $h_j^{\pm}$) have ℓ^∞-distance >1 from $\mathbb{R}^3 \setminus C_{J_0}$, and for $J \geq J_0$ and $(z, w^+, w^-) \in \mathbb{C}^{3N}$ let

$$F_J(z, w^+, w^-) := Z_J^a\Big(\sum_j z_j f_j,\ \sum_j (w_j^+ h_j^+ + w_j^- h_j^-)\Big),$$

entire by (i). *Local bound.* For real $\zeta = (z, w^+, w^-)$ let $k_1 := \sum_j [(w_j^+)_+ h_j^+ + (w_j^-)_+ h_j^-]$ and $k_2 := \sum_j [(w_j^+)_- h_j^+ + (w_j^-)_- h_j^-]$, where $x_\pm = \max(\pm x, 0)$; both are smooth and non-negative, and the total scalar source is $k_1 - k_2$. By Lemma 2.11(f), passed to the limit $k \to \infty$ by (i),

$$F_J(\zeta) \leq Z_J^a\Big(2\sum_j z_j f_j, -2k_2\Big)^{1/2} Z_J^a(0, 2k_1)^{1/2}.$$

The first factor is nonincreasing in J by (ii), hence at most its value at J_0; the second is bounded uniformly in J by $(\mathrm{W2})^{(a)}$ applied to $q = 2k_1$ (whose support lies at distance >1 from $\mathbb{R}^3 \setminus C_{J_0}$ by the choice of J_0). By the convexity of $\ln F_J$ in the real coefficients and Lemma 2.11(d) (both inherited from $Z_{k,J}^a$ through (i)), the family $\{F_J\}_{J \geq J_0}$ is locally bounded on $\mathbb{C}^{3N}$. *Convergence on a set of uniqueness.* On $\Omega_+ := (\mathrm{i}\mathbb{R})^N \times (0,\infty)^{2N}$ the total sources are $(\mathrm{i}f, g)$ with f real and $g = \sum_j (w_j^+ h_j^+ + w_j^- h_j^-) \geq 0$, that is, they lie in $\mathcal{K}$; so $J \mapsto F_J$ is nondecreasing on Ω_+ by (ii) and bounded, hence convergent. Ω_+ is a non-empty open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^N \times \mathbb{R}^{2N}$ of maximal real dimension, hence a set of uniqueness for entire functions on $\mathbb{C}^{3N}$. By Montel's theorem, $F_J \to F_\infty$ locally uniformly on $\mathbb{C}^{3N}$, with F_∞ entire [11]. Setting $w^+ = w$ and $w^- = -w$ gives the claim for the original family, since $\sum_j w_j h_j = \sum_j (w_j h_j^+ - w_j h_j^-)$; the limit is $\lim_J Z_J^a(\sum_j z_j f_j, \sum_j w_j h_j)$ and does not depend on the splitting. Positivity at real sources follows from Lemma 2.11(e), $Z_J^a(f,h) Z_J^a(-f,-h) \geq 1$, in the limit $J \to \infty$ (both factors are finite, so neither vanishes), and the last claim from (ii).

(iv) Φ^a is finite by the positivity in (iii), and convex as a pointwise limit of the convex functions $s \mapsto \ln Z_{k,J}^a(s)$. For J so large that $N_1(\bar{B}_\rho) \subset C_J$, and each k, the function $s \mapsto \ln Z_{k,J}^a(s)$ on $\mathcal{E}_\rho$ depends on s only through the finitely many sampled values $f_{c(p)}(e_\alpha, e_\beta)$, $h(x)$, continuous functionals of s in the C^∞ topology, and the finite-dimensional integral depends continuously on them (dominated convergence); so it is continuous, and $\Phi^a = \lim_J \lim_k$ of continuous functions is Borel. Lemma 2.14, applied on the open convex set $\mathcal{E}_\rho$ itself, gives the continuity.

(v) Let $b \in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty \leq M$. For $k \geq k_0$, τ_b preserves Λ_k and S_{Λ_k}, and the substitution $A'_\alpha(x) = A_\alpha(x+b)$, $\phi'(x) = \phi(x+b)$ of Step 1 of the proof of Proposition 8.1 carries the comparison state on X_J to the comparison state on $(C_J - b) \cap \Lambda_k$ together with both sources, the single-site weights, W_ε, δm_ε^2 and a_k being site-independent; so $Z_{k,X_J}^a(\tau_b \cdot s) = Z_{k,(C_J - b)\cap\Lambda_k}^a(s)$, the right side being the functional of the comparison state on the translated box. As sets of sites, $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$. Let $s \in \mathcal{K}$ have all sources supported at ℓ^∞-distance >1 from $\mathbb{R}^3 \setminus C_{J_0}$ and let $J \geq J_0 + 1$; then, as in (ii), every plaquette and site at which s does not vanish lies in C_{J-1}. The first inequality of Lemma 2.10 (with $X = C_{J-1} \cap \Lambda_k$ in the state on $(C_J - b) \cap \Lambda_k$, then $X = (C_J - b) \cap \Lambda_k$ in the state on $C_{J+1} \cap \Lambda_k$; no site is killed, and the Dirichlet terms of each smaller state are the diagonal parts of the cut bonds) and Lemma 2.11(a) give

$$Z_{k,J-1}^a(s) \leq Z_{k,J}^a(\tau_b \cdot s) \leq Z_{k,J+1}^a(s) \qquad (k \geq k_0,\ J \geq J_0 + 1).$$

Let $k \to \infty$ by (i) (the translated family is as in $(\mathrm{W1})^{(a)}$ at J, since $C_{J-1} + b \subset C_J$ for $|b|_\infty \leq M$) and then $J \to \infty$ by (iii): $Z_\infty^a(\tau_b \cdot s) = Z_\infty^a(s)$ on $\mathcal{K}$. The identity extends to all complex coefficients of a family by the argument of (iii) (both sides are entire in the split coefficients and agree on the set of uniqueness Ω_+), to all $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration over the translated families (each $b \in 5^{-k_0}\mathbb{Z}^3$ is a finite sum of vectors of $5^{-k_0}\mathbb{Z}^3$ with $|\cdot|_\infty \leq M$), and to all $b \in \mathbb{R}^3$ at real coefficients by (iv), since $b \mapsto \tau_b \cdot s \in \mathcal{E}_{\rho'}$ is continuous for $|b| \leq \rho' - \rho$ and $\bigcup_k 5^{-k}\mathbb{Z}^3$ is dense; then to complex coefficients by the identity theorem, $\mathbb{R}^{2N}$ being a set of uniqueness. (King states translation invariance as [16, Thm. 2.3(i)]; his argument is not used.) At $h = 0$, where $(\mathrm{W1})^{(a)}$ is $(\mathrm{E}_C)^{(a)}$, this is Proposition 8.1, whose proof is the one just given with the scalar sources absent.

(vi) Lemma 16.4(a),(b) and (iii). □

Definition 16.7 (the full object in the scheme a)**.** Let the couplings e, λ, m, μ_0 be fixed in King's range, together with the block factor 5 and the large-block integer M; let C_J be the King cubes, $\Lambda_k = \varepsilon_k \mathbb{Z}^3$, $\varepsilon_k = 5^{-k}$, and let $\langle\cdot\rangle^{(a)}_{k,J}$ be the comparison cube of Definition 2.1, with the free-boundary mass counterterm $\delta m^2_{\varepsilon_k}$ of the setting section, one number per spacing. Let a be a W-admissible scheme ($(\mathrm{W})_a$). For a finite family ζ of curvature and scalar sources (Definition 2.3) the *full object in the scheme* a is

$$Z^a_\infty(\zeta) := \lim_{J\to\infty} \lim_{k\to\infty} Z^a_{k,J}(\zeta) :$$

the inner limit is the cube limit along the spacings 5^{-k} at fixed μ_0 and fixed J, taken for the J at which the sources have ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and the outer limit runs along the cube family $(C_J)_J$. Both limits exist, locally uniformly in the coefficients, by Theorem 16.6(i),(iii), and Z^a_∞ is the functional so named among the objects of the setting section, on the comparison convention. It is the object of the Wick-sector theorems of this part (Theorems 18.9 and 18.11) and of Form (A). At $h = 0$ the multiplier of Definition 16.2 is 1, so the object does not depend on the scheme: it is the iterated limit $\lim_J \lim_k Z_{k,J}(\sum_j z_j f_j, 0)$ of the comparison family, which exists under $(\mathrm{E}_C)^{(a)}$ (Lemma 7.7) and then equals King's printed state Z^K_∞ by Corollary 4.4(iii). The existence theorem (E_C) is posed on King's convention, and nothing is asserted about the fixed-cube limits of the comparison family; Corollary 4.4(ii)(γ) gives, at the cone points $z = \mathrm{i}t$,

$$Z^K_J(\mathrm{i}t) \le \liminf_k Z_{k,J}(\mathrm{i}t) \le \limsup_k Z_{k,J}(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t),$$

and both ends tend to $Z^K_\infty(\mathrm{i}t)$ as $J \to \infty$. Throughout this part, *King's printed state* means Z^K_∞ of Theorem 7.6, on King's convention ((E_C)); the curvature-sector statements of the part are posed for it.

Remark 16.8 (scope). The scheme changes Z^a_∞ only by the factor in Theorem 16.6(vi), which affects the one-point function of the Wick square; joint cumulants of order ≥ 2 do not depend on the scheme. The relation of the full object to infinite-volume limits along periodic boxes, King's expected identity [16, (2.24)], is never used.

16.5. **The general-source extension.** For scalar sources with $\int h \neq 0$ King's printed Wick ordering has no finite continuum limit beyond the curvature sector (Remark 16.1), and the object is the full object in a W-admissible scheme (Definition 16.7). For the reference scheme $z = a^\sharp$ the form of $(\mathrm{W})_a$ that carries the extension to such sources is the following.

Remark 16.9 (the reference scheme on the cone). $(\mathrm{W})_{a^\sharp}$ on the cone consists of the two clauses

($\mathrm{W}^\sharp$1) For every King cube C_J, every finite family of real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, sampled at the plaquette centres as in Definition 2.3, and of scalar sources $h_j \in C^\infty_c(\mathbb{R}^3)$, all supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and every real (z, w), the limit $\lim_k Z^{a^\sharp}_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0, \infty)$, locally uniformly in complex (z, w).

($\mathrm{W}^\sharp$2) For every real $0 \le q \in C^\infty_c(\mathbb{R}^3)$ with $\sup q < s_*(e)$, $\sup_{J \ge J_q} \limsup_k Z^{a^\sharp}_{k,J}(0, q) < \infty$; here $s_*(e) > 0$ is a threshold fixed by the bulk bounds of the scheme, positive for small λ outside an exceptional set of at most two values of e^2 for given (λ, μ_0, m).

The first clause is Theorem 3.6 with the threshold on the positive scalar part removed, that is clause (a) of (W2) for the reference scheme; the second is the bound of clause (b) of (W2) below a threshold, in the form proved on the tube in Proposition 6.10 and Corollary 13.3. With these two clauses in place of $(\mathrm{W})_a$, and with $(\mathrm{K})_a$ and $(\mathrm{TIGHT})_a$ for scalar sources (Definitions 17.2 and 18.5), the arguments of this and the following two sections apply on the corresponding cone of sources. None of this enters Theorem 15.1.

17. The torus comparison

The rotation enters the rotation part through a single statement, the torus comparison, posed on King's tori for the two lattices Λ_k and $R^{-1}\Lambda_k$. This section states it, reduces it by an exact

symmetry to one-sided bounds at one lattice spacing, and describes its proof along the lines of [16] in Section 17.3, where the principal statements of the proof are reproduced with their numbers; the complete proof is in the supplemental material.

17.1. **The torus comparison.** Throughout, for a lattice Λ of spacing ε_k descending to a King torus D_J (Definitions 15.4 and 15.6), $Z^a_{\Lambda,D_J}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}\langle\Theta(\zeta)\rangle_{\Lambda,D_J}$ is the functional of the periodic state in the scheme a (Definition 2.3), $\mathsf{Z}_{\Lambda,D_J}(s)$ the unnormalised functional of Definition 15.7, and $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ the walled state of the configuration $\mathfrak{c} = (J', w)$.

Theorem 17.1 ((K_0): the torus comparison, curvature sector)**.** *For every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c} = (J', w)$ of the class of Remark 18.6 (the conditions of Theorem 17.13) and every real smooth $D_{J'}$-periodic 2-form G, the limits $\lim_k \langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.*

The proof is given at the end of the next subsection, which reduces the statement to one-sided bounds at a single lattice spacing; the bounds themselves are treated in Section 17.3, with complete proofs in the supplemental material, and Remark 17.10 of the supplemental material lists the inputs of that proof from the literature. With scalar sources the comparison is a property of a scheme a, which is not proved here (it is listed among the open problems, in the final section) and is used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 17.2 ($(\mathrm{K})_a$: the torus comparison with scalar sources)**.** For every $R \in \mathcal{R}_0$, every King torus D_J, every finite family of real smooth D_J-periodic 2-forms G_j and functions h_j, and every $(z, w) \in \mathbb{R}^{2N}$, the limits

$$\lim_{k\to\infty} Z^a_{\Lambda_k,D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big) \qquad \text{and} \qquad \lim_{k\to\infty} Z^a_{R^{-1}\Lambda_k,D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big)$$

exist in $(0,\infty)$ and are equal.

(K_0) is stated on the sub-class of walled configurations described in Remark 18.6, which is the one on which the results that consume it use it. Here $R^{-1}\Lambda_k$ carries the frame $(R^{-1}e_\alpha)$ and descends to D_J by Lemma 15.3(b); both lattices have spacing ε_k and carry the same a_k, $\delta m^2_{\varepsilon_k}$ and $\mathsf{W}_{\varepsilon_k}$. Read with a single scalar function on all plaquettes, as printed in [16, (2.12), Thm. 2.4], the comparison is false already at $e = 0$ (Proposition H.2 of the supplemental material); the 2-form reading, in which each lattice samples the source in its own frame, is consistent with King's proof (Remark 15.5).

Lemma 17.3. (a) *Under $(\mathrm{K})_a$ the convergence holds locally uniformly on $\mathbb{C}^{2N}$, the two limits are the same entire function, and for every walled configuration $\mathfrak{c}$ and every (f, g) supported in one fundamental cell with $w = 0$ on $\operatorname{supp} g$, $\lim_k T^a_{\mathfrak{c},\Lambda_k}(f, g)$ and $\lim_k T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f, g)$ exist and are equal. Under (K_0) the same holds, for the configurations of its class, for $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ and $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ at complex z.*

(b) *$(\mathrm{K})_a$ implies (K_0), and $(\mathrm{K})_a$ implies $(\mathrm{K})_{a'}$ if $a'_k - a_k$ converges.*

(c) *If the statement of $(\mathrm{K})_a$, respectively (K_0), holds for $R = R_{12}(\theta_0)$, it holds for every $R \in \mathcal{R}_0$.*

(d) *Under $(\mathrm{K})_a$, for every King torus D_J and every $0 \le q \in C^\infty_c$ supported in one fundamental cell, $C_{D_J}(q) := \sup_k Z^a_{\Lambda_k,D_J}(0, q) < \infty$.*

Idea of the proof. (a) At fixed k the functionals are entire with $|Z^a(\zeta)| \le Z^a(\operatorname{Re}\zeta)$ (Lemma 2.11(d)), so convergence on $\mathbb{R}^{2N}$ gives local boundedness on $\mathbb{C}^{2N}$ and, by Vitali's theorem, locally uniform convergence to one entire function; the walled ratios are quotients of such limits. (b) A wall is the scalar source $-w$. (c) The reflection $\sigma_2 = \operatorname{diag}(1,-1,1)$ conjugates $R_{12}(\theta_0)^{-1}$ into $R_{12}(\theta_0)$ and permutation matrices carry the plane $(1,2)$ to the other planes; Lemma 2.12 transports the statement.

17.2. **Reduction to one-sided bounds at one lattice spacing.** King's proof of his Theorem 2.4 compares the lattice of spacing ε with the rotated lattice of spacing 5ε, and obtains the reverse inequality by "switching around the argument" [16, p. 337]. We compare the two lattices at the *same* spacing, and obtain the reverse inequality from an exact symmetry.

Lemma 17.4 (the involution)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$, let $\sigma := \sigma_\nu$ be the reflection $x_\nu \mapsto -x_\nu$, and put $S := R^{-1}\sigma$. Then $S^2 = 1$, $S\Lambda_k = R^{-1}\Lambda_k$ and $S(R^{-1}\Lambda_k) = \Lambda_k$, and S maps $D_J = \mathbb{R}^3/P_J$, $P_J := (50MJ\mathbb{Z})^3$, onto the torus $R^{-1}D_J := \mathbb{R}^3/R^{-1}P_J$. Both lattices descend to both tori, and for all real smooth D_J-periodic sources $s = (G, h)$,*

$$\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s) = \mathsf{Z}_{\Lambda_k, R^{-1}D_J}(S \cdot s), \qquad \mathsf{Z}_{\Lambda_k, D_J}(s) = \mathsf{Z}_{R^{-1}\Lambda_k, R^{-1}D_J}(S \cdot s),$$
$$\mathsf{S}_{R^{-1}\Lambda_k, D_J}(h) = \mathsf{S}_{\Lambda_k, R^{-1}D_J}(S \cdot h).$$

Idea of the proof. $\sigma R \sigma = R^{-1}$ (Lemma 15.3(a)) gives $S^2 = 1$ and the exchange of the two lattices; $R^{-1}P_J \subset 10MJ\,\mathbb{Z}^3$ gives the descent of both lattices to both tori; the frames are carried to the frames of the other lattice up to signs, and Lemma 2.12 with $O = S$ gives the identities.

Definition 17.5 ((F), (V) and (C): one-sided bounds at one spacing)**.** Let $R \in \mathcal{R}_0$ and $J \in \mathbb{N}$. Write $T_1 := D_J$, $T_2 := R^{-1}D_J$. The following are properties of the pair (R, J). (F): for $i = 1, 2$ there are real numbers $v_k(T_i)$ and $c_k(T_i)$, $k \geq k_0(J)$, independent of the fields and the sources, with $|c_k(T_i)| \leq C\varepsilon_k^{-q}$ for some $q < \infty$ and $v_k(T_i)$ bounded, such that for every real smooth T_i-periodic source $s = (G, h)$,

$$\mathsf{Z}_{R^{-1}\Lambda_k, T_i}(s) \leq \big(1 + \delta_k(T_i, s)\big) \exp\big(- v_k(T_i) - c_k(T_i)\, \mathsf{S}_{\Lambda_k, T_i}(h)\big)\, \mathsf{Z}_{\Lambda_k, T_i}(s), \qquad \lim_{k\to\infty} \delta_k(T_i, s) = 0.$$

(V): $v_k(T_1) + v_k(T_2) \to 0$. (C): $c_k(T_1) - c_k(T_2) \to 0$. ($\mathrm{F}_0$) is (F) restricted to sources with $h = 0$ and to the sources $(G, \pm h_0)$, where $h_0 = -w$ is a wall; it involves no Wick scheme.

On each torus, (F) bounds the functional of the rotated lattice from above by that of the axis lattice, the direction of [16, (4.7)–(4.9)]; since smooth periodic functions have rapidly convergent Riemann sums, $|\mathsf{S}_{\Lambda,T}(h) - \int_\Gamma h| \leq C_N(h)\varepsilon_k^N$ for every N, so with $N > q$ the error $c_k(\mathsf{S}_{\Lambda,T}(h) - \int h)$ tends to 0. The proof below uses (F) only for the sources in question and their images under $h \mapsto -h$; under (F_0) and (V) the conclusions (a)–(c) of the next theorem hold for the sources $(G, \pm h_0)$.

Theorem 17.6 (closing at one spacing)**.** *Let $R \in \mathcal{R}_0$, $J \in \mathbb{N}$, and assume* (F)*. Write $v = v_k(T_1)$, $v' = v_k(T_2)$, $c = c_k(T_1)$, $c' = c_k(T_2)$, and $\rho_k(s) := \mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s)/\mathsf{Z}_{\Lambda_k, D_J}(s)$.*

(a) *For every real smooth D_J-periodic source $s = (G, h)$, $e^{v' + c'\int h}(1 + o(1)) \leq \rho_k(s) \leq e^{-v - c\int h}(1 + o(1))$ as $k \to \infty$.*

(b) *$v + v' \leq o(1)$. If* (V) *holds, then $c + c' \to 0$.*

(c) *If* (V) *holds, then $\rho_k(s) = e^{-v - c\int h}(1 + o(1))$ for every s. Consequently, for sources s_0, s_1 with $\int h_0 = \int h_1$,*

$$\frac{\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s_1)}{\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s_0)} = \frac{\mathsf{Z}_{\Lambda_k, D_J}(s_1)}{\mathsf{Z}_{\Lambda_k, D_J}(s_0)}\,(1 + o(1)).$$

(d) *If* (V) *and* (C) *hold, then $c, c' \to 0$, and the identity in (c) holds for all s_0, s_1.*

No convergence of either family and no Wick scheme is assumed.

Idea of the proof. The upper bound in (a) is (F) on T_1; the lower bound is (F) on T_2 applied to the source $S \cdot s$ and pulled back by Lemma 17.4, which exchanges the two lattices. The interval in (a) is non-empty, and applying this to $(G, \pm h)$ and adding gives (b); (c), (d) follow by dividing the two-sided bounds for s_1 and s_0.

Corollary 17.7 (walled curvature ratios)**.** *Assume* (F_0) *and* (V) *for R and J'. Then for every walled configuration $\mathfrak{c} = (J', w)$ and every real smooth periodic 2-form G, $\langle e^{zF(G)}\rangle_{\mathfrak{c}, R^{-1}\Lambda_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c}, \Lambda_k} \to 0$ locally uniformly in $z \in \mathbb{C}$. If moreover $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c}, \Lambda_k}$ exists for real t, then* (K_0) *holds for $\mathfrak{c}$ and G.*

Idea of the proof. $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda}$ is a ratio of two sources with the same $\int h$, so Theorem 17.6(c) gives the convergence of the ratio of the two lattices' values to 1 for real t; both walled states are BFS measures, so $1 \le \langle e^{tF(G)}\rangle \le e^{t^2\sigma_\Lambda/2}$ with σ_Λ bounded uniformly in k (Lemma 2.8), and Vitali's theorem gives the locally uniform convergence on $\mathbb{C}$.

Corollary 17.8 (the full comparison)**.** *Let a be a scheme with $|a_k| \le C(1+k)$. Assume* (F)*,* (V) *and* (C) *for $R = R_{12}(\theta_0)$ and every J, and that for every King torus and every finite family of real smooth periodic sources the limit $\lim_k Z^a_{\Lambda_k,D_J}$ exists in $(0,\infty)$ for real coefficients. Then* $(\mathrm{K})_a$ *holds.*

Idea of the proof. The scheme multipliers of the two lattices differ by $1 + O(k\varepsilon_k^N)$, and Theorem 17.6(d) with $s_0 = 0$ gives the convergence of the ratio of the normalised functionals to 1; Lemma 17.3(c) gives the other generators.

Remark 17.9 (what this subsection supplies)*.* Lemma 17.4, Theorem 17.6 and its corollaries are a reduction of the torus comparison to the one-sided bounds (F), (V), (C); the bounds themselves come from the renormalisation-group analysis. In the curvature sector, what Corollary 17.7 needs is (F_0) and (V) together with the existence of the axis-lattice limits, and these are what the proof of (K_0) along the lines of [16] produces (Remark 17.10 of the supplemental material). In that proof King's torus bound [16, (4.2)] is of no use, because at a non-zero wall it is vacuous; its place is taken by the large-field clause of King's upper bound [15, Thm. 3.1, (3.4)] at blocking factor 5, and at a mass-increasing wall by a polynomial lower bound for the walled partition function. Corollary 17.8 concerns scalar sources and is not used in the curvature sector (Theorem 15.1).

Proof of Theorem 17.1. By Lemma 17.3(c) it suffices to treat $R = R_{12}(\theta_0)$. Fix a walled configuration $\mathfrak{c} = (J', w)$ of the class of Remark 18.6 and a real smooth periodic 2-form G. By Corollary 17.7 the statement follows from (F_0) and (V) for R and J' together with the existence of $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ for real t. The existence of the axis-lattice limit is King's torus theorem [15, Thm. 2.1] with the wall as scalar source, read on the normalised ratio, in which the vacuum and the h-linear constants cancel (Theorem 17.122). The bounds (F_0) on the two tori T_1, T_2, in total form, with (V), come from the comparison, at equal spacing, of the effective actions of King's standard block-spin flow on the rotated lattice and of the hybrid flow on the axis lattice; they are proved in Section 17.3, and the proof is completed there (Theorem 17.13 and the paragraph after Proposition 17.135). □

(K_0) is [16, Thm. 2.4] with the corrections of Proposition H.2 of the supplemental material and Remark 16.1, for arbitrary smooth periodic sources, walls included, and with free-boundary counterterms; by Corollaries 17.7 and 17.8 its two-lattice content is (F) and (V), plus (C) for scalar sources, and its one-lattice content is the existence of the torus limits on the axis lattice. The rotation enters the whole argument of the rotation part only through Theorem 17.1. The inputs of the proof from the literature, including the statements of [1, 2, 3, 4, 15, 16] that are used although printed without an argument and the statements false as printed that are replaced, are listed in Remark 17.10 (see also Remark 17.11) of the supplemental material.

17.3. **Proof of the torus comparison.** This subsection proves the one-sided bounds (F_0) and (V) of Definition 17.5, in the total form in which the vacuum constants are absent and the source-linear constants tend to zero, for the walled configurations of the class of Remark 18.6; with the existence of the axis-lattice limits (Theorem 17.122) and Corollary 17.7 this completes the proof of Theorem 17.1. Here the principal statements are reproduced with their numbers, each proof is replaced by its idea, and every auxiliary statement is cited by its number in the supplemental material, which contains the complete proofs. Throughout, $R \in \mathcal{R}_0$ is fixed, $5R$ and $5R^{-1}$ are integer matrices (Lemma 15.3(a)), and

$$X_k := \Lambda_k = \varepsilon_k\mathbb{Z}^3 \quad \text{(the axis lattice)}, \qquad Y_k := R^{-1}\Lambda_k \quad \text{(the rotated lattice)}, \qquad Y_{k-j} := R^{-1}(5^j\varepsilon_k\mathbb{Z}^3),$$

so that $Y_{k-j} \subset X_k \cap Y_k$ for $j \ge 1$; the frame of X_k is (e_α) and that of Y_k is $(R^{-1}e_\alpha)$; the tori are $T_1 = D_J = \mathbb{R}^3/P_J$, $P_J = 50MJ\,\mathbb{Z}^3$, and $T_2 = R^{-1}D_J$, and $S = R^{-1}\sigma_\nu$ is the involution of Lemma 17.4. Here r denotes the number of block-spin steps.

Theorem 17.13 (the torus comparison at one configuration)**.** *Let the couplings be in King's range and* $\mathfrak{c} = (J, \psi, n)$ *a walled configuration with* $J \in 5^{n_*}\mathbb{N}$, $J \geq J_0(\delta_r)$, $n > 0$, *the wall condition* $n\mathsf{S}_{X_k}(\psi) > \Delta_*|T_1|/2$ *(*Δ_* *the constant of Proposition 17.121(b) of the supplemental material) and block alignment (the source box and the torus unions of blocks of the final scale); here* n_* *is the number of large-field scales of the flow of* Y_k*, fixed by the couplings, and* $J_0(\delta_r)$ *the non-wrapping threshold of Lemma 17.120 of the supplemental material at the coarsest spacing* δ_r *of the flow (a condition void for large* k*, since* $\delta_r \to 0$*). Then for every real smooth* P_J*-periodic 2-form* G*:*

(i) $\langle e^{zF(G)}\rangle_{\mathfrak{c},Y_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},X_k} \to 0$ *as* $k \to \infty$*, locally uniformly in* $z \in \mathbb{C}$*; in particular all moments* $\langle F(G)^m\rangle$ *differ by* $o(1)$*;*

(ii) *both families converge, locally uniformly on* $\mathbb{C}$*, to the same entire function* $\mathcal{W}_{\mathfrak{c}}(zG)$*, and all moments converge to the same limits;*

(iii) *the same holds with* R *replaced by any* $R' \in \mathcal{R}_0$*.*

At $e = 0$ *the statement holds for every walled configuration with no condition:* $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ *exactly (Lemma 17.21), and* $q_{X_k}(G), q_{Y_k}(G) \to q(G)$ *(Lemma 17.22 of the supplemental material).*

The three conditions on $\mathfrak{c}$ are those of Remark 18.6, and the configuration produced by Theorem 19.10 satisfies them (Remark 19.11(iii), Theorem 20.14). The proof is assembled in Proposition 17.134 and the paragraph after it, from the comparison of the effective actions of two block-spin flows at equal spacing (Theorem 17.62), King's upper bound for the rotated lattice (Proposition 17.96), the hybrid lower bound (Proposition 17.113) and a polynomial lower bound at the wall (Theorem 17.116), closed by Theorem 17.6. Both period lattices lie in every block lattice of both flows (Lemma 17.14 of the supplemental material).

The two block-spin flows.

Definition 17.15 (the standard flow of Y_k and the hybrid flow of X_k)**.** Fix k and $r \leq k$; $\delta_r := 5^r\varepsilon_k$ is the physical spacing of the coarse lattice Y_{k-r}. In the unit scale of level r write $\eta := 5^{1-r}$ and $\eta' := \eta/5 = 5^{-r}$. Blocks are centred at every level (Convention 17.17 of the supplemental material).

- *Standard flow on* Y_k (flow Y): Bałaban's and King's block-spin transformation [1, (2.1)–(2.11)], [15, (2.10)–(2.17)] in the frame of Y_k: centred cubes of 125 points, $Y_{k-j+1} \to Y_{k-j}$, with contours made of fine Y_k-bonds in the axial order f_1, f_2, f_3 of the frame inside each block.
- *Hybrid flow on* X_k (flow X): the first step averages over the centred tiles $y + \varepsilon B_0$, $y \in Y_{k-1}$, with
$$B_0 := \mathbb{Z}^3 \cap R^{-1}\big(-\tfrac{5}{2}, \tfrac{5}{2}\big)^3$$
(125 sites, a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$, Lemma 17.19 of the supplemental material), onto Y_{k-1}; this is [16, (4.3)–(4.4)], with the photon average read frame-covariantly, $(\hat{Q}A)_\alpha(y) = \frac{1}{125}\sum_{x\in B(y)} f_\alpha \cdot A(x)$, $f_\alpha = R^{-1}e_\alpha$ (Remark 17.18 of the supplemental material). Steps $2, \dots, r$ are the standard flow's steps $Y_{k-1} \to Y_{k-r}$ with the same blocks, except that every rotated bond of Y_{k-1} occurring in a contour is replaced by a fixed path of fine X_k-bonds ([16, p. 336]: "succeeding steps of the renormalization transformation in the usual way, that is without rotating the blocks").
- *Contours, rotation-paired.* Let $\Gamma^{\mathrm{st}}_{z,w}$ be King's coordinate staircase from z to w [15, (2.12)] and $\Gamma^{(r)}_{y,x}$ his composite contour [15, (2.11)], which the standard flow uses in Y's frame. The legs of a Y-frame staircase between two centres of Y_{k-1}-blocks are concatenations of the η-bonds $(u, u + 5\varepsilon f_\alpha)$, $u \in Y_{k-1}$. The hybrid composite contour from the level-r block centre y to $x' \in X_k$ is the same staircase with every η-bond $(u, u + 5\varepsilon f_\alpha)$ replaced by the *fixed* fine axis staircase $u + \varepsilon\gamma_\alpha$ (and the reversed bond by its reverse), where, for $R = R_{12}(\theta_0)$ and $f_\alpha = R^{-1}e_\alpha$ with $5f_1 = (4,3,0)$, $5f_2 = (-3,4,0)$,

$$\gamma_1 : e_1, e_2, e_1, e_2, e_1, e_2, e_1 \ \text{(to } (4,3,0)\text{)}, \qquad \gamma_2 : e_2, -e_1, e_2, -e_1, e_2, -e_1, e_2 \ \text{(to } (-3,4,0)\text{)}, \qquad \gamma_3 : e_3^5,$$

followed, inside the first-step tile, by a fine axis path of at most 5 bonds (Lemma 17.19(ii) of the supplemental material), chosen once as a function of the displacement only and the same in every tile, so that the block and contour system of the flow is invariant under Y_{k-1}, in particular under the periods of both tori; this is the first-step contour of [16, (4.4)]. (For the other generators conjugate by the permutation and sign matrices of Lemma 17.3(c).) The gauge field along a composite contour is read by King's convention [15, (2.19)] (Convention 17.17(ii) of the supplemental material).

Both flows use King's coefficients $a_j = a(1-5^{-2})/(1-5^{-2j})$ [15, p. 653], take r steps and land on the same coarse lattice Y_{k-r}/P with the same unit-scaled fields (A_r, ϕ_r) and the same small-field indicator χ_r [15, (3.2)]. The effective actions are written so that

$$\mathsf{Z}_{X_k,T}(s) = \int (dA_r)(d\phi_r)\, e^{-\hat{s}_T^{(r)}(A_r,\phi_r;s)}, \qquad \mathsf{Z}_{Y_k,T}(s) = \int (dA_r)(d\phi_r)\, e^{-S_T^{(r)}(A_r,\phi_r;s)}$$

exactly, every c-number kept inside: this is [15, (3.14), (3.90)] with E_0 inserted from [15, (3.89)].

Centred blocks, paired contours, the tile (Lemmas 17.16, 17.19 and 17.20, Convention 17.17, and Remark 17.18 of the supplemental material). Blocks centred at the sites of the next coarse lattice make the inversion about any site of the coarsest lattice an exact symmetry of both flows, which removes every first-moment term from both effective actions; the gauge field along a composite contour is read in King's convention [15, (2.19)], and the photon average frame-covariantly, which lets a constant background pass through the step. Each fixed staircase lies within 0.6ε of the rotated bond it replaces, so every piece of the hybrid contour stays inside its block and the two composite contours bound, with the matching segment, a surface of area $\le 1.5\eta' + 33\eta'^2$; the tile B_0 is a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$ with 125 points, $B_0 = -B_0$, and $\hat{Q}\hat{Q}^* = \mathbf{1}$, $Q(A)Q(A)^* = \mathbf{1}$ for every bond field A, so the first hybrid step is King's transformation with the tile in place of the cube; and by Stokes' theorem the phases $U(A(\Gamma))$ of the two flows at matched endpoints differ by at most $C\eta'\|dA\|_\infty$. The pairing is a condition, not a convenience: if the enclosed loops did not shrink with the scale, the flux would be of the order of the Wilson-line phase at the coarse momenta at which the comparison is made, and Theorem 17.56 would fail.

The Cameron–Martin transfer of the curvature source.

Lemma 17.21 (Cameron–Martin transfer)**.** *Let Λ be a lattice of spacing $s = \varepsilon_k$ with frame (f_α) descending to a torus T, γ_Λ the normalised Gaussian photon measure of the action, with covariance $C_\Lambda = (-\Delta_\Lambda + \mu_0^2)^{-1} \otimes \mathbf{1}_3$ for the pairing $\langle u, v\rangle_\Lambda = \sum_x s^3 \sum_\alpha u_\alpha(x) v_\alpha(x)$, and write $\mathsf{Z}_{\Lambda,T}(G,h) = \int d\gamma_\Lambda(A)\, e^{F_\Lambda(G)} W_\Lambda(A;h)$ with W_Λ the matter integral, which depends on A only through the bond phases. Then $F_\Lambda(G) = \langle A, \ell_\Lambda\rangle_\Lambda$ with $\ell_{\Lambda,\beta} := \sum_{\alpha\neq\beta} \partial^*_\alpha G_{\alpha\beta}$, $G_{\alpha\beta}(x) := G_{c(x;\alpha\beta)}(f_\alpha, f_\beta)$, $\partial^*_\alpha u(x) := s^{-1}(u(x - sf_\alpha) - u(x))$, and for real z*

$$\mathsf{Z}_{\Lambda,T}(zG,h) = e^{z^2 q_\Lambda(G)/2}\, \mathsf{Z}_\Lambda[z a_\Lambda](T;h), \qquad q_\Lambda(G) := \langle \ell_\Lambda, C_\Lambda \ell_\Lambda\rangle_\Lambda, \quad a_\Lambda := C_\Lambda \ell_\Lambda,$$

where $\mathsf{Z}_\Lambda[b](T;h) := \int d\gamma_\Lambda(A)\, W_\Lambda(A+b;h)$. Counterterms, Wick constant and wall are untouched. At $e = 0$, W_Λ does not depend on A, so $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ exactly.

Idea of the proof, and what the transfer does (Lemmas 17.22, 17.23 and 17.25 and Remark 17.24 of the supplemental material). Summation by parts writes $F_\Lambda(G) = \langle A, \ell_\Lambda\rangle$, and completing the square in the photon Gaussian moves the source into the matter coupling. The Gaussian factor is frame-free, $|q_\Lambda(G) - q(G)| \le C(\mu_0) s^2 \|G\|^2_{H^3(T)} + C_N(G) s^N$, and the shift field a_Λ is the sample of one smooth co-exact 1-form $a^{\mathrm{c}} = (-\Delta + \mu_0^2)^{-1} d^* G$ up to $O(s^2)$, so that in King's unit scaling $z a_\Lambda$ and $A^{(r)} + z a_\Lambda$ are regular in the sense of [15, Def. 3.2]. After the transfer the model is King's with a scalar source and a fixed smooth background in the matter coupling only, which enters exactly where King's coarse background enters ([15, (3.82)–(3.87)]), and no frame-paired source vertex [15, (3.20)] appears. Finally $|\mathsf{S}_\Lambda(h) - \int_T h| \le C_N(h) s^N$ for every lattice $\Lambda = O(s\mathbb{Z}^3) \supset P$, with $C_N(h)$ independent of O.

The number of steps, the effective actions, the matching. Summary (Lemmas 17.26, 17.29 and 17.30 and Definitions 17.27 and 17.28 of the supplemental material). The number of steps is $r = r(k)$ with $k - r = \lfloor \varrho k \rfloor$ for a fixed $0 < \varrho < \min\{\gamma/(\gamma+\beta), \frac{1}{9}\}$, γ the smaller of the exponents of [15, Thm. 3.4] and of the gain θ_1 of Theorem 17.39, β the larger of the field-norm exponents; then $\delta_r = 5^r \varepsilon_k \to 0$, $5^{-\gamma r}\delta_r^{-\beta}(1+k)^p \to 0$ for every p, $\varepsilon_k^{-q} e^{-c\delta_r^{-1}} \to 0$, and $p(\delta_r)$ grows at least like k^{p}. The effective actions $S^{(r)}$ (standard flow) and $\hat{s}^{(r)}$ (hybrid flow) are [15, (3.14)] rewritten as [15, (3.90)], with the Taylor polynomial $P^{(r)}$ of order $\bar{n} = 13$ of King's E'_r; a wall enters through the h-vertex [15, (3.21)], in the free-boundary convention in which the subtraction is the same number for both lattices. In the unit scale of level r the first-step cells of the two flows, the tile in $\Lambda_X = \eta'\mathbb{Z}^3$ and the centred cube in $\Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$ around each $x \in \Gamma_1 = R^{-1}(\eta\mathbb{Z}^3)$, have 125 sites each; fixed bijections between them glue to a bijection $\tau : \Lambda_X \to \Lambda_Y$ with $|\tau x' - x'| \le c_\tau \eta'$, $c_\tau < 7.1$, and points $x'' = \tau x'$ are *matched*. The fluctuation covariance of each flow is sliced exactly, $G_r^F = \sum_{j<r} S_j^F$, slice j having length scale $\ell_j = 5^j\eta'$. Every Y-graph is, exactly, a sum over X-vertices with τ-transported kernels; a smooth vertex function sampled at x' and at $\tau x'$ differs by $\le c_\tau \eta' \|\nabla w\|_\infty$; and a counterterm graph of δm^2 or E_1, expanded in the slices of either flow, sums to the same number for both flows, because $\sum_j S_j^F = C^F$ pointwise and C^Y is the image of C^X under the isometry R^{-1}.

The equal-spacing propagator package. In the unit scale of level $r \ge 2$, on the infinite lattice and at $A = 0$, let u_{r-1} be the common symbol of the $(r-1)$-step rotated centred average, u_X the tile symbol and u_Y the cube symbol, A^F the coarse multiplier of flow F, $A^F \le a$, Δ'_F the fine Laplacian of the flow's lattice plus the running mass, and L_2^F the 125 alias classes.

Proposition 17.31 (free core at equal spacing)**.** $A^F(p)^{-1} = a_{r-1}^{-1} + \sum_{l_1 \in \Gamma_r^*/\Gamma_1^*} u_{r-1}(p+l_1)^2 [\eta^2 a^{-1} + S_F(p+l_1)]$ *with* $S_F(q) := \sum_{l \in L_2^F} u_F(q+l)^2/\Delta'_F(q+l)$*, and* $\sup_{V_\eta} |S_X - S_Y| \le 0.198\,\eta^2$*. Hence for all* $r \ge 2$*, all* p *and all* $m_r \ge 0$*,*

$$|A^X(p) - A^Y(p)| \le 0.20\,\eta^2 A^X(p) A^Y(p) \le 0.20\,a\,\eta^2 A^Y(p),$$

and the same for the photon (three copies at mass $\mu_0 \delta_r$*).*

Idea of the proof, and the package (Lemmas 17.32 to 17.35 and Corollaries 17.36 and 17.37 of the supplemental material). The formula is the Woodbury identity for a composite average whose symbol is $u_{r-1}u_F$; the main terms differ by at most 0.0924 by the second moments of the tile and of the cube, and the 124 aliases by at most 0.1047. From the Fourier representation of the lifts, the lifts of the two flows differ at matched points by $\le C_E J_1 \eta$, with a decaying form by the geometric mean with the Combes–Thomas decay [28]; the coarse covariances $\Gamma^F = H^F B^F H^{F*}$, with $C^F = G_r^F + \Gamma^F$ exactly, differ at matched pairs by $\le C_\Gamma \eta^{1/2}$ and are $\frac{1}{2}$-Hölder; the slices $j \ge 2$ differ at matched points by $\le C(5\eta'/\ell_j)^{1/2} \ell_j^{-1} e^{-\delta|x'-y'|/\ell_j}$. The *uniform comparison form*: with $D_j := S_j^X - S_j^Y \circ \tau$ and $\theta = \frac{1}{2}$, the fine slices obey $|D_j(x',y')| \le C_D \eta'^\theta \ell_j^{-1-\theta} e^{-\delta|x'-y'|/\ell_j}$ (for $j \in \{0,1\}$ there is no gain and the factor is paid by a degree loss, King's device at the finest slice [15, p. 675]), the coarse slices obey the same with $\eta^{1/2}$, and, given Proposition 17.72(a) of the supplemental material and [15, Thm. 3.3], the derivative, Hölder and contour versions hold, the contour lines [15, (3.27)] along the paired contour systems differing by the kernel difference along the contour plus the flux of the slice kernel through the sliver. Summing the slices, $|C^X(x'-y') - C^Y(x''-y'')| \le C_6 \varepsilon^{1/2} |u|^{-3/2}$ for matched pairs with $|u| \ge 25\varepsilon$: a quantitative rotation estimate for the massive lattice Green's function.

The comparison principle and the two-flow bound. A *renormalised family* is a finite sum of King's renormalised graphs [15, §3.5], Bałaban's classes of [3, §§2–3] (Definition G.1 of the supplemental material), localised as in [3, (1.33)]. It is in *per-flow normal form* when, for each flow separately and before any element is mixed, every counterterm graph is expanded in the lines of the flow [15, (3.76)], every Taylor subtraction [15, (3.83)] and Ward-identity rewriting has been carried out with that flow's exact identities, and the exact cancellations between members with identical element sets are co-switched. Its *elements* are lines, vertex functions and external fields.

Proposition 17.38 (comparison principle)**.** *Let $\mathcal{R}$ be a renormalised family in per-flow normal form with $n_{\mathcal{R}}$ elements and $E^F(\mathcal{R})$ its value along flow F. Then, with $\theta = \min(\frac{1}{2}, \theta_1)$ and θ_1 of Theorem 17.39,*

$$|E^X(\mathcal{R}) - E^Y(\mathcal{R})| \le C n_{\mathcal{R}}\, p_r^2 \delta_r^{-1/2}\, \eta'^{\theta} (1+k)^p \cdot [\textit{right side of } [3,\ (1.33)] \textit{ for } \mathcal{R}],$$

the prefactor $p_r^2\delta_r^{-1/2}$ being that of the contour elements at $e > 0$ (absent at $e = 0$).

Idea of the proof. Both values are sums over X-vertices; order the elements consistently across members and telescope, switching one element at a time; in each term the difference element, divided by $C\eta'^{\theta}$, obeys the bound of its type weakened by the degree θ (Corollary 17.36, Proposition 17.72(b) of the supplemental material), and the weakened-element bound for each term is Theorem 17.39.

Theorem 17.39 (the two-flow comparison bound)**.** *Let $\alpha \in (0,1)$ be the Hölder exponent of [15, (3.8)] and put $\theta_1 := \alpha^2/(4(1+\alpha)^2)$. There are $p = p(\bar{n})$ and C such that for every renormalised family $\mathcal{R}$ of order $\le \bar{n}$ in per-flow normal form (the h-families of Lemma 17.60 of the supplemental material other than $\mathcal{T}_h$, and the families of step (5) of the proof of Theorem 17.56 in the supplemental material at $e > 0$ after the constant-background reduction of Lemma 17.40 of the supplemental material), for $0 < \theta \le \theta_1$, uniformly in r, k, the torus and the masses, the two-flow difference obeys the bound of Proposition 17.38: in every term of its telescoping, the element replaced by the difference $e_i^X - e_i^Y$, which obeys the bounds of its type weakened by the degree θ, [3, (1.33)] holds with the extra factor $C(1+k)^p\eta'^{\theta}$ and the prefactor stated there. The exponent θ_1 does not depend on $\bar{n}$, and any $\theta > 0$ with any polynomial factor suffices for the consumers (Lemma 17.26 of the supplemental material).*

Idea of the proof (Lemmas 17.42 to 17.45, 17.48 and 17.50 to 17.55, Corollary 17.46, Definition 17.47, and Remark 17.49 of the supplemental material). Inputs: Proposition 17.72 of the supplemental material, [15, Thm. 3.3], [3, Prop. 2.1, (2.8)] with the ordering device (2.7) there, the counterterm conditions [15, (3.33)] through order four with [3, (2.23)–(2.28)], the local limit law [17, Thm. 2.5.6], and the h-census of Lemma 17.60 of the supplemental material. After the constant-background reduction every line is an $A = 0$ line, $G_r^F = C^F - \Gamma^F$, where C^F is *one* lattice function $C = (-\Delta_{\eta'} + m_r^2)^{-1}$ evaluated in each flow's own frame, sliced in the time variable of the simple random walk. Degrees are read per ordering of the scale-summed elements, so that Bałaban's multiscale bound tests only the connected components of initial segments, and a distinguished element carrying $(\eta'/\ell_j)^{\theta}$ keeps the factor η'^{θ} whenever every component containing it has degree $> \theta$. The connected all-C sets of non-positive bare degree are enumerated; each unit (self-energy, vacuum polarisation) is bundled with one of its legs into a *compound*, an own-frame second-order Taylor subtraction whose local terms are common numbers of both flows, and one subtraction rule, Bałaban's integration by parts [3, (2.8)] or a first-order subtraction at a terminal with a common zero moment, raises every newly formed component of degree zero to degree one; afterwards every component of every initial segment has degree ≥ 1, or $n + m\alpha$ with $m \ge 1$, or is an explicit-η' component (a finite case analysis, confirmed by an exhaustive enumeration for every graph with at most two units among the two- and three-point structures through order four). Every element type obeys a two-flow difference law with a positive exponent (the local limit law gives $(\eta'/\ell_j)^{1/2^n}$ for a slice line with n derivatives, by rotation covariance of the isotropic continuum slice; gradients of externals and coarse elements gain η^{κ}, $\kappa = \alpha/(2(1+\alpha))$; compounds gain η'^{θ_U}, $\theta_U \ge \min\{\theta, s/8\}$, $s = \alpha/(1+\alpha)$), and no frame vector ever appears. The telescoping switches compounds first and externals last, uses identities only at the two pure endpoints in each flow's own frame, and bounds every term by the multiscale bound with the switched element weakened by θ; $\theta_1 = \alpha^2/(4(1+\alpha)^2)$ is at most every gain exponent and does not depend on $\bar{n}$.

The constant-background reduction and the one-flow instance (Lemma 17.40 and Corollary 17.41; statements and complete proofs in the supplemental material). King's expansion [15, (3.43)–(3.55)] is carried out for each flow separately on its infinite lattice, in universal-cover form; the constant part of the background is removed exactly by a gauge function, leaving King's

phases [15, (3.46)]; every family at background $A^{(r)}$ becomes a sum of families at $A = 0$ in per-flow normal form, times common explicit coarse-field factors and the phases, plus terms bounded per flow by $C(\delta_r^\sigma + e^{-\delta' p(\delta_r)})|T|$. Corollary 17.41 of the supplemental material is King's weakened-element induction [15, (3.67)–(3.70)] for one flow, standard or hybrid, with zero-degree logarithms $(1+k)^p$; it gives [15, Prop. 3.6], asserted there without a printed argument, for both flows.

The comparison of the effective actions at equal spacing.

Theorem 17.56 (equal-spacing comparison, total form)**.** *Let $T \in \{T_1, T_2\}$, G a real smooth P-periodic 2-form, $z \in \mathbb{R}$, and $r = r(k)$ as in Lemma 17.26 of the supplemental material. Then for $k \geq k_0(zG)$,*

$$\sup_{\operatorname{supp}\chi_r} \left|\hat{s}_T^{(r)}(A_r, \phi_r; zG, 0) - S_T^{(r)}(A_r, \phi_r; zG, 0)\right| \leq \vartheta_k |T|, \qquad \vartheta_k := C\big(5^{-\gamma r}\delta_r^{-\beta}(1+k)^p + \delta_r^\sigma\big) \to 0,$$

every c-number included: there is no vacuum constant and no source-linear constant at $h = 0$.

Idea of the proof (with Theorem 17.57, Proposition 17.58, and Remark 17.59 of the supplemental material). Both effective actions are written in the form of [15, (3.90)] and compared term by term on $\operatorname{supp}\chi_r$, per coarse site, times $|T|\delta_r^{-3}$ sites: the quadratic forms by Proposition 17.31; the normalisations by $\leq 12.5\, a\, 5^{3k-5r}|T|$; the background determinants by $\leq \vartheta_k|T|$ (see the supplemental material; King's [15, (3.94)–(3.98)] is not used, the per-p middle line of (3.98) being false as printed); and the interaction $P^{(r)}$, after the constant-background reduction, by Proposition 17.38 family by family, in per-flow normal form with the same combinatorics in both flows. The vacuum constant $v_k(T) = S_T^{(r)}(0,0;0) - \hat{s}_T^{(r)}(0,0;0)$ tends to 0 on each torus (Theorem 17.57 of the supplemental material): E_1 is the same c-number for both lattices, the torus vacuum graphs are up to $O(\varepsilon^{-q}e^{-c\delta_r^{-1}})$ the infinite-lattice densities, the same for both tori, and the vacuum graphs reduce to families with at least two legs (Proposition 17.58 of the supplemental material for the order λ^2). The comparison must be applied to manifest forms in which every counterterm is expanded in the flow's own lines (Remark 17.59 of the supplemental material).

Walls: the source vertex.

Theorem 17.62 (comparison with a wall)**.** *For $T \in \{T_1, T_2\}$ and $s = (zG, h)$ real, on $\operatorname{supp}\chi_r$: $\left|\hat{s}_T^{(r)}(\cdot; s) - S_T^{(r)}(\cdot; s) + c_k \mathsf{S}_{X_k}(h)\right| \leq \vartheta_k(s)|T| \to 0$, with c_k of Lemma 17.61 of the supplemental material.*

Idea of the proof (Lemmas 17.60 and 17.61 of the supplemental material). In King's degree the h-vertex [15, (3.21)] has $D = 2$, so every non-vanishing field-dependent h-family has $D \geq 1$; the only h-families with $D \leq 0$ are field-independent and linear in h (the free tadpole, the order-λ figure-eight and the order-e^2 tadpoles with their counterterm partners), the family $\mathcal{T}_h$ (Lemma 17.60 of the supplemental material). With f^F the infinite-lattice one-point density of $\mathcal{T}_h$ along the flow F, $\bar{f}^F$ its cell average and $c_k := -(\bar{f}^X - \bar{f}^Y)$, independent of the torus with $|c_k| \leq C\varepsilon^{-q}$, the $\mathcal{T}_h$-part of $\hat{s}_T - S_T$ at zero coarse fields is $-c_k \mathsf{S}_{X_k}(h) + o(1)$ on both tori (Lemma 17.61 of the supplemental material); every other h-family is telescoped with Proposition 17.38 as in the proof of Theorem 17.56.

The source-linear constant vanishes.

Proposition 17.71 (the source-linear constant vanishes)**.** *With c_k as in Lemma 17.61 of the supplemental material and $r = k - \lfloor \varrho k \rfloor$, $\varrho < \frac{1}{9}$, for $k \geq k_0(a, m, \mu_0, \varrho)$*

$$|c_k| \leq N C_\Gamma\, \eta^{1/2}\delta_r^{-1} + C_1 \lambda\, \eta^{1/2} + C_2 e^2\, \eta'^{1/8}(1 + \delta_r^{-1})(1 + \log(1/\eta'))^4 \leq C\, 5^{-(1-9\varrho)k/8}(1+k)^4,$$

with constants depending on a, m, μ_0, N, τ only (not on J, ψ, n, the torus or h). Hence $c_k \to 0$.

Idea of the proof (Lemmas 17.63 to 17.70 of the supplemental material). $|\bar f^X - \bar f^Y| \le \sup_{x'} |f^X(x') - f^Y(\tau x')|$, and the three members of $\mathcal{T}_h$ are compared at matched points: the free tadpole leaves the residue $-N\Gamma^F(x,x)$, with difference $\le NC_\Gamma \eta^{1/2}$; the figure-eight is $-4N(N+2)\lambda_r \sum_y G^F(x,y)^2\Gamma^F(y,y)$, telescoped with the matched-pair bounds of Corollary 17.37 of the supplemental material; and the order-e^2 density is split exactly as $Z^F + A^F(x) + B^F(x)$ with the renormalised one-loop kernel, where the all-C part Z^F cancels exactly between the flows and the parts with external or internal Γ-lines have a positive power-counting margin and gain $\eta'^{1/6}$ up to logarithms. The conversion to physical units costs δ_r^{-1}, and $\varrho < \frac{1}{9}$ makes every term geometric in k; only $c_k \to 0$ is used.

The inputs of the forward bound. Throughout $T \in \{T_1, T_2\}$ and $r = r(k)$ is the choice of Lemma 17.26 of the supplemental material; $p(\delta) = b_0(1+\log \delta^{-1})^{\mathrm{p}}$ is King's small-field function.

The kernel bounds of the hybrid flow at a non-zero gauge field (Proposition 17.72 of the supplemental material). For $r \ge 2$ and a background A regular in the sense of [15, Def. 3.2] with $e_r^\beta \le c_0(a)$: (a) the operators G_r, $\Delta^{(r)}$, $C^{(r)}$ of the hybrid flow with the paired contours obey the bounds of [15, Thm. 3.3, (3.5)–(3.8)] in the periodic form, on the infinite lattice and on tori fitted by both lattices, each in its own quotient distance, with constants depending on a, M, α only, except the Hölder constant of (3.8), which is $C_\alpha \log^{3\alpha}(1/\eta')$; the differences between two such domains with a common fundamental region on which the backgrounds agree carry the boundary factor of [15, Thm. 3.3] in the smaller of the two quotient metrics; the restricted inverses on unions of big blocks obey [16, (5.54)]; the fine-slice lines obey [15, (3.63)–(3.65)]; and the level-r elements H^X, B^X, Γ^X obey their decay and regularity bounds, all uniformly in r, k and the torus, at every level $j \le r$ and, in the form the hybrid lower bound uses, at the levels 0 and 1. (b) After the constant-background reduction the difference elements $\Gamma^X - \Gamma^Y \circ \tau$, ΔH, $\Delta(A^{-1})$ obey the bounds of Corollary 17.36 of the supplemental material with the factor $\eta'^{1/2}$; the contour vertices [15, (3.24)–(3.26)] differ entrywise by a relative $\eta^{1/2}$ against their size $Ce_r p_r^2$; the constant-background phases [15, (3.46)] at matched vertices differ by $\le e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2} \eta'$; and the contour lines, one-ended and double, obey the slice-j bounds in both flows with a relative mismatch $(\eta'/\ell_j)^{1/2}$, a one-ended line with one lattice derivative at its lattice end with $(\eta'/\ell_j)^{\theta_\partial}$, $\theta_\partial = \alpha/(2(2+\alpha)) \ge \theta_1$. At $A = 0$ this is the propagator package; at $A \ne 0$ the proof is the programme of [16, §5.3] written out: coercivity at a regular background by a cover with axis cubes carrying a local gauge for the constant part of the field ([16, (5.59)–(5.60)]); Combes–Thomas decay in L^2 form [28]; the geometric resolvent identity for the two-domain clause; Bałaban's random walk [4, §2] with periodic cells fitted by both lattices [16, (5.55)–(5.56)], the Neumann series summed in the sup norm first so that the smallness condition is uniform in r; the cubic-Hermite cardinal interpolation of the Y kernels; and, for (b), King's constant-background reduction applied to both flows on their infinite lattices, after which every difference element is of one of seven types (an $A = 0$ line; a contour line, with the kernel difference along the contour plus the flux of the slice kernel through the sliver; a double contour line; a contour vertex; an external lifted field; a constant-background phase; a hopping vertex). What would break (b) are mis-paired contours or mis-rooted blocks, both excluded by Definition 17.15 and the centred blocks. The walk-sum bounds are written out in the supplemental material.

King's upper bound for the rotated lattice.

Proposition 17.96 (King's upper bound for the rotated lattice)**.** *For Y_k on T and every real smooth periodic source $s = (zG, h)$, with the free-boundary convention of Definition 15.6 and the centred blocks of Convention 17.17 of the supplemental material,*

$$\mathsf{Z}_{Y_k,T}(s) \le e^{C\delta_r^\sigma |T|} \int \chi_r\, e^{-S_T^{(r)}} + e^{-p(\delta_r)^2 + C|T|}\, \varepsilon^{-q},$$

$C = C(s, |T|)$, $q = q(s) < \infty$, with $q = 0$ when $h = 0$.

Idea of the proof. On T_2, aligned in Y's frame, the passive rotation R maps (Y_k, T_2) to (X_k, D_J), and the statement is King's Theorem 3.1 [15, (3.4)] at block factor 5 (Theorem F.2(iii) of the supplemental material with Lemma 5.1), read with the free-boundary counterterms at a

constant-shifted source (Proposition 17.121(b) of the supplemental material) and with centred blocks; on T_1 the frame torus of Y_k is skew, and the instance is Proposition 17.117. At a wall the stability bound [15, (2.23)] fails as printed (Remark 16.1) and the large-field term is read with the polynomial envelope ε^{-q}, which Proposition 17.134 absorbs.

The cumulant bound for the step Gaussian. Summary (Theorem 17.112 and Lemmas 17.97 to 17.111 of the supplemental material). The fluctuation integral of every block-spin step of both flows has the form $\int \chi_b\, e^{V} d\mu_P$: μ_P is a centred Gaussian on $(\mathbb{R}^n)^\Lambda$, Λ a finite subset of $\mathbb{Z}^3$, of a cubic torus or of one of the skew tori of Proposition 17.117, whose precision $P \geq \gamma_0 > 0$ has exponentially decaying off-diagonal blocks; $V = \sum_{Y \subset J} V_Y$ is a real polynomial of degree at most D with exponentially decaying coefficients, $\mathsf{A} := \|V\|_\varsigma$; and χ_b is a product over the sites of indicators of a fixed symmetric convex body scaled by b. Theorem 17.112 of the supplemental material is the order-$\bar n$ cumulant expansion of such an integral: there are $S < \infty$, $\rho_3 > 0$ and a threshold b_*, depending only on $(n, \bar n, D, \varsigma, \gamma_0, c_P, \delta_P, c_K)$, such that for $b \geq b_*$ and $\mathsf{A} b^{D+6} \leq 1$

$$\Big| \log \int \chi_b\, e^{V} d\mu_P - \sum_{k=1}^{\bar n} \frac{\kappa_{\mu_P}(V;k)}{k!} \Big| \leq S|J| \big[(\mathsf{A} b^{D+6})^{\bar n+1} + e^{-\rho_3 b^{3/2}} \big] + 4n|\Lambda| e^{-\gamma_0 \gamma_K^8 b^2/(2n)},$$

the last term in the lower bound only; nothing depends on Λ, on J, on the torus or on translation invariance. For the step Gaussians of both flows the hypotheses hold with Bałaban's coercivity constant [1, (2.33)] (for the hybrid flow by Proposition 17.72(a) of the supplemental material), the Combes–Thomas decay of the step precision, $\mathsf{A} \leq s_0 O(1) \ell_j^{\beta_0}$ by Corollary 17.41 of the supplemental material, Bałaban's two-component cutoff [1, (3.43), (3.50)] and $b = p_1(\ell_j)$. The printed cluster-expansion lemma of [5], which King cites for this step, uses the Markov property of the nearest-neighbour field in four places; the proof keeps its architecture (cells with corridors, four shifted pavements, the cutoff moved from the corridors into the cells) and replaces those four uses by a Combes–Thomas bound for every principal block, a conditioning lemma, a decoupling lemma across corridors and the comparison of two Gaussian laws.

Proposition 17.113 (the hybrid lower bound)**.** *For the hybrid flow on T and every real smooth periodic source s,*

$$\mathsf{Z}_{X_k,T}(s) \geq e^{-C(1+k)^p \delta_r^\sigma |T|} \int \chi_r\, e^{-\hat s_T^{(r)}}.$$

Idea of the proof. Bałaban's lower-bound induction [1, §§3–4] at block factor 5 run for the hybrid flow on both tori, with the hybrid kernel bounds in place of [1, Props. 2.1–2.3] and [16, Prop. 5.6], the cumulant lower bound, the skew instance Proposition 17.117 on T_1 (whence the constant $C(1+k)^p$ of Corollary 17.41 of the supplemental material) and the counterterm bridge; the induction at block factor 5, the kernel bounds, the cumulant lower bound and the bridge are Theorem F.2, Proposition 17.72(a), Theorem 17.112 and Proposition 17.121 of the supplemental material. The complete proof is in the supplemental material.

Proposition 17.114 (the polynomial lower bound for the rotated lattice)**.** *For Y_k on T and every source $s = (zG, h_0)$ with $h_0 = -n\psi$, $n \geq 0$: $\mathsf{Z}_{Y_k,T}(s) \geq e^{-C_1|T|} \varepsilon^{q(s)}$ for some $q(s) < \infty$.*

Idea of the proof. At $h = 0$ this is Bałaban's lower bound [1, (1.14)] (Theorems F.1 and F.2 of the supplemental material), on T_1 by Proposition 17.117; the source zG is removed by Jensen's inequality, since $\langle F(G) \rangle = 0$ under charge conjugation. At the wall the two-sided ε-uniform bound [16, (4.2)] is false (Remark 16.1) and only the polynomial form is asked for: it is Theorem 17.116 (the alternative through Proposition 17.115 of the supplemental material is not used).

Theorem 17.116 (floor shift: the polynomial lower bound at the wall at every King point)**.** *Let m^2 be any King point and h a real P_J-periodic function (it may depend on k) with $\|h\|_\infty \leq H$ uniformly in k. Write $a_s := s + \delta m^2(s;\varepsilon)$ for the bare scalar coefficient at the King point s and let M^2 be a King point with $\Delta_0(M^2) := \liminf_{\varepsilon \to 0} [a_M - a_m] > 2H$; such an M^2 exists for every coupling and every H, since $\Delta_0(M^2) \to +\infty$ as $M^2 \to \infty$ (the coefficients of $\partial_s \delta m^2$ are non-negative at order λ, $O(s^{-1/2})$ at order e^2 and $O(s^{-1})$ at the two-loop orders; see the*

supplemental material). Then on $\Lambda \in \{Y_k, X_k\}$, $T \in \{T_1, T_2\}$, *in the free-boundary convention of Definition 15.6 and equally in the torus convention of Proposition 17.121 of the supplemental material,*

$$\ln \mathsf{Z}^{m^2}_{\Lambda,T}(0,h) \geq -C|T| - C(1+k)|T| - qk \ln 5, \qquad \textit{hence} \qquad \mathsf{Z}^{m^2}_{\Lambda,T}(tG,h) \geq e^{-C_1|T|} \varepsilon^{\,q'} \quad \textit{for every real } t.$$

This is the polynomial form of Proposition 17.114 at the wall $h_0 = -n\psi$ *for every* $n > 0$*; the sign of* h *is used nowhere.*

Idea of the proof. The *floor shift* $e^{\sum \varepsilon^3 h|\phi|^2} = e^{h_{\min} \sum \varepsilon^3 |\phi|^2} e^{\sum \varepsilon^3 (h - h_{\min})|\phi|^2}$ moves the wall into the bare scalar coefficient, Jensen's inequality bounds the remaining factor below by 1, and the monotonicity of $\langle \sum \varepsilon^3 |\phi|^2 \rangle_\gamma$ in the bare coefficient bounds the shifted integral below by the one at the higher King point M^2, where [1, (1.14)] applies; the ε^{-1}-divergent c-numbers collapse exactly by the integrated counterterm identity, and the perturbative remainder is $\geq -C(1+k)|T|$ by Reisz's theorem [22, Thm. 1] with [15, (3.33)]. The complete proof is in the supplemental material.

The skew-torus instances. A torus $\mathbb{R}^3/\Pi$ is *aligned* in the frame of a lattice $O(s\mathbb{Z}^3)$ if $\Pi = O(\ell_1\mathbb{Z} \times \ell_2\mathbb{Z} \times \ell_3\mathbb{Z})$, and *skew* otherwise; [1, 2, 3] and [15] are written for a torus aligned in the frame of the blocks. T_1 is aligned for X_k and skew for Y_k; T_2 is aligned for Y_k and skew for X_k. On T_1 every block-spin step performed in Y's frame is a step on a torus skew in the frame of its blocks; on T_2 only the first hybrid step is.

Proposition 17.117 (the skew-torus instances)**.** *King's Theorem 3.1* [15, (3.4)] *(Bałaban's upper bound with the renormalisation of* [3]*) and Bałaban's lower-bound machinery* [1, §§2–4, (1.14)] *hold for the standard flow of* Y_k *on* T_1*, the torus* RD_J *in* Y*'s frame, which is skew, with the constants of the aligned case times at most* 125 *in the linear part; and Bałaban's lower-bound argument holds for steps* $2, \dots, r$ *of the hybrid flow on* T_1 *with the constant* $C(1+k)^p$ *of Corollary 17.41 of the supplemental material. In both, the torus enters* [1, 2, 3] *only through the embeddings of the block lattices, the external fields and the image terms of the periodised kernels; no torus symmetry and no torus Fourier analysis is used in the proof of* [3, Prop. 1]*. The statements hold under the provisos* $J \in 5^{n_*}\mathbb{N}$ *and* $J \geq J_0(\delta_r)$ *of Theorem 17.13.*

Idea of the proof (Lemmas 17.118 to 17.120 and the census, all in the supplemental material). The skew torus $T' = RD_J$ in Y's frame is covered by the cube $\tilde{T} = \mathbb{R}^3/250MJ\,\mathbb{Z}^3$, aligned in Y's frame, with a deck group of order 125 that commutes with every block average, so every kernel of a Y-frame flow on T' is the 125-fold periodisation of the kernel on $\tilde{T}$ (*folding*); inequalities between liftable functionals descend from $\tilde{T}$ to T' (*restriction principle*); and every step of [1, §§2–4], [2, §§2–3], [3, §§1–3], [15, §3] is of one of six types, each of which holds verbatim on T' (*type closure*), the local arguments needing sets of sup-diameter below the systole, the proviso $J \geq J_0(\delta_r)$; the cumulant bound is Theorem 17.112 of the supplemental material on the skew tori.

Summary (Proposition 17.121 of the supplemental material; the image-sum estimates are in the supplemental material). [15] and [1, 2, 3] are proved for the model in which the counterterms, the Wick subtraction and the source vertex are defined with torus propagators, while Definition 15.6 uses the free-boundary conventions of [16, p. 330]. With $\Delta_T(s) := \delta m^2(s) - \delta m^2_T(s)$, $\Delta E_T(s) := E_1(s) - E_{1,T}(s)$ and $\Delta c_T(s) := \mathsf{W}_s - c_T(s)$, the free-boundary model is, by exact algebra, the torus-convention model with the bare mass shifted by Δ_T, times a c-number and a shift of the Wick constant, which both block-spin flows carry through unchanged (Lemma F.3 of the supplemental material); the three quantities are bounded uniformly in s, $\Delta_T(0) = \lim_{s\to 0} \Delta_T(s)$ exists and $|\Delta_T(s) - \Delta_T(0)| \leq Cs$. Proposition 17.134, the only consumer of Propositions 17.96, 17.113 and 17.114, is c-number blind, so those bounds, proved for the torus-convention model at the constant-shifted source $h - \frac{1}{2}\Delta_T(s)\mathbf{1}$, hold for the model of Definition 15.6 as consumed; and the walled curvature ratio is the same number in both conventions at the shifted wall $h_0 - \Delta_T(\varepsilon_k)/2$.

The existence of the walled axis-lattice limits.

Theorem 17.122 (the walled torus limit on the axis lattice)**.** *For every walled configuration* $\mathfrak{c} = (J, \psi, n)$ *of the class of Theorem 17.13 and every real smooth* P_J*-periodic 2-form* G *the limit*

$$\mathcal{W}_{\mathfrak{c}}(tG) := \lim_{k\to\infty} \frac{\mathsf{Z}_{X_k,T_1}(tG, h_0)}{\mathsf{Z}_{X_k,T_1}(0, h_0)} = \lim_{k\to\infty} \langle e^{tF(G)} \rangle_{\mathfrak{c},X_k}$$

exists for every real t*; it is the same number in the free-boundary convention of Definition 15.6 and in King's torus convention at the shifted constant source* $h_0 - \Delta_T(\varepsilon_k)/2$ *(Proposition 17.121(c) of the supplemental material), and it equals the limit of the torus-convention ratios* $\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(tG, h_\infty)/\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(0, h_\infty)$ *at the frozen wall* $h_\infty := h_0 - \Delta_T(0)/2$*. The convergence is locally uniform in* $t \in \mathbb{C}$*, the limit is entire, and all moments of* $F(G)$ *converge.*

Idea of the proof (Lemmas 17.123, 17.124, 17.127, 17.128 and 17.131 to 17.133 and Propositions 17.125, 17.126, 17.129 and 17.130 of the supplemental material). This is King's torus theorem [15, Thm. 2.1] with the wall as his scalar source, read on the normalised ratio, in the only form in which it is true at a wall; it is a one-lattice statement. With $\mathsf{Z}^{\mathrm{tor}}_k(t; h)$ the torus-convention functional, $h_k := h_0 - \frac{1}{2}\Delta_T(\varepsilon_k)\mathbf{1}$ the *moving wall* and $h_\infty := h_0 - \frac{1}{2}\Delta_T(0)\mathbf{1}$ the *frozen wall*, $|h_k - h_\infty| \le C\varepsilon_k$, the bridge gives $r_k(t) := \langle e^{tF(G)} \rangle_{\mathfrak{c},X_k} = \mathsf{Z}^{\mathrm{tor}}_k(t; h_k)/\mathsf{Z}^{\mathrm{tor}}_k(0; h_k)$, literally the object of [15, Thms. 3.1, 3.4] with the scalar source h_k; all ratios obey $1 \le \mathsf{Z}^{\mathrm{tor}}_k(t; h)/\mathsf{Z}^{\mathrm{tor}}_k(0; h) \le e^{C_G t^2}$, so convergence for real t gives locally uniform convergence on $\mathbb{C}$. King's upper bound, with the large-field term read with the polynomial envelope $e^{-p(\delta)^2 + C|T|}\varepsilon_k^{-q}$, and Theorem 17.116 show that each ratio differs from its small-field ratio by $o(1)$. King's renormalisation theorem [15, Thm. 3.5] is false as printed on the order-e^2 h-linear field-free family (Remark 16.1), but its convergence clause holds for the field- or source-dependent parts of every graph, because every connected ordered subgraph of their non-vanishing derived graphs has strictly positive degree, so that Corollary 17.41 of the supplemental material applies; hence the effective actions at consecutive k on the common unit lattice differ by a field-free, t- and G-independent constant d_k up to $\eta_k|T|$, King's [15, (3.9)] without the field-free h-sector constant, which is exactly what drifts; d_k cancels in every ratio, and the consecutive differences of the frozen ratios are summable. A constant shift u of the scalar source changes the effective action by at most $C(1+r)|u|p(\delta_r)^{n_1}\delta_r^{-3}|T|$, and the rate $|u_k| \le C\varepsilon_k$ against δ_r^{-3} is geometric for $\varrho < \frac{1}{3}$. No value of d_k and no Wick scheme is used, and the sign of the wall enters nowhere.

The forward bound and the assembly.

Proposition 17.134 (the forward bound)**.** *For* $T \in \{T_1, T_2\}$ *and every real source* $s = (zG, h)$ *with* h *a wall* $-n\psi$ *or its transport,*

$$\mathsf{Z}_{Y_k,T}(s) \le (1 + \delta_k)\, e^{-c_k \mathsf{S}_{X_k}(h)}\, \mathsf{Z}_{X_k,T}(s), \qquad \delta_k = \delta_k(T, s) \to 0,$$

with the constant c_k *of Lemma 17.61 of the supplemental material, independent of* T *and of the source, and* $|c_k| \le C\varepsilon_k^{-q}$*. No bound on* c_k *is used.*

Idea of the proof. By Proposition 17.96, $\mathsf{Z}_Y \le e^{C\delta_r^\sigma |T|} \int \chi_r e^{-S^{(r)}} + B$ with the large-field term $B = e^{-p(\delta_r)^2 + C|T|}\varepsilon^{-q}$; by Proposition 17.114 (at the wall, Theorem 17.116) $B \le b\,\mathsf{Z}_Y$ with $b \to 0$, since $p(\delta_r)^2$ grows at least quadratically in k and $\ln \varepsilon^{-q-q'}$ linearly, so the large-field term is absorbed on the left, which is why no bound on c_k and no lower bound for the axis lattice is needed; on $\operatorname{supp}\chi_r$, Theorem 17.62 gives $e^{-S} \le e^{-\hat{s} - c_k \mathsf{S}_{X_k}(h) + \vartheta_k |T|}$, and Proposition 17.113 completes the chain.

Proposition 17.135 (mass-increasing walls only)**.** *Let* $\mathcal{S}^- := \{(tG, h_0) : t \in \mathbb{R}\}$ *with* $h_0 = -n\psi$*,* $n > 0$*, a family not closed under* $h \mapsto -h$*, and assume the forward bounds of Proposition 17.134 on* T_1 *for* $s \in \mathcal{S}^-$ *and on* T_2 *for their* S*-images, with the common constant* c_k*. Then the conclusions* (a) *and* (c) *of Theorem 17.6 hold for the sources of* $\mathcal{S}^-$*, with* $v = v' = 0$*,* $c = c' = c_k$ *and* $c_k \to 0$ *(Proposition 17.71); the sources* $(tG, +n\psi)$ *do not occur.*

Idea of the proof. Part (a) of Theorem 17.6 uses only the two forward bounds, one source at a time; part (b) is the only place where the pair $(G, \pm h)$ enters, and its conclusion $c + c' \to 0$ is supplied directly by $c = c' = c_k \to 0$; part (c) follows as there.

Proof of Theorem 17.13 (route). Fix $\mathfrak{c} = (J, \psi, n)$ in the class and G, and let $\mathcal{S}^- = \{(tG, h_0) : t \in \mathbb{R}\}$, $h_0 = -n\psi$. *Step 1*: Theorem 17.62 on T_1 for $s \in \mathcal{S}^-$ and on T_2 for $S \cdot s$, with the common constant c_k. *Step 2*: Proposition 17.134 on both tori (inputs: Propositions 17.96, 17.113 and 17.114, hence Theorem F.2 and Proposition 17.121 of the supplemental material, Proposition 17.117 on T_1, Theorem 17.116 at the wall). *Step 3*: Proposition 17.135 with Proposition 17.71 gives conclusion (c) of Theorem 17.6 for the balanced pair (tG, h_0), $(0, h_0)$, i.e. $\langle e^{tF(G)}\rangle_{\mathfrak{c},Y_k} - \langle e^{tF(G)}\rangle_{\mathfrak{c},X_k} \to 0$ for every real t. *Step 4*: by Lemma 2.8 and Lemma 17.21 the difference is bounded on strips uniformly in k, so Vitali's theorem gives (i). *Step 5*: by Theorem 17.122 the axis family converges, and by (i) so does the rotated family, to the same entire limit: (ii). *Step 6*: Lemma 17.3(c) gives (iii). No reverse step ("switching around", [16, p. 337]), no comparison of two spacings and no infinite-volume limit is used. □

Completion of the proof of Theorem 17.1. By Lemma 17.3(c) it suffices to treat $R = R_{12}(\theta_0)$, and by Remark 18.6 the configurations to be treated are those of the class of Theorem 17.13. For such $\mathfrak{c}$ and G, Steps 1–2 of the proof of Theorem 17.13 give (F) on both tori for the sources (tG, h_0), in total form, with $v_k(T_1) = v_k(T_2) = 0$ and $c_k(T_1) = c_k(T_2) = c_k \to 0$; Proposition 17.135 supplies conclusion (c) of Theorem 17.6, Corollary 17.7 gives $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 0$ locally uniformly in z, and with Theorem 17.122 its last clause gives (K_0) for $\mathfrak{c}$ and G. □

18. Walled tori, the closing estimate and rotation invariance

18.1. **Walled tori and the closing estimate.**

Definition 18.1 (admissible walls and walled values)**.** For a bounded set $X \subset \mathbb{R}^3$ let $\mathcal{A}(X)$ be the set of walled configurations $\mathfrak{c} = (J', w)$ of Definition 15.6 with $N_1(X) \subset (-25MJ', 25MJ')^3$ and $w = 0$ on $N_1(X)$. The interior $(-25MJ', 25MJ')^3$ of the cell $\Gamma_{J'}$ is a fundamental domain of $D_{J'}$ for both lattices, $R^{-1}\Lambda_k$ descending to $D_{J'}$ for $R \in \mathcal{R}_0$ by Lemma 15.3(b), and sources supported in X are periodised from it. If $X \subset Y$ then $\mathcal{A}(Y) \subset \mathcal{A}(X)$. Recall the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k \mathsf{S}_\Lambda(g)}\langle \Theta(\mathrm{i}f, g)\rangle_{\mathfrak{c},\Lambda}$ of a lattice Λ of spacing ε_k descending to $D_{J'}$, defined for $(f,g) \in \mathcal{K}$ with $w = 0$ on $\operatorname{supp} g$ (Lemma 15.9(c)). For $g \neq 0$ assume $(\mathrm{K})_a$ (Definition 17.2). For $(f,g) \in \mathcal{K}$ supported in X and $\mathfrak{c} \in \mathcal{A}(X)$ the periodised sources are smooth, and by Lemma 17.3(a) the *walled value*

$$\mathcal{W}^a_{\mathfrak{c}}(f,g) := \lim_{k\to\infty} T^a_{\mathfrak{c},\Lambda_k}(f,g) = \lim_{k\to\infty} T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \qquad (R \in \mathcal{R}_0)$$

exists in $(0,\infty)$. For $b > 0$ and $(f,g) \in \mathcal{K}$ supported in Q_b let $I^a_b(f,g) := \inf_{\mathfrak{c}\in\mathcal{A}(Q_b)} \mathcal{W}^a_{\mathfrak{c}}(f,g)$, which is nondecreasing in b, and

$$\mathcal{W}^{a,\mathrm{env}}(f,g) := \lim_{b\to\infty} I^a_b(f,g) \in (0,\infty].$$

At $g = 0$ these definitions rest on (K_0) alone (Theorem 17.1; Lemma 17.3(a) at the complex coefficient i), and they do not depend on the scheme (Lemma 15.9(c)): $\mathcal{W}_{\mathfrak{c}}(f,0) = \lim_k \langle e^{\mathrm{i}F_{\Lambda_k}(f)}\rangle_{\mathfrak{c},\Lambda_k} \in [-1,1]$, and we write $I_b(f,0)$, $\mathcal{W}^{\mathrm{env}}(f,0)$. King's printed state at a curvature source is written $Z^K_\infty(\mathrm{i}f, 0)$ in this part, the second slot recording the absence of a scalar source; it is $Z^K_\infty(\mathrm{i}f)$ of Theorem 7.6(b), and $Z^K_J(\mathrm{i}f)$ denotes the fixed-cube limit there.

Lemma 18.2 (the free direction)**.** (a) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. If* $(f,g) \in \mathcal{K}$ *is supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$ *and* $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$*, then* $\mathcal{W}^a_{\mathfrak{c}}(f,g) \geq Z^a_J(\mathrm{i}f, g)$*, where* $Z^a_J = \lim_k Z^a_{k,J}$ *is the fixed-cube limit of Theorem 16.6(i).*

(b) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. Then* $\mathcal{W}^{a,\mathrm{env}} \geq Z^a_\infty$ *on* $\mathcal{K}$*.*

(c) *If the real 2-form* $f \in C^\infty_c$ *is supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$ *and* $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$*, then* $\mathcal{W}_{\mathfrak{c}}(f,0) \geq Z^K_J(\mathrm{i}f)$*.*

(d) $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z^K_\infty(\mathrm{i}f, 0) \geq 0$ *for every real 2-form* $f \in C^\infty_c$*.*

Proof. (a) Fix k. The set $X_J = C_J \cap \Lambda_k$ and its nearest neighbours lie in $N_1(\bar{C}_J)$, hence in the interior of the cell, and the wall vanishes there; so Lemma 15.8 applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c},\Lambda_k}$ with $X = X_J$ has as its X-factor the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ of Definition 2.1, and gives $\langle\Theta(\mathrm{i}f,g)\rangle^{(a)}_{k,J} \le \langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda_k}$; every plaquette and site at which (f,g) does not vanish lies in X_J by the support condition. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is the same positive constant on both sides (Lemma 15.9(c), Lemma 2.11(a)), so $Z^a_{k,J}(\mathrm{i}f,g) \le T^a_{\mathfrak{c},\Lambda_k}(f,g)$. Let $k \to \infty$: the left side converges to $Z^a_J(\mathrm{i}f,g)$ by Theorem 16.6(i), the right side to $\mathcal{W}^a_{\mathfrak{c}}(f,g)$ by Definition 18.1.

(b) If $\bar{C}_J \subset Q_b$ then $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$, so $I^a_b(f,g) \ge Z^a_J(\mathrm{i}f,g)$ by (a) for every J at which the support condition of (a) holds; let $b \to \infty$ and then $J \to \infty$ (Theorem 16.6(iii)).

(c) At $g = 0$ no multiplier enters. As in (a), Lemma 15.8 gives $Z_{k,J}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J} \le T_{\mathfrak{c},\Lambda_k}(f,0)$, and the first inequality of Lemma 4.2, whose support condition (S) holds for X_J because f is supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, gives $Z^K_{k,J}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J} \le Z_{k,J}(\mathrm{i}f)$. Hence $Z^K_{k,J}(\mathrm{i}f) \le T_{\mathfrak{c},\Lambda_k}(f,0)$ at every k. Let $k \to \infty$: the left side converges to $Z^K_J(\mathrm{i}f)$ by Theorem 7.6(b) under (E_C), the right side to $\mathcal{W}_{\mathfrak{c}}(f,0)$ by (K_0) (Lemma 17.3(a)).

(d) Fix f and $J \ge J_0(f)$. For $b \ge MJ$ we have $\bar{C}_J = Q_{MJ} \subset Q_b$, so $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$ and $I_b(f,0) \ge Z^K_J(\mathrm{i}f)$ by (c); letting $b \to \infty$, $\mathcal{W}^{\mathrm{env}}(f,0) \ge Z^K_J(\mathrm{i}f)$ for every $J \ge J_0(f)$. By Corollary 4.3 and Theorem 7.6(b), $J \mapsto Z^K_J(\mathrm{i}f)$ is nondecreasing with values in $[0,1]$, and by Corollary 4.4(ii)(β) it converges to $Z^K_\infty(\mathrm{i}f,0)$; let $J \to \infty$. □

Lemma 18.2(a),(c) compare a walled value with a cube of fixed size. They do not assert $\mathcal{W}^a_{\mathfrak{c}} \ge Z^a_\infty$ for an individual configuration, and this need not hold: for instance, at $e = \lambda = 0$, $f = 0$ and $g \ge 0$ small, as $n \to \infty$ the walled value of (J', ψ, n) decreases (Lemma 2.7(c)) towards the value of the Dirichlet state on the set where $\psi = 0$, which is smaller than the infinite-volume value, as the Gaussian formula for $\langle e^{:\phi^2:(g)}\rangle$ shows.

Lemma 18.3 (independence of the orientation)**.** *Assume* $(\mathrm{K})_a$ *if* $g \ne 0$. *For* $R \in \mathcal{R}_0$ *and* $(f,g) \in \mathcal{K}$,

$$\lim_{L\to\infty} \inf_{\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}}(f,g) = \mathcal{W}^{a,\mathrm{env}}(f,g),$$

the infimum being defined as soon as $\operatorname{supp} f \cup \operatorname{supp} g \subset Q_{5ML/7}$. *At* $g = 0$ *the same holds, with no assumption, for* $\mathcal{W}_{\mathfrak{c}}(f,0)$ *and* $\mathcal{W}^{\mathrm{env}}(f,0)$.

Proof. By Lemma 15.3(c), $Q_{5ML/7} \subset R^{-1}Q_{ML} \subset Q_{7ML/5}$, hence $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML}) \subset \mathcal{A}(Q_{5ML/7})$, and $I^a_{5ML/7} \le \inf_{\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}} \le I^a_{7ML/5}$. Both bounds tend to $\mathcal{W}^{a,\mathrm{env}}$ as $L \to \infty$. □

Theorem 18.4 ((TIGHT_0), (TIGHT_2): the closing estimate)**.** (TIGHT_0)*:* $\mathcal{W}^{\mathrm{env}}(f,0) \le Z^K_\infty(\mathrm{i}f,0)$ *for every real* 2*-form* $f \in C^\infty_c$. (TIGHT_2)*: for every real* 2*-form* $f \in C^\infty_c$,

$$\lim_{b\to\infty} \sup_{\mathfrak{c}\in\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \ge S^K_2(f), \qquad \sigma^2_{\mathfrak{c}}(f) := \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{\mathfrak{c},\Lambda_k}, \quad S^K_2(f) := S^K_2(f,f) = \lim_J \lim_k \langle F(f)^2\rangle^K_{k,J}.$$

(TIGHT_0) involves only (K_0), which defines its left side, and King's printed state, which exists by (E_C). It is proved in the heavy-shell sections: Theorem 19.10 derives it from the decoupling property (RES_0) of Definition 19.8, which Theorem 20.14 establishes at one configuration from the heavy-shell estimates (HS); (TIGHT_2) follows from (TIGHT_0) by Lemma 18.7(c). In the Wick sector the closing statement is a property of the scheme, not proved here (it is listed among the open problems, in the final section) and used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 18.5 ($(\mathrm{TIGHT})_a$: the closing property, Wick sector)**.** Let a be W-admissible and assume $(\mathrm{K})_a$. $(\mathrm{TIGHT})_a$: $\mathcal{W}^{a,\mathrm{env}} \le Z^a_\infty$ on $\mathcal{K}$. Equivalently: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}^a_{\mathfrak{c}}(f,g) \le Z^a_\infty(\mathrm{i}f,g) + \eta$.

The equivalence in $(\mathrm{TIGHT})_a$ holds because I^a_b is nondecreasing in b. By Lemma 18.2, $(\mathrm{TIGHT})_a$ is equivalent to $\mathcal{W}^{a,\mathrm{env}} = Z^a_\infty$ on $\mathcal{K}$, and (TIGHT_0) to $\mathcal{W}^{\mathrm{env}}(\cdot,0) = Z^K_\infty(\mathrm{i}\,\cdot,0)$: *King's printed state is the limit of the lower envelopes of walled tori, the continuum limit being taken*

first. The limit defining $\sigma^2_{\mathfrak{c}}$ exists by (K_0), since $t \mapsto \langle e^{\mathrm{i}tF(f)}\rangle_{\mathfrak{c},\Lambda_k}$ converges locally uniformly as an entire function (Lemma 17.3(a)) and second derivatives of locally uniformly convergent entire functions converge; S_2^K is the two-point Schwinger function of Theorem 7.6(c), with the lattice second moments converging to it. The supremum over $\mathcal{A}(Q_b)$ is nonincreasing in b, so the limit in (TIGHT_2) exists.

Remark 18.6 (the configuration class). The proofs below use (K_0) only for the walled configurations that enter them, and they use the infimum over $\mathcal{A}(R^{-1}Q_{ML})$ only through an upper bound by the walled value of one configuration supplied by (TIGHT_0). Hence (K_0) and (TIGHT_0) may be read on any common sub-class of walled configurations which is stable under raising the wall height and contains the single-level configuration produced by Theorem 19.10: the infimum is then taken over that sub-class, (TIGHT_0) is read with the same sub-class, and Lemma 18.3 holds on it, since its proof uses only the inclusions between shrinking families. The sub-class on which the proof of (K_0) along King's lines [16] supplies the comparison consists of tori whose side is a large multiple of a power of 5, of mass-increasing walls whose mean over the torus exceeds a fixed multiple of the torus volume, and of a source box and torus aligned with the blocks of the final scale. The configuration of Theorem 19.10 lies in it: its wall is $(n+K)\psi$ with $n+K \geq 1$, mass-increasing; the condition on the wall mean is linear in the height and holds once K is large at a fixed torus, which Remark 19.11(iii) allows; the torus and the alignment of the source box are choices made in Theorem 20.14.

Lemma 18.7. (a) *If a, a' are* W*-admissible and* $(\mathrm{K})_a$ *holds, then* $(\mathrm{TIGHT})_a$ *and* $(\mathrm{TIGHT})_{a'}$ *are equivalent.*
(b) *Assume* $(\mathrm{W})_a$ *and* (E_C). *Then* $(\mathrm{TIGHT})_a$ *implies* (TIGHT_0).
(c) (TIGHT_0) *implies* (TIGHT_2)*; and* $\limsup_{b} \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \leq S_2^K(f)$ *always, so* (TIGHT_2) *is an equality.*

Proof. (a) By Lemma 16.4(b), $a'_k - a_k \to \kappa_{a'a}$, and $T^{a'}_{\mathfrak{c},\Lambda} = e^{-(a'_k - a_k)\mathsf{S}_\Lambda(g)} T^{a}_{\mathfrak{c},\Lambda}$ by Lemma 15.9(c); since $\mathsf{S}_{\Lambda_k}(g) \to \int g$, $\mathcal{W}^{a'}_{\mathfrak{c}} = e^{-\kappa_{a'a}\int g}\mathcal{W}^{a}_{\mathfrak{c}}$ with a factor independent of $\mathfrak{c}$. The same factor relates $Z^{a'}_\infty$ and Z^a_∞ (Theorem 16.6(vi)).

(b) At $g = 0$ the walled values are scheme-free (Lemma 15.9(c)), so the left sides of $(\mathrm{TIGHT})_a$ and (TIGHT_0) agree, and the right side of $(\mathrm{TIGHT})_a$ is $Z^a_\infty(\mathrm{i}f, 0) = \lim_J \lim_k Z_{k,J}(\mathrm{i}f)$, the iterated limit of the comparison family, which exists under $(\mathrm{W})_a$ (Theorem 16.6(iii)) and equals $Z^K_\infty(\mathrm{i}f, 0)$ under (E_C) by Corollary 4.4(ii)(γ).

(c) Fix f, b and $\eta' > 0$. All functions of t below are real and even. Since $\cos x \geq 1 - \frac{x^2}{2}$ and $\cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$ at every lattice level, and the moments converge in the limits (for the walled torus by Lemma 17.3(a), for King's cube by Theorem 7.6(c)),

$$\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq 1 - \tfrac{t^2}{2}\sigma^2_{\mathfrak{c}}(f), \qquad Z^K_\infty(\mathrm{i}tf, 0) \leq 1 - \tfrac{t^2}{2} S_2^K(f) + \tfrac{t^4}{24} M_4(f),$$

with $M_4(f) := \partial_t^4 Z^K_\infty(tf, 0)|_{t=0} = \lim_J \lim_k \langle F(f)^4\rangle^K_{k,J} < \infty$. By ($\mathrm{TIGHT}_0$) applied to tf, for every $t > 0$ and $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}(tf, 0) \leq Z^K_\infty(\mathrm{i}tf, 0) + \eta$, hence $\sigma^2_{\mathfrak{c}}(f) \geq S_2^K(f) - \frac{t^2}{12} M_4(f) - \frac{2\eta}{t^2}$. Choosing t with $t^2 M_4(f)/12 < \eta'/2$ and then $\eta = t^2\eta'/4$ gives $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \geq S_2^K(f) - \eta'$.

For the second claim, let $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$ with f supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$. By Lemma 18.2(c) applied to tf, $\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq Z^K_J(\mathrm{i}tf)$ for all real t, with equality ($= 1$) at $t = 0$; both sides are even and smooth, so comparing second derivatives at $t = 0$ gives $\sigma^2_{\mathfrak{c}}(f) \leq v_J(f) := \lim_k \langle F(f)^2\rangle^K_{k,J}$, the limit existing by Theorem 7.6(c). Hence $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \leq v_J(f)$ whenever $\bar{C}_J \subset Q_b$. By Corollary 4.3, $\langle e^{\mathrm{i}tF(f)}\rangle^K_{k,J} \leq \langle e^{\mathrm{i}tF(f)}\rangle^K_{k,J+1}$ for all real t, with equality at $t = 0$; comparing second derivatives, the cube variances $\langle F(f)^2\rangle^K_{k,J}$ are nonincreasing in J at every k, hence so are the $v_J(f)$, and they converge to $S_2^K(f)$ (Theorem 7.6(c)). Therefore $\limsup_{b} \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \leq \inf_J v_J(f) = S_2^K(f)$. □

Remark 18.8 (the role of the closing estimate). Apart from the closing estimate, the proofs of Theorems 18.9 and 18.12 below use only: monotonicity of the cube functionals in J (on $\mathcal{K}$

and on the decreasing class), the comparison "cube $\le$ walled torus" on $\mathcal{K}$ and "cube $\ge$ walled torus" on the decreasing class on each lattice, the equality of walled values on the two lattices, passive covariance, and the existence of the limits. All of these hold for the following family of Gaussian functionals, in which the analogue of King's state is not rotation invariant. Let $A = \mathrm{diag}(1,2,3)$, $\mu > 0$, and let C_I, C_R, C_U be the translation-invariant operators on $L^2(\mathbb{R}^3)$ with Fourier symbols $\hat{c}_I(k) = (\langle k, Ak\rangle + \mu^2)^{-1}$, $\hat{c}_R(k) = \hat{c}_I(Rk)$ and $\hat{c}_U = \min(\hat{c}_I, \hat{c}_R)$. Let the "cube of size J" on the lattice $X \in \{I, R\}$ have covariance $C_X + J^{-1}$, let every "walled torus" have covariance C_U on both lattices, and let a covariance C define the functionals $e^{-\frac{1}{2}\langle f, Cf\rangle}$ on the cone and $e^{\frac{1}{2}\langle f, Cf\rangle}$ on the decreasing class. Every relation listed reduces to $C_U \le C_X + J^{-1}$ or to the monotonicity of $C_X + J^{-1}$ in J, and passive covariance holds since $\langle f, C_R f\rangle = \langle R\cdot f, C_I\, R\cdot f\rangle$. But $e^{-\frac{1}{2}\langle R\cdot f, C_I R\cdot f\rangle} \neq e^{-\frac{1}{2}\langle f, C_I f\rangle}$ in general, and the closing estimate fails since $C_U \neq C_I$. Hence the closing estimate is not a consequence of the other properties, and an input asserting "walled torus $\le$ cube + small" is needed. (The same family shows that any placement of the two-lattice comparison on a finite box must also carry an input beyond the correlation inequalities: there the input is the two-lattice comparison itself, which fails for $C_R \neq C_I$.)

18.2. **Rotation invariance.**

Theorem 18.9 (invariance under the generators, Wick sector)**.** *Let a be* W*-admissible (*$(\mathrm{W})_a$*), and assume* $(\mathrm{K})_a$ *and* $(\mathrm{TIGHT})_a$*. Then for every $R \in \mathcal{R}_0$ and every $(f,g) \in \mathcal{K}$,*

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i} f, g),$$

and consequently the same holds for every $R \in G_0$.

Proof. Fix $R \in \mathcal{R}_0$ and $(f,g) \in \mathcal{K}$ supported in Q_ρ.

Step 0 (geometry). Take L with $ML \ge \frac{7}{5}\rho + 2$. By Lemma 15.3(c), $R\cdot(f,g)$ is supported in $Q_{7\rho/5}$, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$, and (f,g) is supported in $Q_{5ML/7} \subset R^{-1}Q_{ML}$. Let $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Then $R^{-1}\Lambda_k$ descends to $D_{J'}$ (Lemma 15.3(b)), and $B'_k := R^{-1}(C_L \cap \Lambda_k)$ is a set of sites of $R^{-1}\Lambda_k$ which, with its nearest neighbours, lies in the interior of the cell, and on which $w = 0$. Since $R\cdot(f,g)$ has distance > 1 from $\mathbb{R}^3 \setminus C_L$, every plaquette and site of $R^{-1}\Lambda_k$ at which (f,g) does not vanish lies in B'_k.

Step 1 (cutting on the rotated lattice). By Lemma 15.8, applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c}, R^{-1}\Lambda_k}$ with $X = B'_k$, whose X-factor is the Neumann–Dirichlet state on B'_k with the scalar alive on all of B'_k, and by the multiplier being the same constant on both sides (Lemma 15.9(c), Lemma 2.11(a)), at every k,

$$Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) \le T^a_{\mathfrak{c}, R^{-1}\Lambda_k}(f,g).$$

Step 2 (passive rotation). By Lemma 15.9(b) with $O = R$, which maps $R^{-1}\Lambda_k$ with its frame $(R^{-1}e_\alpha)$ onto Λ_k with the frame (e_α) and B'_k onto X_L, $Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) = Z^a_{k,L}(\mathrm{i}\, R\cdot f, R\cdot g)$.

Step 3 (continuum limits). As $k \to \infty$ the left side converges by Theorem 16.6(i) (the sources $R\cdot(f,g)$ being at distance > 1 from $\mathbb{R}^3 \setminus C_L$) and the right side by $(\mathrm{K})_a$ (Definition 18.1):

$$Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \le \mathcal{W}^a_{\mathfrak{c}}(f,g) \qquad \text{for every } \mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML}).$$

Step 4 (infimum, volume limit, closing hypothesis). Take the infimum over $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ and let $L \to \infty$. By Theorem 16.6(iii), Lemma 18.3 and $(\mathrm{TIGHT})_a$,

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = \lim_{L\to\infty} Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \le \mathcal{W}^{a,\mathrm{env}}(f,g) \le Z^a_\infty(\mathrm{i} f, g).$$

Step 5 (the inverse). $R^{-1} \in \mathcal{R}_0$ (Lemma 15.3(a)) and $R\cdot(f,g) \in \mathcal{K}$, so Steps 0–4 for $(R^{-1}, R\cdot(f,g))$ give $Z^a_\infty(\mathrm{i} f, g) \le Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g)$. Since $\mathcal{K}$ is G_0-invariant and the transport is a left action, the claim extends to G_0. □

Remark 18.10 (the order of limits). The order of limits is: at fixed R, (f,g), L and $\mathfrak{c}$, first $k \to \infty$; then the infimum over $\mathfrak{c}$; then $L \to \infty$. The continuum limit is always taken first, at fixed volume and fixed wall, as in King's construction [16, p. 336], and the torus comparison is used only as an equality of continuum limits, never at a fixed lattice spacing.

Theorem 18.11 (closure: from the generators to $O(3)$, Wick sector)**.** *Let a be* W*-admissible* ($(\mathrm{W})_a$)*, and suppose $Z^a_\infty(\mathrm{i}\,R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i}f, g)$ for all $(f,g)\in\mathcal{K}$ and $R\in\mathcal{R}_0$. Then for every $R\in O(3)$, every N, all real 2-forms f_j and real h_j in C^∞_c, and all $(z,w)\in\mathbb{C}^{2N}$,*

$$Z^a_\infty\Big(\sum_j z_j\,R\cdot f_j, \sum_j w_j\,R\cdot h_j\Big) = Z^a_\infty\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big).$$

Proof. Complex coefficients. The hypothesis passes to G_0, since $\mathcal{K}$ is invariant under $O(3)$ and $\mathcal{R}_0=\mathcal{R}_0^{-1}$. For $R\in G_0$ and a finite family, split $h_j = h_j^+ - h_j^-$ as in the proof of Theorem 16.6(iii). The difference of the two sides, as a function of $(z,w^+,w^-)\in\mathbb{C}^{3N}$, is entire by Theorem 16.6(iii) and vanishes on $\Omega_+ = (\mathrm{i}\mathbb{R})^N\times(0,\infty)^{2N}$, where all sources lie in $\mathcal{K}$; Ω_+ has non-empty interior within a totally real subspace of maximal dimension, hence a set of uniqueness [11], and the difference vanishes identically.

Closure in $SO(3)$. At real coefficients, $\Phi^a(R\cdot s)=\Phi^a(s)$ for $s\in\mathcal{E}_\rho$ (the pairs of real sources supported in $\bar B_\rho$, Theorem 16.6(iv)) and $R\in G_0$; note $R\cdot s\in\mathcal{E}_\rho$ since $R\bar B_\rho=\bar B_\rho$. The map $R\mapsto R\cdot s\in\mathcal{E}_\rho$ is continuous on $O(3)$, and Φ^a is continuous on $\mathcal{E}_\rho$ by Theorem 16.6(iv). Hence $G:=\{R\in O(3):\Phi^a(R\cdot s)=\Phi^a(s)$ for all ρ and $s\in\mathcal{E}_\rho\}$ is a closed subgroup of $O(3)$ containing G_0, so $G\supset SO(3)$ by Lemma 15.3(d).

Reflection. Let $\sigma_3=\mathrm{diag}(1,1,-1)$ and $(f,g)\in\mathcal{K}$ supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_{J-1}$. Lemma 2.13 gives, at every k, $Z^a_{k,J-1}(\mathrm{i}f,g)\le Z^a_{k,J}(\mathrm{i}\,\sigma_3\cdot f,\sigma_3\cdot g)\le Z^a_{k,J+1}(\mathrm{i}f,g)$. The family $\sigma_3\cdot(f,g)$ is supported at distance >1 from $\mathbb{R}^3\setminus C_J$, the region $\{\mathrm{dist}_\infty(\cdot,\mathbb{R}^3\setminus C_{J-1})>1\}$ being σ_3-symmetric and contained in C_J; so all three terms converge as $k\to\infty$ by Theorem 16.6(i), and then as $J\to\infty$ by Theorem 16.6(iii): $Z^a_\infty(\mathrm{i}\,\sigma_3\cdot f,\sigma_3\cdot g)=Z^a_\infty(\mathrm{i}f,g)$ on $\mathcal{K}$. The first step, with σ_3 in place of R, gives $\sigma_3\in G$, so $G=O(3)=SO(3)\cup\sigma_3 SO(3)$.

Conclusion. For $R\in O(3)$ both sides of the claim are entire in $(z,w)\in\mathbb{C}^{2N}$ (Theorem 16.6(iii)) and agree on $\mathbb{R}^{2N}$, since $R\in G$; hence they agree on $\mathbb{C}^{2N}$. □

Theorem 18.12 (the curvature sector)**.** *King's printed state on curvature sources is $O(3)$-invariant: for every $R\in O(3)$, all real 2-forms $f_1,\dots,f_N\in C^\infty_c$ and all $z\in\mathbb{C}^N$,*

$$Z^K_\infty\Big(\sum_j z_j\,R\cdot f_j,0\Big) = Z^K_\infty\Big(\sum_j z_j f_j,0\Big),$$

and the curvature Schwinger functions S^K_n of Theorem 7.6(c) are $O(3)$-covariant, F transforming as a 2-form. Moreover, for every Euclidean motion $x\mapsto Rx+b$, $Z^K_\infty(\sum_j z_j\,\tau_b R\cdot f_j,0) = Z^K_\infty(\sum_j z_j f_j,0)$.

Proof. The inputs are the existence theorem (E_C) (Theorem 3.3), the torus comparison (K_0) (Theorem 17.1) and the closing estimate (TIGHT_0) (Theorem 18.4), the last two on the class of Remark 18.6.

Invariance under the generators at imaginary points. Fix $R\in\mathcal{R}_0$ and a real 2-form $f\in C^\infty_c$ supported in Q_ρ. At $g=0$ the walled values are those of (K_0) (Definition 18.1). Steps 0–2 of the proof of Theorem 18.9 run unchanged with the source f alone: with L, $\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})$ and B'_k as there, Lemma 15.8 on the rotated lattice and Lemma 15.9(b) give, at every k,

$$Z_{k,L}(\mathrm{i}\,R\cdot f) = \langle e^{\mathrm{i}F(R\cdot f)}\rangle^{(a)}_{k,J}\big|_{J=L} = Z_{R^{-1}\Lambda_k,B'_k}(\mathrm{i}f) \le T_{\mathfrak{c},R^{-1}\Lambda_k}(f,0),$$

the middle term being the functional of the comparison cube X_L. *Step 3.* The right side converges to $\mathcal{W}_\mathfrak{c}(f,0)$ by (K_0) (Lemma 17.3(a)), so $\limsup_k Z_{k,L}(\mathrm{i}\,R\cdot f)\le\mathcal{W}_\mathfrak{c}(f,0)$; and since $R\cdot f$ is supported at distance >1 from $\mathbb{R}^3\setminus C_L$ (Step 0), Corollary 4.4(ii)(γ) under (E_C) gives $Z^K_L(\mathrm{i}\,R\cdot f)\le\liminf_k Z_{k,L}(\mathrm{i}\,R\cdot f)$. Hence

$$Z^K_L(\mathrm{i}\,R\cdot f)\le\mathcal{W}_\mathfrak{c}(f,0)\qquad\text{for every }\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})\text{ of the class.}$$

Step 4. Take the infimum over $\mathfrak{c}$ and let $L\to\infty$: the left side converges to $Z^K_\infty(\mathrm{i}\,R\cdot f,0)$ by Corollary 4.4(ii)(β), the infima converge to $\mathcal{W}^{\mathrm{env}}(f,0)$ by Lemma 18.3 (on the class, Remark 18.6), and (TIGHT_0) gives

$$Z^K_\infty(\mathrm{i}\,R\cdot f,0)\le\mathcal{W}^{\mathrm{env}}(f,0)\le Z^K_\infty(\mathrm{i}f,0).$$

Step 5. $R^{-1} \in \mathcal{R}_0$ and $R \cdot f$ is a real 2-form in C_c^∞, so the same for $(R^{-1}, R \cdot f)$ gives the reverse inequality: $Z_\infty^K(\mathrm{i}\, R \cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for every real 2-form $u \in C_c^\infty$ and $R \in \mathcal{R}_0$, hence for every $R \in G_0$, the transport being a left action.

Closure in $SO(3)$. Fix a real 2-form $u \in C_c^\infty$. The map $R \mapsto R \cdot u$ is continuous from $O(3)$ to $L^2(\mathbb{R}^3; \Lambda^2)$: for $R \to R_0$ the 2-forms $R \cdot u$ converge to $R_0 \cdot u$ uniformly, with supports in one compact set. By the Lipschitz bound of Corollary 8.3, $|Z_\infty^K(\mathrm{i}R \cdot u) - Z_\infty^K(\mathrm{i}R_0 \cdot u)| \le \sqrt{2}\, \|R \cdot u - R_0 \cdot u\|_2$, so $R \mapsto Z_\infty^K(\mathrm{i}\, R \cdot u, 0)$ is continuous on $O(3)$. It is constant on G_0, which is dense in $SO(3)$ (Lemma 15.3(d)); hence $Z_\infty^K(\mathrm{i}\, R \cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for all $R \in SO(3)$.

The reflection. By Theorem 8.5 with $O = \sigma_3 \in O_h$, $Z_\infty^K(\mathrm{i}\, \sigma_3 \cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for every real u. Since $O(3) = SO(3) \cup \sigma_3 SO(3)$ and $(\sigma_3 R) \cdot u = \sigma_3 \cdot (R \cdot u)$, the identity at imaginary points holds for every $R \in O(3)$.

All complex coefficients. Fix $R \in O(3)$ and the family $f_1, \dots, f_N$. By Theorem 7.6(b) both sides of the claim are entire functions of $z \in \mathbb{C}^N$. At $z = \mathrm{i}t$, $t \in \mathbb{R}^N$, we have $\sum_j \mathrm{i}t_j\, R \cdot f_j = \mathrm{i}\, R \cdot u_t$ with $u_t = \sum_j t_j f_j$, so the two sides agree on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal dimension, hence a set of uniqueness [11]; they agree on $\mathbb{C}^N$. Differentiating at $z = 0$ gives $S_n^K(R \cdot f_1, \dots, R \cdot f_n) = S_n^K(f_1, \dots, f_n)$ for every n, the $O(3)$-covariance of the Schwinger functions.

Euclidean motions. By Corollary 8.3, $Z_\infty^K(\sum_j z_j \tau_b R \cdot f_j, 0) = Z_\infty^K(\sum_j z_j R \cdot f_j, 0)$ for every $b \in \mathbb{R}^3$, and the right side equals $Z_\infty^K(\sum_j z_j f_j, 0)$ by the above. □

Corollary 18.13 (the curvature two-point function)**.** *The two-point Schwinger function $S_2^K(f_1, f_2) = \lim_J \lim_k \langle F(f_1)F(f_2)\rangle_{k,J}^K$ of Theorem 7.6(c) satisfies $S_2^K(R \cdot f_1, R \cdot f_2) = S_2^K(f_1, f_2)$ for all $R \in O(3)$ and all real 2-forms $f_1, f_2 \in C_c^\infty$.*

Proof. Of the closing estimate only (TIGHT$_2$) is used. Let $R \in \mathcal{R}_0$ and f a real 2-form. Steps 0–3 of the proof of Theorem 18.12, with the source tf, give $Z_L^K(\mathrm{i}t\, R \cdot f) \le \mathcal{W}_{\mathfrak{c}}(tf, 0)$ for all real t and all $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Both sides are even and smooth in t and equal 1 at $t = 0$; comparing second derivatives, $v_L(R \cdot f) \ge \sigma_{\mathfrak{c}}^2(f)$, where $v_L(u) := \lim_k \langle F(u)^2\rangle_{k,J}^K|_{J=L}$ is the cube variance (Theorem 7.6(c)). Since $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML})$ and $b \mapsto \sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f)$ is nonincreasing,

$$v_L(R \cdot f) \ge \sup_{\mathcal{A}(Q_{7ML/5})} \sigma_{\mathfrak{c}}^2(f) \ge \lim_{b \to \infty} \sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \ge S_2^K(f)$$

by (TIGHT$_2$). The cube variances converge to $S_2^K(R \cdot f)$ as $L \to \infty$ (Theorem 7.6(c)), so $S_2^K(R \cdot f) \ge S_2^K(f)$, and the inverse generator gives equality; hence $S_2^K(u, u)$ is G_0-invariant.

The bound. By Theorem 7.6(a), $\langle e^{tF(f)}\rangle_{k,J}^K \le \exp(\frac{1}{2}t^2\mu_0^{-2}\|d_k^* f\|_{2,k}^2)$ for real t, with equality at $t = 0$; comparing second derivatives, $\langle F(f)^2\rangle_{k,J}^K \le \mu_0^{-2}\|d_k^* f\|_{2,k}^2$, and $\|d_k^* f\|_{2,k} \to \|d^* f\|_{L^2}$ by (1). Letting $k \to \infty$ and then $J \to \infty$, $S_2^K(f) \le \mu_0^{-2}\|d^* f\|_{L^2}^2$; and $S_2^K(f) \ge 0$ since $Z_\infty^K(\mathrm{i}tf, 0) \le 1 = Z_\infty^K(0, 0)$ (Theorem 7.6(b)). (The continuum bound of Theorem 7.6(b) gives the same at once.)

Closure. S_2^K is a positive semidefinite symmetric bilinear form, so $|S_2^K(f, f) - S_2^K(f', f')| = |S_2^K(f - f', f + f')| \le S_2^K(f - f')^{1/2} S_2^K(f + f')^{1/2}$, and with the bound, $R \mapsto S_2^K(R \cdot f, R \cdot f)$ is continuous on $O(3)$, because $d^*(R \cdot f) \to d^*(R_0 \cdot f)$ in L^2 as $R \to R_0$ (uniform convergence of the first derivatives on one compact set). Density of G_0 gives $SO(3)$; Theorem 8.5 with $O = \sigma_3$, differentiated twice at $z = 0$, gives σ_3, hence $O(3)$; and polarisation gives the bilinear form. □

19. The heavy shell I: the photon switch, two-level walls and the dead shell

This section and the next prove the closing estimate (TIGHT$_0$) of Theorem 18.4. Here we prove the photon boundary switch (Lemma 19.1), define the two-level walls and the dead shell, and show that each of the two shell-decoupling properties (SH$_0$) and (RES$_0$) implies (TIGHT$_0$) (Theorems 19.6 and 19.10); the next section proves (RES$_0$) from the heavy-shell estimates.

Throughout this section $\Lambda = \Lambda_k$ is the axis lattice, $\varepsilon = \varepsilon_k \le 1/\mu_0$, and

$$m_\varepsilon := \frac{2}{\varepsilon} \operatorname{arcsinh}\Big(\frac{\varepsilon\mu_0}{2}\Big);$$

since $t \mapsto \operatorname{arcsinh}(t)/t$ is decreasing on $(0,\infty)$ and $\varepsilon\mu_0/2 \le \frac12$, $m_\varepsilon \ge 0.96\,\mu_0$. Let $G_{\mathbb{Z}^3}$ be the Green's function of simple random walk on $\mathbb{Z}^3$ and

$$\mathsf{a} := \sup_{z\in\mathbb{Z}^3\setminus\{0\}} |z|_\infty\, G_{\mathbb{Z}^3}(0,z),$$

which is finite by the asymptotics of the Green's function of simple random walk in three dimensions [17, Chap. 4]. For a set $X \subset \mathbb{R}^3$ put $\mathrm{vol}_\varepsilon(X) := \varepsilon^3 \#(X \cap \Lambda)$, and $\mathrm{vol}(X)$ for the Lebesgue measure. King cubes are $C_L = [-ML, ML)^3$ with $X_L = C_L \cap \Lambda_k$ (Definition 15.2); the comparison cube on X_L is $\langle\cdot\rangle^{(a)}_{k,J}$ with $J = L$ and its functional is $Z^a_{k,L}$ (Definitions 2.1 and 2.3).

19.1. **Photon boundary conditions across a matter-free collar.** The photon in Feynman form consists of three independent scalar Gaussian fields A_α, $\alpha = 1,2,3$, and on a finite graph $\mathcal{G}$ of sites of Λ each has the precision $K_\mathcal{G} = \varepsilon L_\mathcal{G} + \mu_0^2\varepsilon^3$, with $L_\mathcal{G}$ the graph Laplacian of $\mathcal{G}$ with unit conductances and $\langle u, v\rangle = \sum_x u(x)v(x)$; King's Neumann photon on a cube is the case of the induced subgraph [16, (2.5), p. 333], and the periodic photon the case of the torus graph.

Lemma 19.1 (photon boundary switch)**.** *Let $\mathsf{H} = Q_{b'}$ be a closed cube and C_L a King cube with $\mathrm{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) = D + 3$, where $D \ge \max(1, 3/\mu_0)$ and $\varepsilon \le \min(1/\mu_0, 1/4)$. Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be BFS measures with the same matter part: the sites of $\mathsf{H}\cap\Lambda$, BFS single-site measures (arbitrary; wall and killed sites included), and hopping terms only between neighbouring sites of H, coupled to the photon variables of the bonds of H as in King's action. Their photon parts are:*

($\natural$) the variables $A_\alpha(x)$ with $x, x+\varepsilon e_\alpha \in C_L$, with the law of the periodic photon of a torus whose cell contains $N_1(C_L)$, conditioned on $A_\beta(y) = 0$ for all other (β, y);

(N) King's Neumann photon on $C_L \cap \Lambda$.

Let $\Theta = e^{\mathrm{i}F(f)} e^{\sum_x \varepsilon^3 g(x) r_x^2 - c}$ with f supported on plaquettes of H, $g \ge 0$ supported in H, and $c \in \mathbb{R}$. Then

$$0 \le \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \le \langle|\Theta|^2\rangle_N^{1/2} \big(e^\tau - 1\big)^{1/2}, \qquad \tau \le C_{\mathrm{GB}}\, \mathsf{a}\, \mathrm{vol}_\varepsilon(N_2(\mathsf{H}))\, D^{-1} e^{-\frac32 m_\varepsilon D},$$

with an absolute constant C_{GB}.

Proof. Step 1 (marginals and order). Fix a component α. Let $X_\alpha := \{x \in C_L \cap \Lambda : x + \varepsilon e_\alpha \in C_L\}$; conditioning a Gaussian on the vanishing of some coordinates leaves the principal submatrix of the precision on the others, so in ($\natural$) the variables $A_\alpha(x)$, $x \in X_\alpha$, have precision $K_\natural = \varepsilon(6 - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3$, where Adj_{X_α} is the adjacency matrix of X_α. In (N) the variables are $A_\alpha(x)$, $x \in C_L \cap \Lambda$, with precision $K_N = \varepsilon L_{C_L} + \mu_0^2\varepsilon^3$. The principal submatrix of K_N on X_α is $\varepsilon(\deg_{C_L} - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3 \le K_\natural$, and the Schur complement of K_N on X_α is at most its principal submatrix; hence $C_\natural := K_\natural^{-1} \le C_N|_{X_\alpha}$, $C_N := K_N^{-1}$. Let $\hat X := \{(\alpha, x) : x, x + \varepsilon e_\alpha \in \mathsf{H}\}$. The integrands depend on A only through $A_{\hat X}$, whose laws are $N(0, C_\natural|_{\hat X})$ and $N(0, C_N|_{\hat X})$, with $C_\natural|_{\hat X} \le C_N|_{\hat X}$. Put $M := (C_\natural|_{\hat X})^{-1} - (C_N|_{\hat X})^{-1} \ge 0$. The $\natural$-measure is obtained from the N-measure, after integrating out the photon variables off $\hat X$, by adding $\frac12\langle A, MA\rangle$ to the action; writing $M = \sum_j \lambda_j v_j v_j^T$ with $\lambda_j \ge 0$, this adds $\frac{s}{2}\sum_j \lambda_j \langle v_j, A\rangle^2$ at $s = 1$, and by Lemma 2.7(a) along $s \in [0,1]$, $\langle\Theta\rangle_\natural \ge \langle\Theta\rangle_N$.

Step 2 (Cauchy–Schwarz). With $Q := \langle A, MA\rangle \ge 0$ and $\varsigma := e^{-Q/2}$, $\langle\Theta\rangle_\natural = \langle\Theta\varsigma\rangle_N / \langle\varsigma\rangle_N$, so

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\varsigma - \langle\varsigma\rangle_N)\rangle_N}{\langle\varsigma\rangle_N} \le \langle|\Theta|^2\rangle_N^{1/2} \Big(\frac{\langle e^{-Q}\rangle_N}{\langle e^{-Q/2}\rangle_N^2} - 1\Big)^{1/2} \le \langle|\Theta|^2\rangle_N^{1/2} \big(e^{\langle Q\rangle_N} - 1\big)^{1/2},$$

by $\langle e^{-Q}\rangle_N \le 1$ and Jensen's inequality, $\langle e^{-Q/2}\rangle_N \ge e^{-\langle Q\rangle_N/2}$. With $M = \sum_j \lambda_j v_j v_j^T$ as in Step 1, Lemma 15.10(b) in the N-measure, whose Gaussian has covariance C_N, gives $\langle Q\rangle_N = \sum_j \lambda_j \langle\langle v_j, A\rangle^2\rangle_N \le \sum_j \lambda_j \langle v_j, C_N v_j\rangle = \mathrm{Tr}(MC_N|_{\hat X}) =: \tau$.

Step 3 (the trace). Let $\Delta := C_N|_{X_\alpha} - C_\natural \ge 0$, for each α. Then $MC_N|_{\hat X} = (C_\natural|_{\hat X})^{-1}\Delta_{\hat X}$, so $\tau = \mathbb{E}\langle Y, (C_\natural|_{\hat X})^{-1} Y\rangle$ with $Y \sim N(0, \Delta_{\hat X})$. Since the inverse of a principal block of the covariance is the minimum of the precision form over extensions, $\langle a, (C_\natural|_{\hat X})^{-1} a\rangle = \min\{\langle u, K_\natural u\rangle : u|_{\hat X} = a\}$. Let $\tilde Y \sim N(0, \Delta)$ on X_α and $\eta(x) := (1 - \mathrm{dist}_\infty(x, \mathsf{H}))_+$; then $\eta\tilde Y$ is an extension of $\tilde Y|_{\hat X}$, so $\tau \le \sum_\alpha \mathbb{E}\langle \eta\tilde Y, K_\natural \eta\tilde Y\rangle$. Let int be the set of $x \in X_\alpha$ whose six neighbours lie in X_α; it contains

$N_2(\mathsf{H})\cap\Lambda$. At $x\in\mathrm{int}$ the rows of $K_\natural$, K_N and of the infinite-lattice precision K_∞ coincide, so $(K_\infty\Delta)(x,w)=(K_NC_N)(x,w)-(K_\natural C_\natural)(x,w)=0$ for all $w\in X_\alpha$, hence $\mathrm{Var}((K_\infty\tilde Y)(x))=0$ and $K_\infty\tilde Y=0$ on int almost surely. For u with $K_\infty u=0$ on int $\supset\operatorname{supp}\eta$, summation by parts and the identity $(\eta_xu_x-\eta_yu_y)^2-(u_x-u_y)(\eta_x^2u_x-\eta_y^2u_y)=(\eta_x-\eta_y)^2u_xu_y$ give $\langle\eta u,K_\natural\eta u\rangle=\langle\eta u,K_\infty\eta u\rangle=\varepsilon\sum_{\langle xy\rangle}(\eta_x-\eta_y)^2u_xu_y$. Since $|\eta_x-\eta_y|\le\varepsilon$ and $|\Delta(x,y)|\le\frac12(\Delta(x,x)+\Delta(y,y))$,

$$\tau\le\sum_\alpha\varepsilon^3\sum_{\langle xy\rangle\subset N_2(\mathsf{H})}|\Delta(x,y)|\le 9\ \mathrm{vol}_\varepsilon(N_2(\mathsf{H}))\sup_{\alpha,\ x\in N_2(\mathsf{H})}\Delta(x,x).$$

Step 4 (decay of Δ). For $K=\mathcal{D}-\varepsilon\mathrm{Adj}_{\mathcal{G}}$ with $\mathcal{D}$ diagonal, $K^{-1}(x,y)=\sum_\omega\prod_{(u\to v)\in\omega}(\varepsilon/\mathcal{D}(u))\cdot\mathcal{D}(y)^{-1}$, the sum over walks ω from x to y in $\mathcal{G}$. For $K_\natural$ and K_N the weights of steps from sites of int coincide. Decomposing walks from $x\in N_2(\mathsf{H})$ at their first visit to $\Sigma:=X_\alpha\setminus\mathrm{int}$, $K^{-1}(x,y)=G^{\mathrm{int}}(x,y)+\sum_{z\in\Sigma}\mathsf{h}(x,z)K^{-1}(z,y)$ with G^{int} and the hitting kernel h the same for both precisions. Hence

$$\Delta(x,x)=\sum_{z\in\Sigma}\mathsf{h}(x,z)\big(K_N^{-1}(z,x)-K_\natural^{-1}(z,x)\big)\le\Big(\sum_{z\in\Sigma}\mathsf{h}(x,z)\Big)\sup_{z\in\Sigma}K_N^{-1}(z,x).$$

The first factor is the probability that simple random walk from x, killed with probability $\mu_0^2\varepsilon^2/(6+\mu_0^2\varepsilon^2)$ per step, reaches Σ, which lies at ℓ^∞-distance at least D from x. With $\cosh\lambda=1+\mu_0^2\varepsilon^2/2$, that is $\lambda=\varepsilon m_\varepsilon$, $e^{\pm\lambda S_n^\beta}\mathbf{1}_{\{\text{alive at }n\}}$ is a martingale for each coordinate S^β of the walk; optional stopping and a union over the six half-spaces give $\sum_z\mathsf{h}(x,z)\le 6e^{-m_\varepsilon D}$. The same argument, with the decomposition at the first exit from the ℓ^∞-ball of radius $|y|_\infty/2$ and the bound $K^{-1}_{\varepsilon\mathbb{Z}^3,\mu_0}\le K^{-1}_{\varepsilon\mathbb{Z}^3,0}=(6\varepsilon)^{-1}G_{\mathbb{Z}^3}(\cdot/\varepsilon)$ entrywise, gives the free bound

$$G_\varepsilon(y):=K^{-1}_{\varepsilon\mathbb{Z}^3}(0,y)\le 4\mathsf{a}\,|y|_\infty^{-1}e^{-m_\varepsilon|y|_\infty/2}\qquad(|y|_\infty\ge 4\varepsilon).$$

King's Neumann precision on the box $C_L\cap\Lambda$ has the Green's function $K_N^{-1}(z,x)=\sum_{x'}G_\varepsilon(z-x')$, the sum over the images x' of x under the reflections of $\varepsilon\mathbb{Z}^3$ in the planes half a lattice spacing outside the faces of the box (method of images; the mass term is uniform). Every image other than x lies outside C_L at ℓ^∞-distance at least D from every site of C_L, and so does x from $z\in\Sigma$. If $s\ge 2D$ is the side of C_L, at most $8(j+3)^3$ images satisfy $|z-x'|_\infty<D+(j+1)s$; since $m_\varepsilon D\ge 2$,

$$K_N^{-1}(z,x)\le\sum_{j\ge0}8(j+3)^3\,\frac{4\mathsf{a}\,e^{-m_\varepsilon(D+js)/2}}{D+js}\le C_1\,\mathsf{a}\,D^{-1}e^{-m_\varepsilon D/2}$$

with an absolute constant C_1. Combining, $\Delta(x,x)\le 6C_1\mathsf{a}D^{-1}e^{-\frac32 m_\varepsilon D}$, and Step 3 gives the claim with $C_{\mathrm{GB}}=54C_1$. □

The Gaussian measures in Lemma 19.1 are compared only on the bonds of H, where no live matter crosses the region in which the two photon boundary conditions differ; this is what makes the bound uniform in ε. A comparison in which the photon is pinned or cut on a surface that live matter crosses is not of this type: in formal perturbation theory, at one loop, the order-e^2 difference of two such states diverges logarithmically as $\varepsilon\to0$, because the free-boundary mass counterterm compensates the photon self-energy of the matter only where the photon has its bulk covariance.

19.2. **Shell configurations, two-level walls, and shell-kill decoupling.**

Definition 19.2 (shell configurations and shell profiles)**.** Let $b>0$. A *shell configuration* for Q_b is $\mathfrak{s}=(\mathfrak{c},b',L)$, where $\mathfrak{c}=(J',\psi,n)$ with $\psi\in C^\infty(D_{J'};[0,\infty))$ and $n\ge0$ is a walled configuration in $\mathcal{A}(Q_b)$ (so $\psi=0$ on $N_1(Q_b)$; Definitions 15.6 and 18.1), $\mathsf{H}:=Q_{b'}$ with $b'\ge b+3$, and C_L is a King cube with $\mathrm{dist}_\infty(\mathsf{H},\mathbb{R}^3\setminus C_L)=D(\mathfrak{s})+3$, $D(\mathfrak{s})\ge\max(1,3/\mu_0)$, and $N_1(\bar C_L)\subset(-25MJ',25MJ')^3$. Its *shell* is $U:=C_L\setminus\mathsf{H}$, and we require the *plateau condition* $\psi\ge1$ on $N_1(U)$ (compatible with $\psi=0$ on $N_1(Q_b)$, since $\mathrm{dist}_\infty(U,Q_b)\ge3$). Put

$$\tau(\mathfrak{s}):=C_{\mathrm{GB}}\,\mathsf{a}\ \mathrm{vol}(N_3(\mathsf{H}))\,D(\mathfrak{s})^{-1}e^{-1.44\,\mu_0D(\mathfrak{s})},\tag{44}$$

with the constant C_{GB} of Lemma 19.1. A *shell profile* for $\mathfrak{s}$ is $\chi \in C^\infty(D_{J'};[0,1])$ with $\chi = 1$ on U and $\operatorname{supp}\chi \subset N_1(U)$; then $\chi \le \psi$ automatically, and $\chi = 0$ on $N_1(Q_b)$. Its *dead set* is $D_\chi := \{\chi > 0\} \subset N_1(U)$; every site of C_L within ε of $\mathbb{R}^3 \setminus C_L$, and every site of U, lies in D_χ. We call $(\mathfrak{s},\chi)$ a *two-level configuration* for Q_b.

Definition 19.3 (two-level states and the tail)**.** Let $(\mathfrak{s},\chi)$ be a two-level configuration for Q_b, $(f,g) \in \mathcal{K}$ supported in Q_b, a a scheme, and k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$. For $K \in [0,\infty)$ let

$$\Psi_k(n,K) := T^a_{(J',\, n\psi + K\chi),\Lambda_k}(f,g)$$

be the walled ratio (Definition 15.7) of the configuration with the *two-level wall* $n\psi + K\chi$, which lies in $\mathcal{A}(Q_b)$; and let

$$\Psi_k(n,\infty) := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \big\langle \Theta(\mathrm{i}f,g) \big\rangle$$

in the walled torus state of $\mathfrak{c} = (J', n\psi)$ with the matter killed on $D_\chi \cap \Lambda_k$ (Definition 15.6(d)), the *dead-shell state*. Put $\mathrm{tail}_k(K) := \Psi_k(n,K) - \Psi_k(n,\infty)$. At $g = 0$ the multiplier is 1 and nothing depends on the scheme (Lemma 15.9(b)).

Lemma 19.4 (elementary properties)**.** (a) *$K \mapsto \Psi_k(n,K)$ is nonincreasing on $[0,\infty)$, $\Psi_k(n,K) \ge \Psi_k(n,\infty)$, and $\Psi_k(n,K) \to \Psi_k(n,\infty)$ as $K \to \infty$. Hence $\mathrm{tail}_k(K) \ge 0$ is nonincreasing in K and tends to 0 as $K \to \infty$, at every fixed k.*
(b) *(two-level sandwich) For $n, K \ge 0$: $\Psi_k(n+K,0) \le \Psi_k(n,K) \le \Psi_k(n,0)$.*

Proof. All states involved are BFS measures of King type (the paragraph after Definition 15.7), the observable $\Theta(\mathrm{i}f,g)$ is of class $\mathcal{I}$, and the multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for all of them (Lemma 2.11(a), Lemma 15.9(b)). (a) Increasing K to K' multiplies the single-site measures on D_χ by $e^{-(K'-K)\varepsilon^3\chi(x)r_x^2}$, which does not increase the expectation by Lemma 2.7(c). Killing $D_\chi \cap \Lambda_k$ is the limit $K \to \infty$ (Lemma 2.9 with the rates $\chi > 0$ on D_χ; $|\Theta(\mathrm{i}f,g)| \le e^{C\sum_x r_x^2}$), and it does not increase the expectation (Lemma 2.7(c), killing clause, the state being of King type). (b) The wall of $\Psi_k(n+K,0)$ exceeds that of $\Psi_k(n,K)$ by $K(\psi-\chi) \ge 0$, since $\chi \le \psi$; the wall of $\Psi_k(n,K)$ exceeds that of $\Psi_k(n,0)$ by $K\chi \ge 0$. Apply Lemma 2.7(c). □

Part (b) is the two-level sandwich (Lemma 19.7 below in the form used); it is the only place where $\chi \le \psi$ is needed. In words: raising the matter mass on the shell alone, by any amount, gives a state above the one in which the whole plateau is raised by the same amount, and below the one with the shell untouched.

Definition 19.5 ($(\mathrm{SH})_a$ and (SH_0): shell-kill decoupling)**.** $(\mathrm{SH})_a$ is the property: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and

$$\liminf_{k\to\infty} \mathrm{tail}_k(0) = \liminf_{k\to\infty} \big[\Psi_k(n,0) - \Psi_k(n,\infty)\big] \le \eta.$$

(SH_0) is the same statement for $g = 0$; it does not depend on the scheme.

Since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, the condition $\tau(\mathfrak{s}) \le \eta$ only asks the shell to be wide enough, logarithmically in b' and $1/\eta$. The property involves one lattice, a single walled torus and a single dead set, a fixed wall height, and the lim inf over the lattice spacings; it involves no limit $n \to \infty$ and no comparison between lattices.

Theorem 19.6 (shell-kill decoupling implies the closing estimate)**.** *Assume $(\mathrm{W})_a$ and $(\mathrm{K})_a$. Then $(\mathrm{SH})_a$ implies $(\mathrm{TIGHT})_a$. In the curvature sector, (SH_0) implies (TIGHT_0). More precisely, for every two-level configuration $(\mathfrak{s},\chi)$ for Q_b, every $(f,g) \in \mathcal{K}$ supported in Q_b and every k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$,*

$$\Psi_k(n,\infty) \le Z^a_{k,L}(\mathrm{i}f,g) + Z^a_{k,L}(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \tag{45}$$

where $Z^a_{k,L}$ is the functional of the comparison cube on $X_L = C_L \cap \Lambda_k$. Hence, under $(\mathrm{W})_a$, $\limsup_k \Psi_k(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$; and at $g = 0$, under (E_C) alone, $\limsup_k \Psi_k(n,\infty) \le Z^K_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$.

Proof. Fix k with $\varepsilon = \varepsilon_k \le \min(1/\mu_0, 1/4)$ and write $\Theta = \Theta(\mathrm{i}f, g)$. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for every state below (Lemma 2.11(a)); we compare the expectations of Θ and of $|\Theta|^2 = \Theta(0, 2g)$ and restore it at the end. Let E be the set of sites of the torus $D_{J'}$ outside C_L.

Step 1 (pinning). In the dead-shell state, add $s|A_\beta(y)|^2$ to the action for every photon variable on a bond with an endpoint in E, and let $s \to \infty$. By Lemma 2.7(a) the expectation of Θ does not decrease, and in the limit the photon is the conditioned law ($\natural$) of Lemma 19.1, the periodic photon of $D_{J'}$ (whose cell contains $N_1(C_L)$ by Definition 19.2) conditioned on the vanishing of all variables except the $A_\alpha(x)$ with $x, x + \varepsilon e_\alpha \in C_L$. In this state the hopping terms within E do not involve A, and there are no hopping terms between E and live sites: every site of C_L adjacent to E lies in D_χ and is dead. So the matter on E is a factor independent of A and cancels (Lemma 2.7(e)). What remains is the $\natural$-measure of Lemma 19.1, with the matter on H (dead on $D_\chi \cap \mathsf{H}$), the wall $n\psi$ in its single-site measures, and Dirichlet conditions at $\partial\mathsf{H}$, since $U \subset D_\chi$ is dead. Hence $\Psi_k(n, \infty) \le e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \langle\Theta\rangle_\natural$.

Step 2 (photon switch). By Lemma 19.1, $\langle\Theta\rangle_\natural \le \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2} (e^\tau - 1)^{1/2}$ with $|\Theta|^2 = \Theta(0, 2g)$; here f is supported on plaquettes of H and g in H because $b' \ge b + 3$, and $m_\varepsilon \ge 0.96\mu_0$ together with $\mathrm{vol}_\varepsilon(N_2(\mathsf{H})) \le \mathrm{vol}(N_3(\mathsf{H}))$ (for $\varepsilon \le 1$) give $\tau \le \tau(\mathfrak{s})$.

Step 3 (to the comparison cube). The N-measure is the limit $s \to \infty$ of the walled Neumann–Dirichlet state on X_L, with the wall $n\psi$ in its single-site measures, in which the single-site measures on $D_\chi \cap X_L$ are multiplied by $e^{-s\varepsilon^3 \chi(x) r_x^2}$ (Lemma 2.9; the matter alive in the N-measure is that of $X_L \setminus D_\chi \subset \mathsf{H}$, its Dirichlet terms toward the dead sites being the diagonal parts of the bonds into them). By Lemma 2.7(c) the expectations of $\Theta(\mathrm{i}f, g)$ and $\Theta(0, 2g)$ do not decrease as s decreases to 0, nor when the wall is then removed. The result is the comparison cube $\langle\cdot\rangle_{k,J}^{(a)}$ on X_L (Definition 2.1), in which both observables are the same functions of the shared variables, every site with $g \ne 0$ and every plaquette of f lying in H, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$. Restoring the multiplier (Lemma 2.11(a); for $\Theta(0, 2g)$ it is its square), this is (45).

Step 4 (limits). Under $(\mathrm{W})_a$ the cube functionals converge, $Z_{k,L}^a \to Z_L^a$, by Theorem 16.6(i), and $Z_L^a \le Z_\infty^a$ on $\mathcal{K}$ by Theorem 16.6(iii), with $(0, 2g) \in \mathcal{K}$. This gives the lim sup bound. Now let $(\mathfrak{s}, \chi)$ be as in $(\mathrm{SH})_a$ for (f, g) and η. Then $T_{\mathfrak{c},\Lambda_k}^a(f, g) = \Psi_k(n, 0) = \mathrm{tail}_k(0) + \Psi_k(n, \infty)$, the left side converges to $\mathcal{W}_\mathfrak{c}^a(f, g)$ by $(\mathrm{K})_a$ (Lemma 17.3(a), Definition 18.1), and taking $\liminf_k$, with $\liminf_k(x_k + y_k) \le \liminf_k x_k + \limsup_k y_k$,

$$\mathcal{W}_\mathfrak{c}^a(f, g) \le \eta + Z_\infty^a(\mathrm{i}f, g) + Z_\infty^a(0, 2g)^{1/2} \big(e^\eta - 1\big)^{1/2}.$$

Since $\mathfrak{c} \in \mathcal{A}(Q_b)$ and η is arbitrary, $I_b^a(f, g) \le Z_\infty^a(\mathrm{i}f, g)$ for every b, which is $(\mathrm{TIGHT})_a$.

At $g = 0$ the multiplier is 1 and $\langle|\Theta|^2\rangle = 1$, so Step 3 gives $\Psi_k(n, \infty) \le Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$ with $Z_{k,L}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle_{k,J}^{(a)}$ at $J = L$. Under (E_C), since f is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$, Corollary 4.4(ii)($\gamma$) gives $\limsup_k Z_{k,L}(\mathrm{i}f) \le Z_{L+1}^K(\mathrm{i}f)$, and $Z_{L+1}^K(\mathrm{i}f) \le Z_\infty^K(\mathrm{i}f, 0)$ because $J \mapsto Z_J^K(\mathrm{i}f)$ is nondecreasing (Corollary 4.3, Theorem 7.6(b)) and converges to $Z_\infty^K(\mathrm{i}f, 0)$ (Corollary 4.4(ii)(β)). This is the lim sup bound at $g = 0$. With $(\mathfrak{s}, \chi)$ as in (SH_0), $T_{\mathfrak{c},\Lambda_k}(f, 0) = \Psi_k(n, 0) \to \mathcal{W}_\mathfrak{c}(f, 0)$ by (K_0) (Lemma 17.3(a)), and the same lim inf argument gives $\mathcal{W}_\mathfrak{c}(f, 0) \le \eta + Z_\infty^K(\mathrm{i}f, 0) + (e^\eta - 1)^{1/2}$, hence $I_b(f, 0) \le Z_\infty^K(\mathrm{i}f, 0)$ for every b, which is (TIGHT_0). □

The theorem is stated for the cubic shell $U = C_L \setminus \mathsf{H}$ of Definition 19.2, in which C_L is a King cube and H a cube inside it. For the configurations of the next section, whose shell is the exterior of a box in the torus, the three steps of the proof are carried out in Proposition 20.1 ((DS)).

19.3. **Heavy-shell decoupling.**

Lemma 19.7 (two-level sandwich)**.** *For every two-level configuration, every $(f, g) \in \mathcal{K}$ supported in Q_b, every k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$ and $n, K \ge 0$,*

$$\Psi_k(n + K, 0) \;\le\; \Psi_k(n, K) \;=\; \Psi_k(n, \infty) + \mathrm{tail}_k(K) \;\le\; \Psi_k(n, 0), \qquad \mathrm{tail}_k(K) \ge 0.$$

Proof. Lemma 19.4. □

Definition 19.8 ((RES)$_a$ and (RES$_0$): heavy-shell decoupling at one configuration)**.** (RES)$_a$ is the property: for every $b > 0$, every $(f, g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there are a two-level configuration $(\mathfrak{s}, \chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and a shell mass $K \in [0, \infty)$ such that

$$\liminf_{k\to\infty} \mathrm{tail}_k(K) = \liminf_{k\to\infty} \big[\Psi_k(n, K) - \Psi_k(n, \infty)\big] \le \eta.$$

(RES$_0$) is the same statement for $g = 0$; it does not depend on the scheme.

In words: in a walled torus, once the scalar mass squared on a thick shell of the heavy plateau has been raised by $2K$, removing the matter on the shell altogether changes the expectation of the observable by at most η in the continuum limit, for one geometry and one K, with no rate in K and no uniformity in anything. By Lemma 19.4(a) the bracket tends to 0 as $K \to \infty$ at every fixed k; the property says that the heavy-shell limit and the continuum limit may be interchanged at one configuration. The quantifiers are the weakest that Theorem 19.10 accepts: one configuration, one K (with $K = 0$ allowed, which is shell-kill decoupling, Definition 19.5), the lim inf along the spacings, and no uniformity in n, L, $D(\mathfrak{s})$ or the torus.

Proposition 19.9 ((SH)$_a$ implies (RES)$_a$)**.** *For every two-level configuration and every $K \ge 0$, $\mathrm{tail}_k(K) \le \mathrm{tail}_k(0)$. Hence* (SH)$_a$ *implies* (RES)$_a$ *with the same configurations and every K, and likewise* (SH$_0$) *implies* (RES$_0$)*.*

Proof. Lemma 19.4(a). □

Theorem 19.10 ((RES)$_a$ implies (TIGHT)$_a$)**.** *Assume* (W)$_a$ *and* (K)$_a$*. Then* (RES)$_a$ *implies* (TIGHT)$_a$*. In the curvature sector,* (RES$_0$) *implies* (TIGHT$_0$)*. More precisely, if $(\mathfrak{s}, \chi)$ and K are as in* (RES)$_a$ *for (f, g) and η, then the single-level configuration $\mathfrak{c}' := (J', \psi, n+K) \in \mathcal{A}(Q_b)$ satisfies*

$$\mathcal{W}^a_{\mathfrak{c}'}(f, g) \le Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^\eta - 1\big)^{1/2} + \eta,$$

and at $g = 0$, with $(\mathfrak{s}, \chi)$ and K as in (RES$_0$)*, $\mathcal{W}_{\mathfrak{c}'}(f, 0) \le Z^K_\infty(\mathrm{i}f, 0) + (e^\eta - 1)^{1/2} + \eta$.*

Proof. By Lemma 19.7, $T^a_{\mathfrak{c}',\Lambda_k}(f, g) = \Psi_k(n + K, 0) \le \Psi_k(n, \infty) + \mathrm{tail}_k(K)$ at every k. The left side converges to $\mathcal{W}^a_{\mathfrak{c}'}(f, g)$ by (K)$_a$, since $\mathfrak{c}'$ is a walled configuration in $\mathcal{A}(Q_b)$ (Lemma 17.3(a), Definition 18.1); hence, using $\liminf_k(x_k + y_k) \le \limsup_k x_k + \liminf_k y_k$,

$$\begin{aligned}\mathcal{W}^a_{\mathfrak{c}'}(f, g) = \liminf_k \Psi_k(n + K, 0) &\le \limsup_k \Psi_k(n, \infty) + \liminf_k \mathrm{tail}_k(K)\\ &\le Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2} + \eta\end{aligned}$$

by Theorem 19.6 and (RES)$_a$. With $\tau(\mathfrak{s}) \le \eta$ this is the displayed bound; as η is arbitrary and $\mathfrak{c}' \in \mathcal{A}(Q_b)$, $I^a_b(f, g) \le Z^a_\infty(\mathrm{i}f, g)$ for every b, which is (TIGHT)$_a$. At $g = 0$ the same argument runs with the $g = 0$ clause of Theorem 19.6 under (E_C), the convergence of the left side given by (K$_0$), and $Z(0, 0) = 1$; it gives $I_b(f, 0) \le Z^K_\infty(\mathrm{i}f, 0)$ for every b, which is (TIGHT$_0$). □

Remark 19.11 (on the proof). (i) The two-level state is only an intermediate inequality; the configuration that enters (TIGHT)$_a$, respectively (TIGHT$_0$), is the single-level wall $(n + K)\psi$, and the only limit input is the existence of its walled value, which is part of (K)$_a$, respectively (K$_0$) (the existence of the torus limits on the axis lattice, Remark 17.11 of the supplemental material). No existence statement is needed for the two-level state or for the dead-shell state. (ii) The lattice statements of this subsection hold verbatim for the sharp profile $\chi = \mathbf{1}_U$ (then $D_\chi = U$ and $\chi \le \psi$ needs only $\psi \ge 1$ on U); smoothness of χ is used only in Theorem 20.17, where the two-level walled value must exist. (iii) Since $\mathrm{tail}_k(K)$ is nonincreasing in K, (RES)$_a$ for one K implies it for every larger K; so K may be taken as large as one likes, which is used to place $n + K$ in any prescribed window of wall heights (Remark 18.6).

20. The heavy shell II: the heavy-shell estimates and their continuum form

20.1. **The heavy-shell estimates.** In the curvature sector (RES_0) is proved at one configuration, from two estimates on King's walled torus model with the shell mass inside the matter covariance, the heavy-shell estimates (HS) of Theorem 20.2. The shell is now the whole exterior of the source box in the torus: for a two-level configuration $(\mathfrak{s},\chi)$ in the sense of Definitions 19.2 and 19.3, with the walled torus $T := D_{J'}$ of $\mathfrak{c} = (J',\psi,n)$ and the source box $\mathsf{H}' := \mathsf{H} = Q_{b'}$, the shell $C_L \setminus \mathsf{H}$ is replaced by

$$U := T \setminus \mathsf{H}',$$

so that $\partial U = \partial\mathsf{H}'$ is a single convex surface with no re-entrant edge (on a cubic shell the derivative clause of [15, Thm. 3.3] fails at the re-entrant edges); the plateau condition $\psi \geq 1$ on $N_1(U)$ is kept, the profile is the sharp one, $\chi = \mathbf{1}_U$, which Remark 19.11(ii) allows, so that the dead set is $D_\chi = U$, and $\tau(\mathfrak{s})$ is computed, as in Definition 19.2, from H and a King cube C_L with $\mathsf{H}' \subset C_L \subset T$ at ℓ^∞-distance $D(\mathfrak{s})+3$ from H'. For $\varkappa \in [0,\infty)$ let $\Psi_k(n,\varkappa)$ be the two-level expectation of Definition 19.3 with the shell mass $\varkappa\mathbf{1}_U$ added to the wall, so that $\Psi_k(n,K)$, $\Psi_k(n,\infty)$ of Definition 19.8 are the cases $\varkappa = K$ and $\varkappa = \infty$ (the matter killed on U), and let $Z_\varkappa(zf,0)$ be the corresponding unnormalised functional with the curvature source zf (in the convention of Definition 15.7), so that $Z_\varkappa(zf,0)/Z_\varkappa(0,0) = \langle e^{zF_{\Lambda_k}(f)}\rangle_\varkappa$. Finally $n_* := 1$. Here the proofs of this section are replaced by their ideas; the complete proofs are in the supplemental material.

Proposition 20.1 ((DS): the dead-shell chain for the exterior shell)**.** *For the two-level configurations with the exterior shell $U = T \setminus \mathsf{H}'$ the three steps of the proof of Theorem 19.6 hold: pinning the photon outside a King cube $C_L \supset \mathsf{H}'$ all of whose sites outside H' are dead, the photon switch across the matter-free collar between H' and ∂C_L, and the passage to the comparison cube on $X_L = C_L \cap \Lambda_k$. Consequently the dead-shell bound (45) holds for them: at every k with ε_k small, $\Psi_k(n,\infty) \leq Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$ at $g = 0$.*

Idea of the proof. Adding $s|A_\beta(y)|^2$ to the action for every photon variable on a bond with an endpoint outside C_L does not decrease the expectation of $\Theta = e^{\mathrm{i}F(f)}$ (Lemma 2.7(a)), and as $s \to \infty$ the photon law becomes the law ($\natural$) of Lemma 19.1; the collar $C_L \setminus \mathsf{H}'$ carries no live matter, so Lemma 19.1 compares ($\natural$) with King's Neumann photon on C_L at the cost $(e^\tau - 1)^{1/2}$, $\tau \leq \tau(\mathfrak{s})$; and by Lemma 2.9 and Lemma 2.7(c), removing the killing weights on the collar and then the wall factors does not decrease the expectation, whence $\langle\Theta\rangle_N \leq Z_{k,L}(\mathrm{i}f)$.

Theorem 20.2 ((HS): the heavy-shell estimates)**.** *Let $(\mathfrak{s},\chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, with H' and T unions of blocks of the final scale of King's block-spin flow, with H' of side at least two final blocks and at distance at least three final blocks from its periodic images in every direction, and with every corner coordinate of H', counted from the origin in units of the final block, congruent to $M'/2$ modulo M', where the large-block integer is $M = 5M'$ with M' even (the grid-offset condition; King's tori satisfy the clauses on T, and in each direction every interval of M' final blocks contains an admissible corner coordinate) and plateau height $n \geq n_*$, let $g = 0$ and let f be a real 2-form in C_c^∞ supported in Q_b. There are $\eta_1 > 0$ and, for every stopping scale $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \leq \eta_1$: numbers $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \in [0,\infty)$ with $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \to 0$ as $\eta_0 \to 0$; a threshold $\varkappa_0(\eta_0) < \infty$; a spacing bound $\varepsilon_0(\eta_0) > 0$; a function $r(\,\cdot\,;\eta_0)$ on $[\varkappa_0,\infty)$ with $r(\varkappa;\eta_0) \to 0$ as $\varkappa \to \infty$; and, for every k with $\varepsilon_k \leq \varepsilon_0(\eta_0)$, a measure space $(\tilde\chi, db)$ not depending on $\varkappa$ and measurable functions $\tilde S_\varkappa(\,\cdot\,;z)$ on $\tilde\chi$ for $\varkappa \geq \varkappa_0$ and $z \in [-1,1]$ (both may depend on k; the index is suppressed); such that, with*

$$I^{\mathrm{cap}}_\varkappa(z) := \int_{\tilde\chi} e^{-\tilde S_\varkappa(b;z)}\,db, \qquad \Delta_z(b) := \tilde S_{\varkappa'}(b;z) - \tilde S_\varkappa(b;z),$$

for every such k, all $\varkappa_0 \leq \varkappa \leq \varkappa' < \infty$ and all $z \in [-1,1]$:

(i) (the two-sided bound) $e^{-\mathfrak{a}}\, I^{\mathrm{cap}}_\varkappa(z) \;\leq\; Z_\varkappa(zf,0) \;\leq\; e^{\mathfrak{a}}\, I^{\mathrm{cap}}_\varkappa(z) + \mathfrak{b}\, Z_\varkappa(0,0)$;

(ii) (the $\varkappa$-Cauchy property) $\sup_{b,b'\in\tilde\chi} |\Delta_z(b) - \Delta_0(b')| \;\leq\; 2\, r(\varkappa;\eta_0)$.

In words. King's block-spin flow [15, §3] is run for the model in which $\varkappa \mathbf{1}_U$ is part of the matter covariance at every scale, down to a stopping scale η_0 at which the plateau is massive in block units; $\varkappa_0(\eta_0)$, of the order of η_0^{-2}, is the mass above which the shell is heavy in block units at that scale; $I_\varkappa^{\mathrm{cap}}(z)$ is the stopping-scale integral over a capped set of block fields, the *test set*, and $\tilde{S}_\varkappa(b; z)$ the reduced small-field action. Clause (i) is the two-sided bound of King's and Bałaban's type for the model with the shell mass inside the covariance, uniformly in $\varkappa \geq \varkappa_0$ and in the spacing; clause (ii) says that the dependence of the reduced action on the shell mass is, up to a quantity that tends to 0 as the shell gets heavier, the same at every block field and every source coefficient, on one and the same test set. The proof is Propositions 20.3 and 20.13 below; its inputs beyond the printed literature are listed in Remark 20.15 of the supplemental material.

Theorem 20.14 (the heavy-shell estimate from the shell bounds)**.** *Let $g = 0$ and let $(\mathfrak{s}, \chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, plateau height $n \geq n_*$ and H', T satisfying the block-alignment and grid-offset clauses of Theorem 20.2, and let f be a real 2-form in C_c^∞ supported in Q_b. Then for every $\eta > 0$ there are a stopping scale η_0 and a shell mass $K < \infty$ with*

$$\limsup_{k \to \infty} \mathrm{tail}_k(K) \leq \eta;$$

in particular (RES_0) *holds for $(f, 0)$ and η at every such configuration with $\tau(\mathfrak{s}) \leq \eta$, with this K. Consequently the closing estimate* (TIGHT_0) *holds (Theorem 19.10, with Theorem 17.1 and Proposition 20.1), with the single-level configuration $(J', \psi, n + K)$ in the class of Remark 18.6.*

Idea of the proof. Put $\Psi_\varkappa(z) := Z_\varkappa(zf, 0)/Z_\varkappa(0, 0)$, so that $\mathrm{tail}_k(K) = \Psi_K(\mathrm{i}) - \Psi_\infty(\mathrm{i})$, and $\mathcal{J}_\varkappa(z) := I_\varkappa^{\mathrm{cap}}(z)/I_\varkappa^{\mathrm{cap}}(0)$. $z \mapsto \Psi_\varkappa(z)$ is entire with $|\Psi_\varkappa(z)| \leq M_\circ := 2e^{2\langle v, Cv\rangle}$ on $|z| \leq 2$ for every $\varkappa \in [0, \infty]$ (Lemma 2.8, the variance of $F(f)$ under the free periodic photon being bounded uniformly in the spacing), and $\Psi_\varkappa \to \Psi_\infty$ as $\varkappa \to \infty$ at fixed spacing. (HS)(i) at z and at 0 gives $|\Psi_\varkappa - \mathcal{J}_\varkappa| \leq m_1(\eta_0) \to 0$ on $[-1, 1]$; the test set being the same for $\varkappa$ and $\varkappa'$, $\mathcal{J}_\varkappa(z)/\mathcal{J}_{\varkappa'}(z)$ is a ratio of two double integrals over $\tilde{\chi} \times \tilde{\chi}$ whose integrands differ pointwise by $e^{\Delta_z(b) - \Delta_0(b')} \in [e^{-2r}, e^{2r}]$ by (HS)(ii), so $|\Psi_\varkappa(z) - \Psi_{\varkappa'}(z)| \leq m(K, \eta_0) := 2m_1(\eta_0) + \bar{\mathcal{J}}(e^{2\bar{r}(K;\eta_0)} - 1)$ for $\varkappa' \geq \varkappa \geq K$, uniformly in k, and $\varkappa' \to \infty$. The entire function $\Xi := \Psi_K - \Psi_\infty$ is bounded by $2M_\circ$ on $|z| \leq 2$ and by $m(K, \eta_0)$ on the slit $[-1, 1]$, so the two-constants theorem [21, §4.3] gives $|\Xi(\mathrm{i})| \leq (2M_\circ)^{1-\omega} m(K, \eta_0)^\omega$, ω the harmonic measure of the slit at i; by charge conjugation $\Psi_\varkappa(\mathrm{i})$ is real, so $0 \leq \mathrm{tail}_k(K) \leq (2M_\circ)^{1-\omega} m(K, \eta_0)^\omega$. Given η, choose δ, then η_0, then $K \geq \varkappa_0(\eta_0)$, and only then $\varepsilon_k \leq \varepsilon_0(\eta_0)$: this is the bracket of (RES_0) in the quantifier order of Definition 19.8, and K may be enlarged at will. The final clause is Theorem 19.10 with (DS) and (K_0).

Proof of Theorem 15.1. Theorem 20.14 gives, from the heavy-shell estimates (HS) (Theorem 20.2), the decoupling (RES_0) at one two-level configuration for each b, f and η, and its final clause, Theorem 19.10 with the torus comparison (K_0) (Theorem 17.1) and the dead-shell chain (DS) (Proposition 20.1), gives the closing estimate (TIGHT_0). With (E_C) (Theorem 3.3), (K_0) and (TIGHT_0), Theorem 18.12 gives the invariance of $Z_\infty^K(\cdot, 0)$ under $O(3)$ for all complex coefficients, and Corollary 8.3 adds the translations. □

The shell model and the stopping-scale integrals. Fix the exterior-shell configuration of Theorem 20.2. The *shell model* at shell mass $\varkappa \in [0, \infty)$ is King's walled torus model with the wall $n\psi + \varkappa \mathbf{1}_U$, the matter operator $-\Delta_A + m^2 + \delta m_\varepsilon^2 + 2n\psi + 2\varkappa \mathbf{1}_U$ and $\varkappa \mathbf{1}_U$ kept inside the matter covariance of every block-spin step. King's flow is run on Λ_k through r steps down to the *stopping scale* $\eta_0 := 5^{r-k}$; the shell mass in block units there is $w_r := \varkappa \eta_0^2$, the *frozen regime* is $w_r \geq c_0$, and $\varkappa_0 = \varkappa_0(\eta_0) := \max(c_0, 1)/\eta_0^2$. The output is the effective action $S^{(r),\varkappa}(A_r, \phi_r; zf, 0)$ on the block fields, with the small-field indicator χ_r of [15, (3.2)], a *near-shell test* τ_r on the block fields of U, a *first-layer test* λ_r on the first light layer of H', and a weakened gradient family $\chi_r^\flat \geq \chi_r$ with no clause on the bonds inside U or crossing ∂U; $I_\varkappa^\sharp(z) := \int \chi_r \tau_r \lambda_r \, e^{-S^{(r),\varkappa}} \leq I_\varkappa(z) := \int \chi_r^\flat \tau_r \lambda_r \, e^{-S^{(r),\varkappa}}$. For $\varkappa \geq \varkappa_0$ the thresholds of the tests

are dominated by $\varkappa$-free caps $C_7 p_r$ on U and $C_8 p_r$ on the first layer, and the *capped* integral $I_\varkappa^{\mathrm{cap}}(z)$ is $I_\varkappa$ with the tests replaced by the indicators of these caps; with $b := (A_r, \phi_r|_{\mathsf{H}'})$ the *exterior block data*,

$$\tilde{\chi}^{\mathrm{cap}} := \{b : \chi_r^\flat \lambda_r^{\mathrm{cap}}(b) = 1\}, \qquad \tilde{S}_\varkappa^{\mathrm{cap}}(b; z) := -\ln \int \prod_{y \in U^{(r)}} d\phi_r(y)\, \mathbf{1}\{|\phi_r| \le C_7 p_r \text{ on } U^{(r)}\}\, e^{-S^{(r),\varkappa}(A_r, \phi_r; zf, 0)},$$

so that $I_\varkappa^{\mathrm{cap}}(z) = \int_{\tilde{\chi}^{\mathrm{cap}}} e^{-\tilde{S}_\varkappa^{\mathrm{cap}}(b;z)}\, db$ exactly: this is the test set and the reduced action of Theorem 20.2, and $\tilde{\chi}^{\mathrm{cap}}$ is one and the same for every $\varkappa \ge \varkappa_0$.

Proposition 20.3 (the two-sided bound with the shell inside)**.** *Let $L = 5$, the couplings e, λ, m, μ_0 in King's range, (H', T) block-aligned with the grid offset as in Theorem 20.2, $n \ge 1$, and f as in Theorem 20.2. There are $\eta_1 > 0$, $\sigma > 0$, $c > 0$ and, for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_1$, a spacing bound $\varepsilon_0(\eta_0) > 0$ and a constant $C < \infty$ depending on $(\lambda, e, m, \mu_0, |T|, n, f, \mathsf{H}', c_0)$ but not on η_0, $\varkappa$ or ε, such that for all $\varepsilon \le \varepsilon_0(\eta_0)$ and $z \in [-1, 1]$, with*

$$\mathfrak{a}(\eta_0) := C\eta_0^\sigma |T|, \qquad \mathfrak{b}(\eta_0) := e^{-c\,p(\eta_0)^2 + C|T|},$$

$$\begin{aligned}
Z_\varkappa(zf, 0) &\ge e^{-\mathfrak{a}}\, I_\varkappa^\sharp(z) && (\varkappa \in [0, \infty)),\\
Z_\varkappa(zf, 0) &\ge e^{-\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) && (\varkappa \in [\varkappa_0, \infty)),\\
Z_\varkappa(zf, 0) &\le e^{\mathfrak{a}}\, I_\varkappa(z) + \mathfrak{b}\, Z_\varkappa(0, 0) \le e^{\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) + \mathfrak{b}\, Z_\varkappa(0, 0) && (\varkappa \in [0, \infty)).
\end{aligned}$$

In particular $\mathfrak{a}, \mathfrak{b} \to 0$ as $\eta_0 \to 0$ at fixed $(n, |T|)$, uniformly in $\varkappa$ and $\varepsilon \le \varepsilon_0(\eta_0)$, and clause (i) *of Theorem 20.2 holds for $\varkappa \ge \varkappa_0$ with the capped integral.*

Idea of the proof (four layers, each uniform in $\varkappa \in [0, \infty]$; the complete proof is in the supplemental material). (1) The block-spin transformation is a Gaussian identity valid for every positive matter covariance, so with $\varkappa \mathbf{1}_U$ a non-negative diagonal operator the composition law, the slice decomposition and the Schur-complement monotonicity in $\varkappa$ hold verbatim. (2) King's Theorem 3.3 [15, (3.5)–(3.8)] for the shell model, uniformly in $\varkappa$ (coercivity, ℓ^2 decay by conjugation, Loewner monotonicity, Kato's inequality), Hölder bounds on the light side, barrier bounds on the heavy side, and the first- and mixed second-difference bounds of the line kernels on both sides of ∂U, where the convexity of ∂U enters. (3) Bałaban's renormalisation bound [3, Prop. 1, (1.33)] for the renormalised families of order at most 13 with the shell model's propagators and legs, the Ward–Takahashi identities for full step-j families (never for a fixed ordering of the lines), and the bound, uniform in the shell mass, on the first-order coefficient of the kept determinant polynomial on the propagator domains [2, (2.104)–(2.107)], which is Theorem 20.4 of the supplemental material. (4) The lower bound is [1, §§3–4] (Jensen's restriction to the small-field region, King's Appendix class [15, (A.1)–(A.8)] for the shell model, the cumulant bound Theorem 17.112 of the supplemental material), the restriction to the capped set being a positivity step inside the class; the upper bound is [2, §§2–3] (the large/small-field split at every scale, the propagator-domain comparisons in the form Lemma 5.1 with the regrouping Lemma 5.2, the continuation of the large-field terms past the stopping scale), with the remainder $\mathfrak{b}\, Z_\varkappa(0, 0)$ in normalisation-free form by the ultraviolet-stability gap, as in (2).

The first-order determinant term on the sub-domains (Theorems 20.4 and 20.10, Lemmas 20.5 to 20.9, Corollary 20.11, and Remark 20.12 of the supplemental material). In the upper bound [2, §2] the matter determinant of the composed steps on a propagator domain Ω, a union of unit blocks of any shape, is expanded in the photon fluctuation around the regular background; the first-order term vanishes at zero background and its size is governed by a derivative in the background that cannot be taken pointwise near the faces, holes and re-entrant edges. The theorem represents it exactly as a sum over the sites of the domain of second derivatives of the diagonal resolvent elements of the composed step operator, each gauge invariant at its site, so that every bound on the background is replaced by a bound on its first differences, and bounds it per unit block by $Ce_j^2 \sup_{Q_r} |dB| \sup_{Q_r} |da|$ plus exponentially small tails, on every block and uniformly in the shell mass, from the coercivity and the on-diagonal heat-kernel bound

(Lemma F.143 and Lemma A.9 of the supplemental material); both the first- and the second-order coefficient are $O(1)e_j^2p^2$ per unit block, the form in which layer (3) consumes the kept vertices.

Proposition 20.13 (the $\varkappa$-Cauchy property of the reduced action)**.** *In the setting of Proposition 20.3 fix $\alpha_0 \in (0, \frac{1}{2}]$. There is $\eta_2 \le \eta_1$ such that for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_2$ there is $C_{\mathfrak{c}}(\eta_0) < \infty$, depending on $(T, \mathsf{H}', n, \lambda, e, m, \mu_0, \alpha_0, c_0, \eta_0)$ and on f but not on ε, $\varkappa$, z, such that for all $\varepsilon \le \varepsilon_0(\eta_0)$, all $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and all $z \in [-1, 1]$,*

$$\sup_{b,b' \in \tilde{\chi}^{\mathrm{cap}}} \big|\Delta_z(b) - \Delta_0(b')\big| \;\le\; 2R(\varkappa, \varkappa') \;\le\; 2\, r(\varkappa; \eta_0), \qquad R(\varkappa, \varkappa') := \int_{\varkappa\eta_0^2}^{\varkappa'\eta_0^2} \theta(w)\, dw + \tau(\varkappa),$$

$$\theta(w) := C_{\mathfrak{c}}\, w^{-1} w^{-(1-\alpha_0)/2} (1 + \log w)^2, \qquad \tau(\varkappa) := C_{\mathfrak{c}}\, (\varkappa\eta_0^2)^{-\frac{1}{2} + \frac{\alpha_0}{4}},$$

$$r(K; \eta_0) := C'_{\mathfrak{c}}\, (K\eta_0^2)^{-(1-\alpha_0)/2} \big(1 + \log(K\eta_0^2)\big)^2,$$

with $\Delta_z(b) = \tilde{S}^{\mathrm{cap}}_{\varkappa'}(b; z) - \tilde{S}^{\mathrm{cap}}_{\varkappa}(b; z)$ as in Theorem 20.2; $r(K; \eta_0) \to 0$ as $K \to \infty$ at fixed η_0. This is clause (ii) *of Theorem 20.2.*

Idea of the proof. The statement is derived from a bound on King's full explicit action in all block fields, $|S^{(r),\varkappa} - S^{(r),\varkappa'} - (c_\varkappa - c_{\varkappa'})| \le R(\varkappa, \varkappa')$ on the capped support, with a field-free function $c_\varkappa$ independent of z; on the capped set, whose U-section is the same cube for every exterior datum and every $\varkappa$, the reduced actions inherit the bound with the same $c_\varkappa$, which the mixed difference removes. The full-field bound is proved in derivative form: $\partial_\varkappa S^{(r),\varkappa}$ is King's sum of renormalised graphs with one insertion of $\mathbf{2}1_U$ on a line, plus the $\varkappa$-derivative of the block-free part of the background determinant; for an insertion on a line with a scalar connection to the rest of the graph the heavy matter on U pays $w_j^{-1/2}$ per deep end at every frozen scale, and for an insertion whose matter component is closed and connects only through photon lines the members with the same full lines are grouped into one class, to which one gauge identity is applied before the expansion into King's members, the base layer being a first-moment bound, uniform in $\varkappa$, for the two-photon kernel. In the units of the stopping scale the sum over the members less its field-free part is at most $\theta(\hat{\varkappa})$ per unit of $\hat{\varkappa} = \varkappa\eta_0^2$, integrable at infinity, and the block-free part of the determinant has the monotone tail $\tau(\varkappa)$; integrating in $\varkappa$ gives R, and the sum over dyadic intervals gives $r(K; \eta_0)$. The complete proof is in the supplemental material.

Proof of Theorem 20.2. Take η_1 of the theorem to be the η_2 of Proposition 20.13, $\mathfrak{a}, \mathfrak{b}$ and $\varepsilon_0(\eta_0)$ as in Proposition 20.3, $\varkappa_0 = \varkappa_0$, $r(\cdot; \eta_0)$ as in Proposition 20.13, and $(\tilde{\chi}, db) = (\tilde{\chi}^{\mathrm{cap}}, db)$ with $\tilde{S}_\varkappa = \tilde{S}^{\mathrm{cap}}_\varkappa$. Clause (i) is the second and third lines of Proposition 20.3 for $\varkappa \ge \varkappa_0$; clause (ii) is Proposition 20.13; the test set does not depend on $\varkappa$ on $[\varkappa_0, \infty)$, as stated. □

20.2. **The continuum form.** This subsection is not on the chain of Theorem 15.1: it records an equivalent form of $(\mathrm{RES})_a$, which is not used, and the existence statement (E_∞) introduced here is used by nothing. Both sides of the bracket in $(\mathrm{RES})_a$ have continuum limits, the first by $(\mathrm{K})_a$ and the second under one further existence statement; in those terms the property says that two continuum objects coincide up to η. Throughout, $(\mathfrak{s}, \chi)$ is a two-level configuration for Q_b in the sense of Definition 19.2 (smooth profile χ, shell $U = C_L \setminus \mathsf{H}$), and $(f, g) \in \mathcal{K}$ is supported in Q_b.

Proposition 20.16 (the dead-shell state between two cube states)**.** *Let $B_k := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$ in the Neumann–Dirichlet state on $C_L \cap \Lambda_k$ (Neumann photon on C_L, Dirichlet matter outside C_L) whose single-site measures are those of the dead-shell walled torus: the wall $n\psi$, and the matter killed on $D_\chi \cap C_L$. Then at every k with ε_k small*

$$B_k \;\le\; \Psi_k(n, \infty) \;\le\; B_k + Z^a_{k,L}(0, 2g)^{1/2} \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \qquad B_k \le Z^a_{k,L}(\mathrm{i}f, g).$$

Theorem 20.17 (the continuum form)**.** *Assume $(\mathrm{K})_a$ if $g \ne 0$ (at $g = 0$ the torus comparison is Theorem 17.1); for the sandwich in* (d) *at $g \ne 0$ assume also $(\mathrm{W})_a$.*

(a) *For every* $K \in [0,\infty)$ *the limit* $\Psi^0(n,K) := \lim_k \Psi_k(n,K)$ *exists, equals the walled value* $\mathcal{W}^a_{(J',n\psi+K\chi)}(f,g)$*, and is nonincreasing in* K*; so* $\Psi^0(n,\infty^-) := \lim_{K\to\infty} \Psi^0(n,K)$ *exists.*

(b) $\liminf_k \mathrm{tail}_k(K) = \Psi^0(n,K) - \limsup_k \Psi_k(n,\infty)$ *for every* K*, and* $\limsup_k \Psi_k(n,\infty) \le \Psi^0(n,\infty^-)$.

(c) $(\mathrm{RES})_a$ *holds at* $(\mathfrak{s},\chi)$ *with tolerance* η *for some* K *if and only if*

$$\Psi^0(n,\infty^-) = \lim_{K\to\infty}\lim_{k\to\infty} \Psi_k(n,K) \;\le\; \limsup_{k\to\infty} \Psi_k(n,\infty) + \eta.$$

(d) *If the* dead-shell torus value $\Psi^0(n,\infty) := \lim_k \Psi_k(n,\infty)$ *exists, then* (c) *reads* $\lim_{K\to\infty} \Psi^0(n,K) \le \Psi^0(n,\infty) + \eta$*, the reverse inequality* $\lim_K \Psi^0(n,K) \ge \Psi^0(n,\infty)$ *always holds, and*

$$B^0 \le \Psi^0(n,\infty) \le B^0 + Z^a_\infty(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})}-1\big)^{1/2}$$

for every limit point B^0 *of* (B_k)*; in particular* $\Psi^0(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$*. At* $g=0$ *the sandwich reads* $B^0 \le \Psi^0(n,\infty) \le B^0 + (e^{\tau(\mathfrak{s})}-1)^{1/2}$ *and* $\Psi^0(n,\infty) \le Z^K_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$.

Idea of the proofs. The lower bound of Proposition 20.16 is Lemma 15.8 for the dead-shell torus state and $X = C_L \cap \Lambda_k$, its upper bound is Steps 1–2 of the proof of Theorem 19.6 and the last inequality its Step 3. In Theorem 20.17, (a) is Lemma 17.3(a) for the walled configuration $(J', n\psi + K\chi)$ with the monotonicity of Lemma 19.4(a); (b), (c) are manipulations of lim inf; (d) is Proposition 20.16 in the limit along a subsequence, with Theorem 16.6 at $g \neq 0$ and Corollary 4.4(ii)(γ), Theorem 7.6(b) at $g = 0$.

Definition 20.18 ((RES^0): the continuum form)**.** (RES^0) is the property: for every $b > 0$, $(f,g) \in \mathcal{K}$ supported in Q_b and $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ for which the dead-shell torus value $\Psi^0(n,\infty)$ exists and

$$\lim_{K\to\infty} \Psi^0(n,K) \;\le\; \Psi^0(n,\infty) + \eta.$$

By Theorem 20.17, (RES^0) implies $(\mathrm{RES})_a$, and $(\mathrm{RES})_a$ implies (RES^0) whenever the dead-shell torus values exist; the existence of $\Psi^0(n,K)$ is covered by $(\mathrm{K})_a$, that of $\Psi^0(n,\infty)$ is the statement (E_∞), a theorem of King's type that he does not print. Neither (E_∞) nor (RES^0) is consumed by any result of this paper.

20.3. The fixed-spacing statement and the shape of a counterexample.

Proposition 20.19 (fixed spacing)**.** *For every two-level configuration, every* $(f,g) \in \mathcal{K}$ *supported in* Q_b *and every* k, $\mathrm{tail}_k(K) \to 0$ *as* $K \to \infty$*. Consequently, if* $(\mathrm{RES})_a$ *fails for some* (f,g)*,* η*, then for every two-level configuration* $(\mathfrak{s},\chi)$ *for* Q_b *with* $\tau(\mathfrak{s}) \le \eta$ *and every* K, $\mathrm{tail}_k(K) > \eta$ *for all large* k*, while* $\mathrm{tail}_k(K) \le \eta$ *for* $K \ge K_*(k) < \infty$*; hence* $K_*(k) \to \infty$ *as* $k \to \infty$.

Idea of the proof. The first statement is Lemma 19.4(a); the second is its contrapositive form, with $K_*(k) := \inf\{K : \mathrm{tail}_k(K) \le \eta\}$ finite by the first statement and unbounded along k. So any failure of $(\mathrm{RES})_a$ would be carried by shell matter whose mass squared tends to infinity with k: a lattice-scale effect that disappears at every fixed cutoff.

20.4. Why the wall, and why no soft argument.

Proposition 20.20 (no common lattice hyperplanes)**.** *Let* $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$ *and let* ρ *be the third index.*

(a) *There is no lattice* $P \subset \mathbb{R}^3$ *of rank* 3 *with* $RP = P$.

(b) *An affine plane is a lattice hyperplane of both* Λ_k *and* $R^{-1}\Lambda_k$ *(that is, of the form* $\{x_\alpha = c\}$ *and of the form* $\{(R^{-1}e_\beta)\cdot x = c'\}$*) only if it is of the form* $\{x_\rho = c\}$.

(c) *Consequently no bounded region of* $\mathbb{R}^3$ *has all its faces on lattice hyperplanes of both lattices; the only unbounded regions with that property are the slabs* $\{c_1 \le x_\rho \le c_2\}$*. The bounded geometries to which both lattices descend are the tori* $\mathbb{R}^3/P$ *with* $P \subset \Lambda_k \cap R^{-1}\Lambda_k$*, for instance* $P = 50MJ\,\mathbb{Z}^3$*; by* (a)*, no such* P *is* R*-invariant, so the rotated torus* $R^{-1}D_J$ *is never the torus* D_J.

Idea of the proof. (a) In a $\mathbb{Z}$-basis of an invariant lattice R would be an integer matrix, but its characteristic polynomial is $(x-1)(x^2-\frac{8}{5}x+1)$. (b) $e_\alpha = \pm R^{-1}e_\beta$ means $Re_\alpha = \pm e_\beta$, and for $\alpha \in \{\mu,\nu\}$ the vector Re_α has the components $\pm\frac{3}{5}, \pm\frac{4}{5}$. (c) A bounded region with faces on hyperplanes needs face normals spanning $\mathbb{R}^3$.

The ultraviolet analysis of [15, 16] needs boundaries that are lattice hyperplanes of the lattice being analysed ([16, p. 326]: "Clearly the two lattices in Fig. 1 never share such a boundary. Our solution of this problem is to use periodic boundary conditions."). Proposition 20.20 makes this precise: the objects on which the two lattices can be compared by King's method are tori with smooth sources, and smooth walls are the only admissible way to localise a torus; the correlation inequalities give exactly "cube $\le$ walled torus" and never the reverse (Lemma 15.8; Remark 18.8), so the comparison "walled torus $\le$ King's state $+\eta$" must be supplied by an estimate on King's measure, in the curvature sector (HS). At every fixed spacing $(\mathrm{RES})_a$ holds (Proposition 20.19); its content is uniformity in the spacing, a decoupling statement for a heavy, weakly coupled region seen through the interacting plateau, of which there is no instance for the symmetric massive phase of the three-dimensional model in the printed literature ([6] postpone the decoupling estimates); at $e=0$ it holds trivially, but nothing established at $e=0$ has been carried to $e>0$ (Remark 20.21 of the supplemental material).

Remark 20.22 (the chain of the rotation part). Every implication in the chain

$$(\mathrm{HS}) \xrightarrow{\text{Theorem 20.14}} (\mathrm{RES}_0) \xrightarrow{(\mathrm{E}_C),\ (\mathrm{K}_0),\ (\mathrm{DS});\ \text{Theorem 19.10}} (\mathrm{TIGHT}_0),$$

$$(\mathrm{K}_0)+(\mathrm{TIGHT}_0) \xrightarrow{\text{Theorem 18.12}} \text{invariance under } \mathcal{R}_0 \xrightarrow{\text{Theorem 18.12}} O(3) \xrightarrow{\text{Corollary 8.3}} \mathbb{R}^3 \rtimes O(3)$$

is proved in this part, with (E_C) (Theorem 3.3) wherever King's printed state is named (Lemma 18.2, Theorem 18.12); Theorem 15.1 is the composition. Its two ends, the heavy-shell estimates (HS) (Theorem 20.2) and the torus comparison (K_0) (Theorem 17.1), are renormalisation-group statements about King's walled torus model, proved in Propositions 20.3 and 20.13 and in Section 17.3; (DS) (Proposition 20.1) is the dead-shell chain in the exterior-shell geometry.

The main theorem

21. The main theorem for King's printed state

We state the main theorem in full and prove it by assembling Sections 4–12. The object is King's printed state $Z^K_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^K_{k,J}$ on King's RG cube (Definition 2.2), block factor 5, at a fixed photon mass $\mu_0>0$ and all positive couplings e,λ,m (Lemma 22.1). Both sectors are proved on King's convention ND^K, on which the existence theorems Theorems 3.3 and 3.6 are posed. In the Wick sector (Leg W), proved in its tube form, Form (B), the tube statements Theorems 10.7 and 13.10 and Lemma 13.11 are written on the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) as templates; the two of them that the interlacing of Lemma 4.2 does not transfer, the bound (31) and the fixed-cube limits on the tube for the K-family functionals $Z^{a,K}_{k,J}$, are proved directly on King's convention in Theorem 21.3, after which the Wick sector follows on ND^K (Proposition 21.4); the comparison family enters through its LIM states, which coincide with King's tube object (Proposition 21.5). The torus inputs are Proposition 3.9 and Lemma 3.10, and the coupling condition is (T3) (Hypothesis 3.12), the coupling region of which contains the explicit regions of Proposition 22.15; $\mathcal{K} = \{(\mathrm{i}f, g) : g \ge 0\}$ is the cone. Theorem 1.1 is the theorem below.

Theorem 21.1 (the axioms for King's printed state at fixed photon mass)**.** *Let $e>0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$ with King's block integer $M \ge 1$, and let $\mathrm{ND}^K_{k,J}$ be King's cube state.*

A. The object. *For every finite family $f = (f_1, \dots, f_N)$ of real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$ the limit*

$$Z^K_\infty\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty}\lim_{k\to\infty} \langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle^K_{k,J}$$

exists locally uniformly in $z \in \mathbb{C}^N$, is entire, depends on the family only through the total source, is real and ≥ 1 at real z, lies in $[0,1]$ at purely imaginary z, and satisfies $|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$, $g(z) = \sum_j (\operatorname{Re} z_j) f_j$, with no dependence on μ_0, e, λ, m or M.

B. Leg C (the curvature field). *The Schwinger functions $S^K_n(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} Z^K_\infty(\sum_j z_j f_j, 0)|_{z=0}$ satisfy:*

(i) OS0. *$|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$ for all $z \in \mathbb{C}^N$, and also $\leq \exp(\frac{1}{2}\mu_0^{-2}\|\sum_j (\operatorname{Re} z_j) d^* f_j\|_2^2)$; the same two bounds, with the lattice norms $\|\cdot\|_{2,k}$ and d^*_k, hold at every (k, J) for the lattice functionals $\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} \rangle^K_{k,J}$. For real test functions*

$$|S^K_n(f_1, \dots, f_n)| \leq (\sqrt{2}\, e)^n (n!)^{1/2} \prod_{i=1}^n \|f_i\|_2 \leq (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n |f_i|_4;$$

each S^K_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3; \Lambda^2_{\mathbb{C}})^n$, a tempered distribution, the bound carrying the factor 2^n for complex test functions; so E0″ *holds with $s = 4$, $\sigma_n = (e\pi)^n (n!)^{1/2}$ and every $\beta > \frac{1}{2}$.*

(ii) OS1. *The S^K_n are invariant under all translations of $\mathbb{R}^3$ and under the cubic group O_h about every point, F transforming as a 2-form. Moreover Z^K_∞ is invariant under the Euclidean group $\mathbb{R}^3 \rtimes O(3)$ and the S^K_n are Euclidean covariant.*

(iii) OS2. *Reflection positivity about every coordinate plane $\{x_\nu = c\}$, $\nu \in \{1, 2, 3\}$, $c \in \mathbb{R}$: for finite families $\zeta_1, \dots, \zeta_N$ of real 2-form sources supported in $\{x_\nu > c\}$ with arbitrary complex coefficients, $\sum_{i,j} \bar{\gamma}_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \geq 0$ for all $\gamma \in \mathbb{C}^N$ (the matrix form for exponentials), and $S(\theta(P)P) \geq 0$ for every polynomial P in the smeared curvature with sources supported in $\{x_\nu > c\}$ (the polynomial form). The same holds about every plane $\{x \cdot n = c\}$, n a unit vector.*

(iv) OS3. *S^K_n is real for real test functions, symmetric, and zero for odd n.*

(v) The measure. *For every even $r \geq 14$ there is exactly one Borel probability measure $\mu^{K,H}$ on $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2)$ whose moments on C^∞_c are the S^K_n; its characteristic functional is $Z^K_\infty(\mathrm{i}\,\cdot, 0)$, extended by continuity to $\mathcal{S}$; its image μ^K_S on $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3; \Lambda^2)$ is the unique Borel probability measure there with this characteristic functional, and the unique one with these moments. For every real $\chi \in C^\infty_c(\mathbb{R}^3)$ and every box Q containing $\operatorname{supp}\chi$ in its interior, the laws of the localised lattice curvature fields $\chi\Phi^K_{k,J}$ under $\mathrm{ND}^K_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q; \Lambda^2)$ converge weakly, along the full sequence $k \to \infty$ at every fixed $J \geq J_0(\chi)$ and then as $J \to \infty$, to the law of $\chi\Phi$ under μ^K_S. Nothing is asserted about the laws of the unlocalised lattice fields at fixed J.*

(vi) Weak OS4. *μ^K_S is $\mathbb{R}^3 \rtimes O_h$-invariant and is an integral $\int \mu_\xi \, d\nu(\xi)$ of translation-ergodic, reflection-positive, θ_j-invariant components which cluster in the mean and under diagonal separation, have a unique joint vacuum in each coordinate direction, and satisfy the factorial growth bound $|S^\xi_N(u_1, \dots, u_N)| \leq \sigma^\xi_N \prod_i \|u_i\|_{r+2}$, $\sigma^\xi_N \leq \alpha_\beta(\xi)(N!)^\beta$ for every $\beta > 1$, with component constants (no tube bound); there is an O_h-invariant decomposition into mixtures of $k \in \{1, 2, 3, 6\}$ ergodic measures; and ν-almost every component satisfies* OS4 *in the weak form (ergodicity under $\mathbb{R}^3$). Moreover ν-almost every component is $\mathbb{R}^3 \rtimes O(3)$-invariant, reflection positive about every plane, and satisfies* OS4 *in each of its forms for every time direction, with $\ker H_n = \mathbb{C}\Omega_n$ in every direction n; its moments are real and symmetric. The vanishing of a component's odd moments is not claimed.*

C. Leg W, Form (B), on King's convention. *Let the couplings lie in the coupling region of Hypothesis 3.12, the set on which* (T3) *holds, which contains the explicit regions of Proposition 22.15, and fix an admissible scheme a (Definition 3.7; $z = a^\sharp$ the reference scheme). Let $\sigma_+ = s_1$, $H = s_1/16$, $\kappa = C_U/s_1$, K_- be the constants of (30) (K_- built from B_{cell}; for $a \neq z$ with C_U replaced by $C_U + s_1 \sup_k |z_k - a_k|$), $\sigma_0 = H/4$ and $K' = \kappa\pi^2 + K_-$. Then for every finite family $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ of real 2-forms and real scalars the tube object*

$$G^K_\infty(\zeta) = \lim_{J\to\infty} \lim_{k\to\infty} Z^{a,K}_{k,J}\Big(\sum_i z_i f_i, \sum_j w_j h_j\Big), \qquad \zeta = (z, w) \in \mathcal{D}_{\sigma_+,H} = \{\|k_+(\zeta)\|_\infty < \sigma_+/2,\ \|k_-(\zeta)\|_\infty < H/2$$

exists locally uniformly on the tube and is holomorphic there; on the Wick tube of Definition 10.1 the limit $Z^{a,K}_{\infty,T} = \lim_J \lim_k Z^{a,K}_{k,J}$ *exists and equals* G^K_∞ *where both are defined. The joint Schwinger functions* $S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p) = \partial_z^{(n)}\partial_w^{(p)} G^K_\infty(0)$ *of the pair* (F, W^a) *satisfy:*

(i) OS0. $|G^K_\infty(\zeta)| \le \exp(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w)$ *on* $\mathcal{D}_{\sigma_+,H}$*, and the same bound with lattice norms holds for* $Z^{a,K}_{k,J}$ *uniformly in* $k \ge 0$ *and* $J \ge J_1(R)$ *when the supports lie in* Q_R*; for real test functions*

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0}(\sqrt{2}\,e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_{i=1}^n \|f_i\|_2 \prod_{j=1}^p |h_j|_w,$$

with the factor 2^{n+p} *for complex test functions; the* $S_{n,p}$ *are tempered distributions and* E0″ *holds with* $s = 4$ *and every* $\beta > 1$*. The constants* σ_+, H, κ, K_- *depend on* $(e, \lambda, m, \mu_0, M, a)$*, on the fixed auxiliary choices* M_1*,* χ_0 *and the amplitude* $s_1 \in (0, \sigma_1)$ *of* (I2)*, and on nothing else; they are not uniform in* μ_0*.*

(ii) OS1. $G^K_\infty(\tau_b R\cdot\zeta) = G^K_\infty(\zeta)$ *on* $\mathcal{D}_{\sigma_+,H}$ *for every* $b \in \mathbb{R}^3$ *and* $R \in O_h$*,* F *transforming as a 2-form and* W^a *as a scalar (the transformed family has the same tube); hence the* $S_{n,p}$ *are invariant under translations and under* O_h *about every point. No Euclidean clause is asserted.*

(iii) OS2. *For finite families* $\zeta_1,\dots,\zeta_N$ *of real sources supported in* $\{x_\nu > c\}$ *with complex coefficients of amplitude* $\mathsf{a}(\zeta_i) < \sigma_W := \sigma_+/8$ *(in particular all purely imaginary Wick coefficients with arbitrary complex curvature coefficients),* $\sum_{i,j} \bar\gamma_i\gamma_j Z^{a,K}_{\infty,T}(\theta\zeta_i^* \sqcup \zeta_j) \ge 0$ *about every coordinate plane; and the polynomial form* $S(\theta(P)P) \ge 0$ *for polynomials in the smeared* F *and* W^a *with sources in* $\{x_\nu > c\}$*.*

(iv) OS3. *The* $S_{n,p}$ *are symmetric multilinear in the* f*'s and in the* h*'s, real for real test functions, and zero for odd* n*.*

(v),(vi) The measure and weak OS4. *Clauses (v) and (vi) of Leg C hold for the joint field on* $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$ *and* $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$*, every even* $r \ge 14$*, with moments* $S_{n,p}$*, characteristic functional* $(u,v) \mapsto G^K_\infty(\mathrm{i}u, \mathrm{i}v)$ *on real test functions (purely imaginary coefficients have* $k(\zeta) = 0$ *and lie in the tube), the component growth bound with the constants of (i), and the local convergence of the laws of the scheme-shifted lattice fields* $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ *under* $\mathrm{ND}^K_{k,J}$ *along the full sequence* $k \to \infty$ *at every fixed* $J \ge \max(J_0(\chi), J_1(R_\chi))$*, then* $J \to \infty$*. No Euclidean clause is asserted. Two admissible schemes* a, a' *give objects differing by the shift* $W \mapsto W - \kappa_{a'a}\int(\cdot)$ *of the Wick component,* $\kappa_{a'a} = \lim_k(a'_k - a_k)$*; joint cumulants of order* ≥ 2 *are scheme-free.*

No bound at positive scalar sources beyond the cone $\sup q \le s_1$ *of Corollary 13.3 is used.*

D. The comparison family. *Under the conditions of C, every increasing sequence of spacings has a continuum subsequence* $\mathcal{K}$ *for the Wick sector of the comparison family* $\mathrm{ND}^{(a)}$ *(Definition 14.1), and for every such* $\mathcal{K}$ *and every family the LIM state* $Z^{a,\mathcal{K}}_\infty = \lim_J \lim_{k\in\mathcal{K}} Z^a_{k,J}$ *exists on* $\mathcal{D}_{\sigma_+,H}$ *and equals* G^K_∞ *(Proposition 21.5). Hence clauses (i)–(vi) of C hold for the Wick-sector LIM states of the comparison family, which coincide with King's tube object and do not depend on* $\mathcal{K}$*; in (v) the laws of the localised lattice fields* $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ *under* $\mathrm{ND}^{(a)}_{k,J}$ *converge along* $\mathcal{K}$ *at every fixed* J *with* $MJ > R(\chi) + \ell_c$*, then as* $J \to \infty$*, to the law of the joint field under the measure of C(v).*

Proof. A. Theorem 7.6(b): by Theorem 3.3 the two limits exist locally uniformly on $\mathbb{C}^N$ and are entire, $|Z^K_J|, |Z^K_\infty| \le e^{\|g(z)\|_2^2}$, $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing with values in $[0,1]$, and $Z^K_\infty(x) \ge 1$ at real x; realness at real z is the symmetry $(A,\vartheta) \mapsto (-A,-\vartheta)$ of every $\mathrm{ND}^K_{k,J}$, closed under limits. Each lattice functional depends on the family only through its total source, hence so do the limits. The bound is absolute (Remark 7.5).

B. (i) is Theorem 7.6: the finite-volume bound (a) with $\min\{\|g\|^2_{2,k}, \frac12\mu_0^{-2}\|d_k^* g\|^2_{2,k}\}$ at every (k,J); the two limit bounds (b), the second being (b)'s second bound, carried from the lattice by $\|d_k^* g\|_{2,k} \to \|d^* g\|_2$, (1); the Schwinger bounds, the extension to $\mathcal{S}$, the factor 2^n and E0″ (c).

(ii): translations are Corollary 8.3 (the lattice sandwich Lemma 8.2, the two limits, the identity theorem, density); the cubic group about every point is Theorem 8.5; the Euclidean clause is Theorem 8.7, that is Theorem 18.12 (the rotation theorem Theorem 3.14) with Corollary 8.3. (iii): the matrix form about every coordinate plane and the polynomial form are Lemma 10.8 with Proposition 10.10; a plane $\{x \cdot n = c\}$ is the image of a coordinate plane under a Euclidean motion γ, and by (ii) $Z^K_\infty(\theta_\gamma \zeta_i^* \sqcup \zeta_j, 0) = Z^K_\infty(\gamma^{-1} \cdot (\theta_\gamma \zeta_i^* \sqcup \zeta_j), 0)$ is a Gram entry about the coordinate plane for the sources $\gamma^{-1} \cdot \zeta_i$, supported in the corresponding coordinate half-space. (iv) is Proposition 10.9 for $Z^K_\infty(\cdot, 0)$ (through Lemma 10.8; charge conjugation gives the vanishing for odd n). (v) is Theorem 11.1(a), including the local convergence of the lattice laws along the full sequence, with Remark 11.2 for the unlocalised fields; (vi) is Theorem 11.1(b) and (c), whose $O(3)$ input is (ii)'s Euclidean clause; the disclaimer on odd moments is that of Theorem 11.1(c).

C. Under (T3), Proposition 12.6 gives (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U from the torus inputs Proposition 3.9 and Lemma 3.10; for $a \neq z$ the difference $z_k - a_k$ converges by admissibility, so $\sup_k |z_k - a_k| < \infty$ and C_U is replaced as stated (Lemma 14.5); (I3) is Proposition 12.2, the correlation inequalities are Lemma 2.7, and the cell constant $B^{(a)}_{\mathrm{cell}}$ is finite (Lemma 14.4). *(i)*: Theorem 21.3 gives the bound (56) for every $k \geq 0$ and $J \geq J_1(R)$, the fixed-cube limits G^K_J and the cube limit G^K_∞ locally uniformly on $\mathcal{D}_{\sigma_+, H}$, their holomorphy, the bound in the limit, the Schwinger bounds, the factor 2^{n+p} and E0″, with the stated dependence of the constants and no uniformity in μ_0. On the Wick tube $T_{\mathcal{F}'}$ of the enlarged family of Definition 10.1, whose scalar members are nonnegative and whose points have $2\|k_1\|_\infty < \sigma_+/2$, the fixed-cube limits of $Z^{a,K}_{k,J}$ exist locally uniformly by Theorem 6.17(i), whose domain $\{\sup_x 2k_1 < s_W\}$, $s_W = s_1$, contains $T_{\mathcal{F}'}$, for the reference scheme and hence for a (Definition 3.7); their cube limit $Z^{a,K}_{\infty,T}$ is Proposition 10.5(a) run on King's convention (Proposition 21.4(c)), and it equals G^K_∞ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+, H}$ as the same J-limit of the same fixed-cube limits. *(ii)*: Proposition 21.4(b). *(iii)*: Proposition 21.4(c), that is Theorem 10.7 with Proposition 10.5 run on King's convention, and the polynomial form Proposition 10.10. *(iv)*: Proposition 10.9 for $Z^{a,K}_{\infty,T}$ and G^K_∞ (Proposition 21.4(c)). *(v),(vi)*: Proposition 21.4(d). The Gram points of (v)'s characteristic functional have purely imaginary coefficients and lie in the tube, and the reflection positivity consumed there is (iii) at amplitude 0 (Remark 21.6). *The scheme clause*: for admissible a, a', $Z^{a',K}_{k,J}(\zeta) = e^{-(a'_k - a_k)\sum_j w_j \mathsf{S}_{\Lambda_k}(h_j)} Z^{a,K}_{k,J}(\zeta)$ (Definition 2.3), where $a'_k - a_k = (z_k - a_k) - (z_k - a'_k) \to \kappa_{a'a}$ and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$; the factor converges locally uniformly in ζ, so the fixed-cube limits and then the cube limits satisfy $G^{K,a'}_\infty(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} G^{K,a}_\infty(\zeta)$ on the tube, the shift $W \mapsto W - \kappa_{a'a} \int(\cdot)$ of the Wick component; a factor $e^{\ell(w)}$ with ℓ linear changes only the first cumulants in w, so the joint cumulants of order ≥ 2 are scheme-free.

D. Proposition 21.5; the clauses of C then hold for $Z^{a,\mathcal{K}}_\infty = G^K_\infty$, and the convergence of the laws of the localised comparison-cube fields along $\mathcal{K}$ is clause (v) of the Wick sector of Theorem 14.9, whose hypotheses ($\mathrm{B}_{\mathrm{cell}}$) and (SCH) hold for every admissible scheme under (T3) (Lemmas 14.4 and 14.5). □

Remark 21.2 (the Euclidean clause). The Euclidean clauses of B(ii), (iii), (vi) rest on the rotation theorem Theorem 3.14, proved in the rotation part (Theorem 15.1): the walled tori on the two lattices have the same continuum limits (Theorem 17.1), King's printed state is the lower envelope of the walled tori (Theorem 18.4), and that closing estimate comes from the heavy-shell estimates on King's block-spin flow (Theorem 20.2); the Wick sector carries no Euclidean clause (Theorem 23.7).

Theorem 21.3 (the Wick tube on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Let $\sigma_+ = s_1$, H, κ be the constants of* (30) *and let K_- be the constant there, computed from the cell constant $B^{(a)}_{\mathrm{cell}}$ of the comparison convention (Lemma 14.4), which is finite. Then for every finite family of real sources with supports in Q_R, every $k \geq 0$, every $J \geq J_1(R)$ and every $\zeta \in \mathcal{D}_{\sigma_+, H}$,*

$$|Z^{a,K}_{k,J}(\zeta)| \;\leq\; Z^{a,K}_{k,J}(\mathrm{Re}\,\zeta) \;\leq\; \exp\big(\|g(\zeta)\|^2_{2,k} + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\big), \tag{56}$$

the bound (31) *for the K-family functionals with the constants of the comparison convention; and the fixed-cube limits* $G_J^K(\zeta) = \lim_{k\to\infty} Z_{k,J}^{a,K}(\zeta)$ *exist for every* $\zeta \in \mathcal{D}_{\sigma_+,H}$*, locally uniformly, so that Theorem 13.10(i),(ii) hold on* ND^K*:* G_J^K *and* $G_\infty^K = \lim_J G_J^K$ *exist locally uniformly on* $\mathcal{D}_{\sigma_+,H}$ *and are holomorphic, the bound* (56) *passes to the limits, and the Schwinger bounds and* E0″ *of Theorem 21.1C(i) hold for* G_∞^K*; moreover* $G_\infty^K > 0$ *at the real points of* $\mathcal{D}_{\sigma_+,H}$ *and* $G_J^K \le G_\infty^K$ *at its cone points.*

Proof. Throughout, $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ is a family of real sources with supports in Q_R, $J \ge J_1(R)$, so that $MJ \ge R + \ell_c \ge R + 10$: every source lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, the support condition (S) holds for X_J, and $J \ge J_0(\mathcal{F})$. For $\zeta \in \mathcal{D}_{\sigma_+,H}$ write $g = g(\zeta)$ and $k = k(\zeta) = k_+ - k_-$ (Definition 2.5); by Lemma 2.6(E3) the sampled sources satisfy $0 \le 2k_+ \le s_1$ and $0 \le 2k_- \le H$ on Λ_k. The inputs are: (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U, from (T3) and the torus inputs Proposition 3.9 and Lemma 3.10 through Proposition 12.6, for $a \ne z$ with the constant $C_U + s_1 \sup_k |z_k - a_k|$, finite by Definition 3.7 (Lemma 14.5); (I3), which is Proposition 12.2; (I4), which is Lemma 2.7; and the cell constant $B_{\mathrm{cell}}^{(a)} < \infty$ of the comparison convention (Lemma 14.4). Under these, Corollary 13.3 and Proposition 13.9 hold on the comparison convention and Lemma 4.7(d) on King's.

Step 1 (modulus). Lemma 2.11(d) holds for every state of Definition 2.1, killed sites included: $|Z_{k,J}^{a,K}(\zeta)| \le Z_{k,J}^{a,K}(\mathrm{Re}\,\zeta)$. It therefore suffices to bound $Z_{k,J}^{a,K}$ at the real point $\mathrm{Re}\,\zeta$, where $Z_{k,J}^{a,K}(\mathrm{Re}\,\zeta) = N^{a,K}(g,k) := e^{-a_k \mathsf{S}(k)} \langle e^{F(g) + :\phi^2:(k)} \rangle_{k,J}^K$.

Step 2 (curvature off). Let $\langle\cdot\rangle_{K,k}$ be King's cube with each single-site measure at an alive site multiplied by $e^{\varepsilon^3 k(x) r_x^2}$; it is a BFS measure (the factor keeps $\int e^{\alpha r^2} d\rho_x < \infty$ because $\lambda > 0$, and the killed sites are unchanged), and $N^{a,K}(g,k) = N^{a,K}(0,k)\, \langle e^{F(g)} \rangle_{K,k}$. Its Gaussian is the Neumann photon Gaussian of X_J, the same for both conventions, with covariance C as in Lemma 7.1. By (S) every plaquette with $g \ne 0$ has its corners in X_J, so $F(g) = \langle v, A \rangle_k$ with $v = d_k^* g$, and Lemma 2.8 (with $s = 1$) and Lemma 7.1 give $\langle e^{F(g)} \rangle_{K,k} \le e^{\langle v, Cv \rangle_k / 2} \le e^{\|g\|_{2,k}^2}$. This is the chain of Theorem 7.6(a) in the presence of the real scalar source k. Hence $N^{a,K}(g,k) \le e^{\|g\|_{2,k}^2} N^{a,K}(0,k)$.

Step 3 (the split). Lemma 2.11(f) at $f = 0$ with $k_1 = k_+$, $k_2 = k_-$: $N^{a,K}(0, k_+ - k_-) \le N^{a,K}(0, 2k_+)^{1/2} N^{a,K}(0, -2k_-)^{1/2}$.

Step 4 (the positive leg on King's convention). Put $q := 2k_+$, a lattice function with $0 \le q \le s_1$ supported in $Q_R \cap \Lambda_k \subset \mathrm{alive}_J^K$. The observable $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is $\Theta_{0,q'}$ with $q' = \varepsilon^3 q \ge 0$, of class $\mathcal{I}$, and (S) holds for X_J; Step 1 of the proof of Lemma 4.2 (the kill step on $\mathcal{I}$) gives $\langle \Theta_{0,q'} \rangle_{k,J}^K \le \langle \Theta_{0,q'} \rangle_{k,J}^{(a)}$, and the multiplier $e^{-(a_k + 2\mathsf{W}_\varepsilon)\mathsf{S}(q)}$ is one positive constant for both states (Lemma 2.11(a)). Hence

$$N_{k,C_J}^{a,K}(0,q) \;\le\; N_{k,C_J}^{a}(0,q) \;\le\; e^{\kappa \|q\|_{1,k}}$$

by Corollary 13.3, whose hypotheses (I2), (I3) hold (this is the cone instance of Remark 4.8, as in Remark 13.12). So $\frac{1}{2} \ln N^{a,K}(0, 2k_+) \le \kappa \|k_+\|_{1,k}$.

Step 5 (the negative leg on King's convention). Put $q := 2k_-$, a lattice function with $0 \le q \le H$ supported in Q_R, and $J \ge J_1(R)$. Lemma 4.7(d) gives $\ln N_{k,C_J}^{a,K}(0, -q) \le K_- |q|_w$ with the printed constant K_- of (30), built from the cell constant $B_{\mathrm{cell}}^{(a)}$ of the comparison cell, which is finite by Lemma 14.4: by Lemma 4.7(c) every cell on which one of the cutoff sources $8\chi_s q$ does not vanish is a full cell inside C_J containing no killed site, so each factor of the wall insertion of Lemma 4.7(a) that carries a source is a comparison-cell functional, to which Lemmas 13.6 and 13.7 apply as printed, and the sum over cells is Lemma 13.8; the scheme multiplier factorises over the cells (Lemma 2.11(a)). The direction of the kill matters here and is the content of Lemma 4.7(e). Hence $\frac{1}{2} \ln N^{a,K}(0, -2k_-) \le \frac{1}{2} K_- |2k_-|_w = K_- |k_-|_w$.

Step 6 (assembly). Steps 1–5 give, for every $k \ge 0$, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$,

$$|Z_{k,J}^{a,K}(\zeta)| \le Z_{k,J}^{a,K}(\mathrm{Re}\,\zeta) \le \exp\big(\|g(\zeta)\|_{2,k}^2 + \kappa \|k_+(\zeta)\|_{1,k} + K_- |k_-(\zeta)|_w \big),$$

which is (56). The constants are those of (30), with $B^{(a)}_{\mathrm{cell}}$ in K_- and, for $a \neq z$, $C_U + s_1 \sup_k |z_k - a_k|$ in κ.

Step 7 (the fixed-cube limits). Each $Z^{a,K}_{k,J}$ is entire in $\zeta \in \mathbb{C}^{n+p}$: it is a finite-dimensional integral with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda\varepsilon^3 r^4}$ decay in r at every alive site, the killed sites carrying no variables, and differentiation under the integral sign is justified. On a compact $Q \subset \mathcal{D}_{\sigma_+,H}$ the family $\{Z^{a,K}_{k,J}\}_{k\geq 0}$ is uniformly bounded, by (56) and Lemma 2.6(E1)–(E3): $\|g\|^2_{2,k} \leq C_0|g|^2_4$, $\|k_+\|_{1,k} \leq C_0'|k_+|_w$ and $|k_\pm|_w \leq \sum_j |\operatorname{Re} w_j||h_j|_w$, all continuous in ζ. At every real point of $\mathcal{D}_{\sigma_+,H}$ the sequence converges: there $2\|k_+\|_\infty < \sigma_+ = s_1 = s_W$ and the negative part is unrestricted, so Theorem 3.6, which is posed on King's convention (Corollary 6.18(a)), gives the limit in $(0,\infty)$ for the reference scheme; for the admissible scheme a, $Z^{a,K}_{k,J} = e^{-(a_k - z_k)\sum_j w_j \mathsf{S}_k(h_j)} Z^{z,K}_{k,J}$ with $a_k - z_k$ convergent and $\mathsf{S}_k(h_j) \to \int h_j$ (Definition 3.7, Remark 6.19). The real points of $\mathcal{D}_{\sigma_+,H}$ form a non-empty open subset of $\mathbb{R}^{n+p}$ (Definition 2.5), a set of uniqueness in the connected open $\mathcal{D}_{\sigma_+,H}$; by Montel's and Vitali's theorems (the fact (T3) stated before Theorem 13.10),$Z^{a,K}_{k,J} \to G^K_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_J is holomorphic there, and (56) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ is continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one (E3):

$$|G^K_J(\zeta)| \leq \exp\left(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\right) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H},\ J \geq J_1(R)), \tag{57}$$

a bound free of J. At real points $G^K_J > 0$ (Theorem 3.6).

Step 8 (the cube limit on a sub-tube). Split $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$, $\chi_j \in C_c^\infty$, $0 \leq \chi_j \leq 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$, and $h_j^+ := h_j + h_j^- \geq 0$. For the split family $\mathcal{F}'' = (f; h^+, h^-)$ put $F^K_{k,J}(z, w^+, w^-) := Z^{a,K}_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{(z, w^+, w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the scalar source k'' of the split family has $\|k''_\pm\|_\infty \leq \|k''\|_\infty < H/2 \leq \sigma_+/2$, so $\mathcal{D}''$ lies in the tube of $\mathcal{F}''$, whose supports lie in $Q_{R+1/2}$; Steps 1–7 for $\mathcal{F}''$ and $J \geq J_1(R + \frac12)$ give $F^K_{k,J} \to F^K_J$ locally uniformly on $\mathcal{D}''$ with the J-free bound (57) for $\mathcal{F}''$. On $\Omega''_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the curvature coefficients are imaginary and the total scalar source $\sum_j (w_j^+ h_j^+ + w_j^- h_j^-)$ is nonnegative, so the observable is of class $\mathcal{I}$ and (S) holds for X_J ($MJ \geq R + \frac12 + \ell_c$); Corollary 4.3 gives $F^K_{k,J} \leq F^K_{k,J+1}$ at every k (the scheme multiplier is common, Lemma 2.11(a)), and $k \to \infty$ gives $F^K_J \leq F^K_{J+1}$ on Ω''_+. Being nondecreasing and bounded there, $J \mapsto F^K_J$ converges on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$; $\mathcal{D}''$ is convex, hence connected, and the Montel–Vitali fact (T3) stated before Theorem 13.10 gives $F^K_J \to F^K_\infty$ locally uniformly on $\mathcal{D}''$. At $w_j^+ = w_j$, $w_j^- = -w_j$ the total scalar source is $\sum_j w_j(h_j^+ - h_j^-) = \sum_j w_j h_j$, so $F^K_{k,J}(z, w, -w) = Z^{a,K}_{k,J}(\zeta)$ exactly at every k; hence $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}' := \{\zeta : |\operatorname{Re} w_j| < \delta_j\}$, which lies in $\mathcal{D}_{\sigma_+,H}$ because there $\|k_\pm\|_\infty \leq \|k\|_\infty < \sum_j \delta_j \|h_j\|_\infty < H/2$.

Step 9 (the whole tube). The family $\{G^K_J\}_{J \geq J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by (57); by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a holomorphic function, any two such limits agree on the open set $\mathcal{D}'$ by Step 8, and $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_∞ is holomorphic, and (57) passes to G^K_∞. The finitely many indices $J_1(R) \leq J < J_1(R + \frac12)$ do not affect the limit.

Step 10 (positivity and the cone). For a real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$. The function $\varphi_J(t) := \ln G^K_J(t\zeta)$ is finite and convex on $[-\epsilon, 1]$ (a pointwise limit of the convex functions $\ln Z^{a,K}_{k,J}(t\zeta)$, Lemma 2.11(d), with positive limits at real points) and $\varphi_J(0) = 0$, so $\varphi_J(1) \geq -\varphi_J(-\epsilon)/\epsilon \geq -C_\epsilon/\epsilon$, C_ϵ the exponent of (57) at $-\epsilon\zeta$; hence $G^K_\infty(\zeta) \geq e^{-C_\epsilon/\epsilon} > 0$. At the cone points of $\mathcal{D}_{\sigma_+,H}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \geq 0$), Corollary 4.3 at every k and $k \to \infty$ give $G^K_J \leq G^K_{J+1} \leq G^K_\infty$.

Step 11 (the Schwinger bounds and E0″*).* Put $S^K_{n,p}(f; h) := \partial_z^{(n)} \partial_w^{(p)} G^K_\infty(0)$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$. Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $\{|w_j| \leq \rho_j\} \times \mathbb{C}^n$

we have $\|k\|_\infty \le \sum_j |\operatorname{Re} w_j| \|h_j\|_\infty \le \sum_j \rho_j |h_j|_w = \sigma_0 < H/2 \le \sigma_+/2$ (Lemma 2.6(E2)), so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G^K_∞ exists with the bound (57), whose exponent is at most $\|g(z)\|_2^2 + \kappa\pi^2 |k_+|_w + K_- |k_-|_w \le \|g(z)\|_2^2 + K' |k|_w \le \|g(z)\|_2^2 + K'\sigma_0$ (E2). Cauchy's estimate for the mixed first derivative in the w's at $w = 0$, z fixed, gives $|\partial_w^{(p)} G^K_\infty(z,0)| \le \prod_j \rho_j^{-1} \sup |G^K_\infty| \le (p/\sigma_0)^p \prod_j |h_j|_w \, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \le e^p p!$ (E4). The function $z \mapsto \partial_w^{(p)} G^K_\infty(z,0)$ is entire ($\mathbb{C}^n \times \{w\} \subset \mathcal{D}_{\sigma_+,H}$ for $|\operatorname{Re} w_j|$ small) and bounded by $A\, e^{\|g(z)\|_2^2}$ with $A := e^{K'\sigma_0}(e/\sigma_0)^p p! \prod_j |h_j|_w$; on the polydisc of radius $r = (2n)^{-1/2}$ about 0, $\|g(z)\|_2 \le \sum_i |\operatorname{Re} z_i| \le nr$, so Cauchy's estimate for the mixed first z-derivative gives $|S^K_{n,p}| \le A\, r^{-n} e^{n^2 r^2} = A\,(2n)^{n/2} e^{n/2} \le A\,(\sqrt{2}e)^n (n!)^{1/2}$ by (E4). Restoring the normalisation by multilinearity,

$$|S^K_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0} (\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w .$$

By (E2), $\|f_i\|_2 \le \frac{\pi}{2\sqrt{2}} |f_i|_4$ and $|h_j|_w \le |h_j|_4$, and $(n!)^{1/2} p! \le (n+p)!$, so the right side is at most $\sigma_{n+p} \prod_i |f_i|_4 \prod_j |h_j|_4$ with $\sigma_N = e^{K'\sigma_0} C_1^N N!$, $C_1 = 2\max(e\pi/2, e/\sigma_0)$; for complex test functions each argument is split into real and imaginary parts, which gives the factor $2^{n+p} \le 2^N$ absorbed in C_1. The $S^K_{n,p}$ are therefore continuous multilinear forms in the norms $|\cdot|_4$, extend to $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n \times \mathcal{S}(\mathbb{R}^3;\mathbb{C})^p$ and are tempered distributions, as in Theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$: this is E0″ with $s = 4$ and every $\beta > 1$, with the constants of Theorem 21.1C(i). □

Proposition 21.4 (the Wick sector on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Then, with the constants of Theorem 21.1C:*

(a) *$G^K_J := \lim_k Z^{a,K}_{k,J}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ for every $J \ge J_1(R)$, $G^K_\infty := \lim_J G^K_J$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, both are holomorphic, $|G^K_\infty(\zeta)| \le \exp(\|g(\zeta)\|_2^2 + \kappa \|k_+(\zeta)\|_{L^1} + K_- |k_-(\zeta)|_w)$, and the Schwinger bounds and* E0″ *of Theorem 21.1C(i) follow.*

(b) *$G^K_\infty(\tau_b R \cdot \zeta) = G^K_\infty(\zeta)$ on $\mathcal{D}_{\sigma_+,H}$ for all $b \in \mathbb{R}^3$, $R \in O_h$, W^a a scalar.*

(c) *For every family $\mathcal{F}'$ with nonnegative scalar members the Wick-tube object $Z^{a,K}_{\infty,T} := \lim_J \lim_k Z^{a,K}_{k,J}$ exists locally uniformly on $T_{\mathcal{F}'}$ and equals G^K_∞ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$; reflection positivity about every coordinate plane holds in the matrix form at amplitude $< \sigma_W$ and in the polynomial form, and* OS3*, for $Z^{a,K}_{\infty,T}$, G^K_∞ and their Schwinger functions.*

(d) *The measure and weak* OS4 *of Theorem 21.1C(v),(vi) for the joint field under* ND^K*, with the local convergence of the laws of $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ under $\mathrm{ND}^K_{k,J}$ along the full sequence $k \to \infty$ at every fixed $J \ge \max(J_0(\chi), J_1(R_\chi))$, then $J \to \infty$.*

Proof. Parts (a) and (c) are proved below; the complete proofs of parts (b) and (d) are given in the supplemental material; the paragraphs (b) and (d) below summarise them.

(a) is Theorem 21.3, whose proof contains the fixed-cube limits (Step 7), the cube limit on the connected tube with the cube monotonicity of Corollary 4.3 in place of the first inequality of Lemma 2.10 (Steps 8–9), the positivity at real points and the cone monotonicity (Step 10), and the Schwinger bounds by Cauchy's estimates on a polydisc in the tube (Step 11).

(b) The steps of Theorem 8.5 with scalar sources. Coordinate permutations are exact on ND^K for (f,h) by Lemma 8.4. The reflection sandwich is Lemma 4.5(b) at the cone points of the tube: for $(f,g) \in \mathcal{K}$ with $\|g\|_\infty < \sigma_+/2$ the point $(\mathrm{i},1)$ lies in $\mathcal{D}_{\sigma_+,H}$, so (a) supplies the three k-limits and the two J-limits of the two-step sandwich $C^K_{J\mp 2}$. Conjugation and group closure on the cone are unchanged. Off the cone, the identity theorem on the split family's tube of Lemma 13.11, where (a) gives the limits, extends the identity from the cone points to the connected tube; so Lemma 4.5 is needed on $\mathcal{K}$ only, and no class-$\mathcal{D}$ form of it enters (the proof splits $h = h^+ - h^-$ and never applies the sandwich to a negative scalar source). Translations are Lemma 8.2, stated on $\mathcal{K}$ and applied at the cone points of the tube with (a) for the limits, then the continuity route of Corollary 8.3 with the Wick-sector second-moment bound, which Cauchy's estimates give from (31) on ND^K; the translated family needs $J \ge \max(J_0+1, J_1(R+M+\frac{1}{2}))$. Invariance about every point and differentiation concern the limit object.

(c) Theorem 10.7 and Proposition 10.5 on ND^K: in the sandwich of the J-limit the upper box is Lemma 4.6 (the printed boxes C_J, C_{J+2n}) and the lower end carries the kill step of Remark 4.8; the positive-leg bound Corollary 13.3 holds on ND^K by Remark 4.8 prepended to Proposition 13.2 (Remark 13.12); the cube monotonicity in both classes is Corollary 4.3; the fixed-cube limits on $T_{\mathcal{F}'}$ are Theorem 6.17(i), whose domain $\{\sup_x 2k_1 < s_W\}$, $s_W = s_1$, contains $T_{\mathcal{F}'}$ (for the reference scheme, hence for a by Definition 3.7), and on $\mathcal{D}_{\sigma_+,H}$ they are (a); the cube limit on $T_{\mathcal{F}'}$ is then Montel's theorem with the uniqueness set of Definition 10.1(iii), as in Proposition 10.5; the translations to the planes $c \notin D$ are (b). OS3 and the polynomial form use only the existence and holomorphy of the limit near 0 and charge conjugation (Propositions 10.9 and 10.10).

(d) The steps of Theorem 11.1 on $\mathcal{V} = \Lambda^2 \oplus \mathbb{R}$ with the scheme-shifted field: (31) on ND^K in the moment bounds, (a) for the limit functionals and the lattice laws, (b) and (c) for the invariance and the reflection positivity consumed by the abstract chain of [27]. The second moments of the lattice field under $\mathrm{ND}^K_{k,J}$ come from (31) on ND^K by Cauchy's estimates, so tightness and the identification of the limit law run as in the curvature sector, with the fixed-cube limits of (a) along the full sequence, for $J \geq \max(J_0(\chi), J_1(R_\chi))$; the remaining steps concern the limit measure. □

Proposition 21.5 (the comparison family on the Wick tube)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme.*

(a) *Every increasing sequence of spacings has a continuum subsequence $\mathcal{K}$ for the Wick sector of the comparison family (Definition 14.1).*

(b) *For every such $\mathcal{K}$, every family $\mathcal{F}$ of real sources with supports in Q_R and every $\zeta \in \mathcal{D}_{\sigma_+,H}$, the LIM state of Definition 14.2 is King's tube object:*

$$Z^{a,\mathcal{K}}_\infty(\zeta) \;:=\; \lim_{J\to\infty} \lim_{m\to\infty} Z^a_{k_m,J}(\zeta) \;=\; G^K_\infty(\zeta) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}).$$

In particular the Wick-sector LIM state does not depend on $\mathcal{K}$, and the cube limit of every subsequential fixed-cube limit of the comparison family on the tube is G^K_∞.

Proof. (a) By Lemmas 14.4 and 14.5 the scheme has the cell bound and the scheme bound, so Lemma 14.7(b) applies.

(b) Fix $\mathcal{K} = (k_m)$ and $\mathcal{F}$. By Lemma 14.8 and the Wick sector of Lemma 14.7(b), $Z^{a,\mathcal{K}}_\infty$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ and is holomorphic there; by Theorem 21.3 so is G^K_∞. Take the split family $\mathcal{F}'' = (f; h^+, h^-)$ of Step 8 of the proof of Theorem 21.3, with supports in $Q_{R+1/2}$, the domain $\mathcal{D}''$ and the set $\Omega''_+ = \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ defined there; write $F^K_{k,J}$ and $F^{(a)}_{k,J}$ for the functionals of $\mathcal{F}''$ on King's and on the comparison cube. Let J be so large that $MJ > R + \frac{1}{2} + \ell_c$. At a point of Ω''_+ the curvature coefficients are purely imaginary and the total scalar source is nonnegative, so the observable is of class $\mathcal{I}$, and the support condition (S) holds for X_J; Lemma 4.2 (with its clause for the scheme functionals) gives, at every k,

$$F^K_{k,J} \;\leq\; F^{(a)}_{k,J} \;\leq\; F^K_{k,J+1} \qquad \text{on } \Omega''_+.$$

Along $\mathcal{K}$ the two ends converge, by Theorem 21.3 for the family $\mathcal{F}''$ (whose fixed-cube limits exist along the full sequence, hence along $\mathcal{K}$), to the fixed-cube limits F^K_J, F^K_{J+1} of Step 8 of that proof, and the middle term converges to the fixed-cube limit $F^{(a),\mathcal{K}}_J$ of the comparison family along $\mathcal{K}$ (Definition 14.1 for $\mathcal{F}''$, whose tube contains $\mathcal{D}''$ and whose threshold is $MJ > R(\mathcal{F}'') + \ell_c$, $R(\mathcal{F}'') \leq R + \frac{1}{2}$). Letting $J \to \infty$, the two ends tend to F^K_∞ (Steps 8–9 of the proof of Theorem 21.3) and the middle term to the LIM state $F^{(a),\mathcal{K}}_\infty$ of $\mathcal{F}''$ (Lemma 14.8). Hence $F^{(a),\mathcal{K}}_\infty = F^K_\infty$ on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$, and both functions are holomorphic on the connected open set $\mathcal{D}''$; by the identity theorem (the fact (T2) stated before Theorem 13.10) they agree on $\mathcal{D}''$. At $w^+_j = w_j$, $w^-_j = -w_j$ the lattice functionals of $\mathcal{F}''$ coincide exactly with those of $\mathcal{F}$ at ζ, on both conventions and at every k (equal total sources); so $Z^{a,\mathcal{K}}_\infty = G^K_\infty$ on $\mathcal{D}' = \{\zeta : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$. Both are holomorphic on the connected $\mathcal{D}_{\sigma_+,H}$ and agree on the open set $\mathcal{D}'$, hence on $\mathcal{D}_{\sigma_+,H}$. The last clause is

Definition 14.2: every subsequential fixed-cube limit on the tube is the fixed-cube limit along some continuum subsequence. □

Remark 21.6 (the three amplitude constants). Three amplitudes occur in the Wick sector, all built from s_1: the matrix form of OS2 is proved at amplitude $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$ on the Wick tube (Theorem 10.7); the tube $\mathcal{D}_{\sigma_+,H}$ has half-widths $\sigma_+/2$ for the positive and $H/2 = \sigma_+/32$ for the negative part of the real Wick source, and the form of OS2 proved for LIM states (Theorem 14.9) takes $\mathsf{a}(\zeta_i) < H/2 < \sigma_W$, so that every Gram point lies in $\mathcal{D}_{\sigma_+,H}$, where those limits exist; and $\sigma_0 = H/4$ is the radius of the polydisc in Cauchy's estimates for E0″. The consumer of OS2 in clauses (v),(vi), the reflection positivity of exponentials $e^{\mathrm{i}\Phi(u)}$ with real u in the clustering theorems of [27], needs purely imaginary Wick coefficients only, that is amplitude 0 (Remark 10.11).

Remark 21.7 (the photon mass and the couplings). The photon mass $\mu_0 > 0$ is fixed throughout Theorem 21.1. No relation $\mu_0 \le m$ is used or needed: the one input whose printed proof in the torus paper [27] carries the restriction $\mu_0 \in (0, \mu_*]$, the lattice reflection positivity of King's tori, holds for all $\mu_0 > 0$, since that proof uses no bound on μ_0 and the paper's own lemma on the finite-volume photon covariance lifts the scope (Proposition 12.2, Lemma 22.1(d)). Reflection positivity of the cube limit needs the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of (20) inside the upper half of the sandwich (Lemma 9.10) and nowhere in the conclusion. The constants of Leg C are absolute; those of Leg W depend on μ_0 and are not uniform in it. The couplings range over all of $e > 0$, $\lambda, m, \mu_0 > 0$: their quantifier is the sources' (Lemma 22.1(e), Remark 22.16 of the supplemental material), the stopping scale of the lattice constructions behind Theorems 3.3 and 3.6 and Proposition 3.9 being chosen after the point and the datum, with the mass cap of [2, Prop. 3.1] one more one-sided cap on that scale (Remark 22.2 of the supplemental material); so the theorem contains no sentence "e, λ sufficiently small", and the only coupling condition of the Wick sector is the one-sided sign clause (T3) (Lemma 22.1(b)), proved on the explicit regions of Proposition 22.15.

22. The coupling range

This section collects what the sources behind the inputs of Theorem 21.1 say about the couplings and shows that the only coupling condition anywhere in this paper is the one-sided sign clause (T3) of the Wick leg, which Proposition 22.15 proves on explicit regions. Here the proofs are replaced by their ideas; the complete proofs, with the two-loop majorants and the table of numbers, are in the supplemental material.

King's range, units, two letters J. Let $e > 0$ and $\lambda, m, \mu_0 > 0$. Every constant of the cited constructions, including the number J_{K} of renormalisation steps after which the running couplings are small and the stopping scale $5^{-J_{\mathrm{K}}}$, may depend on (e, λ, m, μ_0), on M and on the cube or torus; no numerical restriction is placed on the couplings themselves, and a stopping scale exists at every positive point under the mass cap of [2, Prop. 3.1], a one-sided cap on the stopping scale (Remark 22.2 of the supplemental material). This is what "King's range" means here. The cube index J of C_J is ours; King's stopping index, the "fixed integer J" of [15, Thm. 3.1], is written J_{K}. Couplings are measured in the units in which the coarsest spacing is 1; in $d = 3$ all four of e^2, λ, m, μ_0 have the dimension of a mass.

The relative bare coefficient. Let $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$, $s_k^\infty(M_1) = \frac{1}{2}(\alpha_k^\infty(m) - \alpha_k^\infty(M_1))$ and $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1)$ be as in Lemma 3.10; the theorem is stated through σ_1 alone. Describing *where* (T3) fails uses one further statement about the continuum limit of this coefficient, which is of the same kind as Lemma 3.10 (Remark 3.11) and which we name rather than use tacitly:

(CT_∞) *For every $M > 0$ the limit*

$$\alpha_\infty(M) - \alpha_\infty(m) := \lim_{k\to\infty} \left[\alpha_k^\infty(M) - \alpha_k^\infty(m)\right]$$

exists and is a polynomial of degree ≤ 2 in (e^2, λ) with finite coefficients, differentiable in M^2 at $M = m$.

The divergent parts of the counterterm ($1/\varepsilon$ at one loop, $\ln(1/\varepsilon)$ at two loops) do not depend on the scalar mass and cancel in the difference. (CT_∞) is proved in Theorem F.159(b) of the supplemental material; it is not a hypothesis of Theorem 21.1 and enters only the description of the failure set. Under (CT_∞) put

$$s_\infty(M_1) := \tfrac{1}{2}\big(\alpha_\infty(m) - \alpha_\infty(M_1)\big), \qquad s_* := \sup_{M>0} s_\infty(M) \in [0,\infty],$$

$$E_W := \{s_* = 0\} = \{m \text{ is a global minimiser of } M \mapsto \alpha_\infty(M) \text{ on } (0,\infty)\},$$

$$E_W^< := \{m \text{ minimises } \alpha_\infty \text{ on } (0,m]\} \supseteq E_W,$$

subsets of the coupling space ($s_* \geq 0$ because $M = m$ gives 0; the value $+\infty$ is harmless below). The two sign clauses are

(NE) m is not a global minimiser of α_∞ on $(0,\infty)$, i.e. $s_* > 0$, i.e. $(e,\lambda,m,\mu_0) \notin E_W$;

($\mathrm{NE}^<$) some $M_1 \in (0,m)$ has $\alpha_\infty(M_1) < \alpha_\infty(m)$, i.e. $(e,\lambda,m,\mu_0) \notin E_W^<$ (the one-sided form; it implies (NE)).

Throughout, $'$ denotes d/dM^2, N is the number of real components of the scalar field, and "one-loop order" means the orders e^2 and λ.

Lemma 22.1 (the coupling range)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$.*

(a) The curvature sector. *The existence theorem of the curvature sector, Theorem 3.3 (the existence clause of* [16, Thm. 2.1] *at zero scalar source), carries no condition on e, λ, m, μ_0. The correlation inequalities it uses, Lemma 2.7 (Theorem 6.2(1),(2) of* [7]*), have the standing hypotheses of a finite single-site measure, positive hopping coefficients and a positive photon covariance, which for King's action are $\lambda > 0$ and $\mu_0 > 0$; the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of Lemma 9.10 needs $\mu_0 > 0$ only; and no lemma proved in this paper for the curvature sector carries a condition on the couplings beyond those of its inputs. Hence the curvature sector of Theorem 21.1 holds on all of King's range, with no exceptional set. The one proof-internal proviso in the Bałaban antecedents of Theorem 3.3, the mass clause of* [2, Prop. 3.1]*, is a cap on the dimensionless scalar mass at the stopping scales and no condition on the bare couplings (Remark 22.2 of the supplemental material). Of the bounds of the paper only the absolute bounds of Theorem 7.6 are uniform in μ_0.*

(b) The Wick sector, Form (B). *The inputs of Form (B) beyond Theorem 3.6, Proposition 3.9, and Lemma 3.10 and* (T3) *— Proposition 12.2 and Lemmas 2.7, 12.4, 12.9 and 12.10 and the tube statements Theorems 10.7 and 13.10 and Lemma 13.11, whose conditions on the Wick amplitudes are not conditions on the couplings — print no condition on the bare couplings beyond positivity ($\lambda > 0$, $\mu_0 > 0$); Proposition 3.9 is used at the two masses m and M_1. The only coupling condition of Form (B) is therefore* (T3)*, which Proposition 22.15 proves on explicit regions. Under* (CT_∞)*:*

(i) *$\sigma_1(M_1) = \lim_k s_k^\infty(M_1) = s_\infty(M_1)$ for every $M_1 > 0$. Hence* (T3) *holds if and only if* ($\mathrm{NE}^<$)*: it fails exactly on $E_W^<$. If Proposition 3.9 and Lemma 3.10 are available also at auxiliary masses $M_1 > m$, then by Proposition 12.11 the two-sided clause* (NE) *replaces* (T3)*, with failure set exactly E_W; the branch $M_1 > m$ is needed exactly on $E_W^< \setminus E_W$, for instance when $\alpha_\infty'(m) < 0$ and m minimises α_∞ on $(0,m]$.*

(ii) *To one-loop order, for every $M_1 > 0$,*

$$s_\infty(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + \frac{\lambda(N+2)(m-M_1)}{2\pi} - e^2\,\frac{(m-M_1)\,(m\mu_0 + mM_1 + M_1\mu_0)}{8\pi(\mu_0+M_1)(m+\mu_0)},$$

so that

$$\alpha_\infty'(m) = 1 + \frac{\lambda(N+2)}{2\pi m} - e^2\,\frac{m+2\mu_0}{8\pi(m+\mu_0)^2}, \qquad \frac{m+2\mu_0}{8\pi(m+\mu_0)^2} \leq \frac{1}{4\pi(m+\mu_0)}, \tag{58}$$

and to one-loop order (T3) *holds if and only if*

$$e^2 < e_c^2(\lambda,m,\mu_0) := \max\Big\{\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1+\frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1+\frac{\lambda(N+2)}{\pi m}\Big)\Big\}, \tag{59}$$

the first entry being the larger if and only if $\lambda(N+2)\mu_0 \leq \pi m^2$. So the one-loop failure set of (T3) *is the half-line $e^2 \geq e_c^2$. The first entry is the zero $e_*^2 \geq 4\pi(m+\mu_0)$*

of (58)*; at* $m = \mu_0 = 1$, $\lambda = 0$ *it equals* $32\pi/3 \approx 33.5$ *(for orientation only). Nothing finer than one-loop order is claimed about* $E_W^<$.

(iii) *For the two-sided set,* $E_W \subset \{\alpha'_\infty(m) = 0\}$*; for each* (λ, m, μ_0) *at which the polynomial* $e^2 \mapsto \alpha'_\infty(m)$*, of degree* ≤ 2*, is not identically zero, this is at most two values of* e^2*. At* $\mu_0 = m$ *the window* $0 < \lambda, e^2 < 0.1m$ *of the torus paper* [27] *lies outside* $E_W^<$*, by (d).*

(c) The existence and rotation parts. *The existence theorems Theorems 3.3 and 3.6 and the rotation theorem Theorem 3.14 are proved for* $e > 0$, $\lambda, m, \mu_0 > 0$ *in King's range, and none carries a condition on the bare couplings beyond positivity: every smallness condition of* [1, 2, 3, 4] *that they use (listed in the supplemental material) is a one-sided cap on a stopping scale* $\beta \le 1$*, imposed last in the schedule of constants, and an admissible* β *exists at every positive point.*

(d) Comparison with the torus paper. *The window* $0 < \lambda < 0.1m$, $0 < e^2 < 0.1m$, $\mu_0 \in (0, m]$ *of* [27] *is a hypothesis of its torus theorem only and is not inherited by King's state. Nor is* $\mu_0 \le m$*: the one King-side input printed under it, the torus paper's reflection-positivity proposition behind Proposition 12.2 (stated for* $\mu_0 \in (0, \mu_*]$, $\mu_* = m$*), is lifted to all* $\mu_0 > 0$ *by a proof that uses only site-independent coefficients,* $\lambda > 0$ *and even sides. At* $\mu_0 = m$ *the torus paper's certificate gives* $s_\infty(0.7m) \ge 0.1595\, m^2$ *on the whole window (identifying the two torus counterterm schemes at* $\mu_0 = m$*, which compute the same graphs), so* (T3) *holds there with* $M_1 = 0.7m$ *and an explicit margin; Proposition 22.15 sharpens this and gives explicit regions at every* $\mu_0 \in (0, m]$ *(Remark 22.3 of the supplemental material).*

(e) The quantifier. *For all* $e > 0$, $\lambda, m, \mu_0 > 0$ *and each datum there are an integer* J_{K} *(*[15, Thm. 3.1]*) and a stopping scale* $\beta \le 1$ *such that the running couplings* $(e\beta^{1/2}, \lambda\beta, m\beta, \mu_0\beta)$ *satisfy finitely many one-sided caps on* β *and on three of the four of them (*$\mu_0\beta$ *is uncapped in the sources); no condition is placed on the bare couplings, and the theorems consumed from* [15, 16] *carry no coupling hypothesis. A sentence "*e, λ *sufficiently small,* $m, \mu_0 > 0$ *fixed", read as a hypothesis on the bare couplings, is strictly stronger and is printed in none of the sources (Remark 22.16 of the supplemental material).*

Idea of the proof. (a) Theorem 2.1 of [16] asserts the existence of the cube limits with no hypothesis on the couplings, and its proof rests on [15, Thm. 3.1], stated "for some fixed integer J"; every printed smallness in [1, 2, 3, 4] is a cap on the running charge, a cap on the stopping scale, or the mass clause of [2, Prop. 3.1]; the other inputs of Form (B) carry no coupling condition beyond positivity, so the only coupling condition is the sign clause of Proposition 12.6. (b)(i) By Lemma 12.9 and (CT_∞), $\sigma_1(M_1) = \lim_k s_k^\infty(M_1) = s_\infty(M_1)$. (b)(ii) The closed form is the one-loop evaluation of King's counterterm [15, (3.32)–(3.33)], equal to $\frac{1}{2}\int_{M_1^2}^{m^2} \alpha'_\infty(\mu)\, d\mu^2$; the critical charge $e_c^2(M_1)$ is increasing, or decreasing then increasing, on $(0, m)$, so its supremum is the larger of the two end values, which gives (59). (b)(iii) A global minimum at the interior point m is a critical point. (c) The list of the smallness conditions and the schedule of constants are in the supplemental material. (d) The torus paper's window enters through its counterterm-gap input alone. (e) The quantifier is read off [15, Thm. 3.1]. The complete proof is in the supplemental material. *Two readings* (Remarks 22.2 and 22.3 of the supplemental material). Proposition 3.1 of [2] carries the sentence "we assume $m^2 \le O(1)$ also"; the mass that enters its printed proof is the unit-lattice mass at scale k, so the clause under which it is cited here is that its inequality (3.29) holds at every scale at which $(L^k\varepsilon)^2 m^2 \le c_m(d, a)$, one more one-sided cap on the stopping scale and no condition on the bare mass. The torus paper's two-loop certificate at $\mu_* = m$, identified with King's torus-convention coefficient at $\mu_0 = m$, gives $\sigma_1(0.7m) \ge 0.1595\, m^2 > 0$ on its window $0 < \lambda, e^2 < 0.1m$, so (T3) holds there with an explicit margin; away from $\mu_0 = m$ Proposition 22.15 takes over.

The explicit coupling region. The sign clause (T3) is proved below on an explicit region of the (λ, e^2)-plane at every photon mass $\mu_0 \in (0, m]$, with $N = 2$ and $M_1 = 0.7m$: the three low orders of King's counterterm contribute lattice integrals with exact limits (Lemmas 22.4 and 22.5, with Lemmas 22.7 to 22.9 of the supplemental material), and the two orders $e^2\lambda$ and

e^4 are bounded from below, uniformly in k, through the two-loop majorants of the torus paper's Appendix G [27], read at the photon mass μ_0 (Lemmas 22.11 to 22.14 of the supplemental material). The resulting lower bound σ_{low} on $\sigma_1(0.7m)$, (65), is explicit through Table 1 of the supplemental material.

King's counterterm order by order. With $\varepsilon = \varepsilon_k$, $\Lambda = \Lambda_k$, the Brillouin zone $\mathsf{B} = [-\pi/\varepsilon, \pi/\varepsilon)^3$, $\int_{\mathsf{B}} := \int_{\mathsf{B}} \frac{d^3p}{(2\pi)^3}$, $\hat{p}_\mu := \frac{2}{\varepsilon}\sin\frac{\varepsilon p_\mu}{2}$, the free propagators $G_M(p) = (\hat{p}^2 + M^2)^{-1}$, $\hat{D}(p) = (\hat{p}^2+\mu_0^2)^{-1}$ of King's action on Λ and their position-space kernels C_M, D ($C_m(0) = \mathsf{W}_\varepsilon$), King's free-boundary counterterm [15, (3.32)–(3.33)], [16, p. 330] at scalar mass M is the polynomial

$$\delta m^2_\varepsilon(M) = \sum_{(\alpha,\beta)} e^\alpha \lambda^\beta \, \delta m^2_{(\alpha\beta)}(\varepsilon; M), \qquad (\alpha,\beta) \in \{(0,1),(2,0),(0,2),(2,1),(4,0)\}, \tag{60}$$

over the superficially divergent orders $2 \le \alpha + 2\beta \le 4$, whose coefficients are fixed recursively by the condition that the order-$e^\alpha\lambda^\beta$ part of the amputated one-particle-irreducible two-point function of ϕ at zero external momentum, with the counterterm vertices of lower order inserted, be cancelled by the counterterm vertex of that order: $\delta m^2_{(\alpha\beta)}(\varepsilon; M) = \Sigma^{(\alpha\beta)}(0; M)$ in the convention in which the inverse propagator per component is $\hat{p}^2 + M^2 - \Sigma(p)$. Each coefficient is a finite real number, smooth in M.

Lemma 22.4 (structure)**.** *Put $S_{\alpha\beta,k} := \frac{1}{2}\big[\delta m^2_{(\alpha\beta)}(\varepsilon_k; m) - \delta m^2_{(\alpha\beta)}(\varepsilon_k; M_1)\big]$, numbers free of the couplings. Then*

$$s^\infty_k(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + \lambda S_{01,k} + e^2 S_{20,k} + \lambda^2 S_{02,k} + e^2\lambda S_{21,k} + e^4 S_{40,k}. \tag{61}$$

In $d = 3$, $S_{01,k}$ and $S_{20,k}$ have the dimension of a mass and $S_{02,k}$, $S_{21,k}$, $S_{40,k}$ are dimensionless.

Lemma 22.5 (the three low orders, exact)**.** *For every ε and $M > 0$:*

(i) $\delta m^2_{(01)}(\varepsilon; M) = -4(N+2)\, C_M(0)$;

(ii) $\delta m^2_{(20)}(\varepsilon; M) = \int_{\mathsf{B}} \hat{k}^2 \hat{D}(k) G_M(k)\, dk - 3D(0) = -2D(0) - M^2 \int_{\mathsf{B}} \hat{D} G_M$;

(iii) $\delta m^2_{(02)}(\varepsilon; M) = 32(N+2)\sum_{x\in\Lambda} \varepsilon^3 C_M(x)^3$.

Consequently, with $J_\varepsilon := \int_{\mathsf{B}} G_m G_{M_1}$, $I_\varepsilon := \int_{\mathsf{B}} \hat{k}^2 \hat{D}(k) G_m(k) G_{M_1}(k)\, dk$ and $F_\varepsilon := \sum_{x\in\Lambda} \varepsilon^3 [C_{M_1}(x)^3 - C_m(x)^3]$,

$$\begin{aligned} S_{01,k} &= 2(N+2)\big(C_{M_1}(0) - C_m(0)\big) = 2(N+2)(m^2 - M_1^2) J_{\varepsilon_k}, \\ S_{20,k} &= -\tfrac{1}{2}(m^2 - M_1^2) I_{\varepsilon_k}, \qquad S_{02,k} = -16(N+2) F_{\varepsilon_k}. \end{aligned} \tag{62}$$

Idea of the proof. Lemma 22.4 is (60) and the definition of s^∞_k. For Lemma 22.5, expand the hopping term of King's action into the free action and the vertices: (i) is the λ-tadpole with the $O(N)$ count $2(N+2)$; (ii) is the seagull $-3e^2 D(0)$ plus the photon-exchange graph; (iii) is the sunset, the tadpole-insertion graph being cancelled exactly by the order-λ counterterm on the loop line; (62) follows with the resolvent identity. *Signs, k-uniform bounds and limits* (Lemmas 22.6 to 22.9 of the supplemental material): from $\frac{4}{\pi^2}p^2 \le \hat{p}^2 \le p^2$ on B, the heat-kernel representation of C_M with a positive path expansion, the continuum integrals and dominated convergence, one gets for every k: $S_{01,k} > 0$ with $S_{01,k} \to \frac{(N+2)(m-M_1)}{2\pi}$; $S_{20,k} < 0$ with a μ_0- and k-free bound and $S_{20,k} \to -\frac{1}{2}(m^2 - M_1^2) I_\infty(\mu_0)$, $I_\infty(\mu_0) = \frac{m\mu_0 + mM_1 + M_1\mu_0}{4\pi(m+M_1)(\mu_0+M_1)(m+\mu_0)}$ strictly decreasing in μ_0 with $I_\infty \uparrow I^* = \frac{1}{4\pi(m+M_1)}$ as $\mu_0 \downarrow 0$; and $S_{02,k} < 0$ with a k-free bound and $S_{02,k} \to -\frac{(N+2)\ln(m/M_1)}{\pi^2}$ (a Plancherel argument and Frullani's integral), while each $\delta m^2_{(02)}(\varepsilon; M)$ alone diverges like $\ln(1/\varepsilon)$. Inserted into (61) these limits are the one-loop closed form of Lemma 22.1(b)(ii) and, at $e = 0$, the exact limit of Proposition 22.15(vi). At $N = 2$ the three coefficients are the terms $-16\lambda C_T(M)$, $-2e^2 D_T - e^2 J_T(M)$, $128\lambda^2 S_T(M)$ of the exact two-loop form of the torus coefficient in [27, Lemma G.1], read on Λ; the orders $e^2\lambda$ and e^4 are the terms $16e^2\lambda F_T(M)$ and $e^4 Q_T(M)$ there.

The two-loop orders at every photon mass. From here to the end of the section we use the units $m = 1$, $N = 2$ and $M_1 = 0.7$. The two-loop coefficients $S_{21,k}$, $S_{40,k}$ are bounded from below through the torus paper's Appendix G [27], written on periodic tori of side $\ell_T \geq \ell_0$ with the reference photon mass $\mu_* \in (0, 1]$ as a parameter, read here at $\mu_* = \mu_0$ (Lemmas 22.11 to 22.14 of the supplemental material): the torus coefficient $a_{K,T}(M)$, King's recursion (60) on the torus, which at $\mu_* = \mu_0$ is the bare coefficient of King's torus convention (Definition 12.1(ii)); the one-loop majorants $\mathcal{I}_{\alpha,\beta}(a, b; \ell)$ of [27, Lemma G.15], minima of finitely many closed forms in Γ and B functions, decreasing in ℓ and non-increasing in the masses; the slope bounds [27, Lemma G.2]; the two-loop gap [27, Prop. G.3], order by order; and the cell rule, an upper sum over a partition of $[0.7, 1]$ into $n = 400$ cells. On King's tori T_J with $2MJ \geq \ell_0$ the gap gives $2S^T_{20,k} \geq -\bar{j}_n$, $2S^T_{02,k} \geq -\bar{s}_n$, $2S^T_{21,k} \geq -\bar{B}_{3,n}$, $2S^T_{40,k} \geq -\bar{B}^-_{4,n}$ uniformly in J, and the volume limit $J \to \infty$ at fixed k (Lemma 12.9, order by order) carries closed inequalities to $S_{\alpha\beta,k}$; the torus floor ℓ_0 is a free parameter and $\ell_0 \to \infty$ removes it; every majorant is non-increasing in the photon mass, so an *interval upper sum* evaluated at $\mu_{\rm lo}$ bounds the coefficients for all $\mu_0 \in [\mu_{\rm lo}, \mu_{\rm hi}]$ at once; and replacing every μ-dependent form by a μ-free one gives the closed form $2S_{21,k} \geq -B^*_3$, $2S_{40,k} \geq -(P^* + Q^*/\mu_0)$ with $B^*_3 = 13.187549$, $P^* = 4.7629000$, $Q^* = 0.77120851$.

The numbers (Table 1 of the supplemental material). With $\hat{\mu} = \mu_0/m$, a grid of 35 points from 1 down to 0.0001 defines 34 rows $[\mu_{\rm lo}, \mu_{\rm hi}]$; the *row of* $\hat{\mu}$ is the row containing $\hat{\mu}$ (at a grid point the row with $\mu_{\rm lo} = \hat{\mu}$), $\bar{B}_3(\hat{\mu})$, $\bar{B}^-_4(\hat{\mu})$, $\bar{j}(\hat{\mu})$ denote the interval upper sums of that row, $\bar{s} := \bar{s}_{400} = 6.5231964$, and for $\hat{\mu} < 10^{-4}$ we read $\bar{B}_3(\hat{\mu}) := B^*_3$, $\bar{B}^-_4(\hat{\mu}) := P^* + Q^*/\hat{\mu}$, $\bar{j}(\hat{\mu}) := 0.092610085$. With $A := \frac{1}{2}(1 - M_1^2) = 0.255$, $b := \frac{4(1-M_1)}{2\pi} = 0.1909859$, $L := \frac{4\ln(1/M_1)}{\pi^2} = 0.1445549$, $I_\infty(\hat{\mu}) = \frac{0.7+1.7\hat{\mu}}{4\pi\cdot 1.7\,(\hat{\mu}+0.7)(1+\hat{\mu})}$ and $I^* = \frac{1}{4\pi\cdot 1.7} = 0.0468103$, put

$$\sigma_{\rm low}(\hat{\lambda}, \hat{e}^2; \hat{\mu}) := A\big(1 - \hat{e}^2 I_\infty(\mu_{\rm lo})\big) + b\hat{\lambda} - L\hat{\lambda}^2 - \tfrac{1}{2}\bar{B}_3(\hat{\mu})\,\hat{e}^2\hat{\lambda} - \tfrac{1}{2}\bar{B}^-_4(\hat{\mu})\,\hat{e}^4 \qquad (\hat{\mu} \in [10^{-4}, 1]), \tag{65}$$

$$\sigma^*(\hat{\lambda}, \hat{e}^2; \hat{\mu}) := A\big(1 - \hat{e}^2 I^*\big) + b\hat{\lambda} - L\hat{\lambda}^2 - \tfrac{1}{2}B^*_3\,\hat{e}^2\hat{\lambda} - \tfrac{1}{2}\big(P^* + Q^*/\hat{\mu}\big)\hat{e}^4 \qquad (\hat{\mu} \in (0, 1]), \tag{66}$$

and $\sigma_{\rm low}(\cdot; \hat{\mu}) := \sigma^*(\cdot; \hat{\mu})$ for $\hat{\mu} < 10^{-4}$. Table 1 of the supplemental material lists the three upper sums (rounded upward to eight significant digits) and the derived columns $e^2_{\max}$, the side c of the largest box $[0, c]^2$ on which $\sigma_{\rm low} \geq 0$, and $\sigma_{\rm low}(0.1, 0.1; \hat{\mu})$ for every row; its first row $[0.99, 1]$ reads $\bar{j} = 0.085313936$, $\bar{B}_3 = 8.5786210$, $\bar{B}^-_4 = 1.5854190$, $e^2_{\max} = 0.5618$, $c = 0.2390$, and its last row $[0.0001, 0.0002]$ reads $\bar{B}_3 = 13.186961$, $\bar{B}^-_4 = 7716.1679$, $e^2_{\max} = c = 0.0081$. The upper sums are finite sums of closed forms in Γ and B, evaluated with 50 significant digits and rounded upward, and the same evaluation at $\hat{\mu} = 1$, $\ell_0 = 96$ reproduces the four numbers of [27, Cor. G.4] to every printed digit.

Proposition 22.15 (the explicit coupling region of (T3) at every photon mass)**.** *Let $N = 2$, $M_1 = 0.7m$ and $\mu_0 \in (0, m]$; write $\hat{\lambda} = \lambda/m$, $\hat{e}^2 = e^2/m$, $\hat{\mu} = \mu_0/m$. Let $\delta m^2_k(M)$ be King's counterterm (60) on Λ_k at the photon mass μ_0, $S_{\alpha\beta,k}$ the coefficients of Lemma 22.4, $\sigma_1(M_1) = \liminf_k s^\infty_k(M_1)$ as in Lemma 3.10, and let $\sigma_{\rm low}$, σ^*, $\bar{j}(\hat{\mu})$, $\bar{s}$, $\bar{B}_3(\hat{\mu})$, $\bar{B}^-_4(\hat{\mu})$, A, b, L, I_∞, I^* be as defined before Table 1 of the supplemental material, B^*_3, P^*, Q^* as in Lemma 22.14 of the supplemental material, and $\lambda_c := \frac{b+\sqrt{b^2+4AL}}{2L} = 2.14398\ldots$.*

(i) (The five orders.) *For every k with $\varepsilon_k \leq 1/m$, in units $m = 1$: $S_{01,k} > 0$; $2S_{20,k} \geq -\bar{j}(\hat{\mu}) \geq -0.092611$; $2S_{02,k} \geq -\bar{s} = -6.5231964$; $2S_{21,k} \geq -\bar{B}_3(\hat{\mu})$ and $2S_{21,k} \geq -B^*_3$; $2S_{40,k} \geq -\bar{B}^-_4(\hat{\mu})$ and $2S_{40,k} \geq -(P^* + Q^*/\hat{\mu})$.*

(ii) (The lower bound.) $\sigma_1(0.7m) \geq m^2 \max\big\{\sigma_{\rm low}(\hat{\lambda}, \hat{e}^2; \hat{\mu}),\ \sigma^*(\hat{\lambda}, \hat{e}^2; \hat{\mu})\big\}$.

(iii) (The region.) (T3) *holds on $R(\mu_0) := \{(\lambda, e^2) : \sigma_{\rm low}(\hat{\lambda}, \hat{e}^2; \hat{\mu}) > 0\}$, with $\sigma_1(0.7m) \geq m^2\sigma_{\rm low} > 0$ there. $R(\mu_0)$ contains the box $[0, c(\hat{\mu})m]^2$ and the segment $\{\lambda = 0,\ e^2 < e^2_{\max}(\hat{\mu})m\}$, with c and $e^2_{\max}$ from Table 1 of the supplemental material for $\hat{\mu} \in [10^{-4}, 1]$, and the segment $\{e = 0,\ \lambda < \lambda_c m\}$ for every $\hat{\mu}$. The map $\hat{\mu} \mapsto \sigma_{\rm low}(\hat{\lambda}, \hat{e}^2; \hat{\mu})$ is non-decreasing (on $[10^{-4}, 1]$ a step function of $\hat{\mu}$, constant on each row of the table; below 10^{-4} the closed form σ^*), so $R(\mu_0) \supseteq R(\mu_0')$ for $\mu_0 \geq \mu_0'$. In particular $[0, 0.2m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.5m, m]$; $[0, 0.15m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.1m, m]$; $[0, 0.1m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in$*

$[0.03m, m]$, *with* $\sigma_1(0.7m) \geq 0.0582\, m^2$ *at* $(\lambda, e^2) = (0.1m, 0.1m)$; $[0, 0.075m]^2 \subseteq R(\mu_0)$ *for* $\mu_0 \in [0.01m, m]$; *and at* $\mu_0 = m$, $R(m)$ *contains the box* $[0, 0.239m]^2$ *and the segment* $\{\lambda = 0,\ e^2 < 0.5618\, m\}$, *and* $\sigma_1(0.7m) \geq 0.2209\, m^2$ *on* $[0, 0.1m]^2$.

(iv) (Uniform in the photon mass; closed form.) (T3) *holds whenever* $\sigma^*(\hat\lambda, \hat e^2; \hat\mu) > 0$, *with* $\sigma_1(0.7m) \geq m^2\sigma^*$; *in particular, for every* $\mu_0 \in (0, m]$, *on each of the boxes* $\{\lambda \leq \lambda_1 m,\ e^2 \leq c_1\sqrt{\mu_0 m}\}$ *with* $(\lambda_1, c_1) \in \{(0, 0.3014), (0.05, 0.2533), (0.1, 0.2152), (0.2, 0.1616), (0.5, 0.0884), (1, 0.0447)\}$.

(v) (Every k.) *For every* k *with* $\varepsilon_k \leq 1/m$: $s_k^\infty(0.7m) \geq m^2\big[A - \frac{1}{2}\bar\jmath(\hat\mu)\hat e^2 - \frac{1}{2}\bar s\hat\lambda^2 - \frac{1}{2}\bar B_3(\hat\mu)\hat e^2\hat\lambda - \frac{1}{2}\bar B_4^-(\hat\mu)\hat e^4\big]$.

(vi) ($e = 0$; exact.) *At* $e = 0$, *for every* $M_1 \in (0, m)$, *every* $\mu_0 > 0$ *and every* $N \geq 1$: $\lim_k s_k^\infty(M_1) = \frac{1}{2}(m^2 - M_1^2) + \frac{(N+2)\lambda(m-M_1)}{2\pi} - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ *exactly, and* (T3) *holds if and only if* $\lambda < \frac{\pi}{4}\big(1 + \sqrt{1 + \frac{16}{N+2}}\big)m$, *which is* $\frac{\pi}{4}(1+\sqrt5)\, m = 2.5416\ldots\, m$ *for* $N = 2$; *at* $M_1 = 0.7m$, $N = 2$, *the exact window is* $\lambda < \lambda_c m$, $\lambda_c = 2.14398\ldots$, *in particular* $\lambda \leq 2.1439\, m$.

(vii) (Two-sided.) *On each region of (iii), (iv) and (vi) the two-sided clause* (NE) *holds as well, with no use of Proposition 12.11.*

Outside these regions the sign clause (T3) is not decided here. The region is explicit at every photon mass but not uniform in it: by (66), $e^2_{\max}(\hat\mu) \sim \sqrt{2A/Q^*}\sqrt{\hat\mu} = 0.81\sqrt{\hat\mu}$ as $\hat\mu \to 0$, and $\bigcap_{\mu_0 \in (0,m]} R(\mu_0) = \{e = 0,\ \lambda < \lambda_c m\}$, since for $e > 0$ the term $-\frac{1}{2}(Q^*/\hat\mu)\hat e^4$ of σ^* dominates as $\hat\mu \to 0$. The $1/\hat\mu$ sits in the order-e^4 majorant alone: the low orders and B_3^* are uniform in the photon mass.

Idea of the proof. (i) is Lemma 22.7(i) and the four bounds of Lemmas 22.12 to 22.14 of the supplemental material, applied to the row of $\hat\mu$. (ii) By (61) and the limits of the first four coefficients, $\sigma_1(0.7) = A + b\hat\lambda - \hat e^2 A I_\infty(\hat\mu) - L\hat\lambda^2 + \liminf_k(\hat e^2\hat\lambda S_{21,k} + \hat e^4 S_{40,k})$, and (i) gives the two lower bounds, neither of which dominates the other on a whole row. (iii) σ_{low} is concave in each variable separately, so its minimum over a box is attained at a corner, which is how c and $e^2_{\max}$ are read off; the monotonicity in $\hat\mu$ holds because the three row values that enter with negative coefficients do not decrease down the table. (iv) At a corner $(\hat\lambda, c_1\sqrt{\hat\mu})$, $\sigma^* \geq \sigma^*(\hat\lambda, c_1; 1)$, and the listed c_1 were determined by bisection. (vi) At $e = 0$ the limit is $g_\lambda(M_1) = (m - M_1)[\frac{m+M_1}{2} + \frac{(N+2)\lambda}{2\pi}] - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ exactly, strictly concave in M_1 with $g_\lambda(m) = 0$, so (T3) holds iff $g'_\lambda(m) < 0$. (vii) By Theorem F.159(b) of the supplemental material, $\sigma_1(0.7m) > 0$ says $\alpha_\infty(0.7m) < \alpha_\infty(m)$, which is (NE$^<$). The complete proof is in the supplemental material.

The theorems of this paper print the sources' quantifier of Lemma 22.1(e); a sentence "e, λ sufficiently small, $m, \mu_0 > 0$ fixed", read as a hypothesis on the bare couplings, is strictly stronger than what the sources need and names no set exactly (Remark 22.16 of the supplemental material).

23. Open problems

Throughout, K denotes King's printed state of Theorem 21.1: the Neumann–Dirichlet cube limit on King's RG cube at block factor 5, fixed photon mass $\mu_0 > 0$, with an admissible scheme a in the Wick-square sector; S denotes a torus limit state of the authors' earlier paper [27] (block factor 2, torus counterterm scheme with reference photon mass m, $0 < \lambda, e^2 < 0.1m$, $\mu_0 \in (0, m]$). No identification between the two is proved. "Symmetric phase" is, for both, the expectation that the states lie above the surface predicted by one-loop perturbation theory; it is neither a hypothesis nor a conclusion of any theorem here, and Theorem 21.1 holds on all of King's coupling range (Lemma 22.1), the surface included. The problems below are extensions of the main theorem, not conditions on it.

23.1. **Relation to the torus states.** Three hypotheses bear on the relation of K to S: (FAM$_L$), King's cube construction exists on the block-factor-2 family in the torus counterterm scheme;

(KI_L), the fixed-torus continuum limit exists in that family; (CE), a convergent unit-scale cluster expansion with sources on the cubic tori, their tilted copies and their cuts along King's cubes, uniform in the lattice spacing. Under (FAM_L) and a sign or decay hypothesis on the difference of the two counterterms, every torus state dominates the King state at a matched bare mass on the observables $e^{\mathrm{i}F(f)+:\phi^2:(k)}$, $k \geq 0$, a one-sided inequality carrying no information about rotations. Under (CE) at a point of a massive window ($\lambda m^2 \geq K_0 e^6$, $K_0 \geq 1/(18\pi^2)$, small e^2, λ) every torus limit state equals the cube limit of the same family and scheme in every order of limits; with (FAM_L) or (KI_L) it is unique, exponentially clustering and ergodic; with King's cube limits on both families it equals K at an explicitly shifted bare mass. Whether (CE) holds anywhere at $e > 0$ is open. Two obstacles are structural: the torus paper computes its mass counterterm with the photon propagator frozen at m while King's free-boundary counterterm uses the model's own μ_0, a difference of order $e^2|m - \mu_0|$, not small (Remark 22.3 of the supplemental material); and King's construction needs block factor 5 for the rotation by $\arctan(3/4)$, while no lattice family admits both a dyadic structure and a rotation of infinite order. On the one-loop surface $\lambda m^2 = e^6/(72\pi^2)(1 + O(e^2/m))$ the two constructions are expected to select different phases (periodic tori the broken one, Dirichlet faces the symmetric one); this is a heuristic, not a theorem, and no statement of this paper uses the phase.

23.2. **Uniqueness and ergodicity of K.** Uniqueness of King's cube limit is part of the construction (monotonicity in the cube), but uniqueness of the infinite-volume state, in the sense of King's expected identity between cube limits and periodic-box limits, is not proved, and neither is ergodicity of K under translations: Theorem 21.1 gives clustering only on the ergodic components. The usual route, a cluster expansion with a mass gap, is (CE). This is distinct from the independence of a LIM state from its continuum subsequence, (UNIQ), which holds (Proposition 14.6, from Theorem 3.14 and from Theorem 3.3; Proposition 14.12, Corollary 4.4). Cubic symmetry of K is a theorem (Theorem 8.5): it needs only the uniqueness of the monotone cube limit and the reflection sandwich (Remark 8.6).

23.3. **Non-triviality.** For the torus states the truncated correlation of $F(g)^2$ with the Wick square is non-positive, and strictly negative at small couplings under the two hypotheses of [27, Thm. 12.9]: the one-loop statement (C) on the continuum coefficient and a remainder bound on it, which that paper's uniform estimates do not provide. The analogous statements for King's cube, and a non-vanishing two-point function, are left to later work. For K nothing is proved here beyond the sign inequalities that hold on every lattice. Whether K is non-Gaussian, or even not decoupled, is open.

23.4. **The limit of vanishing photon mass.** Every constant of the Wick leg of Theorem 21.1 depends on μ_0 and is not uniform in it (Remark 21.7); the absolute bounds of the curvature leg are. The limit $\mu_0 \downarrow 0$, taken along a frozen family (counterterm and scheme frozen at a reference mass, so that the members are monotone in μ on the cone by the correlation inequalities), is left to later work: in the curvature sector it needs the existence theorem for every member and nothing else; in the Wick sector it needs a cell bound uniform in the photon mass (the analogue, along the family, of the cell bound ($\mathrm{B}_{\mathrm{cell}}$) of Lemma 14.4, which at a fixed mass is a lemma; the clause "several models at once" of Lemma 14.7 is formulated for this purpose), together with the torus inputs at the reference mass and the transfer of Theorem 21.3 for the members on King's convention; the identification of the members with King's models at shifted scalar masses, masslessness proper (the vanishing of the photon mass in the limit theory) and the two-point function are further questions there. None of these enters the present paper.

23.5. **The optional extension: (W2) at arbitrary amplitudes, Form (A), and the Euclidean clause of the Wick sector.** Form (B) of Theorem 21.1 lives on the tube $\mathcal{D}_{\sigma_+,H}$: the only positive-source bound it uses is the one proved on the cone $\sup q \leq s_1 < \sigma_1(M_1)$ (Corollary 13.3), and the only existence statement is Theorem 3.6, whose positive scalar part is below the same threshold. Above the threshold, the existence of the fixed-cube limits and the uniform positive-source bound are the hypothesis (W2) below, which is open; the Euclidean clause of Form (A) and its King-convention form are the further hypotheses (R_W) and (FA^K).

Hypothesis 23.1 ((W2): the Wick sector at arbitrary amplitudes)**.** Let a be an admissible scheme. (a) For every finite family of real 2-forms and real functions, every $J \geq J_0(f,h)$ and *every* real coefficient vector (z,w), the limit $\lim_k Z^a_{k,J}(z,w)$ exists and lies in $(0,\infty)$. (b) For every $q \in C^\infty_c(\mathbb{R}^3)$ with $q \geq 0$,

$$\sup_{J \geq J_q} \lim_{k\to\infty} Z^a_{k,J}(0,q) < \infty,$$

where J_q is the least J with $\operatorname{supp} q$ at ℓ^∞-distance greater than 1 from $\mathbb{R}^3 \setminus C_J$.

Remark 23.2 (on (W2))*.* (i) Clause (a) is Theorem 3.6 with the threshold on the positive scalar part removed. On the cone of amplitudes $\sup q \leq s_1 < \sigma_1(M_1)$, clause (b) is proved in this paper in L^1 form with $\lim_k$ replaced by $\limsup_k$ (Corollary 13.3, on the comparison convention; on ND^K by Remark 4.8); with the limit itself it is proved on the comparison convention for $2\sup q < s_1$, under the convergence of Definition 3.2. Above the threshold both clauses are open; the exact obstruction is the free-energy bound $(\mathrm{B}^-)[u]$ at amplitude u above the threshold (Hypothesis 23.9 below): the ε-uniform stability of King's torus model with its bare coefficient shifted by $-2u$. (ii) No clause of Theorem 21.1 uses (W2). The Wick result of the main theorem is Form (B) on the tube $\mathcal{D}_{\sigma_+,H}$, where only the proved cone instance enters, through Theorem 13.10; (W2) is a hypothesis of Form (A) (Theorem 23.7) and of (R_W). In the rotation part, (W2) for a scheme a is the W-admissibility $(\mathrm{W})_a$ of Definition 16.3.

Hypothesis 23.3 ((R_W): rotation invariance of the Wick sector at arbitrary amplitudes)**.** For an admissible scheme a satisfying (W2), the full object $Z^a_\infty = \lim_J \lim_k Z^a_{k,J}$ of the comparison family (whose existence and entireness follow from (W2) as in Lemma 10.2) is invariant under every Euclidean motion for all complex coefficients, F transforming as a 2-form and W^a as a scalar.

Remark 23.4 (on (R_W))*.* (R_W) enters only in the Euclidean clause of Form (A) (Theorem 23.7). Form (B) has no Euclidean clause and does not assume it. The rotation part proves the curvature sector; its Wick-sector theorems (Theorems 18.9 and 18.11) give the $O(3)$ clause of (R_W) for a scheme a (its translations are Theorem 23.7(ii)) under the torus comparison with scalar sources and the closing hypothesis for that scheme (Definitions 17.2 and 18.5).

Hypothesis 23.5 ((FA^K): Form (A) on King's convention)**.** On King's convention ND^K, under (W2) read for ND^K: the limit functional $Z^{a,K}_\infty$ is entire in all coefficients; the lattice symmetries O_h about every point hold for the joint field at all amplitudes; reflection positivity holds at arbitrary complex coefficients; the measure exists off the tube; and the Euclidean clause of the Wick sector holds for the King-family object.

Remark 23.6 (on (FA^K))*.* (FA^K) is the King-convention counterpart of Form (A) as the comparison-convention results give it (Theorem 23.7, last paragraph). The main theorem makes no use of it: its Wick leg on King's convention is Form (B) on the tube (Theorem 21.3, Proposition 21.4).

Theorem 23.7 (Form (A): the Wick leg at arbitrary amplitudes)**.** *Let the couplings satisfy* (T3)*, and let a be an admissible scheme satisfying* (W2) *for the comparison convention* $\mathrm{ND}^{(a)}$*. Then for every finite family $(f_1,\dots,f_n;h_1,\dots,h_p)$ of real 2-forms and real scalars the full object $Z^a_\infty(\zeta) = \lim_J \lim_k Z^a_{k,J}(\sum_i z_i f_i, \sum_j w_j h_j)$ exists for every $\zeta = (z,w) \in \mathbb{C}^{n+p}$, locally uniformly, Z^a_∞ is entire, and $Z^a_\infty = G_\infty$ on $\mathcal{D}_{\sigma_+,H}$. Its Schwinger functions are those of Form (B), and:*

(i) OS0 *as in Form (B)(i): the tube bound on $\mathcal{D}_{\sigma_+,H}$ and the bounds on the $S_{n,p}$; no quantitative bound on Z^a_∞ off the tube is asserted.*

(ii) OS1*: $Z^a_\infty(\tau_b R\cdot\zeta) = Z^a_\infty(\zeta)$ for every $\zeta \in \mathbb{C}^{n+p}$, $b \in \mathbb{R}^3$, $R \in O_h$ (F a 2-form, W^a a scalar).*

(iii) OS2 *in matrix form about every coordinate plane for exponentials with arbitrary complex coefficients (Theorem 23.8), and in polynomial form (Proposition 10.10).*

(iv) OS3 *as in Form (B)(iv).*

(v),(vi) *As in Form (B), the characteristic functional $(u,v) \mapsto Z^a_\infty(\mathrm{i}u,\mathrm{i}v)$ being the restriction of the entire Z^a_∞ to the imaginary axes.*

Euclidean clause. *If in addition* (R_W) *holds for the comparison family, Z^a_∞ is invariant under every Euclidean motion, its mixed Schwinger functions are $\mathbb{R}^3 \rtimes O(3)$-covariant, and ν-almost every ergodic component of the measure of (v) is $\mathbb{R}^3 \rtimes O(3)$-invariant and reflection positive about every plane (no vanishing of odd moments is asserted for a component). This is the only Euclidean clause of the Wick sector.*

On King's convention. *The same statements for the K-family object $Z^{a,K}_\infty$ under* (W2) *read for* ND^K *are the content of* (FA^K) *(Hypothesis 23.5); nothing is derived for it from the comparison-family statements, since* (W2) *on* ND^K *does not give* (W2) *for the comparison family (Corollary 4.4(iv)).*

Indications. The complete proof is given in the supplemental material; in outline: existence at all complex coefficients, locally uniform convergence and entireness follow from (W2) by the argument of Lemma 10.2 (version (A_∞) of Lemma 10.3), and $Z^a_\infty = G_\infty$ on $\mathcal{D}_{\sigma_+,H}$ as the same J-limit of the same Z^a_J (Theorem 13.10). (i) is Form (B)(i). (ii): Form (B)(ii) is the identity on $\mathcal{D}_{\sigma_+,H}$, an open neighbourhood of the origin of $\mathbb{C}^{n+p}$, and both sides are entire. (iii) is Theorem 23.8 with Proposition 10.10; (iv) is Proposition 10.9 for Z^a_∞; (v),(vi): the measure of Form (B) is built from G_∞ on the imaginary axes, which lie in $\mathcal{D}_{\sigma_+,H}$. The Euclidean clause is (R_W) itself; the component clause follows through the ergodic decomposition of Form (B)(vi) as in the curvature sector (Theorem 11.1(c)). □

For a LIM state of a continuum subsequence (Theorem 14.9) the corresponding Form (A) requires both clauses of (W2): the existence of the limits at every real coefficient vector and the uniform bound at every amplitude.

Theorem 23.8 (reflection positivity at arbitrary complex coefficients)**.** *Let a be an admissible scheme satisfying* (W2)*, let $\nu \in \{1,2,3\}$, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real sources $f_{il} \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C^\infty_c(\mathbb{R}^3)$ supported in $\{x_\nu > c\}$, with arbitrary complex coefficients z_{il}, w_{il}. Then*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \, Z^a_\infty\big(\theta\zeta_i^* \sqcup \zeta_j\big) \;\geq\; 0 \qquad \text{for all } \gamma \in \mathbb{C}^N.$$

Refinement. *Let B_1, B'', the vertex sources q_v ($v \in \mathrm{vert}(B'')$) and q_ζ be those of Definition 10.1 for the plane $\{x_\nu = c\}$. (a) If $c \in D$, the proof uses* (W2)*(b) only at the finitely many sources q_v, and it suffices to assume it at the single source $(1+\eta)q_\zeta$ for any one $\eta > 0$ (Remark 10.4, one direction only); all other steps use* (W2)*(a) and the inequalities of Lemma 2.7, and the tube machinery is not needed. (b) If $c \notin D$, with $c' \in D$ and $b := (c'-c)e_\nu$: the finitely many instances q_v and $\tau_b q_v$, $v \in \mathrm{vert}(B'')$, together with the tube inputs* (I2), (I3) *of Theorem 10.7. Such a plane needs a non-lattice translation, which finitely many instances of* (W2)*(b) do not supply; it is carried by the translation invariance of the tube object (Proposition 10.5(b)).*

Proof. Let $c \in D$ and let $\mathcal{F}$, Φ_k, ζ^*_{ij}, $\mathcal{T}_1$, $T_{\mathcal{F}}$, U be as in Definition 10.1. By Lemma 10.2, under (W2)(a) and (W2)(b) at the q_v, $\{Z^a_J\}_{J\geq J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z^a_J \to Z^a_\infty$ locally uniformly on $\mathcal{T}_1$, Z^a_∞ is holomorphic there and $Z^a_J \leq Z^a_\infty$ on $U \cap \mathcal{T}_1$. By Lemma 10.3(A), $\limsup_k \sup_{\mathcal{T}_1} |\Phi_k| \leq C$ with C independent of $J \geq J_0$, and at fixed J the Φ_k are locally bounded on $\mathbb{C}^m$ ((W2)(a) only). Montel's theorem gives a subsequence along which $\Phi_k \to \Phi^J$ locally uniformly, Φ^J entire; the matrices $(\Phi_k(\zeta^*_{ij}))_{ij}$ are positive semidefinite for large k (Lemma 9.3(b)), hence so is their limit, and $\sup_{\mathcal{T}_1} |\Phi^J| \leq C$ uniformly in J. On U every lattice value is real and the scheme multipliers of the two states agree, so the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10 give $Z^a_{k,J} \leq \Phi_k \leq Z^a_{k,J+2n}$, hence $Z^a_J \leq \Phi^J \leq Z^a_{J+2n}$ on U. On $U \cap \mathcal{T}_1$, $Z^a_J \uparrow Z^a_\infty$ (Lemma 10.2), so $\Phi^J \to Z^a_\infty$ there; by the uniform bound on $\mathcal{T}_1$, Montel's theorem and the uniqueness set $U \cap \mathcal{T}_1$ (Definition 10.1), $\Phi^J \to Z^a_\infty$ locally uniformly on the connected $\mathcal{T}_1$, in particular at the ζ^*_{ij}, and positivity passes to the limit. The only instances of (W2)(b) used are the q_v of Lemmas 10.2 and 10.3; Remark 10.4 replaces them by $(1+\eta)q_\zeta$.

Let $c \notin D$, $c' \in D$, $b := (c'-c)e_\nu$. The $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$ and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, so the display at c follows from the display at c' for the translated families once Z^a_∞ is translation

invariant at the points in question. The enlarged family of the translated problem at c' is $\tau_b\mathcal{F}$, with the same coefficient space, matrix points, B_1, B'', and vertex sources $\tau_b q_v$; the argument above for $\tau_b\mathcal{F}$ gives the positivity of $(Z^a_\infty(\tau_b\cdot\zeta^*_{ij}))_{ij}$, and $\zeta\mapsto Z^a_\infty(\tau_b\cdot\zeta)$, $\zeta\mapsto Z^a_\infty(\zeta)$ are holomorphic on $\mathcal{T}_1$ by Lemma 10.2. On the non-empty open set $\mathcal{T}_1\cap T_{\mathcal{F}}$ both equal the tube limits $Z^a_{\infty,T}(\tau_b\cdot\zeta)$, $Z^a_{\infty,T}(\zeta)$, which agree by Proposition 10.5(b); by the identity theorem they agree on $\mathcal{T}_1$, in particular at the ζ^*_{ij}. □

Hypothesis 23.9 ((B$^-$)[u]: stability of the shifted torus model)**.** Let $u>0$. In the free convention and the scheme a,

$$\sup_{k\ge0}\ \limsup_{J\to\infty}\ \alpha^a_{k,T_J}(u)<\infty,\qquad \alpha^a_{k,T}(u)=|T|^{-1}\ln N^a_{k,T}(0,u\mathbf{1}),$$

the free-energy density of the torus functional at the constant scalar source u; that is, (I2) of Definition 12.5 at the amplitude $s_1=u$.

For u below the threshold s_* of Lemma 22.1, (B$^-$)[u] is Proposition 12.6 (or Proposition 12.11) at an auxiliary mass M_1 with $\sigma_1(M_1)>u$, so Form (B), whose constants are built from (I2) at one such amplitude, already contains every amplitude at which the positive-source bound is known. For $u\ge s_*$ the hypothesis is the exact obstruction. By Lemma 12.7, exact at fixed (k,T), $\alpha^z_{k,T}(u)$ is, up to explicit c-numbers, $|T|^{-1}$ times the difference of the free energies of King's torus model with bare scalar coefficient $\alpha^{\mathrm{tor}}_{k,T}(m)-2u$ and with $\alpha^{\mathrm{tor}}_{k,T}(m)$, both with the counterterms of the mass m: the second is Proposition 3.9 at m, the first the ultraviolet stability of the shifted model. For $u<s_*$ the shifted coefficient is that of King's model at a positive mass M_1; for $u\ge s_*$ it lies, asymptotically in k, below $\inf_{M>0}\alpha^\infty_k(M)$, and no positive scalar mass realises it. The printed proofs of [1, 2] expand about the Gaussian at the positive scalar mass m and use its mass term in the small-field bounds; no printed theorem covers a bare coefficient below every positive-mass value. Conversely (B$^-$)[u] is sufficient for (W2)(b): Proposition 13.2 at $s_1=u$, whose steps do not depend on the amplitude, gives $\ln N^a_{k,X}(0,q)\le(C_U/u)\|q\|_{1,k}$ for $0\le q\le u$ on every Neumann–Dirichlet box, hence (W2)(b) in L^1 form on $\sup q\le u$. It is not asserted that (W2) implies (B$^-$)[u]: the inequalities available bound a constant-source density from below by local sources, never from above, so (B$^-$)[u] is sufficient and not shown necessary. Also open: an L^1 bound for the negative part of the Wick source on cubes, as on the torus paper's tori (the tube bound (31) carries the weighted norm $|k_-|_w$).

23.6. **Rotation invariance of the Wick sector.** Theorem 21.1 gives invariance under $\mathbb{R}^3\rtimes O(3)$ for the curvature sector; for the Wick sector it gives invariance under $\mathbb{R}^3\rtimes O_h$ on the tube and no Euclidean clause. Euclidean invariance of the joint field is the hypothesis (R$_W$) of the optional extension (Hypothesis 23.3, Theorem 23.7), and on King's convention (FA^K) (Hypothesis 23.5); whether the heavy-shell comparison of the rotation part extends to the Wick square, on the tube or at arbitrary amplitudes, is open. Cubic symmetry is a theorem in every case. The two statements the rotation part would need for the Wick sector are open: the torus comparison with scalar sources (K)$_a$ (Definition 17.2) and the closing property (TIGHT)$_a$ (Definition 18.5) for a W-admissible scheme; given both, Theorems 18.9 and 18.11 give (R$_W$). In the curvature sector both are theorems (Theorems 17.1 and 18.4).

23.7. **The coupling condition.** The sign clause (T3) is proved on the explicit regions of Proposition 22.15 (at $\mu_0=m$ on a region containing $0\le\lambda,e^2\le0.1m$; at every $\mu_0\in(0,m]$ on explicit boxes; exactly for $\lambda<\frac{\pi}{4}(1+\sqrt5)\,m$ at $e=0$), and the Wick leg is stated there. Outside those regions the exact position of the failure set $E^<_W$ is not determined; the two-sided clause that m is not a global minimiser of α_∞, with the auxiliary mass on either side of m (Proposition 12.11, Lemma 22.1(b)), would enlarge the region. The cell bound (B$_{\mathrm{cell}}$), the scheme bound (SCH) and the independence of the subsequence (UNIQ) of the statements on continuum subsequences are proved (Lemmas 14.4 and 14.5 and Proposition 14.6, Proposition 14.12); a LIM state of a scheme that is not admissible, or a LIM state along a family of photon masses, would require the first two to be re-established for it. The mass clause of [2, Prop. 3.1], a cap on the dimensionless scalar mass at the stopping scales where that proposition is used, belongs to the statement of

the proposition, which is cited under it, and imposes nothing on the bare mass (Remark 22.2 of the supplemental material).

References

[1] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. I. A lower bound, Commun. Math. Phys. **85** (1982), 603–626. DOI 10.1007/BF01403506.

[2] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. II. An upper bound, Commun. Math. Phys. **86** (1982), 555–594. DOI 10.1007/BF01214890.

[3] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. III. Renormalization, Commun. Math. Phys. **88** (1983), 411–445. DOI 10.1007/BF01213217.

[4] T. Bałaban, Regularity and decay of lattice Green's functions, Commun. Math. Phys. **89** (1983), 571–597. DOI 10.1007/BF01214744.

[5] G. Benfatto, M. Cassandro, G. Gallavotti, F. Nicolò, E. Olivieri, E. Presutti, E. Scacciatelli, Some probabilistic techniques in field theory, Commun. Math. Phys. **59** (1978), 143–166. DOI 10.1007/BF01614247.

[6] T. Bałaban, J. Imbrie, A. Jaffe, Effective action and cluster properties of the abelian Higgs model, Commun. Math. Phys. **114** (1988), 257–315. DOI 10.1007/BF01225037.

[7] D. Brydges, J. Fröhlich, E. Seiler, On the construction of quantized gauge fields. I. General results, Ann. Physics **121** (1979), 227–284.

[8] P. Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley, New York, 1999.

[9] J. Fröhlich, R. Israel, E. H. Lieb, B. Simon, Phase transitions and reflection positivity. I. General theory and long range lattice models, Commun. Math. Phys. **62** (1978), 1–34.

[10] J. Glimm, A. Jaffe, *Quantum Physics. A Functional Integral Point of View*, 2nd ed., Springer, New York, 1987.

[11] L. Hörmander, *An Introduction to Complex Analysis in Several Variables*, 3rd ed., North-Holland, Amsterdam, 1990.

[12] S. Janson, *Gaussian Hilbert Spaces*, Cambridge Tracts in Mathematics **129**, Cambridge University Press, 1997.

[13] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics **156**, Springer, New York, 1995.

[14] C. G. Khatri, On certain inequalities for normal distributions and their applications to simultaneous confidence bounds, Ann. Math. Statist. **38** (1967), 1853–1867. DOI 10.1214/aoms/1177698618.

[15] C. King, The $U(1)$ Higgs model. I. The continuum limit, Commun. Math. Phys. **102** (1986), 649–677. DOI 10.1007/BF01221651.

[16] C. King, The $U(1)$ Higgs model. II. The infinite volume limit, Commun. Math. Phys. **103** (1986), 323–349. DOI 10.1007/BF01206942.

[17] G. F. Lawler, V. Limic, *Random Walk: A Modern Introduction*, Cambridge Studies in Advanced Mathematics **123**, Cambridge University Press, Cambridge, 2010. DOI 10.1017/CBO9780511750854.

[18] E. Lukacs, *Characteristic Functions*, 2nd ed., Griffin, London, 1970.

[19] K. Osterwalder, R. Schrader, Axioms for Euclidean Green's functions, Commun. Math. Phys. **31** (1973), 83–112.

[20] K. Osterwalder, R. Schrader, Axioms for Euclidean Green's functions. II, Commun. Math. Phys. **42** (1975), 281–305.

[21] T. Ransford, *Potential Theory in the Complex Plane*, London Mathematical Society Student Texts **28**, Cambridge University Press, Cambridge, 1995. DOI 10.1017/CBO9780511623776.

[22] T. Reisz, A power counting theorem for Feynman integrals on the lattice, Commun. Math. Phys. **116** (1988), 81–126. DOI 10.1007/BF01239027.

[23] T. Reisz, Renormalization of Feynman integrals on the lattice, Commun. Math. Phys. **117** (1988), 79–108. DOI 10.1007/BF01228412.

[24] M. Reed, B. Simon, *Methods of Modern Mathematical Physics. I. Functional Analysis*, revised and enlarged ed., Academic Press, New York, 1980.

[25] E. Seiler, *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, Lecture Notes in Physics **159**, Springer, Berlin, 1982.

[26] Z. Šidák, Rectangular confidence regions for the means of multivariate normal distributions, J. Amer. Statist. Assoc. **62** (1967), 626–633. DOI 10.1080/01621459.1967.10482935.

[27] The symmetric phase of the three-dimensional abelian Higgs model: torus limits and the Osterwalder–Schrader axioms with cubic symmetry, companion paper (2026).

[28] J. M. Combes, L. Thomas, Asymptotic behaviour of eigenfunctions for multiparticle Schrödinger operators, Commun. Math. Phys. **34** (1973), 251–270. DOI 10.1007/BF01646473.

[29] E. A. Carlen, S. Kusuoka, D. W. Stroock, Upper bounds for symmetric Markov transition functions, Ann. Inst. H. Poincaré Probab. Statist. **23** (1987), no. 2, suppl., 245–287.

[30] E. B. Davies, Large deviations for heat kernels on graphs, J. London Math. Soc. (2) **47** (1993), 65–72. DOI 10.1112/jlms/s2-47.1.65.